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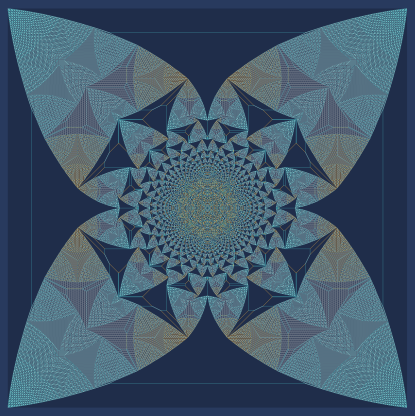
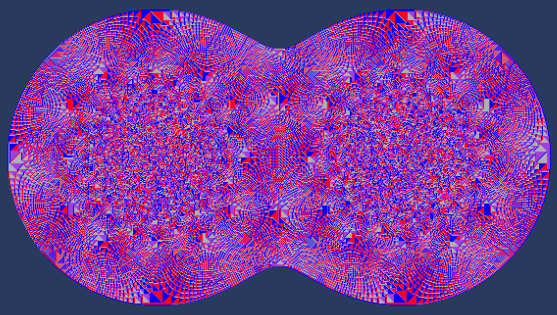
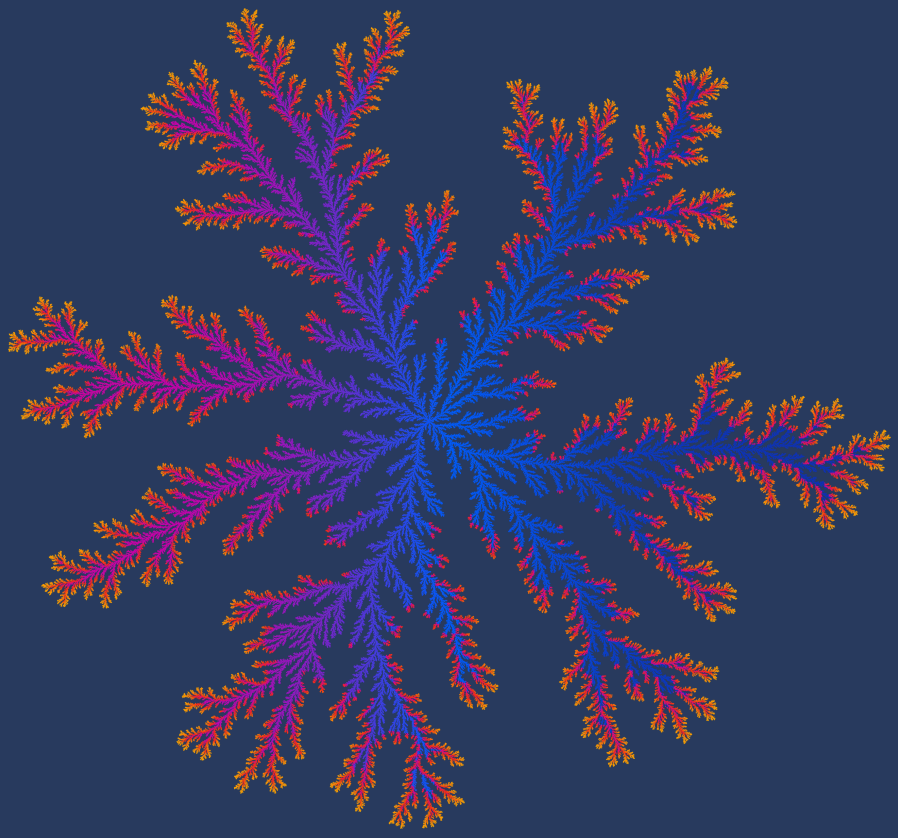
Discrete Laplacian Growth

Sandpiles, Rotors, and DLA







Ahmed Bou-Rabee, Lionel Levine, Yuval Peres

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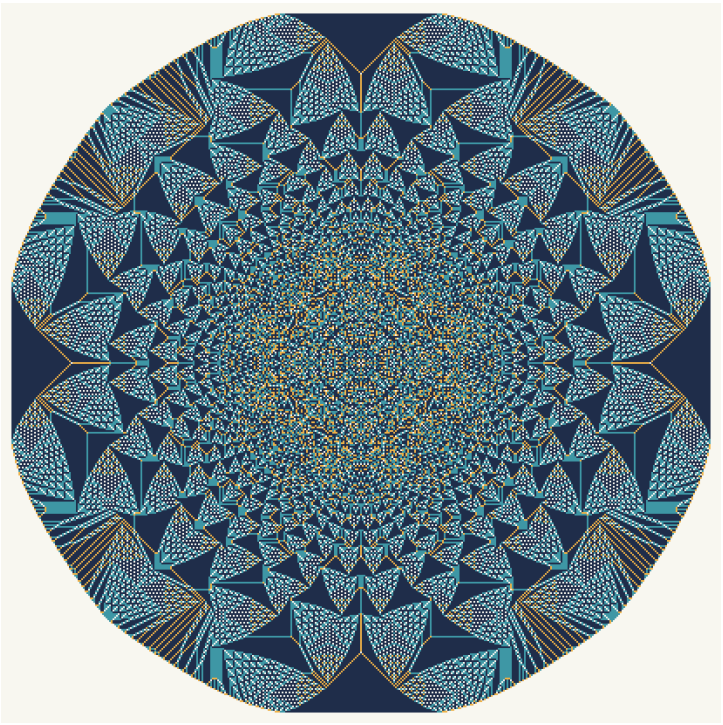


The abelian sandpile formed by stabilizing 3×10^5 grains at the origin of $\mathbb{Z}^2$. Each site is colored by the number of grains it retains: off-white for 0, orange for 1, teal for 2, and navy for 3.

Contents

Preface vii

1. Models of Laplacian Growth 1
- 1.1 The Abelian Sandpile 3
- 1.2 The Divisible Sandpile 4
- 1.3 Internal DLA 6
- 1.4 Rotor Walk and Rotor-Router Aggregation 7
- 1.5 External DLA 10
- 1.6 Universality 11

PART 1. POTENTIAL THEORY 17

2. Potential Theory in $\mathbb{R}^d$ 18
- 2.1 Harmonic Functions 18
- 2.2 Superharmonic Functions 21
- 2.3 Poisson Kernel: From Small Balls to All Balls 28
- 2.4 Harnack Inequality 38
- 2.5 Superharmonic Potentials 40
- 2.6 Harmonic Measure and the Beurling Estimate* 49

3. Potential Theory in $\mathbb{Z}^d$ 73
- 3.1 Discrete Harmonic Functions 73
- 3.2 Random Walk and Martingales 76
- 3.3 Green Function 79
- 3.4 Approximating Discrete Harmonic Functions 87
- 3.5 Discrete Harmonic Polynomials 96
- 3.6 Asymptotics of the Green Function* 105
- 3.7 Harmonic Measure from Infinity in the Plane 111

PART 2. LATTICE GROWTH 128

4. Divisible Sandpile in $\mathbb{Z}^d$: Single Source 129
- 4.1 The Divisible Sandpile and Least Action 130
- 4.2 Ball Asymptotics for a Point Source 137
- 4.3 Ball Bounds for the Abelian Sandpile 144

5. Internal DLA in $\mathbb{Z}^d$: Single Source 156
5.1 Instruction Stacks and Mass Balance 157
5.2 Inner Bound 159
5.3 Outer Bound 164

6. Rotor-Router Aggregation in $\mathbb{Z}^d$: Single Source 175
6.1 Bounded Stack Discrepancy 176
6.2 Inner Bound 178
6.3 Outer Bound 184

7. Internal Aggregation in $\mathbb{Z}^2$: Two Sources 191
7.1 The Shape Theorem 191
7.2 The Potential of the Limit Shape 194
7.3 The Divisible Sandpile 198
7.4 Rotor-Router Aggregation 201
7.5 Internal DLA 204
7.6 Separation and Coalescence 207

8. External DLA in $\mathbb{Z}^d$ 209
8.1 Radius from Attachment Probabilities 210
8.2 The Discrete Beurling Estimate 215
8.3 The Radius of the Cluster 222
8.4 External DLA in Higher Dimensions* 222

PART 3. SANDPILES AND ROTOR WALKS 241

9. Combinatorics of the Abelian Sandpile 242
9.1 Toppling and the Abelian Property 242
9.2 Stabilization Time with Sink 246
9.3 Stabilization Time Without Sink 249
9.4 Sandpile Group 252
9.5 Recurrent Sandpiles 256
9.6 Eulerian Graphs and the Burning Algorithm 261
9.7 Least Action Principle 264
9.8 Monotonicity of the Odometer on the Cubic Lattice 267
9.9 Growth Rates 270
9.10 Robust and Explosive Backgrounds 276
9.11 Bootstrap Percolation* 283
9.12 Explosive Backgrounds in Higher Dimensions* 285

10. The Divisible Sandpile 293
10.1 Discrete Quadrature 293
10.2 General Toppling Procedures 295
10.3 Random Walk Representation 303
10.4 Random Sandpiles on Bounded-Degree Graphs 308
10.5 Random Sandpiles on Transitive Graphs 315

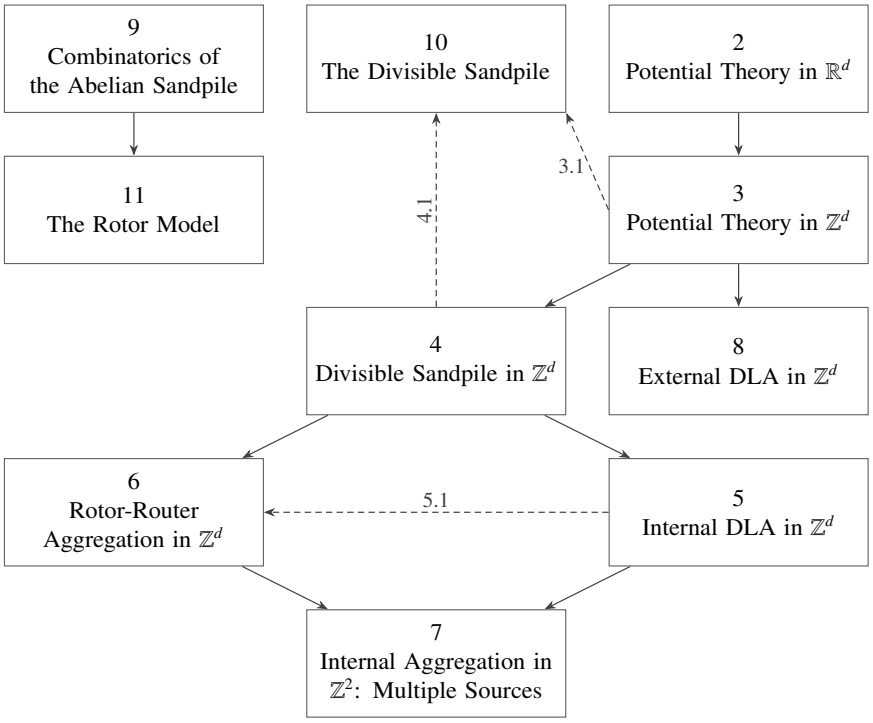


Figure 1: Prerequisites for the main text of Chapters 2–11: an arrow points from a chapter to a later one that uses it, and a prerequisite implied by a chain of solid arrows is not drawn. A dashed arrow lists the sections that suffice. Historical comparisons in the Notes are not prerequisites.

11. The Rotor Model 327
11.1 Rotor Walk and the Abelian Property 328
11.2 Relating Rotor Walk and Sandpiles 332
11.3 Recurrence of Rotor Walk 338
11.4 Circuits in Eulerian Digraphs 343
11.5 Range of Rotor Walk 348
11.6 Rotor Walk on Directed Lattices 351
11.7 Escape Rate of Rotor Walk 355
11.8 Live Paths and Recurrence 360
11.9 Recurrence of Uniform Rotor Walk 362
11.10 Uniform Rotor Walk on the Comb 365

A Preview of Volume II 381

Solutions to Exercises 385

Bibliography 395

Index 408

Preface

Inject a viscous fluid between two closely spaced glass plates, and the region it fills grows at each point of its boundary at a rate given by harmonic measure, the distribution of the point where a Brownian motion started at the injection point first reaches the boundary. Growth of this kind is called Laplacian growth, and its mathematics is potential theory. This book studies lattice models of Laplacian growth, in which particles move on $\mathbb{Z}^d$ by simple local rules, and asks what shapes they build.

Chapter 1 introduces the models of the book, the abelian and divisible sandpiles, internal DLA, rotor walk and rotor-router aggregation, and external DLA, and states their main properties in words. The rest of the book has three parts. The first develops potential theory from the mean value property, on $\mathbb{R}^d$ in Chapter 2 and on $\mathbb{Z}^d$ in Chapter 3, where the central objects are the Green function of simple random walk and, in the plane, the harmonic measure of a finite set from infinity.

The second part proves the shape theorems. Chapter 4 defines the divisible sandpile, proves its least action principle, shows by way of an obstacle problem that a point source fills a ball up to a bounded error in the radius, and deduces that an abelian sandpile from a point source topples a set between two Euclidean balls (Section 4.3). Chapters 5 and 6 compare internal DLA and rotor-router aggregation with the divisible sandpile. Chapter 7 starts the three models from two equal point sources in the plane and proves that they have the same limit shape: two disks or Neumann's oval, depending on the distance between the sources. Chapter 8 proves the bound of Kesten (1987b, p. L29) that a planar external DLA cluster of n particles has radius at most of order $n^{2/3}$, almost surely for all large n, from a discrete Beurling estimate and the harmonic measure from infinity of Chapter 3; a starred section proves the corresponding bounds of order $n^{2/(d+1)}$ in dimensions $d \geq 4$ and $n^{1/2}(\log n)^{1/4}$ in $\mathbb{Z}^3$, where the capacity of the cluster replaces the Beurling estimate.

The third part studies sandpiles and rotor walk on general graphs. Chapter 9 develops the combinatorics of the abelian sandpile, including the sandpile group and the least action principle. Chapter 10 characterizes the final divisible sandpile by quadrature inequalities, relates its odometer to optimal stopping for random walk, and asks when a random initial configuration stabilizes. Chapter 11 studies rotor walk, whose abelian property leads to a

proof of the matrix tree theorem, and asks for which initial rotors it keeps returning to its starting point. A preview of Volume II ends the volume.

We assume familiarity with measure-theoretic probability (conditional expectation, martingales and stopping times), linear algebra and multivariable calculus. No previous knowledge of potential theory or of the models is needed. A few passages use more, and each says what it uses:

- Section 9.4 uses quotients of abelian groups;
- the planar arguments of Subsection 2.2.1 and Section 7.2 use basic complex analysis, and Section 2.6 uses planar Brownian motion together with its strong Markov property and the facts that it almost surely hits every disk and never hits a given point;
- Exercises 2.49–2.53 use Brownian motion;
- the solution of Exercise 2.42 uses the identity theorem for real analytic functions;
- Subsection 8.2.3 uses recurrence and the potential kernel of planar random walks with bounded steps.

A star after the title of a section, or after the number of an exercise, marks material that can be skipped on a first reading. We write $\mathbb{N} := \{0, 1, 2, \ldots\}$.

After Chapter 1, the chapters can be read in order. Figure 1, before this preface, shows prerequisites for the main text. A reader who wants the shape theorems can read Chapters 2 and 3, omitting the harmonic measure from infinity in Section 3.7, and then Chapters 4 to 7. The Green function is radial to leading order (Theorem 3.9); the proof in Section 3.6 may be skipped. A reader interested mainly in external DLA can pass from Chapters 2 and 3, including Section 3.7, directly to Chapter 8, omitting Chapters 4 to 7. The general-graph chapters can be read afterward.

Each chapter ends with exercises and Notes. Some exercises supply arguments that the text points to, and others give examples and generalizations. The book solves the exercises it uses, and a separate document solves them all. The Notes give the sources of the results and suggest further reading.

The research of Yuval Peres was supported by National Natural Science Foundation of China grant RFIS-W2531011.

Chapter One

Models of Laplacian Growth

Pile n grains of sand on the origin of the square grid $\mathbb{Z}^2$. A site that holds four grains or more is unstable, and we let it topple: it sends one grain to each of its four neighbors (North, East, South and West). A toppling can make its neighbors unstable in turn, so we keep toppling until every site holds at most three grains. The frontispiece shows what this rule does with 3×10^5 grains. The rule treats the four directions alike, yet the pile is far from round, and it is covered with self-similar patterns that are still only partly understood.

Change the rule a little and the patterns disappear. If the sand is replaced by a liquid that a site can split as finely as it likes, or if each grain in turn wanders at random from the origin until it finds an empty site, then the occupied region is very nearly a disk (Figure 1.1). Why should a cluster grown on the square lattice be round? The answer, which takes up much of this book, is that harmonic functions describe these processes on large scales, and the harmonic functions of the lattice are nearly as symmetric as those of the plane.

Processes of this kind are lattice models of **Laplacian growth**, in which a region expands at each point of its boundary at a rate given by harmonic measure, the distribution of the point where a Brownian motion started at the source first reaches the boundary. In the continuum, Laplacian growth describes the Hele-Shaw flow of a viscous fluid injected between two plates, the growth of bacterial colonies, and the melting of ice in the quasistatic Stefan problem. Its mathematical home is potential theory, which grew out of gravitation and electrostatics. This chapter introduces the abelian sandpile, the divisible sandpile, internal DLA, rotor walk and rotor-router aggregation, and external DLA, one section each, and states their main properties in words, leaving the proofs to the later chapters. First, in the abelian and divisible sandpiles the outcome does not depend on the order in which the sites topple, and the actual evolution does the least work at every site among all ways of reaching a stable state (Sections 1.1 and 1.2); internal DLA has the same freedom in the order of its particles (Section 1.3). Second, in the scaling limit, internal DLA, rotor-router aggregation and the divisible sandpile, started from n particles at the origin of $\mathbb{Z}^d$ with $d \geq 2$, fill a Euclidean ball, and from several sources in the plane they still have one common shape; the abelian sandpile is the exception (Section 1.6, which explains why the clusters are

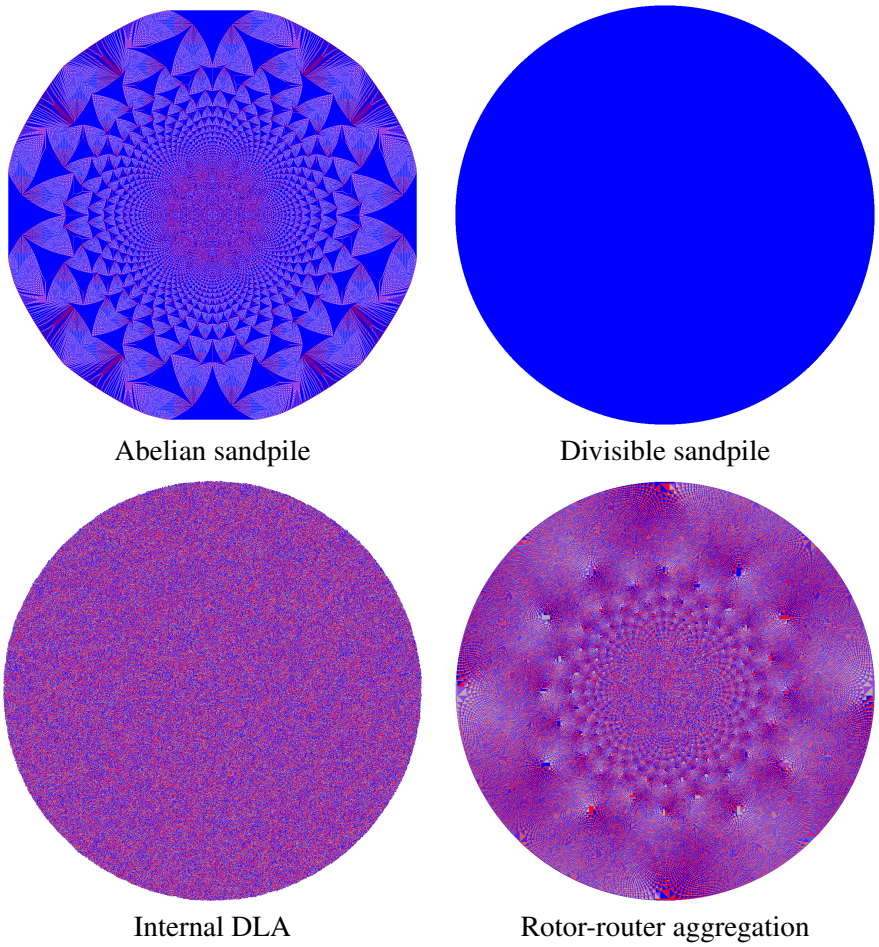


Figure 1.1: The abelian sandpile of 2.3×10^6 grains, the divisible sandpile of mass 10^6, and internal DLA and rotor-router aggregation of 10^6 particles each, with all rotors initially North, started at the origin of $\mathbb{Z}^2$ and drawn at the same scale; the abelian sandpile is colored by the number of grains at each site, and the two particle models by the direction of the last step taken from each site, or as North if there was none. Only the abelian sandpile is far from a disk.

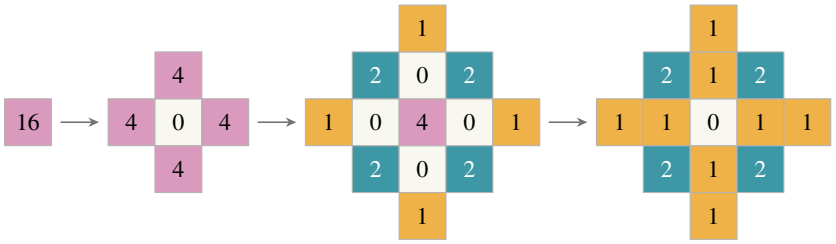


Figure 1.2: Sixteen grains at the origin of $\mathbb{Z}^2$ (left) reach the stable sandpile (right) in three rounds, in each of which every unstable site (pink) topples as many times as its grains allow. The numbers count the grains at each site.

balls). Third, in external DLA, where particles arrive from far away, the radius of a planar cluster of n particles is almost surely at most of order $n^{2/3}$ for all large n (Section 1.5). Part I of the book develops the potential theory behind these results, Part II proves the growth theorems, and Part III returns to sandpiles and rotor walk on general graphs.

1.1 THE ABELIAN SANDPILE

Our main setting is the lattice $\mathbb{Z}^d$, in which two sites are neighbors when they differ by one in exactly one coordinate. We write $x \sim y$ when x and y are neighbors, $\mathbf{0}$ for the origin and $|x|$ for the Euclidean norm. Every site has $2d$ neighbors. A site of the **abelian sandpile** on $\mathbb{Z}^d$ is **unstable** when it holds at least $2d$ grains, and then it **topples** by sending one grain to each of its neighbors; a site with fewer grains is **stable**. Physicists study the model as an example of self-organized criticality, and combinatorialists, who found it independently, call it **chip-firing**. Start with 16 grains at the origin of $\mathbb{Z}^2$ and topple unstable sites until none is left: the origin topples four times, each of its neighbors then topples once and returns a grain to the origin, the origin topples once more, and the grains come to rest on 12 sites (Figure 1.2).

Does it matter which unstable site we topple first? It does not: on a finite connected graph with a sink, the final sandpile is the same for every order of legal topplings, and so is the number of times each site topples, as Chapter 9 proves. This **abelian property** explains the name of the model. The number of topplings at x is the **odometer** $u(x)$. Write

$$\Delta^{::} f(x) := \sum_{y \sim x} (f(y) - f(x))$$

for the **discrete Laplacian** of a function f on $\mathbb{Z}^d$, and write σ for the initial sandpile and σ° for the stable sandpile that it reaches. A site that topples $u(x)$ times sends $u(x)$ grains to each neighbor and receives $u(y)$ grains from

each neighbor y, so

$$\sigma^\circ = \sigma + \Delta^{::} u \,.$$

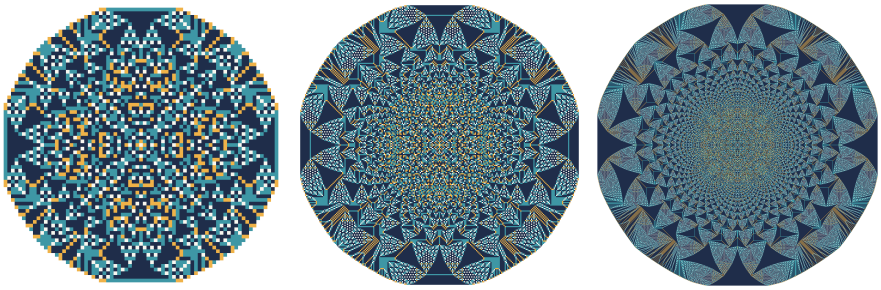
The odometer is also minimal. If we allow illegal topplings as well, of sites holding fewer than $2d$ grains, then every sequence of topplings that ends in a stable sandpile topples each site at least as many times as the sandpile itself does. This **least action principle** is stronger than the abelian property, and it is the tool we use again and again: it bounds a complicated evolution from above by every simple procedure that we can guess.

On a finite connected graph the sand needs somewhere to go, so one vertex is a **sink**, where arriving grains disappear. Adding grains one at a time and stabilizing after each addition leads from one stable sandpile to another. The sandpiles that can be reached in this way from every sandpile are called **recurrent**, and under addition followed by stabilization they form a finite abelian group, the **sandpile group** (which is the second reason for the name of the model). Its order is the determinant of the reduced Laplacian, the Laplacian matrix of the graph with the row and column of the sink deleted. The burning algorithm of Dhar (1990, p. 1615) recognizes a recurrent sandpile by a single further stabilization. Chapter 9 proves these facts, the last one on every finite strongly connected directed graph in which each vertex has as many edges entering as leaving. It also gives criteria for two kinds of infinite backgrounds on $\mathbb{Z}^d$: **robust** ones, which absorb every finite addition of grains, and **explosive** ones, on which a finite addition can make every site topple. On three grains at every site of $\mathbb{Z}^2$, a single added grain makes every site topple infinitely often (Section 9.10). Section 4.3 uses the divisible sandpile of the next section to show that, for $d \geq 2$, the sites toppled by n grains at the origin of $\mathbb{Z}^d$ lie between two Euclidean balls of volume of order n (Figure 4.7 draws them for $n = 10^5$).

These bounds do not determine the shape (Figure 1.3). Pegden and Smart (2013, Theorem 1.1 and Remark 4.3) proved that the sandpile of n grains at the origin of $\mathbb{Z}^d$, rescaled by $n^{1/d}$, converges in the sense of local averages as $n \to \infty$, and Volume II proves that the limit is not a ball (see the preview on page 381). What happens if the grains can be divided as finely as we like?

1.2 THE DIVISIBLE SANDPILE

Now replace the grains by a liquid. Pour $n > 1$ grams of maple syrup onto the middle square of a very large waffle, each square of which holds one gram before it overflows and shares the excess equally among its four neighbors. Which squares fill up? On $\mathbb{Z}^d$, with real amounts of mass, this process is the **divisible sandpile**: a site with mass $s > 1$ topples by keeping one unit and



$n = 10^4$ $n = 10^5$ $n = 10^6$

Figure 1.3: The stable sandpiles reached from n grains at the origin of $\mathbb{Z}^2$, scaled to the same width and colored by the number of grains as in Figure 1.2. The same large patches appear at each size, with finer patterns between them as n grows.

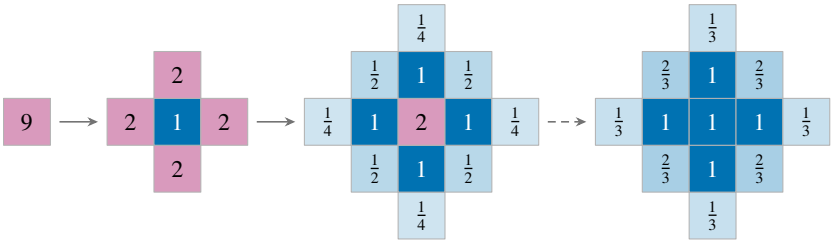


Figure 1.4: Mass 9 at the origin of $\mathbb{Z}^2$ (left), after one and after two rounds in which every site with mass greater than one (pink) topples, and the limit (right), with the full sites dark blue. The second round returns mass to the origin, and the dashed arrow stands for infinitely many further rounds.

giving $(s-1)/(2d)$ to each of its $2d$ neighbors. A site can receive mass again after it has toppled, so the process usually needs infinitely many topplings (Figure 1.4), but the masses converge, and in the limit every partially filled site is a neighbor of a fully occupied one.

The odometer $u(x)$ is now the total mass that x sends to each of its neighbors, and the bookkeeping of the abelian sandpile carries over: the final configuration is $\sigma + \Delta^{\cdot\cdot} u$. The least action principle singles out the odometer as the smallest nonnegative function f with $\sigma + \Delta^{\cdot\cdot} f \leq 1$. The odometer of the abelian sandpile is characterized in the same way among nonnegative integer-valued functions, with the threshold 1 replaced by $2d-1$, the largest number of grains that a stable site holds. So, after all masses are divided by $2d-1$, the divisible sandpile is the abelian sandpile without its integrality constraint, and it is much easier to analyze. How close to a ball are the fully occupied sites? Write ω_d for the volume of the unit ball in $\mathbb{R}^d$. If n units of mass fill a ball at density one, then the ball has radius $r = (n/\omega_d)^{1/d}$. For $d \geq 2$, the fully occupied sites contain every lattice point at distance less than $r - C$ from the origin and none at distance more than $r + C$, where C depends

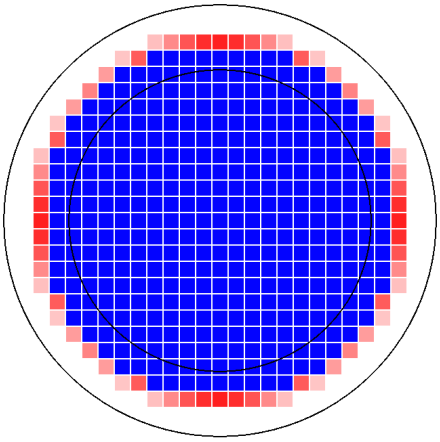


Figure 1.5: The divisible sandpile in $\mathbb{Z}^2$ started from mass 400 at the origin: a square is blue if its site is full, and red if it is partially filled, the deeper the red the larger the mass. Every red square lies between the circles of radii $r-2$ and $r+2$, where $\pi r^2 = 400$.

only on d (Figure 1.5). So the error in the radius stays bounded however much syrup we pour.

On an infinite graph a new question appears: does the redistribution stop at all? Start from mass one at every vertex of a locally finite connected graph, and add an extra mass at one vertex. This configuration stabilizes exactly when simple random walk on the graph is transient (Figure 10.2 in Chapter 10 shows the extra mass spreading like the walk), so on $\mathbb{Z}^2$ the extra mass cannot be absorbed, while on $\mathbb{Z}^3$ it can. Take independent, identically distributed, bounded masses on an infinite connected graph of bounded degree. If their mean is less than one, then they stabilize almost surely. If their mean is greater than one, or if their mean is one and the masses are not almost surely equal to one, then they almost surely fail to stabilize. Chapter 4 proves the least action principle (Section 4.1) and the ball, and Chapter 10 proves the results on infinite graphs. What if the liquid is replaced by particles that move at random?

1.3 INTERNAL DLA

Internal DLA replaces the liquid by particles that move at random. Let each of n particles in turn start at the origin of $\mathbb{Z}^d$, where $d \geq 2$, and perform simple random walk until it reaches a site that no earlier particle occupies, where it stays (Figure 1.6). This process is **internal diffusion-limited aggregation**, or **internal DLA**. The pictures suggest, and Lawler, Bramson, and Griffeath (1992, Theorem 1) proved, that the cluster is random but its shape on large

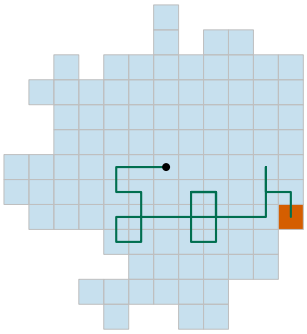


Figure 1.6: Internal DLA in $\mathbb{Z}^2$: the 101st particle walks from the origin (dot) through the cluster of the first 100 particles (light blue) until it reaches an empty site (vermilion), where it settles.

scales is not: after rescaling by $n^{1/d}$ it converges almost surely to a Euclidean ball. The cluster of 10^6 particles in Figure 1.1 (bottom left) contains every site at distance less than 560.7 from the origin and no site at distance more than 567.5, while the disk of area 10^6 has radius 564.2.

All of the randomness can be drawn in advance: put an infinite stack of random arrows at each site, each pointing to a neighbor, and let the kth particle to leave a site follow the kth arrow there. An abelian property for these stacks says that every complete sequence of legal moves produces the same cluster from the same stacks, so we may release the particles in any order, or pause some of them and restart them later, without changing the law of the cluster. With this freedom we show that, with $r = (n/\omega_d)^{1/d}$ as in Section 1.2, the cluster of n particles contains the ball of radius r minus an error of order $r^{1/3}$ and lies in the ball of radius r plus an error of order $r^{1-2/(3d)}$, both up to logarithmic factors, with high probability and also almost surely for all large n. In the plane the fluctuations are in fact at most logarithmic in the radius (Figure 1.9, left), as Jerison, Levine, and Sheffield (2012, Theorem 1) and Asselah and Gaudillière (2013b, Remark 1.2) proved. Chapter 5 proves the abelian property, and the inclusions with polynomial errors by comparison with the divisible sandpile. On $\mathbb{Z}$ the cluster is an interval, and Exercise 1.3 shows that it is off center by order $\sqrt{n}$ sites in root mean square. Can the randomness be taken out of the walks altogether?

1.4 ROTOR WALK AND ROTOR-ROUTER AGGREGATION

Place an arrow, called a **rotor**, at each site of $\mathbb{Z}^2$, pointing to one of its four neighbors. A walker at a site first turns the rotor there a quarter turn clockwise, and then steps to the neighbor that the rotor now points to. This is **rotor walk**, which has been introduced several times: as a model of self-

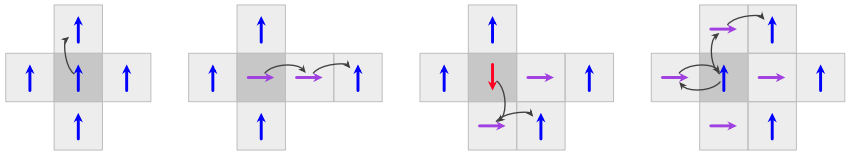


Figure 1.7: Rotor-router aggregation of 5, 6, 7 and 8 particles started at the origin (dark square) of $\mathbb{Z}^2$, with rotors turning clockwise and all initially North. Each arrow is the rotor at its site, blue for North, purple for East, red for South and gray for West, and the curved arrows trace the walk of the last particle.

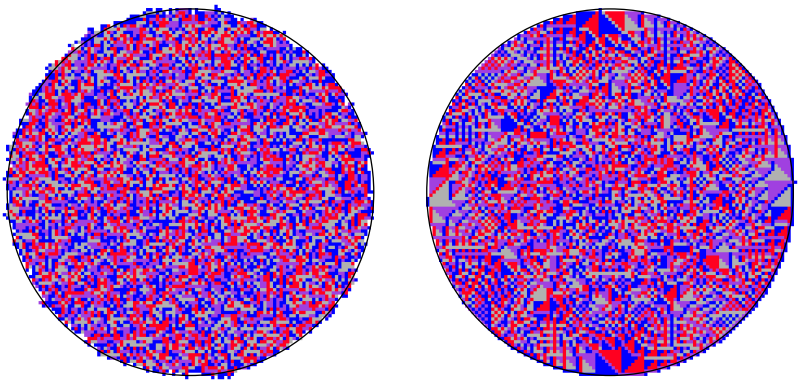


Figure 1.8: Internal DLA (left) and rotor-router aggregation (right) of 10^4 particles, started and colored as in Figure 1.1 (by the last step from each site: North blue, East purple, South red, West gray), with the circle of area 10^4. The internal DLA cluster contains every site within distance 54.5 of the origin and none beyond 58.9; for the rotor-router aggregate these radii are 55.7 and 57.3.

organized criticality, under the name Eulerian walkers; as a way for robots to explore an unknown network; and as a way to take the randomness out of random walk. A site that the walker has left m times has sent it to each neighbor either $\lfloor m/4 \rfloor$ or $\lceil m/4 \rceil$ times, which is what a random walk does on average. On $\mathbb{Z}^d$ the rotor at each site runs through the $2d$ neighbors in a fixed cyclic order.

In **rotor-router aggregation**, each of n particles in turn starts at the origin and performs rotor walk until it reaches a site that no earlier particle occupies, where it stays. The rotors are not reset between walks, so each site sends the particles of the whole process to its neighbors in strict rotation. For example, if all rotors initially point North, then the first particle stays at the origin, the second turns the rotor at the origin to East and settles at $(1, 0)$, and the third turns it to South and settles at $(0, -1)$. After five particles every rotor points North again, and the sixth particle turns the rotor at the origin to East, moves to $(1, 0)$, turns the rotor there to East as well, and settles at $(2, 0)$ (Figure 1.7).

Simulations show that these aggregates are extraordinarily close to circular (Figure 1.8), and it is natural to ask why. In the simulations of Figure 1.9,

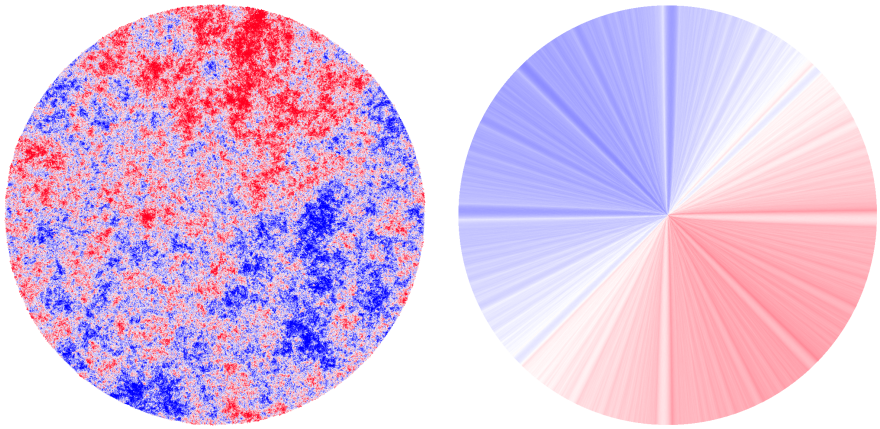


Figure 1.9: The internal DLA cluster (left) and the rotor-router aggregate (right) of 10^6 particles of Figure 1.1, with the site of the jth particle colored by its distance from the origin minus the radius of the disk of area j (white for 0, the deepest blue for -2 or less, the deepest red for 2 or more). The rotor-router panel is blue in its upper left half and red in its lower right half because that aggregate is very nearly a disk centered at $(1/2, -1/2)$ rather than at the origin.

with 10^6 particles and all rotors initially North, the distance of the jth particle from the origin minus the radius of the disk of area j ranges from -0.95 to 0.96 for rotor-router aggregation, and from -5.44 to 5.34 for internal DLA. Part of the answer is that the rotors imitate, site by site, the averaging of the divisible sandpile, with an error of less than one particle per edge. Chapter 6 makes this precise: for $d \geq 2$ and all sufficiently large $r = (n/\omega_d)^{1/d}$ (with ω_d as in Section 1.2), the aggregate of n particles contains the ball of radius $r - C \log r$ and lies in the ball of radius $r + C(\log r)^2$, where C depends only on d, whatever the initial rotors and the cyclic orders in which they turn.

What happens if a single rotor walk runs forever? Every time the walker returns to a site, the rotor it left there decides its next step, so the initial rotors can have a lasting effect. On $\mathbb{Z}^d$ in every dimension, a suitable choice of initial rotors makes rotor walk return to the origin infinitely often, while a chain of rotors pointing back toward the start along an infinite path sends it to infinity (Figure 11.6 in Chapter 11 shows both on $\mathbb{Z}^2$). Whatever the initial rotors, rotor walk on $\mathbb{Z}^d$ visits at least a constant times $t^{d/(d+1)}$ sites in its first t steps. The clockwise rotor walk on $\mathbb{Z}^2$ with independent uniform initial rotors is almost surely recurrent (Figure 11.15). Rotor walk also proves a classical theorem: on a finite connected graph with a sink, the sandpile group acts freely and transitively on the spanning trees oriented toward the sink, so the number of these trees is the order of the group, the determinant of the reduced Laplacian. This is the matrix tree theorem. Chapter 11 studies rotor walk on general graphs and proves all of these statements about rotor walk.

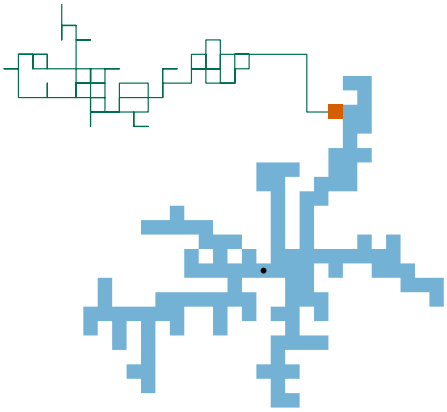


Figure 1.10: External DLA in $\mathbb{Z}^2$: a walk arriving from far away (green, last steps only) first reaches a site next to a cluster of 120 sites (light blue) at the vermilion site, near the end of an arm, which joins the cluster.

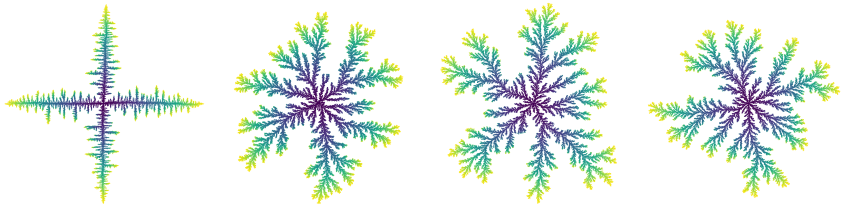


Figure 1.11: External DLA clusters of 10^9 sites on four planar lattices, each site colored from dark to light by the time at which it joined the cluster. On the square lattice the arms follow the two axes, and on the other three lattices they spread in more directions.

In all of the models so far the particles start inside the cluster; what if they arrive from outside?

1.5 EXTERNAL DLA

In external DLA the particles come from outside. Start with a single occupied site at the origin of $\mathbb{Z}^2$, release a particle very far away, and let it perform simple random walk until it first reaches a site next to the cluster, where it stays; then release the next particle (Figure 1.10). This process is **external diffusion-limited aggregation**, usually just **DLA**. It is a model of irreversible aggregation driven by diffusion, and its clusters grow long branching arms. An exposed tip is much easier for an incoming walk to reach than a site deep between two arms, and growth at a tip exposes it further, so the arms keep growing while the sites between them are screened. The lattice shapes

very large clusters: on $\mathbb{Z}^2$ their arms grow along the two axes (Figure 1.11; Figure 8.4 in Chapter 8 shows a cluster of 10^{10} sites). Of all the models in this book it is the one we understand least: off-lattice simulations suggest that a cluster of n particles has radius of order $n^{0.585}$, but it is not even known that the radius grows like a power of n.

The probability that a walk arriving from far away first hits a finite set K at a site x is the **harmonic measure from infinity** $\mathcal{H}_K(x)$, and the next site of a DLA cluster is chosen according to the harmonic measure of its outer boundary. To grow quickly, a cluster must put much of its harmonic measure on a few exposed tips. The **discrete Beurling estimate** limits how much a single site can carry: there is an absolute constant C such that, for every finite connected set $K \subset \mathbb{Z}^2$ of diameter $\mathrm{diam}(K)$ and every $x \in K$,

$$\mathcal{H}_K(x) \le \frac{C}{\sqrt{1+\mathrm{diam}(K)}}\,.$$

The exponent $\frac{1}{2}$ is sharp: the endpoint of a straight segment of n sites has harmonic measure of order $n^{-1/2}$ (Figure 3.16 in Chapter 3 shows harmonic measure on a segment of 11 sites). Kesten (1987b) deduced from it that the radius of the DLA cluster of n particles is at most a constant times $n^{2/3}$ for all large n, almost surely, and the bound that we prove has an exponential tail. This does not identify the radius exponent (simulated clusters grow more slowly), but it shows how an estimate valid for every connected set constrains an irregular random growth process. Chapter 8 proves the Beurling estimate and the bound on the radius. Why do the other models, unlike this one, produce balls?

1.6 UNIVERSALITY

The divisible sandpile, internal DLA and rotor-router aggregation move mass by three different rules, one by exact averaging, one at random and one by rotation, yet from one source each fills a Euclidean ball (Sections 1.2–1.4). The agreement goes further. Start n particles at each of two sites $(\pm a, 0)$ of $\mathbb{Z}^2$, where a is of order $\sqrt{n}$. Each source alone would build a disk of area about n. In the scaling limit, if the sources are far apart, then the two disks stay disjoint, and if they are close, then the disks merge and the merged shape is no longer round (Figure 1.12). The three models still have the same limit shape (Figure 1.13). Chapter 7 proves this for two equal sources, where the limit is either two disks or a domain bounded by a quartic curve, Neumann's oval, depending on the distance between the sources (Figure 7.2 in that chapter shows internal DLA from two sources with the quartic boundary of its limit). Levine and Peres (2010, Theorem 1.4) proved that the three

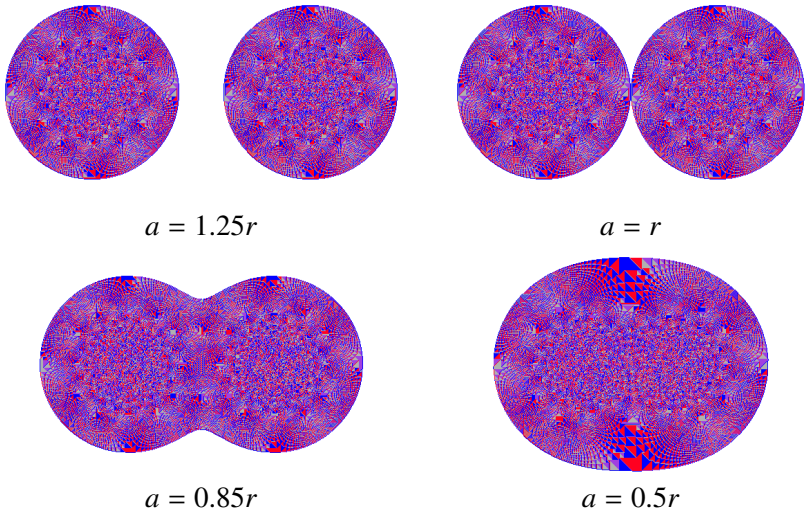


Figure 1.12: Rotor-router aggregation of $n = 5 \times 10^4$ particles from each of the sites $(\pm a, 0)$, with all rotors initially North, at one scale and colored as in Figure 1.1; here $r = \sqrt{n/\pi}$ is the radius of a disk of area n. The disks touch for $a = r$, and for $a < r$ the cluster lies within about one site of Neumann's oval, bounded by the quartic $(x^2 + y^2)^2 - 2r^2(x^2 + y^2) - 2a^2(x^2 - y^2) = 0$ in lattice coordinates.

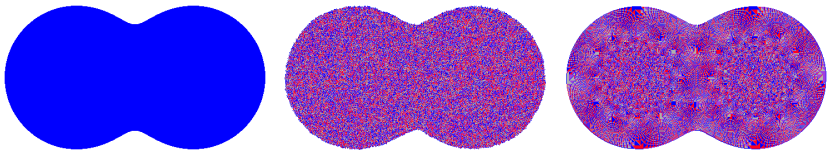


Figure 1.13: The divisible sandpile of mass 5×10^4, and internal DLA and rotor-router aggregation of 5×10^4 particles, at each of the sites $(\pm a, 0)$ with $a = 0.85r$ as in Figure 1.12, drawn at one scale and colored as in Figure 1.1. The three clusters have the same shape, Neumann's oval.

models have a common limit for every finite collection of point sources, and for distinct sources its boundary lies on an algebraic curve of degree twice the number of sources (Gustafsson and Shapiro, 2005, Section 4).

To see why the three models agree and why their clusters are balls, we start with harmonic functions. A function f on $\mathbb{Z}^d$ is **discrete harmonic** at x if $\Delta^{::} f(x) = 0$, that is, if $f(x)$ is the average of f over the $2d$ neighbors of x. Harmonic functions keep track of where the mass of a divisible sandpile goes. If h is harmonic at x, then a toppling at x leaves the total weight $\sum_y h(y)\sigma(y)$ of a mass σ unchanged, and if h is **discrete superharmonic** at x, that is, if $\Delta^{::} h(x) \le 0$, then a toppling at x cannot increase this weight. Hence, if σ is a finitely supported nonnegative initial mass with final configuration σ_∞, and

$h : \mathbb{Z}^d \to \mathbb{R}$ is discrete superharmonic at every site, then

$$\sum_{x\in\mathbb{Z}^d} h(x)\sigma_\infty(x) \le \sum_{x\in\mathbb{Z}^d} h(x)\sigma(x)\,, \tag{1.1}$$

with equality when h is harmonic at every site (Exercise 4.2 asks for the details). The continuum analogue is the mean value property: spreading a point mass $\omega_d r^d$, the volume of the ball of radius r, uniformly over this ball leaves its integral against a function harmonic in a neighborhood of the closed ball unchanged, and it cannot increase its integral against a function superharmonic there. The divisible sandpile performs the same spreading on the lattice, which already suggests that the region it fills should be nearly a ball. The random walks of internal DLA spread the particles in the same way on average, and the rotors do so up to the error described in Section 1.4. From two equal sources the limit shape is a quadrature domain: the integral over it of every function that is harmonic near its closure equals the sum of the values of the function at the two sources (Chapter 7).

The abelian sandpile is the exception (Figure 1.1). Its odometer is characterized by the same least action principle, but only among integer-valued functions (Section 1.2), and the integrality matters on large scales: the scaling limit of Section 1.1 is not a ball, as Volume II proves.

To prove that the three models fill balls we need functions with a prescribed Laplacian, such as the odometer, and these are built from the **Green function**, whose Laplacian vanishes everywhere except at the origin. On $\mathbb{Z}^d$ with $d \ge 3$ it is the expected number of visits to x of simple random walk started at the origin, divided by $2d$, and it decays like $|x|^{2-d}$; on $\mathbb{Z}^2$ its role is played by the potential kernel, which grows like $\frac{1}{2\pi}\log|x|$. These leading terms are rotationally symmetric, although the lattice is not, and the agreement is already close at small distances (Figure 3.5 in Chapter 3 compares level sets of the potential kernel with circles): at every site $x \ne \mathbf{0}$ of $\mathbb{Z}^2$ with $|x| \le 10$, the potential kernel differs from $\frac{1}{2\pi}\log|x| + \kappa$, where $\kappa \approx 0.257$ is the constant term in its asymptotics, by less than 0.008. This near symmetry is why the three models fill a Euclidean ball, and why with several sources they build the same quadrature domain. Chapters 2 and 3 develop potential theory in the continuum and on the lattice, from the mean value property to the Harnack inequality, the asymptotics of the Green functions, including the value of κ, and the Beurling estimate for planar Brownian motion, and the preface describes routes through the rest of the book.

EXERCISES

Exercise 1.1. In the setting of Section 1.1, let $d \geq 1$ be an integer, and let $\sigma : \mathbb{Z}^d \to \mathbb{N}$ be a sandpile with finitely many grains. Prove that every legal toppling sequence started from σ topples only finitely many sites.

Exercise 1.2. In the setting of Section 1.1, let $d \geq 1$ be an integer, let $\sigma : \mathbb{Z}^d \to \mathbb{N}$ be a sandpile that stabilizes, and let σ° be the stable sandpile that it reaches. Let $A \subset \mathbb{Z}^d$ be a finite set of sites that all topple. Prove that $\sum_{x \in A} \sigma^\circ(x)$ is at least the number of lattice edges with both endpoints in A.

Exercise 1.3. Consider internal DLA as in Section 1.3, but on $\mathbb{Z}$ and started from the origin. After $n \geq 1$ particles have settled it occupies an interval $A_n = \{-a_n, \ldots, b_n\}$, where $a_n + b_n = n - 1$.

(a) Show that, given A_n, the next particle settles at $-a_n - 1$ with probability $(b_n + 1)/(n + 1)$.

(b) Show that $\mathbb{E}[(a_n - b_n)^2] = (n + 1)/3$ for every $n \geq 2$.

Thus the cluster is off center by order $\sqrt{n}$ sites in root mean square, whereas the divisible sandpile on $\mathbb{Z}$ is symmetric about the origin, and rotor-router aggregation with all rotors initially pointing West stays within one site of symmetric (Exercise 6.4).

Exercise 1.4. Section 1.3 says that rescaled internal DLA clusters converge to a ball. Here is one precise way to say that rescaled lattice regions converge to an open set. Let $d \geq 1$ be an integer. For each real number $r \geq 1$, let $D_r \subset r^{-1}\mathbb{Z}^d$. For an open set $\Omega \subset \mathbb{R}^d$, say that $D_r \to \Omega$ if, for every $\epsilon > 0$ and every sufficiently large r, the set D_r contains every point $x \in r^{-1}\mathbb{Z}^d$ such that the closed ball $\overline{\mathrm{B}}(x, \epsilon)$ lies in Ω, and every point of D_r lies at distance less than ϵ from Ω. Let Ω_1 and Ω_2 be open subsets of $\mathbb{R}^d$. Show that if $D_r \to \Omega_1$ and $D_r \to \Omega_2$, then $\bar{\Omega}_1 = \bar{\Omega}_2$. The limit need not be unique: if D_r is the set of points of $r^{-1}\mathbb{Z}^2$ in the open unit disk, then D_r converges both to this disk and to the disk with the segment $[0, 1) \times \{0\}$ removed.

NOTES

Bak, Tang, and Wiesenfeld (1987, pp. 381–384) proposed the two-dimensional sandpile model as an example of self-organized criticality, and Dhar (1990, pp. 1613–1614) formulated it on arbitrary finite systems and proved that stabilization does not depend on the toppling order; Dhar (1990, p. 1615) introduced the burning test. The same toppling rule had appeared earlier in the probabilistic abacus of Engel (1975, 1976), who

used it to compute absorption probabilities of finite Markov chains exactly, and combinatorialists later studied it independently as the chip-firing games of Björner, Lovász, and Shor (1991) and Björner and Lovász (1992). Fey, Levine, and Peres (2010, Section 1) defined robust and explosive backgrounds and observed that the background of three grains at every site of $\mathbb{Z}^2$ is explosive; their Lemma 2.3 is the least action principle for sandpiles on $\mathbb{Z}^d$ that stabilize, and their Theorems 4.1 and 4.2 give criteria for explosion and for robustness. Pegden and Smart (2013, Theorem 1.1 and Remark 4.3) proved that the abelian sandpile formed by n grains at the origin of $\mathbb{Z}^d$, rescaled by $n^{1/d}$, converges in the sense of local averages as $n \to \infty$, and identified the limit as the Laplacian of the solution of an obstacle problem.

Scott Sheffield suggested the divisible sandpile to Levine and Peres in a conversation around 2006, and Levine and Peres (2009) introduced the model. For the waffle we thank Sara Romik; the waffle follows Section 2 of the survey of Levine and Peres (2017). The stabilization criterion for one extra mass on a graph with mass one at every vertex is Lemma 2.9 of Levine, Murugan, Peres, and Ugurcan (2016a). The phase transition for independent initial masses on graphs of bounded degree is a special case of Theorems 1.2–1.3 of Bou-Rabee, Peres, and Sava-Huss (2026), which assume less: below mean one, a finite pth moment of the positive part for at least one $p > 3$, and at mean one, either a positive finite variance or a nonconstant law that is symmetric about one. The ball inclusions for the divisible sandpile are Theorem 1.3 of Levine and Peres (2009).

Internal DLA appears on pp. 96–97 of Diaconis and Fulton (1991), and the stacks on pp. 97–98. The shape theorem for internal DLA from a point source is Theorem 1 of Lawler, Bramson, and Griffeath (1992, p. 2119). In the plane, Theorem 1 of Jerison, Levine, and Sheffield (2012) shows that, for every $\gamma > 0$, the probability that the fluctuations of internal DLA about the disk of radius r exceed a constant times $\log r$ is at most $r^{-\gamma}$, where the constant depends on γ; this bound also holds almost surely for all large n. By another method, Theorem 1.2 of Asselah and Gaudillière (2013a) bounds the inner error by a constant times $\log r$ and the outer error by a constant times $(\log r)^2$ in every dimension $d \geq 2$, and Remark 1.2 of Asselah and Gaudillière (2013b) gives a constant times $\log r$ on both sides in the plane; the Notes to Chapter 5 give the precise statements.

Priezzhev, Dhar, Dhar, and Krishnamurthy (1996) introduced rotor walk as Eulerian walkers in connection with self-organized criticality, and with one walker and fixed tie-breaking, ANT-WALK-1 in Section 2.1 of Wagner, Lindenbaum, and Bruckstein (1996), an algorithm for a robot exploring an unknown network, is the same walk. The Introduction of Holroyd, Levine, Mészáros, Peres, Propp, and Wilson (2008) records further independent appearances, including Propp's derandomization of random walk, and Levine

and Peres (2009, p. 2) record that Propp introduced rotor-router aggregation, observed from simulations that its aggregates are extraordinarily close to circular, and asked why. Theorem 1.1 of Levine and Peres (2009) gives the inner inclusion of the shape theorem for rotor-router aggregation (Theorem 6.6) and an outer error of order $r^{1-1/d} \log r$; the Notes to Chapter 6 describe the later improvements of the outer error. The lower bound $t^{d/(d+1)}$ on the range is the volume-growth form of Theorem 1.1 of Florescu, Levine, and Peres (2016), whose Theorem 1.2 gives the transience from an infinite chain of rotors directed toward the starting point (theorem numbers refer to its long version, arXiv:1408.5533). The recurrence of the clockwise rotor walk on $\mathbb{Z}^2$ with independent uniform initial rotors (Theorem 11.42) is the recurrence part of the square-lattice case of Theorem 1.1 of Bou-Rabee and Peres (2026). The action of the sandpile group on oriented spanning trees, and the resulting proof of the matrix tree theorem, follow Sections 3–4 of Holroyd, Levine, Mészáros, Peres, Propp, and Wilson (2008).

Diffusion-limited aggregation was introduced by Witten and Sander (1981, 1983). The discrete Beurling estimate is equivalent, by translation and comparison of radius with diameter, to the planar part of the theorem on page 167 of Kesten (1987a), and the growth bound for planar DLA is due to Kesten (1987b), who proved that, almost surely, the cluster of n particles lies in the ball of radius $Cn^{2/3}$ about the origin for all large n, for an absolute constant C. The Notes to Chapter 8 discuss the numerical estimates of the fractal dimension.

The multiple-source shape theorem for all three models, in every dimension $d \geq 2$, is Theorem 1.4 of Levine and Peres (2010); the Notes to Chapter 7 discuss the planar case, Neumann's oval, the algebraic boundary and the quadrature domains. The values of the potential kernel behind the bound 0.008 in Section 1.6 are determined by its exact values on the diagonal (Lemma 3.7), its value $\frac{1}{4}$ at the neighbors of the origin, its symmetries and its harmonicity away from the origin.

The simulations in the figures of this chapter and in Section 1.4 follow the rules in the text, with one exception: in the external DLA simulations, as is customary, each walk starts at a uniform point of a circle around the cluster, a walk far from the cluster jumps to a uniform point of a circle around it that does not meet the cluster, and a walk that wanders past a larger circle starts again on the first one. The first two shortcuts are exact for Brownian motion and approximate for random walk, and the third is an approximation for both.

For further reading, the survey of Levine and Peres (2017) treats the growth models of this book and their scaling limits, Holroyd, Levine, Mészáros, Peres, Propp, and Wilson (2008) develop the algebra of chip-firing and rotor-routing on directed graphs, and Lawler (1996) and Lawler and Limic (2010) develop the estimates for random walk that Chapters 3 and 8 use.

PART 1

Potential Theory

Chapter Two

Potential Theory in $\mathbb{R}^d$

Potential theory is the study of harmonic functions, the solutions of Laplace's equation $\Delta u = 0$. It grew out of the theory of gravitation and of electrostatics, where the potential of a mass or a charge distribution is harmonic away from the sources. Potential theory is the natural language for the growth models of this book, whose scaling limits are described by harmonic and superharmonic functions.

There are two equivalent ways to define a harmonic function. The analytic definition asks that its Laplacian vanish. The geometric one, which we take as our starting point, asks that its average over every small ball equal its value at the center. The geometric definition is the more useful of the two: it never mentions derivatives, yet from it alone we will deduce that a harmonic function is automatically infinitely differentiable (Theorem 2.3). It also gives at once the minimum principle: a harmonic function on a bounded domain that is continuous on its closure attains its minimum on the boundary (Lemma 2.12). The minimum principle lets boundary values determine a harmonic function throughout a region.

From the mean value property we develop the potential theory of $\mathbb{R}^d$: the regularity of harmonic functions, the minimum principle for harmonic and superharmonic functions, the Poisson integral formula that recovers a harmonic function from its boundary values (Section 2.3), the Harnack inequality that controls how much a positive harmonic function can vary (Section 2.4). We then construct the Newtonian potential of a mass distribution, which inverts the negative Laplacian (Section 2.5), and prove the Beurling estimate, which bounds the chance that planar Brownian motion started near the origin reaches a distant circle without meeting a connected set that joins the origin to that circle (Section 2.6). The next chapter develops discrete analogues on $\mathbb{Z}^d$, where the same results become tools for random walks and for the discrete growth models.

2.1 HARMONIC FUNCTIONS

The simplest definition of a harmonic function is a function u satisfying

$$\Delta u = 0\,,$$

where $\Delta = \sum_i \partial^2/\partial x_i^2$ is the Laplacian. The differential definition requires taking two derivatives. For this reason we will define harmonic functions via the **mean value property**: the average value of a harmonic function on a ball equals the function's value at the center. In Section 2.2 we prove the equivalence of the two definitions.

Definition 2.1 (Averaging Operator) Let u be a locally integrable function on an open set $\Omega \subset \mathbb{R}^d$. If $\mathrm{B}(x,r) \subset \Omega$ and $u \in L^1(\mathrm{B}(x,r))$, then the **averaging operator** is

$$\mathbf{A}_r u(x) := \fint_{\mathrm{B}(x,r)} u(y)\,\mathrm{d}y := \frac{1}{\omega_d r^d} \int_{\mathrm{B}(x,r)} u(y)\,\mathrm{d}y\,,$$

where ω_d is the volume of the unit ball in $\mathbb{R}^d$.

Remark. Since u is locally integrable, $\mathbf{A}_r u$ is continuous on Ω_r, the set of points x with $\overline{\mathrm{B}}(x,r) \subset \Omega$. Indeed, if $x_n \to x$ then

$$|\mathbf{A}_r u(x_n) - \mathbf{A}_r u(x)| \le \frac{1}{\omega_d r^d} \int |u(y)| \left|\mathbf{1}_{\mathrm{B}(x_n,r)} - \mathbf{1}_{\mathrm{B}(x,r)}\right| \mathrm{d}y\,,$$

and the right side tends to 0 as $n \to \infty$ by the dominated convergence theorem.

Definition 2.2 (Harmonic Function) Let Ω be an open set in $\mathbb{R}^d$. A function u defined on Ω is **harmonic** at a point $x \in \Omega$ if there exists an open neighborhood $U \subset \Omega$ of x such that $u \in L^1(U)$ and

$$u(y) = \mathbf{A}_r u(y) \qquad \text{for all } \mathrm{B}(y,r) \subset U\,.$$

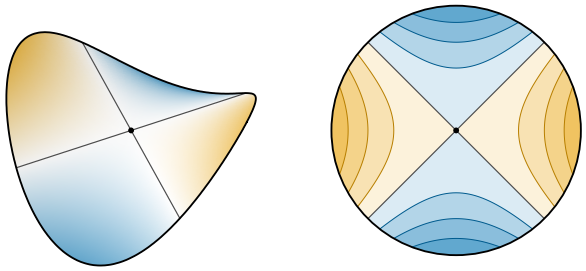
We say that u is harmonic on Ω if it is harmonic at x for all $x \in \Omega$.

For example, $x^2 - y^2$ is harmonic on $\mathbb{R}^2$: its Laplacian vanishes, and Section 2.2 shows that for twice differentiable functions this is equivalent to the mean value property; Figure 2.1 shows its graph over the unit disk.

The first thing we will prove about harmonic functions is that they are infinitely differentiable. We prove smoothness by convolving with a **mollifier**: a nonnegative, spherically symmetric, infinitely differentiable function vanishing outside the unit ball $\mathrm{B}(\mathbf{0},1)$ with integral 1. An example is

$$\eta(x) := \begin{cases} C_d e^{1/(|x|^2-1)}, & |x| < 1 \\ 0, & |x| \ge 1\,, \end{cases} \tag{2.1}$$

with the normalizing constant C_d chosen so that $\int_{\mathbb{R}^d} \eta\,\mathrm{d}x = 1$. A visualization is given in Figure 2.2.



(a) (b)

Figure 2.1: The harmonic function $u(x, y) = x^2 - y^2$ on the unit disk: its graph (a), and the regions where $u > c$ (orange) and $u < -c$ (blue) for $c = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$ (b), darker for larger c. The rotation by a right angle about the origin turns u into $-u$, so the average of u over every disk centered at the origin is $0 = u(\mathbf{0})$.

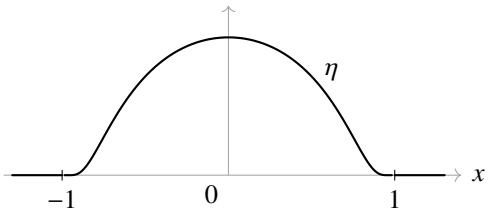


Figure 2.2: The mollifier η in (2.1) when $d = 1$. Observe that $\eta^{(k)}(\pm 1) = 0$ for all $k \geq 0$.

The **convolution** of two functions u and v on $\mathbb{R}^d$ is the function

$$(u * v)(x) := \int_{\mathbb{R}^d} u(y)v(x - y)\,\mathrm{d}y\,,$$

defined when the integrand is in $L^1(\mathbb{R}^d)$. The averaging operator (Definition 2.1) can be expressed as a convolution, namely

$$\mathbf{A}_r u = u * \beta_r\,,$$

where $\beta_r := \frac{1}{\omega_d r^d}\mathbf{1}_{\mathrm{B}(\mathbf{0},r)}$. The advantage of considering more general averages defined by convolution is that $u * v$ is at least as smooth as v. Indeed, let u be locally integrable and v be n times continuously differentiable with bounded support. For x in a fixed compact neighborhood the translated supports of v all lie in one compact set K, and the successive difference quotients in x are bounded by $C|u(y)|\mathbf{1}_K(y)$, which is integrable. By the dominated convergence theorem we may therefore bring the derivative inside the integral, for every differential operator $D = \partial^n/\partial x_{i_1} \cdots \partial x_{i_n}$, to obtain

$$D(u * v)(x) = \int_{\mathbb{R}^d} u(y)Dv(x - y)\,\mathrm{d}y = (u * Dv)(x)\,.$$

Now let $\epsilon > 0$, and let $v(y) = V(|y|)$ be a nonnegative smooth radial function with integral 1, where V is weakly decreasing and vanishes for $r \geq \epsilon$. The fundamental theorem of calculus gives

$$v(y) = \int_0^\epsilon (-V'(r))\mathbf{1}_{\mathrm{B}(\mathbf{0},r)}(y)\,\mathrm{d}r\,.$$

For locally integrable u, Fubini's theorem therefore gives

$$(u * v)(x) = \int_0^\epsilon (-V'(r))\omega_d r^d \mathbf{A}_r u(x)\,\mathrm{d}r\,. \tag{2.2}$$

Taking $u \equiv 1$ shows that the nonnegative weights in this integral have total mass 1. This mixture of ball averages proves smoothness.

THEOREM 2.3 *Let $\Omega \subset \mathbb{R}^d$ be open. If u is harmonic on Ω, then $u \in C^\infty(\Omega)$.*

Proof. Fix $x_0 \in \Omega$. By Definition 2.2, choose $\epsilon > 0$ so that $u \in L^1(U)$ for $U := \mathrm{B}(x_0, 2\epsilon)$ and $\mathbf{A}_r u(x) = u(x)$ whenever $\mathrm{B}(x,r) \subset U$. Let $v(y) := \epsilon^{-d}\eta(y/\epsilon)$, with η as in (2.1), and extend u by zero outside U. For $x \in \mathrm{B}(x_0, \epsilon)$, the convolution uses only values in U, so (2.2) and the normalization of its weights give $(u * v)(x) = u(x)$. Since convolution with v is smooth and x_0 was arbitrary, $u \in C^\infty(\Omega)$. □

Next we weaken equality to inequality.

2.2 SUPERHARMONIC FUNCTIONS

Superharmonic functions are defined by replacing the mean value equality $\mathbf{A}_r u(x) = u(x)$ by the inequality $\mathbf{A}_r u(x) \leq u(x)$. In the previous section we deduced from the mean value equality that harmonic functions are infinitely differentiable. Superharmonic functions need not be differentiable: the function $u(x) = -|x|$ on $\mathbb{R}$ is superharmonic but not differentiable. A superharmonic function can even be discontinuous (Example 2.41).

DEFINITION 2.4 (Lower Semicontinuity) Let $\Omega \subset \mathbb{R}^d$ be an open set, and u a function on Ω taking values in $\mathbb{R} \cup \{+\infty\}$. We say that u is **lower semicontinuous** at a point $x \in \Omega$ if for all sequences $x_n \to x$, we have $u(x) \leq \liminf u(x_n)$. We say that u is lower semicontinuous in Ω if it is lower semicontinuous at every point of Ω.

Exercise 2.1 gives an equivalent definition in the case that u takes values in $\mathbb{R}$.

For continuous functions, the preimage of every open set is open. For lower semicontinuous functions, only the preimages of open half-lines need

be open: the set $\{x : u(x) > a\}$ is open for every real a. The next lemma characterizes lower semicontinuity by the open-half-line condition; Exercise 2.2 asks you to prove it.

Lemma 2.5 *Let $\Omega \subset \mathbb{R}^d$ be open, and let $u : \Omega \to \mathbb{R} \cup \{+\infty\}$. For $a \in \mathbb{R}$, let*

$$L_a := \{x \in \Omega \mid u(x) > a\}.$$

Then u is lower semicontinuous in Ω if and only if L_a is open for every $a \in \mathbb{R}$.

Exercise 2.4 shows that a pointwise supremum of functions that are lower semicontinuous at a point is lower semicontinuous there, and Exercise 2.5 shows conversely that every lower semicontinuous function is a pointwise supremum of continuous functions.

We say that u is **upper semicontinuous** if $-u$ is lower semicontinuous. Note that if both u and $-u$ are lower semicontinuous, then u is continuous.

Definition 2.6 (Superharmonic Function) Let $\Omega \subset \mathbb{R}^d$ be an open set, and u a function on Ω taking values in $\mathbb{R} \cup \{+\infty\}$. We say that u is **superharmonic** at $x \in \Omega$ if there is a neighborhood W of x such that $u \in L^1(W)$ and

$$u(y) \geq \mathbf{A}_r u(y) \qquad \text{for all } \mathrm{B}(y, r) \subset W. \tag{2.3}$$

We say that u is **subharmonic** if $-u$ is superharmonic. Subharmonic functions take values in $\mathbb{R} \cup \{-\infty\}$. Note that if u is both superharmonic and subharmonic, then u takes values in $\mathbb{R}$ and is harmonic (Definition 2.2).

We do not include lower semicontinuity in the definition of superharmonicity, so $\mathbf{1}_{\{\mathbf{0}\}}$, which is nonnegative with vanishing ball averages, is superharmonic on $\mathbb{R}^d$ although it is not lower semicontinuous at $\mathbf{0}$. It attains its minimum without being constant, so the strong minimum principle (Lemma 2.8) needs lower semicontinuity as a separate hypothesis.

Example 2.7 The function $|x| = \left(\sum_{i=1}^d x_i^2\right)^{1/2}$ is subharmonic in $\mathbb{R}^d$. Indeed,

$$\mathbf{A}_r|x| = \fint_{\mathrm{B}(x,r)} |y| \, \mathrm{d}y \geq \left| \fint_{\mathrm{B}(x,r)} y \, \mathrm{d}y \right| = |x|,$$

where the inequality is the triangle inequality for integrals. Exercise 2.6 uses Jensen's inequality to show that $f \circ h$ is subharmonic for every convex $f : \mathbb{R} \to \mathbb{R}$ and harmonic h.

Note that (2.3) asserts the mean value inequality only for "small balls," namely balls contained in a neighborhood of x, centered at x or not. In Section 2.3 we will prove the stronger "all balls" mean value inequality: if u is superharmonic on Ω, then $\mathbf{A}_r u(y) \leq u(y)$ whenever $\overline{\mathrm{B}}(y, r) \subset \Omega$

(Lemma 2.23 for continuous u, and Corollary 2.27 in general). Along the way we prove that spherical and ball averages preserve superharmonicity. The advantage of the "small balls" definition is that it is local: it makes clear that if u is superharmonic on both Ω_1 and Ω_2, then u is superharmonic on $\Omega_1 \cup \Omega_2$.

If u is lower semicontinuous on $\Omega \subset \mathbb{R}^d$, then u is bounded below on each compact subset $K \subset \Omega$ (Exercise 2.3). For lower semicontinuous u, the integral $\mathbf{A}_r u$ is therefore defined in $\mathbb{R} \cup \{+\infty\}$ whenever $\overline{\mathrm{B}}(x, r) \subset \Omega$.

The most important property of superharmonic functions is the minimum principle. Next we prove several versions of this principle.

Lemma 2.8 (Strong Minimum Principle) *Let Ω be a connected open set in $\mathbb{R}^d$. If u is superharmonic and lower semicontinuous in Ω, and the infimum $\inf_{x\in\Omega} u(x)$ is attained, then u is constant in Ω.*

Proof. Let $m := \inf_{x\in\Omega} u(x)$, and let

$$\Omega_{\min} := \{x \in \Omega \mid u(x) = m\}.$$

We will argue that $\Omega_{\min}$ is both closed and open in the induced topology on Ω. Since Ω is connected, if the infimum is attained, so that $\Omega_{\min}$ is nonempty, then $\Omega_{\min} = \Omega$.

By Lemma 2.5, since u is lower semicontinuous in Ω, the set

$$\Omega^c_{\min} = \{x \in \Omega \mid u(x) > m\}$$

is open, so $\Omega_{\min}$ is closed.

To argue that $\Omega_{\min}$ is open, fix $x \in \Omega_{\min}$. By Definition 2.6 the mean value inequality (2.3) holds throughout a neighborhood of x; choose $\epsilon > 0$ with $\mathrm{B}(x, \epsilon)$ contained in that neighborhood. We claim $\mathrm{B}(x, \epsilon) \subset \Omega_{\min}$. Otherwise, there exists $y \in \mathrm{B}(x, \epsilon)$ with $u(y) > m$; choose $\delta > 0$ with $u(y) > m + \delta$. Since u is lower semicontinuous, $u > m + \delta$ on a neighborhood of y contained in $\mathrm{B}(x, \epsilon)$, as in Figure 2.3. The inequality $u \geq m$ throughout Ω then forces $\mathbf{A}_\epsilon u(x) > m = u(x)$, contradicting the mean value inequality at x for the ball $\mathrm{B}(x, \epsilon)$. □

A corollary of the strong minimum principle is that a lower semicontinuous superharmonic function has no strict local minima.

The following topological fact is useful when applying Lemma 2.8.

Lemma 2.9 *Let Y be a connected component of an open set $\Omega \subset \mathbb{R}^d$. Then $\partial Y \subset \partial\Omega$.*

Proof. Since $\overline{Y} \cap \Omega = Y$, we have $(\partial Y) \cap \Omega = \emptyset$, hence $\partial Y \subset \overline{\Omega} \setminus \Omega = \partial\Omega$. □

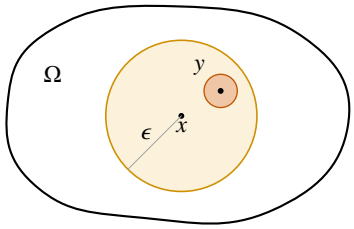


Figure 2.3: In the proof of Lemma 2.8, if $u(x) = m$ and some $y \in \mathrm{B}(x, \epsilon)$ had $u(y) > m$, then $u > m + \delta$ on a small ball about y (shaded) and $u \geq m$ on the rest of $\mathrm{B}(x, \epsilon)$, so $\mathbf{A}_\epsilon u(x) > m = u(x)$.

Lemma 2.10 (Unbounded Minimum Principle) *Let $\Omega \subset \mathbb{R}^d$ be open. If u is superharmonic and lower semicontinuous in Ω, then there is a sequence $(x_n)_{n\geq 0}$ in Ω that converges either to a point of $\partial\Omega$ or to ∞ and satisfies $u(x_n) \downarrow \inf_\Omega u$.*

Proof. Let $m = \inf_\Omega u$, and let $(y_n)_{n\geq 0}$ be a sequence in Ω such that $u(y_n) \downarrow m$. If y_n has a subsequence converging either to a point in $\partial\Omega$ or to ∞, then we may take x_n to be that subsequence. Otherwise, the sequence y_n has a limit point $y \in \Omega$. Lower semicontinuity gives $u(y) = m$. By Lemma 2.8, u is constant on the connected component Y of Ω containing y. By Lemma 2.9, $\partial Y \subset \partial\Omega$, so we may take x_n to be a sequence in Y converging either to a point in ∂Y or to ∞. □

Corollary 2.11 *Let $\Omega \subset \mathbb{R}^d$ be open, let u be superharmonic and lower semicontinuous in Ω, and let h be subharmonic and upper semicontinuous in Ω. If $\liminf (u(x_n) - h(x_n)) \geq 0$ for every sequence $(x_n)_{n\geq 0}$ in Ω converging either to a point in $\partial\Omega$ or to ∞, then $u \geq h$ in Ω.*

Proof. Apply Lemma 2.10 to the function $u - h$. □

Next we state the most commonly used version of the minimum principle.

Lemma 2.12 (Minimum Principle) *Let Ω be a bounded open set in $\mathbb{R}^d$ and let $u : \overline{\Omega} \to \mathbb{R} \cup \{+\infty\}$ be lower semicontinuous on $\overline{\Omega}$ and superharmonic in Ω. Then the following hold.*

(a) $\min_{\overline{\Omega}} u = \min_{\partial\Omega} u$.

(b) If h is harmonic in Ω, continuous in $\overline{\Omega}$, and $u \geq h$ on $\partial\Omega$, then $u \geq h$ in $\overline{\Omega}$.

Proof. Since $\overline{\Omega}$ is compact, the infimum $m := \inf_{x\in\overline{\Omega}} u(x)$ is attained (Exercise 2.3). If m is attained by a point $x \in \partial\Omega$, then the proof is complete. Otherwise, m is attained by a point $x \in \Omega$. By Lemma 2.8, we have $u \equiv m$ on the connected component X of Ω containing x. By lower semicontinuity,

$u \equiv m$ on $\overline{X}$. By Lemma 2.9 we have $\partial X \subset \partial\Omega$, which completes the proof of (a). To prove (b), apply (a) to $u - h$. □

A consequence of the minimum principle is that a harmonic function on a bounded domain is determined by its boundary values: if two functions u and h are harmonic on Ω, continuous on $\overline{\Omega}$, and $u = h$ on $\partial\Omega$, then $u = h$ on Ω.

The following gives two canonical examples of harmonic functions.

Lemma 2.13 *Let $b_i, M_{ij} \in \mathbb{R}$ for $1 \le i, j \le d$. Let*

$$L(x) := b^T x = \sum_{i=1}^{d} b_i x_i \qquad \text{and} \qquad Q(x) := x^T M x = \sum_{i=1}^{d}\sum_{j=1}^{d} M_{ij} x_i x_j \,.$$

Then the following hold.

(a) The function L is harmonic on $\mathbb{R}^d$.

(b) The function Q is harmonic on $\mathbb{R}^d$ if and only if $\sum_{i=1}^d M_{ii} = 0$.

(c) The function Q is superharmonic on $\mathbb{R}^d$ if and only if $\sum_{i=1}^d M_{ii} \le 0$.

Proof. Replacing M by $\frac{1}{2}(M + M^T)$ does not change Q, so we may assume that M is symmetric. Translation invariance and symmetry of the ball give

$$\mathbf{A}_r L(y) = L(y) \qquad \text{and} \qquad \mathbf{A}_r Q(y) = Q(y) + \frac{r^2}{d+2}\sum_{i=1}^{d} M_{ii} \,.$$

(a) The first identity shows that L has the mean value property on every ball.

(b) The second identity shows that Q has the mean value property exactly when $\sum_i M_{ii} = 0$.

(c) The second identity also shows that Q satisfies the mean value inequality exactly when $\sum_i M_{ii} \le 0$. □

We have defined harmonic functions by the mean value property in balls $\mathrm{B}(x, r)$. Next we will show that they also enjoy a mean value property on spheres $\partial\mathrm{B}(x, r)$.

The surface measure $S(A)$ of a Borel set $A \subset \partial\mathrm{B}(\mathbf{0}, r) \subset \mathbb{R}^d$ can be defined using the cone $\mathrm{Cone}(A) := \{Rx : R \ge 0 \text{ and } x \in A\}$. For $t > 0$, write

$$f_A(t) := \mathcal{L}(\mathrm{Cone}(A) \cap \mathrm{B}(\mathbf{0}, t)) \,,$$

where $\mathcal{L}$ denotes Lebesgue measure on $\mathbb{R}^d$. Define $S(A) := f_A'(r)$. Scaling gives $S(A) = dr^{d-1} f_A(1)$.

Definition 2.14 (Spherical Averaging Operator) Let u be a Borel function defined on an open set $\Omega \subset \mathbb{R}^d$. The **spherical averaging operator** is

$$\mathbf{a}_r u(x) := \fint_{\partial \mathrm{B}(x,r)} u(y)\,\mathrm{d}S(y) := \frac{1}{d\omega_d r^{d-1}} \int_{\partial \mathrm{B}(x,r)} u(y)\,\mathrm{d}S(y)\,,$$

whenever these integrals are defined.

The ball average $\mathbf{A}_R u$ can be expressed in terms of the spherical averages $\mathbf{a}_r u$ for $0 < r < R$ by integrating over $\mathrm{B}(x,R)$ in spherical shells:

$$\begin{aligned}\mathbf{A}_R u(x) &= \frac{1}{\omega_d R^d} \int_0^R \int_{\partial \mathrm{B}(x,r)} u(y)\,\mathrm{d}S(y)\,\mathrm{d}r \\ &= \frac{d}{R^d} \int_0^R r^{d-1} \mathbf{a}_r u(x)\,\mathrm{d}r\,. \end{aligned} \tag{2.4}$$

Now we prove that Definition 2.6 is equivalent to the differential characterization when u is twice continuously differentiable. Recall that

$$\Delta = \sum_{i=1}^d \frac{\partial^2}{\partial x_i^2}\,.$$

Lemma 2.15 *Let $\Omega \subset \mathbb{R}^d$ be open, and let $u \in C^2(\Omega)$. Then u is superharmonic in Ω if and only if $\Delta u \le 0$ in Ω.*

Proof. Let $\mathbf{n}$ denote the outward unit normal to a sphere. For $\overline{\mathrm{B}}(x,r) \subset \Omega$, differentiating the spherical average gives

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}r}\mathbf{a}_r u(x) &= \frac{\mathrm{d}}{\mathrm{d}r} \fint_{\partial \mathrm{B}(\mathbf{0},1)} u(x+ry)\,\mathrm{d}S(y) \\ &= \fint_{\partial \mathrm{B}(\mathbf{0},1)} \frac{\mathrm{d}}{\mathrm{d}r}[u(x+ry)]\,\mathrm{d}S(y) \\ &= \fint_{\partial \mathrm{B}(\mathbf{0},1)} (\nabla u)(x+ry)\cdot y\,\mathrm{d}S(y) \\ &= \fint_{\partial \mathrm{B}(x,r)} \frac{\partial u}{\partial \mathbf{n}}(z)\,\mathrm{d}S(z)\,, \end{aligned} \tag{2.5}$$

and by the divergence theorem the right side equals $\frac{1}{d\omega_d r^{d-1}} \int_{\mathrm{B}(x,r)} \Delta u(y)\,\mathrm{d}y$. Moreover, $\mathbf{a}_r u(x) \to u(x)$ as $r \downarrow 0$, since u is continuous.

Suppose first that u is superharmonic in Ω, and suppose for contradiction that there is $x \in \Omega$ with $\Delta u(x) > 0$. Since Δu is continuous, there is $s > 0$ such that $\overline{\mathrm{B}}(x,s) \subset \Omega$ and $\Delta u > 0$ on $\overline{\mathrm{B}}(x,s)$. Then $\mathbf{a}_r u(x)$ is strictly increasing for $0 < r \le s$, so $\mathbf{a}_r u(x) > u(x)$ for these r, and (2.4) gives $\mathbf{A}_r u(x) > u(x)$ for

$0 < r \le s$. This contradicts the mean value inequality (2.3) at x for small r. Hence $\Delta u \le 0$ in Ω.

Conversely, suppose that $\Delta u \le 0$ in Ω. Then $\mathbf{a}_r u(x)$ is weakly decreasing in r whenever $\overline{\mathrm{B}}(x,r) \subset \Omega$. In particular, $\mathbf{a}_r u(x) \le u(x)$, which implies $\mathbf{A}_r u(x) \le u(x)$ by (2.4). Thus u is superharmonic in Ω. □

An alternative proof that avoids the divergence theorem is described in Exercise 2.29.

The normal derivative in (2.5) comes from averaging over the sphere $\partial\mathrm{B}(x,r)$. Exercises 2.16–2.21 replace the ball average $\mathbf{A}_r u$ by an average against a general probability measure.

Combining Theorem 2.3 with Lemma 2.15, we see that u is harmonic on Ω if and only if $\Delta u = 0$ on Ω.

2.2.1 Interlude: Harmonic Functions in Two Dimensions

Identify $\mathbb{R}^2$ with the complex plane $\mathbb{C}$, and let Ω be a bounded connected open set in $\mathbb{C}$. A holomorphic function $\varphi = u + \mathbf{i}v : \Omega \to \mathbb{C}$ satisfies the **Cauchy–Riemann equations**

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \qquad \text{and} \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}\,.$$

Holomorphic functions are infinitely differentiable, so $u, v \in C^2(\Omega)$, and the Cauchy–Riemann equations give

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y} = -\frac{\partial^2 u}{\partial y^2}\,,$$

so $\Delta u = 0$ in Ω. A similar calculation shows that $\Delta v = 0$. Thus, the real and imaginary parts of a holomorphic function are harmonic.

The mean value property for u and v also follows from the Cauchy integral formula: if $\varphi : \Omega \to \mathbb{C}$ is holomorphic, then for every ball $\overline{\mathrm{B}}(a,r) \subset \Omega$

$$\varphi(a) = \frac{1}{2\pi\mathbf{i}} \int_{\partial\mathrm{B}(a,r)} \frac{\varphi(w)}{w-a}\,\mathrm{d}w = \frac{1}{2\pi}\int_0^{2\pi} \varphi(a + re^{\mathbf{i}\theta})\,\mathrm{d}\theta = \mathbf{a}_r\varphi(a)\,.$$

Taking real and imaginary parts we see that $\mathbf{a}_r u = u$ and $\mathbf{a}_r v = v$.

Conversely, if Ω is simply connected, then every harmonic function on Ω is the real part of a holomorphic function φ on Ω. Indeed, if $\Delta u = 0$, then the function

$$g := \frac{\partial u}{\partial x} - \mathbf{i}\frac{\partial u}{\partial y}$$

satisfies the Cauchy–Riemann equations. Now fix a base point $z_0 \in \Omega$ and define

$$\varphi(z) := u(z_0) + \int_{z_0}^{z} g(w)\,\mathrm{d}w\,,$$

by integrating along a path in Ω from z_0 to z. Since Ω is simply connected and g is holomorphic in Ω, the integral of $g\,\mathrm{d}z$ around every closed path is zero, which shows that φ is well-defined.

By using a path that ends with a horizontal or vertical segment, we deduce that $\nabla\varphi = (g, \mathbf{i}g)$. Hence, $\operatorname{Re}\nabla\varphi = (\operatorname{Re} g, -\operatorname{Im} g) = \nabla u$. Since $u - \operatorname{Re}\varphi$ vanishes at z_0 and its gradient is identically zero, we conclude that $u = \operatorname{Re}\varphi$.

The representation $u = \operatorname{Re}\varphi$ shows that harmonicity is preserved under holomorphic changes of variable.

Lemma 2.16 *Let $\Omega_1, \Omega_2 \subset \mathbb{C}$ be open. If u is harmonic on Ω_2 and $\psi : \Omega_1 \to \Omega_2$ is holomorphic, then $u \circ \psi$ is harmonic on Ω_1.*

Proof. Fix $z \in \Omega_1$. Choose a ball $B \subset \Omega_2$ about $\psi(z)$ and a neighborhood U of z such that $\psi(U) \subset B$. On B there is a holomorphic function φ with $u = \operatorname{Re}\varphi$. Hence $u \circ \psi = \operatorname{Re}(\varphi \circ \psi)$ is harmonic on U. Since z was arbitrary, $u \circ \psi$ is harmonic on Ω_1. □

Exercise 2.25 proves that composition with a nonconstant holomorphic map preserves superharmonicity as well.

In the next section we show that on a ball every continuous function on the boundary is the boundary value of a harmonic function, and we write this function as an explicit integral, the Poisson integral.

2.3 POISSON KERNEL: FROM SMALL BALLS TO ALL BALLS

Given a bounded open set $\Omega \subset \mathbb{R}^d$ and a continuous function F on $\partial\Omega$, the **Dirichlet problem** asks for a continuous function h in $\overline{\Omega}$ satisfying

$$\begin{aligned} \Delta h &= 0 && \text{in } \Omega \\ h &= F && \text{on } \partial\Omega\,. \end{aligned}$$

By the minimum principle, two solutions differ by the zero function, so the Dirichlet problem has at most one solution. A solution need not exist for every Ω. For example, on the punctured disk $\mathrm{B}(\mathbf{0}, 1) \setminus \{\mathbf{0}\}$ in $\mathbb{R}^2$, no solution takes the value 0 on the unit circle and 1 at the origin. Indeed, let h be such a solution, let $\epsilon > 0$, and choose $0 < \delta < 1$ with $\epsilon \log(1/\delta) \geq \max h$. The harmonic function $\epsilon \log(1/|x|)$ is then at least h on both boundary circles of the annulus $\{\delta < |x| < 1\}$ (Figure 2.4), so the minimum principle gives

$h(x) \le \epsilon \log(1/|x|)$ in the annulus. Letting $\delta \downarrow 0$ and then $\epsilon \downarrow 0$ gives $h \le 0$ off the origin, so $h(\mathbf{0}) \le 0$ by continuity, a contradiction. This section constructs a solution when Ω is a ball.

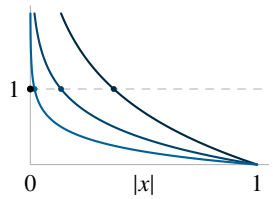


Figure 2.4: Radial profiles of the barriers $\epsilon \log(1/|x|)$ for $\epsilon = 1, \frac{1}{2}, \frac{1}{4}$ (darker for smaller ϵ): each exceeds the level $\max h$, drawn as 1 (dashed), inside a circle of radius δ (dots), and they tend to 0 at every $x \neq \mathbf{0}$ as $\epsilon \downarrow 0$.

Consider first the case when F is a polynomial, and Ω is the unit ball. Write V_m for the vector space of polynomials in $\mathbb{R}[x_1, \dots, x_d]$ of total degree at most m, with $V_m = \{0\}$ when $m < 0$. Then Δ maps V_m to V_{m-2}, and multiplication by $|x|^2 = \sum x_i^2$ maps V_{m-2} to V_m.

For polynomial boundary data, we will show that the harmonic extension is also a polynomial. For example, when $F(x, y) = 2x^2$ on the unit circle, the harmonic extension of F to the unit disk is $h(x, y) = x^2 - y^2 + 1$.

Lemma 2.17 *Let $F \in \mathbb{R}[x_1, \dots, x_d]$ be a polynomial of total degree m. Then there are polynomials g of total degree at most $m - 2$ and h of total degree at most m such that $\Delta h = 0$ on $\mathbb{R}^d$ and*

$$F = (1 - |x|^2)g + h\,.$$

Proof. Let $B \coloneqq \mathrm{B}(\mathbf{0}, 1)$. Note that if $g(x)$ is a polynomial such that $(1 - |x|^2)g$ is harmonic in B, then $(1 - |x|^2)g$ is identically zero in B by the minimum principle, so g is the zero polynomial. Thus $g \mapsto \Delta((1 - |x|^2)g)$ defines an injective linear map from V_{m-2} to itself. Since V_{m-2} is finite-dimensional, this map is also surjective. In particular, there exists $g \in V_{m-2}$ such that $\Delta((1 - |x|^2)g) = \Delta F$. Then $h \coloneqq F - (1 - |x|^2)g$ satisfies $\Delta h = 0$. □

Since $1 - |x|^2$ vanishes on the unit sphere, the harmonic polynomial h of Lemma 2.17 agrees with F on the unit sphere. Exercise 2.28 asks for the corresponding statement with the unit sphere replaced by a paraboloid.

We now show that on the unit ball the harmonic extension of continuous boundary data is an explicit integral of the data against a kernel. For $x \in \mathrm{B}(\mathbf{0}, 1)$ and $y \in \partial\mathrm{B}(\mathbf{0}, 1)$, let

$$P(x, y) \coloneqq \frac{1 - |x|^2}{|x - y|^d}\,. \tag{2.6}$$

The function P is called the **Poisson kernel** for the unit ball. In general, a "kernel" is a function $P : X \times Y \to \mathbb{R}$, where X is a set and Y is a measure space.

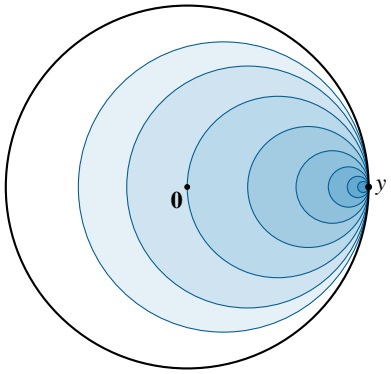


Figure 2.5: Level sets of the Poisson kernel (2.6) of the unit disk for a fixed point y of the unit circle. The shaded disks are the sets where $P(\cdot, y) > c$ for $c = 2^k$ and $-2 \le k \le 5$, darker for larger c; each is the disk of radius $1/(1+c)$ tangent to the unit circle at y. Points approaching y with $1 - |x|$ at most a constant times $|x-y|^2$ stay outside the small disks, so the kernel stays bounded along such paths.

It acts on a function $f : Y \to \mathbb{R}$ to produce the function $x \mapsto \int_Y P(x,y) f(y)\,\mathrm{d}y$ on X. Figure 2.5 shows $P(\cdot, y)$ for fixed y.

Theorem 2.18 (Poisson Integral Formula) *Let $d \ge 1$, let $B = \mathrm{B}(\mathbf{0}, 1) \subset \mathbb{R}^d$, and let $P : B \times \partial B \to \mathbb{R}$ be given by*

$$P(x,y) := \frac{1-|x|^2}{|x-y|^d} \qquad \text{for } x \in B \text{ and } y \in \partial B\,.$$

Let $F : \partial B \to \mathbb{R}$ be integrable with respect to the surface measure $\mathrm{d}S$ (counting measure on $\partial B = \{-1, 1\}$ when $d = 1$), and let $h : B \to \mathbb{R}$ be given by

$$h(x) := \fint_{\partial B} P(x,y) F(y)\,\mathrm{d}S(y)\,.$$

Then h is harmonic in B, and for every point $y_0 \in \partial B$ at which F is continuous,

$$\lim_{x \to y_0} h(x) = F(y_0)\,.$$

Proof. We start with harmonicity. Fix $y \in \partial B$. Since $x \ne y$ for $x \in B$, differentiation gives

$$\nabla_x |x-y|^{-d} = -d\,(x-y)\,|x-y|^{-d-2} \qquad \text{and} \qquad \Delta_x |x-y|^{-d} = 2d\,|x-y|^{-d-2}\,.$$

Since $\nabla_x(1-|x|^2) = -2x$ and $\Delta_x(1-|x|^2) = -2d$, the product rule gives

$$\Delta_x P(x,y) = 2d\,|x-y|^{-d-2}(1-|x|^2+2x\cdot(x-y)-|x-y|^2) = 2d\,|x-y|^{-d-2}(1-|y|^2)\,,$$

where the second equality follows by expanding $|x-y|^2$. The right side vanishes because $|y|=1$, in every dimension $d \geq 1$. (For $d \geq 2$, Exercise 2.48 proves the harmonicity of $P(\cdot, y)$ without second derivatives.) On each compact subball of B the distance $|x-y|$ is bounded below uniformly over $y \in \partial B$, so the first two x-derivatives of $P(x,y)F(y)$ are dominated by $C|F(y)|$, which is integrable. By the dominated convergence theorem we may differentiate twice under the integral sign, so $h \in C^2(B)$ with $\Delta h = 0$, and h is harmonic by Lemma 2.15.

Next we show that the kernel has mean one. By the first step with $F \equiv 1$, the function $h_1(x) := \fint_{\partial B} P(x,y)\,dS(y)$ is harmonic in B. Since $P(Qx, Qy) = P(x,y)$ for every orthogonal matrix Q, and Q preserves surface measure on ∂B, we have $h_1(Qx) = h_1(x)$, so h_1 is constant on each sphere $\partial \mathrm{B}(\mathbf{0}, r)$ with $0 < r < 1$. The minimum principle, Lemma 2.12(a), applied on $\mathrm{B}(\mathbf{0}, r)$ to h_1 and to $-h_1$, shows that h_1 takes this constant value on all of $\overline{\mathrm{B}}(\mathbf{0}, r)$, in particular at the origin, where $h_1(\mathbf{0}) = 1$ because $P(\mathbf{0}, y) = 1$ for every $y \in \partial B$. As $r < 1$ was arbitrary, $\fint_{\partial B} P(x,y)\,dS(y) = 1$ for every $x \in B$.

Finally we prove the boundary limit. Suppose that F is continuous at y_0, and fix $\epsilon > 0$. Choose $\delta > 0$ such that $|F(y) - F(y_0)| < \epsilon$ whenever $y \in \partial B$ and $|y - y_0| < \delta$. Let $|x - y_0| < \delta/2$. Every $y \in \partial B$ with $|y - y_0| \geq \delta$ then satisfies $|x-y| \geq \delta/2$, as in Figure 2.6, so $P(x,y) \leq (1-|x|^2)(2/\delta)^d$. Since the kernel is positive with mean one, and the points y with $|y - y_0| < \delta$ contribute at most ϵ to the integral below, we obtain

$$\begin{aligned} |h(x) - F(y_0)| &\leq \fint_{\partial B} P(x,y)\,|F(y) - F(y_0)|\,dS(y) \\ &\leq \epsilon + (1-|x|^2)\Big(\frac{2}{\delta}\Big)^d \fint_{\partial B} |F(y) - F(y_0)|\,dS(y) \end{aligned}.$$

The last integral is finite because F is integrable, and $1 - |x|^2 \to 0$ as $x \to y_0$, so $\limsup_{x \to y_0} |h(x) - F(y_0)| \leq \epsilon$. As ϵ was arbitrary, $h(x) \to F(y_0)$. □

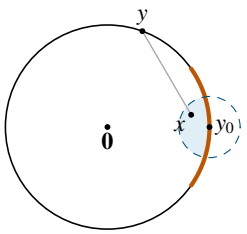


Figure 2.6: In the proof of Theorem 2.18, for x within $\delta/2$ of y_0 (dashed circle), every $y \in \partial B$ outside the cap $|y - y_0| < \delta$ (vermilion) satisfies $|x - y| \geq \delta/2$.

Corollary 2.19 *Let $B := \mathrm{B}(\mathbf{0}, 1) \subset \mathbb{R}^d$, and let $F \in C(\partial B)$. Then there is a continuous function $h : \overline{B} \to \mathbb{R}$ such that $\Delta h = 0$ on B and $h = F$ on ∂B.*

Proof. Let h be the Poisson integral of F, extended to ∂B by $h = F$. By Theorem 2.18, h is harmonic in B and $h(x) \to F(y_0)$ as $x \to y_0$ for every $y_0 \in \partial B$, so h is continuous on $\overline{B}$. □

In the plane, identify $\mathbb{R}^2$ with $\mathbb{C}$ and parametrize the unit circle by $e^{\mathbf{i}\theta}$. The normalized surface measure becomes $\mathrm{d}\theta/(2\pi)$, and the Poisson integral of F becomes

$$h(z) = \frac{1}{2\pi} \int_0^{2\pi} \frac{1 - |z|^2}{|e^{\mathbf{i}\theta} - z|^2} F(e^{\mathbf{i}\theta}) \, \mathrm{d}\theta \qquad \text{for } |z| < 1 \,. \tag{2.7}$$

Every function u that is harmonic in the unit ball $B = \mathrm{B}(\mathbf{0}, 1)$ and continuous on $\overline{B}$ is the Poisson integral of its boundary values. Indeed, by Theorem 2.18 the Poisson integral h of $u|_{\partial B}$ is harmonic in B and extends continuously to $\overline{B}$ with $h = u$ on ∂B, so Lemma 2.12(b), applied to $u - h$ and to $h - u$, gives $u = h$.

If u is a continuous function on $\Omega \subset \mathbb{R}^d$ and $\overline{\mathrm{B}}(x_0, r) \subset \Omega$, let $W := \mathrm{B}(x_0, r)$ and write $\mathbf{P}_W u$ for the function on W that extends $u|_{\partial W}$ harmonically. Then $\mathbf{P}_W = T^{-1} \circ \mathbf{P}_B \circ T$, where $Tu(x) := u(x_0 + rx)$ and $B := \mathrm{B}(\mathbf{0}, 1)$.

Note that if, for each $x \in \Omega$, there is $r_0 > 0$ such that $u \geq \mathbf{P}_W u$ for all balls $W = \mathrm{B}(y, r)$ with $r < r_0$ and $|y - x| < r_0$, and with $\overline{\mathrm{B}}(y, r) \subset \Omega$, then $u(y) \geq \mathbf{P}_W u(y) = \mathbf{a}_r u(y)$ for these balls, so u is superharmonic in Ω by (2.4).

Recall that Definition 2.6 requires the mean value inequality only for sufficiently small balls. We now prove that a superharmonic function satisfies the inequality for every ball in its domain, for continuous functions in Lemma 2.23 and in general in Corollary 2.27. We begin with the case of a harmonic function.

Lemma 2.20 (Mean Value Equality for All Spheres and Balls) *If h is harmonic in an open set Ω and $\overline{\mathrm{B}}(x, R) \subset \Omega$, then for every $r \in (0, R]$ we have*

$$h(x) = \mathbf{a}_r h(x) = \mathbf{A}_r h(x) \,.$$

Proof. By Theorem 2.3 and Lemma 2.15, h is smooth and $\Delta h = 0$. Hence, by the calculation (2.5) and the divergence theorem, $\mathbf{a}_r h(x)$ has derivative zero in r for $0 < r \leq R$. It tends to $h(x)$ as $r \downarrow 0$, so $\mathbf{a}_r h(x) = h(x)$, and integrating in spherical shells by (2.4) gives $\mathbf{A}_r h(x) = h(x)$. □

The next lemma gives a uniform radius δ for which the mean value inequality holds on every ball of radius at most δ. Exercise 2.30 asks for the proof.

Lemma 2.21 (Mean Value Inequality for All Small Balls) *If Ω is a bounded domain and u is superharmonic in a neighborhood of $\overline{\Omega}$, then there exists $\delta > 0$ such that $\mathbf{A}_r u \le u$ on $\overline{\Omega}$ for all $0 < r \le \delta$.*

Now we show that if u is superharmonic, then $\mathbf{a}_r u$ and $\mathbf{A}_r u$ are themselves superharmonic.

Lemma 2.22 *Let $x \in \mathbb{R}^d$ and $r > 0$, let $\Omega \subset \mathbb{R}^d$ be an open set that contains $\overline{\mathrm{B}}(x, r)$, and let u be superharmonic in Ω. Then the following hold.*

(a) The function $\mathbf{A}_r u$ is superharmonic at x.

(b) If u is continuous, then $\mathbf{a}_r u$ is superharmonic at x.

Proof. By compactness there exists $r_1 > r$ such that $\overline{\mathrm{B}}(x, r_1) \subset \Omega$. By Lemma 2.21 there exists $\delta > 0$ such that $\mathbf{A}_\rho u \le u$ on $\mathrm{B}(x, r_1)$ for all $\rho \le \delta$. Let U be the open ball of radius $\min(\delta, r_1 - r)$ centered at x.

(a) By Fubini's theorem, for all balls $\mathrm{B}(y, \rho) \subset U$,

$$\mathbf{A}_\rho \mathbf{A}_r u(y) = \mathbf{A}_r \mathbf{A}_\rho u(y) \le \mathbf{A}_r u(y)\,,$$

so $\mathbf{A}_r u$ is superharmonic at x.

(b) If u is continuous, then its spherical averages are defined pointwise and vary continuously with the center. Fubini's theorem and the mean value inequality give

$$\mathbf{A}_\rho \mathbf{a}_r u(y) = \mathbf{a}_r \mathbf{A}_\rho u(y) \le \mathbf{a}_r u(y)\,,$$

so $\mathbf{a}_r u$ is superharmonic at x. □

Our next lemma shows that if u is superharmonic and continuous on Ω, then u satisfies the mean value inequality on every ball whose closure lies in Ω, and $\mathbf{A}_r u(x)$ is weakly decreasing in the radius r.

Lemma 2.23 *Let u be superharmonic and continuous on an open set $\Omega \subset \mathbb{R}^d$. Let $x \in \Omega$ and $0 < r < R$ satisfy $\overline{\mathrm{B}}(x, R) \subset \Omega$. Then*

$$\mathbf{a}_R u(x) \le \mathbf{a}_r u(x) \le u(x)\,,$$

and

$$\mathbf{A}_R u(x) \le \mathbf{A}_r u(x) \le u(x)\,.$$

Proof. Let $B := \mathrm{B}(x, R)$. By Theorem 2.18 there is a continuous function h on $\overline{B}$ that is harmonic on B and agrees with u on ∂B. By Lemma 2.12(b) we have $h \le u$ on $\overline{B}$. Since h is the Poisson integral of $u|_{\partial B}$ and the Poisson

kernel at the center of the ball is identically 1, we have $h(x) = \mathbf{a}_R u(x)$. The mean value property for spheres, Lemma 2.20, applied in B, gives

$$\mathbf{a}_R u(x) = h(x) = \mathbf{a}_r h(x) \le \mathbf{a}_r u(x)\,,$$

which proves $\mathbf{a}_R u(x) \le \mathbf{a}_r u(x)$. Applying the minimum principle to the harmonic extension of $u|_{\partial \mathrm{B}(x,r)}$ gives $\mathbf{a}_r u(x) \le u(x)$.

By (2.4) the inequalities for $\mathbf{A}$ follow from the inequalities for $\mathbf{a}$. □

Next we show that superharmonicity is preserved under locally uniform limits.

Lemma 2.24 *Let $\Omega_n \uparrow \Omega$ be open sets. If u_n is superharmonic and continuous in Ω_n, and $u_n \to u$ uniformly on compact subsets of Ω, then u is superharmonic and continuous in Ω.*

Proof. Since u is a uniform limit of continuous functions on compact subsets of Ω, the function u is continuous. To prove that u is superharmonic, it suffices by Definition 2.6 to verify the mean value inequality for every ball $\overline{\mathrm{B}}(y,\rho) \subset \Omega$. Fix such a ball. For all sufficiently large n, the ball $\overline{\mathrm{B}}(y,\rho)$ lies in Ω_n. Lemma 2.23 gives $\mathbf{A}_\rho u_n(y) \le u_n(y)$, and uniform convergence on $\overline{\mathrm{B}}(y,\rho)$ gives

$$\mathbf{A}_\rho u(y) = \lim_{n\to\infty} \mathbf{A}_\rho u_n(y) \le \lim_{n\to\infty} u_n(y) = u(y)\,,$$

as required. □

Bounded pointwise limits of harmonic functions are harmonic, and the convergence is automatically locally uniform.

Theorem 2.25 (Bounded Limits of Harmonic Functions) *Let $(u_n)_{n\ge 1}$ be uniformly bounded harmonic functions that converge pointwise in an open set $\Omega \subset \mathbb{R}^d$. Then the convergence is uniform on compact subsets of Ω, and the limit is harmonic in Ω.*

Proof. Let $|u_n| \le M$ for every n, and let $u := \lim_n u_n$. Then u is Borel and $|u| \le M$. If $\overline{\mathrm{B}}(y,\rho) \subset \Omega$, then Lemma 2.20 gives $u_n(y) = \mathbf{A}_\rho u_n(y)$, and the dominated convergence theorem gives $\mathbf{A}_\rho u_n(y) \to \mathbf{A}_\rho u(y)$, so $u(y) = \mathbf{A}_\rho u(y)$. Thus u has the mean value property on every ball whose closure lies in Ω, and hence u is harmonic in Ω.

Now fix a compact set $K \subset \Omega$, and choose $\rho > 0$ such that $K_\rho := \{z : \operatorname{dist}(z,K) \le \rho\}$ is contained in Ω. For $y \in K$ we have $u_n(y) - u(y) = \mathbf{A}_\rho(u_n - u)(y)$, so

$$\sup_{y\in K} |u_n(y) - u(y)| \le \frac{1}{\omega_d \rho^d} \int_{K_\rho} |u_n - u|\,\mathrm{d}z \longrightarrow 0\,,$$

by the dominated convergence theorem, since $|u_n - u| \le 2M$ on the bounded set K_ρ. □

So far we have used superharmonic functions only through the mean value inequality. The next subsection looks more closely at the definition itself: every superharmonic function has a lower semicontinuous representative, and for continuous functions, superharmonicity is equivalent to the distributional inequality $\Delta u \le 0$.

2.3.1 Fine Properties of Superharmonic Functions

We now show that each superharmonic function u admits a lower semicontinuous representative $\tilde{u}$ that agrees with u almost everywhere.

Theorem 2.26 *Let u be superharmonic on an open set $\Omega \subset \mathbb{R}^d$. Let $R > 0$, and let*

$$\Omega_R := \{x \in \Omega : \overline{\mathrm{B}}(x, R) \subset \Omega\}.$$

Then the following hold.

(a) For every $0 < r < R$,

$$\mathbf{A}_R u(x) \le \mathbf{A}_r u(x) \qquad \text{for all } x \in \Omega_R.$$

(b) The function

$$\tilde{u}(x) := \lim_{\epsilon \downarrow 0} \mathbf{A}_\epsilon u(x)$$

is an increasing limit by (a), is lower semicontinuous and superharmonic in Ω, and equals u almost everywhere in Ω.

Proof. (a) By the remark after Definition 2.1, the function $\mathbf{A}_\epsilon u$ is continuous in Ω_ϵ for every $\epsilon > 0$. Lemma 2.22(a) shows that $\mathbf{A}_\epsilon u$ is superharmonic in Ω_ϵ. Hence, for $0 < r < R$, Lemma 2.23 gives

$$\mathbf{A}_R \mathbf{A}_\epsilon u \le \mathbf{A}_r \mathbf{A}_\epsilon u \qquad \text{on } \Omega_{R+\epsilon}. \tag{2.8}$$

By Fubini's theorem, the two sides of (2.8) equal $\mathbf{A}_\epsilon(\mathbf{A}_R u)$ and $\mathbf{A}_\epsilon(\mathbf{A}_r u)$. Since $\mathbf{A}_R u$ and $\mathbf{A}_r u$ are continuous, letting $\epsilon \downarrow 0$ proves $\mathbf{A}_R u \le \mathbf{A}_r u$ on Ω_R.

(b) As an increasing limit of continuous functions, $\tilde{u}$ is lower semicontinuous (Exercise 2.4). The mean value inequality (2.3) gives $\tilde{u} \le u$. Every compact subset of Ω has a neighborhood on which, for a suitable $\delta > 0$, $\mathbf{A}_\delta u$ is continuous and $\mathbf{A}_\delta u \le \tilde{u} \le u$; hence $\tilde{u}$ is locally integrable.

Next we show that $\tilde{u} = u$ almost everywhere. Let $\overline{\mathrm{B}}(x, R) \subset \Omega$. Averaging $\mathbf{A}_\epsilon u \le \tilde{u} \le u$ over $\mathrm{B}(x, R)$ and commuting the two averages by Fubini's theorem gives, for all sufficiently small $\epsilon > 0$,

$$\mathbf{A}_\epsilon(\mathbf{A}_R u)(x) = \mathbf{A}_R(\mathbf{A}_\epsilon u)(x) \le \mathbf{A}_R \tilde{u}(x) \le \mathbf{A}_R u(x)\,.$$

Since $\mathbf{A}_R u$ is continuous at x, the left side tends to the right side as $\epsilon \downarrow 0$, so $\mathbf{A}_R \tilde{u}(x) = \mathbf{A}_R u(x)$. Hence the nonnegative difference $u - \tilde{u}$, defined almost everywhere, vanishes almost everywhere on $\mathrm{B}(x, R)$, and countably many such balls cover Ω.

Finally, fix $R > 0$. Since $u = \tilde{u}$ almost everywhere, $\mathbf{A}_R \tilde{u} = \mathbf{A}_R u$, and for every $0 < r < R$ we have $\mathbf{A}_R u \le \mathbf{A}_r u$. Letting $r \downarrow 0$ gives $\mathbf{A}_R \tilde{u} \le \tilde{u}$ on Ω_R, so $\tilde{u}$ is superharmonic. □

Part (a) of Theorem 2.26 gives the mean value inequality on every ball, without assuming continuity.

Corollary 2.27 *Let u be superharmonic on an open set $\Omega \subset \mathbb{R}^d$, and let $\overline{\mathrm{B}}(x, R) \subset \Omega$. Then $\mathbf{A}_R u(x) \le u(x)$.*

Proof. By Definition 2.6, $\mathbf{A}_r u(x) \le u(x)$ for all sufficiently small $r > 0$. Since $x \in \Omega_R$, Theorem 2.26(a) gives $\mathbf{A}_R u(x) \le \mathbf{A}_r u(x)$. □

By Theorem 2.26(b), every superharmonic function agrees almost everywhere with a lower semicontinuous superharmonic function. Next we approximate each lower semicontinuous superharmonic function by continuous superharmonic functions.

Lemma 2.28 *Let u be a lower semicontinuous superharmonic function on an open set $\Omega \subset \mathbb{R}^d$. Then*

$$u(x) = \lim_{r \downarrow 0} \mathbf{A}_r u(x) \qquad \text{for all } x \in \Omega\,. \tag{2.9}$$

Proof. Fix $x \in \Omega$ and a real number $a < u(x)$. By lower semicontinuity (Lemma 2.5), $u > a$ on a neighborhood of x. For all sufficiently small $r > 0$, this bound and the mean value inequality (2.3) give

$$a \le \mathbf{A}_r u(x) \le u(x)\,.$$

Letting $a \uparrow u(x)$ proves (2.9), also when $u(x) = +\infty$. □

By Theorem 2.26(a), the convergence in (2.9) is monotone. Hence, if $K \subset \Omega$ is compact and $r_n \downarrow 0$ satisfies $\overline{\mathrm{B}}(x, r_n) \subset \Omega$ for every $x \in K$ and every n, then the continuous superharmonic functions $\mathbf{A}_{r_n} u$ increase pointwise to u on K, as Figure 2.7 shows for $u(x) = -|x|$.

We conclude this section by giving one more characterization of superharmonic functions in the continuous case.

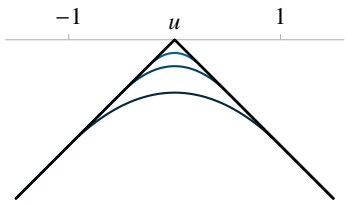


Figure 2.7: The ball averages $\mathbf{A}_\epsilon u$ of the superharmonic function $u(x) = -|x|$ on $\mathbb{R}$, for $\epsilon = 1, \frac{1}{2}, \frac{1}{4}$ (darker for smaller ϵ): they are continuous, lie below u, and increase to u as $\epsilon \downarrow 0$.

Theorem 2.29 *Let u be continuous on an open set $\Omega \subset \mathbb{R}^d$. Then u is superharmonic in Ω if and only if*

$$\int u\, \Delta\psi \, \mathrm{d}x \le 0\,, \tag{2.10}$$

for all nonnegative C^∞ compactly supported functions ψ on Ω.

Proof. We mollify u. For $\epsilon > 0$, let $\varphi_\epsilon(z) := \epsilon^{-d}\eta(z/\epsilon)$ with η the mollifier (2.1), and let $u_\epsilon := u * \varphi_\epsilon$ on the set Ω_ϵ of points x with $\overline{\mathrm{B}}(x,\epsilon) \subset \Omega$. By the regularity of convolutions in Section 2.1, u_ϵ is infinitely differentiable on Ω_ϵ and $\Delta u_\epsilon = u * \Delta\varphi_\epsilon$. Since $|u_\epsilon(y) - u(y)| \le \sup_{|z|<\epsilon} |u(y-z) - u(y)|$ and u is uniformly continuous on compact sets, $u_\epsilon \to u$ uniformly on compact subsets of Ω as $\epsilon \downarrow 0$.

Suppose first that u is superharmonic in Ω. By Lemma 2.23, $\mathbf{A}_r u \le u$ on Ω_r for every $r > 0$, so by Fubini's theorem $\mathbf{A}_r u_\epsilon = (\mathbf{A}_r u) * \varphi_\epsilon \le u_\epsilon$ on $\Omega_{r+\epsilon}$. Hence u_ϵ is superharmonic in Ω_ϵ, so $\Delta u_\epsilon \le 0$ there by Lemma 2.15. Let ψ be a nonnegative C^∞ function with compact support in Ω. For all sufficiently small ϵ the support of ψ lies in Ω_ϵ, and integrating by parts twice, with no boundary terms since ψ has compact support, gives

$$\int u_\epsilon\, \Delta\psi \, \mathrm{d}x = \int \psi\, \Delta u_\epsilon \, \mathrm{d}x \le 0\,.$$

Since $u_\epsilon \to u$ uniformly on the support of ψ, letting $\epsilon \downarrow 0$ proves (2.10).

Conversely, suppose that u satisfies (2.10). For $x \in \Omega_\epsilon$, the function $\psi(y) := \varphi_\epsilon(x-y)$ is nonnegative and C^∞ with support in $\overline{\mathrm{B}}(x,\epsilon) \subset \Omega$, so

$$\Delta u_\epsilon(x) = \int u(y)\, (\Delta\varphi_\epsilon)(x-y)\, \mathrm{d}y = \int u\, \Delta\psi \, \mathrm{d}y \le 0\,.$$

By Lemma 2.15, u_ϵ is superharmonic and continuous in the open set Ω_ϵ. These sets increase to Ω as $\epsilon \downarrow 0$, and $u_\epsilon \to u$ uniformly on compact subsets of Ω, so Lemma 2.24, applied along a sequence $\epsilon_n \downarrow 0$, shows that u is superharmonic in Ω. □

We now return to harmonic functions with a quantitative question. If a nonnegative harmonic function is large at one point of a ball, how large must it be at every other point of that ball?

2.4 HARNACK INEQUALITY

The Harnack inequality bounds the oscillation of a nonnegative harmonic function on a small ball in terms of its value at the center. The Poisson formula gives the sharp constants. We then give an elementary mean-value proof with a larger constant.

Theorem 2.30 *Let $d \geq 1$ and $0 \leq r < 1$. Let h be a nonnegative harmonic function on $\mathrm{B}(\mathbf{0}, 1) \subset \mathbb{R}^d$. Then for every $x, y \in \overline{\mathrm{B}}(\mathbf{0}, r)$,*

$$h(x) \leq \left(\frac{1+r}{1-r}\right)^d h(y)\,.$$

Proof of Theorem 2.30 using the Poisson kernel. Fix $w \in \overline{\mathrm{B}}(\mathbf{0}, r)$, and choose ρ with $|w| < \rho < 1$. The function $z \mapsto h(\rho z)$ is harmonic in $\mathrm{B}(\mathbf{0}, 1)$ and continuous on its closure, so, as shown after the proof of Theorem 2.18, it is the Poisson integral of its boundary values. Scaling the surface measure, we obtain

$$h(w) = \fint_{\partial \mathrm{B}(\mathbf{0},\rho)} \rho^{d-2} \frac{\rho^2 - |w|^2}{|w - v|^d} h(v)\, \mathrm{d}S(v)\,.$$

The spherical average of h equals $h(\mathbf{0})$. For $|v| = \rho$ we have the bounds $\rho - |w| \leq |w - v| \leq \rho + |w|$, as in Figure 2.8, and therefore

$$\rho^{d-2} \frac{\rho^2 - |w|^2}{(\rho + |w|)^d} h(\mathbf{0}) \leq h(w) \leq \rho^{d-2} \frac{\rho^2 - |w|^2}{(\rho - |w|)^d} h(\mathbf{0})\,.$$

Letting $\rho \uparrow 1$ and taking the worst case over $|w| \leq r$ gives

$$\frac{1-r}{(1+r)^{d-1}} h(\mathbf{0}) \leq h(w) \leq \frac{1+r}{(1-r)^{d-1}} h(\mathbf{0})\,. \tag{2.11}$$

Applying the upper bound to x and the lower bound to y proves the theorem. □

2.4.1 A Mean-Value Proof

For every $0 < r < 1$, the mean value property bounds $h(x)$ by a finite multiple of $h(y)$ depending only on d and r, for every nonnegative function h harmonic in $\mathrm{B}(\mathbf{0}, 1)$ and all $x, y \in \mathrm{B}(\mathbf{0}, r)$. The resulting constant is not sharp.

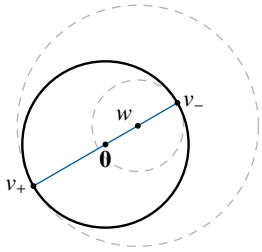


Figure 2.8: For $|w| < \rho$, every point v of the sphere of radius ρ lies between the circles about w of radii $\rho - |w|$ and $\rho + |w|$ (dashed), which touch it at the points v_- and v_+ nearest to and farthest from w. This gives the two bounds on the Poisson kernel in the proof of Theorem 2.30.

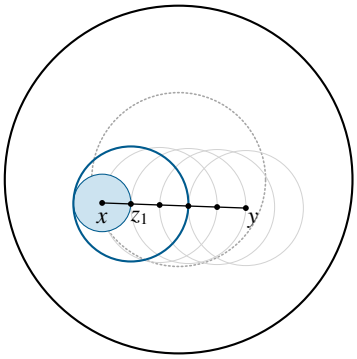


Figure 2.9: The chain $x = z_0, z_1, \dots, z_N = y$ of equally spaced points in $\mathrm{B}(\mathbf{0}, r)$ (dotted circle), drawn for $r = 1/2$, so that $N = 5$. The shaded ball $\mathrm{B}(z_0, a)$ lies in $\mathrm{B}(z_1, 2a)$ (blue), and every ball $\overline{\mathrm{B}}(z_i, 2a)$ lies in the unit ball, so each step along the chain costs at most a factor 2^d.

Proof of a weaker form of Theorem 2.30. The idea is to join x to y by equally spaced points and to compare consecutive values by two nested balls, as in Figure 2.9. Let $N := 1 + \lfloor 4r/(1-r) \rfloor$. If $x = y$, then there is nothing to prove. Otherwise, let $a := |x - y|/N$, and let $z_i := (1 - i/N)x + (i/N)y$ for $0 \le i \le N$. Each point z_i lies in $\mathrm{B}(\mathbf{0}, r)$, consecutive points are at distance a, and $2a < 4r/N < 1 - r$. Hence $\mathrm{B}(z_{i-1}, a) \subset \mathrm{B}(z_i, 2a)$, and $\overline{\mathrm{B}}(z_i, 2a)$ lies in $\mathrm{B}(\mathbf{0}, 1)$. For $1 \le i \le N$, Lemma 2.20 and $h \ge 0$ give

$$h(z_{i-1}) = \frac{1}{\omega_d a^d} \int_{\mathrm{B}(z_{i-1},a)} h(v)\,\mathrm{d}v \le \frac{1}{\omega_d a^d} \int_{\mathrm{B}(z_i,2a)} h(v)\,\mathrm{d}v = 2^d h(z_i)\,.$$

Iterating along the chain gives $h(x) \le 2^{dN} h(y)$, and N depends only on r. □

Exercise 2.35 shows that the two center bounds in the Poisson-kernel proof are optimal, and Exercise 2.36 extends the result to other domains. An important corollary of Theorem 2.30, which Exercise 2.37 asks you to prove, is the following.

Corollary 2.31 *Every nonnegative harmonic function on $\mathbb{R}^d$ is constant.*

The Harnack inequality has another consequence: a nonnegative function on $\mathbb{R}^d$ with bounded Laplacian can grow at most quadratically.

Proposition 2.32 *Let $d \geq 1$, let $R > 0$, let $1/2 \leq \beta < 1$, and let $C := (1+\beta)/(1-\beta)^{d-1}$. Let f be a nonnegative continuous function on $\mathrm{B}(\mathbf{0}, R)$, let $\lambda > 0$, and let $f_\pm(x) := f(x) \pm \lambda|x|^2$. If f_+ is subharmonic and f_- is superharmonic in $\mathrm{B}(\mathbf{0}, R)$, then, for every $x \in \mathrm{B}(\mathbf{0}, \beta R)$,*

$$f(x) \leq C(f(\mathbf{0}) + 8\lambda|x|^2)\,.$$

Proof. The case $x = \mathbf{0}$ is immediate since $C \geq 1$, so assume $x \neq \mathbf{0}$, and let $r := |x|/\beta < R$. Let h be the harmonic function on $B := \mathrm{B}(\mathbf{0}, r)$, continuous on $\overline{B}$, which agrees with f on ∂B. Since $f \geq 0$, the minimum principle gives $h \geq 0$ in B. The harmonic functions $h \pm \lambda r^2$ agree with $f_\pm$ on ∂B, so Lemma 2.12(b), applied to the superharmonic functions f_- and $-f_+$, gives

$$|f(z) - h(z)| \leq \lambda(r^2 - |z|^2) \qquad \text{for all } z \in B\,. \tag{2.12}$$

Since $|x|/r = \beta$, the upper bound in (2.11), after rescaling, gives $h(x) \leq Ch(\mathbf{0})$. By (2.12) at x and at $\mathbf{0}$,

$$f(x) \leq Ch(\mathbf{0}) + \lambda r^2 \leq Cf(\mathbf{0}) + (C+1)\lambda r^2 \leq C(f(\mathbf{0}) + 8\lambda|x|^2)\,,$$

where the last step uses $r = |x|/\beta \leq 2|x|$ and $C \geq 1$. □

We remark that in dimensions $d \geq 2$, the upper and lower bounds on the Laplacian are both necessary: see Exercise 2.39. In dimension 1, the upper bound alone suffices; and in all dimensions, an upper bound on the Hessian suffices (Exercise 2.40).

So far we have built harmonic functions from their boundary values. The next section builds superharmonic functions from a mass distribution: the potential of a nonnegative density is superharmonic, and its Laplacian, in the distributional sense, is minus the density.

2.5 SUPERHARMONIC POTENTIALS

Throughout this section we assume $d \geq 2$. We construct superharmonic potentials from the Green function $\mathcal{G}(x-y)$, the fundamental solution of the negative Laplacian. The Green function is radial and harmonic away from y, and we derive its explicit form in Exercise 2.41.

For a bounded measurable function σ on $\mathbb{R}^d$ with compact support, write

$$\mathbf{G}\sigma(x) := \int_{\mathbb{R}^d} \mathcal{G}(x-y)\sigma(y)\,\mathrm{d}y\,, \tag{2.13}$$

where

$$\mathcal{G}(x) := \begin{cases} -\frac{1}{2\pi}\log|x|, & d = 2, \\ a_d|x|^{2-d}, & d \geq 3\,. \end{cases} \tag{2.14}$$

We use the convention $\mathcal{G}(\mathbf{0}) = +\infty$. Here $a_d := \frac{1}{d(d-2)\omega_d}$, where ω_d is the volume of the unit ball in $\mathbb{R}^d$.

The operator $\mathbf{G}$ has both physical and mathematical significance. If $\sigma \geq 0$ and we interpret $\sigma(y)\,\mathrm{d}y$ as the density of mass near $y \in \mathbb{R}^d$, then $-\mathbf{G}\sigma$ is the **gravitational potential** of σ, and an observer located at x feels an attractive gravitational force proportional to $\nabla(\mathbf{G}\sigma)(x)$. We could also interpret $\sigma(y)\,\mathrm{d}y$ as a charge density, in which case the requirement $\sigma \geq 0$ is not needed since charge may be negative; $\mathbf{G}\sigma$ is then called the **Coulomb potential** of σ.

The mathematical significance of $\mathbf{G}$ derives from the identity

$$\mathbf{G} = (-\Delta)^{-1}\,,$$

made precise by Theorem 2.38. Given a function σ, formula (2.13) tells us how to construct a function whose Laplacian is $-\sigma$.

Our first result, Newton's shell theorem, has a simple interpretation in terms of gravity. The gravitational potential produced by a uniform spherical shell ∂B, where B is a ball centered at the origin, is constant in the interior of B, so the gravitational field there is zero, and on $\mathbb{R}^d \setminus \overline{B}$ the field agrees with that produced by a point mass at the center of B. Thus an observer inside B feels no net gravitational force from the mass on ∂B, while an observer outside $\overline{B}$ cannot distinguish the uniform spherical shell from a point mass at its center by measuring the gravitational field.

Lemma 2.33 (Newton's Shell Theorem) *Let $d \geq 2$, let $r > 0$, let $B = \mathrm{B}(\mathbf{0}, r) \subset \mathbb{R}^d$, and let $\mathbf{e}_1$ be the first standard basis vector. Then*

$$\fint_{\partial B} \mathcal{G}(x-y)\,\mathrm{d}S(y) = \begin{cases} \mathcal{G}(r\mathbf{e}_1), & |x| < r, \\ \mathcal{G}(x), & |x| \geq r\,. \end{cases}$$

Proof. Since $\mathcal{G}(x-y)$ depends only on $|x-y|$ and surface measure is rotation invariant, the spherical average $u(x) := \fint_{\partial B} \mathcal{G}(x-y)\,\mathrm{d}S(y)$ depends only on $|x|$. On each compact subball of B the Green function $\mathcal{G}(x-y)$ is bounded uniformly over $y \in \partial B$, so Fubini's theorem lets us average in x before integrating in y; since $\mathcal{G}(\cdot - y)$ is harmonic on B for each $y \in \partial B$, the function u is harmonic on B. For $|x| < r$ we have by the mean value property for spheres, Lemma 2.20,

$$u(x) = \mathbf{a}_{|x|}u(\mathbf{0}) = u(\mathbf{0}) = \mathcal{G}(r\mathbf{e}_1)\,.$$

Now fix $x \in \mathbb{R}^d$ with $|x| > r$. Since $y \mapsto \mathcal{G}(x-y)$ is harmonic on $\mathbb{R}^d \setminus \{x\}$, we have $u(x) = \mathbf{a}_r[\mathcal{G}(x-\cdot)](\mathbf{0}) = \mathcal{G}(x)$.

Finally, suppose $|x| = r$. As $t \downarrow 1$ with $t > 1$, the functions $\mathcal{G}(tx - y)$ increase pointwise to $\mathcal{G}(x - y)$. They are nonnegative when $d \geq 3$. When $d = 2$, adding a constant independent of y makes them nonnegative for $1 < t \leq 2$. The monotone convergence theorem therefore gives the boundary value as well. □

Corollary 2.34 *Let $d \geq 2$ and $x \in \mathbb{R}^d$. Then the function $y \mapsto \mathcal{G}(x - y)$ is superharmonic in $\mathbb{R}^d$.*

Proof. For every ball $\mathrm{B}(z, r) \subset \mathbb{R}^d$, translation invariance gives

$$\mathbf{a}_r[\mathcal{G}(x - \cdot)](z) = \fint_{\partial \mathrm{B}(\mathbf{0}, r)} \mathcal{G}(x - z - y)\, \mathrm{d}S(y)\,.$$

By Lemma 2.33, the right side is at most $\mathcal{G}(x - z)$, since $\mathcal{G}(w)$ decreases with $|w|$. From (2.4) we conclude that $\mathbf{A}_r[\mathcal{G}(x - \cdot)](z) \leq \mathcal{G}(x - z)$. □

In dimensions $d \geq 3$, the Green function $\mathcal{G}(x)$ is positive and superharmonic on $\mathbb{R}^d$. In dimension $d = 2$, every nonnegative superharmonic function on $\mathbb{R}^2$ is constant almost everywhere. Figure 2.10 shows the planar Green function, which is superharmonic and is negative outside the unit circle.

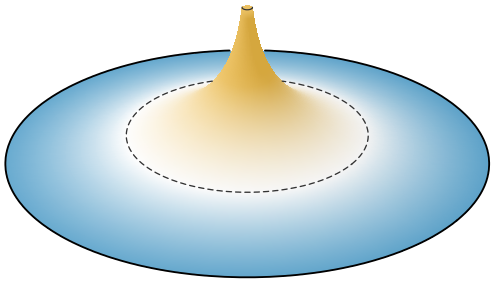


Figure 2.10: The graph of $\min(\mathcal{G}(x), \frac{1}{2})$ over the disk $\mathrm{B}(\mathbf{0}, 2)$ in the plane, where $\mathcal{G}(x) = -\frac{1}{2\pi} \log|x|$ is positive (orange) inside the unit circle (dashed) and negative (blue) outside it. The Green function must take negative values: by Lemma 2.35, a nonnegative superharmonic function on $\mathbb{R}^2$ is constant almost everywhere.

Lemma 2.35 *If $u \geq 0$ is superharmonic in $\mathbb{R}^2$, then u is constant almost everywhere.*

Proof. By Theorem 2.26(b), replacing u by its lower semicontinuous representative changes u only on a null set, and the representative is superharmonic and nonnegative, being a limit of averages of u. So we may assume that u is lower semicontinuous. Fix $x \neq y$. If $u(y) = 0$, then $u(x) \geq u(y)$ trivially; otherwise fix a real number M with $0 \leq M < u(y)$. By lower semicontinuity, choose $0 < \delta < |x - y|$ with $u > M$ on $\overline{\mathrm{B}}(y, \delta)$. For $R > |x - y|$, let

$$h_R(z) := M \frac{\log(R/|z - y|)}{\log(R/\delta)}\,.$$

The function h_R is harmonic in the annulus $A_R := \mathrm{B}(y,R) \setminus \overline{\mathrm{B}}(y,\delta)$, equals $M < u$ on the inner circle and equals $0 \le u$ on the outer circle, and $x \in A_R$. Lemma 2.12(b) gives $u \ge h_R$ in A_R, and letting $R \to \infty$ gives $u(x) \ge M$, as in Figure 2.11. Hence $u(x) \ge u(y)$ for all x and y, so u is constant. The constant is finite because u is locally integrable. □

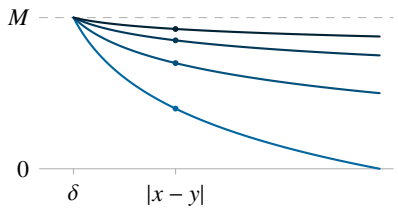


Figure 2.11: The barriers h_R of the proof of Lemma 2.35 as functions of $|z-y|$, for $\delta = M = 1$ and $R = 10, 10^2, 10^4, 10^8$ (lighter for larger R): each equals M on the circle of radius δ about y, and at a fixed point x (dots) they increase to M as $R \to \infty$.

For an alternative proof using Brownian motion, see Exercise 2.49.

Next we show that the potential of a nonnegative source is superharmonic, and harmonic away from the source.

Lemma 2.36 *Let $d \ge 2$, and let σ be a bounded measurable function on $\mathbb{R}^d$ with compact support.*

(a) If $\sigma \ge 0$ on an open set $\Omega \subset \mathbb{R}^d$, then $\mathbf{G}\sigma$ is superharmonic in Ω.

(b) The potential $\mathbf{G}\sigma$ is harmonic off the closed support of σ.

Proof. Choose $L > 0$ so that σ vanishes outside $\mathrm{B}(\mathbf{0}, L)$. For every $M > 0$ and $|z| \le M$, local integrability of $\mathcal{G}$ gives

$$\int_{\mathbb{R}^d} |\sigma(y)\mathcal{G}(z-y)|\,\mathrm{d}y \le \|\sigma\|_\infty \int_{\mathrm{B}(\mathbf{0},L+M)} |\mathcal{G}(w)|\,\mathrm{d}w < \infty\,.$$

Thus $\mathbf{G}\sigma$ is locally bounded, and Fubini's theorem applies absolutely to the integrals below.

(a) Fix $\overline{\mathrm{B}}(x,r) \subset \Omega$. By Fubini's theorem,

$$\mathbf{G}\sigma(x) - \mathbf{A}_r(\mathbf{G}\sigma)(x) = \int_{\mathbb{R}^d} \sigma(y)\left(\mathcal{G}(x-y) - \fint_{\mathrm{B}(x,r)} \mathcal{G}(z-y)\,\mathrm{d}z\right)\mathrm{d}y\,.$$

By Corollary 2.34, the bracket is nonnegative. By Newton's shell theorem, Lemma 2.33, and (2.4), it vanishes when $|x-y| \ge r$. Since $\sigma \ge 0$ on $\mathrm{B}(x,r)$, the integral is nonnegative. Hence $\mathbf{G}\sigma$ is superharmonic in Ω.

(b) Off the closed support of σ, both σ and $-\sigma$ vanish. Applying (a) to both shows that $\mathbf{G}\sigma$ is superharmonic and subharmonic there, and hence harmonic. □

Next we give an explicit formula for the potential of a ball.

Lemma 2.37 *Let $d \geq 2$, let $r > 0$, and let $B = \mathrm{B}(\mathbf{0}, r)$. In dimensions $d \geq 3$,*

$$\mathbf{G}\mathbf{1}_B(x) = \begin{cases} \frac{r^2}{2(d-2)} - \frac{1}{2d}|x|^2, & |x| < r, \\ \frac{r^2}{d(d-2)}\left(\frac{|x|}{r}\right)^{2-d}, & |x| \geq r\,. \end{cases} \tag{2.15}$$

In dimension $d = 2$,

$$\mathbf{G}\mathbf{1}_B(x) = \begin{cases} \frac{r^2}{4}(1 - 2\log r) - \frac{1}{4}|x|^2, & |x| < r, \\ -\frac{r^2}{2}\log|x|, & |x| \geq r\,. \end{cases} \tag{2.16}$$

Proof. The case $d \geq 3$ is Exercise 2.44, and the case $d = 2$ is Exercise 2.45. □

Figure 2.12 compares the potential of a uniform ball with those of its boundary sphere and of a point mass at its center.

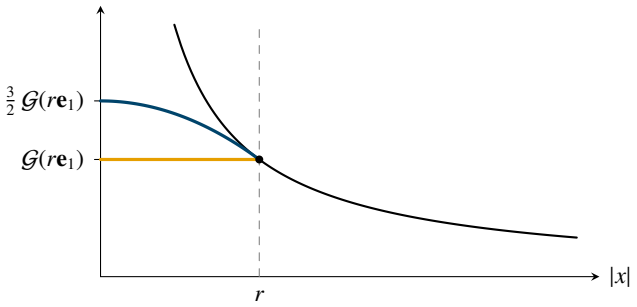


Figure 2.12: Radial profiles in $d = 3$ of the potentials of unit mass at the origin (black), spread uniformly on $\mathrm{B}(\mathbf{0}, r)$ (blue), and spread uniformly on the sphere $\partial\mathrm{B}(\mathbf{0}, r)$ (orange). Outside the ball the three agree, as Newton's shell theorem asserts; inside, the shell potential is constant and the ball potential is a parabola that joins the outer curve with the same slope at $|x| = r$.

We now make precise the identity $\mathbf{G} = (-\Delta)^{-1}$: the potential of a continuous source with compact support inverts the negative Laplacian.

Theorem 2.38 (Green Function and Inverse Laplacian) *Let $d \geq 2$, and let σ be a continuous function on $\mathbb{R}^d$ with compact support. If the convolution $\mathbf{G}\sigma$ is C^2, then*

$$\Delta(\mathbf{G}\sigma) = -\sigma\,. \tag{2.17}$$

Proof. We compare σ locally with constants. Fix $z \in \mathbb{R}^d$ and $\epsilon > 0$. By continuity, choose $\delta > 0$ so that $|\sigma - \sigma(z)| \le \epsilon$ on $\mathrm{B}(z,\delta)$. Let $q := \mathbf{G}\mathbf{1}_{\mathrm{B}(z,\delta)}$. Lemma 2.37, translated to z, gives $q \in C^2$ and $\Delta q = -1$ on this ball. Both

$$\mathbf{G}\sigma - (\sigma(z) - \epsilon)q \qquad \text{and} \qquad (\sigma(z) + \epsilon)q - \mathbf{G}\sigma$$

are C^2 there and have nonnegative sources on the ball, so they are superharmonic by Lemma 2.36(a). By Lemma 2.15, their Laplacians are nonpositive:

$$-\sigma(z) - \epsilon \le \Delta\mathbf{G}\sigma(z) \le -\sigma(z) + \epsilon\,.$$

Since $\epsilon > 0$ was arbitrary, $\Delta\mathbf{G}\sigma(z) = -\sigma(z)$. □

Remark. If σ is C^2 with compact support, then we may write

$$\mathbf{G}\sigma(x) = \int_{\mathbb{R}^d} \mathcal{G}(z)\sigma(x - z)\,\mathrm{d}z\,.$$

The derivatives of σ are bounded and, locally in x, the functions $z \mapsto \sigma(x-z)$ have a common compact support. Since $\mathcal{G}$ is locally integrable, dominated convergence permits differentiating twice under the integral, so $\mathbf{G}\sigma$ is C^2. A bounded measurable source σ that vanishes outside $\mathrm{B}(\mathbf{0}, R)$ has a locally bounded potential. For $d \ge 3$ this follows from $\mathcal{G} > 0$, since $|\mathbf{G}\sigma| \le \|\sigma\|_\infty \mathbf{G}\mathbf{1}_{\mathrm{B}(\mathbf{0},R)}$, which Lemma 2.37 computes and which is globally bounded. For $d = 2$, the identity $|\mathcal{G}(w)| = \mathcal{G}(w) + \frac{1}{\pi}\log^+|w|$ gives

$$|\mathbf{G}\sigma(x)| \le \|\sigma\|_\infty\Big(\mathbf{G}\mathbf{1}_{\mathrm{B}(\mathbf{0},R)}(x) + R^2\log^+(|x| + R)\Big)\,.$$

The potential is in fact continuous, because

$$|\mathbf{G}\sigma(x) - \mathbf{G}\sigma(x')| \le \|\sigma\|_\infty \int_{\mathrm{B}(\mathbf{0},R)} |\mathcal{G}(x - y) - \mathcal{G}(x' - y)|\,\mathrm{d}y$$

and, for $|x' - x| < 1$, the right side is at most $\|\sigma\|_\infty$ times the L^1 distance between the integrable function $\mathcal{G}\mathbf{1}_{\mathrm{B}(\mathbf{0},|x|+R+1)}$ and its translate by $x' - x$. This distance tends to 0 as $x' \to x$, because translation is continuous in L^1.

Corollary 2.39 *Let $d \ge 2$, and let $\psi : \mathbb{R}^d \to \mathbb{R}$ be a C^2 function with compact support. Then $\mathbf{G}(\Delta\psi) = -\psi$.*

Proof. By the remark after Theorem 2.38, differentiating twice under the integral in $\mathbf{G}\psi(x) = \int_{\mathbb{R}^d} \mathcal{G}(z)\psi(x - z)\,\mathrm{d}z$ shows that $\mathbf{G}\psi$ is C^2 and that $\Delta\mathbf{G}\psi = \mathbf{G}(\Delta\psi)$. Theorem 2.38 gives $\Delta\mathbf{G}\psi = -\psi$. □

If $\mathbf{G}\sigma$ is not C^2, then we cannot interpret (2.17) in the classical sense. Let f and u be locally integrable functions on $\mathbb{R}^d$. We call u a **distributional solution** of $\Delta u = f$ when, for every compactly supported C^∞ function ψ,

$$\int_{\mathbb{R}^d} u(x)\Delta\psi(x)\,\mathrm{d}x = \int_{\mathbb{R}^d} f(x)\psi(x)\,\mathrm{d}x\,.$$

The next corollary shows that $\mathbf{G}\sigma$ is a distributional solution of $\Delta u = -\sigma$ when σ is only bounded, measurable, and compactly supported rather than differentiable. The distributional formulation transfers the differentiability requirement from the solution u to the test function ψ through integration by parts.

Corollary 2.40 *Let $d \geq 2$, and let $\sigma : \mathbb{R}^d \to \mathbb{R}$ be bounded, measurable, and compactly supported. Then for every compactly supported C^2 function $\psi : \mathbb{R}^d \to \mathbb{R}$,*

$$\int_{\mathbb{R}^d} (\mathbf{G}\sigma)(x)\Delta\psi(x)\,\mathrm{d}x = -\int_{\mathbb{R}^d} \sigma(x)\psi(x)\,\mathrm{d}x\,.$$

Proof. By Fubini's theorem,

$$\int_{\mathbb{R}^d}\left(\int_{\mathbb{R}^d} \mathcal{G}(x-y)\sigma(y)\,\mathrm{d}y\right)\Delta\psi(x)\,\mathrm{d}x = \int_{\mathbb{R}^d} \mathbf{G}(\Delta\psi)(y)\sigma(y)\,\mathrm{d}y\,.$$

Now use Corollary 2.39. □

Example 2.41 We construct a superharmonic function that is not continuous. In dimensions $d \geq 2$, let $y_n \in \mathbb{R}^d \setminus \{\mathbf{0}\}$ be a sequence converging to $\mathbf{0}$, and let m_n be a sequence of positive real numbers satisfying

$$\sum_{n=1}^{\infty} m_n \mathcal{G}(y_n) < 1\,.$$

The hypothesis implies $\sum_n m_n < \infty$, because $\mathcal{G}(y_n) \to \infty$. Let

$$u(x) := \min\left(1, \sum_{n=1}^{\infty} m_n \mathcal{G}(x - y_n)\right).$$

The function u is superharmonic on $\mathbb{R}^d$ by Exercise 2.46. The function u is discontinuous at $\mathbf{0}$: $u(y_n) = 1$ for every n, whereas $u(\mathbf{0}) = \sum_n m_n \mathcal{G}(y_n) < 1$.

On the other hand, Exercise 2.47 shows that every lower semicontinuous superharmonic function is continuous in dimension $d = 1$.

2.5.1 The Arcsine Law

We end the section with one potential that we can compute exactly. It gives the square root in the Beurling estimates of Section 2.6 and Chapter 8, which bound the probability that Brownian motion, or random walk, avoids a long connected set.

For real numbers $\alpha < \beta$, the **arcsine law** on $[\alpha,\beta]$ is the probability measure

$$\text{Arcsine}_{\alpha,\beta}(\,\mathrm{d}t) := \frac{\mathrm{d}t}{\pi\sqrt{(t-\alpha)(\beta-t)}} \qquad \text{for } \alpha < t < \beta\,.$$

Its density is symmetric about the midpoint of the interval and is of order the inverse square root of the distance to the nearer endpoint. Exercise 2.52 shows that the arcsine law is the harmonic measure of the segment from infinity for planar Brownian motion. The next lemma computes its logarithmic potential. Figure 2.13 shows the arcsine law on $[0,1]$ as the vertical projection of the uniform distribution on a circle, the description that the proof of the next lemma uses.

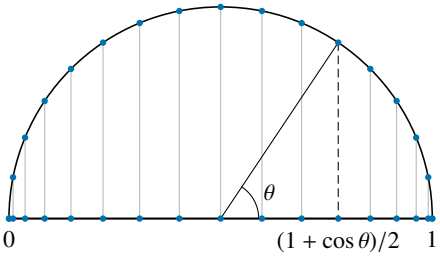


Figure 2.13: Vertical projection carries the uniform distribution on the upper half of the circle with diameter $[0,1]$ to the arcsine law $\text{Arcsine}_{0,1}$: if θ is uniform on $[0,\pi]$, then $(1+\cos\theta)/2$ has law $\text{Arcsine}_{0,1}$. The points at angles $\pi k/16$ cut the half-circle into arcs of equal length, so their projections cut $[0,1]$ into 16 intervals of equal mass, which crowd near the endpoints.

Lemma 2.42 *Let $\alpha < \beta$ be real numbers, and let $\text{Arcsine}_{\alpha,\beta}$ be the arcsine law on $[\alpha,\beta]$.*

(a) For every real number s,

$$\int \log|s-t|\,\text{Arcsine}_{\alpha,\beta}(\,\mathrm{d}t) \geq \log\frac{\beta-\alpha}{4}\,,$$

with equality when $\alpha \leq s \leq \beta$.

(b) For every $q \geq 0$,

$$\int \log\Bigl(1+\frac{q}{t-\alpha}\Bigr)\,\text{Arcsine}_{\alpha,\beta}(\,\mathrm{d}t) \leq 2\sqrt{\frac{q}{\beta-\alpha}}\,.$$

Since $\log(1+q/(t-\alpha)) = \log|\alpha-q-t| - \log|\alpha-t|$, part (b) compares the logarithmic potential at the point $\alpha-q$, at distance q from the interval, with its constant value on the interval. In other words, the potential rises only

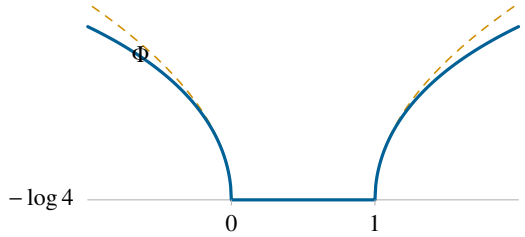


Figure 2.14: The logarithmic potential $\Phi(s) = \int \log|s-t|\,\mathrm{Arcsine}_{0,1}(\mathrm{d}t)$ of the arcsine law on $[0,1]$: it equals $-\log 4$ on the interval and, outside it, stays below $-\log 4 + 2\sqrt{\mathrm{dist}(s,[0,1])}$ (dashed), which it follows near the endpoints.

like the square root of the distance as the point leaves the interval through an endpoint, as Figure 2.14 shows.

Proof of Lemma 2.42. The map $t \mapsto \alpha + (\beta-\alpha)t$ carries the arcsine law on $[0,1]$ to the arcsine law on $[\alpha,\beta]$. It adds $\log(\beta-\alpha)$ to the potential in (a) and replaces q by $q/(\beta-\alpha)$ in (b), so it suffices to treat $\alpha = 0$ and $\beta = 1$.

We first write the potential as an average over a circle. If θ is uniformly distributed on $[0, 2\pi]$, then $(1+\cos\theta)/2$ has the law $\mathrm{Arcsine}_{0,1}$: each $t \in (0,1)$ has two preimages, and $|\,\mathrm{d}t/\,\mathrm{d}\theta| = |\sin\theta|/2 = \sqrt{t(1-t)}$. Let s be real. The roots of $w^2 - (4s-2)w + 1 = 0$ have product 1, so one of them, w, satisfies $|w| \geq 1$; when $0 \leq s \leq 1$, both roots have modulus 1. Expanding the product shows that

$$s - \frac{1+\cos\theta}{2} = \frac{(w-e^{\mathrm{i}\theta})(w-e^{-\mathrm{i}\theta})}{4w}\,.$$

Identify $\mathbb{R}^2$ with $\mathbb{C}$. In the plane, $\mathcal{G}$ is a negative multiple of the logarithm, so Newton's shell theorem, Lemma 2.33, for the unit circle gives $\frac{1}{2\pi}\int_0^{2\pi} \log|w - e^{\pm\mathrm{i}\theta}|\,\mathrm{d}\theta = \log|w|$, because $|w| \geq 1$. Hence

$$\int \log|s-t|\,\mathrm{Arcsine}_{0,1}(\mathrm{d}t) = \log|w| - \log 4\,. \tag{2.18}$$

(a) Since $|w| \geq 1$, with equality when $0 \leq s \leq 1$, equation (2.18) proves part (a).

(b) Let $s = -q$. The root of modulus at least one is $w = -(\sqrt{1+q} + \sqrt{q})^2$, whose reciprocal is $-(\sqrt{1+q} - \sqrt{q})^2$. Since $\log(1+q/t) = \log|-q-t| - \log|t|$, subtracting (2.18) at $s = 0$ from (2.18) at $s = -q$ gives

$$\begin{aligned}\int \log\Bigl(1+\frac{q}{t}\Bigr)\,\mathrm{Arcsine}_{0,1}(\mathrm{d}t) &= 2\log(\sqrt{1+q} + \sqrt{q}) \\ &= 2\int_0^{\sqrt{q}} \frac{\mathrm{d}b}{\sqrt{1+b^2}} \leq 2\sqrt{q}\,.\end{aligned}$$

The integral identity follows by differentiating $\log(b + \sqrt{1+b^2})$; the final inequality proves part (b). □

This is all we will need about superharmonic potentials. The last section of the chapter uses planar Brownian motion, which the rest of the book does not need, to prove the Beurling estimate. Chapter 8 proves the corresponding estimate for random walk on $\mathbb{Z}^2$ from Lemma 2.42, without Brownian motion.

2.6 HARMONIC MEASURE AND THE BEURLING ESTIMATE*

How likely is planar Brownian motion, started near the origin, to reach a distant circle without meeting a set that joins the origin to that circle? If the set is a radial segment and the starting point lies on the opposite ray, the probability is of order the square root of the ratio of the two radii. The Beurling estimate says that, up to a constant factor, no connected set is easier to avoid. The square root comes from the arcsine law on an interval (Subsection 2.5.1), whose logarithmic potential is constant on the interval and whose density blows up like the inverse square root of the distance to an endpoint. We apply Lemma 2.43 on exit distributions to the logarithm of the distance from one point of the set at each radius, use the triangle inequality, and average over the radii with the arcsine law. Subsection 2.6.1 proves a stronger comparison, the Beurling projection theorem: among all compact sets with the same distances from the origin, the easiest to avoid is the set of these distances placed on the ray opposite the starting point.

Throughout this section $Z = (Z_t)_{t\ge 0}$ is planar Brownian motion, and $\tau_F := \inf\{t \ge 0 : Z_t \in F\}$ is its hitting time of a Borel set $F \subset \mathbb{R}^2$. Write $\mathfrak{C}_r := \{z : |z| = r\}$ for the circle of radius r, and abbreviate $\tau_r := \tau_{\mathfrak{C}_r}$. Because planar Brownian motion is neighborhood-recurrent, it hits every circle almost surely, so the hitting times τ_r are almost surely finite. The **harmonic measure** of a boundary set $A \subseteq \partial D$ of a domain D, seen from a point $x \in D$, is the probability $\mathbb{P}_x\{Z_{\tau_{\partial D}} \in A\}$ that the Brownian motion from x first leaves D through A, defined whenever ∂D is hit almost surely.

Bounded harmonic functions determine Brownian exit distributions: if a bounded harmonic function on a domain tends to given values at the boundary, then its value at a point is the expected boundary value at the place where Brownian motion from that point first leaves the domain. The next lemma proves this with a sequence of circles, using only the strong Markov property and rotation invariance.

Lemma 2.43 *Let $D \subset \mathbb{R}^2$ be open, and suppose that $\mathbb{P}_z\{\tau_{\partial D} < \infty\} = 1$ for every $z \in D$. Let $E \subset \partial D$ be finite, let $f : \partial D \to \mathbb{R}$ be bounded and Borel measurable, and let h be a bounded harmonic function on D such that*

$h(w) \to f(y)$ as $w \to y$ within D, for every $y \in \partial D \setminus E$. Then, for every $z \in D$,

$$h(z) = \mathbb{E}_z\big[f(Z_{\tau_{\partial D}})\big]\,.$$

Proof. Fix $z \in D$, and write $\tau := \tau_{\partial D}$. Let $T_0 := 0$, and for $k \geq 0$ let T_{k+1} be the first time after T_k at which Z reaches the circle of radius $\frac{1}{2}\mathrm{dist}(Z_{T_k}, \partial D)$ about Z_{T_k}, as in Figure 2.15. These circles bound closed disks contained in D, so $T_k < \tau$ for every k. By the strong Markov property at T_k and rotation invariance, $Z_{T_{k+1}}$ is uniformly distributed on that circle given the path up to time T_k, so the mean value property of h gives $\mathbb{E}_z h(Z_{T_{k+1}}) = \mathbb{E}_z h(Z_{T_k})$. Hence $\mathbb{E}_z h(Z_{T_k}) = h(z)$ for every k.

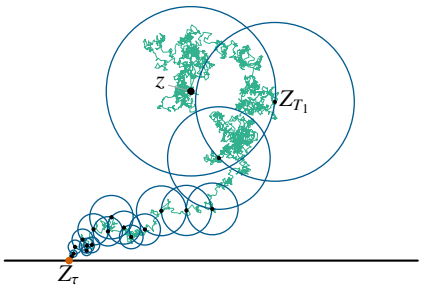


Figure 2.15: The first eighteen circles in the proof of Lemma 2.43, for D the upper half-plane and a simulated Brownian path from z: the circle about Z_{T_k} has radius $\frac{1}{2}\mathrm{dist}(Z_{T_k}, \partial D)$, and the path first reaches it at $Z_{T_{k+1}}$. The points Z_{T_k} converge to the exit point Z_τ.

The times T_k increase to a limit $T \leq \tau$, and $Z_{T_k} \to Z_T$ by continuity of the path. If Z_T were in D, then the steps $|Z_{T_{k+1}} - Z_{T_k}| = \frac{1}{2}\mathrm{dist}(Z_{T_k}, \partial D)$ would tend to $\frac{1}{2}\mathrm{dist}(Z_T, \partial D) > 0$, which is impossible for a convergent sequence. So $Z_T \in \partial D$, hence $T = \tau$ and $Z_{T_k} \to Z_\tau$. Planar Brownian motion almost surely never hits a given point, so $Z_\tau \notin E$ almost surely, and then $h(Z_{T_k}) \to f(Z_\tau)$. Since h is bounded, the bounded convergence theorem gives $h(z) = \mathbb{E}_z f(Z_\tau)$. □

For the unit disk, Lemma 2.43 identifies the harmonic measure, which Exercises 2.50 and 2.51 use. Let f be continuous on $\mathfrak{C}_1$, and let h be its Poisson integral (2.7). By Theorem 2.18, h is harmonic in the unit disk and $h(z) \to f(y)$ as $z \to y \in \mathfrak{C}_1$. Brownian motion leaves the disk almost surely, so Lemma 2.43 gives $\mathbb{E}_z f(Z_{\tau_1}) = h(z)$ for $|z| < 1$. Two finite measures on $\mathfrak{C}_1$ that give every continuous function the same integral are equal, so for every Borel set $A \subseteq \mathfrak{C}_1$,

$$\mathbb{P}_z\{Z_{\tau_1} \in A\} = \frac{1}{2\pi}\int_0^{2\pi} \mathbf{1}_A(e^{\mathrm{i}\theta})\, P(z, e^{\mathrm{i}\theta})\,\mathrm{d}\theta\,, \tag{2.19}$$

where P is the Poisson kernel (2.6). By scaling, Brownian motion started at z with $|z| < s$ leaves the disk of radius s with density $(s^2 - |z|^2)/|y - z|^2$ at

$y \in \mathfrak{C}_s$ with respect to normalized arc length. Figure 2.16 shows how the law (2.19) concentrates near the point of the circle closest to z: the other ends of the chords through z from equally spaced points of the circle cut the circle into arcs of equal harmonic measure. Indeed, if a chord through z has ends at distances a and b from z, then $ab = 1 - |z|^2$ (the power of the point z) and, by symmetry about the perpendicular bisector of the chord, the chord meets the circle at the same angle at both ends, so when the chord turns about z its end at distance b moves along the circle b/a times as fast as its end at distance a, whereas the density of $\mathbb{P}_z\{Z_{\tau_1} \in \cdot\}$ with respect to normalized arc length at the end at distance b is $(1 - |z|^2)/b^2 = a/b$, so the end at distance b sweeps out harmonic measure at the same rate at which the end at distance a sweeps out normalized arc length.

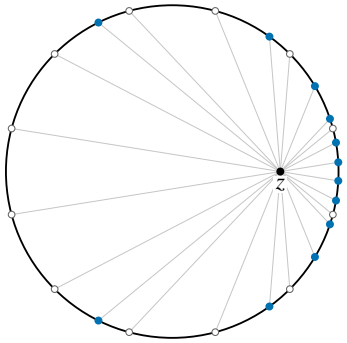


Figure 2.16: The chords through $z = 0.65$ from 12 equally spaced points of the unit circle (open dots) end at points (blue) that cut the circle into 12 arcs of equal harmonic measure seen from z. The arcs are shortest near the point of the circle closest to z, where Brownian motion from z is most likely to leave the disk.

We can now prove the Beurling estimate, using the logarithmic potential of the arcsine law computed in Lemma 2.42. It needs only one point of the set at each distance from the origin between the two radii. Recall that τ_F is the hitting time of a set F, and τ_R that of the circle $\mathfrak{C}_R$ of radius R.

Theorem 2.44 (Beurling Estimate) *Let $0 < r < R$, and let $K \subset \mathbb{R}^2$ be a compact set with $[r, R] \subseteq \{|y| : y \in K\}$. Then*

$$\mathbb{P}_z\{\tau_R < \tau_K\} \le 4\sqrt{\frac{r}{R}} \qquad \textit{for every } z \textit{ with } |z| \le r\,.$$

The set K need not be connected, and it may wind around the origin many times. Only its set of distances from the origin enters the hypothesis.

Proof. If $r > R/8$, then $4\sqrt{r/R} > 1$ and there is nothing to prove. If $z \in K$, then the probability in the theorem is zero. So we assume that $r \le R/8$ and $z \notin K$. The idea is to apply Lemma 2.43 to $\log|w - y|$ for one point $y \in K$ at each distance t from the origin, to eliminate y by the triangle inequality, and

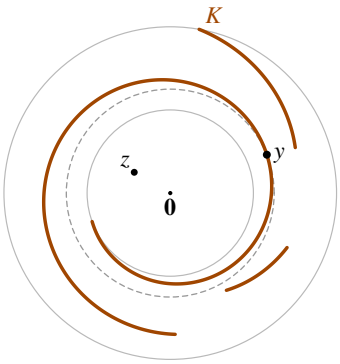


Figure 2.17: The part of a compact set K between the circles of radii r and $R/4$ (gray), for $R = 8r$: three disjoint pieces, one winding around the origin, whose distances from the origin cover $[r, R/4]$. For each t in $[r, R/4]$, Step 1 of the proof of Theorem 2.44 uses only one point y of K on the circle of radius t (dashed); the starting point z has $|z| \le r$.

only then to average over t with respect to the arcsine law $\mathrm{Arcsine}_{r,R/4}$. Let

$$\Phi(s) := \int \log|s-t| \,\mathrm{Arcsine}_{r,R/4}(\,\mathrm{d}t) \qquad \text{for } s \in \mathbb{R}\,,$$

and let $c := \log\frac{R/4-r}{4}$. By Lemma 2.42(a), $\Phi \ge c$ on $\mathbb{R}$. For each integer $n > 1/\mathrm{dist}(z,K)$, let T_n be the exit time of Z from the open set $\{w : |w| < R,\ \mathrm{dist}(w,K) > 1/n\}$, which contains z; then $T_n \le \tau_R$.

Step 1: We fix one distance $t \in [r, R/4]$ and $y \in K$ with $|y| = t$, as in Figure 2.17. Since the closure of the set defining T_n does not meet K, Lemma 2.43 with $E = \emptyset$ applies to the bounded harmonic function $w \mapsto \log|w-y|$ and gives $\mathbb{E}_z \log|Z_{T_n} - y| = \log|z-y|$. Since $\big||w|-t\big| \le |w-y|$ for every w and $|z-y| \le r+t$, it follows that

$$\mathbb{E}_z \log\big||Z_{T_n}| - t\big| \le \log(r+t)\,. \tag{2.20}$$

The point y no longer appears.

Step 2: We average over the distances. For $0 \le s \le R$ and $r \le t \le R/4$ the function $\log R - \log|s-t|$ is nonnegative, and $|Z_{T_n}| \le R$. Tonelli's theorem therefore lets us integrate (2.20) against the arcsine law $\mathrm{Arcsine}_{r,R/4}$ and exchange the expectation and the integral, which gives

$$\mathbb{E}_z \Phi(|Z_{T_n}|) \le \int \log(r+t)\,\mathrm{Arcsine}_{r,R/4}(\,\mathrm{d}t)\,.$$

Lemma 2.42(a) at $s = r$ gives $c = \int \log(t-r)\,\mathrm{Arcsine}_{r,R/4}(\,\mathrm{d}t)$, so Lemma 2.42(b), with $q = 2r$, gives

$$\mathbb{E}_z[\Phi(|Z_{T_n}|) - c] \le \int \log\Big(1+\frac{2r}{t-r}\Big)\,\mathrm{Arcsine}_{r,R/4}(\,\mathrm{d}t) \le 2\sqrt{\frac{2r}{R/4-r}} \le 8\sqrt{\frac{r}{R}}\,,$$

where the last step uses $R/4 - r \geq R/8$.

Step 3: We let $n \to \infty$. The random variable $\Phi(|Z_{T_n}|) - c$ is nonnegative. On the event $\{T_n = \tau_R\}$ we have $|Z_{T_n}| = R$, and $R - t \geq 3R/4$ for every $t \leq R/4$, so

$$\Phi(|Z_{T_n}|) - c \geq \log \frac{3R}{4} - \log \frac{R}{16} = \log 12 \,.$$

Hence $(\log 12)\, \mathbb{P}_z\{T_n = \tau_R\} \leq 8\sqrt{r/R}$. On the event $\{\tau_R < \tau_K\}$, the compact path $\{Z_t : 0 \leq t \leq \tau_R\}$ has positive distance from K, so $T_n = \tau_R$ for all large n. Fatou's lemma therefore gives

$$(\log 12)\, \mathbb{P}_z\{\tau_R < \tau_K\} \leq \liminf_{n\to\infty} (\log 12)\, \mathbb{P}_z\{T_n = \tau_R\} \leq 8\sqrt{\frac{r}{R}} \,.$$

Since $\log 12 > 2$, this proves the theorem. □

For a connected set the hypothesis of Theorem 2.44 holds automatically, because the distance from the origin is a continuous function, and the continuous image of a connected set is connected. This gives the Beurling estimate in its usual form.

Corollary 2.45 *Let $R > 0$, and let $K \subset \mathbb{R}^2$ be a compact connected set that contains the origin and meets the circle $\mathfrak{C}_R$. Then for every $z \in \mathbb{R}^2$,*

$$\mathbb{P}_z\{\tau_R < \tau_K\} \leq 4\sqrt{\frac{|z|}{R}} \,.$$

Proof. If $z = \mathbf{0}$, then $z \in K$ and the probability is zero. If $|z| \geq R/16$, then the right side is at least 1. Otherwise $0 < |z| < R$, and the set $\{|y| : y \in K\}$ is a connected subset of the real line that contains 0 and R, so it contains the interval $[|z|, R]$. Theorem 2.44 with $r = |z|$ therefore gives the bound. □

The exponent $\frac{1}{2}$ cannot be improved. Let $0 < r < R$, let K be the segment $[0, R]$ on the positive real axis, and start the Brownian motion at $-r$. Write the points of $\mathrm{B}(\mathbf{0}, R) \setminus K$ as $w = \rho e^{\mathbf{i}\theta}$ with $0 < \rho < R$ and $0 < \theta < 2\pi$. The function

$$w \mapsto \sqrt{\frac{\rho}{R}}\, \sin\frac{\theta}{2}$$

is the imaginary part of the branch of $\sqrt{w/R}$ with argument $\theta/2$, so it is harmonic on $\mathrm{B}(\mathbf{0}, R) \setminus K$. It extends continuously by zero to K, it takes values between 0 and 1 on $\mathfrak{C}_R$, and it equals $\sqrt{r/R}$ at $-r$ (Figure 2.18). Lemma 2.43 therefore gives $\mathbb{P}_{-r}\{\tau_R < \tau_K\} \geq \sqrt{r/R}$, so a radial segment attains the bound of Theorem 2.44 up to the factor 4.

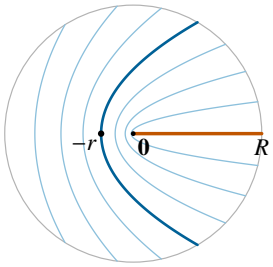


Figure 2.18: Level curves of $\sqrt{\rho/R}\,\sin(\theta/2) = \operatorname{Im}\sqrt{w/R}$ on the disk of radius R slit along $K = [0, R]$ (vermilion), at the values $1/8, \ldots, 7/8$: they are parabolas with focus $\mathbf{0}$. The level $\sqrt{r/R}$ (dark) passes through $-r$, drawn for $r = R/4$.

Chapter 8 proves the discrete Beurling estimate for random walk on $\mathbb{Z}^2$ by the same argument. The potential kernel of the walk replaces the logarithm, and the masses that the arcsine law gives to unit intervals replace its density. The new work is to show that this discretization changes the potential by at most the mass at an endpoint.

2.6.1 The Beurling Projection Theorem*

The proof of Theorem 2.44 uses only one point of K at each distance from the origin. The Beurling projection theorem compares the whole set with its circular projection, which places all of these distances on a single ray, and shows that the projection is the easiest of the sets with these distances to avoid. We prove it for planar Brownian motion started at a point a of the negative real axis: replacing a compact set by its circular projection, placed on the positive real axis opposite a, can only increase the probability of reaching a small circle about the origin before the set. Figure 2.19 shows the set and its circular projection.

For a compact set $K \subset \mathbb{R}^2$, the **circular projection** $K^* := \{|z| : z \in K\}$ collects the distances from the origin to points of K, placed on the positive real axis. Recall that τ_F is the hitting time of a set F by the planar Brownian motion Z, and τ_r that of the circle $\mathfrak{C}_r$ of radius r.

Theorem 2.46 (Beurling Projection Theorem) *Let $r > 0$ and let $K \subset \{|z| \geq r\}$ be compact with circular projection K^*. Then for every real $a < -r$,*

$$\mathbb{P}_a\{\tau_r < \tau_K\} \leq \mathbb{P}_a\{\tau_r < \tau_{K^*}\}\,.$$

The set K in Theorem 2.46 need not be connected. For a compact connected set that contains the origin and meets the unit circle, the theorem yields a square-root estimate.

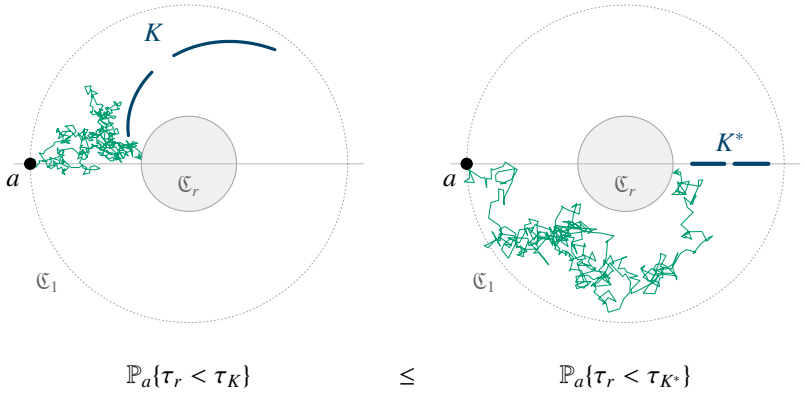


Figure 2.19: A Brownian path from a that reaches the circle $\mathfrak{C}_r$ of radius r, the boundary of the shaded disk, before the set K (left), and the circular projection $K^* = \{|z| : z \in K\}$, placed on the positive real axis (right). Projecting can only increase the chance of reaching $\mathfrak{C}_r$ first: $\mathbb{P}_a\{\tau_r < \tau_K\} \leq \mathbb{P}_a\{\tau_r < \tau_{K^*}\}$, where τ_F denotes the hitting time of a set F and τ_r that of $\mathfrak{C}_r$.

Corollary 2.47 *Let $E \subset \mathbb{R}^2$ be compact and connected, with $\mathbf{0} \in E$ and $E \cap \{|z| \geq 1\} \neq \emptyset$. Then for every $0 < r < 1$ and every starting point a with $|a| \geq 1$,*

$$\mathbb{P}_a\{\tau_r < \tau_E\} \leq \frac{4\sqrt{2}}{\pi}\, r^{1/2}\,.$$

Corollary 2.47 is the counterpart of Corollary 2.45 for a Brownian motion that starts outside the unit circle and must reach a small circle, now with the explicit constant $4\sqrt{2}/\pi \approx 1.80$, which cannot be improved. To deduce it, stop the Brownian motion at its first visit to the unit circle, rotate that point to -1, and project the set E. The circular projection of E contains the segment joining $\mathfrak{C}_r$ to the unit circle, so it suffices to bound the chance of reaching $\mathfrak{C}_r$ before that segment. Two conformal maps compute this probability exactly, and it is of order $r^{1/2}$.

We prove Theorem 2.46 by successively folding K across lines through the origin.

Let $\mathcal{R}z := \bar{z}$ denote reflection across the real axis, acting on sets by $\mathcal{R}K := \{\mathcal{R}z : z \in K\}$. Lemma 2.48 below reflects the part of K below the real axis into the upper half-plane, as in Figure 2.20, and shows that this fold can only increase the chance of reaching the inner circle before K. Since every circle centered at the origin is symmetric across every line through the origin, folding across the real axis is enough: the fold across a general line follows by a rotation.

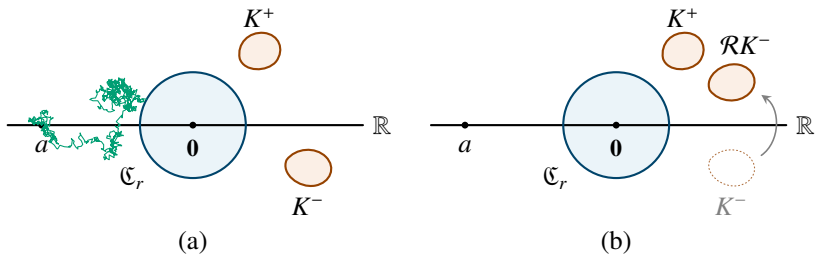


Figure 2.20: (a) The set $K = K^+ \cup K^-$ and a simulated Brownian path from a that realizes the event $\{\tau_r < \tau_K\}$. (b) Reflecting K^-, dotted in its original position, yields $\widetilde{K} = K^+ \cup \mathcal{R}K^-$, which lies in the closed upper half-plane. Here $K^+ = K \cap \{\operatorname{Im} z > 0\}$, $K^- = K \cap \{\operatorname{Im} z \le 0\}$, and $\mathcal{R}$ is reflection in the real axis. Folding can only increase the chance of reaching $\mathfrak{C}_r$ before the set.

Lemma 2.48 *Let $r > 0$, let $K \subset \{|z| \ge r\}$ be compact, let $K^+ := K \cap \{\operatorname{Im} z > 0\}$ and $K^- := K \cap \{\operatorname{Im} z \le 0\}$, and let $\widetilde{K} := K^+ \cup \{\bar{z} : z \in K^-\}$ be the set obtained by folding K^- across the real axis. Then for every real a with $|a| > r$,*

$$\mathbb{P}_a\{\tau_r < \tau_K\} \le \mathbb{P}_a\{\tau_r < \tau_{\widetilde{K}}\}\,.$$

Proof. We first compare the two hitting probabilities at reflected pairs of points of $K \cup \mathcal{R}K$. We then use reflection symmetry on the real axis to show that folding cannot increase the probability of hitting the set before the circle. For $|z| \ge r$, let

$$h(z) := \mathbb{P}_z\{\tau_K \le \tau_r\} \qquad \text{and} \qquad \widetilde{h}(z) := \mathbb{P}_z\{\tau_{\widetilde{K}} \le \tau_r\}\,,$$

so that $h = \mathbf{1}_K$ and $\widetilde{h} = \mathbf{1}_{\widetilde{K}}$ on $\mathfrak{C}_r$. Since τ_r is finite almost surely, the lemma states that $\widetilde{h}(a) \le h(a)$. Write $g := \widetilde{h} - h$, let $S := K \cup \mathcal{R}K$, which contains K and $\widetilde{K}$, and let

$$\delta := \max\Big\{0,\ \sup\{g(x) : x \in \mathbb{R},\ |x| \ge r\}\Big\} \le 1\,.$$

Step 1: We show that $g(z) + g(\bar{z}) \le \delta$ for every $z \in S$ with $\operatorname{Im} z > 0$ and $|z| > r$. From z or $\bar{z}$ the path stays outside the disk of radius r until time τ_r, so at the hitting time $\tau_{\mathbb{R}}$ of the real axis, $Z_{\tau_{\mathbb{R}}}$ is a real point with $|Z_{\tau_{\mathbb{R}}}| \ge r$ on the event $\{\tau_{\mathbb{R}} \le \tau_r\}$. Reflection across the real axis maps the Brownian motion from z to a Brownian motion from $\bar{z}$, as in Figure 2.21, and it fixes the real axis and $\mathfrak{C}_r$, so the law of $Z_{\tau_{\mathbb{R}}}$ on this event is the same from z and from $\bar{z}$. The point z lies in $\widetilde{K}$, because either $z \in K$ or $\bar{z} \in K$, so $\widetilde{h}(z) = 1$. The set $\widetilde{K}$ lies in the closed upper half-plane, so a path from $\bar{z}$ can reach it only at or after time $\tau_{\mathbb{R}}$, and the strong Markov property gives

$$\widetilde{h}(z) + \widetilde{h}(\bar{z}) = 1 + \mathbb{E}_{\bar{z}}\big[\widetilde{h}(Z_{\tau_{\mathbb{R}}});\ \tau_{\mathbb{R}} \le \tau_r\big]\,.$$

At least one of z and $\bar{z}$ lies in K, and h equals 1 there. From the other point, reaching K after time $\tau_{\mathbb{R}}$ but by time τ_r is one way to hit K by time τ_r, so, by the strong Markov property and the symmetry of the law of $Z_{\tau_{\mathbb{R}}}$,

$$h(z) + h(\bar{z}) \geq 1 + \mathbb{E}_{\bar{z}}\big[h(Z_{\tau_{\mathbb{R}}});\ \tau_{\mathbb{R}} \leq \tau_r\big]\,.$$

This inequality also applies when both points lie in K, because $h \leq 1$. Subtracting the two displays gives $g(z) + g(\bar{z}) \leq \mathbb{E}_{\bar{z}}[g(Z_{\tau_{\mathbb{R}}});\ \tau_{\mathbb{R}} \leq \tau_r] \leq \delta$.

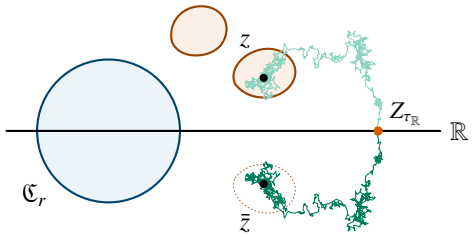


Figure 2.21: Step 1 of the proof of Lemma 2.48, for the set K of Figure 2.20 and a point $z \in \mathcal{R}K^-$, so that $\bar{z} \in K^-$ (dotted). Reflection in the real axis maps the Brownian path from $\bar{z}$ (dark), up to time $\tau_{\mathbb{R}}$, to a path from z (light) with the same endpoint $Z_{\tau_{\mathbb{R}}}$, and the path from $\bar{z}$ cannot meet $\widetilde{K}$ (shaded) before $\tau_{\mathbb{R}}$.

Step 2: We show that $\delta = 0$. Let a be real with $|a| > r$, and let T be the first time at which the path from a meets $S \cup \mathfrak{C}_r$; then $T \leq \tau_r < \infty$. Both K and $\widetilde{K}$ lie in S, so the strong Markov property at time T gives $g(a) = \mathbb{E}_a g(Z_T)$. We bound g at the possible values of Z_T. At a real point x of S, both $h(x)$ and $\widetilde{h}(x)$ equal 1, because the real points of S, of K and of $\widetilde{K}$ coincide; at a real point of $\mathfrak{C}_r$, both equal $\mathbf{1}_K(x)$. So $g = 0$ at the real values of Z_T. For a point w of $\mathfrak{C}_r$ with $\operatorname{Im} w > 0$, we have $\mathbf{1}_{\widetilde{K}}(w) = \max\{\mathbf{1}_K(w), \mathbf{1}_K(\bar{w})\}$ and $\mathbf{1}_{\widetilde{K}}(\bar{w}) = 0$, so $g(w) + g(\bar{w}) \leq 0$. Every other value of Z_T lies in a pair as in Step 1. The law of Z_T is symmetric under reflection across the real axis, so it gives probability at most $1/2$ to the upper half-plane, and pairing each point there with its reflection gives

$$g(a) = \mathbb{E}_a\big[g(Z_T) + g(\overline{Z_T});\ \operatorname{Im} Z_T > 0\big] \leq \frac{\delta}{2}\,.$$

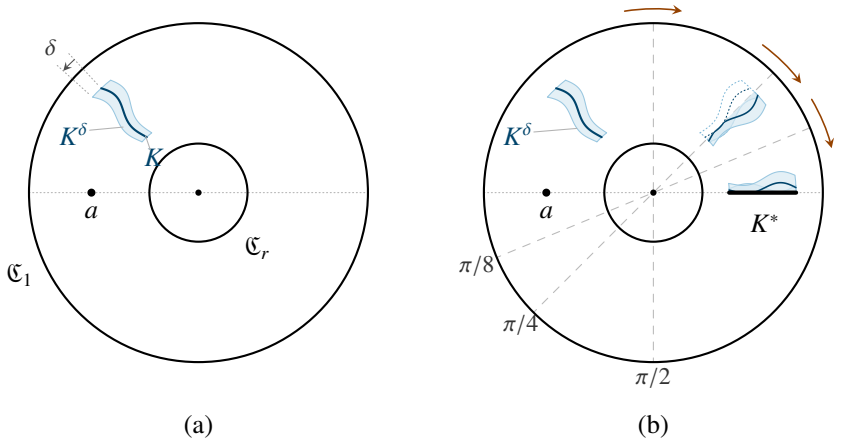
Since also $g = 0$ at the two real points of $\mathfrak{C}_r$, taking the supremum over real a with $|a| \geq r$ gives $\delta \leq \delta/2$, so $\delta = 0$. Hence $\widetilde{h}(a) \leq h(a)$ for every real a with $|a| > r$, which proves the lemma. □

Lemma 2.48 treats the fold across the real axis. In general, for a line ℓ through the origin other than the real axis, let $\mathbb{H}_\ell$ be the open half-plane bounded by ℓ that contains the negative real ray, and let $\mathcal{R}_\ell$ be the reflection in ℓ. The **fold** across ℓ maps a compact set E to the compact set

$$\mathcal{F}_\ell E := (E \setminus \mathbb{H}_\ell) \cup \mathcal{R}_\ell(E \cap \mathbb{H}_\ell)\,,$$

which lies in the closed half-plane opposite $\mathbb{H}_\ell$ and satisfies

$$\{|z| : z \in \mathcal{F}_\ell E\} = \{|z| : z \in E\}\,.$$



(a) (b)

Figure 2.22: (a) A compact set K and its angular thickening K^δ, which contains the point at angle $3\pi/4$ at each distance in K^*. (b) The folds across the dashed lines at angles $\pi/2$, $\pi/4$ and $\pi/8$ map the ray at angle $3\pi/4$ onto the positive real axis, so the image of K^δ contains K^* (heavy segment); the folds at angles $-\pi/4$ and $-\pi/8$ do not move the pictured sets.

The next lemma is geometric: finitely many folds map the angular thickening K^δ of K, which consists of the rotations of points of K by angles at most δ, to a set containing K^*.

Lemma 2.49 *Let K be a compact set in the plane, let $\delta > 0$, and let $K^\delta := \{e^{i\varphi} z : z \in K,\ |\varphi| \le \delta\}$ be its angular thickening. Then there exist an integer $N \ge 0$ and lines $\ell_1, \dots, \ell_{2N+1}$ through the origin, none equal to the real axis, with*

$$K^* \subseteq \mathcal{F}_{\ell_{2N+1}} \circ \cdots \circ \mathcal{F}_{\ell_1} K^\delta\,.$$

Proof. Figure 2.22 illustrates the proof. Choose an integer $N \ge 0$ with $\pi/2^{N+1} < \delta$. Let ℓ_1 be the imaginary axis. For $1 \le j \le N$, let ℓ_{2j} and ℓ_{2j+1} be the lines at angles $\pi/2^{j+1}$ and $-\pi/2^{j+1}$, respectively.

We prove by induction on m that the first $2m+1$ folds send a ray of angle θ to a ray whose absolute angle is

$$\operatorname{dist}\left(\theta, \frac{\pi}{2^m}\mathbb{Z}\right).$$

The case $m = 0$ is the fold across the imaginary axis. For the inductive step, the angle after the first $2m+1$ folds lies in $[-\pi/2^{m+1}, \pi/2^{m+1}]$. Folding

across the lines at angles $\pi/2^{m+2}$ and $-\pi/2^{m+2}$ replaces the absolute angle s by $\min\{s, \pi/2^{m+1} - s\}$, which equals the distance from θ to $(\pi/2^{m+1})\mathbb{Z}$ and proves the inductive step. In particular, after the first $2m+1$ folds every ray lies in $|\arg z| \le \pi/2^{m+1}$, and every ray with angle $k\pi/2^m$ is mapped to the positive real axis.

The point $\mathbf{0}$, if it lies in K^*, lies in K^δ and is fixed by every fold. Now let $x \in K^*$ be positive and choose θ with $xe^{\mathbf{i}\theta} \in K$. There is an angle $\psi = k\pi/2^N$ whose circular distance from θ is at most $\pi/2^{N+1} < \delta$. Hence $xe^{\mathbf{i}\psi} \in K^\delta$, and the induction shows that the first $2N+1$ folds map $xe^{\mathbf{i}\psi}$ to x. Therefore

$$K^* \subseteq (\mathcal{F}_{\ell_{2N+1}} \circ \cdots \circ \mathcal{F}_{\ell_1})(K^\delta)\,.$$

□

The proof of the projection theorem is now a chain of inequalities between hitting probabilities.

Proof of Theorem 2.46. For a point x and a compact set $E \subset \{|z| \ge r\}$, write $h_x(E) := \mathbb{P}_x\{\tau_E \le \tau_r\}$ for the probability that the Brownian motion from x meets E by the time it reaches $\mathfrak{C}_r$. Each h_x is increasing in E, and since $\mathbb{P}_a\{\tau_r < \tau_E\} = 1 - h_a(E)$ the assertion is $h_a(K) \ge h_a(K^*)$. If $a \in K$ then $\tau_K = 0$, so $h_a(K) = 1$ and the inequality is immediate; we therefore assume $a \notin K$.

We first show that, for every line ℓ through the origin other than the real axis,

$$h_a(\mathcal{F}_\ell E) \le h_a(E)\,.$$

Since a lies on the negative real ray, $a \in \mathbb{H}_\ell$, while $\mathcal{F}_\ell E$ lies in the closed half-plane $\mathbb{C}\setminus\mathbb{H}_\ell$; hence the Brownian motion from a can meet $\mathcal{F}_\ell E$ only at or after time τ_ℓ. Let μ be the law of Z_{τ_ℓ} on the event $\{\tau_\ell \le \tau_r\}$, a sub-probability measure on $\ell \cap \{|z| \ge r\}$. By the strong Markov property at τ_ℓ,

$$h_a(\mathcal{F}_\ell E) = \int_\ell h_y(\mathcal{F}_\ell E)\ \mathrm{d}\mu(y)\,.$$

We bound $h_a(E)$ from below. On $\{\tau_\ell \le \tau_r\}$, the strong Markov property at τ_ℓ gives

$$\mathbb{P}_a(\tau_E \le \tau_r \,\big|\, (Z_{t\wedge\tau_\ell})_{t\ge 0}) = \mathbf{1}\{\tau_E \le \tau_\ell\} + \mathbf{1}\{\tau_E > \tau_\ell\}\, h_{Z_{\tau_\ell}}(E) \ge h_{Z_{\tau_\ell}}(E)\,.$$

Hence

$$h_a(E) \ge \int_\ell h_y(E)\ \mathrm{d}\mu(y)\,.$$

For each $y \in \ell$ with $|y| > r$, rotate coordinates so that $\ell = \mathbb{R}$ and $\mathbb{H}_\ell$ becomes the lower half-plane; then $\mathcal{F}_\ell E$ is exactly the fold of Lemma 2.48,

so $h_y(E) \geq h_y(\mathcal{F}_\ell E)$. If $|y| = r$, then $h_y(E) = \mathbf{1}_E(y) = \mathbf{1}_{\mathcal{F}_\ell E}(y) = h_y(\mathcal{F}_\ell E)$, because ℓ meets neither $\mathbb{H}_\ell$ nor its reflection. Integrating against μ proves the inequality.

Now fix $\delta > 0$ small enough that $a \notin K^\delta$. The thickening is compact, satisfies $(K^\delta)^* = K^*$, and lies in $\{|z| \geq r\}$, as does every folded image of K^δ, since thickening and folds preserve the modulus. Apply Lemma 2.49 to choose N and lines $\ell_1, \dots, \ell_{2N+1}$. Successive applications of $h_a(\mathcal{F}_\ell E) \leq h_a(E)$ give

$$h_a(K^\delta) \geq h_a(\mathcal{F}_{\ell_{2N+1}} \circ \cdots \circ \mathcal{F}_{\ell_1} K^\delta)\,.$$

The inclusion from Lemma 2.49 and monotonicity of h_a give

$$h_a(\mathcal{F}_{\ell_{2N+1}} \circ \cdots \circ \mathcal{F}_{\ell_1} K^\delta) \geq h_a(K^*)\,.$$

We now remove the angular thickening. As $\delta \downarrow 0$ the compact sets K^δ decrease, with $\bigcap_{\delta>0} K^\delta = K$. Almost surely the stopped path $\{Z_t : 0 \leq t \leq \tau_r\}$ is compact. The stopped trace meets every K^δ exactly when the stopped trace meets K: a sequence of meeting points has a subsequential limit on the stopped trace lying in every K^δ, hence in K. Thus the decreasing events $\{\tau_{K^\delta} \leq \tau_r\}$ have intersection $\{\tau_K \leq \tau_r\}$, and $h_a(K^\delta) \downarrow h_a(K)$ by continuity from above of $\mathbb{P}_a$. Letting $\delta \downarrow 0$ in $h_a(K^\delta) \geq h_a(K^*)$ gives $h_a(K) \geq h_a(K^*)$, proving Theorem 2.46. □

The next lemma computes the probability of reaching $\mathfrak{C}_r$ before the segment $[r, 1]$. A Möbius map sends the outside of $\mathfrak{C}_r$ into the unit disk and the segment onto a radius, and the square root maps the image into a half-disk, where the probability is a sum of two arctangents. Figure 2.23 shows the three domains.

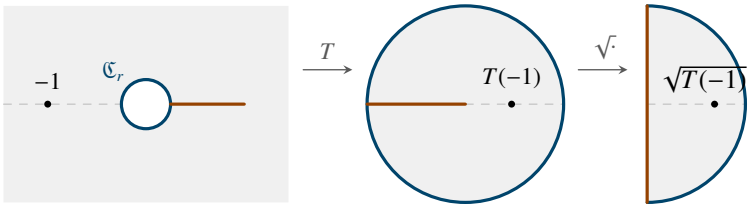


Figure 2.23: The map T in the proof of Lemma 2.50 sends the outside of $\mathfrak{C}_r$ slit along $[r, 1]$ (a) into the unit disk slit along $(-1, 0]$ (b), and the square root opens this onto the right half-disk (c); drawn for $r = 1/4$. The circle (blue) becomes the arc, both sides of the slit (vermilion) become the diameter, and the starting point -1 moves to $\sqrt{T(-1)} = \sqrt{2r/(1+r^2)}$.

Lemma 2.50 *For every* $0 < r < 1$,

$$\mathbb{P}_{-1}\{\tau_r < \tau_{[r,1]}\} = \frac{4}{\pi} \arctan \sqrt{\frac{2r}{1+r^2}}\,.$$

Proof. Let $D := \{|z| > r\} \setminus [r, 1]$. Brownian motion started at -1 hits $\mathfrak{C}_r$ almost surely, so it leaves D almost surely, and the point r is hit with probability zero. We construct a bounded harmonic function on D with boundary values 1 on $\mathfrak{C}_r \setminus \{r\}$ and 0 on $(r, 1]$.

We first map D into a half-disk. Let

$$T(z) := r\,\frac{z-1}{z-r^2} \qquad \text{for } z \in D\,.$$

The pole r^2 lies outside D, and

$$|z-r^2|^2 - r^2|z-1|^2 = (1-r^2)(|z|^2-r^2) \qquad \text{and} \qquad \operatorname{Im} T(z) = \frac{r(1-r^2)\operatorname{Im} z}{|z-r^2|^2}\,.$$

Hence $|T| < 1$ on D. The value $T(z)$ is real only for real z, and for real z it lies in $(-\infty, 0]$ only when $r^2 < z \le 1$, so T takes no value in $(-\infty, 0]$ on D. On $\mathbb{C} \setminus (-\infty, 0]$, let $\sqrt{w}$ be the square root with positive real part, $\sqrt{\rho e^{\mathbf{i}\theta}} = \sqrt{\rho}\, e^{\mathbf{i}\theta/2}$ for $\rho > 0$ and $-\pi < \theta < \pi$. It is continuous, and it is holomorphic, because $(\sqrt{w+\eta} - \sqrt{w})/\eta = 1/(\sqrt{w+\eta} + \sqrt{w}) \to 1/(2\sqrt{w})$ as $\eta \to 0$. Hence $\sqrt{T}$ is holomorphic on D, with $\operatorname{Re}\sqrt{T} > 0$ and $|\sqrt{T}| < 1$.

Next we define the harmonic function on the half-disk. For $x > 0$ and $x^2 + y^2 < 1$, let

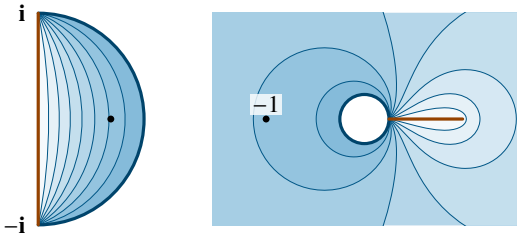
$$v(x + \mathbf{i}y) := \frac{2}{\pi}\Big(\arctan\frac{x}{1-y} + \arctan\frac{x}{1+y}\Big)\,.$$

The function $(s, t) \mapsto \arctan(s/t)$ is harmonic for $t > 0$, since its second derivatives in s and in t are $\mp 2st/(s^2 + t^2)^2$, so v is harmonic. The two arctangents are positive, and the product of their tangents is $x^2/(1 - y^2) < 1$, so their sum lies in $(0, \pi/2)$ and $0 < v < 1$. By Lemma 2.16, $h := v \circ \sqrt{T}$ is a bounded harmonic function on D. Figure 2.24 shows the level curves of v and of h.

We now check the boundary values. On $\mathfrak{C}_r$ we have $|T| = 1$, and $T = -1$ only at $z = r$. Hence $\sqrt{T(z)}$ tends to a point of the unit circle with positive real part as z tends to a point of $\mathfrak{C}_r \setminus \{r\}$. There the product of the two tangents equals 1, so the sum of the arctangents is $\pi/2$, and $h(z) \to 1$. On $(r, 1]$ the map T takes values in $(-1, 0]$, so as z tends to a point of $(r, 1]$, from either side of the segment, every limit point of $\sqrt{T(z)}$ lies on the imaginary axis strictly between $-\mathbf{i}$ and $\mathbf{i}$, where v tends to 0, and $h(z) \to 0$. Lemma 2.43, with $E = \{r\}$ and f the indicator function of $\mathfrak{C}_r$, therefore gives

$$\mathbb{P}_{-1}\{\tau_r < \tau_{[r,1]}\} = h(-1) = \frac{4}{\pi}\arctan\sqrt{\frac{2r}{1+r^2}}\,,$$

because $T(-1) = 2r/(1+r^2)$, so $\sqrt{T(-1)} = \sqrt{2r/(1+r^2)}$ is real and the two arctangents in v coincide there. □



(a) (b)

Figure 2.24: Level curves of v on the right half-disk (a) and of $h = v \circ \sqrt{T}$ outside $\mathfrak{C}_r$ slit along $[r, 1]$ (b) at the values $1/8, \dots, 7/8$, for $r = 1/4$; $h(z)$ is the probability that Brownian motion from z reaches $\mathfrak{C}_r$ (blue) before $[r, 1]$ (vermilion). All level curves of v pass through $\pm\mathbf{i}$, and all those of h pass through r, where the boundary values jump.

Since $\arctan s \le s$ for $s \ge 0$ and $1 + r^2 \ge 1$, Lemma 2.50 gives

$$\mathbb{P}_{-1}\{\tau_r < \tau_{[r,1]}\} \le \frac{4\sqrt{2}}{\pi}\, r^{1/2} \qquad \text{for every } 0 < r < 1\,, \tag{2.21}$$

and since $\arctan s / s \to 1$ as $s \downarrow 0$, the ratio of the two sides of (2.21) tends to 1 as $r \downarrow 0$. For the segment $E = [0, 1]$ and the starting point $a = -1$, the probability in Corollary 2.47 equals the probability of Lemma 2.50, so the constant $4\sqrt{2}/\pi$ there cannot be replaced by a smaller number.

The Beurling estimate, Corollary 2.47, now follows by projecting E and shrinking the projection to the segment $[r, 1]$.

Proof of Corollary 2.47. If $a \in E$ the probability is zero, so assume $a \notin E$. Since $|a| \ge 1$ and $r < 1$, reaching $\mathfrak{C}_r$ requires first crossing $\mathfrak{C}_1$; on $\{\tau_r < \tau_E\}$ the path reaches $\mathfrak{C}_1$, at time τ_1, before $\mathfrak{C}_r$ and before E, so the strong Markov property at τ_1 gives

$$\mathbb{P}_a\{\tau_r < \tau_E\} \le \sup_{y \in \mathfrak{C}_1} \mathbb{P}_y\{\tau_r < \tau_E\}\,.$$

For each $y \in \mathfrak{C}_1$, multiplication by $-\bar{y}$ is a rotation that maps y to -1. Rotational invariance gives $\mathbb{P}_y\{\tau_r < \tau_E\} = \mathbb{P}_{-1}\{\tau_r < \tau_{-\bar{y}E}\}$, and $-\bar{y}E$ is compact and connected, contains $\mathbf{0}$, and meets $\{|z| \ge 1\}$. It therefore suffices to prove the estimate for $a = -1$. Let $K := E \cap \{|z| \ge r\}$, a compact subset of $\{|z| \ge r\}$. The part of E inside $\{|z| < r\}$ cannot be met before $\mathfrak{C}_r$, so $\mathbb{P}_{-1}\{\tau_r < \tau_E\} = \mathbb{P}_{-1}\{\tau_r < \tau_K\}$. Since E is connected, contains $\mathbf{0}$, and meets $\{|z| \ge 1\}$, its circular projection contains $[0, 1]$, so K^* contains the segment $[r, 1]$. Theorem 2.46 and $K^* \supseteq [r, 1]$ give

$$\mathbb{P}_{-1}\{\tau_r < \tau_K\} \le \mathbb{P}_{-1}\{\tau_r < \tau_{K^*}\} \le \mathbb{P}_{-1}\{\tau_r < \tau_{[r,1]}\}\,,$$

and (2.21) bounds the last probability by $(4\sqrt{2}/\pi)\, r^{1/2}$. □

Chapter 3 turns to the lattice $\mathbb{Z}^d$.

EXERCISES

Exercise 2.1. Let $\Omega \subset \mathbb{R}^d$ be open, $u : \Omega \to \mathbb{R}$, and $x \in \Omega$. Show that u is lower semicontinuous at x (Section 2.2) if and only if for every $\epsilon > 0$ there is a neighborhood $U \subset \Omega$ of x such that

$$u(y) \geq u(x) - \epsilon \qquad \text{for every } y \in U\,.$$

Exercise 2.2. Prove Lemma 2.5.

Exercise 2.3. Let $K \subset \mathbb{R}^d$ be nonempty and compact, and let $u : K \to \mathbb{R} \cup \{+\infty\}$ be lower semicontinuous in the relative topology (Section 2.2). Show that $\inf_K u$ is attained.

Exercise 2.4. Let $\Omega \subset \mathbb{R}^d$ be open, $x_0 \in \Omega$, and $\mathcal{F}$ a nonempty family of functions $\Omega \to \mathbb{R} \cup \{+\infty\}$, each lower semicontinuous at x_0 (Section 2.2). Show that the pointwise supremum $s(x) := \sup_{f \in \mathcal{F}} f(x)$ is lower semicontinuous at x_0.

Exercise 2.5. (a) Let $K \subset \mathbb{R}^d$ be compact and $u : K \to \mathbb{R}$ lower semicontinuous in the relative topology (Section 2.2). Show that there is a family $\mathcal{F}$ of continuous functions $K \to \mathbb{R}$ such that for every $x \in K$

$$u(x) = \sup_{f \in \mathcal{F}} f(x)\,.$$

(b) Show that the conclusion remains true for every set $K \subset \mathbb{R}^d$ and every lower semicontinuous function $u : K \to \mathbb{R} \cup \{+\infty\}$, where K has the relative topology.

Exercise 2.6. Let $\Omega \subset \mathbb{R}^d$ be open and $h : \Omega \to \mathbb{R}$ harmonic. Prove that $|h|$ and h^2 are subharmonic on Ω (Definition 2.6). More generally, prove that $f \circ h$ is subharmonic for every convex $f : \mathbb{R} \to \mathbb{R}$.

Exercise 2.7. Prove that a superharmonic function on an open set $\Omega \subset \mathbb{R}^d$, in the sense of Definition 2.6 and without assuming lower semicontinuity, is bounded below on every compact subset $K \subset \Omega$. Show by example that it need not be bounded above on K, even if it is real-valued.

Exercise 2.8. Let $d \geq 2$. Show that each of the functions

$$x_1, \quad x_1 x_2, \quad x_1^2 - x_2^2$$

is harmonic in $\mathbb{R}^d$. Use this to give a second proof of Lemma 2.13.

Exercise 2.9. Consider the function

$$f(x) = \begin{cases} 1, & x \geq 0, \\ 0, & x < 0\,. \end{cases}$$

Is f superharmonic on $\mathbb{R}$ (Definition 2.6)? Explain.

Exercise 2.10. Let μ be a Borel probability measure on $\partial \mathrm{B}(\mathbf{0}, 1) \subset \mathbb{R}^d$. For $r > 0$ and every Borel set A on the sphere, let

$$\mu_r(A) := \int_A \frac{\mu(\mathrm{B}(x, r))}{S\,(\mathrm{B}(x, r) \cap \partial \mathrm{B}(\mathbf{0}, 1))}\, \mathrm{d}S\,(x)\,.$$

Equivalently,

$$\mu_r(A) = \int_A \frac{1}{S\,(\mathrm{B}(x, r) \cap \partial \mathrm{B}(\mathbf{0}, 1))} \int \mathbf{1}_{\{|x-y|<r\}}\, \mathrm{d}\mu(y)\, \mathrm{d}S\,(x)\,,$$

where S is the surface measure on the unit sphere defined in Section 2.2. Show that μ_r converges weakly to μ as $r \to 0$.

Exercise 2.11. Let $d \geq 2$ and let μ be a rotationally invariant Borel probability measure on $\partial \mathrm{B}(\mathbf{0}, 1)$. Show that μ is the normalized surface measure on the sphere (Section 2.2).

Exercise 2.12. Let $\Omega \subset \mathbb{R}^d$ be open, let $x \in \Omega$, and let $u : \Omega \to \mathbb{R}$ be locally integrable and twice differentiable at x. Prove or disprove the following local version of Lemma 2.15: the function u is superharmonic at x if and only if $\Delta u(x) \leq 0$.

Exercise 2.13. Let $\Omega \subset \mathbb{R}^d$ be a bounded domain and let $u \in C^2(\Omega) \cap C(\overline{\Omega})$ satisfy $\Delta u \leq 0$. Show without using Lemmas 2.12 and 2.15 that $\min_{\overline{\Omega}} u = \min_{\partial\Omega} u$.

Hint: first reduce to the case where $\Delta u < 0$ by considering $u(x) - \epsilon|x|^2$.

Exercise 2.14. Consider the semi-infinite strip

$$\Omega := \{(x, y) : |x| < 1 \text{ and } y > 0\}\,.$$

Suppose that u is harmonic on Ω, bounded above, and extends continuously to the open boundary edges, with value 1 on the bottom edge and 0 on the two vertical edges. Assume also that the limit superior of u at each bottom corner is at most 1. Using Corollary 2.11, show that $u(x, y) \leq C \exp(-y)$ for some $C < \infty$. Hint: compare u with a multiple of $\cos(x)e^{-y}$. The bounded-above hypothesis cannot be dropped.

Exercise 2.15. In two dimensions, the maximum principle for subharmonic functions (Lemma 2.12 applied to $-h$) extends to unbounded domains. Let $\Omega \subset \mathbb{R}^2$ be a domain not equal to $\mathbb{R}^2$, and suppose h is subharmonic on Ω,

upper semicontinuous on $\overline{\Omega}$, and satisfies $h(x) = o(\log|x|)$ as $x \to \infty$. Show that $\sup_\Omega h \le \sup_{\partial\Omega} h$.

Hint: fix $y \in \partial\Omega$ and compare with $\log(|x-y|/r)$ outside a small circle about y.

The next exercises replace ball averages by averages against a general measure. Let μ be a compactly supported probability measure on $\mathbb{R}^d$. For an open set $\Omega \subset \mathbb{R}^d$, a point $x \in \Omega$, and a radius $r > 0$ satisfying $x + r\operatorname{supp}\mu \subset \Omega$, write

$$\mathbf{A}_r^\mu u(x) := \int_{\mathbb{R}^d} u(x + ry)\, d\mu(y)\,.$$

A lower semicontinuous function $u : \Omega \to \mathbb{R} \cup \{+\infty\}$ is **μ-superharmonic** if

$$\mathbf{A}_r^\mu u(x) \le u(x) \qquad \text{whenever } x + r\operatorname{supp}\mu \subset \Omega\,. \tag{2.22}$$

A continuous function is **μ-harmonic** if equality holds for every such x and r. When μ is uniform on $\mathrm{B}(\mathbf{0}, 1)$, the operator $\mathbf{A}_r^\mu$ equals the ball average $\mathbf{A}_r$ of Definition 2.1.

Exercise 2.16. Extend Lemma 2.12: suppose $\operatorname{supp}\mu$ contains a neighborhood of $\mathbf{0}$. Let $\Omega \subset \mathbb{R}^d$ be bounded and open, and let u be lower semicontinuous on $\overline{\Omega}$ and μ-superharmonic in Ω. Prove that $\min_{\overline{\Omega}} u = \min_{\partial\Omega} u$.

Exercise 2.17. Let $d \ge 2$. Suppose that μ is invariant under coordinate reflections and interchanges of coordinates. Extending Lemma 2.13, show that x_i is μ-harmonic for $1 \le i \le d$, and that $x_i x_j$ and $x_i^2 - x_j^2$ are μ-harmonic for $1 \le i < j \le d$.

Exercise 2.18. For $d\mu = 2^{-2}\mathbf{1}_{[-1,1]^2}\, dx$, extend Lemma 2.13: show that all harmonic polynomials of degree at most 3 are μ-harmonic. Find a μ-harmonic polynomial of degree 4.

Exercise 2.19. Let $\mu * \mu$ be the law of a sum of two independent vectors with law μ. Show that if h is μ-harmonic on $\mathbb{R}^d$ (compare Definition 2.2), then it is $\mu * \mu$-harmonic on $\mathbb{R}^d$.

Exercise 2.20. Let $(X_n)_{n\ge0}$ be a random walk in $\mathbb{R}^d$ with $X_0 = \mathbf{0}$ and independent increments with law μ. Let T be a bounded integer-valued stopping time, and let μ_T be the law of X_T. Show that every μ-harmonic function on $\mathbb{R}^d$ (compare Definition 2.2) is μ_T-harmonic.

Exercise 2.21*. Suppose that $\int x_i\, d\mu = 0$ and $\int x_i x_j\, d\mu = c\delta_{i,j}$ for $1 \le i, j \le d$, where $c > 0$. Show that if u is μ-harmonic on an open set Ω, then its spherical averages satisfy

$$\mathbf{a}_r u(x) = u(x) \qquad \text{whenever } \overline{\mathrm{B}}(x, r) \subset \Omega\,.$$

Deduce that every μ-harmonic function is harmonic. Hint: differentiate the mean value identity in the distributional sense; use Theorem 2.29.

Exercise 2.22. In Subsection 2.2.1, prove that 1 and the real and imaginary parts of $(x+\mathbf{i}y)^k$ for $k \geq 1$ form a basis for the space of harmonic polynomials in $\mathbb{R}^2$.

Exercise 2.23. In Subsection 2.2.1, find a function that is harmonic in the punctured disk $\mathrm{B}(\mathbf{0}, 1) \setminus \{\mathbf{0}\} \subset \mathbb{R}^2$ but is not the real part of a holomorphic function there.

Exercise 2.24. Extend Lemma 2.16. Let $\Omega_1, \Omega_2 \subset \mathbb{C}$ be open, let $u \in C^2(\Omega_2)$, and let $\psi : \Omega_1 \to \Omega_2$ be holomorphic. Show that

$$\Delta(u \circ \psi)(z) = |\psi'(z)|^2 (\Delta u)(\psi(z)) \,.$$

Exercise 2.25. Extend Lemma 2.16. Let $\Omega_2 \subset \mathbb{C}$ be open and $\Omega_1 \subset \mathbb{C}$ a domain. Show that if u is superharmonic on Ω_2 and $\psi : \Omega_1 \to \Omega_2$ is holomorphic and nonconstant, then $u \circ \psi$ is superharmonic on Ω_1.

Exercise 2.26. Let $d \geq 2$ and $p \in \mathbb{R}[x_1, \ldots, x_d]$ be nonzero. Show that $p(x)|x|^2$ is not harmonic. Hint: compare with the harmonic polynomial extension of p from the unit sphere (Lemma 2.17).

Exercise 2.27. Let $d \geq 2$ and $m \geq 0$ be integers. Let $\mathcal{P}_m(d)$ denote the vector space of homogeneous polynomials of degree m in $\mathbb{R}^d$ and $\mathrm{Harm}_m(d)$ the subspace of $\mathcal{P}_m(d)$ which consists of harmonic polynomials. Use the convention $\mathcal{P}_m(d) = \{0\}$ when $m < 0$.

(a) Show that $\mathcal{P}_m(d) = \mathrm{Harm}_m(d) \oplus |x|^2 \mathcal{P}_{m-2}(d)$.

(b) Show that $\dim(\mathrm{Harm}_m(d)) = \dim(\mathcal{P}_m(d)) - \dim(\mathcal{P}_{m-2}(d))$.

(c) Show that $\dim(\mathcal{P}_m(d)) = \binom{d+m-1}{d-1}$.

(d) For $m \geq 1$, deduce that $\mathrm{Re}\,((x + \mathbf{i}y)^m)$ and $\mathrm{Im}\,((x + \mathbf{i}y)^m)$ form a basis for $\mathrm{Harm}_m(2)$, which re-derives Exercise 2.22.

Exercise 2.28. Extend Lemma 2.17 from spheres to paraboloids. Let $d \geq 2$ and let $p(x) = x_d - \sum_{i=1}^{d-1} x_i^2$. Consider the superlevel set $W := \{x \in \mathbb{R}^d : p(x) > 0\}$. Let $F \in \mathbb{R}[x_1, \ldots, x_d]$ be a polynomial of total degree $m \geq 0$. Show that there is a polynomial h of total degree at most m such that $\Delta h = 0$ on $\mathbb{R}^d$ and $h|_{\partial W} = F|_{\partial W}$. Moreover, establish that $F = pg + h$ for a polynomial g of total degree at most $m - 2$, with $g = 0$ when $m < 2$.

(a) For $M > 0$, let $W_M := W \cap \{x_d < M\}$. Suppose that u is harmonic in W_M, continuous on $\overline{W_M}$, nonpositive on $\partial W \cap \{x_d \leq M\}$, and at most 1 on $W \cap \{x_d = M\}$. Show that

$$u(x) \leq \frac{1}{\cos 1} \exp\left(-\frac{M - x_d}{\sqrt{M}}\right) \qquad \text{for } x \in W_M \,,$$

by comparing u with a harmonic function depending only on x_1 and x_d.

(b) Let u be harmonic in W and continuous on $\overline{W}$. Suppose that $u = 0$ on ∂W and $|u(x)| \le C(1 + |x|)^k$ for constants $C < \infty$ and $k \ge 0$. Use part (a) to prove that $u = 0$ in W.

(c) Conclude by a suitable modification of the proof of Lemma 2.17.

Exercise 2.29. This exercise gives an alternative proof of the converse implication in Lemma 2.15 without using the divergence theorem. Let $\Omega \subset \mathbb{R}^d$ be a bounded domain and let $u \in C^2(\Omega)$ satisfy $\Delta u \le 0$. We will verify the mean value inequality $\mathbf{a}_r u(x) \le u(x)$ for spherical averages whenever $\overline{\mathrm{B}}(x, r) \subset \Omega$. Translation and scaling reduce the proof to the unit ball $B := \mathrm{B}(\mathbf{0}, 1)$.

(a) Let w be the Poisson integral $\mathbf{P}_B u$ of $u|_{\partial B}$ on B (Theorem 2.18), and $w := u$ on ∂B. Show that $w \in C(\overline{B})$ and $\Delta w = 0$ in B.

(b) Apply Exercise 2.13 to $u - w$ to deduce $\mathbf{a}_1 u(\mathbf{0}) = w(\mathbf{0}) \le u(\mathbf{0})$.

Hint for (a): to avoid Lemma 2.15, show that the Poisson kernel has mean one by applying Exercise 2.13 to the Poisson integral h_1 of the constant function 1 from the proof of Theorem 2.18.

Exercise 2.30. Use a compactness argument to prove Lemma 2.21.

Exercise 2.31. This exercise refines Lemma 2.23, which shows that the spherical averages of a continuous superharmonic function decrease with the radius.

(a) Let $d \ge 1$ and $R > 0$. For $t > 0$ when $d \ne 2$, and for $t \in \mathbb{R}$ when $d = 2$, write

$$r(t) = \begin{cases} t, & d = 1, \\ e^t, & d = 2, \\ t^{-1/(d-2)}, & d \ge 3\,. \end{cases}$$

Prove that if u is upper semicontinuous and subharmonic in $\mathrm{B}(\mathbf{0}, R)$, then $\mathbf{a}_{r(t)} u(\mathbf{0})$ is convex on the interval where $0 < r(t) < R$.

(b) Let $0 < R_0 < R_1$. Prove the same statement when u is upper semicontinuous and subharmonic in $\mathrm{B}(\mathbf{0}, R_1) \setminus \overline{\mathrm{B}}(\mathbf{0}, R_0)$, on the interval where $R_0 < r(t) < R_1$.

Hint: $r(t)$ solves a differential equation of the form

$$\frac{\mathrm{d}r}{\mathrm{d}t} = c_d r^{d-1}\,,$$

where c_d is positive when $d = 1, 2$ and negative when $d \ge 3$. Assume first that u is smooth, express $\frac{\mathrm{d}}{\mathrm{d}r}\mathbf{a}_r u(\mathbf{0})$ as an integral involving ∇u and apply the divergence theorem.

Exercise 2.32. By Lemma 2.24, continuity and superharmonicity pass to locally uniform limits. This exercise asks about pointwise infima. Let $d \geq 2$. Let $\Omega \subset \mathbb{R}^d$ be a bounded open set and $F : \partial\Omega \to \mathbb{R}$ a bounded function. Consider the family of continuous superharmonic functions u on Ω with $\liminf_{x \to z} u(x) \geq F(z)$ for every $z \in \partial\Omega$. Let (u_n) be a sequence in this family. Is its pointwise infimum again in the family?

Exercise 2.33. Let u be superharmonic on an open set $\Omega \subset \mathbb{R}^d$.

(a) Prove or disprove a uniform version of Lemma 2.28: $\mathbf{A}_r u \to u$ uniformly on compact subsets of Ω.

(b) Suppose that u is also lower semicontinuous. Does $\mathbf{A}_r u \to u$ uniformly on compact subsets of Ω? Consider $d = 1$ and $d \geq 2$ separately.

Exercise 2.34. Let $\Omega \subset \mathbb{R}^d$ be open, and let u be superharmonic in Ω. Recall that Ω_ϵ consists of the points of Ω at distance $> \epsilon$ from the boundary of Ω. Show that there exist C^∞ superharmonic functions u_n in $\Omega_{1/n}$ such that $u_n \leq u$ in $\Omega_{1/n}$ and $u_n \to u$ almost everywhere in Ω. Show moreover that if u is continuous, then the convergence is uniform on compact subsets of Ω.

Hint: use mollifiers as in Theorem 2.3.

Exercise 2.35. Let $d \geq 1$ and $0 \leq r < 1$. Show that the constants $(1-r)/(1+r)^{d-1}$ and $(1+r)/(1-r)^{d-1}$ in the center bounds (2.11) from the Poisson-kernel proof of Theorem 2.30 are optimal.

Exercise 2.36. Let $d \geq 2$. Let Ω_1 and Ω_2 be bounded domains in $\mathbb{R}^d$ with $\overline{\Omega}_1 \subset \Omega_2$. Show, as in Theorem 2.30, that there is a constant C depending only on Ω_1 and Ω_2 such that for every positive harmonic function $h : \Omega_2 \to \mathbb{R}$ we have $\sup_{\Omega_1} h \leq C \inf_{\Omega_1} h$.

Exercise 2.37. Deduce from Theorem 2.30 that every nonnegative harmonic function in $\mathbb{R}^d$ is constant.

Exercise 2.38. Let $d \geq 1$ and $k \geq 0$, and let h be harmonic in $\mathbb{R}^d$. Show that if $h(x) = O(|x|^k)$ as $|x| \to \infty$, then h is a polynomial in $x_1, \ldots, x_d$ of degree at most $\lfloor k \rfloor$.

Hint: by Lemma 2.20, h equals its convolution with every smooth radial averaging kernel of radius R; differentiate the kernel.

Exercise 2.39. Let $d \geq 2$. Drop the subharmonicity assumption from Proposition 2.32. Must a nonnegative continuous function f on $\mathrm{B}(\mathbf{0}, 2)$ with $f(\mathbf{0}) = 0$ and $f(x) - |x|^2$ superharmonic in $\mathrm{B}(\mathbf{0}, 2)$ satisfy $f(x) \leq 100|x|^2$ in $\mathrm{B}(\mathbf{0}, 1)$?

Exercise 2.40. Let $d \geq 1$, let I be the $d \times d$ identity matrix, and write $D^2 f$ for the Hessian matrix of a function f. Compare Proposition 2.32. True or false: If $f \in C^2(\mathrm{B}(\mathbf{0}, 1))$ is nonnegative with $f(\mathbf{0}) = 0$, and $I - D^2 f$ is nonnegative definite in $\mathrm{B}(\mathbf{0}, 1)$, then $f(x) \leq \frac{1}{2}|x|^2$ for every $x \in \mathrm{B}(\mathbf{0}, 1)$.

Exercise 2.41. To derive (2.14), let $d \geq 2$ and let $f : (0, \infty) \to \mathbb{R}$ be twice continuously differentiable. Show that $x \mapsto f(|x|^2)$ is harmonic in $\mathbb{R}^d \setminus \{\mathbf{0}\}$ if and only if

$$2sf''(s) + df'(s) = 0 \qquad \text{for all } s > 0\,.$$

Check that the solution to this ODE is $f'(s) = cs^{-d/2}$ and deduce that

$$f(s) = \begin{cases} c_1 \log s + c_2, & d = 2, \\ \frac{c_3}{s^{(d-2)/2}} + c_4, & d \geq 3\,. \end{cases}$$

Exercise 2.42. Let $d \geq 2$ and $n \geq 2$. Let $q_1, \ldots, q_n$ be real numbers, and let $y_1, \ldots, y_n$ be distinct points in $\mathbb{R}^d$. Fix an integer i with $1 \leq i \leq n$. Prove Earnshaw's theorem: if $q_i \neq 0$ and $q_j \neq 0$ for at least one $j \neq i$, then $q_i \sum_{j \neq i} q_j \mathcal{G}(x - y_j)$ cannot have a local minimum at $x = y_i$. You may use Lemma 2.8 and the identity theorem: a real analytic function that is constant on a nonempty open subset of a connected open set is constant.

Exercise 2.43. Let n be a positive integer. (a) Prove that for every set of points $z_1, \ldots, z_n$ on the unit circle in $\mathbb{R}^2$, there exists a point z on the unit circle such that $\prod_{k=1}^n |z - z_k| \geq 1$.

(b) Show that for every n, the lower bound can be improved to 2.

(c) Show that the lower bound of 2 is sharp.

Hint for (a): by Lemma 2.33 with $d = 2$, the average of $\log|z - z_k|$ over z on the unit circle is 0.

Exercise 2.44. Prove the case $d \geq 3$ of Lemma 2.37.

Exercise 2.45. Prove the case $d = 2$ of Lemma 2.37.

Exercise 2.46. Verify that the discontinuous function $u(x)$ of Example 2.41 is superharmonic in $\mathbb{R}^d$ for $d \geq 2$.

Exercise 2.47. In contrast with Example 2.41, prove that every lower semicontinuous superharmonic function on $\mathbb{R}$ is continuous and differentiable outside an at most countable set. Show by example that such a function need not be differentiable.

Exercise 2.48. Let $d \geq 2$ and let $B := \mathrm{B}(\mathbf{0}, 1) \subset \mathbb{R}^d$. For a fixed $y \in \partial\mathrm{B}(\mathbf{0}, 1)$ let $p(x) := P(x, y)$ where P is the Poisson kernel for the unit ball. Show, without computing a second derivative, that $\Delta p = 0$ in $\mathrm{B}(\mathbf{0}, 1)$.

Hint: for $0 < r < 1$, the function $x \mapsto \mathcal{G}(x - ry) - \mathcal{G}(rx - y)$ is harmonic on $B \setminus \{ry\}$, and it vanishes identically when $r = 1$; show that its derivative in r at $r = 1$ is $-p/(d\omega_d)$.

Exercise 2.49. Let $D \subset \mathbb{R}^d$ be a bounded domain, and let $Z = (Z_t)_{t \geq 0}$ be Brownian motion started at a point $x \in D$. Let $\tau_{\partial D}$ be its exit time. Suppose that f is continuous and superharmonic in a neighborhood of $\overline{D}$. Show that

$f(Z_{t\wedge\tau_{\partial D}})$ is a supermartingale, and deduce Lemma 2.35 from neighborhood recurrence of planar Brownian motion. Hint: mollify f, and differentiate the Gaussian density to show that $g(Z_t) - \frac{1}{2}\int_0^t \Delta g(Z_s)\,\mathrm{d}s$ is a martingale for every $g \in C_c^2(\mathbb{R}^d)$; stop it at $t \wedge \tau_{\partial D}$.

Exercise 2.50*. Let $s > 0$ and let x be a real number with $0 < x < s$. Show that planar Brownian motion started at x hits the circle of radius s centered at the origin before hitting the imaginary axis with probability

$$\frac{4}{\pi}\arctan\frac{x}{s}\,,$$

and deduce that this probability is at most $4x/(\pi s)$. Hint: reflecting the path across the imaginary axis after its first visit to the imaginary axis shows that the half-disk exit probability equals the difference between the probabilities of exiting $\mathrm{B}(\mathbf{0}, s)$ through the right and the left half-circles; compute both with the harmonic measure (2.19) of the disk.

Exercise 2.51*. Let $Z = (Z_t)_{t\geq 0}$ be planar Brownian motion, with hitting times τ_F as in Section 2.6. Let $K \subset \mathbb{R}^2$ be compact, and suppose that $\mathbb{P}_z\{\tau_K < \infty\} = 1$ for every $z \in \mathbb{R}^2 \setminus K$. Show that, for every continuous function f on K, the following limit exists:

$$\lim_{|z|\to\infty} \mathbb{E}_z[f(Z_{\tau_K})]\,.$$

The limit, as a function of f, is given by integration against a probability measure on K, called **harmonic measure from infinity** on K. Hint: first prove convergence of the harmonic measure of a disk from z, by inversion and (2.19). Then stop Brownian motion on entering a disk that contains K.

Exercise 2.52*. Show that harmonic measure from infinity (Exercise 2.51) on $[-1, 1]$ has density

$$\frac{1}{\pi\sqrt{1-x^2}} \qquad \text{for } -1 < x < 1\,.$$

Hint: use $\varphi(w) = \frac{1}{2}(w + w^{-1})$.

Exercise 2.53*. In the notation of Section 2.6, let $0 < r < R$, and let $K \subset \mathbb{R}^2$ be a compact set with $[r, R] \subseteq \{|y| : y \in K\}$. Show that for every z with $|z| \leq r$,

$$\mathbb{P}_z\{\tau_R < \tau_K\} \leq \frac{4}{\pi}\arctan\sqrt{\frac{2rR}{r^2+R^2}} \leq \frac{4\sqrt{2}}{\pi}\sqrt{\frac{r}{R}}\,,$$

with equality in the first inequality when $K = [r, R]$ and $z = -r$. Deduce that Theorem 2.44 holds with $4\sqrt{2}/\pi$ in place of 4, and that the constant $4\sqrt{2}/\pi$ cannot be replaced by a smaller number. Hint: use the inversion $w \mapsto r/w$ and approximate compact sets from outside by finite unions of disks.

NOTES

Potential theory grew out of Newton's theory of gravitation and Coulomb's electrostatics. For an elementary exposition of harmonic functions, see Axler, Bourdon, and Ramey (2001). Harmonic and superharmonic functions are treated together in (Lieb and Loss, 2001, §9). For the origin of the term "harmonic," see (Axler et al., 2001, p. 25). The Poisson kernel of the disk first appears in §4 of Poisson (1820), pp. 422–423: Poisson sums the series $\frac{1}{2} + \sum_{i\ge 1} r^i \cos i\theta$ in closed form, obtaining one half of the kernel (2.6) for $d = 2$, and shows that integrating it against a function f recovers f as $r \uparrow 1$. Harnack (1887) proved the center bounds (2.11) for $d = 2$ in §19, pp. 61–62, by the argument of our Poisson-kernel proof of Theorem 2.30, and deduced the comparison of Exercise 2.36 for planar domains by a chain of disks. In §20, pp. 67–68, he showed that if a series of harmonic functions of one sign converges at one point of a domain, then it converges at every point of the domain to a harmonic function; this monotone form is usually called Harnack's principle.

The most comprehensive reference on potential theory is Doob (1984). Other classical texts are Helms (2009) and Wermer (1974).

For more on mollifiers, see (Evans, 1998, Appendix C.4).

Definitions of superharmonicity differ on whether lower semicontinuity is included. We follow Lieb and Loss (2001) in omitting lower semicontinuity. By Theorem 2.26(b) every superharmonic function has a lower semicontinuous representative. Helms (2009) discusses several variations on the definition, among them one that asks for the mean value inequality only for small balls centered at each point, together with local integrability and lower semicontinuity; see (Helms, 2009, §2.3).

The Green function (2.14) is the fundamental solution of the negative Laplacian, normalized so that $-\Delta_x \mathcal{G}(x - y) = \delta_y$.

Exercises 2.16–2.21 explore μ-superharmonicity, the variant of Definition 2.6, introduced just before those exercises, in which the ball average is replaced by a general average $\mathbf{A}_r^\mu$.

For another proof of Theorem 2.26, see (Lieb and Loss, 2001, Theorem 9.3).

For $d \ge 2$, the spherical averages $\mathbf{a}_r u$ of a continuous superharmonic function are decreasing in r by Lemma 2.23. They are concave as functions of $\log r$ when $d = 2$ and of r^{2-d} when $d \ge 3$; see (Doob, 1984, §I.II.10).

Exercise 2.28 gives a paraboloid version of Lemma 2.17. Flatto, Newman, and Shapiro (1966) study level curves of harmonic functions in two dimensions.

Exercise 2.51 introduces harmonic measure from infinity as the limiting Brownian hitting distribution seen from points tending to infinity; see also

(Mörters and Peres, 2010, Theorem 3.46). Exercise 2.52 computes this distribution on a line segment, giving the arcsine density; this is the interval example in (Mörters and Peres, 2010, Example 7.24), after an affine change of variables.

The Beurling estimate and the Beurling projection theorem, Theorems 2.44 and 2.46, are due to Beurling. Our proof of the projection theorem follows the proofs of Lemma 1 and Theorem 1 of Øksendal (1983). Werner (1996) gave another proof, based on estimates for the times at which one-dimensional Brownian motion hits moving boundaries, for compact sets that are countable unions of closed path-connected sets (Section 4 of his paper). The proof of Theorem 2.44 rests on the arcsine law, which is the equilibrium measure of a segment for the logarithmic kernel: its logarithmic potential is constant on the segment, as in Lemma 2.42, and it is the harmonic measure of the segment from infinity, as in Exercise 2.52. Chapter 8 carries the same argument over to random walk on $\mathbb{Z}^2$.

The Brownian inputs of Theorem 2.44 are neighborhood recurrence (Theorem 3.20 of Mörters and Peres (2010)), the strong Markov property and rotation invariance, which prove Lemma 2.43 when $E = \emptyset$.

The argument of Subsection 2.6.1 uses in addition the invariance of planar Brownian motion under reflection across a line, which maps Brownian motion from a point to Brownian motion from the reflected point, and the fact that planar Brownian motion almost surely never hits a given point. In the computation of Lemma 2.50, Lemmas 2.16 and 2.43 take the place of the conformal invariance of Brownian motion and harmonic measure, Theorems 7.20 and 7.23 of Mörters and Peres (2010). The solution of Exercise 2.53, in the separate document of solutions, uses the same two lemmas to carry a hitting probability through an inversion, after approximating the image of the compact set under the inversion from outside by finite unions of disks.

Chapter Three

Potential Theory in $\mathbb{Z}^d$

The previous chapter developed potential theory on $\mathbb{R}^d$. We now turn to the lattice $\mathbb{Z}^d$. A function is discrete harmonic when its value at each site is the average of its values at the $2d$ nearest neighbors. Equivalently, a discrete harmonic function evaluated along simple random walk is a martingale. Random-walk identities therefore replace the continuum's integral formulas.

We first prove the discrete minimum principle, Pólya's theorem that simple random walk is recurrent exactly in dimensions one and two, and the existence of the Green function, the fundamental solution of the negative discrete Laplacian. The starred Section 3.6 compares the discrete and continuum Green functions to prove the precise asymptotic: logarithmic growth in dimension two and decay like $|x|^{2-d}$ in dimensions $d \geq 3$. Combined with the continuum theory, this asymptotic also yields the discrete Harnack inequality (Lemma 3.16, proved in Section 3.4 before the starred section), which bounds the variation of a nonnegative discrete harmonic function on a lattice ball over the concentric ball of half the radius.

The chapter's other main result concerns discrete harmonic functions of polynomial growth. In Section 3.5, we prove that every such function is a polynomial and compute the dimension of the space of discrete harmonic polynomials of degree at most k. The proof uses a reflection coupling of two random walks started at neighboring sites.

We finish in the plane, where simple random walk is recurrent and so still hits every nonempty finite set from far away. Section 3.7 proves that the hitting distribution on a nonempty finite set K of a walk started at y converges as $|y| \to \infty$ to a probability measure on K, the harmonic measure of K from infinity, and that this limit is the unique probability measure on K whose potential, formed with the potential kernel, is constant on K. It gives the attachment probabilities of external DLA in Chapter 8. The next chapter begins the study of lattice growth models with the divisible sandpile, and uses the Green function asymptotics of this chapter to show that it fills a ball.

3.1 DISCRETE HARMONIC FUNCTIONS

We begin with the discrete Laplacian and the discrete minimum principle, which together show that a function on a finite set of sites is determined

by its discrete Laplacian and its values on the boundary. Throughout this chapter, $d \geq 1$ is an integer, and two sites $x, y \in \mathbb{Z}^d$ are neighbors, written $x \sim y$, when they differ by one in exactly one coordinate. For a subset A of $\mathbb{Z}^d$, the **outer boundary** of A is

$$\partial A := \{x \in \mathbb{Z}^d \setminus A : x \sim y \text{ for some } y \in A\}.$$

The **closure** of A is $\bar{A} := A \cup \partial A$. For $x \in \mathbb{R}^d$ and real r, the **discrete ball** is $\mathrm{B}^{::}(x, r) := \{y \in \mathbb{Z}^d : |y - x| < r\}$, and we abbreviate $\mathrm{B}^{::}_r := \mathrm{B}^{::}(\mathbf{0}, r)$. Figure 3.1 shows $\mathrm{B}^{::}_5$ and its outer boundary in $\mathbb{Z}^2$.

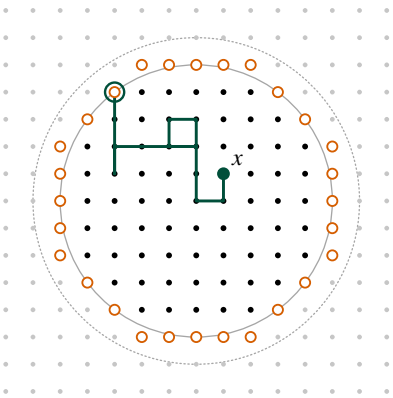


Figure 3.1: The discrete ball $\mathrm{B}^{::}_5$ in $\mathbb{Z}^2$ (black dots) and its outer boundary (open dots), which lies in the annulus $5 \leq |x| < 6$ between the two circles. A simple random walk from x is stopped at its first visit to the outer boundary, as in Lemmas 3.4 and 3.5.

For a function $f : \mathbb{Z}^d \to \mathbb{R}$, define the discrete Laplacian by

$$\Delta^{::} f(x) := \sum_{y \sim x} (f(y) - f(x)). \tag{3.1}$$

The function f is **discrete harmonic** at a site $x \in \mathbb{Z}^d$ if $\Delta^{::} f(x) = 0$, **discrete superharmonic** at x if $\Delta^{::} f(x) \leq 0$, and **discrete subharmonic** at x if $\Delta^{::} f(x) \geq 0$.

On a locally finite undirected simple graph $G = (V, E)$, we use the same normalization: for $f : V \to \mathbb{R}$,

$$\Delta f(v) := \sum_{w \sim v} (f(w) - f(v)), \tag{3.2}$$

where $w \sim v$ means $\{v, w\} \in E$. On $\mathbb{Z}^d$ this is $\Delta^{::}$. A function f on V is harmonic at v if $\Delta f(v) = 0$, superharmonic at v if $\Delta f(v) \leq 0$, and subharmonic at v if $\Delta f(v) \geq 0$.

We will also use the maximum principle on finite subsets of a graph. We record it here in the form needed below.

Lemma 3.1 *Let $G = (V, E)$ be a locally finite connected graph, let A be a nonempty finite proper subset of V, and let $h : V \to \mathbb{R}$ satisfy $\Delta h \geq 0$ on A. Then*

$$\max_{x \in A} h(x) \leq \max_{\substack{x \notin A \\ x \sim y \text{ for at least one } y \in A}} h(x) .$$

Proof. Suppose the maximum on A exceeds every value at a neighboring vertex outside A. At a vertex where this maximum is attained, $\Delta h \geq 0$ forces every neighbor to have the same value. Repeating along a path from that vertex to $V \setminus A$ gives a neighboring vertex outside A with the same value, a contradiction. □

A superharmonic function cannot have a strict minimum away from the outer boundary. Lemma 3.1, applied to the negative of a superharmonic function, gives the following version on $\mathbb{Z}^d$.

Lemma 3.2 (Discrete Minimum Principle) *Let $d \geq 1$, let $A \subset \mathbb{Z}^d$ be finite and nonempty, and let $f : \bar{A} \to \mathbb{R}$ be discrete superharmonic on A. Then*

$$\min_{\bar{A}} f = \min_{\partial A} f .$$

Proof. Suppose that f attains its minimum on $\bar{A}$ at $x_0 \in A$. Superharmonicity forces every neighbor of x_0 to have value $f(x_0)$. Repeating this along a nearest-neighbor path $x_0, x_1, \ldots, x_m$ whose first vertex outside A is x_m gives $f(x_m) = f(x_0)$, as in Figure 3.2. Since $x_m \in \partial A$, the minimum is attained on ∂A. □

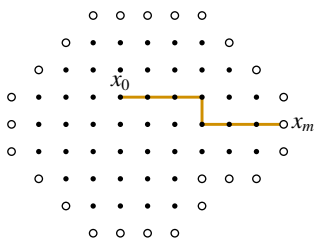


Figure 3.2: The proof of Lemma 3.2: along a nearest-neighbor path from a minimum point $x_0 \in A$ (dots) to its first vertex x_m in ∂A (open dots), superharmonicity forces f to keep the value $f(x_0)$.

Lemma 3.2 shows that the map $u \mapsto \Delta^{::} u|_A$ from functions on $\bar{A}$ that vanish on ∂A to functions on A is injective, hence surjective because both spaces have dimension $|A|$. Thus, for every $F : A \to \mathbb{R}$ and every $b : \partial A \to \mathbb{R}$, the boundary-value problem

$$\begin{aligned} \Delta^{::} u &= F && \text{on } A, \\ u &= b && \text{on } \partial A \end{aligned} \tag{3.3}$$

has a unique solution. The next section gives a random walk formula for the solution when $F = 0$; Exercise 3.20 treats the general case, and Exercise 3.5 shows that the solution with $F = 0$ has the least energy among functions with its boundary values.

3.2 RANDOM WALK AND MARTINGALES

A sequence of random variables $(X_n)_{n\ge 0}$ is a **random walk** if the **increments** $X_{n+1} - X_n$ are independent and identically distributed (**i.i.d.**). A **simple random walk** in $\mathbb{Z}^d$ has each increment uniformly distributed on $\{\pm\mathbf{e}_1, \dots, \pm\mathbf{e}_d\}$. For $x \in \mathbb{Z}^d$, write $\mathbb{P}_x$ for the law of simple random walk started at x, and write $\mathbb{E}_x$ for expectation under $\mathbb{P}_x$.

A martingale has conditionally mean-zero increments, but its increments need not be independent. We recall the definitions. A **filtration** $(\mathcal{F}_n)_{n\ge 0}$ is an increasing sequence of σ-algebras. A sequence of random variables $(M_n)_{n\ge 0}$ such that each M_n is $\mathcal{F}_n$-measurable and integrable is an $(\mathcal{F}_n)$-**martingale** if $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$ almost surely for every $n \ge 0$. Replacing this equality by $\le$ defines an $(\mathcal{F}_n)$-**supermartingale**, and replacing it by $\ge$ defines an $(\mathcal{F}_n)$-**submartingale**. Taking expectations shows that $\mathbb{E}M_n$ is nonincreasing in n for a supermartingale and constant for a martingale. When no filtration is specified, we use the **natural filtration** of the process, in which $\mathcal{F}_n$ is the σ-algebra generated by $M_0, \dots, M_n$. Exercise 3.7 shows that a supermartingale for some filtration is also a supermartingale for its natural filtration.

A random variable T with values in $\{0, 1, 2, \dots\} \cup \{\infty\}$ is an $(\mathcal{F}_n)$-**stopping time** if $\{T \le n\} \in \mathcal{F}_n$ for every $n \ge 0$, so that whether $T \le n$ is determined by the information available at time n. If $(M_n)_{n\ge 0}$ is a supermartingale and T is a stopping time, then the stopped process $(M_{T\wedge n})_{n\ge 0}$ is again a supermartingale, because $M_{T\wedge(n+1)} - M_{T\wedge n} = (M_{n+1} - M_n)\mathbf{1}\{T > n\}$ and $\{T > n\} \in \mathcal{F}_n$. Applying this to a martingale and to its negative gives

$$\mathbb{E}[M_{T\wedge n}] = \mathbb{E}[M_0] \qquad \text{for every } n \ge 0\,.$$

If $(X_n)_{n\ge 0}$ is simple random walk and $h : \mathbb{Z}^d \to \mathbb{R}$ is discrete harmonic, then $h(X_n)$ is a martingale. If h is discrete superharmonic, then $h(X_n)$ is a supermartingale. For a random walk $(X_t)_{t\ge 0}$ and a set $B \subset \mathbb{Z}^d$, the **hitting time** of B is

$$\tau_B := \inf\{t \ge 0 : X_t \in B\}\,. \tag{3.4}$$

Doob's optional stopping theorem gives conditions under which we may let $n \to \infty$ in the identity $\mathbb{E}[M_{T\wedge n}] = \mathbb{E}[M_0]$.

Theorem 3.3 (Doob's Optional Stopping Theorem) *Let $(M_n)_{n\geq 0}$ be a martingale with respect to a filtration $(\mathcal{F}_n)_{n\geq 0}$, and let T be a stopping time that is finite almost surely. Suppose that at least one of the following conditions holds:*

(a) $|M_{n\wedge T}| \leq Z$ almost surely for every $n \geq 0$, where Z is an integrable random variable;

(b) $\mathbb{E}[T] < \infty$ and there is $K > 0$ such that $\sup_{n\geq 1} |M_n - M_{n-1}| \leq K$ almost surely.

Then M_T is integrable and

$$\mathbb{E}[M_T] = \mathbb{E}[M_0]\,.$$

Proof. For every $n \geq 0$, we have $\mathbb{E}[M_{T\wedge n}] = \mathbb{E}[M_0]$. Under condition (a), dominated convergence permits $n \to \infty$. Under condition (b),

$$|M_{T\wedge n} - M_0| \leq \sum_{k=1}^{T\wedge n} |M_k - M_{k-1}| \leq KT\,.$$

The random variable KT is integrable, so dominated convergence again permits $n \to \infty$. □

Exercise 3.8 applies Theorem 3.3 to one-dimensional random walk.

The integrability hypotheses cannot be omitted. For simple random walk on $\mathbb{Z}$ started at zero, the time $\tau_{\{1\}}$ is finite almost surely; this follows by letting $\ell \to \infty$ in Exercise 3.8(a). We have $\mathbb{E}X_n = 0$ for every n, but $X_{\tau_{\{1\}}} = 1$. Exercise 3.10 constructs a stopping time T for which X_T is not integrable.

Stopping first at $T \wedge n$ and then letting $n \to \infty$ often works even when Theorem 3.3 does not apply directly. We now apply the stopped martingale identity to the exit time from a ball.

Lemma 3.4 *Let $d \geq 1$, let $r > 0$, and let $B := \mathrm{B}^{\cdot\cdot}(\mathbf{0}, r)$. Then, for every $x \in B$,*

$$r^2 - |x|^2 \leq \mathbb{E}_x[\tau_{\partial B}] < (r+1)^2 - |x|^2\,.$$

Proof. Since $|X_t|^2 - t$ is a martingale, optional stopping at the bounded time $\tau_{\partial B} \wedge n$ gives, for every $n \in \mathbb{N}$,

$$|x|^2 = \mathbb{E}_x[|X_{\tau_{\partial B}\wedge n}|^2] - \mathbb{E}_x[\tau_{\partial B} \wedge n]\,.$$

Since $|X_{\tau_{\partial B}\wedge n}| < r + 1$, the stopped identity gives

$$\mathbb{E}_x[\tau_{\partial B} \wedge n] \leq (r+1)^2 - |x|^2\,.$$

Monotone convergence yields $\mathbb{E}_x[\tau_{\partial B}] < \infty$, so $\tau_{\partial B} < \infty$ almost surely. Dominated convergence now applies to the stopped positions. Since $X_{\tau_{\partial B}} \in \partial B$, we have $r \le |X_{\tau_{\partial B}}| < r + 1$, which proves both bounds. □

Simple random walk also gives discrete harmonic functions with prescribed boundary values.

Lemma 3.5 *Let $d \ge 1$, let $A \subset \mathbb{Z}^d$ be finite, and let $h_0 : \partial A \to \mathbb{R}$. Let $h : \bar{A} \to \mathbb{R}$ be given by*

$$h(x) := \mathbb{E}_x[h_0(X_{\tau_{\partial A}})] .$$

Then $h = h_0$ on ∂A, and h is discrete harmonic in A.

Proof. Choose $r > 0$ such that $\bar{A} \subset \mathrm{B}^{::}(\mathbf{0}, r)$. A walk started in $\bar{A}$ must hit ∂A before $\partial \mathrm{B}^{::}(\mathbf{0}, r)$, so Lemma 3.4 gives $\tau_{\partial A} \le \tau_{\partial \mathrm{B}^{::}(\mathbf{0},r)} < \infty$ almost surely. By the Markov property, conditioning on the first step gives, for every $x \in A$,

$$h(x) = \mathbb{E}_x[h_0(X_{\tau_{\partial A}})] = \frac{1}{2d} \sum_{y \sim x} \mathbb{E}_y[h_0(X_{\tau_{\partial A}})] = \frac{1}{2d} \sum_{y \sim x} h(y) .$$

Thus h is discrete harmonic in A. For $x \in \partial A$, the stopping time $\tau_{\partial A}$ equals zero, so $h(x) = h_0(x)$. □

For a finite set $A \subset \mathbb{Z}^d$, a site $x \in \bar{A}$ and $B \subset \partial A$, the **harmonic measure** of B seen from x is the probability $\mathbb{P}_x\{X_{\tau_{\partial A}} \in B\}$ that the walk from x first leaves A through B. Thus Lemma 3.5 says that the harmonic function with boundary values h_0 is the average of h_0 against harmonic measure.

3.2.1 Conditioned Random Walks

Harmonic measure tells us where a walk leaves a finite set. How does the walk move if we know where it will leave?

Let $A \subset \mathbb{Z}^d$ be finite and connected, and let B be a nonempty subset of ∂A. The exit probability $h(x) := \mathbb{P}_x\{X_{\tau_{\partial A}} \in B\}$ for $x \in \bar{A}$ is discrete harmonic in A by Lemma 3.5, equals $\mathbf{1}_B$ on ∂A, and is positive on A, since A is connected. The **Doob h-transform** is the walk on $\bar{A}$, stopped on reaching ∂A, with transition probabilities

$$p^h(x, y) := \frac{h(y)}{2d\, h(x)} \qquad \text{for } x \in A \text{ and } y \sim x . \tag{3.5}$$

They sum to one because h is harmonic at x, and they vanish when $y \in \partial A \setminus B$. Along a nearest-neighbor path $x_0, \dots, x_n$ with $x_0, \dots, x_{n-1} \in A$, the product

telescopes:

$$\prod_{j=1}^{n} p^h(x_{j-1}, x_j) = (2d)^{-n} \frac{h(x_n)}{h(x_0)} . \tag{3.6}$$

By the Markov property at time n, the right side is also the probability that simple random walk from x_0 follows this path, conditional on $X_{\tau_{\partial A}} \in B$. Thus the h-transform is simple random walk conditioned to leave A through B (see Figure 3.3). Exercises 3.13 and 3.31 condition a walk never to leave a half-space or never to hit the origin.

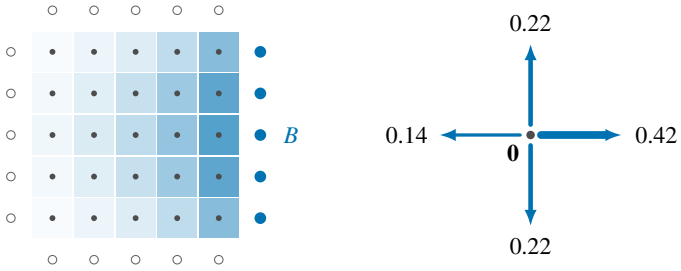


Figure 3.3: The probability h of leaving $A = \{-2, \dots, 2\}^2$ through its right side B (left), and the transition probabilities $p^h(\mathbf{0}, y) = h(y)$ from the center, where $h(\mathbf{0}) = \frac{1}{4}$ by symmetry (right). The step toward B is about three times as likely as the step away.

Harmonic measure solves the Dirichlet problem on a finite set. What replaces it when the equation carries a source term, or when the set is all of $\mathbb{Z}^d$?

3.3 GREEN FUNCTION

To solve $\Delta^{::} F = f$ while controlling the growth of F, we use the **Green function**, also called the **fundamental solution**, whose convolution formula inherits its growth from a single kernel. (Prescribing F on two adjacent parallel hyperplanes determines a solution recursively, but gives no useful growth bound; Exercise 3.14 asks you to prove the first claim.)

We seek a function $g^{::} : \mathbb{Z}^d \to \mathbb{R}$ satisfying

$$\Delta^{::} g^{::}(x) = -\delta_{\mathbf{0}}(x) = \begin{cases} -1, & x = \mathbf{0}, \\ 0, & x \neq \mathbf{0} . \end{cases} \tag{3.7}$$

If $f : \mathbb{Z}^d \to \mathbb{R}$ has finite support, then $F(x) := -\sum_{y \in \mathbb{Z}^d} f(y) g^{::}(x - y)$ satisfies $\Delta^{::} F(x) = \sum_{y \in \mathbb{Z}^d} f(y) \delta_{\mathbf{0}}(x - y) = f(x)$.

Let $(X_k)_{k\ge 0}$ be simple random walk in $\mathbb{Z}^d$. For $n \ge 0$, write

$$g_n^{::}(x) := \frac{1}{2d}\mathbb{E}_{\mathbf{0}} \sum_{k=0}^{n} \mathbf{1}\{X_k = x\}\,.$$

Conditioning on the first step gives

$$\begin{aligned} 2d g_{n+1}^{::}(x) &= \delta_{\mathbf{0}}(x) + \mathbb{E}_{\mathbf{0}} \sum_{k=1}^{n+1} \mathbf{1}\{X_k = x\} \\ &= \delta_{\mathbf{0}}(x) + \sum_{z\sim\mathbf{0}} g_n^{::}(x-z)\,. \end{aligned}$$

Therefore, for every $x \in \mathbb{Z}^d$,

$$\Delta^{::} g_n^{::}(x) = -\delta_{\mathbf{0}}(x) + \mathbb{P}_{\mathbf{0}}\{X_{n+1} = x\}\,. \tag{3.8}$$

Whether $g_n^{::}(\mathbf{0})$ stays bounded as $n \to \infty$ is the question of recurrence. Write $b_m := 4^{-m}\binom{2m}{m}$ for the probability that one-dimensional simple random walk is back at its starting point at time $2m$. Stirling's formula gives

$$b_m \sim \frac{1}{\sqrt{\pi m}} \qquad \text{as } m \to \infty\,. \tag{3.9}$$

In the plane, the diagonal coordinates $U_k := X_k\cdot(\mathbf{e}_1+\mathbf{e}_2)$ and $V_k := X_k\cdot(\mathbf{e}_1-\mathbf{e}_2)$ change by ± 1 at each step, and the four pairs of signs of $(U_{k+1}-U_k, V_{k+1}-V_k)$ are equally likely (Figure 3.4(a)). Hence U and V are independent one-dimensional simple random walks started at 0, and

$$\mathbb{P}_{\mathbf{0}}\{X_k = x\} = \mathbb{P}\{U_k = x_1 + x_2\}\,\mathbb{P}\{V_k = x_1 - x_2\} \qquad \text{for } x \in \mathbb{Z}^2\,. \tag{3.10}$$

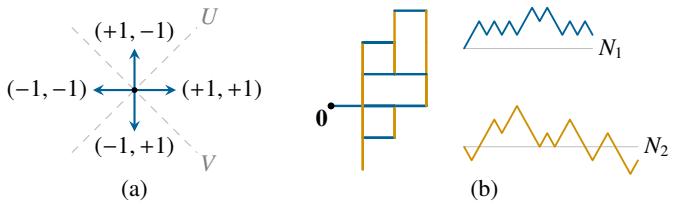


Figure 3.4: (a) The four steps of simple random walk in $\mathbb{Z}^2$, labeled by the increments of $U = x_1 + x_2$ and $V = x_1 - x_2$ (dashed diagonals), as in (3.10). (b) A walk of 40 steps with its horizontal (blue) and vertical (orange) steps, and each coordinate as a one-dimensional walk run for its own number N_i of steps, as in the proof of Lemma 3.6.

Lemma 3.6 *Simple random walk in $\mathbb{Z}^d$ is recurrent in dimensions $d = 1, 2$ and transient in dimensions $d \ge 3$.*

Proof. By the strong Markov property, the probability of another return is the same after each visit to the origin, so the number of visits is geometric and its expectation $2d \lim_{n\to\infty} g_n^{::}(\mathbf{0})$ is infinite exactly when the return probability equals one.

The return probability at time $2m$ is b_m for $d = 1$ and, by (3.10), b_m^2 for $d = 2$. By (3.9) these are of order $m^{-1/2}$ and m^{-1}, so their sums diverge, and the walk is recurrent.

For $d \geq 3$, we bound the probability of each site at time $n \geq 1$. At time k, every probability of one-dimensional simple random walk is at most $b_{\lfloor k/2 \rfloor}$, by the unimodality of binomial coefficients and, for odd k, Pascal's rule; by (3.9) it is therefore at most $C(k+1)^{-1/2}$. Let N_i be the number of the first n steps that change the ith coordinate. Given $N_1, \dots, N_d$, the coordinates of X_n are independent, and the ith has the law of one-dimensional simple random walk at time N_i (Figure 3.4(b)). If every N_i is at least $n/(2d)$, the conditional probability that $X_n = x$ is therefore at most $C_d n^{-d/2}$; it is always at most $C_d n^{-1/2}$, because some N_i is at least n/d. Each N_i is binomial with parameters n and $1/d$, so Chebyshev's inequality gives $\mathbb{P}\{N_i < n/(2d)\} \leq 4d/n$. Hence

$$\sup_{x\in\mathbb{Z}^d} \mathbb{P}_{\mathbf{0}}\{X_n = x\} \leq C_d n^{-d/2} + C_d n^{-1/2} \cdot \frac{4d^2}{n} \leq C_d n^{-3/2} . \tag{3.11}$$

These bounds are summable, so $g_n^{::}(\mathbf{0})$ is bounded, and the walk is transient. □

For $d \geq 3$, Lemma 3.6 allows us to define the Green function by

$$g^{::}(x) := \lim_{n\to\infty} g_n^{::}(x) = \frac{1}{2d} \sum_{k\geq 0} \mathbb{P}_{\mathbf{0}}\{X_k = x\} .$$

The terms of this convergent series tend to zero, so (3.8) gives $\Delta^{::} g^{::} = -\delta_{\mathbf{0}}$.

In dimensions $d = 1, 2$, recurrence makes $g_n^{::}(x)$ diverge. Define the **potential kernel** by

$$a(x) := \lim_{n\to\infty} \left(g_n^{::}(\mathbf{0}) - g_n^{::}(x)\right) . \tag{3.12}$$

In dimension one, a direct calculation gives $a(x) = |x|/2$; in dimension two, the next two lemmas prove that the limit exists. Once it exists, equation (3.8) gives $\Delta^{::} a = \delta_{\mathbf{0}}$, because $\mathbb{P}_{\mathbf{0}}\{X_{n+1} = x\} \to 0$ by (3.10) and the one-dimensional bound in the proof of Lemma 3.6. We write $g^{::} := -a$ for $d = 1, 2$, so that $\Delta^{::} g^{::} = -\delta_{\mathbf{0}}$ in every dimension. We focus on dimensions $d \geq 2$.

We first compute the potential kernel along the diagonal, where the walks U and V of (3.10) make the sums explicit. For integers $m \geq 0$ and j, let $p_m(j) := 4^{-m}\binom{2m}{m+j}$, with $p_m(j) = 0$ when $|j| > m$, be the probability that

one-dimensional simple random walk is at $2j$ at time $2m$; thus $p_m(0) = b_m$. For integers k, l, the site $x = (k + l, k - l)$ has $x_1 + x_2 = 2k$ and $x_1 - x_2 = 2l$. The walk visits such a site only at even times, so by (3.10),

$$g^{::}_{2M}(\mathbf{0}) - g^{::}_{2M}(x) = \frac{1}{4} \sum_{m=0}^{M} [b_m^2 - p_m(k)p_m(l)] , \tag{3.13}$$

and the same identity is true with $2M + 1$ in place of $2M$. Every summand is nonnegative, because $p_m(j) \le b_m$ by the unimodality of binomial coefficients.

Lemma 3.7 *In dimension two, the limit* (3.12) *exists at the diagonal sites* (n, n)*, and for every integer* $n \ge 0$*,*

$$a(n, n) = \frac{1}{\pi} \sum_{j=1}^{n} \frac{1}{2j - 1} .$$

Proof. The site (n, n) corresponds to $k = n$ and $l = 0$ in (3.13). For $j, M \ge 0$, write

$$D_M(j) := \frac{1}{4} \sum_{m=0}^{M} b_m [b_m - p_m(j)] ,$$

so that $D_M(n) = g^{::}_{2M}(\mathbf{0}) - g^{::}_{2M}(n, n)$. We compute each increment $D_M(j) - D_M(j - 1)$ by telescoping. Fix $j \ge 1$, and let $t_m := b_m p_m(j - 1)$. For $m \ge j - 1$, the ratios of binomial coefficients are

$$\frac{p_m(j)}{p_m(j - 1)} = \frac{m - j + 1}{m + j} \qquad \text{and} \qquad \frac{t_{m+1}}{t_m} = \frac{(2m + 1)^2}{4(m + j)(m - j + 2)} .$$

The first ratio gives $b_m[p_m(j - 1) - p_m(j)] = (2j - 1)t_m/(m + j)$, and the second, together with the identity $(2m+1)^2 - 4(m-j+1)(m+j) = (2j-1)^2$, gives

$$b_m[p_m(j - 1) - p_m(j)] = \frac{4}{2j - 1} [(m - j + 2)t_{m+1} - (m - j + 1)t_m] .$$

Both sides vanish for $m < j - 1$, and the last term vanishes at $m = j - 1$, so summing over $m \le M$ gives

$$D_M(j) - D_M(j - 1) = \frac{(M - j + 2)\, t_{M+1}}{2j - 1} \qquad \text{for } M \ge j - 1 .$$

For fixed j, the ratio $p_m(j - 1)/b_m = \prod_{i=1}^{j-1} (m - i + 1)/(m + i)$ tends to 1, so $t_m \sim b_m^2 \sim 1/(\pi m)$ by (3.9). Hence $D_M(j) - D_M(j - 1) \to 1/(\pi(2j - 1))$ as $M \to \infty$. Since $D_M(0) = 0$, summing over $1 \le j \le n$ shows that $D_M(n)$

converges to the stated value. The odd truncations give the same sums, so the limit (3.12) exists at (n, n). □

The sums $D_M(j)$ increase with M, and so do their increments $D_M(j+1) - D_M(j)$, which are sums of nonnegative terms by the proof above; hence

$$D_M(j) \le a(j,j) \qquad \text{and} \qquad D_M(j+1) - D_M(j) \le \frac{1}{\pi(2j+1)}\,. \tag{3.14}$$

The diagonal values control the potential kernel everywhere.

Lemma 3.8 *In dimension two, the limit* (3.12) *exists at every* $x \in \mathbb{Z}^2$. *The potential kernel vanishes at the origin and satisfies* $\Delta^{\cdot\cdot} a = \delta_{\mathbf{0}}$ *and* $a(\mathbf{e}_1) = \frac{1}{4}$. *There is a constant* C *such that*

$$\Big| a(x) - \frac{1}{2\pi} \log(1 + |x|) \Big| \le C \qquad \textit{for every } x \in \mathbb{Z}^2\,.$$

Proof. We first bound a coordinate increment of the sums (3.13) by a diagonal increment. For integers k, l and $M \ge 0$, write

$$A_M(k,l) := \frac{1}{4} \sum_{m=0}^{M} [b_m^2 - p_m(k) p_m(l)]\,.$$

Let $k, l \ge 0$ and $r := \max\{k, l\}$. For $m \ge j \ge 0$, the binomial ratios give

$$p_m(j) - p_m(j+1) = p_m(j)\, \frac{2j+1}{m+j+1}\,,$$

and the fraction increases with j. If $r = k$, then we use $p_m(l) \le b_m$; if $r = l$, then we use $p_m(k) \le b_m$ and compare the fractions at k and l. In both cases

$$0 \le p_m(l)[p_m(k) - p_m(k+1)] \le b_m[p_m(r) - p_m(r+1)] \qquad \text{for } m \ge r\,,$$

and for $m < r$ the middle term vanishes. Summing over $m \le M$ and using (3.14) gives

$$0 \le A_M(k+1, l) - A_M(k, l) \le D_M(r+1) - D_M(r) \le \frac{1}{\pi(2r+1)}\,.$$

By symmetry, $A_M(k, l)$ depends only on $|k|$ and $|l|$ and is symmetric in them. Let $r := \max\{|k|, |l|\}$ and $s := \min\{|k|, |l|\}$. Since $A_M(r, 0) = D_M(r)$, adding this bound along the s steps from $(r, 0)$ to (r, s) and using (3.14) gives

$$D_M(r) \le A_M(k,l) \le a(r,r) + \frac{s}{\pi(2r+1)} \le a(r,r) + \frac{1}{2\pi}\,. \tag{3.15}$$

Next we pass to the limit. At the site $x = (k+l, k-l)$, the sums (3.13) are $A_M(k,l)$, which increase with M and are bounded by (3.15). Hence the limit (3.12) exists at every site x with $x_1 + x_2$ even. Every other site x is not the origin, and its four neighbors are of this kind. At such a site, equation (3.8) reads

$$4(g_n^{::}(\mathbf{0}) - g_n^{::}(x)) = \sum_{y \sim x} (g_n^{::}(\mathbf{0}) - g_n^{::}(y)) + \mathbb{P}_{\mathbf{0}}\{X_{n+1} = x\}\,,$$

and the last term tends to zero, so the limit exists there as well. The identity $\Delta^{::} a = \delta_{\mathbf{0}}$ was shown above, and $a(\mathbf{0}) = 0$ by definition. At the origin, $\Delta^{::} a(\mathbf{0}) = 1$ is the sum of the four values $a(\pm \mathbf{e}_i)$, which are equal by symmetry, so $a(\mathbf{e}_1) = \frac{1}{4}$.

Finally we compare a with the logarithm. Comparing the sum in Lemma 3.7 with an integral gives $|a(r,r) - \frac{1}{2\pi}\log(1+r)| \le C$ for every $r \ge 0$. At the site $x = (k+l, k-l)$ we have $|x|^2 = 2(k^2 + l^2)$, so $\sqrt{2}\, r \le |x| \le 2r$. Thus (3.15), in the limit $M \to \infty$, gives the stated bound at every site with $x_1 + x_2$ even. At every other site, $a(x)$ is the average of its four neighbors, whose norms differ from $|x|$ by at most one, which gives the bound with a larger constant. □

We next determine the growth of $g^{::}(x)$ as $|x| \to \infty$.

Theorem 3.9 *Let $d \ge 2$. If $d \ge 3$, then, as $|x| \to \infty$,*

$$g^{::}(x) = a_d |x|^{2-d} + O(|x|^{-d})\,, \tag{3.16}$$

where $a_d = 1/(d(d-2)\omega_d)$ and ω_d is the volume of $\mathrm{B}(\mathbf{0}, 1) \subset \mathbb{R}^d$.

If $d = 2$, then, as $|x| \to \infty$,

$$g^{::}(x) = -a(x) = -\frac{1}{2\pi}\log|x| - \kappa + O(|x|^{-2})\,, \tag{3.17}$$

where $\kappa = (2\gamma_E + \log 8)/(4\pi)$ and γ_E is Euler's constant.

The leading term in Theorem 3.9 is radial, although the lattice is not (Figure 3.5). When $d \ge 3$ the error is two powers of $|x|$ smaller than the leading term; when $d = 2$ it is $O(|x|^{-2})$, two powers below the constant term κ. Section 3.6 proves the theorem by comparing $g^{::}$ with a corrected continuum Green function. Exercise 3.29 uses the error term in (3.16) to compare the level sets of $g^{::}$ with concentric Euclidean balls.

Let $A \subset \mathbb{Z}^d$ be such that simple random walk killed on leaving A is transient, as it is for every finite A and, when $d \ge 3$, for every A. The killed Green function is

$$g_A^{::}(x,y) := \frac{1}{2d} \sum_{n \ge 0} \mathbb{P}_x\{X_n = y,\ n < \tau_{A^c}\}\,. \tag{3.18}$$

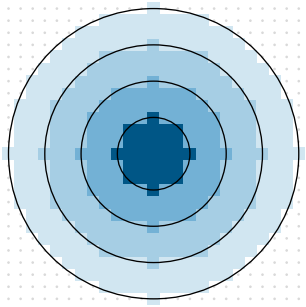


Figure 3.5: The sets $\{x \in \mathbb{Z}^2 : a(x) < \frac{1}{2\pi}\log\rho + \kappa\}$ for $\rho = 3, 6, 9, 12$ (darker for smaller ρ), against the circles of radius ρ: already at these radii the potential kernel is nearly radial, as Theorem 3.9 predicts.

By definition, $g_A^{::}(x,y) = 0$ whenever $x \notin A$ or $y \notin A$. For each $n \geq 0$, the number $(2d)^n \mathbb{P}_x\{X_n = y,\ n < \tau_{A^c}\}$ counts the paths of length n in A from x to y. Reversing each path gives $g_A^{::}(x,y) = g_A^{::}(y,x)$. Conditioning on the first step and using this symmetry gives the Poisson equation

$$\Delta_y^{::} g_A^{::}(x,y) = -\delta_x(y) \qquad \text{for } x \in \mathbb{Z}^d \text{ and } y \in A\,. \tag{3.19}$$

When A is finite, Exercise 3.18 asks you to extend this identity to all of $\mathbb{Z}^d$ by including the exit distribution on ∂A.

In the plane, simple random walk killed at the origin is transient, and its killed Green function gives a probabilistic interpretation of the potential kernel.

Lemma 3.10 *For all $x, y \in \mathbb{Z}^2 \setminus \{\mathbf{0}\}$,*

$$g_{\mathbb{Z}^2\setminus\{\mathbf{0}\}}^{::}(x,y) = a(x) + a(y) - a(x-y)\,.$$

Proof. Fix $y \neq \mathbf{0}$, and let $f(x) := a(x) + a(y) - a(x-y)$. By Lemma 3.8, f is bounded, since $\log(1+|x|) - \log(1+|x-y|)$ is bounded for fixed y. The walk is symmetric, so a is even, $f(\mathbf{0}) = 0$, and $\Delta^{::} f = \delta_{\mathbf{0}} - \delta_y$. Start the walk at $x \neq \mathbf{0}$. Since $\Delta^{::} f = -\delta_y$ off the origin, the process

$$f(X_{n\wedge\tau_{\mathbf{0}}}) + \frac{1}{4}\sum_{k<n\wedge\tau_{\mathbf{0}}} \mathbf{1}\{X_k = y\}$$

is a martingale, so

$$f(x) = \mathbb{E}_x f(X_{n\wedge\tau_{\mathbf{0}}}) + \frac{1}{4}\mathbb{E}_x \sum_{k<n\wedge\tau_{\mathbf{0}}} \mathbf{1}\{X_k = y\}\,.$$

The walk hits the origin almost surely: by Lemma 3.6 it returns to x infinitely often, its excursions from x are independent by the strong Markov property,

and each of them visits the origin with the same positive probability. As $n \to \infty$, the bounded convergence theorem sends the first term to $f(\mathbf{0}) = 0$, and the monotone convergence theorem sends the second to $\frac{1}{4}$ times the expected number of visits to y before $\tau_{\mathbf{0}}$ (Figure 3.6), which is $g^{\cdot\cdot}_{\mathbb{Z}^2\setminus\{\mathbf{0}\}}(x, y)$ by (3.18). □

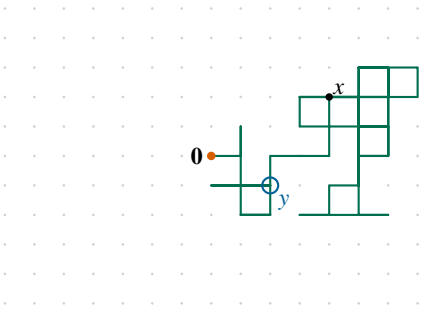


Figure 3.6: A walk from x stopped when it hits the origin (vermilion), with its visits to y (circled). By Lemma 3.10, the expected number of such visits is $4g^{\cdot\cdot}_{\mathbb{Z}^2\setminus\{\mathbf{0}\}}(x, y) = 4(a(x) + a(y) - a(x - y))$.

The next lemma collects the preliminary bounds from which the proof of Theorem 3.9 in Section 3.6 starts.

Lemma 3.11 *For each $d \geq 3$, the Green function $g^{\cdot\cdot}$ is positive and bounded, and $g^{\cdot\cdot}(x) \to 0$ as $|x| \to \infty$. In dimension $d = 2$, there is a constant $C < \infty$ such that*

$$0 \leq a(x) \leq C \log(1 + |x|) \qquad \text{for every } x \in \mathbb{Z}^2 .$$

Proof. For $d \geq 3$, if $|x_1| + \cdots + |x_d| > N \geq 0$, then the walk is not at x before time $N + 1$, so (3.11) gives $g^{\cdot\cdot}(x) \leq C_d \sum_{n>N} n^{-3/2}$, which is finite for $N = 0$ and tends to zero as $N \to \infty$. Every site is reached with positive probability, so $g^{\cdot\cdot} > 0$. For $d = 2$, Lemma 3.10 with $y = x$ and $a(\mathbf{0}) = 0$ gives $2a(x) = g^{\cdot\cdot}_{\mathbb{Z}^2\setminus\{\mathbf{0}\}}(x, x) \geq 0$ for $x \neq \mathbf{0}$, and the upper bound follows from Lemma 3.8. □

Theorem 3.9 and the minimum principle determine the killed Green function of a large discrete ball with pole at its center; Exercise 3.24 gives a probabilistic derivation of the planar leading term.

Lemma 3.12 *Let $d \geq 2$, let $r \geq 2$ be real, and let B be the discrete ball $\mathrm{B}^{\cdot\cdot}(\mathbf{0}, r)$ in $\mathbb{Z}^d$.*

(a) For every $x \in B \setminus \{\mathbf{0}\}$,

$$g^{\cdot\cdot}_B(\mathbf{0}, x) = \begin{cases} \dfrac{1}{2\pi} \log \dfrac{r}{|x|} + O(r^{-1} + |x|^{-2}), & d = 2; \\ a_d(|x|^{2-d} - r^{2-d}) + O(r^{1-d} + |x|^{-d}), & d \geq 3 , \end{cases}$$

where $a_d = 1/(d(d-2)\omega_d)$ *for* $d \geq 3$ *and* ω_d *is the volume of* $\mathrm{B}(\mathbf{0}, 1) \subset \mathbb{R}^d$.

(b) At the center,

$$g_B^{::}(\mathbf{0}, \mathbf{0}) = \begin{cases} \dfrac{1}{2\pi} \log r + \kappa + O(r^{-1}), & d = 2; \\ g^{::}(\mathbf{0}) - a_d r^{2-d} + O(r^{1-d}), & d \geq 3\,, \end{cases}$$

where $\kappa = (2\gamma_E + \log 8)/(4\pi)$ *and* γ_E *is Euler's constant.*

Proof. Let $v_r := \lceil r \rceil \mathbf{e}_1 \in \partial B$. Let

$$f(x) := g^{::}(x) - g_B^{::}(\mathbf{0}, x) - g^{::}(v_r)\,.$$

Every vertex of ∂B has norm in $[r, r+1)$, so Theorem 3.9 gives $f = O(r^{1-d})$ on ∂B. Since f is discrete harmonic in B, the minimum principle gives, for every $x \in B$,

$$g_B^{::}(\mathbf{0}, x) = g^{::}(x) - g^{::}(v_r) + O(r^{1-d})\,. \tag{3.20}$$

(a) Substituting the two asymptotics from Theorem 3.9 into (3.20) proves the formulas for $x \neq \mathbf{0}$.

(b) Taking $x = \mathbf{0}$ in (3.20), and using $g^{::}(\mathbf{0}) = 0$ when $d = 2$, proves the formulas at the origin. □

Every formula above is a continuum Green function with a lattice correction, which suggests that on a fine lattice discrete harmonic functions should be close to harmonic ones.

3.4 APPROXIMATING DISCRETE HARMONIC FUNCTIONS

We now compare discrete harmonic functions on $\delta\mathbb{Z}^d$ with harmonic functions on $\mathbb{R}^d$ as $\delta \downarrow 0$. For a set $A \subset \delta\mathbb{Z}^d$, write

$$A_\delta^{\square} := \left(A + \left[-\frac{\delta}{2}, \frac{\delta}{2}\right]^d\right)^o\,.$$

For $x \in \delta\mathbb{Z}^d$, write $x_\delta^{\square} := x + \left[-\frac{\delta}{2}, \frac{\delta}{2}\right]^d$ for the closed cube centered at x. Thus $A_\delta^{\square}$ is the interior of the union of the cubes $x_\delta^{\square}$ for $x \in A$. For a domain A in $\mathbb{R}^d$, write $A_\delta^{::} := A \cap \delta\mathbb{Z}^d$. For $x \in \mathbb{R}^d$, let $x_\delta^{::}$ be the unique point satisfying

$$x_\delta^{::} \in \left(x + \left(-\frac{\delta}{2}, \frac{\delta}{2}\right]^d\right) \cap \delta\mathbb{Z}^d\,.$$

For a function f on $\delta\mathbb{Z}^d$, write

$$f_\delta^{\square}(x) := f(x_\delta^{\cdots}).$$

For a function f on $\mathbb{R}^d$, write $f_\delta^{\cdots} := f|_{\delta\mathbb{Z}^d}$. Table 3.1 summarizes the notation, and Figure 3.7 draws the sets for an ellipse.

On $\delta\mathbb{Z}^d$, two sites are neighbors when they differ by δ in exactly one coordinate, and outer boundaries and discrete harmonic functions are defined as in Section 3.1 with this neighbor relation. The scaled discrete Laplacian of a function f on $\delta\mathbb{Z}^d$ is

$$\Delta_\delta^{\cdots} f(x) := \delta^{-2} \sum_{y \sim x} (f(y) - f(x)).$$

For an open set $\Omega \subset \mathbb{R}^d$, write Ω_δ^+ for the set of points of $\Omega_\delta^{\cdots}$ whose segments to their $2d$ neighbors lie in Ω:

$$\Omega_\delta^+ := \left\{ x \in \Omega_\delta^{\cdots} : x + t\mathbf{e}_i \in \Omega \text{ for every } t \in [-\delta, \delta] \text{ and } 1 \le i \le d \right\}. \quad (3.21)$$

The scaled discrete Laplacian of a function $f : \Omega \to \mathbb{R}$ is defined on Ω_δ^+.

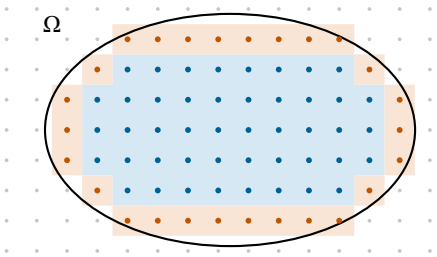


Figure 3.7: An ellipse Ω and the lattice $\delta\mathbb{Z}^2$ for $\delta = 1/6$: the points of Ω_δ^+ (blue) and of $\Omega_\delta^{\cdots} \setminus \Omega_\delta^+$ (vermilion), each with its cube $x_\delta^{\square}$ shaded, so that the shading is the closure of $(\Omega_\delta^{\cdots})_\delta^{\square}$. The vermilion points are the outer boundary of Ω_δ^+ in $\delta\mathbb{Z}^2$, so $U_\delta = \Omega_\delta^+$ in Theorem 3.14 has $U_\delta \cup \partial U_\delta \subset \Omega$.

Taylor's theorem compares the scaled discrete Laplacian with the continuum Laplacian. Two derivatives give convergence, four derivatives give an error of order δ^2, and six derivatives identify the leading error.

Lemma 3.13 *Let $d \ge 1$, and let $\Omega \subset \mathbb{R}^d$ be an open set.*

(a) If $f \in C^2(\Omega)$ has uniformly continuous second derivatives in Ω, then

$$\lim_{\delta \to 0} \sup_{\Omega_\delta^+} \left| \Delta_\delta^{\cdots} f - \Delta f \right| = 0.$$

(b) If $f \in C^4(\Omega)$ has uniformly bounded fourth derivatives in Ω, then

$$\sup_{\Omega_\delta^+} \left| \Delta_\delta^{\cdots} f - \Delta f \right| \le \frac{\delta^2}{12} \sum_{i=1}^{d} \sup_{\Omega} \left| \frac{\partial^4 f}{\partial x_i^4} \right|.$$

	$\delta\mathbb{Z}^d$	$\mathbb{R}^d$
points	$x_\delta^{\vdots} \in \left(x + \left(-\frac{\delta}{2}, \frac{\delta}{2}\right]^d\right) \cap \delta\mathbb{Z}^d$	$x_\delta^{\square} := x + \left[-\frac{\delta}{2}, \frac{\delta}{2}\right]^d$
sets	$A_\delta^{\vdots} := A \cap \delta\mathbb{Z}^d$	$A_\delta^{\square} := \left(A + \left[-\frac{\delta}{2}, \frac{\delta}{2}\right]^d\right)^o$
	$\Omega_\delta^+ \subset \Omega_\delta^{\vdots}$, see (3.21)	
functions	$f_\delta^{\vdots} := f\|_{\delta\mathbb{Z}^d}$	$f_\delta^{\square}(x) := f(x_\delta^{\vdots})$

Table 3.1: Notation for passing between $\delta\mathbb{Z}^d$ and $\mathbb{R}^d$.

(c) If $f \in C^6(\Omega)$ has uniformly bounded sixth derivatives in Ω, then

$$\sup_{x\in\Omega_\delta^+} \left| \Delta_\delta^{\vdots} f(x) - \Delta f(x) - \frac{\delta^2}{12} \sum_{i=1}^d \frac{\partial^4 f}{\partial x_i^4}(x) \right| \le \frac{2\delta^4}{6!} \sum_{i=1}^d \sup_\Omega \left| \frac{\partial^6 f}{\partial x_i^6} \right| .$$

Proof. We prove the three estimates in order. Write $f_{ii} = \partial^2 f/\partial x_i^2$. For part (a), define

$$\omega(\delta) := \max_{1\le i\le d} \sup_{\substack{x,y\in\Omega \\ |x-y|\le\delta}} |f_{ii}(x) - f_{ii}(y)| .$$

Uniform continuity gives $\omega(\delta) \to 0$ as $\delta \downarrow 0$.

Applying Taylor's theorem in the directions $\pm\mathbf{e}_i$ gives, for each $1 \le i \le d$,

$$f(x+\delta\mathbf{e}_i) - 2f(x) + f(x-\delta\mathbf{e}_i) = \frac{\delta^2}{2}\left(f_{ii}(x + t_{i,+}\mathbf{e}_i) + f_{ii}(x - t_{i,-}\mathbf{e}_i)\right),$$

where $t_{i,+}, t_{i,-} \in [0,\delta]$. Summing over i and dividing by δ^2 yields

$$\left|\Delta_\delta^{\vdots} f - \Delta f\right| \le d\,\omega(\delta) .$$

Since $d\omega(\delta) \to 0$ as $\delta \downarrow 0$, part (a) follows.

For part (b), carry the Taylor expansion to fourth order. The terms of degree three cancel, leaving

$$\begin{aligned} f(x+\delta\mathbf{e}_i) - 2f(x) + f(x-\delta\mathbf{e}_i) - \delta^2 f_{ii}(x) \\ = \frac{\delta^4}{24}\left(\frac{\partial^4 f}{\partial x_i^4}(x + a_{i,+}\mathbf{e}_i) + \frac{\partial^4 f}{\partial x_i^4}(x - a_{i,-}\mathbf{e}_i)\right), \end{aligned} \tag{3.22}$$

where $a_{i,+}, a_{i,-} \in [0,\delta]$. Summing over i and dividing by δ^2 proves part (b).

For part (c), carry the Taylor expansion to sixth order. The terms of odd

degree cancel, leaving

$$f(x+\delta\mathbf{e}_i) - 2f(x) + f(x-\delta\mathbf{e}_i) - \delta^2 f_{ii}(x) - \frac{\delta^4}{12}\frac{\partial^4 f}{\partial x_i^4}(x)$$
$$= \frac{\delta^6}{6!}\left(\frac{\partial^6 f}{\partial x_i^6}(x+\alpha_{i,+}\mathbf{e}_i) + \frac{\partial^6 f}{\partial x_i^6}(x-\alpha_{i,-}\mathbf{e}_i)\right),$$

where $\alpha_{i,+}, \alpha_{i,-} \in [0,\delta]$. Summing over i and dividing by δ^2 proves part (c). □

The Taylor estimates control the difference between discrete and continuum Dirichlet solutions. The next theorem turns Lemma 3.13 into uniform convergence and will later give the scaling limit of killed Green functions. Recall from (3.21) that Ω_δ^+ is the set of points of $\Omega \cap \delta\mathbb{Z}^d$ whose segments to their $2d$ neighbors lie in Ω.

Theorem 3.14 *Let $d \geq 1$, let $x_0 \in \mathbb{R}^d$, let $R > 0$, and let $\Omega \subset \mathrm{B}(x_0, R) \subset \mathbb{R}^d$ be a bounded open set. Let $h : \Omega \to \mathbb{R}$ be harmonic. For every $\delta > 0$, let $U_\delta \subset \Omega_\delta^+$, and let h_δ be discrete harmonic in U_δ with boundary values $h|_{\partial U_\delta}$.*

(a) If h has uniformly continuous second derivatives in Ω, then

$$\max_{U_\delta} |h_\delta - h| \longrightarrow 0 \qquad \text{as } \delta \downarrow 0\,.$$

(b) If h has uniformly bounded fourth derivatives in Ω, then

$$\max_{U_\delta} |h_\delta - h| \leq \frac{R^2\delta^2}{24d} \sum_{i=1}^{d} \sup_{\Omega} \left|\frac{\partial^4 h}{\partial x_i^4}\right|.$$

(c) If h has uniformly bounded sixth derivatives in Ω and there is $\beta > 0$ such that

$$\frac{1}{12}\sum_{i=1}^{d} \frac{\partial^4 h}{\partial x_i^4} \geq 2\beta \qquad \text{in } \Omega\,,$$

then, for all sufficiently small $\delta > 0$,

$$h_\delta(x) - h(x) \geq \frac{\beta}{2d}\delta^2 \mathrm{dist}(x, \partial U_\delta)^2 \qquad \text{for every } x \in U_\delta\,.$$

Proof. We compare $h_\delta - h$ with explicit quadratic functions and apply the discrete minimum principle, Lemma 3.2, after scaling $\delta\mathbb{Z}^d$ to $\mathbb{Z}^d$. By translation, take $x_0 = \mathbf{0}$. The assertions are vacuous when U_δ is empty, so suppose it is not. Since $U_\delta \subset \Omega_\delta^+$, the finite set $U_\delta \cup \partial U_\delta$ lies in $\Omega \subset \mathrm{B}(\mathbf{0}, R)$, as in

Figure 3.7. By definition, $h_\delta - h = 0$ on ∂U_δ. Since h_δ is discrete harmonic in U_δ, we have $\Delta_\delta^{::}(h_\delta - h) = -\Delta_\delta^{::} h$ there. The scaled Laplacian of $|z|^2$ equals $2d$ at every lattice point, because the two neighbors $z \pm \delta \mathbf{e}_i$ change $|z|^2$ by $\pm 2\delta z_i + \delta^2$.

For parts (a) and (b), let

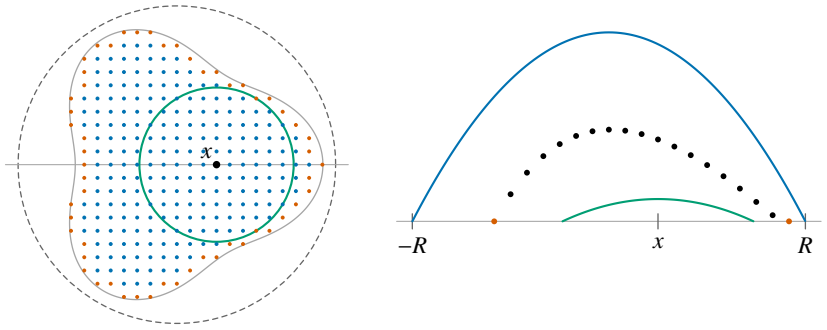
$$b(z) := \frac{R^2 - |z|^2}{2d} \max_{U_\delta} \bigl|\Delta_\delta^{::} h\bigr| .$$

Then

$$\Delta_\delta^{::}\bigl(b \pm (h_\delta - h)\bigr) = -\max_{U_\delta} \bigl|\Delta_\delta^{::} h\bigr| \mp \Delta_\delta^{::} h \le 0 \qquad \text{in } U_\delta .$$

On ∂U_δ both functions equal $b \ge 0$. The minimum principle therefore gives $|h_\delta - h| \le b$ in U_δ (Figure 3.8), and since $b \le R^2 \max_{U_\delta} |\Delta_\delta^{::} h|/(2d)$, we obtain

$$\max_{U_\delta} |h_\delta - h| \le \frac{R^2}{2d} \max_{U_\delta} \bigl|\Delta_\delta^{::} h\bigr| . \tag{3.23}$$



(a) (b)

Figure 3.8: The proof of Theorem 3.14 for $h = \mathrm{Re}\,((x_1 + \mathbf{i}x_2)^4)$, $R = 1$ and $\delta = 1/12$: (a) $U_\delta = \Omega_\delta^+$ (blue) and its outer boundary (vermilion) inside Ω (gray) and $\mathrm{B}(\mathbf{0}, R)$ (dashed), and the disk of radius $\mathrm{dist}(x, \partial U_\delta)$ about x (green). Along the horizontal axis (b), the error $h_\delta - h$ (dots) vanishes on ∂U_δ and lies below the barrier b of parts (a) and (b) (blue) and above the barrier b_x of part (c) (green), all three divided by δ^2.

It remains to insert the Taylor estimates. Since h is harmonic, $\Delta h = 0$, so Lemma 3.13(a) makes $\max_{U_\delta} |\Delta_\delta^{::} h|$ tend to zero as $\delta \downarrow 0$, and (3.23) proves part (a). Lemma 3.13(b) gives

$$\max_{U_\delta} \bigl|\Delta_\delta^{::} h\bigr| \le \frac{\delta^2}{12} \sum_{i=1}^{d} \sup_{\Omega} \left|\frac{\partial^4 h}{\partial x_i^4}\right| ,$$

and (3.23) proves part (b).

For part (c), Lemma 3.13(c) and the hypothesis on the fourth derivatives give

$$\min_{\Omega_\delta^+} \Delta_\delta^{\vdots} h \geq 2\beta\delta^2 - \frac{2\delta^4}{6!} \sum_{i=1}^{d} \sup_{\Omega} \left| \frac{\partial^6 h}{\partial x_i^6} \right| \geq \beta\delta^2$$

for all sufficiently small δ. Fix $x \in U_\delta$, and let

$$b_x(z) := \frac{\beta\delta^2}{2d} (\operatorname{dist}(x, \partial U_\delta)^2 - |z - x|^2) .$$

Then $\Delta_\delta^{\vdots}(h_\delta - h - b_x) = -\Delta_\delta^{\vdots} h + \beta\delta^2 \leq 0$ in U_δ, while $h_\delta - h - b_x = -b_x \geq 0$ on ∂U_δ, since $|z - x| \geq \operatorname{dist}(x, \partial U_\delta)$ there. The minimum principle gives $h_\delta - h \geq b_x$ in U_δ, and at $z = x$ this reads

$$h_\delta(x) - h(x) \geq \frac{\beta}{2d} \delta^2 \operatorname{dist}(x, \partial U_\delta)^2 ,$$

which proves part (c). □

The polynomial

$$f(x) = \operatorname{Re}\left((x_1 + \mathbf{i}x_2)^4\right) = x_1^4 - 6x_1^2x_2^2 + x_2^4$$

is harmonic. Both pure fourth derivatives equal 24, so

$$\frac{1}{12} \sum_{i=1}^{2} \frac{\partial^4 f}{\partial x_i^4} = 4 .$$

Thus f satisfies the hypothesis of Theorem 3.14(c) with $\beta = 2$.

We next prove the discrete counterpart of the Harnack inequality from Theorem 2.30. Its proof compares killed Green functions of lattice balls with the continuum Green function $\mathcal{G}$. Recall from (2.14) in Chapter 2 that $\mathcal{G}(x) = -\frac{1}{2\pi} \log|x|$ for $d = 2$ and $\mathcal{G}(x) = a_d |x|^{2-d}$ for $d \geq 3$.

We first identify the scaling limit of a killed Green function when its two arguments stay apart and away from the boundary. The proof combines the whole-space asymptotic from Theorem 3.9, the method of images for the ball, and Theorem 3.14(b).

Lemma 3.15 *Let $d \geq 2$ and $\varpi > 0$, and let $B := \mathrm{B}(\mathbf{0}, \varpi)$. For each $y \in B$, let h_y be the harmonic function in B with boundary values $\mathcal{G}(\cdot - y)$, and let*

$$\mathcal{G}_B(x, y) := \mathcal{G}(x - y) - h_y(x) .$$

Let $K \subset B \times B$ be compact and disjoint from the diagonal. In $d = 2$,

$$\lim_{R \to \infty} \sup_{\substack{x,y \in \mathrm{B}_{\varpi R}^{\vdots} \\ (x/R, y/R) \in K}} \left| g_{\mathrm{B}_{\varpi R}}^{\vdots}(x, y) - \mathcal{G}_B(x/R, y/R) \right| = 0 ,$$

whereas for $d \geq 3$,

$$\lim_{R\to\infty} \sup_{\substack{x,y\in \mathrm{B}^{::}_{\varpi R} \\ (x/R,y/R)\in K}} \left| R^{d-2} g^{::}_{\mathrm{B}_{\varpi R}}(x,y) - \mathcal{G}_B(x/R, y/R) \right| = 0\,.$$

We interpret a supremum over an empty set of lattice pairs as zero.

Proof. The functions $g^{::}(\cdot - y) - g^{::}_{\mathrm{B}_{\varpi R}}(\cdot, y)$ and $\mathbb{E}_{\cdot} g^{::}(X_{\tau_{\partial \mathrm{B}^{::}_{\varpi R}}} - y)$ agree on $\partial \mathrm{B}^{::}_{\varpi R}$ and are discrete harmonic in $\mathrm{B}^{::}_{\varpi R}$. Lemma 3.2 therefore gives

$$g^{::}_{\mathrm{B}_{\varpi R}}(x,y) = g^{::}(x-y) - \mathbb{E}_x g^{::}(X_{\tau_{\partial \mathrm{B}^{::}_{\varpi R}}} - y)\,. \tag{3.24}$$

Theorem 3.9, with the constant κ of (3.17), gives, uniformly for $|u| \geq cR$ with fixed $c > 0$,

$$g^{::}(u) + \frac{1}{2\pi} \log R + \kappa = \mathcal{G}(u/R) + O(R^{-2}) \qquad \text{in } d = 2\,, \tag{3.25}$$

and

$$R^{d-2} g^{::}(u) = \mathcal{G}(u/R) + O(R^{-2}) \qquad \text{for } d \geq 3\,. \tag{3.26}$$

Choose $\epsilon \in (0, \varpi/4)$ with $|\eta| \leq \varpi - 2\epsilon$ and $|\xi - \eta| \geq 2\epsilon$ for all $(\xi, \eta) \in K$. For $\eta \in B \setminus \{\mathbf{0}\}$, let $\eta^* := \varpi^2 \eta/|\eta|^2$ and $h_\eta(z) := \mathcal{G}(|\eta|(z - \eta^*)/\varpi)$, and let $h_{\mathbf{0}} \equiv \mathcal{G}(\varpi \mathbf{e}_1)$. If $|z| = \varpi$, then $|\eta|\,|z - \eta^*|/\varpi = |z - \eta|$, as in Figure 3.9. Since $\mathcal{G}$ is harmonic off the origin (Section 2.5), h_η is harmonic off η^*, so it is the function of the statement. If $|\eta| \leq \varpi - 2\epsilon$, then $|\eta^*| \geq \varpi + 2\epsilon$. Hence h_η is harmonic in $\mathrm{B}(\mathbf{0}, \varpi + \epsilon)$, and there its partial derivatives of orders 1 to 4 are bounded by a constant $C = C(d, \varpi, \epsilon)$, uniformly in η.

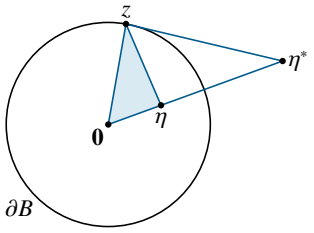


Figure 3.9: The image point $\eta^* = \varpi^2 \eta/|\eta|^2$ of $\eta \in B = \mathrm{B}(\mathbf{0}, \varpi)$ in the proof of Lemma 3.15: for $z \in \partial B$ the triangles $\mathbf{0}\eta z$ (shaded) and $\mathbf{0} z \eta^*$ are similar with ratio $|\eta|/\varpi$, so $|z - \eta| = |\eta|\,|z - \eta^*|/\varpi$.

Fix $x, y \in \mathrm{B}^{::}_{\varpi R}$ with $(\xi, \eta) := (x/R, y/R) \in K$, write $\tau := \tau_{\partial \mathrm{B}^{::}_{\varpi R}}$, and let $R > 1/\epsilon$. Each $v \in \partial \mathrm{B}^{::}_{\varpi R}$ satisfies $\varpi R \leq |v| < \varpi R + 1$, so $|v - y| \geq 2\epsilon R$; also $|x - y| \geq 2\epsilon R$. In $d = 2$, with $c_R := \frac{1}{2\pi} \log R + \kappa$, (3.24) and (3.25) give

$$\begin{aligned} g^{::}_{\mathrm{B}_{\varpi R}}(x,y) &= [g^{::}(x-y) + c_R] - \mathbb{E}_x[g^{::}(X_\tau - y) + c_R] \\ &= \mathcal{G}(\xi - \eta) - \mathbb{E}_x \mathcal{G}(X_\tau/R - \eta) + O(R^{-2})\,. \end{aligned}$$

For $d \geq 3$, (3.24) and (3.26) give the same expansion of $R^{d-2} g^{::}_{\mathrm{B}_{\varpi R}}(x, y)$.

For $v \in \partial \mathrm{B}^{::}_{\varpi R}$, let $v' := \varpi v/|v| \in \partial B$, so that $|v/R - v'| < 1/R$. The segment $[v', v/R]$ lies in $\{\varpi \leq |z| < \varpi + \epsilon\}$, where $|z - \eta| \geq 2\epsilon$. Enlarging C, both $\mathcal{G}(\cdot - \eta)$ and h_η have gradient at most C there. Since $\mathcal{G}(v' - \eta) = h_\eta(v')$, this gives $|\mathcal{G}(v/R - \eta) - h_\eta(v/R)| \leq 2C/R$. Finally, $u(z) := h_\eta(z/R)$ is harmonic in $\Omega := \mathrm{B}(\mathbf{0}, (\varpi + \epsilon)R)$, with fourth derivatives at most CR^{-4}, and $\mathrm{B}^{::}_{\varpi R} \subset \Omega^+_1$, the set (3.21) with $\delta = 1$. Theorem 3.14(b) with $\delta = 1$ gives $|\mathbb{E}_x u(X_\tau) - u(x)| \leq C(\varpi + \epsilon)^2 R^{-2}$. Since $u(x) = h_\eta(\xi)$, we conclude that in $d = 2$

$$g^{::}_{\mathrm{B}_{\varpi R}}(x, y) = \mathcal{G}(\xi - \eta) - h_\eta(\xi) + O(R^{-1}) = \mathcal{G}_B(\xi, \eta) + O(R^{-1}),$$

and that the same holds for $R^{d-2} g^{::}_{\mathrm{B}_{\varpi R}}(x, y)$ when $d \geq 3$, uniformly over $(\xi, \eta) \in K$. □

Lemma 3.15 and the continuous Harnack inequality compare killed Green functions from different starting points. Counting the last step before exit converts this comparison into a comparison of exit distributions, which gives the discrete Harnack inequality.

Lemma 3.16 (Discrete Harnack Inequality) *For each $d \geq 2$ there is $R_0(d) < \infty$ such that the following holds. Let $z \in \mathbb{Z}^d$, let $R \geq R_0(d)$, and let $h : \mathrm{B}^{::}(z, R) \to \mathbb{R}_{\geq 0}$ be discrete harmonic at every lattice point whose nearest neighbors all lie in $\mathrm{B}^{::}(z, R)$. Then*

$$\sup_{\mathrm{B}^{::}(z, R/2)} h \leq 4^d \inf_{\mathrm{B}^{::}(z, R/2)} h .$$

Proof. By translation, take $z = \mathbf{0}$. Let $R \geq 32$, and let

$$A := \mathrm{B}^{::}(\mathbf{0}, 7R/8) \qquad \text{and} \qquad B := \mathrm{B}^{::}(\mathbf{0}, 15R/16) .$$

Write $U := \mathrm{B}(\mathbf{0}, 15/16)$, so that $\mathcal{G}_U$ is the kernel of Lemma 3.15 with $\varpi = 15/16$. Then $\bar{A} \subset B$ and $\bar{B} \subset \mathrm{B}^{::}(\mathbf{0}, R)$, so h is discrete harmonic on B. Figure 3.10 shows these balls.

We first compare the killed Green functions. If $q \in A$ has a neighbor outside A, then

$$\frac{27}{32} \leq \frac{7}{8} - \frac{1}{R} \leq \frac{|q|}{R} < \frac{7}{8} .$$

For $27/32 \leq |p| \leq 7/8$, the function $\mathcal{G}_U(\cdot, p)$ is positive and harmonic in $\mathrm{B}(\mathbf{0}, 27/32)$. Positivity follows from the minimum and strong minimum principles, Lemmas 2.12 and 2.8: the kernel vanishes on ∂U and tends to $+\infty$ at its pole. The continuous Harnack inequality, Theorem 2.30, gives

$$\mathcal{G}_U(\xi, p) \leq \left(\frac{27/32 + 1/2}{27/32 - 1/2} \right)^d \mathcal{G}_U(\eta, p) = \left(\frac{43}{11} \right)^d \mathcal{G}_U(\eta, p) ,$$

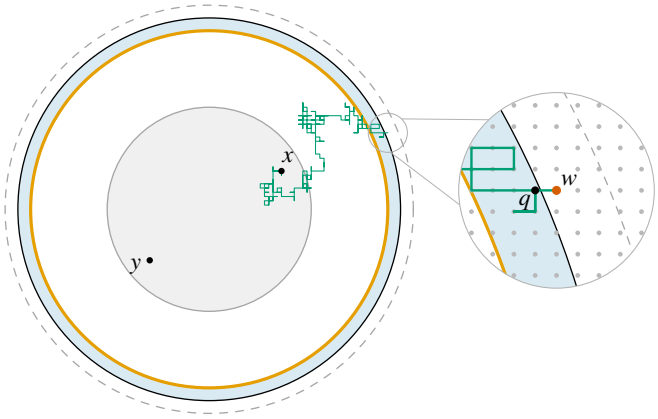


Figure 3.10: The balls in the proof of Lemma 3.16 in $\mathbb{Z}^2$ for $R = 48$: x and y lie in $\mathrm{B}^{::}(\mathbf{0}, R/2)$ (gray), the thick circle bounds $A = \mathrm{B}^{::}(\mathbf{0}, 7R/8)$, the thin one $B = \mathrm{B}^{::}(\mathbf{0}, 15R/16)$, and the dashed one $\mathrm{B}^{::}(\mathbf{0}, R)$. A walk from x leaves B at w after a last step from a site q of the shaded shell $B \setminus A$ (magnified on the right), where the minimum principle extends the bound $g_B^{::}(x, q) \le 4^d g_B^{::}(y, q)$ from the sites of A next to the thick circle.

for $|\xi| \le 1/2$ and $|\eta| \le 1/2$. The formula for h_η in the proof of Lemma 3.15 gives joint continuity of the kernel. It therefore has a positive minimum on the compact set

$$\overline{\mathrm{B}}(\mathbf{0}, 1/2) \times \{v \in \mathbb{R}^d : 27/32 \le |v| \le 7/8\} \subset U \times U\,.$$

This set avoids the diagonal. Since $(43/11)^d < 4^d$, uniform convergence in Lemma 3.15 gives, for all sufficiently large R,

$$g_B^{::}(x, q) \le 4^d g_B^{::}(y, q)\,, \tag{3.27}$$

for every $x, y \in \mathrm{B}^{::}(\mathbf{0}, R/2)$ and every such q.

Fix these x and y. The function

$$v(q) := 4^d g_B^{::}(y, q) - g_B^{::}(x, q)\,,$$

extended by zero outside B, is discrete harmonic in $B \setminus A$ by (3.19). It is nonnegative on the part of $\partial(B \setminus A)$ in A by (3.27), and zero on the rest. The discrete minimum principle, Lemma 3.2, gives $v \ge 0$ on $B \setminus A$.

We now compare exit distributions. For $u \in B$ and $w \in \partial B$, write $\mu_u(w) := \mathbb{P}_u\{X_{\tau_{\partial B}} = w\}$. Counting the last step before exit and using the normalization in (3.18) gives

$$\mu_u(w) = \sum_{\substack{q \in B \\ q \sim w}} g_B^{::}(u, q)\,.$$

Since $\bar{A} \subset B$, every q in this sum lies in $B \setminus A$. Thus $\mu_x(w) \le 4^d \mu_y(w)$. Lemma 3.5 and uniqueness of the Dirichlet problem give

$$h(x) = \sum_{w \in \partial B} \mu_x(w) h(w) \le 4^d \sum_{w \in \partial B} \mu_y(w) h(w) = 4^d h(y),$$

because $h \ge 0$ on ∂B. Taking the supremum over x and the infimum over y proves the lemma. □

The Harnack inequality controls a nonnegative discrete harmonic function on a fixed ball. We turn now to global questions. Which functions are harmonic on all of $\mathbb{Z}^d$, and how fast can they grow?

3.5 DISCRETE HARMONIC POLYNOMIALS

A polynomial on $\mathbb{Z}^d$ is the restriction of a polynomial in $x_1, \dots, x_d$. The monomial $x_1^{k_1} \cdots x_d^{k_d}$ has total degree $k_1 + \cdots + k_d$. We first prove that every discrete harmonic function of polynomial growth is a polynomial. We then construct discrete harmonic polynomials and compute the dimension of the space of discrete harmonic polynomials of degree at most k.

The proof of the polynomial-growth theorem begins with discrete harmonic functions that vanish at infinity.

Lemma 3.17 *Let $d \ge 1$, and let $h : \mathbb{Z}^d \to \mathbb{R}$ be discrete harmonic. If $h(x) \to 0$ as $|x| \to \infty$, then $h = 0$.*

Proof. Fix $x \in \mathbb{Z}^d$ and $\epsilon > 0$. Choose R so large that $x \in \mathrm{B}^{::}(\mathbf{0}, R)$ and $|h(y)| < \epsilon$ for every $y \in \partial \mathrm{B}^{::}(\mathbf{0}, R)$. Applying Lemma 3.2 to h and $-h$ gives $|h(x)| < \epsilon$. Since $\epsilon > 0$ was arbitrary, $h(x) = 0$. □

The next theorem generalizes the fact that a discrete harmonic function of sublinear growth must be constant. We use the convention that nonzero constants are polynomials of degree 0 and that the zero polynomial has degree -1.

Theorem 3.18 *Let $d \ge 2$, let $n \ge 0$ be an integer, and let $h : \mathbb{Z}^d \to \mathbb{R}$ be discrete harmonic. If $|h(x)|/|x|^n \to 0$ as $|x| \to \infty$, then h is a polynomial of degree at most $n - 1$ in the coordinates $x_1, \dots, x_d$.*

The case $n = 1$ includes the discrete Liouville theorem: every bounded discrete harmonic function on $\mathbb{Z}^d$ is constant. To prove Theorem 3.18, we characterize polynomials on $\mathbb{Z}^d$ using discrete derivatives, starting with the one-dimensional case.

Lemma 3.19 *Let $n \geq 0$ be an integer, and let $f : \mathbb{Z} \to \mathbb{R}$. Then f is a polynomial of degree at most n if and only if $Df(x) := f(x+1) - f(x)$ is a polynomial of degree at most $n-1$, where a polynomial of degree at most -1 means the zero polynomial.*

Proof. The case $n = 0$ is immediate. Assume $n \geq 1$, and let $\mathcal{P}_n$ be the vector space of real polynomials of degree at most n. Write $\mathcal{P}_n^0 := \{f \in \mathcal{P}_n : f(0) = 0\}$.

Suppose first that $f \in \mathcal{P}_n$. The binomial theorem gives $Df \in \mathcal{P}_{n-1}$.

Conversely, suppose $Df \in \mathcal{P}_{n-1}$. The map $D : \mathcal{P}_n^0 \to \mathcal{P}_{n-1}$ is injective: if $Dg = 0$, then g is constant on $\mathbb{Z}$, and $g(0) = 0$ gives $g = 0$. Both vector spaces have dimension n, so the map is surjective. Hence $Df = Dh$ for some $h \in \mathcal{P}_n^0$. The difference $f - h$ is constant on $\mathbb{Z}$, and therefore $f \in \mathcal{P}_n$. □

Applying Lemma 3.19 along each coordinate gives the multivariable characterization.

Lemma 3.20 *Let $d \geq 1$ and $n \geq 0$ be integers, and let $f : \mathbb{Z}^d \to \mathbb{R}$. Then f is a polynomial of degree at most n if and only if, for every $1 \leq i \leq d$, the function $D_i f(x) := f(x + \mathbf{e}_i) - f(x)$ is a polynomial of degree at most $n-1$, where a polynomial of degree at most -1 means the zero polynomial.*

Proof. If f is a polynomial of degree at most n, then the binomial theorem shows that each $D_i f$ is a polynomial of degree at most $n-1$.

Conversely, suppose that each $D_i f$ is a polynomial of degree at most $n-1$. The proof of Lemma 3.19 shows that, for every integer $k \geq 0$, there is a polynomial g_k of degree at most $k+1$ with $g_k(0) = 0$ and $Dg_k(t) = t^k$ for $t \in \mathbb{Z}$. Fix $1 \leq i \leq d$, and let P_i be the polynomial obtained from $D_i f$ by replacing x_i^k with $g_k(x_i)$ in each monomial. Then P_i has degree at most n, vanishes when $x_i = 0$, and satisfies $D_i P_i = D_i f$, so $f - P_i$ does not depend on x_i. Changing one coordinate at a time along the path from $\mathbf{0}$ to x therefore gives

$$f(x) = f(\mathbf{0}) + \sum_{i=1}^{d} P_i(x_1, \ldots, x_i, 0, \ldots, 0),$$

which is a polynomial of degree at most n. (For $n = 0$ all the P_i vanish, and f is constant.) □

3.5.1 Coupling

To prove Theorem 3.18, we must show that taking one discrete difference lowers the growth order by one. We compare $h(x + \mathbf{e}_i)$ with $h(x)$ by coupling

walks started at the two neighboring vertices. The walks agree after meeting, so only paths that leave a large ball before meeting contribute to the difference.

To avoid periodicity, use a **lazy random walk** $(X_t)_{t\ge 0}$ in $\mathbb{Z}^d$. At each time $t \ge 1$, choose $j(t)$ uniformly from $\{1, \dots, d\}$ and let

$$X_t - X_{t-1} = \begin{cases} \mathbf{e}_{j(t)}, & \text{with probability } 1/4; \\ -\mathbf{e}_{j(t)}, & \text{with probability } 1/4; \\ 0, & \text{with probability } 1/2\,. \end{cases} \tag{3.28}$$

The coordinate choices and increments at distinct times are independent.

A **coupling** of probability measures μ and ν is a probability measure on the product space whose marginals are μ and ν. Thus a coupling of random variables consists of a pair (U, V) on one probability space such that U and V have the prescribed marginal laws.

Let X be a lazy random walk started at $X_0 = x$, and set $Y_0 = x + \mathbf{e}_i$. We define the increments of Y from those of X by three rules, applied from time $t = 1$ onward. If $X_{t-1} = Y_{t-1}$, then set

$$Y_t - Y_{t-1} = X_t - X_{t-1}\,.$$

Otherwise, write $Y_{t-1} - X_{t-1} = m\mathbf{e}_i$ with $m \ge 1$. If $j(t) \ne i$, then again set

$$Y_t - Y_{t-1} = X_t - X_{t-1}\,.$$

If $j(t) = i$ and $m > 1$, then give the two walks opposite increments,

$$Y_t - Y_{t-1} = X_{t-1} - X_t\,.$$

Finally, if $j(t) = i$ and $m = 1$, then choose

$$(X_t - X_{t-1},\, Y_t - Y_{t-1}) = \begin{cases} (\mathbf{e}_i, 0), & \text{with probability } 1/4; \\ (0, -\mathbf{e}_i), & \text{with probability } 1/4; \\ (-\mathbf{e}_i, \mathbf{e}_i), & \text{with probability } 1/4; \\ (0, 0), & \text{with probability } 1/4\,. \end{cases} \tag{3.29}$$

Since $m = 1$ at time $t = 1$, the last rule applies at the first step when $j(1) = i$. In (3.29), the first two outcomes make the walks meet, while the last two change m by 2 or leave it unchanged. Each marginal increment in (3.29) equals $\mathbf{e}_i$ with probability $1/4$, equals $-\mathbf{e}_i$ with probability $1/4$, and equals $\mathbf{0}$ with probability $1/2$. Consequently, both X and Y have the law (3.28) of lazy random walk. We call (X, Y) the **reflection coupling**.

Every step that does not make the walks meet changes m by 0 or ± 2 and changes the ith coordinates of X and Y by opposite amounts. Hence, until the

walks meet, m is a positive odd integer, and the sum of the ith coordinates stays at its initial value $2x_i + 1$. Equivalently, until the walks meet, Y is the reflection of X across the fixed hyperplane

$$\Pi = \{z \in \mathbb{R}^d : z_i = x_i + \tfrac{1}{2}\}\,.$$

In particular, the ith coordinate of X never exceeds x_i before the walks meet.

The walks coalesce at the **meeting time**

$$T_{\text{meet}} := \inf\{t \geq 0 : X_t = Y_t\}\,.$$

Figure 3.11 shows a pair of coupled paths before and after T_{meet}.

The reflection coupling bounds one discrete derivative of a discrete harmonic function. Fix $R > 1$, and let $h : \mathrm{B}^{::}(x,R) \cup \partial\mathrm{B}^{::}(x,R) \to \mathbb{R}$ be discrete harmonic in $\mathrm{B}^{::}(x,R)$. Write τ_R^X and τ_R^Y for the first exit times of X and Y from $\mathrm{B}^{::}(x,R)$. Since h is discrete harmonic in $\mathrm{B}^{::}(x,R)$, it is also harmonic there for lazy random walk, so optional stopping gives

$$h(x) = \mathbb{E}h(X_{\tau_R^X}) \qquad \text{and} \qquad h(x + \mathbf{e}_i) = \mathbb{E}h(Y_{\tau_R^Y})\,.$$

Let $\tau_R := \tau_R^X \wedge \tau_R^Y$. On the event $\{T_{\text{meet}} \leq \tau_R\}$, the walks have coalesced by time τ_R, so $\tau_R^X = \tau_R^Y$ and $X_{\tau_R^X} = Y_{\tau_R^Y}$. Therefore,

$$D_i h(x) = \mathbb{E}\left[(h(Y_{\tau_R^Y}) - h(X_{\tau_R^X}))\mathbf{1}\{T_{\text{meet}} > \tau_R\}\right]\,.$$

Consequently,

$$|D_i h(x)| \leq \left(\max_{\partial\mathrm{B}^{::}(x,R)} h - \min_{\partial\mathrm{B}^{::}(x,R)} h\right)\mathbb{P}\{T_{\text{meet}} > \tau_R\}\,. \tag{3.30}$$

The next lemma bounds the probability in (3.30).

Lemma 3.21 *Let $d \geq 2$, let $x \in \mathbb{Z}^d$, and let $1 \leq i \leq d$. Let (X, Y) be the reflection coupling of lazy random walks started at x and $x + \mathbf{e}_i$. Let T_{meet} be their meeting time. Let $R > 0$, and let τ_R be the first time either walk exits $\mathrm{B}^{::}(x,R)$. Then*

$$\mathbb{P}\{\tau_R < T_{\text{meet}}\} \leq \frac{4d}{R}\,.$$

The proof bounds the probability that the walks have not met by the probability that a walk exits a ball before leaving a half-space. A **half-space** in $\mathbb{Z}^d$ has the form

$$\mathbb{H} = \{x \in \mathbb{Z}^d : b \cdot x \leq r\}\,,$$

where $b \in \mathbb{R}^d$ is a unit vector and $r \in \mathbb{R}$. The reflection coupling uses $b = \mathbf{e}_i$. Chapter 6 uses the estimate for general unit vectors.

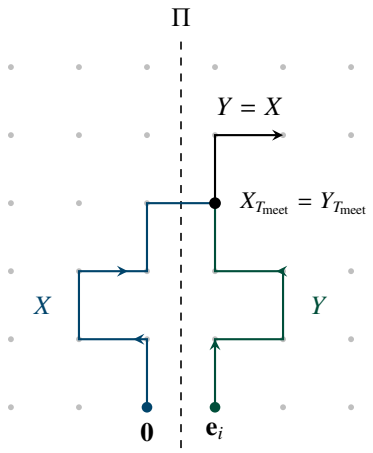


Figure 3.11: The reflection coupling started at $X_0 = \mathbf{0}$ and $Y_0 = \mathbf{e}_i$. Until T_{meet}, the paths are reflections of each other across the fixed hyperplane $\Pi = \{z_i = 1/2\}$; after T_{meet}, the two walks coincide.

Lemma 3.22 *Let $d \geq 1$, let $R > 0$ and $r \geq 0$, and let $b \in \mathbb{R}^d$ be a unit vector. Let $\mathbb{H} = \{x \in \mathbb{Z}^d : b \cdot x \leq r\}$, and let T_R and $T_{\mathbb{H}}$ be the first exit times of simple random walk from $\mathrm{B}^{::}(\mathbf{0}, R)$ and $\mathbb{H}$, respectively. Then*

$$\mathbb{P}_{\mathbf{0}}\{T_R \leq T_{\mathbb{H}}\} \leq \frac{d(r+1)(R+1)}{R^2}\,.$$

Proof. For simple random walk $(X_t)_{t \geq 0}$ started at $\mathbf{0}$, let

$$M_t = (r + 1 - b \cdot X_t)(R + 1 + b \cdot X_t) + \frac{1}{d}|X_t|^2\,.$$

Since b is a unit vector, M is a martingale. Let $T = T_{\mathbb{H}} \wedge T_R$. The stopped process is bounded, so optional stopping gives

$$\begin{aligned}(r+1)(R+1) = M_0 &= \mathbb{E}M_T \\ &= \mathbb{E}\left[M_{T_{\mathbb{H}}}\mathbf{1}\{T_{\mathbb{H}} < T_R\} + M_{T_R}\mathbf{1}\{T_R \leq T_{\mathbb{H}}\}\right]\,.\end{aligned}$$

On $\{T_{\mathbb{H}} < T_R\}$, we have $r < b \cdot X_{T_{\mathbb{H}}} \leq r + 1$, so $M_{T_{\mathbb{H}}} > 0$. On $\{T_R \leq T_{\mathbb{H}}\}$, we have $-R - 1 < b \cdot X_{T_R} \leq r + 1$ and $|X_{T_R}| \geq R$, as in Figure 3.12. The product in M_{T_R} is therefore nonnegative, and $M_{T_R} \geq R^2/d$. Hence

$$(r+1)(R+1) \geq \mathbb{P}_{\mathbf{0}}\{T_R \leq T_{\mathbb{H}}\}\frac{R^2}{d}\,.$$

Rearranging the final inequality proves the lemma. □

We now prove Lemma 3.21.

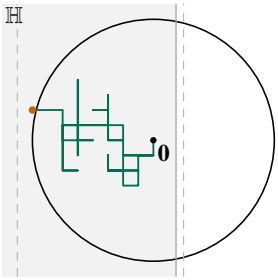


Figure 3.12: A walk from $\mathbf{0}$ that leaves the disk of radius $R = 8$ before the half-plane $\mathbb{H} = \{x_1 \le r\}$ with $r = 1$ (shaded), the event bounded in Lemma 3.22 for $b = \mathbf{e}_1$. The product $(r + 1 - x_1)(R + 1 + x_1)$ in the martingale vanishes on the dashed lines and is nonnegative between them.

Proof of Lemma 3.21. Since $d \ge 2$, the bound is trivial when $R \le 8$, because then $4d/R \ge 1$; so we may assume that $R > 8$. By translation, take $X_0 = \mathbf{0}$ and $Y_0 = \mathbf{e}_i$, so that the reflecting hyperplane is $\{z_i = 1/2\}$. Before the walks meet, the ith coordinate of X is at most 0, so X stays in the half-space $\mathbb{H} := \{z \in \mathbb{Z}^d : z_i \le 0\}$. Reflection across $\{z_i = 1/2\}$ also gives $|Y_t| \le |X_t| + 1$ before the walks meet, because the walks agree in every coordinate except the ith, and the ith coordinate of Y_t exceeds that of X_t by one in absolute value.

Suppose $\tau_R < T_{\text{meet}}$. Whichever of the two walks exits $\mathrm{B}^{::}(\mathbf{0}, R)$ at time τ_R, the bound $|Y_{\tau_R}| \le |X_{\tau_R}| + 1$ gives $|X_{\tau_R}| \ge R - 1$. Thus X exits $\mathrm{B}^{::}(\mathbf{0}, R - 1)$ before leaving $\mathbb{H}$.

Holding times do not affect which of the ball and half-space boundaries is hit first. Lemma 3.22, with $b = \mathbf{e}_i$, half-space parameter $r = 0$, and ball radius $R - 1$, therefore gives

$$\mathbb{P}\{\tau_R < T_{\text{meet}}\} \le \frac{dR}{(R-1)^2} \le \frac{4d}{R},$$

where the last inequality holds because $R \ge 2$. □

Combining Lemma 3.21 with (3.30) shows that the discrete derivative $D_i h(x) = h(x + \mathbf{e}_i) - h(x)$ of Lemma 3.20 grows one order more slowly.

Lemma 3.23 *Let $d \ge 2$ and $n \ge 1$ be integers, and let $1 \le i \le d$. Let $h : \mathbb{Z}^d \to \mathbb{R}$ be discrete harmonic and satisfy $|h(x)|/|x|^n \to 0$ as $|x| \to \infty$. Then $|D_i h(x)|/|x|^{n-1} \to 0$ as $|x| \to \infty$.*

Proof. Fix $x \in \mathbb{Z}^d$ with $|x| \ge 2$, and let $R = 2|x|$. Every $y \in \partial \mathrm{B}^{::}(x, R)$ satisfies $|x| \le |y| < 3|x| + 1$. The growth hypothesis therefore gives

$$\max_{y \in \partial \mathrm{B}^{::}(x,R)} |h(y)| = o(|x|^n).$$

degree k	p_k	q_k
0	1	–
1	x	y
2	$x^2 - y^2$	$2xy$
3	$x^3 - 3xy^2 - \frac{1}{4}x$	$3x^2y - y^3 + \frac{1}{4}y$
4	$x^4 - 6x^2y^2 + y^4 - x^2 - y^2 + \frac{3}{4}$	$4x^3y - 4xy^3$
	$\dots$	$\dots$

Table 3.2: A basis for the discrete harmonic polynomials in $\mathbb{Z}^2$. For each $k \geq 1$, the highest-degree homogeneous parts of p_k and q_k are nonzero scalar multiples of the real and imaginary parts of $(x + \mathbf{i}y)^k$.

Lemma 3.21 and (3.30) now give

$$|D_i h(x)| \leq \frac{8d}{R} \max_{y \in \partial \mathrm{B}^{::}(x,R)} |h(y)| = o(|x|^{n-1}) .$$

□

Exercise 3.36 uses the same bound to show that locally bounded discrete harmonic functions on finer and finer lattices have harmonic subsequential limits. Lemma 3.23 completes the induction for Theorem 3.18.

Proof of Theorem 3.18. We proceed by induction on n. For $n = 0$, Lemma 3.17 gives $h = 0$.

For the inductive step, let $n \geq 1$ and assume the theorem at $n - 1$. For every $1 \leq i \leq d$, the function $D_i h$ is discrete harmonic. Lemma 3.23 gives $|D_i h(x)|/|x|^{n-1} \to 0$. The induction hypothesis shows that $D_i h$ is a polynomial of degree at most $n - 2$. Since the conclusion holds for every coordinate, Lemma 3.20 shows that h is a polynomial of degree at most $n - 1$. □

The first planar examples appear in Table 3.2. We next give a construction in every degree.

We now transform a harmonic polynomial on $\mathbb{R}^d$ into a discrete harmonic polynomial with the same leading homogeneous part. For a polynomial g and $1 \leq i \leq d$, define the centered difference

$$\mathcal{D}_i g(x) := g\left(x + \frac{1}{2}\mathbf{e}_i\right) - g\left(x - \frac{1}{2}\mathbf{e}_i\right) .$$

Then

$$\mathcal{D}_i^2 g(x) = g(x + \mathbf{e}_i) - 2g(x) + g(x - \mathbf{e}_i) .$$

For every integer $k \geq 0$, define the monic polynomial

$$P_k(x) := \prod_{j=0}^{k-1} \left(x + j - \frac{k-1}{2} \right).$$

The empty product gives $P_0 = 1$. For $k \geq 1$,

$$P_k(x) = \begin{cases} \displaystyle\prod_{j=1}^{k/2} \left(x^2 - \frac{(2j-1)^2}{4} \right), & k \text{ even}; \\ \displaystyle x \prod_{j=1}^{(k-1)/2} (x^2 - j^2), & k \text{ odd}. \end{cases}$$

The products for $P_k(x + 1/2)$ and $P_k(x - 1/2)$ share $k - 1$ factors. Factoring the common terms gives

$$\mathcal{D} P_k = k P_{k-1}. \tag{3.31}$$

Here $\mathcal{D}$ denotes the centered difference in one variable.

Define the linear map $\Xi : \mathbb{R}[x_1, \ldots, x_d] \to \mathbb{R}[x_1, \ldots, x_d]$ on monomials by

$$\Xi \left(\prod_{j=1}^{d} x_j^{k_j} \right) = \prod_{j=1}^{d} P_{k_j}(x_j).$$

Since every P_k is monic of degree k, the map Ξ is triangular with ones on its diagonal in the monomial basis, and hence is invertible.

Equation (3.31) shows that Ξ intertwines the continuum and discrete Laplacians.

Lemma 3.24 *Let $d \geq 1$ and $k \geq 0$ be integers, and let $f \in \mathbb{R}[x_1, \ldots, x_d]$ be harmonic with total degree k.*

(a) The polynomial $\Xi(f)$ is discrete harmonic.

(b) If $k = 0$, then $f - \Xi(f) = 0$; if $k \geq 1$, then $f - \Xi(f)$ has total degree at most $k - 2$.

Proof. (a) By (3.31),

$$\begin{aligned}\mathcal{D}_i\Xi\left(\prod_{j=1}^{d} x_j^{k_j}\right) &= \mathcal{D}_i P_{k_i}(x_i) \prod_{\substack{1\le j\le d\\ j\neq i}} P_{k_j}(x_j)\\ &= k_i P_{k_i-1}(x_i) \prod_{\substack{1\le j\le d\\ j\neq i}} P_{k_j}(x_j)\\ &= \Xi\left(k_i x_i^{k_i-1} \prod_{\substack{1\le j\le d\\ j\neq i}} x_j^{k_j}\right).\end{aligned}$$

Thus, for every polynomial $f \in \mathbb{R}[x_1, \dots, x_d]$,

$$\mathcal{D}_i\Xi(f) = \Xi\left(\frac{\partial f}{\partial x_i}\right).$$

Applying $\mathcal{D}_i$ to both sides gives

$$\mathcal{D}_i^2\Xi(f) = \Xi\left(\frac{\partial^2 f}{\partial x_i^2}\right). \tag{3.32}$$

Summing (3.32) over $1 \le i \le d$ gives

$$\Delta^{::}\Xi(f) = \Xi(\Delta f).$$

Since Ξ is invertible, $\Xi(f)$ is discrete harmonic exactly when f is harmonic.

(b) The polynomial P_k has the same parity as k. Since P_k is monic, $P_k(x) - x^k$ vanishes when $k = 0$ and has degree at most $k - 2$ when $k \ge 1$. Applying the one-variable conclusion to each factor of every monomial proves the degree statement for $f - \Xi(f)$. □

In dimension $d = 2$, the real and imaginary parts of $(x + \mathbf{i}y)^k$ are harmonic for every $k \ge 1$. Their images under Ξ give the polynomials in Table 3.2. Exercise 3.37 shows that the two images in each positive degree, together with the constant function, form a basis of the discrete harmonic polynomials on $\mathbb{Z}^2$.

We finish by computing the dimension of the space of discrete harmonic polynomials of degree at most k.

Lemma 3.25 *Let $d \ge 1$ and $k \ge 0$ be integers. Then the vector space of discrete harmonic polynomials of degree at most k in $x_1, \dots, x_d$ has dimension*

$$\binom{k+d}{d} - \binom{k+d-2}{d},$$

where the second binomial coefficient is understood to be zero when $k < 2$.

Proof. For every integer j, let $\mathcal{P}_j^d$ be the real vector space of polynomials of degree at most j in $x_1, \dots, x_d$, with $\mathcal{P}_j^d = \{0\}$ when $j < 0$. Let $\mathrm{Harm}_{\le k}$ and $\mathrm{Harm}_{\le k}^{\dddot{}}$ be the spaces of harmonic and of discrete harmonic polynomials in $\mathcal{P}_k^d$. By Lemma 2.17, every $F \in \mathcal{P}_k^d$ can be written as $F = (1 - |x|^2)g + h$ with $g \in \mathcal{P}_{k-2}^d$ and $h \in \mathrm{Harm}_{\le k}$. The decomposition is unique: a harmonic polynomial $(1-|x|^2)g$ vanishes on $\partial \mathrm{B}(\mathbf{0}, 1)$, hence on $\mathrm{B}(\mathbf{0}, 1)$ by Lemma 2.12 applied to it and to its negative, so $g = 0$. Thus $\dim \mathrm{Harm}_{\le k} = \binom{k+d}{d} - \binom{k+d-2}{d}$, because $\dim \mathcal{P}_j^d = \binom{j+d}{d}$ for $j \ge 0$. The map Ξ is a bijection from $\mathcal{P}_j^d$ to itself for every $j \ge 0$, and the proof of Lemma 3.24 gives $\Delta^{\dddot{}}\Xi = \Xi\Delta$. Hence $\Xi(\mathrm{Harm}_{\le k}) = \mathrm{Harm}_{\le k}^{\dddot{}}$, which proves the dimension formula. □

For every integer $k \ge 0$, Lemma 3.25 and rank–nullity show that $\Delta^{\dddot{}}$ maps the polynomials of degree at most k onto those of degree at most $k-2$.

The preceding sections use the asymptotic estimates stated in Theorem 3.9, but not their proof. The next section proves those estimates; a reader willing to take the theorem on faith may skip to Section 3.7, which uses them to construct the harmonic measure from infinity in the plane.

3.6 ASYMPTOTICS OF THE GREEN FUNCTION*

Theorem 3.9 compares the discrete Green function with the rotationally invariant continuum Green function: the error is $O(|x|^{-d})$, two powers below the leading term, when $d \ge 3$; when $d = 2$ the difference is the constant $-\kappa$ plus $O(|x|^{-2})$. The proof corrects the continuum Green function by a term bounded by $C|x|^{-d}$. The negative discrete Laplacian of the corrected function is an even function $q : \mathbb{Z}^d \to \mathbb{R}$ of total mass one, with $|q(y)| \le C(1+|y|)^{-d-4}$. For $d \ge 3$, convolving the discrete Green function with q reproduces the corrected function; in the plane, the equality has an additive constant. After accounting for this constant, the error is a weighted sum of the differences $g^{\dddot{}}(x) - g^{\dddot{}}(x-y)$. Since q is even, we can pair these differences into second differences and apply the coupling estimate from Section 3.5. The constant κ then comes from the diagonal values of Lemma 3.7.

Throughout this section, $\mathcal{G}$ is the continuum Green function of (2.14) in Chapter 2, namely $\mathcal{G}(x) = a_d|x|^{2-d}$ for $d \ge 3$ and $\mathcal{G}(x) = -\frac{1}{2\pi}\log|x|$ for $d = 2$, and we write $\partial_i := \partial/\partial x_i$.

The first lemma constructs the corrected continuum Green function. The correction cancels the fourth-order term in the Taylor expansion of $\Delta^{\dddot{}}\mathcal{G}$, which would otherwise make the source of order $|y|^{-d-2}$, too large for the absolute second moment to be finite.

Lemma 3.26 *Let $d \geq 2$, and let $K : \mathbb{Z}^d \to \mathbb{R}$ be given by $K(\mathbf{0}) := 0$ and*

$$K(x) := \mathcal{G}(x) - \frac{1}{24} \sum_{i=1}^{d} x_i \, \partial_i^3 \mathcal{G}(x) \qquad \text{for } x \neq \mathbf{0} \,.$$

Then the function $q := -\Delta^{::} K$ is even, $\sum_{y \in \mathbb{Z}^d} q(y) = 1$, and there is a constant C such that

$$|q(y)| \leq C(1 + |y|)^{-d-4} \qquad \text{and} \qquad |K(x) - \mathcal{G}(x)| \leq C|x|^{-d}$$

for every $y \in \mathbb{Z}^d$ and every $x \in \mathbb{Z}^d \setminus \{\mathbf{0}\}$.

Proof. For $x \in \mathbb{R}^d \setminus \{\mathbf{0}\}$, write

$$\psi(x) := -\frac{1}{24} \sum_{i=1}^{d} x_i \, \partial_i^3 \mathcal{G}(x) \,.$$

At every nonzero lattice point, $\psi(x) = K(x) - \mathcal{G}(x)$. Every derivative of $\mathcal{G}$ of order $j \geq 1$ is homogeneous of degree $2 - d - j$, also when $d = 2$, so ψ is homogeneous of degree $-d$ and its derivatives of order j are $O(|x|^{-d-j})$. In particular $|K - \mathcal{G}| \leq C|x|^{-d}$. The function $\mathcal{G}$ is even and its third derivatives are odd, so K is even, and hence so is q.

Next we bound q. Since $\mathcal{G}$ is harmonic off the origin, so is $\partial_i^3 \mathcal{G}$, and the product rule gives $\Delta(x_i \, \partial_i^3 \mathcal{G}) = 2\partial_i^4 \mathcal{G}$. Hence

$$\Delta\psi = -\frac{1}{12} \sum_{i=1}^{d} \partial_i^4 \mathcal{G} \,.$$

Let $|x| \geq 2$, and apply Lemma 3.13 with $\delta = 1$ on the ball $\mathrm{B}(x, 3/2)$, on which every point has norm at least $|x|/4$. Part (c), applied to $\mathcal{G}$, whose sixth derivatives are $O(|x|^{-d-4})$ there, and part (b), applied to ψ, whose fourth derivatives are $O(|x|^{-d-4})$ there, give

$$\Delta^{::} \mathcal{G}(x) = \frac{1}{12} \sum_{i=1}^{d} \partial_i^4 \mathcal{G}(x) + O(|x|^{-d-4}) \,,$$

$$\text{and} \qquad \Delta^{::} \psi(x) = \Delta\psi(x) + O(|x|^{-d-4}) \,.$$

Adding these, the fourth derivatives cancel, so $|q(x)| \leq C|x|^{-d-4}$ for $|x| \geq 2$. Enlarging C to cover the finitely many remaining sites gives the stated bound.

Finally we compute the total mass of q. For an integer $N \geq 2$, sum $\Delta^{::} K$ over the lattice cube $[-N, N]^d \cap \mathbb{Z}^d$. Each edge with both endpoints in the

cube contributes two terms that cancel, so only the edges leaving the cube remain (Figure 3.13):

$$\sum_{x\in[-N,N]^d\cap\mathbb{Z}^d} \Delta^{::} K(x) = \sum_{j=1}^{d} \sum_{\pm} \sum_{\substack{x\in[-N,N]^d\cap\mathbb{Z}^d \\ x_j=\pm N}} \bigl(K(x\pm\mathbf{e}_j) - K(x)\bigr)\,.$$

There are $O(N^{d-1})$ edges leaving the cube. The first derivatives of ψ are $O(N^{-d-1})$ near the faces, so the differences of ψ contribute $O(N^{-2})$ in total. The gradient of $\mathcal{G}$ is homogeneous of degree $1-d$, also when $d=2$, and the second derivatives of $\mathcal{G}$ are $O(N^{-d})$ within distance one of the faces. Taylor's theorem therefore gives

$$\mathcal{G}(x+\mathbf{e}_j) - \mathcal{G}(x) = N^{1-d}\,\partial_j\mathcal{G}(x/N) + O(N^{-d})$$

uniformly for x in the cube with $x_j = N$, and the same expansion with $-\partial_j\mathcal{G}$ on the face $x_j = -N$. The errors total $O(N^{-1})$, and the main terms are Riemann sums, with mesh $1/N$, for the flux of $\nabla\mathcal{G}$ out of the cube $[-1,1]^d$. Since $\Delta^{::}K = -q$ is absolutely summable, letting $N\to\infty$ gives

$$-\sum_{y\in\mathbb{Z}^d} q(y) = \int_{\partial[-1,1]^d} \nabla\mathcal{G}\cdot n\,\mathrm{d}S\,,$$

where n is the outward unit normal. For $0<s<1$, the function $\mathcal{G}$ is harmonic between the sphere $\partial\mathrm{B}(\mathbf{0},s)$ and the boundary of the cube, so the divergence theorem shows that the flux of $\nabla\mathcal{G}$ out of the cube equals its flux out of the sphere. By (2.14), the outward normal derivative of $\mathcal{G}$ on the sphere is $-1/(d\omega_d s^{d-1})$, and the sphere has area $d\omega_d s^{d-1}$. Hence the flux is -1, and $\sum_y q(y) = 1$. □

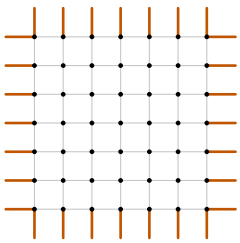


Figure 3.13: Summing $\Delta^{::}K$ over the cube $[-N,N]^2\cap\mathbb{Z}^2$, here for $N=3$: the two terms of each edge inside the cube (gray) cancel, and only the edges leaving it (vermilion) remain, as in the proof of Lemma 3.26.

The second lemma bounds second differences of a discrete harmonic function by its oscillation, as the coupling bounds first differences.

For a unit vector $v \in \{\pm\mathbf{e}_1, \dots, \pm\mathbf{e}_d\}$ and a function f on $\mathbb{Z}^d$, write $D_v f(w) := f(w+v) - f(w)$.

Lemma 3.27 *Let $d \geq 2$. There is a constant C, depending only on d, with the following property. Let $z \in \mathbb{Z}^d$, let $R \geq 2$ and $M \geq 0$, and let h be a real-valued function on the closure of $\mathrm{B}^{::}(z, 4R)$, discrete harmonic in $\mathrm{B}^{::}(z, 4R)$. Suppose that $|h(w) - h(w')| \leq M$ for all $w, w' \in \mathrm{B}^{::}(z, 4R) \cup \partial\mathrm{B}^{::}(z, 4R)$. Then*

$$\bigl|h(z+y) + h(z-y) - 2h(z)\bigr| \leq \frac{CM|y|^2}{R^2} \qquad \text{for every } y \in \mathbb{Z}^d \text{ with } |y| \leq R\,.$$

Proof. We apply the coupling estimate first to h and then to its first differences, and telescope along a lattice path to obtain the stated bound. By (3.30) and Lemma 3.21, a function discrete harmonic in a ball $\mathrm{B}^{::}(w, \rho)$ with $\rho > 1$ has first differences at w bounded in absolute value by $4d/\rho$ times its oscillation on $\partial\mathrm{B}^{::}(w, \rho)$. For radius one, we obtain the same estimate directly from the minimum principle: the ball consists of its center, whose value lies between the minimum and maximum of the boundary values. For $w \in \mathrm{B}^{::}(z, 3R)$, the ball $\mathrm{B}^{::}(w, R)$ lies in $\mathrm{B}^{::}(z, 4R)$, as in Figure 3.14, and its boundary lies in $\mathrm{B}^{::}(z, 4R) \cup \partial\mathrm{B}^{::}(z, 4R)$, so $|D_{\mathbf{e}_i} h(w)| \leq 4dM/R$. Each first difference $D_{\mathbf{e}_i} h$ is discrete harmonic in $\mathrm{B}^{::}(z, 3R)$, because differences commute with $\Delta^{::}$. For $w \in \mathrm{B}^{::}(z, 2R)$, the ball $\mathrm{B}^{::}(w, R/2)$ and its boundary lie in $\mathrm{B}^{::}(z, 3R)$, where the oscillation of $D_{\mathbf{e}_i} h$ is at most $8dM/R$. Hence

$$|D_{\mathbf{e}_j} D_{\mathbf{e}_i} h(w)| \leq \frac{64d^2 M}{R^2} \qquad \text{for } w \in \mathrm{B}^{::}(z, 2R)\,.$$

For $v, v' \in \{\pm\mathbf{e}_1, \dots, \pm\mathbf{e}_d\}$, the second difference $D_v D_{v'} h(w)$ uses four sites: w and $w+v$ and their translates by v'. It equals, up to sign, a forward second difference $D_{\mathbf{e}_j} D_{\mathbf{e}_i} h(w')$ at the point w' among them with the smallest coordinates.

Now let $|y| \leq R$, and let $p_0 = \mathbf{0}, p_1, \dots, p_L = y$ be a path of length $L = |y_1| + \cdots + |y_d|$ in which every coordinate moves monotonically, with steps $v_k := p_k - p_{k-1}$. Telescoping h along the path and its translate by $-y$ gives

$$\begin{aligned} &h(z+y) + h(z-y) - 2h(z) \\ &\qquad = \sum_{k=1}^{L} \bigl[D_{v_k} h(z + p_{k-1}) - D_{v_k} h(z - y + p_{k-1})\bigr]\,. \end{aligned}$$

Now telescope each bracket along the same path translated to $z - y + p_{k-1}$. Since differences commute, this gives

$$h(z+y) + h(z-y) - 2h(z) = \sum_{k=1}^{L} \sum_{l=1}^{L} D_{v_k} D_{v_l} h(z - y + p_{k-1} + p_{l-1})\,.$$

The four points used by each term lie in the coordinate box between $z-y$ and $z+y$, so the corresponding point w' lies within distance $|y| \le R$ of z. Since $L^2 \le d|y|^2$, the sum is at most $64d^3M|y|^2/R^2$ in absolute value. □

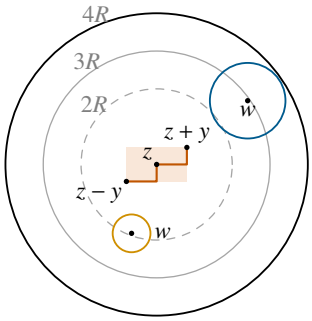


Figure 3.14: The balls in the proof of Lemma 3.27: for $w \in \mathrm{B}^{::}(z, 3R)$ the ball $\mathrm{B}^{::}(w, R)$ (blue) lies in $\mathrm{B}^{::}(z, 4R)$, and for $w \in \mathrm{B}^{::}(z, 2R)$ the ball $\mathrm{B}^{::}(w, R/2)$ (orange) lies in $\mathrm{B}^{::}(z, 3R)$. The second difference at z is telescoped along a monotone path from $z-y$ through z to $z+y$, inside the box between them (shaded).

We record how the potential kernel varies. By the nonnegativity of the killed Green function in Lemma 3.10, applied with the pair $(u, -v)$, and the evenness of a,

$$a(u+v) \le a(u) + a(v) \qquad \text{for all } u, v \in \mathbb{Z}^2 ,$$

where the cases $u = \mathbf{0}$ and $v = \mathbf{0}$ are immediate. Since $a(\pm\mathbf{e}_i) = \frac{1}{4}$, it follows that $a(y) \le \frac{1}{4}(|y_1| + |y_2|)$, and

$$|a(x) - a(x-y)| \le a(y) \le \frac{1}{4}(|y_1| + |y_2|) \qquad \text{for all } x, y \in \mathbb{Z}^2 . \tag{3.33}$$

Proof of Theorem 3.9. Let K and q be as in Lemma 3.26. Counting lattice points in shells, its bound on q gives

$$\sum_{y \in \mathbb{Z}^d} |q(y)|(1+|y|^2) < \infty \qquad \text{and} \qquad \sum_{|y|>R} |q(y)|(1+|y|) \le CR^{-3} \quad \text{for } R \ge 1 .$$

By Lemma 3.11 and (3.33), every convolution below converges absolutely. Fix x, let $r := |x|$ be large, and let $R := r/12$. Every point of $\mathrm{B}^{::}(x, 4R) \cup \partial\mathrm{B}^{::}(x, 4R)$ has norm at least $r/2$, so the ball does not contain the origin (Figure 3.15).

Step 1: We prove the transient case $d \ge 3$. By Lemma 3.11, dominated convergence with majorant $C|q(y)|$ gives $(g^{::} * q)(w) \to 0$ as $|w| \to \infty$. Also $K(w) \to 0$ by Lemma 3.26. Since $\Delta^{::}(K - g^{::} * q) = 0$, Lemma 3.17 gives $K = g^{::} * q$. Using $\sum_y q(y) = 1$, we obtain

$$g^{::}(x) - K(x) = \sum_{y \in \mathbb{Z}^d} q(y)\big[g^{::}(x) - g^{::}(x-y)\big] .$$

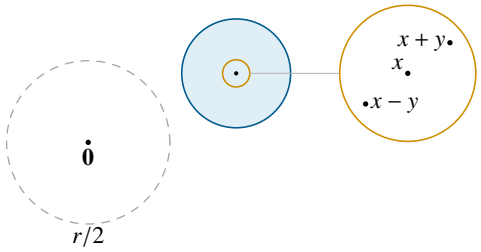


Figure 3.15: The proof of Theorem 3.9 for $|x| = r$ and $R = r/12$: the ball $\mathrm{B}^{::}(x, 4R)$ (shaded) misses the disk of radius $r/2$ (dashed), and the ball $\mathrm{B}^{::}(x, R)$ (orange, enlarged on the right) contains the pairs $x \pm y$ whose second differences are bounded by Lemma 3.27.

Suppose that $g^{::}(v) \le C(1 + |v|)^{-\beta}$ for every v, where $0 \le \beta \le d - 2$; Lemma 3.11 gives this for $\beta = 0$. We split the sum at $|y| = R$. On $\mathrm{B}^{::}(x, 4R)$ and its boundary, the function $g^{::}$ is discrete harmonic, with oscillation at most $Cr^{-\beta}$. Pairing y with $-y$, which q allows because it is even, and applying Lemma 3.27, bound the part with $|y| \le R$ by

$$\frac{1}{2} \sum_{|y| \le R} |q(y)| \left| 2g^{::}(x) - g^{::}(x - y) - g^{::}(x + y) \right|$$
$$\le \frac{Cr^{-\beta}}{R^2} \sum_{y \in \mathbb{Z}^d} |q(y)||y|^2 \le Cr^{-\beta-2} .$$

In the part with $|y| > R$, the terms containing $g^{::}(x)$, or $g^{::}(x - y)$ with $|x - y| > r/2$, total at most $Cr^{-\beta}R^{-3}$; the others have $|y| \ge r/2$ and total at most $Cr^{-d-4} \sum_{|v| \le r/2} g^{::}(v) \le Cr^{-\beta-4}$. Hence $g^{::}(x) - K(x) = O(r^{-\beta-2})$. Since $K(x) = O(r^{2-d})$, this gives the supposition with $\min\{\beta + 2, d - 2\}$ in place of β. Repeating, we reach $\beta = d - 2$, where $g^{::}(x) - K(x) = O(r^{-d})$, and Lemma 3.26 gives $g^{::}(x) = a_d|x|^{2-d} + O(|x|^{-d})$.

Step 2: We prove the planar case. The function $h := K + a * q$ has discrete Laplacian $-q + q = 0$. Since $\sum_y q(y) = 1$, equation (3.33) gives

$$|(a * q)(w) - a(w)| \le \sum_{y \in \mathbb{Z}^2} |q(y)|a(y) \le \frac{1}{4} \sum_{y \in \mathbb{Z}^2} |q(y)|(|y_1| + |y_2|) < \infty ,$$

uniformly in w. The logarithmic estimate in Lemma 3.8 and Lemma 3.26 make $K + a$ bounded. Hence h is bounded and is constant by Theorem 3.18 with $n = 1$. Write κ_0 for this constant. Since $\sum_y q(y) = 1$,

$$a(x) + K(x) - \kappa_0 = \sum_{y \in \mathbb{Z}^2} q(y)[a(x) - a(x - y)] .$$

On $\mathrm{B}^{::}(x, 4R)$ and its boundary, the function a is discrete harmonic, and by Lemma 3.8 its oscillation is bounded, since $\log(1 + |w|)$ varies

by a bounded amount there. As in Step 1, Lemma 3.27 bounds the part with $|y| \le R$ by CR^{-2}, and (3.33) bounds the part with $|y| > R$ by $\frac{1}{4}\sum_{|y|>R} |q(y)|(|y_1| + |y_2|) \le CR^{-3}$. Since $-K(x) = \frac{1}{2\pi}\log|x| + O(|x|^{-2})$ by Lemma 3.26, we conclude that

$$a(x) = \frac{1}{2\pi}\log|x| + \kappa_0 + O(|x|^{-2}).$$

Step 3: We identify κ_0. Lemma 3.7 and Exercise 3.16 give

$$a(n,n) = \frac{1}{2\pi}\log n + \frac{\log 2}{\pi} + \frac{\gamma_E}{2\pi} + O(n^{-2}).$$

Since $|(n,n)| = \sqrt{2}\,n$, comparison with Step 2 gives

$$\kappa_0 = \frac{2\gamma_E + 3\log 2}{4\pi} = \frac{2\gamma_E + \log 8}{4\pi},$$

which is the constant κ in (3.17). Negating the asymptotic for $a(x)$ completes the proof. □

Since $g^{::} > 0$, and $a(x) > 0$ for $x \neq \mathbf{0}$ by Lemma 3.10 with $y = x$, Theorem 3.9 gives $c(1+|x|)^{2-d} \le g^{::}(x) \le C(1+|x|)^{2-d}$ for $d \ge 3$ and the lower bound $a(x) \ge c\log(1+|x|)$ matching Lemma 3.11 for $d = 2$. With $y = x$, Lemma 3.10 and the planar bounds also show that, for a walk started at x, the expected number of visits to x before hitting the origin is of order $\log(1+|x|)$.

These asymptotics let the lattice models imitate their continuum counterparts. Chapter 4 uses them to place the divisible sandpile within a bounded distance of a Euclidean ball, and Chapter 5 uses them again for internal DLA, where the boundary fluctuations are no longer bounded: there the inner width is at most of order $r^{1/3}$, up to logarithmic factors. The next section applies them to random walk in the plane started far from a nonempty finite set, whose hitting distribution is the attachment rule of external DLA in Chapter 8.

3.7 HARMONIC MEASURE FROM INFINITY IN THE PLANE

How does a random walk started far from a nonempty finite set $K \subset \mathbb{Z}^2$ first hit K? Simple random walk in the plane is recurrent (Lemma 3.6), so the walk hits K almost surely, and one expects the hitting distribution to depend less and less on the starting point as it recedes. We prove that it converges and describe the limit through the potential kernel. This limit gives meaning to releasing a particle from infinity in external DLA, the model of Chapter 8 that adds to a cluster the first site of its outer boundary that such a walk visits.

We first define the limiting hitting distribution on the finite set and then on its outer boundary.

Throughout this section, $X = (X_n)_{n\ge 0}$ is simple random walk on $\mathbb{Z}^2$, and $\mathbb{P}_y$ and $\mathbb{E}_y$ denote its law and expectation started at y. For $E \subset \mathbb{Z}^2$, let τ_E be the hitting time (3.4) of E, and recall the outer boundary ∂A of $A \subset \mathbb{Z}^2$ from Section 3.1.

Definition 3.28 Let $K \subset \mathbb{Z}^2$ be finite and nonempty. The **harmonic measure of K from infinity** is

$$\mathcal{H}_K(x) := \lim_{|y|\to\infty} \mathbb{P}_y\{X_{\tau_K} = x\} \qquad \text{for every } x \in K\,.$$

For a finite nonempty set $A \subset \mathbb{Z}^2$, we write

$$H_A(z) := \mathcal{H}_{\partial A}(z) \qquad \text{for every } z \in \partial A\,,$$

for the harmonic measure of its outer boundary.

The limit in the definition exists for every nonempty finite set, and Figure 3.16 shows it for two small sets. We prove this with the potential kernel a of Section 3.3, defined in (3.12).

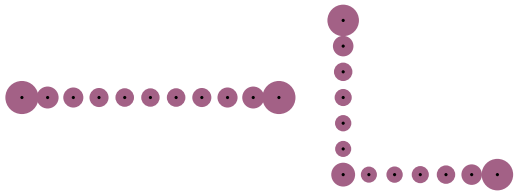


Figure 3.16: Harmonic measure from infinity $\mathcal{H}_K$ (dot areas proportional to mass) on a segment of 11 sites and on an L of 13 sites in $\mathbb{Z}^2$, computed from the potential kernel by Lemma 3.29(a). It is largest at the ends and at the outer corner, like the arcsine law of Figure 2.13.

Recall that $\Delta^{\cdot\cdot} a = \delta_{\mathbf{0}}$ for the discrete Laplacian $\Delta^{\cdot\cdot} f(x) = \sum_{y\sim x}(f(y)-f(x))$ of (3.1), that $a(\mathbf{0}) = 0$, that a is even by the symmetry of the walk, and that, by (3.17),

$$a(x) = \frac{1}{2\pi}\log|x| + \kappa + O(|x|^{-2}) \qquad \text{as } |x| \to \infty\,, \tag{3.34}$$

where κ is a constant. In particular, $a(x-v) - a(x) \to 0$ as $|x| \to \infty$, for every $v \in \mathbb{Z}^2$.

We also use the following consequence of recurrence. Let $A \subset \mathbb{Z}^2$ be finite and nonempty, and let $g : \mathbb{Z}^2 \to \mathbb{R}$ be bounded and harmonic on $\mathbb{Z}^2 \setminus A$. For $y \in \mathbb{Z}^2$, the process $g(X_{k\wedge\tau_A})$ is a bounded martingale under $\mathbb{P}_y$. Moreover, $\tau_A < \infty$ almost surely: by Lemma 3.6, the walk returns to y infinitely often, its excursions from y are independent by the strong Markov property, and each of

them visits a fixed point of A with the same positive probability. The optional stopping theorem and the bounded convergence theorem therefore give

$$g(y) = \mathbb{E}_y[g(X_{\tau_A})] \qquad \text{for every } y \in \mathbb{Z}^2 . \tag{3.35}$$

The next lemma proves that the limit exists and describes harmonic measure through a potential that vanishes on the set. Chapter 8 uses this potential to prove the discrete Beurling estimate (Section 8.2), which bounds the harmonic measure of a single site of a connected set.

Lemma 3.29 *Let $A \subset \mathbb{Z}^2$ be finite and nonempty.*

(a) For every $z \in A$, the limit $\mathcal{H}_A(z) = \lim_{|y|\to\infty} \mathbb{P}_y\{X_{\tau_A} = z\}$ exists, and $\mathcal{H}_A$ is the unique probability measure ν on A whose potential $x \mapsto \sum_{v\in A} \nu(v)\, a(x-v)$ is constant on A.

(b) Let U_A be the potential of $\mathcal{H}_A$ minus its value on A. Then U_A is nonnegative, vanishes on A, is harmonic on $\mathbb{Z}^2 \setminus A$, satisfies $\Delta^{::} U_A(v) = \mathcal{H}_A(v)$ for every $v \in A$, and satisfies $U_A(x) = a(x) + O(1)$ as $|x| \to \infty$.

(c) Let $F : \mathbb{Z}^2 \to \mathbb{R}$ be harmonic on $\mathbb{Z}^2 \setminus A$ and such that $F - a$ is bounded. Then $F(x) - U_A(x) = \mathbb{E}_x[F(X_{\tau_A})]$ for every $x \in \mathbb{Z}^2$.

Proof. All three parts rest on (3.35).

(a) Fix $v, w \in A$. By (3.34), the function $x \mapsto a(x-v) - a(x-w)$ tends to zero as $|x| \to \infty$, so it is bounded, and it is harmonic on $\mathbb{Z}^2 \setminus \{v, w\}$. By (3.35), for every $y \in \mathbb{Z}^2$,

$$a(y-v) - a(y-w) = \sum_{z\in A} \mathbb{P}_y\{X_{\tau_A} = z\}[a(z-v) - a(z-w)] . \tag{3.36}$$

Since $\tau_A < \infty$ almost surely, each $\mathbb{P}_y\{X_{\tau_A} = \cdot\}$ is a probability measure on the finite set A, so every sequence y_k with $|y_k| \to \infty$ has a subsequence along which these measures converge to a probability measure ν on A. Letting y run along this subsequence in (3.36) gives $\sum_{z\in A} \nu(z)[a(z-v)-a(z-w)] = 0$. Since a is even, the potential of ν takes the same value at v and at w. Thus every subsequential limit is a probability measure on A with constant potential on A.

We show that there is only one such measure. Let ν_1 and ν_2 be two of them, and let $P(x) := \sum_{v\in A}(\nu_1 - \nu_2)(v)\, a(x-v)$. Since $\nu_1 - \nu_2$ has total mass zero, $P(x) = \sum_{v\in A}(\nu_1 - \nu_2)(v)[a(x-v) - a(x)]$ tends to zero as $|x| \to \infty$, so P is bounded. It is harmonic off A and constant on A, so (3.35) shows that P is constant on $\mathbb{Z}^2$. Hence $\nu_1 - \nu_2 = \Delta^{::} P = 0$. All subsequential limits therefore coincide, which proves (a).

(b) Since $\Delta^{::} a = \delta_{\mathbf{0}}$, we have $\Delta^{::} U_A = \sum_{v \in A} \mathcal{H}_A(v) \delta_v$, which gives the harmonicity off A and the Laplacian on A. Since $\mathcal{H}_A$ is a probability measure and $a(x - v) - a(x) \to 0$ as $|x| \to \infty$, the function $U_A - a$ is bounded at infinity. Since $a(x) \to \infty$, it follows that $U_A(x) \to \infty$ as $|x| \to \infty$. Fix $x \notin A$, and choose $R > |x|$ so large that $U_A \geq 0$ outside $\mathrm{B}^{::}(\mathbf{0}, R)$. The minimum principle, Lemma 3.2, applied on the finite set $\mathrm{B}^{::}(\mathbf{0}, R) \setminus A$, whose boundary values are those on A and on $\partial \mathrm{B}^{::}(\mathbf{0}, R)$, gives $U_A(x) \geq 0$.

(c) The function $F - U_A$ is bounded, because $F - a$ and $U_A - a$ are, and it is harmonic off A and equal to F on A. Hence (3.35), applied to $F - U_A$, proves (c). □

Let F be as in part (c). Since $X_{\tau_A} \in A$, part (c) gives

$$\min_A F \leq F - U_A \leq \max_A F \qquad \text{on } \mathbb{Z}^2 . \tag{3.37}$$

This compares U_A with other potentials in both directions, and the proof of the Beurling estimate in Section 8.2 uses it both ways. Taking $A = \{\mathbf{0}\}$ in part (b) gives $U_{\{\mathbf{0}\}} = a$, so $a \geq 0$. Part (a) also shows that $\mathcal{H}_K$ is a probability measure on K for every finite nonempty set K.

On every recurrent graph, the Laplacian of a bounded function that is harmonic off a finite set sums to zero over that set; Exercise 3.43 asks you to prove this. In dimensions $d \geq 3$, the unconditioned hitting probabilities tend to zero; Exercise 3.46 obtains a probability measure by conditioning the walk to hit the set, and Section 8.4 of Chapter 8 uses it as the attachment rule of external DLA in $\mathbb{Z}^d$.

The harmonic measure of the outer boundary of a growing cluster is the attachment rule of external DLA, to which Chapter 8 returns. First, however, Chapter 4 turns from potential theory to lattice growth, beginning with the divisible sandpile, whose odometer the Green function of this chapter controls.

EXERCISES

Exercise 3.1. In the setting of Section 3.1, let $A \subset \mathbb{Z}^d$ be finite, and let $h, g : \bar{A} \to \mathbb{R}$ be functions such that h is discrete harmonic on A and g is discrete superharmonic on A. Prove that if $g \geq h$ on ∂A, then $g \geq h$ on A.

Exercise 3.2. In the setting of Section 3.1, let $g : \mathbb{Z}^d \to \mathbb{R}$ be discrete superharmonic, and let $A \subset \mathbb{Z}^d$ be finite. Let $h : \mathbb{Z}^d \to \mathbb{R}$ be the unique function that equals g outside A and is discrete harmonic on A. Prove that h is discrete superharmonic on $\mathbb{Z}^d$.

Exercise 3.3. In the setting of Section 3.1, let $f, g : \mathbb{Z}^d \to \mathbb{R}$ be discrete

superharmonic. Determine whether $\min(f, g)$ and $\max(f, g)$ must be discrete superharmonic.

Exercise 3.4. Let $L^2(\mathbb{Z}^d)$ be the space of square-summable functions $\mathbb{Z}^d \to \mathbb{R}$, with the inner product $(f, g) := \sum_{x\in\mathbb{Z}^d} f(x)g(x)$. Prove that the discrete Laplacian (3.1) is self-adjoint on $L^2(\mathbb{Z}^d)$, that is, $(f, \Delta^{::} g) = (\Delta^{::} f, g)$ for all $f, g \in L^2(\mathbb{Z}^d)$.

Exercise 3.5. Let $A \subset \mathbb{Z}^d$ be finite and nonempty. The energy of $u : \bar{A} \to \mathbb{R}$ is the sum of $(u(x) - u(y))^2$ over the edges $\{x, y\}$ of $\mathbb{Z}^d$ that meet A. Show that if h is discrete harmonic in A and v vanishes on ∂A, then the energy of $h + v$ is that of h plus that of v. Deduce that h is the unique function of least energy with its values on ∂A.

Exercise 3.6. Let $G = (V, E)$ be a finite connected simple graph, and let Δ be the Laplacian (3.2) of G.

(a) Show that the image of Δ is the space of functions on V whose sum is zero.

(b) Let $v, w \in V$. Find a function $f : V \to \mathbb{R}$ satisfying $\Delta f = \delta_v - \delta_w$.

Exercise 3.7. Show that a supermartingale $(X_n)_{n\geq 0}$ with respect to a filtration, in the sense of Section 3.2, is also a supermartingale with respect to its natural filtration.

Exercise 3.8. For positive integers ℓ and r, let $(X_t)_{t\geq 0}$ be a random walk on $\mathbb{Z}$ started at zero, and let τ be its hitting time of $\{-\ell, r\}$. Apply Theorem 3.3.

(a) Compute $\mathbb{P}_0\{X_\tau = r\}$ when $(X_t)_{t\geq 0}$ is simple random walk.

(b) Compute $\mathbb{P}_0\{X_\tau = r\}$ when the increments are independent, equal to 1 with probability p, and equal to -1 with probability $1 - p$, where $1/2 < p < 1$.

Hint for (b): Check that $((1 - p)/p)^{X_{t\wedge\tau}}$ is a martingale.

Exercise 3.9. Let $(U_k)_{k\geq 0}$ be simple random walk from zero on $\mathbb{Z}$, and let τ_b be its hitting time (3.4) of an integer $b \geq 1$. Prove that, for $0 < s < 1$,

$$\mathbb{E}[s^{\tau_b}] = \left(\frac{1 - \sqrt{1 - s^2}}{s}\right)^b .$$

Hint: The function $u(j) := \mathbb{E}_j[s^{\tau_b}]$ satisfies $u(j) = s(u(j - 1) + u(j + 1))/2$ for $j < b$, equals 1 at b, and is bounded as $j \to -\infty$.

Exercise 3.10. For simple random walk $(X_n)_{n\geq 0}$ on $\mathbb{Z}$ from zero, find a stopping time T for its natural filtration such that X_T is not integrable, in contrast to Theorem 3.3.

Hint: Use twice the first positive return time to zero.

Exercise 3.11. Let $d \geq 1$ and $R \geq 1$. Strengthen Lemma 3.4 by proving that, for every $r > 0$,

$$\mathbb{P}_{\mathbf{0}}\{\tau_{\partial \mathrm{B}_R^{\vdots}} \leq rR^2\} \leq 2de^{-1/(2dr)} \qquad \text{and} \qquad \sup_{x \in \mathrm{B}_R^{\vdots}} \mathbb{P}_x\{\tau_{\partial \mathrm{B}_R^{\vdots}} > rR^2\} \leq 2^{1-r/9}\,.$$

Deduce that $\mathbb{E}_x e^{\theta \tau_{\partial \mathrm{B}_R^{\vdots}}/R^2}$ is bounded uniformly in $R \geq 1$ and $x \in \mathrm{B}_R^{\vdots}$ if $\theta < (\log 2)/9$.

Exercise 3.12. Show that Lemma 3.5 can fail for an infinite set A, even if $\mathbb{P}_x\{\tau_{\partial A} < \infty\} = 1$ for every $x \in A$. Can the lemma fail when $\mathbb{E}_x \tau_{\partial A} < \infty$ for every $x \in A$?

Exercise 3.13. Let $d \geq 2$ and $h(x) := x_d$. Show that (3.5) defines transition probabilities from every x with $x_d \geq 1$: to $x \pm \mathbf{e}_j$ with probability $1/(2d)$ for $j < d$, and to $x \pm \mathbf{e}_d$ with probability $(x_d \pm 1)/(2dx_d)$. Show that, on every finite path in $\{x_d \geq 1\}$, simple random walk conditioned to reach height R before height 0 agrees with this h-transform for all large integers R.

Exercise 3.14. For $\Delta^{\vdots}$ in (3.1), let $d \geq 1$ and $f : \mathbb{Z}^d \to \mathbb{R}$, and prescribe real values on the two layers $\{x \in \mathbb{Z}^d : x_1 \in \{0, 1\}\}$. Prove that there is a unique function $F : \mathbb{Z}^d \to \mathbb{R}$ that takes these values on the two layers and satisfies $\Delta^{\vdots} F = f$.

Exercise 3.15. Let $0 \leq k < d$. Use Lemma 3.6 to show that simple random walk in $\mathbb{Z}^d$ visits $\{x : x_{k+1} = \cdots = x_d = 0\}$ infinitely often almost surely if $d - k \leq 2$, and finitely often almost surely if $d - k \geq 3$. Thus a walk in $\mathbb{Z}^3$ visits a coordinate axis infinitely often.

Exercise 3.16. Prove that the sum in Lemma 3.7 satisfies, as $n \to \infty$,

$$\sum_{j=1}^{n} \frac{1}{2j-1} = \frac{1}{2} \log n + \log 2 + \frac{\gamma_E}{2} + O(n^{-2})\,,$$

with Euler's constant γ_E.

Hint: Subtract half the nth harmonic number from the $2n$th.

Exercise 3.17. Give a second proof of Lemma 3.7 by reflection. Let $n \geq 1$ be an integer, let U and V be the independent one-dimensional walks of (3.10), let $b := 2n - 1$, and let τ_b be the first time U reaches b.

(a) Show that $\mathbb{P}\{U_{2m} = b - 1\} - \mathbb{P}\{U_{2m} = b + 1\} = 2\mathbb{P}\{\tau_b = 2m + 1\}$ for every $m \geq 0$.

(b) Show that $\sum_{m \geq 0} \mathbb{P}\{V_{2m} = 0\}\mathbb{P}\{\tau_b = 2m + 1\} = 2/(\pi b)$, and deduce that $a(n, n) - a(n - 1, n - 1) = 1/(\pi b)$.

Hint for (b): Use the identity $\mathbb{P}\{V_{2m} = 0\} = \frac{2}{\pi}\int_0^{\pi/2} \cos^{2m} t \, \mathrm{d}t$.

Exercise 3.18. Let $d \geq 1$, let $A \subset \mathbb{Z}^d$ be finite, and let $x \in A$. Define $\mu_x : \mathbb{Z}^d \to [0,1]$ by

$$\mu_x(z) := \begin{cases} \mathbb{P}_x\{X_{\tau_{\partial A}} = z\}, & z \in \partial A, \\ 0, & z \notin \partial A\,. \end{cases}$$

Show that the killed Green function (3.18) satisfies

$$\Delta^{::} g_A^{::}(x, \cdot) = \mu_x - \delta_x\,.$$

Exercise 3.19. Let $A \subset \mathbb{Z}^d$ be finite, and let $x \in A$. Show that the number of visits to x before $\tau_{\partial A}$ by the walk from x is geometric on $\{1, 2, \ldots\}$ with mean $2d\, g_A^{::}(x,x)$, and independent of $X_{\tau_{\partial A}}$.

Exercise 3.20. Let $d \geq 2$, let $A \subset \mathbb{Z}^d$ be finite, and let $F : A \to \mathbb{R}$ and $b : \partial A \to \mathbb{R}$. Let $u : A \cup \partial A \to \mathbb{R}$ be the unique solution of the problem (3.3) from Section 3.1, so that $\Delta^{::} u = F$ in A and $u = b$ on ∂A.

(a) Show that

$$u(x) = g(x) - \sum_{z \in A} g^{::}(x - z) F(z)\,,$$

where g is discrete harmonic in A and $g^{::}$ is the Green function from (3.7).

(b) Show that

$$u(x) = h(x) - \sum_{z \in A} g_A^{::}(x, z) F(z)\,,$$

where h is discrete harmonic in A and equals b on ∂A. Here $g_A^{::}$ is the killed Green function from (3.18).

(c) Give a probabilistic representation of $u(x)$.

Exercise 3.21. For $d \geq 2$ and $z \in \mathbb{Z}^d$, write $h_z(x) := g^{::}(x - z)$, so that $\Delta^{::} h_z = -\delta_z$ by (3.7). For finitely supported $f : \mathbb{Z}^d \to \mathbb{R}$, write

$$(f, h_z) := \sum_{x \in \mathbb{Z}^d} f(x) h_z(x)\,.$$

Show that the numbers (f, h_z) for $z \in \mathbb{Z}^d$ determine f uniquely.

Exercise 3.22. For $d \geq 2$ and $z \in \mathbb{Z}^d$, write $h_z(x) := g^{::}(x - z)$ with $g^{::}$ from (3.7). For finitely supported $f : \mathbb{Z}^d \to \mathbb{R}$, write

$$(f, h_z) := \sum_{x \in \mathbb{Z}^d} f(x) h_z(x)\,.$$

Prove that the following are equivalent:

(a) There is a finitely supported function F such that $f = \Delta^{::} F$.

(b) The number (f, h_z) vanishes for all but finitely many $z \in \mathbb{Z}^d$.

Exercise 3.23. Let $G = (V, E)$ be a finite connected simple graph, and let Δ be the Laplacian (3.2) of G. Let $g : V \times V \to \mathbb{R}$ be the symmetric function with zero row sums and

$$\Delta_y \, g(x, y) = |V|^{-1} - \mathbf{1}\{x = y\} \, .$$

For distinct vertices $u, v \in V$, let $R(u, v) := g(u, u) + g(v, v) - 2g(u, v)$. Let $(X_t)_{t \geq 0}$ be simple random walk on G. For $w \in V$, write

$$\tau_w := \inf\{t \geq 0 : X_t = w\} \qquad \text{and} \qquad \tau_w^+ := \inf\{t \geq 1 : X_t = w\} \, .$$

Prove the escape identity

$$R(u, v) = \frac{1}{\deg(u) \, \mathbb{P}_u\{\tau_v < \tau_u^+\}} \, ,$$

and deduce that

$$\deg(u) \mathbb{P}_u\{\tau_v < \tau_u^+\} = \deg(v) \mathbb{P}_v\{\tau_u < \tau_v^+\} \, .$$

Exercise 3.24. Give a probabilistic proof of the leading term in Lemma 3.12(b) for a disk in $\mathbb{Z}^2$ as follows.

(a) For simple random walk $(X_n)_{n \geq 0}$ in $\mathbb{Z}^2$ started at $\mathbf{0}$, use Stirling's formula, or Wallis' integral $\binom{2k}{k} 2^{-2k} = \frac{2}{\pi} \int_0^{\pi/2} \cos^{2k} t \, \mathrm{d}t$, to prove that, for $k \geq 1$,

$$\mathbb{P}_{\mathbf{0}}\{X_{2k} = \mathbf{0}\} = \frac{1}{\pi k} + O(k^{-2}) \, .$$

Hint: Rotating by 45° makes the two coordinate processes independent.

(b) Deduce that

$$\sum_{k=0}^{2n} \mathbb{P}_{\mathbf{0}}\{X_k = \mathbf{0}\} = \frac{1}{\pi} \log n + O(1) \, . \tag{3.38}$$

(c) Let

$$\tau_r := \inf\{n \geq 0 : |X_n| \geq r\} \, .$$

Show that, for every $\epsilon > 0$,

$$\mathbb{P}_{\mathbf{0}}\{r^{2-\epsilon} \le \tau_r \le r^{2+\epsilon}\} \longrightarrow 1 \qquad \text{as } r \to \infty\,.$$

Conclude from (3.38) that the expected number of visits to $\mathbf{0}$ before time τ_r is $(2/\pi + o(1)) \log r$. Since $g^{::}_{\mathrm{B}(\mathbf{0},r)}(\mathbf{0},\mathbf{0})$ is one fourth of this expectation by (3.18), we obtain

$$g^{::}_{\mathrm{B}(\mathbf{0},r)}(\mathbf{0},\mathbf{0}) = \Big(\frac{1}{2\pi} + o(1)\Big) \log r\,.$$

Compare the result with Lemma 3.12(b). Hint: reversing a path of length $2n$ from $\mathbf{0}$ to $\mathbf{0}$ turns an exit from the disk after time n into an exit before time n.

Exercise 3.25. For simple random walk in $\mathbb{Z}^2$, denote by $\tau^+_{\mathbf{0}}$ the first positive hitting time of zero. For $r \ge 2$, use Theorem 3.9 to prove that, as $r \to \infty$,

$$\mathbb{P}_{\mathbf{0}}\{\tau_{\partial \mathrm{B}^{::}(\mathbf{0},r)} < \tau^+_{\mathbf{0}}\} = \frac{\pi/2}{\gamma_E + \frac{3}{2}\log 2 + \log r} + O\left(\frac{1}{r \log^2 r}\right),$$

where γ_E is Euler's constant.

Exercise 3.26. For $d \ge 2$ and real $r \ge 2$, use Lemma 3.12 with $B := \mathrm{B}^{::}(\mathbf{0}, r)$ to prove that, for $x \in B \setminus \mathrm{B}^{::}(\mathbf{0}, r/2)$,

$$g^{::}_B(\mathbf{0}, x) \asymp \frac{r + 1 - |x|}{r^{d-1}}\,,$$

where $\asymp$ denotes bounds by positive constant multiples depending only on d.

Exercise 3.27. Let $d \ge 2$, let $R \ge 2$, and let $B := \mathrm{B}^{::}(\mathbf{0}, R)$. For $x \in B \setminus \{\mathbf{0}\}$, prove that

$$\mathbb{P}_x\{\tau_{\mathbf{0}} < \tau_{\partial B}\} = \frac{g^{::}_B(\mathbf{0}, x)}{g^{::}_B(\mathbf{0}, \mathbf{0})}\,.$$

Use Lemma 3.12 to estimate this probability explicitly in dimensions $d = 2$ and $d \ge 3$.

Exercise 3.28. Let $A \subset \mathbb{Z}^d$ be finite, let $z \in A$, and let $x, y \in A \setminus \{z\}$. Prove that

$$g^{::}_{A\setminus\{z\}}(x, y) = g^{::}_A(x, y) - \frac{g^{::}_A(x, z)\, g^{::}_A(z, y)}{g^{::}_A(z, z)}\,,$$

and interpret the last term. Prove that, for $b \in \partial A$, the walk from x leaves $A \setminus \{z\}$ at b with probability $\mathbb{P}_x\{X_{\tau_{\partial A}} = b\} - \mathbb{P}_z\{X_{\tau_{\partial A}} = b\} g^{::}_A(x, z)/g^{::}_A(z, z)$.

Exercise 3.29. For $d \geq 3$, use Theorem 3.9 to prove that every sufficiently small $\ell > 0$ admits a radius $r > 0$ such that the level set of the Green function

$$L_\ell := \{x \in \mathbb{Z}^d : g^{::}(x) \geq \ell\}$$

satisfies

$$\mathrm{B}^{::}(\mathbf{0}, r - c/r) \subset L_\ell \subset \overline{\mathrm{B}}(\mathbf{0}, r + c/r)\,,$$

where c is a constant depending only on d.

Exercise 3.30. For simple random walk $(X_n)_{n\geq 0}$ in $\mathbb{Z}^2$, let $z \in \mathbb{Z}^2$ satisfy $z \cdot \mathbf{e}_2 > 0$, and let

$$\tau := \inf\{n \geq 0 : X_n \cdot \mathbf{e}_2 = 0\}\,,$$

and write $p(z) := \mathbb{P}_z\{X_\tau \cdot \mathbf{e}_1 = 0\}$.

(a) Find a formula for $p(z)$ in terms of the potential kernel (3.12).

(b) What is $p(0, 1)$?

(c) Let $K \subset \mathbb{R} \times (0, \infty)$ be compact. Show that, uniformly for $(x, y) \in K \cap n^{-1}\mathbb{Z}^2$,

$$\mathbb{P}_{(0,ny)}\{X_\tau = (nx, 0)\} = \frac{y}{\pi n(x^2 + y^2)} + O_K(n^{-2})\,.$$

(d) Let w lie in the upper half-plane, let $w_n \in n^{-1}\mathbb{Z}^2$ converge to w, and let $[t_1, t_2]$ be an interval on the real axis. Show that the probability that simple random walk on $n^{-1}\mathbb{Z}^2$, started at w_n and stopped on first hitting the real axis, lands in $[t_1, t_2]$ converges to $\angle t_1 w t_2/\pi$.

Hint for (a): Use the method of images.

Exercise 3.31*. Show that (3.5), with h the potential kernel a, defines a walk on $\mathbb{Z}^2 \setminus \{\mathbf{0}\}$, that its expected number of visits to its starting point x is $8a(x)$, and that it is transient. Show that, on every finite path avoiding $\mathbf{0}$, it is the limit as $R \to \infty$ of simple random walk conditioned to leave $\mathrm{B}^{::}_R$ before hitting $\mathbf{0}$.

Exercise 3.32. Let $d \geq 3$ and $\alpha > 0$. Use the Green-function asymptotic in Theorem 3.9 to show that

$$\sum_{x \in \mathbb{Z}^d} g^{::}(x)^\alpha = \infty \quad \text{if and only if} \quad \alpha \leq \frac{d}{d-2}\,.$$

Conclude that $\sum_x g^{::}(x)^\alpha = \infty$ on $\mathbb{Z}^5$ when $\alpha \leq 5/3$ and on $\mathbb{Z}^6$ when $\alpha \leq 3/2$.

Exercise 3.33*. Let $d \geq 2$. Extend Exercise 3.11 by showing that there are $C, c > 0$, depending only on d, such that every finite nonempty $A \subset \mathbb{Z}^d$ satisfies

$$\mathbb{E}_x \tau_{\partial A} \leq C|A|^{2/d} \qquad \text{and} \qquad \mathbb{P}_x\{\tau_{\partial A} > t\} \leq 2e^{-ct/|A|^{2/d}}$$

for $x \in A$ and $t \geq 0$, and that balls attain this scale.

Hint: For $d \geq 3$, use Theorem 3.9; for $d = 2$, use (3.10).

Exercise 3.34. Lemma 3.13(b) bounds the difference between the discrete and the continuum Laplacian by the pure fourth derivatives. This exercise identifies the polynomials for which that bound is zero. Let $d \geq 1$, and let $f \in \mathbb{R}[x_1, \ldots, x_d]$ be a linear combination of monomials $x_1^{n_1} \cdots x_d^{n_d}$ with $0 \leq n_i \leq 3$ for $1 \leq i \leq d$. Prove that $\Delta^{::} f = \Delta f$.

Exercise 3.35. Let $d \geq 2$ and $\epsilon > 0$. Show that there is $R_0 = R_0(d, \epsilon) < \infty$ such that the following holds. Let $z \in \mathbb{Z}^d$, let $R \geq R_0$, and let $h : \mathrm{B}^{::}(z, R) \to \mathbb{R}_{\geq 0}$ be discrete harmonic at every lattice point whose nearest neighbors all lie in $\mathrm{B}^{::}(z, R)$. Then

$$\sup_{\mathrm{B}^{::}(z,R/2)} h \leq (3 + \epsilon)^d \inf_{\mathrm{B}^{::}(z,R/2)} h \,.$$

Hint: In the proof of Lemma 3.16, replace $27/32$, $7R/8$ and $15R/16$ by t, sR and ρR, respectively, where $1/2 < t < s < \rho < 1$ and

$$\frac{t + 1/2}{t - 1/2} < 3 + \epsilon \,.$$

Exercise 3.36*. Let $d \geq 2$, let $\Omega \subset \mathbb{R}^d$ be open, and let $\delta_n \downarrow 0$. Let $h_n : \Omega \cap \delta_n \mathbb{Z}^d \to \mathbb{R}$ have vanishing scaled discrete Laplacian $\Delta^{::}_{\delta_n} h_n$ on the set $\Omega^+_{\delta_n}$ of (3.21), and be bounded on each compact set uniformly in n. Show that a subsequence of their multilinear interpolations converges uniformly on compact subsets of Ω to a harmonic function.

Hint: Use (3.30), the Arzelà–Ascoli theorem and Theorem 2.29.

Exercise 3.37. Let Ξ be the map of Lemma 3.24. Using that lemma and Lemma 3.25, prove that 1 and the polynomials

$$\Xi(\mathrm{Re}\,(x + \mathbf{i}y)^k) \qquad \text{and} \qquad \Xi(\mathrm{Im}\,(x + \mathbf{i}y)^k) \qquad \text{for } k \geq 1 \,,$$

form a basis for the discrete harmonic polynomials on $\mathbb{Z}^2$.

The next three exercises lead to Exercise 3.40, a periodic analogue of Theorem 3.18.

Exercise 3.38. For $d \geq 1$ and $\Delta^{::}$ in (3.1), let $f : \mathbb{Z}^d \to \mathbb{R}$ have a positive integer period m, so $f(x + my) = f(x)$ for all $x, y \in \mathbb{Z}^d$. Suppose that

$$\sum_{x \in \{1,\ldots,m\}^d} f(x) = 0 \,.$$

Prove that there is a periodic function F such that $f = \Delta^{::} F$.

Exercise 3.39. For $d \geq 1$ and $\Delta^{::}$ in (3.1), let $f : \mathbb{Z}^d \to \mathbb{R}$ have a positive integer period m, so $f(x + my) = f(x)$ for all $x, y \in \mathbb{Z}^d$. Prove that there are a constant $c \in \mathbb{R}$ and a periodic function F such that

$$f = \Delta^{::}(F + c|x|^2)\,.$$

Exercise 3.40. For $d \geq 2$ and a positive integer m, let $u : \mathbb{Z}^d \to \mathbb{R}$ have polynomial growth and satisfy $\Delta^{::} u(x + my) = \Delta^{::} u(x)$ for all $x, y \in \mathbb{Z}^d$. Prove, as a periodic analogue of Theorem 3.18, that $u = F + P$, where F is periodic and P is a polynomial. Moreover, prove that

$$P(x) = c|x|^2 + h(x)\,,$$

where $c \in \mathbb{R}$ and h is a discrete harmonic polynomial.

Exercise 3.41. Using Lemma 3.25, prove that for $d \geq 2$ and $f : \mathbb{Z}^d \to \mathbb{R}$ of polynomial growth, if $\Delta^{::} f$ is a polynomial, then so is f.

Exercise 3.42. Let $d \geq 2$.

(a) Use Theorem 3.18 to show that $g^{::}$ is the unique function $u : \mathbb{Z}^d \to \mathbb{R}$ satisfying $\Delta^{::} u = -\delta_{\mathbf{0}}$ and $u(x)/|x| \to 0$ as $|x| \to \infty$, together with the normalization $u(x) \to 0$ as $|x| \to \infty$ when $d \geq 3$ and $u(\mathbf{0}) = 0$ when $d = 2$.

(b) Let $K \subset \mathbb{Z}^d$ be finite and nonempty, and let $u : \mathbb{Z}^d \to \mathbb{R}$ satisfy $u(x)/|x| \to 0$ as $|x| \to \infty$ and $\Delta^{::} u = 0$ off K. Show that $u(x) = b - \sum_{y \in K} g^{::}(x - y)\Delta^{::} u(y)$ for a constant b, so that $u(x) \to b$ if $d \geq 3$, and $u(x) - sa(x) \to b$ if $d = 2$, where $s := \sum_{y \in K} \Delta^{::} u(y)$.

(c) In part (b), show that $u(x) = \mathbb{E}_x[u(X_{\tau_K})\mathbf{1}\{\tau_K < \infty\}] + b\mathbb{P}_x\{\tau_K = \infty\}$ if $d \geq 3$, and $u(x) = sU_K(x) + \mathbb{E}_x[u(X_{\tau_K})]$ if $d = 2$, with the function U_K of Lemma 3.29(b), which vanishes on K and differs from a by a bounded function.

Exercise 3.43. This exercise extends the optional stopping argument behind (3.35) to a general recurrent graph.

Let $G = (V, E)$ be a connected recurrent locally finite graph with the Laplacian Δ of (3.2), and let $(X_n)_{n \geq 0}$ denote the simple random walk on G.

(a) Suppose $v \in V$ and $h : V \to \mathbb{R}$ is bounded and harmonic on $V \setminus \{v\}$. Show that h is constant.

(b) Suppose $K \subset V$ is finite and nonempty, and $h : V \to \mathbb{R}$ is bounded and harmonic on $V \setminus K$. Show that

$$\sum_{w \in K} \Delta h(w) = 0\,.$$

Hint: Let $\tau_K^+ := \inf\{n \geq 1 : X_n \in K\}$ and show

$$\Delta h(w) = \deg(w)(\mathbb{E}_w[h(X_{\tau_K^+})] - h(w))\,.$$

Then show

$$\deg(w)\,\mathbb{P}_w\{X_{\tau_K^+} = x\} = \deg(x)\,\mathbb{P}_x\{X_{\tau_K^+} = w\} \qquad \text{for all } w, x \in K\,.$$

Exercise 3.44. This exercise computes hitting probabilities of Section 3.7 for a set of two neighboring sites. Let $K = \{\mathbf{0}, \mathbf{e}_1\} \subset \mathbb{Z}^2$, and for $y \in \mathbb{Z}^2$ let

$$h(y) := \mathbb{P}_y\{X_{\tau_K} = \mathbf{0}\}\,.$$

Show that every neighbor of $2\mathbf{e}_1$ outside K has h-value strictly larger than $h(2\mathbf{e}_1)$:

$$h(2\mathbf{e}_1) < h(3\mathbf{e}_1) \qquad \text{and} \qquad h(2\mathbf{e}_1) < h(2\mathbf{e}_1 + \mathbf{e}_2) = h(2\mathbf{e}_1 - \mathbf{e}_2)\,.$$

The only neighbor of $2\mathbf{e}_1$ with smaller h-value is $\mathbf{e}_1 \in K$, so the restriction of h to $\mathbb{Z}^2 \setminus K$ has a strict local minimum at $2\mathbf{e}_1$.

Hint: The potential kernel satisfies $a(\mathbf{e}_1) = \frac{1}{4}$. Show that the function

$$g(y) := \frac{1}{2} + 2[a(y - \mathbf{e}_1) - a(y)]$$

is bounded, harmonic off K, and equal to h on K, and deduce $h = g$ from (3.35). Then use the harmonicity of a at $\mathbf{e}_1 + \mathbf{e}_2$ and at $2\mathbf{e}_1$, the value $a(\mathbf{e}_1 + \mathbf{e}_2) = \frac{1}{\pi}$ from Lemma 3.7, the value $a(2\mathbf{e}_1) = 1 - \frac{2}{\pi}$ that follows from it by the harmonicity of a at $\mathbf{e}_1$, and the bounds $3 < \pi < 16/5$.

Exercise 3.45. Let $A \subset B \subset \mathbb{Z}^2$ be finite and nonempty, and let U_A be as in Lemma 3.29(b). Prove that the value of the potential of $\mathcal{H}_B$ on B exceeds that of $\mathcal{H}_A$ on A by $\sum_{y \in B} \mathcal{H}_B(y) U_A(y) \geq 0$, which is $\mathcal{H}_B(z) U_A(z)$ if $B = A \cup \{z\}$.

Exercise 3.46*. This exercise identifies the analogue in dimensions $d \geq 3$ of the harmonic measure from infinity of Definition 3.28. Proposition 8.9 in Chapter 8 gives another proof of part (b), through the last-exit decomposition.

Let $d \geq 3$, let X be simple random walk on $\mathbb{Z}^d$, with $\mathbb{P}_y$ and the hitting times τ_E defined as in Section 3.7, let $K \subset \mathbb{Z}^d$ be finite and nonempty, and let

$$\tau_K^+ := \inf\{n \geq 1 : X_n \in K\}$$

denote the first return time to K.

(a) Show that $\lim_{|y| \to \infty} \mathbb{P}_y\{X_{\tau_K} = x\} = 0$ for every $x \in K$.

(b) Show that, for every $x \in K$,

$$\lim_{|y|\to\infty} \mathbb{P}_y\{X_{\tau_K} = x \mid \tau_K < \infty\} = \frac{\mathrm{Es}_K(x)}{\mathrm{cap}(K)},$$

where $\mathrm{Es}_K(x) := \mathbb{P}_x\{\tau_K^+ = \infty\}$ is the escape probability from x and $\mathrm{cap}(K) := \sum_{x\in K} \mathrm{Es}_K(x)$ is the capacity of K.

Hint for (b): Reverse the paths from y that first hit K at x.

NOTES

For an introduction to martingales in discrete time, see (Williams, 1991, Chapter 10, pp. 93–101).

Lemma 3.6 is due to Pólya; see (Spitzer, 1976, p. 52, Problems 4–5). For treatments of simple random walk in $\mathbb{Z}^d$, see Spitzer (1976) and Chapter 1 of Lawler (1996). For finite-range, symmetric, irreducible random walks, see (Lawler and Limic, 2010, § 1.1). The proof given here needs no Fourier analysis.

The planar case of Theorem 3.9, with the constant κ and the error $O(|x|^{-2})$, is due to Stöhr in 1950, as Kozma and Schreiber (2004) report on p. 2 and Fukai and Uchiyama (1996) in their Remark 2. For $d \geq 3$, Uchiyama obtained the asymptotic expansion of the Green function in 1998; see p. 2 of Kozma and Schreiber (2004). Their Theorem 1 expands, to arbitrary order, the function $x \mapsto \sum_{n\geq 0}(\mathbb{P}_{\mathbf{0}}\{X_n = \mathbf{0}\} - \mathbb{P}_{\mathbf{0}}\{X_n = x\})$ of every balanced bounded random walk on a lattice; for simple random walk this function is $2d(g^{::}(\mathbf{0}) - g^{::}(x))$ when $d \geq 3$ and $4a(x)$ when $d = 2$. Their Theorem 5 shows that for a reversible walk, apart from the additive constant, only inverse powers whose order has the parity of the dimension appear. For simple random walk this gives the errors $O(|x|^{-d})$ and $O(|x|^{-2})$ of Theorem 3.9. In dimension two, identify $(x, y) \in \mathbb{Z}^2$ with the Gaussian integer $z = x + \mathbf{i}y$. Then the potential kernel admits the asymptotic expansion

$$a(z) = \frac{1}{2\pi} \log |z| + \kappa - \frac{\mathrm{Re}\,(z^4)}{24\pi|z|^6} + O(|z|^{-4}),$$

with κ as in Theorem 3.9. See Theorem 2 and Remark 2 on p. 1980 of Fukai and Uchiyama (1996), proved in their Section 4, and Theorem 3 of Kozma and Schreiber (2004) for the coefficients in the arbitrary-order expansion. In Spitzer's unnormalized convention, the potential kernel is $4a = -4g^{::}$. Its values are integers plus rational multiples of $1/\pi$; for example, $-4g^{::}(1) = 1$ and $-4g^{::}(1 + \mathbf{i}) = 4/\pi$. Sections 7–8 of McCrea and Whipple (1940) give a procedure for computing these values; see also (Spitzer, 1976, §15).

Claim 2.1 of Nachmias and Peres (2026) identifies $g^{::}_{\mathbb{Z}^2\setminus\{\mathbf{0}\}}(\cdot, y)$ as the unique nonnegative function u with $\Delta^{::}u = \delta_{\mathbf{0}} - \delta_y$ on $\mathbb{Z}^2$ that vanishes at the origin. Their Remark 3.3 expresses the potential kernel through the diagonal values $g^{::}_{\mathbb{Z}^2\setminus\{\mathbf{0}\}}(x, x)$. Lemma 3.10 records the corresponding off-diagonal identity in our normalization.

On a finite connected graph, the kernel of Δ is the space of constants and its range is the space of mean-zero functions, so $\Delta u = -\delta_x$ has no solution u. Projecting δ_x onto the mean-zero functions and inverting $-\Delta$ there gives the x-column of the **Moore–Penrose pseudoinverse**, which annihilates constants. If $g(x, y)$ are the entries of the pseudoinverse, then the effective resistance between x and y is $g(x, x) + g(y, y) - 2g(x, y)$, a formula proved in Theorem A and Corollary A on pp. 87–88 of Klein and Randić (1993). Exercise 3.23 expresses the resistance through random-walk escape probabilities. The potential kernel in $\mathbb{Z}^2$ plays the analogous role, replacing the Green function, which is infinite by recurrence. The potential kernel still satisfies $\Delta^{::}a = \delta_{\mathbf{0}}$. Among the solutions u of $\Delta^{::}u = \delta_{\mathbf{0}}$ with $u(x)/|x| \to 0$ as $|x| \to \infty$, it is unique up to an additive constant, which the normalization $a(\mathbf{0}) = 0$ fixes; Exercise 3.42 asks for the proof, in the equivalent form for $g^{::} = -a$.

For background on coupling, see (Levin and Peres, 2017, Chapter 5) and (Lindvall, 1992, Chapter II, Sections 12–18).

The polynomial-growth theorem is due to Heilbronn. Theorem 6 on p. 199 of Heilbronn (1949), proved on p. 206, shows that a discrete harmonic function of growth $O(|x|^n)$ is a polynomial of degree at most n. The little-o form in Theorem 3.18 follows because a polynomial of degree n cannot be $o(|x|^n)$. Theorem 3 on p. 196 of Heilbronn (1949) computes the dimension of the space of discrete harmonic polynomials.

The case $n = 1$ of Theorem 3.18 includes the discrete analogue of Liouville's theorem that a bounded harmonic function is constant. For constructions of discrete harmonic polynomials in two variables, see Sections 3 and 5 of Duffin (1956), who uses discrete contour integration, and pp. 97–99 of Zeilberger (1977). Sections 4.1 and 4.3 of Lovász (2004) describe discrete analytic polynomials whose real and imaginary parts are discrete harmonic on the even and odd diagonal sublattices. Section 2.2 of Jerison, Levine, and Sheffield (2014) defines the map Ξ used in Lemma 3.24. Their modification of Lovász's construction makes $f - \Xi(f)$ have degree at most $k - 2$, rather than $k - 1$, when f has degree k. In their Section 2.3, the map Ξ enters the proof that internal DLA fluctuations, tested against discrete harmonic polynomials, converge to the augmented Gaussian free field. Sections 2–5 of Smirnov (2010) survey discrete harmonic and discrete holomorphic functions and their applications in probability.

The proof of Theorem 3.9 in Section 3.6 compares the discrete Green

function with the continuum Green function $\mathcal{G}$ corrected by the term $-\frac{1}{24}\sum_i x_i\,\partial_i^3\mathcal{G}$. Section 1 of Kozma and Schreiber (2004) sketches a related planar approach that corrects the sampled continuum logarithm by repeated convolution. Lemma 3.15 combines Theorem 3.9 with the method of images and Theorem 3.14(b). The proof of Lemma 3.16 compares killed Green functions on an annulus, then exit distributions. The discrete Harnack inequality with an unspecified constant is Theorem 1.7.2 of Lawler (1996); for another proof, see Theorem 6.3.9 of Lawler and Limic (2010).

Theorem 3.9 can also be proved by Fourier analysis. Write $\varphi(\theta)$ for the average of $\cos\theta_1,\dots,\cos\theta_d$. Extracting Fourier coefficients from the generating function of the transition probabilities gives, for $d \ge 3$,

$$g^{::}(x) = \frac{1}{2d}\left(\frac{1}{2\pi}\right)^d \int_{(-\pi,\pi)^d} \frac{e^{-\mathbf{i}x\cdot\theta}}{1-\varphi(\theta)}\,\mathrm{d}\theta\,,$$

and, for $d = 2$, the same formula for $-a(x)$ with $e^{-\mathbf{i}x\cdot\theta}-1$ in place of $e^{-\mathbf{i}x\cdot\theta}$ and $\frac{1}{4}$ in place of $\frac{1}{2d}$. The Fourier proof estimates these integrals after replacing $1-\varphi(\theta)$ near $\theta = \mathbf{0}$ by $|\theta|^2/(2d)$. In the plane, P12.3 of Spitzer (1976), on pp. 124–127, treats random walks whose steps generate $\mathbb{Z}^2$ and that have mean zero, a covariance matrix proportional to the identity and a finite moment of order $2+\delta$ for some $\delta > 0$. Simple random walk is one of them. For these walks P12.3 shows that the potential kernel minus a multiple of $\log|x|$ converges as $|x| \to \infty$, by replacing the reciprocal of $1-\varphi$, where φ is now the characteristic function of a step, by the reciprocal of its quadratic part.

Spitzer proved the existence of the harmonic measure from infinity of a finite set in the plane, the first assertion of part (a) of Lemma 3.29: Theorem (B′) on p. 595 of Spitzer (1962) treats every recurrent random walk on $\mathbb{Z}^2$ whose steps generate $\mathbb{Z}^2$, and p. 603 derives the existence from the convergence of $a(x-v)-a(x)$ to zero. Theorem 14.1 on pp. 141–142 of Spitzer (1976) expresses the limit through the inverse of the matrix of values $a(x-v)$ with x and v in the set, so that its potential is constant on the set, as in part (a). Other proofs are Theorem 2.1.3 of Lawler (1996), which adds an error estimate, and Proposition 6.6.1 with equation (6.47) of Lawler and Limic (2010).

The h-transform of Subsection 3.2.1 is the h-process of Section 11.1 of Lawler and Limic (2010), pp. 245–248. Their Proposition 11.1.1 on p. 246 sums (3.6) over paths, and pp. 247–248 treat the case $d = 2$ of Exercise 3.13 and describe the walk of Exercise 3.31 as a limit of conditioned walks; its transience is part (i) of their Exercise 11.2 on p. 258. Several other exercises come from the same book. Exercise 3.5 is their Exercise 6.8 on p. 163. Exercise 3.11 is their Proposition 2.4.5 on p. 49 for simple random walk,

with explicit constants; their proof of the second bound uses the central limit theorem instead of Lemma 3.4. Exercise 3.42(c) is their Proposition 6.4.6 on p. 134 when $d \geq 3$, and extends their Proposition 6.4.8 on p. 135, which assumes $u = 0$ on K, when $d = 2$; they estimate exit distributions instead of using Theorem 3.18. Exercise 3.45 is their Lemma 6.6.6 on p. 149: four times the value of the potential of $\mathcal{H}_A$ on A is their capacity of A in two dimensions, defined on p. 146. Exercise 3.36 extends their Proposition 8.3.2 on p. 193, for functions bounded by one on a ball. Their Exercise 6.9 on p. 163 and Proposition 8.1.1 on pp. 181–182 give the formula of Exercise 3.30(a) by the method of images, and their Exercise 4.2 on p. 100 estimates how long a walk takes to return to a line through its starting point.

For systematic background on the Green function, the potential kernel, harmonic measure and capacity of random walk, see Chapters 1 and 2 of Lawler (1996) and Chapters 4 and 6 of Lawler and Limic (2010).

PART 2

Lattice Growth

Chapter Four

Divisible Sandpile in $\mathbb{Z}^d$: Single Source

This chapter studies the divisible sandpile, a model in which real mass spreads over a graph. A vertex holding more than one unit of mass topples by dividing its excess equally among its neighbors, and we keep toppling until every vertex is stable. (Pour a large amount of maple syrup onto the center of a flat griddle: it spreads into a disk, and the divisible sandpile is a lattice model of exactly this kind of spreading.) We first develop the model on a general graph and then specialize to a point source on $\mathbb{Z}^d$ for $d \geq 2$.

The first result, in Section 4.1, is the least action principle (Theorem 4.3): when stabilization occurs, the final configuration and the odometer, which records the cumulative mass sent from a vertex to each neighbor, do not depend on the order of topplings, provided every vertex is chosen infinitely often, and the odometer is the pointwise smallest nonnegative function whose Laplacian, added to the initial configuration, leaves at most one unit of mass at every vertex. For a point source on $\mathbb{Z}^d$, the odometer is also the gap between an obstacle and its least superharmonic majorant. This connects the model to the obstacle problem, the simplest free boundary problem in partial differential equations, in which the unknown is both a function and the region where it touches a prescribed obstacle.

The second result, in Section 4.2, says where the mass comes to rest when we place mass m at the origin of $\mathbb{Z}^d$: the fully occupied sites differ from a Euclidean ball by at most a constant in radius (Theorem 4.7). The same theorem determines the odometer up to a bounded additive error. The proof compares the odometer with a quadratic function plus the discrete Green function: the maximum principle gives upper and lower comparisons, a path of increasing odometer values bounds the outer radius, and Green function asymptotics control the error. The obstacle representation is the divisible counterpart of the obstacle problem for the abelian sandpile described in the Notes to Chapter 1, where an integer constraint produces fractal grain patterns. The obstacle bounds proved here also supply the estimates behind the inner bounds for internal DLA in Chapter 5 and for rotor-router aggregation in Chapter 6, and Proposition 4.9 shows that the odometer grows quadratically inside the ball near its boundary. Section 4.3 uses the divisible sandpile to place Euclidean balls inside and outside the set of sites that the abelian sandpile topples.

4.1 THE DIVISIBLE SANDPILE AND LEAST ACTION

Throughout this section, $G = (V, E)$ is undirected, locally finite, connected and simple, and has at least two vertices (so that every vertex has a neighbor).

The graph Laplacian of Chapter 3 describes how mass moves when a vertex spills over. In the **divisible sandpile**, a configuration $\sigma : V \to \mathbb{R}$ assigns real mass to each vertex, and a vertex holding more than one unit topples by dividing its excess equally among its neighbors. We define stabilization by sequential topplings and prove the least action principle (Theorem 4.3): when stabilization occurs, the final configuration and the odometer (the cumulative mass that each vertex sends to each of its neighbors) do not depend on the order of topplings, provided every vertex occurs in it infinitely often, and the odometer is the pointwise smallest nonnegative function whose Laplacian, added to σ, leaves at most one unit at every vertex. We then show that a finitely supported configuration on an infinite graph stabilizes, with only finitely many vertices ever emitting mass, and relate the odometer to an obstacle problem.

Write $a_+ := \max\{a, 0\}$ for the positive part of a real number a. Given a configuration σ, to **topple** a vertex v means to remove the mass $(\sigma(v) - 1)_+$ from v and divide it equally among its neighbors. Recall the graph Laplacian (3.2). Toppling v changes σ to

$$\sigma + \frac{(\sigma(v) - 1)_+}{\deg(v)} \Delta\delta_v \,.$$

A vertex $v \in V$ is **unstable** for σ when $\sigma(v) > 1$; otherwise it is **stable**. Thus toppling v leaves σ unchanged exactly when v is stable.

We allow negative masses, which represent deficits.

We want to topple until every vertex is stable. For example, if neighboring vertices $v \sim w$ satisfy $\sigma(v) > 1$ and $\sigma(w) \geq 1$, then toppling v makes w unstable. A vertex may therefore need to topple more than once. Let $v_1, v_2, \ldots$ be a sequence of vertices in V, which we call a **toppling sequence**. Write $\sigma_0 := \sigma$, and for $k \geq 1$ let σ_k be the configuration obtained from σ_{k-1} by toppling v_k:

$$\sigma_k := \sigma_{k-1} + \frac{(\sigma_{k-1}(v_k) - 1)_+}{\deg(v_k)} \Delta\delta_{v_k} \,. \tag{4.1}$$

Since $(\sigma_{j-1}(v_j) - 1)_+$ is the mass emitted at step j, define

$$u_k := \sum_{j=1}^{k} \frac{(\sigma_{j-1}(v_j) - 1)_+}{\deg(v_j)} \delta_{v_j} \,.$$

Then $u_k(v)$ is the cumulative mass emitted from v to each neighbor during

the first k topplings, and we call u_k the **odometer** after k topplings. The net change in mass at v is

$$\sigma_k(v) - \sigma_0(v) = \Delta u_k(v) = \sum_{w\sim v} \big(u_k(w) - u_k(v)\big)\,.$$

Each toppling raises the odometer by an amount that the current odometer determines. Write

$$\zeta(x) \coloneqq \frac{\sigma(x) - 1}{\deg(x)} \qquad \text{for } x \in V\,.$$

If $u : V \to [0,\infty)$ is the odometer so far, so that the current configuration is $\sigma + \Delta u$, then toppling a vertex x raises the odometer at x from $u(x)$ to

$$\max\Big\{u(x),\ \frac{1}{\deg(x)} \sum_{y\sim x} u(y) + \zeta(x)\Big\} \tag{4.2}$$

and leaves it unchanged elsewhere. Indeed, $((\sigma + \Delta u)(x) - 1)/\deg(x)$ is the second entry of the maximum minus $u(x)$, and $a+(b-a)_+ = \max\{a,b\}$ for real a and b. Two properties of (4.2) will do most of the work: it is nondecreasing in each value of u, and it equals $u(x)$ exactly when $(\sigma + \Delta u)(x) \le 1$.

Since u_k is nondecreasing, it converges pointwise to a function u_∞ with values in $[0,\infty]$. We call u_∞ the odometer of σ for the toppling sequence $(v_k)_{k\ge1}$. The simplest example is a point source on the path $\mathbb{Z}$: started from $7\delta_{\mathbf{0}}$, the final mass is 1 on each of the sites $-3,\dots,3$, and the odometer takes the values 6, 3, 1 and 0 at distances 0, 1, 2 and 3 from the source.

The odometer may be infinite. A toppling sequence $(v_k)_{k\ge1}$ is **thorough** when every vertex occurs infinitely often.

A finite odometer at one vertex forces finiteness everywhere. This reduces stabilization to a pointwise question.

Lemma 4.1 *Let $\sigma : V \to \mathbb{R}$ be a configuration, and let u_∞ be the odometer of a thorough toppling sequence started from σ. Then either $u_\infty(v) < \infty$ for every $v \in V$, or $u_\infty \equiv \infty$ on V.*

Proof. Suppose there is $v \in V$ with $u_\infty(v) = \infty$, and let w be a neighbor of v. Assume for contradiction that $u_\infty(w) < \infty$. Then

$$\Delta u_k(w) = \sum_{y\sim w} \big(u_k(y) - u_k(w)\big) \xrightarrow[k\to\infty]{} \infty\,,$$

since the contribution from v tends to $+\infty$, while every other term is bounded below by $-u_\infty(w)$. Hence $\sigma_k(w) = \sigma(w) + \Delta u_k(w) \to \infty$. But thoroughness makes w topple infinitely often, and its mass is at most 1 after each such

toppling. This contradiction proves $u_\infty(w) = \infty$. Connectivity now gives $u_\infty \equiv \infty$. □

By Lemma 4.1, the odometer of a thorough toppling sequence is either finite at every vertex or infinite at every vertex. This dichotomy leads to the following definition.

Definition 4.2 A configuration σ on V is **stabilizable** when there exists a thorough toppling sequence whose odometer is finite on V. We then also say that σ **stabilizes**; otherwise we say that σ **explodes**.

Suppose that σ is stabilizable. Fix a thorough toppling sequence with finite odometer. Local finiteness gives the final configuration

$$\sigma_\infty := \lim_{k\to\infty} \sigma_k = \lim_{k\to\infty} (\sigma + \Delta u_k) = \sigma + \Delta u_\infty .$$

Since the toppling sequence is thorough, $\sigma_k(v) \le 1$ for infinitely many k, so $\sigma_\infty \le 1$. Thus stabilization reaches a stable configuration while every vertex emits finite mass. If σ is not stabilizable, then Lemma 4.1 shows that every thorough toppling sequence emits infinite mass from every vertex; its configurations need not converge.

We now show that the odometer and final configuration do not depend on the thorough toppling sequence.

Theorem 4.3 (Least Action Principle)
Let $\sigma : V \to \mathbb{R}$ be a configuration, and let

$$\mathcal{F}_\sigma := \{f : V \to \mathbb{R} \mid f \ge 0 \text{ and } \sigma + \Delta f \le 1\} .$$

Then the following statements hold.

(i) If $f \in \mathcal{F}_\sigma$, then the odometer u_∞ of every toppling sequence started from σ satisfies

$$u_\infty(v) \le f(v) \qquad \text{for every } v \in V .$$

(ii) If σ stabilizes, then the odometer u_∞ of every thorough toppling sequence started from σ belongs to $\mathcal{F}_\sigma$ and satisfies

$$u_\infty(v) = \min_{f\in\mathcal{F}_\sigma} f(v) \qquad \text{for every } v \in V .$$

(iii) The configuration σ stabilizes if and only if $\mathcal{F}_\sigma$ is nonempty.

Proof. The idea is to follow the odometer one toppling at a time: the monotone update rule (4.2) never pushes it above a given $f \in \mathcal{F}_\sigma$, which proves part (i). Parts (ii) and (iii) then follow from part (i).

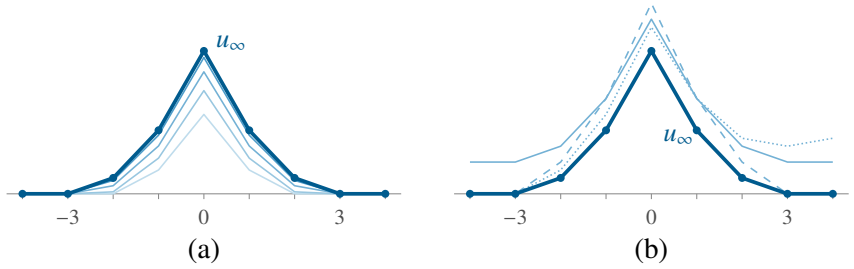


Figure 4.1: The least action principle on $\mathbb{Z}$ for $\sigma = 6\delta_{\mathbf{0}}$, whose odometer u_∞ (thick) takes the values 4.5, 2, 0.5 and 0 at distances 0, 1, 2 and 3 from the source. (a) The odometers after 1, 2, 4 and 8 sweeps of the toppling sequence $0, 1, -1, 2, -2, 3, -3$ (light to dark) increase to u_∞ from below; (b) three other members of $\mathcal{F}_\sigma$ lie above it: $u_\infty + 1$ (solid), the odometer of $7\delta_{\mathbf{0}}$ (dashed) and $u_\infty(x) + \frac{1}{4}(x+3)_+$ (dotted).

(i) Fix $f \in \mathcal{F}_\sigma$ and a toppling sequence $v_1, v_2, \ldots$. We prove by induction on k that $u_k \le f$ for every $k \ge 0$. The case $k = 0$ follows from $u_0 = 0 \le f$. For the inductive step, suppose $u_{k-1} \le f$. By the inductive hypothesis, the update rule (4.2) and its monotonicity give

$$u_k(v_k) \le \max\Big\{f(v_k),\ \frac{1}{\deg(v_k)} \sum_{y \sim v_k} f(y) + \zeta(v_k)\Big\} = f(v_k)\,,$$

where the equality follows from $(\sigma + \Delta f)(v_k) \le 1$. The functions u_k and u_{k-1} agree away from v_k, so $u_k \le f$. Since $u_k \uparrow u_\infty$ pointwise (Figure 4.1(a)) and $u_k \le f$ for every k, it follows that $u_\infty \le f$.

(ii) First consider a thorough toppling sequence with finite odometer. As shown after Definition 4.2, its final configuration $\sigma + \Delta u_\infty$ is at most 1, so $u_\infty \in \mathcal{F}_\sigma$.

Now fix a thorough toppling sequence with odometer u_∞. Since σ stabilizes, there is a thorough toppling sequence whose odometer g is finite, and $g \in \mathcal{F}_\sigma$ by the previous paragraph. Part (i), applied with $f = g$, gives $u_\infty \le g$. Hence u_∞ is finite and belongs to $\mathcal{F}_\sigma$. A second application of part (i) gives $u_\infty(v) \le f(v)$ for every $f \in \mathcal{F}_\sigma$, which proves the minimum formula.

(iii) If σ stabilizes, then part (ii) gives an element of $\mathcal{F}_\sigma$. Conversely, if $f \in \mathcal{F}_\sigma$, then part (i) bounds the odometer of every thorough toppling sequence by f. A thorough toppling sequence exists because V is countable, being connected and locally finite, so σ stabilizes. □

Part (ii) gives the abelian property: all thorough toppling sequences started from a stabilizable configuration have the same odometer and the same final configuration.

The name "least action principle" comes from the following interpretation. Let $f : V \to [0, \infty)$ be a function. Starting from σ, send mass $f(v)$ from each vertex v to each of its neighbors, for a total emission of $\deg(v) f(v)$,

without requiring that each vertex emit only its current excess. The condition $\sigma + \Delta f \le 1$ says that the resulting configuration is stable. The least action principle says that the odometer is the pointwise smallest nonnegative function with this property. Figure 4.1(b) illustrates the comparison on $\mathbb{Z}$.

For a finite source, we also need to know that only finitely many vertices emit mass and that every vertex that emits mass finishes with mass one. The second fact holds because a vertex that has toppled never again holds less than mass one, and counting such vertices gives the first.

Lemma 4.4 *Let $\sigma : V \to \mathbb{R}$ be a configuration, and assume that V is infinite. Then the following statements hold.*

(i) If σ is stabilizable with odometer u_∞, then the final configuration $\sigma_\infty = \sigma + \Delta u_\infty$ satisfies $\{u_\infty > 0\} \subseteq \{\sigma_\infty = 1\}$.

(ii) If σ has finite support, then σ is stabilizable and its odometer has finite support.

Proof. Fix a thorough toppling sequence $v_1, v_2, \ldots$, and let $\sigma_k = \sigma + \Delta u_k$ be its configurations. Once a vertex emits positive mass, its mass stays at least 1: the toppling that emits the mass leaves mass exactly 1, between topplings the vertex only receives mass, and every later toppling again leaves mass 1, as at the origin in Figure 1.4. Thus

$$\sigma_j(v) \ge 1 \qquad \text{whenever } u_k(v) > 0 \text{ and } j \ge k\,. \tag{4.3}$$

(i) Suppose σ is stabilizable. By Theorem 4.3(ii), the odometer of the fixed thorough toppling sequence is finite, so $u_k \uparrow u_\infty < \infty$ and $\sigma_k \to \sigma_\infty$. Fix v with $u_\infty(v) > 0$. Then there is k with $u_k(v) > 0$, so (4.3) gives $\sigma_j(v) \ge 1$ for every $j \ge k$. Since $\sigma_j(v) \to \sigma_\infty(v)$, letting $j \to \infty$ gives $\sigma_\infty(v) \ge 1$. Since $\sigma_\infty \le 1$, it follows that $\sigma_\infty(v) = 1$.

(ii) Now suppose σ has finite support, and write $M := \sum_{v \in V} \sigma(v)_+$, which is finite. We divide the proof into three steps: the total positive mass never exceeds M, only finitely many vertices ever emit mass, and the odometer is finite.

Step 1. We start by showing that toppling never increases the total positive mass. Indeed, a toppling that emits mass $a > 0$ from a vertex lowers the positive part there by exactly a, while each neighbor receives $a/\deg(v)$ and the inequality $(b + c)_+ \le b_+ + c$ for $c \ge 0$ bounds the resulting increase at the neighbors by a in total. Hence $\sum_{v \in V} \sigma_k(v)_+ \le M$ for every k.

Step 2. Next we show that only finitely many vertices ever emit mass. Write $A_k := \{u_k > 0\}$ for the set of vertices that have emitted positive mass by time k. Every vertex of A_k contributes at least 1 to the total positive mass

by (4.3), so

$$|A_k| \le \sum_{v\in V} \sigma_k(v)_+ \le M \qquad \text{for every } k\,.$$

Write $A := \bigcup_{k\ge 0} A_k$ for the union of the sets A_k. The sets A_k increase with k and have uniformly bounded cardinality, so A is finite.

Step 3. It remains to show that u_∞ is finite. If u_∞ were infinite at one vertex, then Lemma 4.1 would make $u_\infty \equiv \infty$, forcing $A = V$ and contradicting the assumption that V is infinite. Therefore u_∞ is finite on V, which makes σ stabilizable, and $\operatorname{supp}(u_\infty) = A$ is finite. □

Harmonic functions keep track of the mass after stabilization. Finite support lets us pair the two orientations of each edge without a limiting argument.

Lemma 4.5 *Let $\sigma : V \to \mathbb{R}$ be a finitely supported stabilizable configuration with odometer u_∞ and final configuration σ_∞, and assume that u_∞ has finite support. Let $h : V \to \mathbb{R}$ be harmonic on $\{u_\infty > 0\}$. Then*

$$\sum_{v\in V} (\sigma_\infty(v) - \sigma(v))h(v) = 0\,.$$

Proof. The supports of u_∞ and Δu_∞ are finite, by local finiteness. Pairing the two orientations of each edge gives

$$\sum_{v\in V} (\Delta u_\infty)(v)h(v) = \sum_{v\in V} u_\infty(v)\Delta h(v) = 0\,.$$

The last equality follows because $u_\infty = 0$ outside $\{u_\infty > 0\}$ and $\Delta h = 0$ on $\{u_\infty > 0\}$. Since $\sigma_\infty - \sigma = \Delta u_\infty$, this proves the identity. □

In particular, in the setting of the lemma, if $\sigma = m\delta_o$ for a vertex $o \in V$ and a real number $m > 0$, then $\sum_{z\in V} \sigma(z)h(z) = mh(o)$, so the lemma gives

$$\frac{1}{m}\sum_{z\in V} \sigma_\infty(z)h(z) = h(o)\,. \tag{4.4}$$

For finite graphs the support hypothesis on the odometer is automatic; for infinite graphs it follows from Lemma 4.4(ii).

We can also translate the least action principle into an obstacle problem. By Theorem 4.3(ii), the odometer of a stabilizable configuration σ is the pointwise minimum of the family $\mathcal{F}_\sigma$ of that theorem. Given an **obstacle** $\gamma : V \to \mathbb{R}$, consider the family

$$\mathcal{S}_\gamma := \{\varphi : V \to \mathbb{R} \mid \varphi \ge \gamma \text{ and } \Delta\varphi \le 0\}\,.$$

The **least superharmonic majorant** of γ is

$$s(v) := \inf_{\varphi \in \mathcal{S}_\gamma} \varphi(v)\,. \tag{4.5}$$

We adopt the convention that $s \equiv \infty$ when $\mathcal{S}_\gamma$ is empty.

Subtracting the obstacle identifies the two minimizing families.

Lemma 4.6 *Let $\sigma, \gamma : V \to \mathbb{R}$ satisfy $\Delta\gamma = \sigma - 1$, and let*

$$\mathcal{F}_\sigma := \{f : V \to \mathbb{R} \mid f \geq 0 \text{ and } \sigma + \Delta f \leq 1\}\,,$$
$$\mathcal{S}_\gamma := \{\varphi : V \to \mathbb{R} \mid \varphi \geq \gamma \text{ and } \Delta\varphi \leq 0\}\,.$$

Then

$$\mathcal{F}_\sigma = \mathcal{S}_\gamma - \gamma\,.$$

Proof. For a function $\varphi : V \to \mathbb{R}$, let $f := \varphi - \gamma$. Then $\varphi \geq \gamma$ exactly when $f \geq 0$, and $\Delta\varphi \leq 0$ exactly when $\Delta f \leq 1 - \sigma$. Hence $\varphi \in \mathcal{S}_\gamma$ exactly when $f \in \mathcal{F}_\sigma$, which proves $\mathcal{F}_\sigma = \mathcal{S}_\gamma - \gamma$. □

If $\Delta\gamma = \sigma - 1$ and σ stabilizes, then Theorem 4.3(ii) identifies the odometer u_∞ of σ with the pointwise minimum of $\mathcal{F}_\sigma$, so Lemma 4.6 gives

$$u_\infty = s - \gamma\,,$$

where s is the least superharmonic majorant (4.5) of γ. In other words, the odometer is the gap between the obstacle γ and its least superharmonic majorant, as Figure 4.2 shows on $\mathbb{Z}$, so finding it is an obstacle problem. The next section analyzes this problem for a point source $\sigma = m\delta_{\mathbf{0}}$ on $\mathbb{Z}^d$ with $d \geq 2$, taking for γ the sum of a negative quadratic function and a multiple of the Green function of Chapter 3, and shows that the sites that end with mass one fill a Euclidean ball up to an error of at most a constant in the radius.

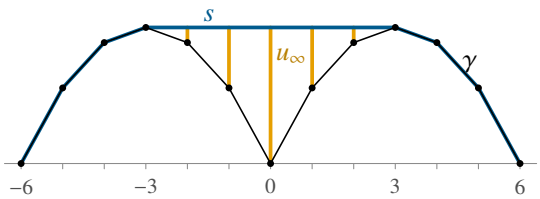


Figure 4.2: The obstacle $\gamma(x) = 3|x| - x^2/2$ on $\mathbb{Z}$ (black), which satisfies $\Delta\gamma = \sigma - 1$ for $\sigma = 6\delta_{\mathbf{0}}$, and its least superharmonic majorant s (blue), which on $\mathbb{Z}$ is its least concave majorant. The gaps $s - \gamma$ (orange) are the odometer u_∞ of Figure 4.1, with values 4.5, 2, 0.5 and 0 at distances 0, 1, 2 and 3 from the source.

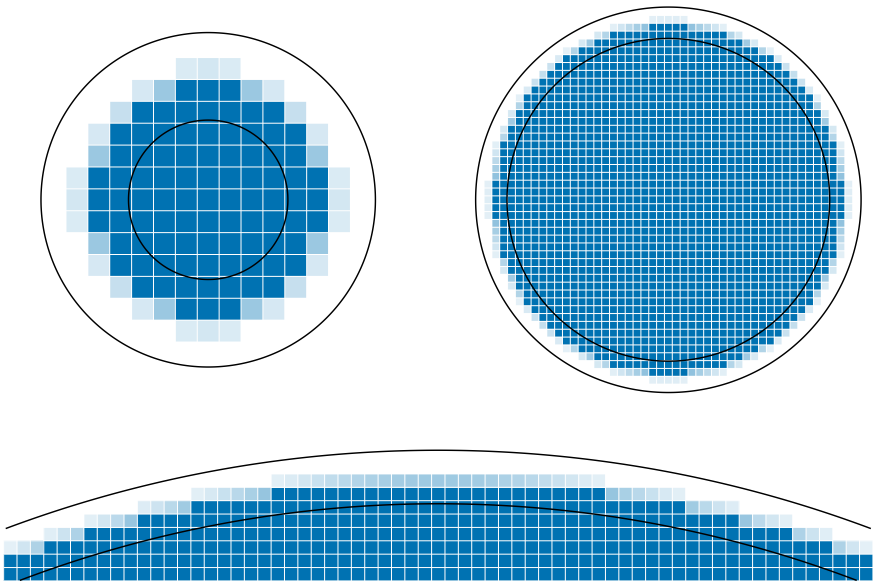


Figure 4.3: Divisible sandpiles in $\mathbb{Z}^2$ started from mass m at the origin, for $m = 100$ (top left) and $m = 1600$ (top right), and part of the boundary for $m = 25{,}600$ (bottom), with black circles of radii $r - 2$ and $r + 2$, where $r = \sqrt{m/\pi}$. Dark squares carry mass 1, and light squares carry mass strictly between 0 and 1, darker for larger mass.

4.2 BALL ASYMPTOTICS FOR A POINT SOURCE

Figure 4.3 shows the stable configuration for several values of m, and Figure 4.4 shows one in $\mathbb{Z}^3$. The fully occupied sites fill a lattice ball up to a boundary layer of bounded thickness.

Since $m\delta_{\mathbf{0}}$ has finite support, the divisible sandpile started from it stabilizes, by Lemma 4.4(ii). The next theorem shows that the fully occupied sites differ from a Euclidean ball by at most an additive constant in the radius, and determines the odometer up to a bounded additive error. Recall the abbreviation $\mathrm{B}^{::}_r = \mathrm{B}^{::}(\mathbf{0}, r)$ from Section 3.1; it is empty when $r \le 0$. We use $\Delta^{::}$ for the discrete Laplacian in $\mathbb{Z}^d$ defined by (3.1). Recall that $\mathcal{G}$ is the continuum Green function defined in (2.14). We write $g^{::}$ for the Green function (3.16) when $d \ge 3$ and for minus the potential kernel (3.17) when $d = 2$. Thus $\Delta^{::} g^{::} = -\delta_{\mathbf{0}}$ for every $d \ge 2$. Equations (3.16) and (3.17) give

$$g^{::}(x) = \begin{cases} \mathcal{G}(x) - \kappa + O(|x|^{-2}), & d = 2, \\ \mathcal{G}(x) + O(|x|^{-d}), & d \ge 3, \end{cases} \qquad \text{for } x \in \mathbb{Z}^d \setminus \{\mathbf{0}\}, \tag{4.6}$$

where κ is the constant in the potential kernel asymptotics of Theorem 3.9.

Figure 4.4: The divisible sandpile in $\mathbb{Z}^3$ started from mass $m = 10^6$ at the origin, with the octant $\{x_1, x_2, x_3 > 0\}$ removed: full sites are blue, and the partially filled sites, which form a layer one site thick, are red. On the cut faces, alternating shades mark eight bands of values of the odometer, which are nearly spherical shells.

Theorem 4.7 *Let $d \geq 2$ and $m > 0$. Let ω_d be the volume of the unit ball in $\mathbb{R}^d$, and let $r := (m/\omega_d)^{1/d}$. Let $\sigma_\infty : \mathbb{Z}^d \to [0,1]$ and $u_\infty : \mathbb{Z}^d \to [0,\infty)$ be the stable configuration and the odometer of the divisible sandpile started from $m\delta_{\mathbf{0}}$, and let $D_m := \{x \in \mathbb{Z}^d : \sigma_\infty(x) = 1\}$. Then there is a constant $c = c(d) > 0$ such that the following statements hold.*

(i) The odometer and the set D_m satisfy

$$\mathrm{B}^{::}_{r-c} \subset \{u_\infty > 0\} \subset D_m \subset \mathrm{B}^{::}_{r+c}\,.$$

(ii) For every $x \in \mathrm{B}^{::}_{r+c}$, the odometer satisfies

$$u_\infty(x) = mg^{::}(x) - m\mathcal{G}(r\mathbf{e}_1) + \frac{|x|^2 - r^2}{2d} + \begin{cases} m\kappa, & d = 2, \\ 0, & d \geq 3, \end{cases} + O(1)\,, \quad (4.7)$$

where the constant in the $O(1)$ depends only on d.

The set D_m consists of the fully occupied sites. Recall that ∂D_m is the outer boundary of D_m. For $x \in \mathbb{Z}^d \setminus \{\mathbf{0}\}$, positive final mass must have arrived from a neighbor. Thus if $\sigma_\infty(x) > 0$, then there exists $y \sim x$ with $u_\infty(y) > 0$, and consequently $\sigma_\infty(y) = 1$ by Lemma 4.4(i). The stable configuration is therefore supported in $\{\mathbf{0}\} \cup D_m \cup \partial D_m$.

Equation (4.7) approximates the odometer by a quadratic function plus m times the discrete Green function, which is not radial. At distances comparable to r from the origin, however, the error term in the asymptotics of $g^{::}$ is multiplied by m and is still $O(1)$, so there the odometer is radial up to a bounded error.

We compare $u_\infty + \gamma$ with a constant, choosing γ so that $\Delta^{::}\gamma = m\delta_{\mathbf{0}} - 1$. Mass balance then makes $u_\infty + \gamma$ superharmonic everywhere and harmonic wherever the odometer is positive. Since $x \mapsto |x|^2$ and $g^{::}$ have discrete Laplacians $2d$ and $-\delta_{\mathbf{0}}$, respectively, we let

$$\gamma(x) := -\frac{|x|^2}{2d} - m g^{::}(x) \qquad \text{for } x \in \mathbb{Z}^d\,. \tag{4.8}$$

Equation (4.6) shows that replacing $g^{::}$ by $\mathcal{G}$ when $d \geq 3$, or by $\mathcal{G} - \kappa$ when $d = 2$, changes $\gamma(x)$ by $O(m|x|^{-d})$ for $x \neq \mathbf{0}$. Since $m = \omega_d r^d$, the difference is $O(1)$ when $|x| \geq r/4$. The function obtained by this replacement is radial, and its value on the sphere of radius r is

$$\gamma_r := \begin{cases} -r^2/4 - m(\mathcal{G}(r\mathbf{e}_1) - \kappa), & d = 2, \\ -r^2/(2d) - m\mathcal{G}(r\mathbf{e}_1), & d \geq 3\,. \end{cases} \tag{4.9}$$

Equation (4.7) is equivalent to $u_\infty(x) = \gamma_r - \gamma(x) + O(1)$. We will prove this by bounding $u_\infty + \gamma$ above and below by constants close to γ_r.

There is also an obstacle interpretation. Lemma 4.6 and Theorem 4.3 identify u_∞ with $s - \gamma$, where s is the least superharmonic majorant (4.5) of γ. Figure 4.5 illustrates the continuum version. This interpretation is optional for the estimates below; Exercise 4.5 explores it on a general graph.

We now compute γ, in every dimension at once. Write

$$\psi(t) := \frac{t^2}{2d} + m\,\mathcal{G}(t\mathbf{e}_1) \qquad \text{for } t > 0\,.$$

Although the continuum Green function $\mathcal{G}$ of (2.14) is logarithmic when $d = 2$ and a power when $d \geq 3$, differentiating $\mathcal{G}(t\mathbf{e}_1)$ in t gives the single formula $-1/(d\omega_d t^{d-1})$ in both cases. Since $m = \omega_d r^d$, it follows that

$$\psi'(t) = \frac{t}{d}\left[1 - \left(\frac{r}{t}\right)^d\right] \qquad \text{and} \qquad \psi''(t) = \frac{1}{d}\left[1 + (d-1)\left(\frac{r}{t}\right)^d\right].$$

Thus ψ is convex and has its minimum at r.

Comparing (4.8) with (4.9) and using (4.6) to replace $g^{::}$ by the continuum Green function, we obtain

$$\gamma_r - \gamma(x) = \psi(|x|) - \psi(r) + O(m|x|^{-d}) \qquad \text{for } x \in \mathbb{Z}^d \setminus \{\mathbf{0}\}\,.$$

In dimension two the constant κm occurs in both γ_r and $\gamma(x)$ and cancels. For $|x| > r/4$ the error term is at most a constant depending only on d, because $m = \omega_d r^d$. Since $\psi'(r) = 0$, Taylor's theorem therefore gives a number ξ between $|x|$ and r such that

$$\gamma_r - \gamma(x) = \frac{1}{2d}\left[1 + (d-1)\left(\frac{r}{\xi}\right)^d\right](r - |x|)^2 + O(1) \qquad \text{for } |x| > r/4\,. \tag{4.10}$$

When $|x| = r$ we take $\xi = r$.

The next lemma records the bounds on the obstacle used in the proof of the ball asymptotics.

Lemma 4.8 *Let $d \geq 2$ and $m > 0$. Let ω_d be the volume of the unit ball in $\mathbb{R}^d$, and let $r := (m/\omega_d)^{1/d}$. Let $\gamma : \mathbb{Z}^d \to \mathbb{R}$ be given by*

$$\gamma(x) := -\frac{|x|^2}{2d} - m g^{::}(x) \qquad \text{for } x \in \mathbb{Z}^d\,.$$

Let

$$\gamma_r := \begin{cases} -r^2/4 - m(\mathcal{G}(r\mathbf{e}_1) - \kappa), & d = 2, \\ -r^2/(2d) - m\mathcal{G}(r\mathbf{e}_1), & d \geq 3\,. \end{cases}$$

Then there is a constant $C = C(d) < \infty$ with the following properties.

(i) For every $x \in \mathbb{Z}^d$ with $|x| > r/4$,

$$\gamma_r - \gamma(x) \geq \frac{(r - |x|)^2}{2d} - C\,.$$

(ii) For every $b > 0$, the absolute value of $\gamma_r - \gamma(x)$ is bounded by a constant depending only on b and d when $x \in \mathrm{B}^{::}_{r+b} \setminus \mathrm{B}^{::}_{r-b}$.

(iii) For every $x \in \mathbb{Z}^d$,

$$\gamma_r - \gamma(x) \geq \frac{(r - |x|)^2}{8d} - C\,.$$

Proof. (i) Equation (4.10) gives part (i) after increasing C.

(ii) Suppose first that $r \geq 2b$. Every point in $\mathrm{B}^{::}_{r+b} \setminus \mathrm{B}^{::}_{r-b}$ has norm at least $r - b \geq r/2$, so (4.10) proves part (ii). Suppose next that $r < 2b$. Then $m < \omega_d(2b)^d$, the annulus lies in $\mathrm{B}^{::}_{3b}$, and, for $x \in \mathrm{B}^{::}_{3b}$,

$$\gamma_r - \gamma(x) = \frac{|x|^2 - r^2}{2d} + m g^{::}(x) - m\mathcal{G}(r\mathbf{e}_1) + \begin{cases} m\kappa, & d = 2, \\ 0, & d \geq 3\,. \end{cases}$$

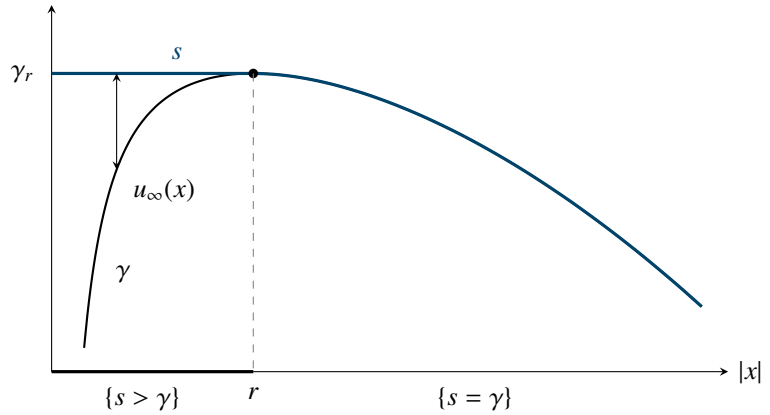


Figure 4.5: Radial profiles in $d = 3$ of the obstacle (4.8) with $g^{::}$ replaced by $\mathcal{G}$ (black) and of its least superharmonic majorant (blue), both shifted by a common constant. The majorant equals γ_r for $|x| \le r$ and the obstacle for $|x| \ge r$, and the vertical difference of the two curves is the continuum analogue of $u_\infty = s - \gamma$.

Each term is bounded by a constant depending only on b and d. Indeed, $|x|^2$ and r^2 are less than $9b^2$, and $g^{::}$ is bounded on the finite set $\mathrm{B}^{::}_{3b}$. For $d \ge 3$ we have $m\mathcal{G}(r\mathbf{e}_1) = r^2/(d(d-2))$, and for $d = 2$ we have $m\mathcal{G}(r\mathbf{e}_1) = -\frac{1}{2}r^2 \log r$, which lies between $-2b^2 \max\{\log(2b), 0\}$ and $1/(4e)$. This proves part (ii).

(iii) For $|x| \ge r/3$, part (i) gives (iii), since $r/3 > r/4$. For $x \in \mathrm{B}^{::}_{r/3}$ we compare with the boundary of this ball. The function $\gamma(x) + |x|^2/(2d) = -mg^{::}(x)$ is subharmonic, since its discrete Laplacian is $m\delta_{\mathbf{0}}$. The discrete minimum principle (Lemma 3.2), applied to its negative, shows that it attains its maximum over $\mathrm{B}^{::}_{r/3} \cup \partial\mathrm{B}^{::}_{r/3}$ at a vertex $z \in \partial\mathrm{B}^{::}_{r/3}$. Hence, for $x \in \mathrm{B}^{::}_{r/3}$, dropping the term $|x|^2/(2d) \ge 0$ and applying part (i) at z, where $r/3 \le |z| < r/3 + 1$, we obtain

$$\gamma_r - \gamma(x) \ge \gamma_r - \gamma(z) - \frac{|z|^2}{2d} \ge \frac{r^2 - 2r|z|}{2d} - C \ge \frac{r^2}{6d} - \frac{r}{d} - C\,.$$

Since $r^2/24 - r = (r-12)^2/24 - 6$ and $(r - |x|)^2 \le r^2$, the right side is at least $(r-|x|)^2/(8d) - C - 6/d$. Replacing C by $C + 6/d$, which preserves parts (i) and (ii), completes the proof. □

We now prove the ball asymptotics.

Proof of Theorem 4.7. We compare $u_\infty + \gamma$ with a constant, then follow increasing odometer values to prove the outer inclusion. The odometer estimate will follow from the same comparisons. By Lemma 4.4, the set $A := \{u_\infty > 0\}$ is finite and

$$\Delta^{::}(u_\infty + \gamma) = \sigma_\infty - 1 \le 0\,,$$

with equality on A.

Step 1: We prove the comparisons and the inner inclusion in part (i). Lemma 4.8(iii) gives $\gamma \le \gamma_r + C$ everywhere. Off A the odometer vanishes, so $u_\infty + \gamma \le \gamma_r + C$ there. If A is nonempty, then the maximum principle, Lemma 3.2 applied to $-u_\infty - \gamma$, gives the same bound on A. Thus

$$u_\infty(x) \le \gamma_r + C - \gamma(x) \qquad \text{for } x \in \mathbb{Z}^d\,. \tag{4.11}$$

On $\partial \mathrm{B}_r^{::}$ we have $r \le |x| < r+1$, so Lemma 4.8(ii) and $u_\infty \ge 0$ give $u_\infty + \gamma \ge \gamma_r - C$, after increasing C. The minimum principle therefore gives

$$u_\infty(x) \ge \gamma_r - C - \gamma(x) \qquad \text{for } x \in \mathrm{B}_r^{::}\,.$$

Choose $c = c(d) > 0$ with $c^2 > 16dC$. For $x \in \mathrm{B}_{r-c}^{::}$, Lemma 4.8(iii) now gives

$$u_\infty(x) \ge \gamma_r - C - \gamma(x) \ge \frac{(r-|x|)^2}{8d} - 2C > 0\,.$$

Hence $\mathrm{B}_{r-c}^{::} \subset \{u_\infty > 0\}$, and $\{u_\infty > 0\} \subset D_m$ by Lemma 4.4(i). When $r \le c$, the first inclusion is vacuous.

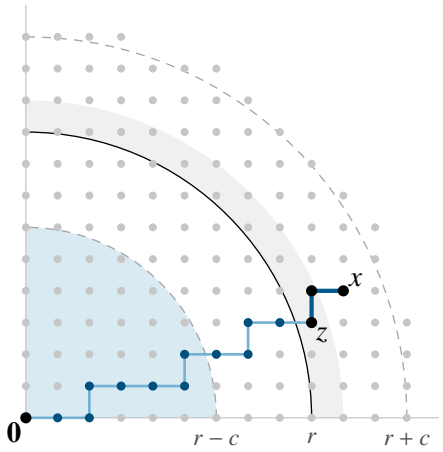


Figure 4.6: Step 2 of the proof of Theorem 4.7, drawn in the quarter plane of $\mathbb{Z}^2$: the odometer is positive on $\mathrm{B}_{r-c}^{::}$ (blue) by Step 1 and at most C_1 on the band $\mathrm{B}_{r+1}^{::} \setminus \mathrm{B}_r^{::}$ (gray). Along the path from a site $x \in D_m \setminus \mathrm{B}_r^{::}$ to $\mathbf{0}$ it increases by at least $1/(2d)$ at each step, so at most $2dC_1$ steps (thick) lead from x to the vertex z before the first entrance into $\mathrm{B}_r^{::}$, and $|x| < r + c$.

Step 2: We prove the outer inclusion in part (i), as illustrated in Figure 4.6. Equation (4.11) and Lemma 4.8(ii) give $u_\infty \le C_1$ on $\mathrm{B}_{r+1}^{::} \setminus \mathrm{B}_r^{::}$, for a constant $C_1 = C_1(d)$. At every $z \in D_m \setminus \{\mathbf{0}\}$,

$$\sum_{y \sim z} \bigl(u_\infty(y) - u_\infty(z)\bigr) = 1\,.$$

Thus a neighbor y satisfies $u_\infty(y) \geq u_\infty(z) + 1/(2d) > 0$, so $y \in D_m$. Iterating produces a path to $\mathbf{0}$: otherwise its odometer values would increase without bound, contrary to finite support. For $x \in D_m \setminus \mathrm{B}_r^{::}$, follow this path to its first entrance in $\mathrm{B}_r^{::}$. Let z be the preceding vertex and let k be the number of steps from x to z. Then $r \leq |z| < r + 1$ and

$$\frac{k}{2d} \leq u_\infty(z) - u_\infty(x) \leq C_1 .$$

Consequently $|x| \leq |z| + k < r + 1 + 2dC_1$. Increasing c to at least $1 + 2dC_1$ proves the outer inclusion and preserves the inner inclusion.

Step 3: We prove part (ii). The two comparisons give

$$u_\infty(x) = \gamma_r - \gamma(x) + O(1) \qquad \text{for } x \in \mathrm{B}_{r+c}^{::}, \tag{4.12}$$

first on $\mathrm{B}_r^{::}$. On $\mathrm{B}_{r+c}^{::} \setminus \mathrm{B}_r^{::}$, the same estimate follows from $u_\infty \geq 0$, (4.11), and Lemma 4.8(ii). Substituting the definitions of γ and γ_r gives (4.7). □

Theorem 4.7(ii) also controls the odometer near the sphere of radius r: inside that sphere, the odometer grows at least quadratically with the distance to it.

Proposition 4.9 *Let $d \geq 2$ and $m > 0$. Let $u_\infty : \mathbb{Z}^d \to \mathbb{R}_{\geq 0}$ be the divisible sandpile odometer for the initial configuration $m\delta_{\mathbf{0}}$ in $\mathbb{Z}^d$. Let ω_d be the volume of the unit ball in $\mathbb{R}^d$, and let $r := (m/\omega_d)^{1/d}$. Then there is a constant $C = C(d) < \infty$ such that*

$$u_\infty(x) \geq \frac{(r - |x|)^2}{2} - C \qquad \textit{for } x \in \mathrm{B}_r^{::} \setminus \mathrm{B}_{r/3}^{::} .$$

Proof. By Theorem 4.7(ii), in the form (4.12), we have $u_\infty(x) \geq \gamma_r - \gamma(x) - C$ for $x \in \mathrm{B}_r^{::}$. For $x \in \mathrm{B}_r^{::} \setminus \mathrm{B}_{r/3}^{::}$, the number ξ in (4.10) satisfies $\xi \leq r$, so $(r/\xi)^d \geq 1$ and the coefficient of $(r - |x|)^2$ there is at least $1/2$ (compare the continuum profile of Figure 4.5). Hence $u_\infty(x) \geq (r - |x|)^2/2 - C$, after increasing C. □

The same two estimates give a matching quadratic upper bound on $\mathrm{B}_{r+c}^{::} \setminus \mathrm{B}_{r/3}^{::}$, where $\xi \geq r/3$. It is the discrete counterpart of Proposition 2.32, by which a nonnegative function on $\mathbb{R}^d$ with bounded Laplacian grows at most quadratically away from a point where it vanishes. Exercise 4.11 asks you to prove the sharper expansion

$$u_\infty(x) = \frac{(r - |x|)^2}{2} + O\left(1 + \frac{\left|r - |x|\right|^3}{r}\right) \qquad \text{for } x \in \mathrm{B}_{r+c}^{::} \setminus \mathrm{B}_{r/3}^{::} ,$$

where the implied constant depends only on d.

We now apply the ball asymptotics to the abelian sandpile, where grains move in integer amounts.

4.3 BALL BOUNDS FOR THE ABELIAN SANDPILE

Start with n grains at the origin of $\mathbb{Z}^d$ and topple every site that holds at least $2d$ grains, sending one grain to each of its $2d$ neighbors, until every site is stable. (Figure 1.3 shows the sandpiles reached from 10^4, 10^5 and 10^6 grains in $\mathbb{Z}^2$.) Using the divisible sandpile, we will show that the sites that topple contain a Euclidean ball of volume about $n/(2d-1)$ and lie in a Euclidean ball whose volume is as close to n/d as we like, up to a boundary layer of bounded thickness (Figure 4.7).

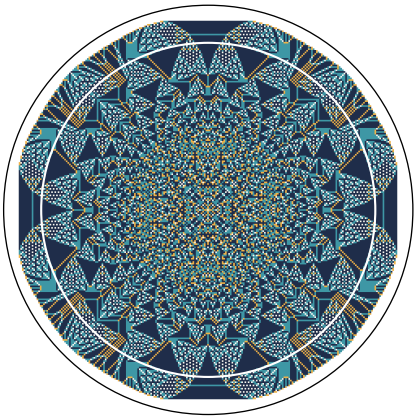


Figure 4.7: The stable sandpile reached from $n = 10^5$ grains at the origin of $\mathbb{Z}^2$, colored by the number of grains as in Figure 1.3, and the circles of radii $(2d-1)^{-1/d}r$ (white) and $d^{-1/d}r$ (black) for $d = 2$, where $\pi r^2 = n$. These are the leading radii in Proposition 4.11, the second as the limit of $a^{-1/d}r$ as $a \uparrow d$, and the toppled set S_n lies between them.

Recall from Section 1.1 that the odometer $u(x)$ is the number of times that x topples, and that the stable sandpile σ° reached from the initial sandpile $n\delta_{\mathbf{0}}$ satisfies

$$\sigma^\circ = n\delta_{\mathbf{0}} + \Delta^{::} u\,, \tag{4.13}$$

where $\Delta^{::}$ is the discrete Laplacian (3.1). Stability of σ° means $\sigma^\circ \le 2d-1$. We call a toppling legal when it takes place at a site that holds at least $2d$ grains. (Chapter 9 develops the abelian sandpile on general graphs; here we prove only what this section needs.) The next lemma proves three facts about stabilization on the infinite lattice: a set of sites that all topple keeps at least one grain for each of its internal edges, every sequence of legal topplings stops, and the order of the topplings does not matter.

Lemma 4.10 *Let $d \geq 1$ and $n \geq 1$ be integers. Start the abelian sandpile on $\mathbb{Z}^d$ with n grains at $\mathbf{0}$ and no grains elsewhere. Then the following statements hold.*

(i) Let A be a finite set of sites, and let a finite sequence of legal topplings topple every site of A at least once. Then at the end of the sequence the sites of A hold in total at least as many grains as there are edges of $\mathbb{Z}^d$ with both endpoints in A.

(ii) Every sequence of legal topplings is finite.

(iii) Every two finite sequences of legal topplings that end in stable sandpiles topple each site the same number of times.

Proof. We prove (i) by recording, for each edge, which endpoint topples last; we prove (ii) by combining (i) with the second moment of the sandpile, which grows by $2d$ at every toppling; and we prove (iii) by comparing two sequences one toppling at a time.

(i) Heights are never negative: the initial heights are nonnegative, a legal toppling leaves at least 0 grains at its site, and it only adds grains elsewhere. Let x and y be neighbors in A such that the last toppling of x comes before the last toppling of y. (Two topplings never occur at the same time, so the last topplings of x and y are distinct.) When y topples for the last time, it sends a grain to x, and from then on x only receives grains. Hence at the end of the sequence each site $x \in A$ holds at least as many grains as it has neighbors in A whose last toppling is later than its own. Each edge with both endpoints in A is counted at exactly one of its endpoints, the one whose last toppling comes first (Figure 4.8(a)). Summing over $x \in A$ proves (i).

(ii) Consider a finite sequence of legal topplings, and let W be the set of sites that topple. If W is empty, then no grain moves. Otherwise $\mathbf{0} \in W$, because $\mathbf{0}$ is the only site that holds grains at the start. Every other site of W holds no grains at the start and at least $2d$ grains at its first toppling, so it has received a grain from a neighbor in W that toppled earlier. The first topplings of the sites along a chain of such neighbors occur at strictly earlier times, so the chain ends at $\mathbf{0}$ (Figure 4.8(b)). Thus every site of W is joined to $\mathbf{0}$ by a path in W, and in particular at least $|W| - 1$ edges have both endpoints in W. By (i) the sites of W hold at least $|W| - 1$ grains at the end. There are only n grains, so $|W| \leq n + 1$. A shortest path in W from a site of W to $\mathbf{0}$ has at most $|W| - 1 \leq n$ steps, and therefore $|x| \leq n$ for every $x \in W$. A grain that moves goes from a site of W to a neighbor, and a grain that does not move stays at $\mathbf{0}$, so every grain ends within distance $n + 1$ of $\mathbf{0}$.

This holds for every prefix of a legal sequence. Write σ_t for the sandpile

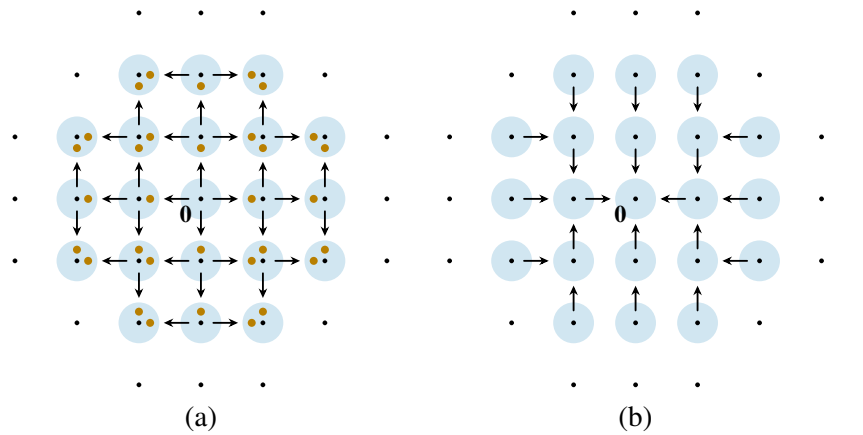


Figure 4.8: Lemma 4.10 for $n = 64$ grains in $\mathbb{Z}^2$, toppled one at a time in first-in, first-out order, with the 21 sites that topple shaded. (a) Each of the 32 edges between them points to the endpoint whose last toppling comes first, and that endpoint keeps the grain sent along the edge (orange); (b) each of them other than $\mathbf{0}$ points to the neighbor that sent it its first grain, which toppled before it, so the arrows lead to $\mathbf{0}$.

after t topplings of a legal sequence. Then

$$\sum_{x\in\mathbb{Z}^d} |x|^2 \sigma_t(x) \le n(n+1)^2 \,.$$

A toppling at x changes this sum by $\sum_{i=1}^d (|x+\mathbf{e}_i|^2 + |x-\mathbf{e}_i|^2 - 2|x|^2) = 2d$, and the sum is 0 when $t = 0$. Hence it equals $2dt$ after t topplings, and $t \le n(n+1)^2/(2d)$. This proves (ii).

(iii) Let f be the odometer of a finite sequence of legal topplings that ends in the stable sandpile $\sigma^\circ = n\delta_{\mathbf{0}} + \Delta^{::} f$. Consider a second sequence of legal topplings, and let g_t be its odometer after t topplings. We prove by induction on t that $g_t \le f$. The case $t = 0$ is clear. For the inductive step, suppose that $g_{t-1} \le f$, and let x be the site of the tth toppling. If $g_t(x) > f(x)$, then $g_{t-1}(x) = f(x)$. Just before the tth toppling the number of grains at x is

$$n\delta_{\mathbf{0}}(x) + \sum_{y\sim x} g_{t-1}(y) - 2d\, g_{t-1}(x) \le \sigma^\circ(x) \le 2d - 1 \,,$$

where the first inequality holds because $g_{t-1} \le f$ by the inductive hypothesis, $g_{t-1}(x) = f(x)$, and $\sigma^\circ(x) = n\delta_{\mathbf{0}}(x) + \sum_{y\sim x} f(y) - 2d\, f(x)$. This contradicts the legality of the tth toppling. Hence $g_t \le f$. If the second sequence is finite and ends in a stable sandpile, with final odometer g, then $g \le f$, and exchanging the roles of the two sequences gives $f \le g$. Thus $f = g$. □

We now compare the odometer of the abelian sandpile with two divisible odometers. For the inner ball, the number of topplings at each site, divided

by $2d-1$, is a nonnegative function that stabilizes the divisible sandpile with mass $n/(2d-1)$ at the origin, so by the least action principle it dominates the divisible odometer. For the outer ball, the count in Lemma 4.10(i) shows that a cube of toppled sites holds nearly d grains per site, and we compare the number of topplings, averaged over such cubes, with a divisible odometer. By Lemma 4.10, the set of sites that topple is finite and does not depend on the order of the topplings, so the next statement is well posed.

Proposition 4.11 *Let $d \geq 2$ and $n \geq 1$ be integers, and let $0 < a < d$. Let S_n be the set of sites that topple when the abelian sandpile on $\mathbb{Z}^d$ that starts with n grains at $\mathbf{0}$ and no grains elsewhere is stabilized by legal topplings. Let ω_d be the volume of the unit ball in $\mathbb{R}^d$, and let $r := (n/\omega_d)^{1/d}$. Then there are constants $c = c(d) > 0$ and $C = C(d,a) > 0$, independent of n, such that*

$$\mathrm{B}^{::}_{(2d-1)^{-1/d}r-c} \subseteq S_n \subseteq \mathrm{B}^{::}_{a^{-1/d}r+C}\,.$$

Proof. The idea is to compare the odometer u of the abelian sandpile with the odometer of a divisible sandpile, directly for the inner ball and after averaging over cubes for the outer ball. We divide the proof into three steps: the inner inclusion, a comparison of averaged odometers, and the outer inclusion.

Let u be the odometer of a stabilization by legal topplings, and let σ° be the stable sandpile that it reaches. By Lemma 4.10, the function u does not depend on the order of the topplings, it has finite support, and $S_n = \{u > 0\}$. By (4.13) and stability,

$$0 \leq n\delta_{\mathbf{0}} + \Delta^{::}u = \sigma^\circ \leq 2d-1\,.$$

Step 1: We prove the inner inclusion. The function $u/(2d-1)$ is nonnegative, and

$$\frac{n}{2d-1}\delta_{\mathbf{0}} + \Delta^{::}\Big(\frac{u}{2d-1}\Big) = \frac{\sigma^\circ}{2d-1} \leq 1\,.$$

By the least action principle, Theorem 4.3(i), the odometer of the divisible sandpile started from $\frac{n}{2d-1}\delta_{\mathbf{0}}$ is at most $u/(2d-1)$. The radius that Theorem 4.7 assigns to this mass is $\big(n/((2d-1)\omega_d)\big)^{1/d} = (2d-1)^{-1/d}r$. Let $c = c(d)$ be the constant of that theorem. By part (i) of the theorem, the odometer of the divisible sandpile is positive on $\mathrm{B}^{::}_{(2d-1)^{-1/d}r-c}$. Hence u is positive there, which proves the inner inclusion.

Step 2: We compare the odometer averaged over cubes with the odometer of a divisible sandpile. Fix an integer $k \geq 1$ with $2dk/(2k+1) \geq a$; this is possible because $2dk/(2k+1)$ tends to $d > a$ as $k \to \infty$. Write $Q^{::}(x,k) := \{y \in \mathbb{Z}^d : |y-x|_\infty \leq k\}$ for the cube of side $2k+1$ centered at

$x \in \mathbb{Z}^d$, and write

$$f^{(k)}(x) := \frac{1}{(2k+1)^d} \sum_{y \in Q^{::}(x,k)} f(y)$$

for the average of a function $f : \mathbb{Z}^d \to \mathbb{R}$ over this cube. Let

$$A := \{x \in \mathbb{Z}^d : Q^{::}(x,k) \subseteq S_n\},$$

the set of centers of cubes all of whose sites topple (Figure 4.9). Then $A \subseteq S_n$ is finite. The cube $Q^{::}(x,k)$ has $2k$ edges on each of its $(2k+1)^{d-1}$ lines in each of the d coordinate directions, which gives $2dk(2k+1)^{d-1}$ edges in total. By Lemma 4.10(i) applied at the end of the stabilization, for every $x \in A$,

$$\frac{1}{(2k+1)^d} \sum_{y \in Q^{::}(x,k)} \sigma^\circ(y) \geq \frac{2dk}{2k+1} \geq a\,. \tag{4.14}$$

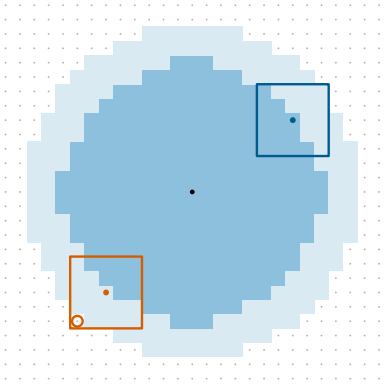


Figure 4.9: Step 2 of the proof of Proposition 4.11 for $n = 1000$ grains in $\mathbb{Z}^2$ and $k = 2$: the toppled set S_n (light blue) and the set A of centers of cubes $Q^{::}(x,2) \subseteq S_n$ (dark blue). A cube centered in A (blue) holds at least $2dk/(2k+1) = 8/5$ grains per site on average, and a cube centered outside A (vermilion) contains a site where u vanishes (circled), so $u \leq C_0$ on it.

Let v be a times the odometer of the divisible sandpile started from $\frac{n}{a}\delta_{\mathbf{0}}$. Then $v \geq 0$. The final divisible mass is at most 1 at every site, so $\Delta^{::} v \leq a - n\delta_{\mathbf{0}}$. The operator $f \mapsto f^{(k)}$ preserves inequalities, maps constants to themselves, and commutes with $\Delta^{::}$. Averaging (4.13) and the inequality for v and subtracting, we find that the averaged sources $n\delta_{\mathbf{0}}^{(k)}$ cancel. Together with (4.14), this gives

$$\Delta^{::}(u^{(k)} - v^{(k)})(x) \geq \frac{1}{(2k+1)^d} \sum_{y \in Q^{::}(x,k)} \sigma^\circ(y) - a \geq 0 \qquad \text{for } x \in A\,.$$

Thus $u^{(k)} - v^{(k)}$ is discrete subharmonic on A.

Next we bound $u^{(k)} - v^{(k)}$ outside A. Let x and y be neighbors. Since $u \geq 0$, the identity (4.13) and stability give

$$u(y) \leq \sum_{z \sim x} u(z) = 2d\, u(x) + \sigma^{\circ}(x) - n\delta_{\mathbf{0}}(x) \leq 2d\, u(x) + 2d - 1\,,$$

that is, $u(y) + 1 \leq 2d(u(x) + 1)$. If $x \notin A$, then $Q^{::}(x, k)$ contains a site where u vanishes, and every site of the cube is joined to it by a path of at most $2dk$ steps. Iterating the last inequality along this path gives $u \leq C_0$ on $Q^{::}(x, k)$, where $C_0 := (2d)^{2dk} - 1$. Hence $u^{(k)}(x) \leq C_0$, and since $v \geq 0$,

$$u^{(k)} - v^{(k)} \leq C_0 \qquad \text{outside } A\,.$$

If A is empty, then this bound holds on all of $\mathbb{Z}^d$. Otherwise $v^{(k)} - u^{(k)}$ is discrete superharmonic on the finite set A and is at least $-C_0$ on the outer boundary ∂A, which lies outside A. The discrete minimum principle, Lemma 3.2, shows that $v^{(k)} - u^{(k)} \geq -C_0$ on A as well. In both cases

$$u^{(k)} \leq v^{(k)} + C_0 \qquad \text{on } \mathbb{Z}^d\,.$$

Step 3: We prove the outer inclusion. The divisible sandpile in the definition of v has mass n/a at the origin, so the radius in Theorem 4.7 is $a^{-1/d}r$. By part (i) of that theorem, with the same constant c as in Step 1, the odometer of this divisible sandpile vanishes outside $\mathrm{B}^{::}_{a^{-1/d}r+c}$. Write $R := a^{-1/d}r + c + \sqrt{d}k$. Every site of $Q^{::}(x, k)$ lies within distance $\sqrt{d}k$ of x, so $v^{(k)}(x) = 0$ for $|x| \geq R$. Write $C_1 := (2k+1)^d C_0$. Since $u \geq 0$, we have $u(x) \leq (2k+1)^d u^{(k)}(x)$, and Step 2 gives

$$u(x) \leq C_1 \qquad \text{for } |x| \geq R\,. \tag{4.15}$$

If S_n is empty, then there is nothing to prove. Otherwise let $x \in S_n$ have maximal Euclidean norm; we show that $|x| < R + C_1 + 1$. If $x = \mathbf{0}$ or $|x| < R$, there is nothing to show, so suppose that $x \neq \mathbf{0}$ and $|x| \geq R$. Changing a nonzero coordinate of x by one unit away from zero gives a neighbor of larger norm, which lies outside S_n by the maximality of $|x|$, and $u = 0$ there. At every site $z \neq \mathbf{0}$, identity (4.13) gives $\Delta^{::}u(z) = \sigma^{\circ}(z) \geq 0$. Hence if a neighbor of z has smaller odometer than z, then another neighbor has larger odometer, and so odometer at least $u(z) + 1$, because u is integer-valued. Start at $x_0 := x$, which has a neighbor of smaller odometer. As long as $x_j \neq \mathbf{0}$, the site x_j has a neighbor of smaller odometer, namely the neighbor outside S_n when $j = 0$ and the site x_{j-1} when $j \geq 1$, so we can choose a neighbor x_{j+1} with $u(x_{j+1}) \geq u(x_j) + 1$. Since u is bounded, this path stops, and it stops only at $\mathbf{0}$, so there is an index m with $x_m = \mathbf{0}$, as for the divisible sandpile in Figure 4.6.

We now count the steps of this path outside $\mathrm{B}_R^{\vdots}$. Let j be the largest index with $|x_j| \geq R$. Then $j < m$ and $|x_{j+1}| < R$, so $|x_j| < R + 1$. By (4.15),

$$j \leq u(x_j) - u(x_0) \leq C_1 \,,$$

and therefore $|x| \leq |x_j| + j < R + C_1 + 1$. By the maximality of $|x|$, every site of S_n has norm less than $R + C_1 + 1 = a^{-1/d} r + C$, where $C := c + \sqrt{d}k + C_1 + 1$. This proves the outer inclusion, and with Step 1 the proposition. □

In other words, the sites that topple contain a Euclidean ball of volume about $n/(2d-1)$ and lie in a Euclidean ball of volume about n/a, up to a boundary layer of bounded thickness. The volume n/a can be made as close to n/d as we like, at the price of a larger constant C, because the cube side $2k+1$ must grow to bring $2dk/(2k+1)$ close to d. The two leading radii differ because the proposition does not determine how the grains are distributed among the toppled sites: it uses only that a stable site holds at most $2d-1$ grains and that cubes of toppled sites hold nearly d grains per site on average.

The gap disappears when the number of particles per occupied site is known. In Chapter 5 we study internal DLA, a random model in which every occupied site holds exactly one particle, and its cluster fills a Euclidean ball up to an error of lower order, with the divisible sandpile as a yardstick. Chapter 9 returns to the abelian sandpile and bounds its toppled set by combinatorial means alone, from inside by a diamond and from outside by a cube, on a family of backgrounds.

EXERCISES

Exercise 4.1. Let $n \geq 1$ be an integer. Find the odometer of the divisible sandpile on $\mathbb{Z}$ with initial mass $\sigma = 2\mathbf{1}_{[-n,n]}$. As in Section 4.1, the odometer records the mass emitted from each site to each of its neighbors.

Exercise 4.2. In the setting of Section 4.1, let $h : \mathbb{Z}^d \to \mathbb{R}$ be discrete superharmonic, and let σ be a finitely supported nonnegative initial mass whose divisible sandpile has final configuration σ_∞. Prove that a single toppling cannot increase the total weight $\sum_x h(x)\mu(x)$ of the current configuration μ. Then prove the quadrature inequality (1.1) of Chapter 1: the final configuration σ_∞ has total weight at most that of σ. Deduce that the two total weights are equal when h is discrete harmonic.

Exercise 4.3. Let $\sigma : \mathbb{Z}^d \to \mathbb{R}$ be a finitely supported nonnegative function. By Lemma 4.4(ii), its divisible sandpile odometer has finite support. Use the discrete minimum principle, Lemma 3.2, to prove that this odometer is the

only finitely supported function $u : \mathbb{Z}^d \to \mathbb{R}$ that satisfies the three conditions

$$\begin{aligned} u &\geq 0\,, \\ \sigma + \Delta^{::} u &\leq 1\,, \\ \sigma(x) + \Delta^{::} u(x) &= 1 \qquad \text{if } u(x) > 0\,. \end{aligned} \tag{4.16}$$

Exercise 4.4. In the setting of Section 4.1, let $\sigma \geq 0$ be a finitely supported initial mass on $\mathbb{Z}^d$. Exercise 4.3 shows that the odometer is the only finitely supported function that satisfies the three conditions in (4.16). The third condition cannot be dropped from this characterization. When $\sigma = 0$ the odometer vanishes, while the function $\mathbf{1}_{\{\mathbf{0}\}}$ is finitely supported and satisfies the first two conditions but not the third, since its discrete Laplacian at the origin is $-2d$. This exercise gives further examples on $\mathbb{Z}$. Each is obtained by sampling a continuous function, satisfies the first two conditions, and fails the third at only two sites. For $a \geq 0$, define $\nu_a : \mathbb{R} \to \mathbb{R}$ by

$$\nu_a(x) := \begin{cases} 1 + a - \frac{1}{2}x^2, & |x| \leq 1, \\ \frac{1}{2}(2 - |x|)^2 + a(2 - |x|), & 1 < |x| \leq 2, \\ 0, & |x| > 2\,. \end{cases} \tag{4.17}$$

Each ν_a has second derivative -1 on $(-1, 1)$ and 1 on $(-2, -1) \cup (1, 2)$; these are the values that a continuum odometer should have when mass density 2 on $[-1, 1]$ spreads to density 1 on $[-2, 2]$. Only ν_0 has a continuous derivative. For an integer $n \geq 1$, take

$$\sigma = 2\mathbf{1}_{[-n+1,n-1]} + \mathbf{1}_{\{-n,n\}} \qquad \text{and} \qquad \nu_{n,a}(x) := n^2 \nu_a(x/n) \quad \text{for } x \in \mathbb{Z}\,.$$

(a) Show that $\nu_{n,0}$ is the divisible sandpile odometer for the initial mass σ.

(b) Let $0 < a \leq 1/(2n)$. Show that $\nu_{n,a} \geq 0$ and $\sigma + \Delta^{::}\nu_{n,a} \leq 1$, but that $\nu_{n,a}(n) > 0$ and

$$\sigma(n) + \Delta^{::}\nu_{n,a}(n) = 1 - an < 1\,.$$

Conclude that $\nu_{n,a}$ satisfies the first two conditions in (4.16) but is not the odometer.

Exercise 4.5*. Let $G = (V, E)$ be an infinite locally finite connected graph, let $\sigma : V \to \mathbb{R}$, and let $\gamma : V \to \mathbb{R}$ satisfy $\Delta\gamma = \sigma - 1$ and have a superharmonic majorant. Let s be the least superharmonic majorant of γ, let $u_\infty := s - \gamma$, and let $\sigma_\infty := \sigma + \Delta u_\infty$.

(a) Show directly from the definition (4.5) of s that s is superharmonic and that $\Delta s(x) = 0$ whenever $s(x) > \gamma(x)$.

(b) Deduce that $\sigma_\infty \le 1$ on V and that $\sigma_\infty = 1$ on $\{u_\infty > 0\}$.

By Lemma 4.6 and Theorem 4.3, σ stabilizes and u_∞ is its odometer, so part (b) recovers Lemma 4.4(i) without the toppling dynamics.

Exercise 4.6. In the setting of Lemma 4.5 with $\sigma = m\delta_{\mathbf{0}}$ on $\mathbb{Z}^d$, where $d \ge 2$ and $m > 0$, let $(X_k)_{k\ge 0}$ be simple random walk started at $\mathbf{0}$. Prove that, after adjoining independent randomness, there is a stopping time τ such that $\mathbb{E}\tau < \infty$ and

$$\mathbb{P}\{X_\tau = z\} = \frac{\sigma_\infty(z)}{m} \qquad \text{for } z \in \mathbb{Z}^d .$$

Show that τ can be chosen so that $X_n \in \operatorname{supp}(\sigma_\infty)$ for $n \le \tau$ and $X_n \in \{u_\infty > 0\}$ for $n < \tau$.

Exercise 4.7. In the setting of Theorem 4.7, show that, after increasing $c = c(d)$,

$$\mathrm{B}^{::}_{r-c} \subset \{u_\infty > 0\} \subset \{\sigma_\infty > 0\} \subset \mathrm{B}^{::}_{r+c} .$$

Exercise 4.8. In the setting of Exercise 4.7, let $D_m := \{\sigma_\infty = 1\}$. Show that for every $x \in D_m \setminus \{\mathbf{0}\}$ there is a simple path $x = x_0, \dots, x_k = \mathbf{0}$ in D_m such that

$$u_\infty(x_{i+1}) \ge u_\infty(x_i) + \frac{1}{2d} \qquad \text{for } 0 \le i < k .$$

Conclude that D_m is empty or nearest-neighbor connected.

Exercise 4.9. Let $d \ge 2$, and let $c = c(d)$ be large enough that the conclusion of Exercise 4.7 holds. Let $R > 0$, let $0 \le r < R - 3c$, and let $h : \mathbb{Z}^d \to \mathbb{R}_{\ge 0}$ be discrete harmonic on $\mathrm{B}^{::}_R$. Prove that

$$h(x) \le \left(\frac{R-c}{R-r-3c}\right)^d h(\mathbf{0}) \qquad \text{for } x \in \mathrm{B}^{::}_r .$$

This is a second proof of a discrete Harnack inequality, independent of Lemma 3.16 and with a constant close to the continuum one.

Exercise 4.10. (Exact Mean Value on the Lattice) Let $d \ge 2$. Deduce from Exercise 4.7 and the mean value identity (4.4) that there is a constant $c = c(d)$ such that for every $r > 0$ there is a function $\sigma : \mathbb{Z}^d \to [0, 1]$ satisfying

(i) $\sigma(x) = 1$ for every $x \in \mathrm{B}^{::}_{r-c}$;

(ii) $\sigma(x) = 0$ for every $x \notin \mathrm{B}^{::}_r$;

(iii) for every $h : \mathbb{Z}^d \to \mathbb{R}$ that is discrete harmonic on $\mathrm{B}^{::}_r$,

$$\sum_{x\in\mathbb{Z}^d} \sigma(x)(h(x) - h(\mathbf{0})) = 0 .$$

Exercise 4.11. Let $d \geq 2$ and $m > 0$. Let ω_d be the volume of the unit ball in $\mathbb{R}^d$, let $r := (m/\omega_d)^{1/d}$, and let u_∞ be the divisible sandpile odometer for initial mass $m\delta_{\mathbf{0}}$ in $\mathbb{Z}^d$. Let $c = c(d)$ be the constant in Theorem 4.7. Use equation (4.10), together with (4.12), to verify that

$$u_\infty(x) = \frac{(r - |x|)^2}{2} + O\left(1 + \frac{\left||x| - r\right|^3}{r}\right) \qquad \text{for } x \in \mathrm{B}^{::}_{r+c} \setminus \mathrm{B}^{::}_{r/3}\,,$$

where the implied constant depends only on d.

The final three exercises relate inner and outer neighborhoods of bounded open sets to Hausdorff distance, in which Chapters 5 and 6 state the convergence of rescaled internal DLA clusters and rotor-router aggregates to the closed unit ball.

Exercise 4.12. Extending Ω_r of Section 2.1, for $A \subset \mathbb{R}^d$ and $r > 0$ the **inner r-neighborhood** and **outer r-neighborhood** of A (Figure 4.10) are

$$A_r := \{x \in A : \overline{\mathrm{B}}(x, r) \subset A\} \qquad \text{and} \qquad A^r := \{x \in \mathbb{R}^d : \mathrm{B}(x, r) \cap A \neq \emptyset\}\,.$$

Show that if A is open, then A_r and A^r are open.

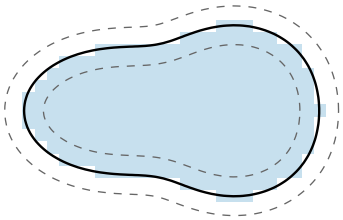


Figure 4.10: A bounded open set $A \subset \mathbb{R}^2$ (black), its inner and outer r-neighborhoods A_r and A^r for $r = 0.16$ (dashed), and the union B of the squares of side $\delta = 0.1$ centered at the points of $\delta\mathbb{Z}^2$ in A (shaded). Since $\delta/\sqrt{2} < r$, we have $A_r \subset B \subset A^r$.

Exercise 4.13. The **Hausdorff distance** between nonempty compact sets K and L in $\mathbb{R}^d$ is

$$d_H(K, L) := \inf\{r > 0 : K \subset L^r \text{ and } L \subset K^r\}\,.$$

The **inner-outer Hausdorff metric** on nonempty bounded open sets A and B in $\mathbb{R}^d$ is

$$d_{\mathrm{io}}(A, B) := \inf\{r > 0 : A_r \subset B \subset A^r \text{ and } B_r \subset A \subset B^r\}\,,$$

where A_r and A^r are as in Exercise 4.12. Show that d_H is a metric and that, for every compact set F containing $A \cup B$,

$$d_{\mathrm{io}}(A, B) = \max\{d_H(\overline{A}, \overline{B}), d_H(F \setminus A, F \setminus B)\}\,.$$

Conclude that d_{io} is a metric.

Exercise 4.14. For the metrics of Exercise 4.13, find nonempty bounded open sets $A, B \subset \mathbb{R}^d$ such that $d_H(\overline{A}, \overline{B}) = 0$ but $d_{\mathrm{io}}(A, B) = 1$.

NOTES

The divisible sandpile of Section 4.1 was introduced by Levine and Peres (2009). The identity $u_\infty = s - \gamma$ after Lemma 4.6 is (Levine and Peres, 2009, Lemma 3.2), in other sign and normalization conventions, for nonnegative finitely supported configurations on $\mathbb{Z}^d$, and Levine and Peres (2010, Lemma 3.2) reformulated it as the least action principle. Our odometer is the mass that a vertex sends to each neighbor, as in Levine, Murugan, Peres, and Ugurcan (2016a); in Levine and Peres (2009) and Levine and Peres (2017) it is the total mass emitted, which is $2d$ times larger on $\mathbb{Z}^d$. Theorem 4.3, for real configurations on locally finite graphs, follows Levine, Murugan, Peres, and Ugurcan (2016a, Proposition 2.5), which allows general toppling procedures. Our "thorough" is stronger than the condition in Levine and Peres (2017, Section 2) that a site that attains mass above 1 is eventually toppled.

The ball inclusions in Theorem 4.7 are Theorem 1.3 of Levine and Peres (2009), whose Lemma 3.2 gives the obstacle representation (Exercise 4.5) and whose Section 2 and the proof of Theorem 3.3 give the odometer bounds. Proposition 4.9 is the point-source analogue of the non-degeneracy estimate in Lemma 5 of Caffarelli (1998), page 388.

Proposition 4.11 is Theorem 1.4 of Levine and Peres (2009), with their $d - \epsilon$ in place of our a. They state it for the set of sites visited by the grains, which lies in the set of toppled sites together with its outer boundary and the origin, and their Section 4 bounds the set of toppled sites from inside and outside by the same radii. They compare these bounds in dimension two with the earlier diamond and square of Le Borgne and Rossin (2002, Theorem 5), and they credit to Le Borgne and Rossin (2002) the count in Lemma 4.10(i) of the grains held by toppled sites, which Le Borgne and Rossin (2002, Section 3.2) attribute to a thesis of Rossin. The proof given here differs from theirs. Levine and Peres (2009) obtain the inner bound by an explicit comparison function, a quadratic function plus the Green function, and the outer bound by averaging a second such function over cubes. Here the inner bound is the least action principle for a divisible sandpile, and the outer bound compares the odometer averaged over cubes with the odometer of a divisible sandpile, so that Theorem 4.7 replaces further Green function estimates. Both proofs finish the outer bound by following a path along which the odometer increases (Lemma 4.2 of Levine and Peres (2009)). Their Theorem 4.1 also treats a constant background of h grains at every site, with $2d - 1$ and $d - \epsilon$ replaced by $2d - 1 - h$ and $d - \epsilon - h$, for $h \le 2d - 2$ in the inner bound and $h \le d - 1$ and $0 < \epsilon < d - h$ in the outer bound, and it allows holes, that is, negative h. Section 9.9 gives a combinatorial proof of bounds of the same kind, by a diamond and a cube, on backgrounds.

On lattices whose spacing tends to zero, Theorem 3.3 of Levine and Peres (2010) proves uniform convergence of the rescaled odometers under their conditions (27)–(31): the initial densities take values between 0 and a common constant and are supported in a common ball, and they converge at every continuity point of a limiting density whose discontinuities form a null set. Their Theorem 3.9 proves convergence of the fully occupied sets under four further conditions, (48)–(51): the limiting density takes no values strictly between some $\lambda < 1$ and 1; the set where it is at least 1 is the closure of its interior; and, for every $\epsilon > 0$ and fine enough lattices, the densities are at least 1 on the inner ϵ-neighborhood of that set and at most λ outside its outer ϵ-neighborhood. Their Lemma 3.10, used in the proof of Theorem 3.9, is the discrete quadratic-growth estimate, and their Section 6 introduces the inner-outer Hausdorff metric of Exercise 4.13.

Aleksanyan and Shahgholian (2019) introduce a perturbation of the divisible sandpile that depends on a threshold: a site topples its mass in excess of the threshold, and topples all of its mass once its odometer exceeds a level inversely proportional to the threshold. The final mass therefore lies on a sub-level set of the odometer, which links sandpile dynamics to quadrature surfaces. For a point source and a fixed threshold, their Theorem 4.1 and Remark 4.2 give the scaling limits of the odometer and of the visited sets. Their Theorem 4.3 shows that, for some threshold tending to infinity slowly enough as the initial mass grows, the rescaled odometers converge to a spherically symmetric function, which the discussion following Remark 4.2 and Appendix A identify as a ball with the final mass uniformly distributed on its boundary. These limits connect the discrete quadrature inequality (1.1) to free-boundary problems of Bernoulli type.

On $\mathbb{Z}^2$ and for $1 < \alpha < 2$, Frómeta and Jara (2018) study a divisible sandpile in which mass is redistributed according to the transition probabilities of their equation (1.1), those of a truncated α-stable random walk with a nearest-neighbor correction. Their Theorems 1.2 and 1.3 give the scaling limit of this sandpile started from a compactly supported initial density of class C^2 and identify the limiting odometer through an obstacle problem for a truncated fractional Laplacian. Their Theorem 1.4 expands the rescaled Green functions of this walk asymptotically.

Chapter Five

Internal DLA in $\mathbb{Z}^d$: Single Source

Internal diffusion-limited aggregation is a model of random growth driven by diffusion. Start with $A_0 = \emptyset$. Given A_{n-1}, run a simple random walk from $\mathbf{0}$ until it first reaches a site outside A_{n-1}, and add that site to obtain A_n. The random set A_n is the **internal diffusion-limited aggregation (IDLA)** cluster. As the pictures suggest, after rescaling the cluster fills a Euclidean ball. This chapter proves a sharper statement, with an explicit bound on how far the boundary strays from a sphere.

The divisible sandpile of Chapter 4 replaces the particles by continuous mass, and its fully occupied sites fill a ball up to a bounded error in the radius (Theorem 4.7). We use that deterministic model as a yardstick. Figure 5.1 shows a simulated cluster.

Throughout the chapter, $d \geq 2$ is fixed, ω_d is the volume of the unit ball in $\mathbb{R}^d$, and $\mathrm{B}_r^{::} = \mathrm{B}^{::}(\mathbf{0}, r)$ is the discrete ball of Section 3.1. We write c and C for positive constants that depend only on d, unless we say otherwise, and whose values may change from one occurrence to the next. For $r := (n/\omega_d)^{1/d}$, the rescaled cluster A_n/r converges almost surely in Hausdorff distance to the closed unit ball. Theorem 5.9 in Section 5.3 quantifies this convergence. Up to logarithmic factors, the cluster contains $\mathrm{B}_{r-\varpi}^{::}$ for ϖ of order $r^{1/3}$ and is contained in $\mathrm{B}_{r+m}^{::}$ with m^d of order ϖr^{d-1}; explicit bounds control the probability that either containment fails. By the Borel–Cantelli lemma, Corollary 5.10 deduces that almost surely, for all large n, both containments hold with widths of these orders, which gives the almost sure convergence. In Figure 5.1, both containments hold with $\varpi = m = 4$.

Section 5.1 encodes IDLA by instruction stacks, whose abelian property lets us move particles in whichever order is convenient. For the inner bound, we compare IDLA with the divisible sandpile and follow a product of positive function values over occupied sites. For the outer bound, we divide a cubical shell into thinner shells. Passage through each shell reduces the particle count by a fixed factor, except with probability exponentially small in the incoming count.

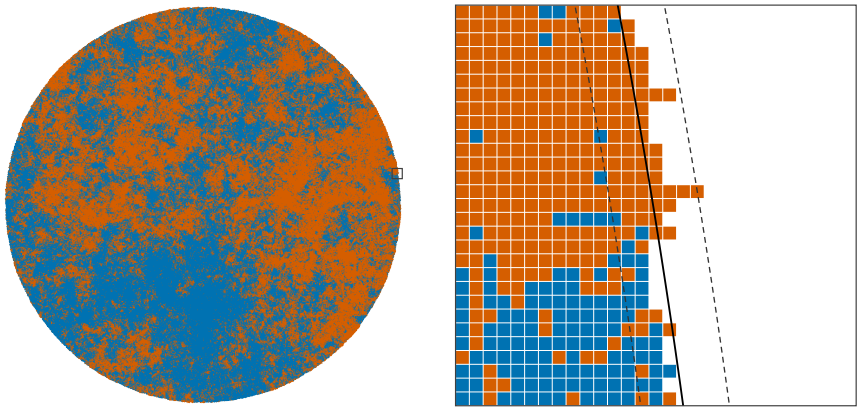


Figure 5.1: An internal DLA cluster of $n = 10^6$ particles in $\mathbb{Z}^2$, with each site x orange if $\pi|x|^2$ exceeds the number of the particle that settled there (it joined early) and blue otherwise (late). Right: the marked square, with the circle of radius $r = (n/\pi)^{1/2}$ (solid) and the circles through the nearest empty site and the farthest occupied site (dashed).

5.1 INSTRUCTION STACKS AND MASS BALANCE

We encode IDLA by placing an infinite stack of arrows at each site. By the abelian property, Corollary 5.2 below, we may release, pause and restart the particles in every convenient order without changing the cluster, as the proof of the inner bound in Section 5.2 does. At each $x \in \mathbb{Z}^d$, let $\rho_1(x), \rho_2(x), \ldots$ be a sequence of **arrows**, each a directed edge from x to one of its neighbors. We call this sequence the **instruction stack** at x. A **legal move** at a site containing at least two particles sends one particle along the next unused arrow in its stack, and that arrow is then discarded. The requirement of two particles means that a site never sends away its last particle. A finite sequence of legal moves is **complete** if it leaves at most one particle at each site (Figure 5.2).

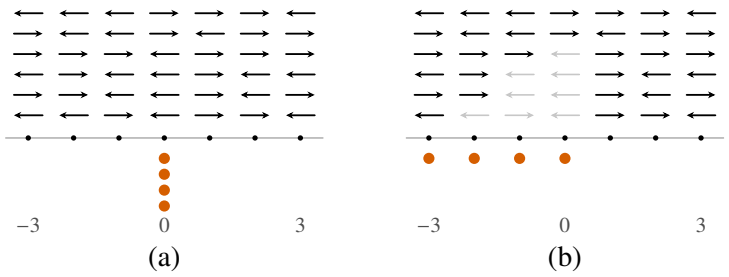


Figure 5.2: (a) Four particles at 0 in $\mathbb{Z}$, and the instruction stack at each site, read from the bottom up. (b) Every complete sequence of legal moves uses the same arrows (gray), four at 0, three at -1 and one at -2, and leaves one particle at each of the sites $-3, -2, -1, 0$.

Independent uniform arrows give IDLA: releasing the particles one at a time and letting each one walk until it reaches an unoccupied site uses every arrow at most once, so the particles perform independent simple random walks. Periodic instruction stacks that cycle once through the $2d$ neighbors give rotor-router aggregation, the subject of Chapter 6. The next lemma shows that every complete sequence makes at least as many moves at each site as every finite legal sequence. Its corollary shows that all complete sequences have the same outcome.

Lemma 5.1 *Let $d \geq 1$ be an integer. At each site of $\mathbb{Z}^d$, fix an infinite stack of arrows, each a directed edge from that site to one of its neighbors, and fix an initial configuration of finitely many particles. Then every finite sequence of legal moves makes at most as many moves at each site as every complete sequence of legal moves.*

Proof. Fix a complete legal sequence and a finite legal sequence. We show that, throughout the finite sequence, the number of moves it has made at each site is at most the number of moves the complete sequence makes there. Suppose, for contradiction, that this fails, and let x be the site of the first move of the finite sequence that breaks it. Just before that move, the finite sequence has made exactly as many moves at x as the complete sequence, and at most as many at every other site. Both sequences read the stacks from the top, so the arrows that have sent particles into x so far are among the arrows used by the complete sequence. Thus the finite sequence has sent exactly as many particles out of x as the complete sequence, and at most as many into x. It follows that x contains at most as many particles as at the end of the complete sequence, namely at most one, so the move at x is not legal. This contradiction proves the lemma. □

Applying the lemma in both directions to two complete sequences gives the abelian property.

Corollary 5.2 (Abelian Property) *Let $d \geq 1$ be an integer. At each site of $\mathbb{Z}^d$, fix an infinite stack of arrows, each a directed edge from that site to one of its neighbors, and fix an initial configuration of finitely many particles. Then every two complete sequences of legal moves make the same number of moves at each site and produce the same final configuration.*

Proof. Lemma 5.1, applied with each of the two sequences in the role of the complete one, shows that they make the same number of moves at each site. These numbers determine the final configuration: the final number of particles at a site x is the initial number, minus the number of moves at x, plus

the number of arrows pointing to x among those used at the neighbors of x. Hence the two complete sequences produce the same final configuration. □

For independent uniform stacks, almost surely each walk of the sequential release leaves the finite occupied set, so sequential release is a complete legal sequence, and by Corollary 5.2 every other complete schedule produces the same cluster from the same stacks. The proof of Lemma 5.1 never uses that the moves of the complete sequence are legal, only that they read the stacks from the top and leave at most one particle at each site. Thus comparison with a complete sequence also bounds legal moves when the moves of the complete sequence are allowed to be illegal.

The balance of either type of stack is measured by its discrepancy. For $x \in \mathbb{Z}^d$, a neighbor $y \sim x$, and an integer $m \geq 0$, let

$$\mathcal{D}_\rho(x, y, m) := \#\{1 \leq j \leq m : \rho_j(x) = (x, y)\} - \frac{m}{2d} \tag{5.1}$$

be the **discrepancy** of the stack at x. It records how far y is ahead of its fair share of the first m particles to leave x, or behind it.

We next express mass conservation through the odometer and the discrepancies. Let $\sigma : \mathbb{Z}^d \to \mathbb{N}$ be a finitely supported initial configuration of particles for which there is a finite complete sequence of legal moves. Let A be the set of sites that such a sequence leaves occupied, and let $v(x)$ be the number of moves it makes at x. By the abelian property, Corollary 5.2, neither depends on the choice of complete legal sequence, and v is the odometer; it counts all moves at a site, so $v/(2d)$ corresponds to the odometer of Chapter 4. The number of particles sent from x to a neighbor y is $v(x)/(2d) + \mathcal{D}_\rho(x, y, v(x))$. Mass balance at x, the number of particles received minus the number sent, therefore gives

$$\mathbf{1}_A(x) - \sigma(x) = \frac{1}{2d}\Delta^{::} v(x) + \sum_{y \sim x} \mathcal{D}_\rho(y, x, v(y)) \,.$$

Thus the divisible-sandpile term $\frac{1}{2d}\Delta^{::} v$ is the main term, and the discrepancies record the error caused by routing whole particles. Chapter 6 sums this identity against a test function (Proposition 6.1), for sequences of moves that need not be complete.

We turn to the inner bound, where the abelian property lets us pause the particles at the edge of a ball and restart them later.

5.2 INNER BOUND

We bound A_n from below by comparing its growth to the divisible sandpile. The comparison uses the solution of a Poisson problem in a lattice ball slightly

smaller than the Euclidean ball of volume n, and the next lemma gives the quadratic lower bound for it that we need. It compares this solution with the obstacle (4.8) of Chapter 4, and Chapter 6 uses it again for rotor-router aggregation.

Lemma 5.3 *Let $d \geq 2$ be an integer, let $\eta > 0$, and let ω_d be the volume of the unit ball in $\mathbb{R}^d$. There is a constant $C = C(\eta, d) < \infty$ with the following property. Let $n \geq 1$ be a real number, let $r := (n/\omega_d)^{1/d} > \eta$, and let $B := \mathrm{B}^{::}_{r-\eta}$. If $f : B \cup \partial B \to \mathbb{R}$ satisfies*

$$\Delta^{::} f = 1 - n\delta_{\mathbf{0}} \quad \text{in } B \qquad \text{and} \qquad f = 0 \quad \text{on } \partial B\,,$$

then

$$f(z) \geq \frac{(r - |z|)^2}{8d} - C \qquad \text{for } z \in B\,.$$

Proof. The idea is to compare f with the obstacle, measured from its value at radius r, which has the same Laplacian on all of $\mathbb{Z}^d$. Recall the obstacle $\gamma(x) = -|x|^2/(2d) - ng^{::}(x)$ of (4.8), with mass n, and its value γ_r at radius r of (4.9), from Chapter 4. Since $\Delta^{::}|x|^2 = 2d$ and $\Delta^{::} g^{::} = -\delta_{\mathbf{0}}$, the function $F := \gamma_r - \gamma$ satisfies $\Delta^{::} F = 1 - n\delta_{\mathbf{0}}$ on $\mathbb{Z}^d$. Every $x \in \partial B$ satisfies $r - \eta \leq |x| < r - \eta + 1$, so Lemma 4.8(ii) with $b = \eta + 1$ gives $|F| \leq C$ on ∂B. Hence $f - F$ is discrete harmonic in B and at least $-C$ on ∂B, where $f = 0$. The minimum principle (Lemma 3.2) gives $f \geq F - C$ in B, and Lemma 4.8(iii) gives $F(z) \geq (r - |z|)^2/(8d) - C$ for every $z \in \mathbb{Z}^d$. This proves the lemma. □

We prove the main step of the inner bound for arbitrary starting sites, since Chapter 7 uses it again for internal DLA from two sources. Recall from (3.18) the Green function $g^{::}_B$ of simple random walk killed on leaving B. While a site z stays unoccupied, the function $1 - \lambda g^{::}_B(\cdot, z)$ is harmonic at every occupied site.

Lemma 5.4 *Let $d \geq 2$ and $N \geq 1$ be integers, let $B \subset \mathbb{Z}^d$ be finite, and let $z, s_1, \ldots, s_N \in B$. Release N particles in order, the ith starting at s_i and performing simple random walk, independently of the others. A particle settles at the first unoccupied site of B it reaches, and is paused if it leaves B first. Let I_N be the set of sites where particles settle. Let $\varphi := g^{::}_B(\cdot, z)$, let $L \geq \varphi(z)$ be a positive real number, and let $0 < \lambda \leq 1/(2L)$. Then*

$$\mathbb{P}\{z \notin I_N\} \leq \exp\Big\{-\lambda\Big(\sum_{i=1}^{N} \varphi(s_i) - \sum_{y \in B} \varphi(y)\Big) + \lambda^2 L \sum_{y \in B} \varphi(y)\Big\}\,.$$

Proof. **Step 1:** We bound the expected product of the values of $h := 1 - \lambda\varphi$ over the occupied sites. The function φ is nonnegative, vanishes off B, and satisfies $\Delta^{::}\varphi = -\delta_z$ on B by symmetry and (3.19). The minimum principle (Lemma 3.2) applied to $\varphi(z) - \varphi$ on $B \setminus \{z\}$ gives $0 \le \varphi \le \varphi(z) \le L$, trivially if $B = \{z\}$. Hence $1/2 \le h \le 1$, $h = 1$ off B, and h is harmonic on $B \setminus \{z\}$. For $0 \le i \le N$, let I_i be the set of sites where the first i particles settle, let $\mathcal{F}_i$ be generated by their walks, and let $W_i := \mathbf{1}\{z \notin I_i\} \prod_{y \in I_i} h(y)$. We claim that $\mathbb{E}[W_i \mid \mathcal{F}_{i-1}] \le h(s_i) W_{i-1}$ for $1 \le i \le N$. If $z \in I_{i-1}$, then both sides vanish. Otherwise $I_{i-1} \subset B \setminus \{z\}$, so h is harmonic on I_{i-1} (Figure 5.3). Let Y_i be the first site outside I_{i-1} visited by particle i, where it settles or is paused. We have $\mathbb{E}[h(Y_i) \mid \mathcal{F}_{i-1}] = h(s_i)$: this is clear if $s_i \notin I_{i-1}$, and otherwise $x \mapsto \mathbb{E}_x h(X_{\tau_{\partial I_{i-1}}})$ is harmonic on I_{i-1} and equals h on ∂I_{i-1} by Lemma 3.5, hence equals h on I_{i-1} by the minimum principle applied to the difference and its negative. If particle i settles at a site other than z, then $W_i = h(Y_i) W_{i-1}$; if it is paused, then $h(Y_i) = 1$ and $W_i = W_{i-1}$; and if it settles at z, then $W_i = 0$. Thus $W_i \le h(Y_i) W_{i-1}$, which proves the claim. Taking expectations in the claim and inducting on i from $W_0 = 1$ gives $\mathbb{E} W_N \le \prod_{i=1}^{N} h(s_i)$.

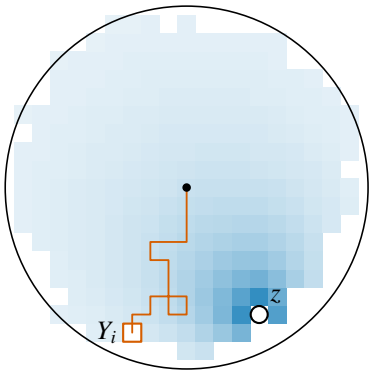
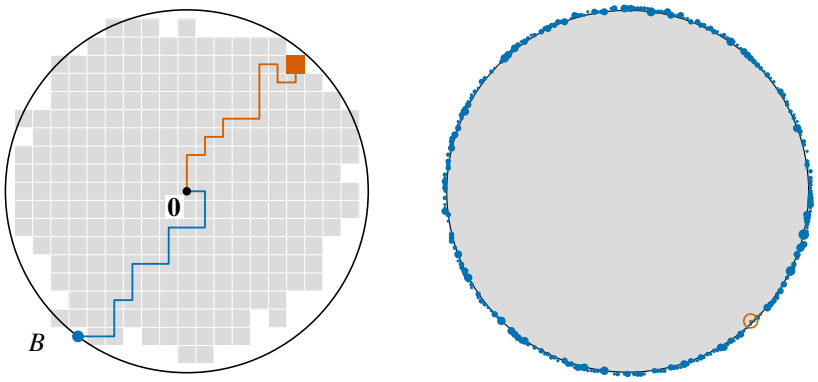


Figure 5.3: Step 1 of the proof of Lemma 5.4 in $B = \mathrm{B}_{10}^{::}$: each occupied site y is shaded by $g_B^{::}(y, z)$, where z (circled) is unoccupied, so $h = 1 - \lambda g_B^{::}(\cdot, z)$ is harmonic at every occupied site. The next walk from $\mathbf{0}$ stops at Y_i, where it first leaves the occupied sites.

Step 2: We bound $\mathbb{P}\{z \notin I_N\}$. Since $I_N \subset B$ and $0 < h \le 1$, we have $W_N \ge \mathbf{1}\{z \notin I_N\} \prod_{y \in B} h(y)$, so Step 1 gives

$$\mathbb{P}\{z \notin I_N\} \le \frac{\prod_{i=1}^{N} (1 - \lambda\varphi(s_i))}{\prod_{y \in B} (1 - \lambda\varphi(y))}.$$

We use $\log(1 - t) \le -t$ and $-\log(1 - t) \le t + t^2$ for $0 \le t \le 1/2$, with $t = \lambda\varphi \le 1/2$. The first inequality bounds the logarithm of the numerator by $-\lambda \sum_{i=1}^{N} \varphi(s_i)$. The second, with $\varphi^2 \le L\varphi$, bounds minus the logarithm of the denominator by $\lambda \sum_{y \in B} \varphi(y) + \lambda^2 L \sum_{y \in B} \varphi(y)$. This proves the lemma. □



(a) (b)

Figure 5.4: (a) The stopped construction in the proof of Theorem 5.5 for $r = 12$, with $B = \mathrm{B}^{::}_{r-2}$ inside the circle: the next particle walks through the occupied sites (gray) until it settles at an unoccupied site of B (orange) or is paused where it first leaves B (blue). (b) The end of the construction for $n = 10^4$: one site of B is still empty (ringed), and the 696 paused particles wait where they first left B (dot areas proportional to their number).

By Lemma 3.12(b), the Green function of a ball of radius $r \geq 2$ at its center is at most a constant times

$$\ell(r) := \begin{cases} \log(e + r), & d = 2, \\ 1, & d \geq 3\,, \end{cases} \tag{5.2}$$

and this factor enters the inner bound. We pause each particle that leaves $\mathrm{B}^{::}_{r-2}$ before settling, and apply Lemma 5.4 in this ball to bound the probability that a site of $\mathrm{B}^{::}_{r-\varpi}$ remains unoccupied.

Theorem 5.5 (IDLA Inner Bound) *Let $d \geq 2$ be an integer. There are positive constants c and C depending only on d with the following property. Let n be a positive integer, let $r := (n/\omega_d)^{1/d} \geq 4$, and let $B := \mathrm{B}^{::}_{r-2}$. Release n particles from $\mathbf{0}$ in order, letting them perform independent simple random walks. A particle settles when it first reaches an unoccupied site of B; if it leaves B first, then it is paused at the exit site. Let I_n be the set occupied by the particles that settle in B. Then, for every integer ϖ satisfying $2 \leq \varpi \leq r/2$,*

$$\mathbb{P}\{\mathrm{B}^{::}_{r-\varpi} \not\subset I_n\} \leq C r^d \exp\left(-c \frac{\varpi^3}{r\ell(r)}\right).$$

Proof of Theorem 5.5. The idea is to apply Lemma 5.4 with every starting site at $\mathbf{0}$ and the Green function of B with pole at a fixed site $z \in B$, and to bound its exponent above using Lemma 5.3.

Step 1: We set up the stopped aggregate (Figure 5.4(a)). Restarting the paused particles in order, each continuing its own walk until it first reaches an unoccupied site, produces an aggregate with the law of A_n. Indeed, every move of this schedule uses fresh randomness, so the schedule can equally be run with the instruction stacks of Section 5.1, where the abelian property, Corollary 5.2, identifies its outcome with the outcome of sequential release. We use this coupling to identify the restarted aggregate with A_n, so $I_n \subset A_n$.

Step 2: We bound the probability that a fixed site stays unoccupied. Fix $z \in B$, and let $s := r - |z|$, so that $2 < s \le r$. We first record three facts about $g^{::}_B$. First, there is a constant $C \ge 1$ such that $L := C\ell(r)$ satisfies $g^{::}_B(z,z) \le L$. Indeed, B lies in $\mathrm{B}^{::}(z, 2r)$, and the killed Green function (3.18) increases with the domain, so translation invariance and Lemma 3.12(b) give $g^{::}_B(z,z) \le g^{::}_{\mathrm{B}_{2r}}(\mathbf{0},\mathbf{0}) \le C\ell(r)$.

Second, let

$$f(x) := n g^{::}_B(\mathbf{0}, x) - \sum_{y \in B} g^{::}_B(y, x) \qquad \text{for } x \in B \cup \partial B\,.$$

Then $\Delta^{::} f = 1 - n\delta_{\mathbf{0}}$ in B and $f = 0$ on ∂B, so Lemma 5.3 with $\eta = 2$ gives $f(z) \ge s^2/(8d) - C$. Third, symmetry, normalization (3.18) and the exit-time estimate (Lemma 3.4) give

$$\sum_{y \in B} g^{::}_B(y, z) = \frac{1}{2d}\mathbb{E}_z \tau_{\partial B} \le \frac{(r-1)^2 - |z|^2}{2d} \le \frac{rs}{d}\,.$$

Now let $\lambda := s/(16rL)$, which is at most $1/(2L)$ since $s \le r$. Lemma 5.4 applies with $N = n$, every $s_i = \mathbf{0}$, and $\varphi = g^{::}_B(\cdot, z)$, since $\varphi(z) \le L$ by the first fact, and the definition of f gives $\sum_{i=1}^n \varphi(s_i) - \sum_{y \in B} \varphi(y) = f(z)$. Since $\lambda^2 L r s/d = \lambda s^2/(16d)$, the second and third facts give

$$\mathbb{P}\{z \notin I_n\} \le \exp\Big\{-\lambda\Big(\frac{s^2}{8d} - C\Big) + \frac{\lambda s^2}{16d}\Big\}\,.$$

The exponent equals $\lambda C - s^3/(256d\,rL)$. Since $\lambda \le 1/16$ and L is a constant times $\ell(r)$, we conclude that

$$\mathbb{P}\{z \notin I_n\} \le \exp\left(C - c\frac{s^3}{r\ell(r)}\right)\,.$$

Step 3: We take a union bound over $\mathrm{B}^{::}_{r-\varpi}$. Since $\varpi \ge 2$, every $z \in \mathrm{B}^{::}_{r-\varpi}$ lies in B and satisfies $r - |z| > \varpi$. The ball $\mathrm{B}^{::}_{r-\varpi}$ has at most $(2r+1)^d \le (3r)^d$ sites, so Step 2 gives

$$\mathbb{P}\{\mathrm{B}^{::}_{r-\varpi} \not\subset I_n\} \le (3r)^d e^C \exp\left(-c\frac{\varpi^3}{r\ell(r)}\right)\,,$$

which proves the theorem. □

In particular, fix $\epsilon \in (0, 1/2]$ and take $\varpi := \lfloor \epsilon r \rfloor$, which is at least $\epsilon r/2$ for large r. The inclusion $I_n \subset A_n$ and Theorem 5.5, after the factor Cr^d is absorbed into the exponent by decreasing c, give

$$\mathbb{P}\left\{\mathrm{B}^{\ddot{}}_{(1-\epsilon)r} \not\subset A_n\right\} \le \exp\left(-c\frac{\epsilon^3 r^2}{\ell(r)}\right) \tag{5.3}$$

for all sufficiently large r. In dimensions $d \ge 3$, Exercise 5.3 gives a lower bound showing that the power r^2 in the exponent is best possible.

To obtain an outer bound, we must control the particles paused on the boundary of the ball.

5.3 OUTER BOUND

The particles paused in the proof of the inner bound start near the boundary of the occupied region. We show that if at most $b\varpi^d$ particles, where $b > 0$ depends only on d, enter a cubical shell of width comparable to ϖ, then the probability that one of them crosses the shell decays exponentially in ϖ.

Recall the hitting time (3.4) and, from the proof of Proposition 4.11, the cube $Q^{\ddot{}}(x, h) = \{y \in \mathbb{Z}^d : |y - x|_\infty \le h\}$ of side $2h + 1$ centered at $x \in \mathbb{Z}^d$. Next we bound where simple random walk X first hits a hyperplane.

Lemma 5.6 *Let $d \ge 2$ be an integer. There is a constant $c = c(d) > 0$ with the following property. Let h be a positive integer, let $y \in \mathbb{Z}^d$ satisfy $y_1 = h$, and let $\Pi := \{x \in \mathbb{Z}^d : x_1 = 0\}$. Then*

$$\mathbb{P}_y\{X_{\tau_\Pi} = z\} \ge ch^{1-d} \qquad \text{for every } z \in \Pi \cap Q^{\ddot{}}(y, h)\,.$$

Proof. The idea is to write the hitting probability as a difference of two Green function values by reflecting paths across Π (Figure 5.5), and to estimate it with Theorem 3.9. Fix $z \in \Pi \cap Q^{\ddot{}}(y, h)$. Translation by $-z$ maps Π to itself, so we may assume that $z = \mathbf{0}$, and then $y_1 = h$ and $|y_i| \le h$ for $2 \le i \le d$.

Step 1: We prove the identity

$$\mathbb{P}_y\{X_{\tau_\Pi} = \mathbf{0}\} = g^{\ddot{}}(y - \mathbf{e}_1) - g^{\ddot{}}(y + \mathbf{e}_1)\,.$$

Let $k \ge 0$. Reflecting across Π the part of a path after its first visit to Π is a bijection from the paths of length k from y to $\mathbf{e}_1$ that visit Π onto the paths of length k from y to $-\mathbf{e}_1$, all of which visit Π (Figure 5.5(a)). The remaining paths from y to $\mathbf{e}_1$, followed by the step to $\mathbf{0}$, are exactly the paths that first

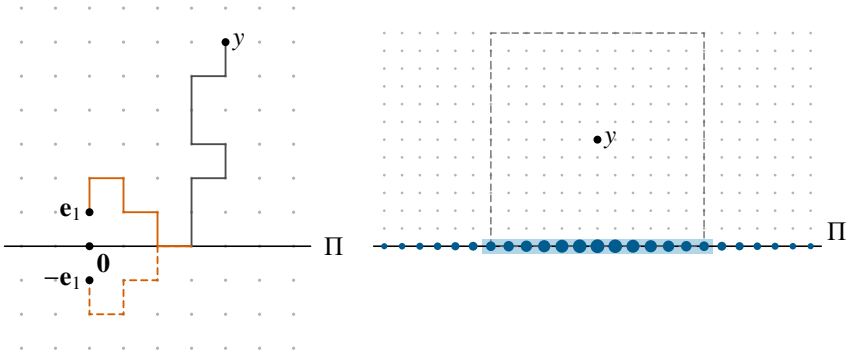


Figure 5.5: (a) Step 1 of the proof of Lemma 5.6: reflecting across Π the part after the first visit to Π (dashed) turns a path from y to $\mathbf{e}_1$ that visits Π into a path from y to $-\mathbf{e}_1$. (b) For $h = 6$ in $\mathbb{Z}^2$, the probability that the walk from y first hits Π at each of its sites (dot areas), computed exactly from the identity in Step 1; on the face of $Q^{::}(y, h)$ (dashed) in Π (shaded), every value is at least $0.15/h$.

hit Π at $\mathbf{0}$ at time $k+1$. Each path of length k has probability $(2d)^{-k}$, so

$$\mathbb{P}_y\{\tau_\Pi = k+1,\ X_{\tau_\Pi} = \mathbf{0}\} = \frac{1}{2d}\big(\mathbb{P}_y\{X_k = \mathbf{e}_1\} - \mathbb{P}_y\{X_k = -\mathbf{e}_1\}\big)\,.$$

By the symmetry $x \mapsto -x$ of the walk, $\mathbb{P}_y\{X_k = \pm\mathbf{e}_1\} = \mathbb{P}_{\mathbf{0}}\{X_k = y \mp \mathbf{e}_1\}$. Summing over $0 \le k \le t$ and recalling the truncated Green function $g_t^{::}$ of Section 3.3, we obtain

$$\mathbb{P}_y\{\tau_\Pi \le t+1,\ X_{\tau_\Pi} = \mathbf{0}\} = g_t^{::}(y - \mathbf{e}_1) - g_t^{::}(y + \mathbf{e}_1)\,.$$

As $t \to \infty$, the left side increases to $\mathbb{P}_y\{X_{\tau_\Pi} = \mathbf{0}\}$. If $d \ge 3$, then $g_t^{::} \to g^{::}$. If $d = 2$, then the right side equals $(g_t^{::}(\mathbf{0}) - g_t^{::}(y + \mathbf{e}_1)) - (g_t^{::}(\mathbf{0}) - g_t^{::}(y - \mathbf{e}_1))$, which tends to $a(y + \mathbf{e}_1) - a(y - \mathbf{e}_1)$ by the definition (3.12) of the potential kernel a, and $a = -g^{::}$ in dimension two. This proves the identity, which Figure 5.5(b) evaluates for $h = 6$.

Step 2: We bound the right side of this identity from below for large h. Recall the constants a_d and κ of Theorem 3.9. That theorem gives $g^{::}(x) = \varphi(|x|^2) + O(|x|^{-d})$ as $|x| \to \infty$, where $\varphi(u) := a_d u^{(2-d)/2}$ when $d \ge 3$ and $\varphi(u) := -\frac{1}{4\pi}\log u - \kappa$ when $d = 2$. In both cases $\varphi'(u) = -u^{-d/2}/(2d\omega_d)$, since $\omega_2 = \pi$. For $h \ge 2$, the squared norms of $y \pm \mathbf{e}_1$ lie between $(h-1)^2 \ge h^2/4$ and $(h+1)^2 + (d-1)h^2 \le (d+2)h^2$, and they differ by $4h$. Since $|\varphi'|$ is decreasing, the mean value theorem gives

$$\mathbb{P}_y\{X_{\tau_\Pi} = \mathbf{0}\} = \varphi(|y - \mathbf{e}_1|^2) - \varphi(|y + \mathbf{e}_1|^2) + O(h^{-d}) \ge \frac{2h^{1-d}}{d\omega_d(d+2)^{d/2}} - Ch^{-d}\,.$$

Hence there is $h_0 = h_0(d) \ge 2$ such that $\mathbb{P}_y\{X_{\tau_\Pi} = \mathbf{0}\} \ge (d+2)^{-d/2}h^{1-d}/(d\omega_d)$ for $h \ge h_0$.

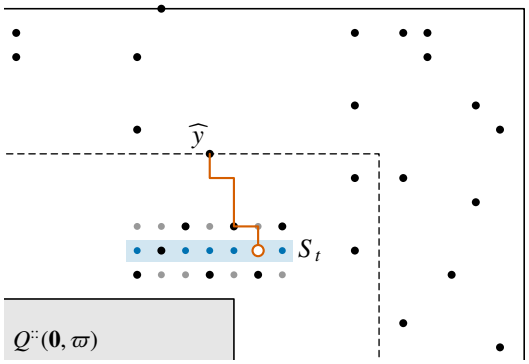


Figure 5.6: The proof of Lemma 5.7 near the entrance point $\widehat{y}$, with the sites of O in black. The walk crosses the levels of all the patches before it reaches $Q^{::}(\mathbf{0}, \varpi)$, and with probability bounded below it first meets the level of the patch S_t with fewest sites of O (blue) outside O.

Step 3: We treat the remaining values of h. If $h < h_0$, then a path of at most $(d-1)h$ steps parallel to Π leads from y to $h\mathbf{e}_1$, and h further steps toward Π first hit Π at $\mathbf{0}$. Hence $\mathbb{P}_y\{X_{\tau_\Pi} = \mathbf{0}\} \geq (2d)^{-dh_0} \geq (2d)^{-dh_0} h^{1-d}$, and $c := \min\{(d+2)^{-d/2}/(d\omega_d), (2d)^{-dh_0}\}$ proves the lemma. □

We apply the hyperplane estimate to a cubical shell: if it contains few occupied sites compared with its volume, then a walk that enters it reaches an unoccupied site before crossing it with probability bounded below.

For the rest of this section, each released particle follows an independent infinite simple random walk and settles at the first unoccupied site it reaches.

Lemma 5.7 *Let $d \geq 2$ be an integer. There is a constant $p = p(d) \in (0,1)$ with the following property. Let h and ϖ be positive integers with $h \geq 16$, and let $O \subset \mathbb{Z}^d$ satisfy $\#(O \cap Q^{::}(\mathbf{0}, \varpi + h)) \leq 2^{-2d-4} h^d$. Then, for every $x \in \mathbb{Z}^d \setminus Q^{::}(\mathbf{0}, \varpi + h)$,*

$$\mathbb{P}_x\{\tau_{Q^{::}(\mathbf{0},\varpi)} < \infty \text{ and } \tau_{Q^{::}(\mathbf{0},\varpi)} \leq \tau_{\mathbb{Z}^d \setminus O}\} \leq p\,.$$

Proof. The idea is to find, between the two cubes, a patch of a coordinate hyperplane with few sites of O, on which Lemma 5.6 gives a uniform probability of reaching $\mathbb{Z}^d \setminus O$ (Figure 5.6).

Let $w := \lfloor h/4 \rfloor$, so that $h/8 \leq w \leq h/4$. If the walk never enters $Q^{::}(\mathbf{0}, \varpi + 2w)$, or reaches $\mathbb{Z}^d \setminus O$ before or at its first entrance, then it does not visit $Q^{::}(\mathbf{0}, \varpi)$ by time $\tau_{\mathbb{Z}^d \setminus O}$. Otherwise let $\widehat{y}$ be its first entrance point. The permutations of the coordinates and the reflections in the coordinate hyperplanes preserve the cubes and the hypothesis on O, so we may assume that $\widehat{y}_1 = \varpi + 2w$. For $1 \leq t \leq w$, consider the patch

$$S_t := \{x \in \mathbb{Z}^d : x_1 = \varpi + t \text{ and } |x_i - \widehat{y}_i| \leq w \text{ for } 2 \leq i \leq d\}\,.$$

The patches lie in $Q^{::}(\mathbf{0}, \varpi + 3w) \subset Q^{::}(\mathbf{0}, \varpi + h)$, are pairwise disjoint, and each contains $(2w+1)^{d-1} \geq (h/4)^{d-1}$ sites. Since O has at most $2^{-2d-4}h^d$ sites in $Q^{::}(\mathbf{0}, \varpi + h)$ and $w \geq h/8$, there is a patch S_t with

$$\#(S_t \cap O) \leq \frac{2^{-2d-4}h^d}{w} \leq \frac{1}{2}\left(\frac{h}{4}\right)^{d-1},$$

so at least half of its sites lie outside O.

The hyperplane containing S_t lies at distance $2w - t$ from $\widehat{y}$, which is between w and h. After a translation and a reflection, Lemma 5.6 shows that the walk from $\widehat{y}$ first hits this hyperplane at each site of S_t with probability at least ch^{1-d}. The walk must hit this hyperplane before it visits $Q^{::}(\mathbf{0}, \varpi)$, and if it first hits the hyperplane at a site outside O, then it reaches $\mathbb{Z}^d \setminus O$ before visiting $Q^{::}(\mathbf{0}, \varpi)$. The strong Markov property at the first entrance time therefore proves the lemma with $p := 1 - c/(2 \cdot 4^{d-1})$. □

Applying Lemma 5.7 across nested cubical shells shows that the innermost cube is reached only with exponentially small probability.

Lemma 5.8 *Let $d \geq 2$ be an integer. There are positive constants b, c_0, c_1 depending only on d with the following property. Let ϖ and k be positive integers satisfying $k \leq b\varpi^d$. Let $O \subset \mathbb{Z}^d \setminus Q^{::}(\mathbf{0}, 3\varpi)$, and place one settled particle at each site of O. Release k additional particles one at a time from specified sites in $\mathbb{Z}^d \setminus Q^{::}(\mathbf{0}, 3\varpi)$, letting them perform independent simple random walks until they first reach unoccupied sites. A particle that never reaches an unoccupied site adds no occupied site. Then the probability that at least one additional particle visits $Q^{::}(\mathbf{0}, \varpi)$ is at most $c_0 e^{-c_1\varpi}$.*

Proof. The idea is to attach to each particle count j a cube of radius 2ϖ plus a constant times $j^{1/d}$. By Lemma 5.7, it is exponentially unlikely in j that at most j particles visit the cube attached to j while many visit the cube attached to a fixed fraction of j. A union bound over j then proves the lemma.

Let $p \in (0,1)$ be the constant in Lemma 5.7, and let $q := (1+p)/2$ and $b := ((1-q^{1/d})/18)^d$. For every integer $0 \leq j \leq k$, let $R_j := 2\varpi + \lfloor (j/b)^{1/d} \rfloor$, and let V_j be the number of additional particles that visit $Q^{::}(\mathbf{0}, R_j)$. Since $k \leq b\varpi^d$, we have $2\varpi \leq R_j \leq 3\varpi$. The counts V_j are nondecreasing in j, and $V_k \leq k$.

Step 1: Let $c_1 := (1-q)(1-\log 2) > 0$. We show that, for all integers j and i with $1 \leq j \leq k$ and $0 \leq i \leq qj$,

$$\mathbb{P}\{V_j \leq j \text{ and } V_i > qj\} \leq e^{-c_1 j}. \tag{5.4}$$

Since each floor costs at most one and $i \leq qj$,

$$R_j - R_i - 1 \geq b^{-1/d}(1-q^{1/d})j^{1/d} - 2 = 18j^{1/d} - 2 \geq 16j^{1/d},$$

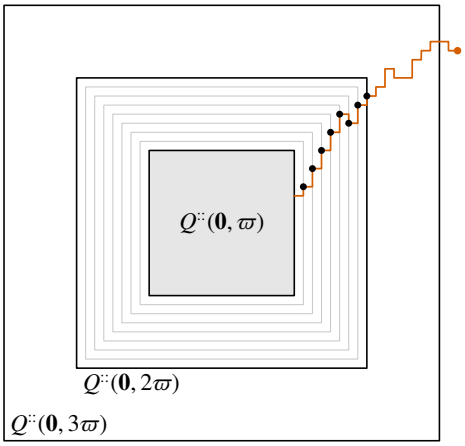


Figure 5.7: Step 2 of the proof of Lemma 5.8. A particle that reaches $Q^{\cdot\cdot}(\mathbf{0}, \varpi)$ from outside $Q^{\cdot\cdot}(\mathbf{0}, 3\varpi)$ passes over an occupied site at each level between 2ϖ and ϖ (dots), and each of these was settled by an earlier particle that visited $Q^{\cdot\cdot}(\mathbf{0}, 2\varpi)$.

so Lemma 5.7 applies with R_i in place of ϖ and $h := R_j - R_i - 1$, because $h \geq 16$ and $2^{-2d-4}h^d \geq 2^{2d-4}\, j \geq j$.

Expose the particles in order, retaining the first j that enter $Q^{\cdot\cdot}(\mathbf{0}, R_j)$. This cube is initially empty, so only earlier retained particles can have settled in it. Hence fewer than j sites of the cube are occupied when a retained particle enters, and its entrance point lies outside $Q^{\cdot\cdot}(\mathbf{0}, R_j - 1) = Q^{\cdot\cdot}(\mathbf{0}, R_i + h)$. By the strong Markov property at the entrance time, Lemma 5.7 bounds by p the conditional probability that the particle visits $Q^{\cdot\cdot}(\mathbf{0}, R_i)$ by its settling time (infinite if it never settles), given everything that happened before its entrance.

For $0 \leq t \leq j$, let M_t be the number of the first t retained particles that visit $Q^{\cdot\cdot}(\mathbf{0}, R_i)$, where a retained particle that does not exist counts as not visiting. Given everything before the entrance of the tth retained particle, M_{t-1} is determined and $M_t - M_{t-1}$ equals 1 with probability at most p, so $\mathbb{E}\, 2^{M_t} \leq (1+p)\mathbb{E}\, 2^{M_{t-1}}$, and hence $\mathbb{E}\, 2^{M_j} \leq (1+p)^j$. On $\{V_j \leq j\}$, every particle that visits $Q^{\cdot\cdot}(\mathbf{0}, R_i)$ first enters $Q^{\cdot\cdot}(\mathbf{0}, R_j)$ and is retained, so $M_j = V_i$. Hence $2^{M_j} > 2^{qj}$ on $\{V_j \leq j \text{ and } V_i > qj\}$, and Markov's inequality applied to 2^{M_j} gives

$$\mathbb{P}\{V_j \leq j \text{ and } V_i > qj\} \leq 2^{-qj}(1+p)^j = (q2^{1-q})^j \leq e^{-c_1 j}\,,$$

where we used $1 + p = 2q$ and $\log q \leq q - 1$.

Step 2: We show that $V_0 > \varpi$ if an additional particle visits $Q^{\cdot\cdot}(\mathbf{0}, \varpi)$ (Figure 5.7). Such a particle starts outside $Q^{\cdot\cdot}(\mathbf{0}, 3\varpi)$, and each step changes $|x|_\infty$ by at most one, so before its first visit to $Q^{\cdot\cdot}(\mathbf{0}, \varpi)$ it visits a site at each of the sup-norm distances $\varpi + 1, \ldots, 2\varpi$ from the origin. These ϖ

sites lie in $Q^{::}(\mathbf{0}, 2\varpi)$, so they are initially unoccupied, and they are occupied when visited, since the particle has not settled yet. Hence ϖ distinct earlier particles settled at them. These particles and the particle itself visit $Q^{::}(\mathbf{0}, 2\varpi) = Q^{::}(\mathbf{0}, R_0)$, so $V_0 \geq \varpi + 1$.

Step 3: We conclude with a union bound. Since $V_k \leq k$, there is a least integer j with $0 \leq j \leq k$ and $V_j \leq j$. If an additional particle visits $Q^{::}(\mathbf{0}, \varpi)$, then Step 2 and the monotonicity of V_j give $j \geq V_j \geq V_0 > \varpi$, so $\lfloor qj \rfloor < j$, and minimality gives $V_{\lfloor qj \rfloor} > \lfloor qj \rfloor$, that is, $V_{\lfloor qj \rfloor} > qj$. Thus this event is contained in the union over $\varpi < j \leq k$ of the events in (5.4) with $i = \lfloor qj \rfloor$, and its probability is at most

$$\sum_{j > \varpi} e^{-c_1 j} = \frac{e^{-c_1 \varpi}}{e^{c_1} - 1} .$$

This proves the lemma with $c_0 := 1/(e^{c_1} - 1)$. □

We apply Lemma 5.8 to the particles paused when they first leave the inner ball. Recall the factor $\ell(r)$ of (5.2).

Theorem 5.9 *Let $d \geq 2$ be an integer. There are positive constants c and C depending only on d with the following property. Let n be a positive integer, let A_n be the IDLA cluster formed by n particles started at $\mathbf{0}$, and let $r := (n/\omega_d)^{1/d}$. Let ϖ and m be positive integers satisfying*

$$\varpi^3 \geq C r \ell(r) \log(e + r) , \qquad \varpi \leq r/2 \qquad \text{and} \qquad m^d \geq C \varpi r^{d-1} .$$

Then

$$\mathbb{P}\{(\mathrm{B}^{::}_{r-\varpi} \subset A_n \subset \mathrm{B}^{::}_{r+m})^c\} \leq \exp\left(-c \frac{\varpi^3}{r\ell(r)}\right) + C(r+m)^{d-1} e^{-cm} .$$

Proof. The idea is to bound the probability that $\mathrm{B}^{::}_{r-\varpi} \not\subset I_n$ by Theorem 5.5, and otherwise to apply Lemma 5.8 to the paused particles around each site of $\partial \mathrm{B}^{::}_{r+m}$ (Figure 5.8).

Step 1: We bound the probability that the inner containment fails. Let $B := \mathrm{B}^{::}_{r-2}$, and let I_n be the stopped aggregate of Theorem 5.5. We take the constant C in the hypotheses large enough that they force $\varpi \geq 2$, hence $r \geq 2\varpi \geq 4$, and $m \geq 12\sqrt{d}$. Theorem 5.5 bounds the probability that $\mathrm{B}^{::}_{r-\varpi} \not\subset I_n$ by a constant times $r^d \exp(-c\varpi^3/(r\ell(r)))$, where c is the constant of that theorem. If the constant in the hypothesis $\varpi^3 \geq C r \ell(r) \log(e + r)$ is large, then the factor in front of the exponential is at most $\exp(c\varpi^3/(2r\ell(r)))$, so this probability is at most $\exp(-c\varpi^3/(2r\ell(r)))$.

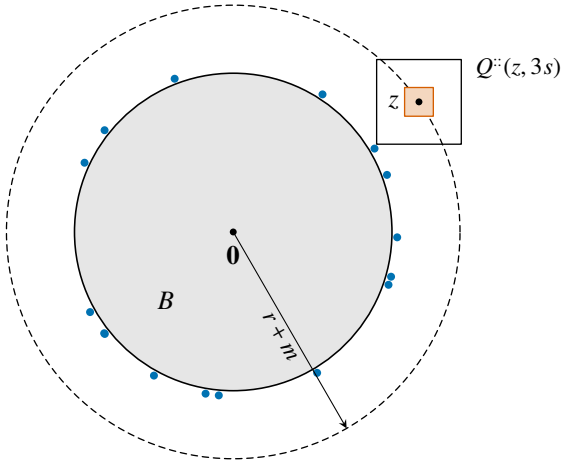


Figure 5.8: Step 2 of the proof of Theorem 5.9. The paused particles (dots) start on the boundary of $B = \mathrm{B}^{::}_{r-2}$, outside the cube $Q^{::}(z,3s)$ around a site z on the outer boundary of $\mathrm{B}^{::}_{r+m}$, and Lemma 5.8 shows that they rarely reach $Q^{::}(z,s)$.

Step 2: We prove the outer containment when $\mathrm{B}^{::}_{r-\varpi} \subset I_n$. In the coupling of Step 1 of the proof of Theorem 5.5, restarting the paused particles produces A_n. When $\mathrm{B}^{::}_{r-\varpi} \subset I_n$, the number of paused particles (Figure 5.4(b)) satisfies

$$n - \#I_n \le n - \#\mathrm{B}^{::}_{r-\varpi} \le C\varpi r^{d-1}\,,$$

as we see by comparing the unit cubes centered at lattice points with the Euclidean shell between radii $r - \varpi - \sqrt{d}$ and r.

Fix $z \in \partial\mathrm{B}^{::}_{r+m}$, and let

$$s := \left\lfloor \frac{m-1}{3\sqrt{d}} \right\rfloor .$$

We have $s \ge m/(6\sqrt{d})$ and $Q^{::}(z,3s) \subset \mathrm{B}^{::}(z,m)$. The discrete cube $Q^{::}(z,3s)$ is disjoint from B and from the paused positions: every paused position has norm less than $r-1$, while $|z| \ge r+m$ and every point of $Q^{::}(z,3s)$ lies within distance $m-1$ of z. Conditional on the stopped aggregate and the paused positions, the strong Markov property shows that the restarted particles are independent walks from the paused positions. Since $s^d \ge m^d/(6\sqrt{d})^d$, increasing the constant in the hypothesis on m gives $n - \#I_n \le bs^d$, where b is the constant of Lemma 5.8. That lemma, translated to the center z and applied with the initially occupied set I_n, shows that the conditional probability that $z \in A_n$ is at most $c_0 e^{-c_1 s} \le c_0 e^{-c_1 m/(6\sqrt{d})}$, where c_0 and c_1 are its constants.

The cluster A_n is connected and contains the origin. Hence, if $A_n \not\subset \mathrm{B}^{::}_{r+m}$, then it contains a site of $\partial\mathrm{B}^{::}_{r+m}$, a set of at most $C(r+m)^{d-1}$ sites. A union

bound, followed by averaging over the stopped construction, gives

$$\mathbb{P}\{A_n \not\subset \mathrm{B}^{::}_{r+m} \text{ and } \mathrm{B}^{::}_{r-\varpi} \subset I_n\} \le C c_0 (r+m)^{d-1} e^{-c_1 m/(6\sqrt{d})}.$$

Since $I_n \subset A_n$, combining this with Step 1 proves the theorem, with c the smaller of $c_1/(6\sqrt{d})$ and half the constant c of Theorem 5.5. □

Theorem 5.9 and the Borel–Cantelli lemma give almost sure inclusions with explicit widths.

Corollary 5.10 *Let $d \ge 2$ be an integer. For every positive integer n, let A_n be the IDLA cluster formed by n particles started at $\mathbf{0}$, and let $r_n := (n/\omega_d)^{1/d}$. There is a constant $C = C(d) < \infty$ such that, almost surely, for all sufficiently large n,*

$$\min_{x \notin A_n} |x| \ge r_n - C(r_n \ell(r_n) \log(e + r_n))^{1/3}$$

and

$$\max_{x \in A_n} |x| < r_n + C r_n^{1-2/(3d)} (\log(e + r_n))^{2/(3d)}.$$

Proof. Let $K \ge 1$ be a constant, depending only on d, to be chosen large, and apply Theorem 5.9 with

$$\varpi = \lceil K(r_n \ell(r_n) \log(e + r_n))^{1/3} \rceil \qquad \text{and} \qquad m = \lceil (K \varpi r_n^{d-1})^{1/d} \rceil.$$

If K is at least the constant C of the theorem, then for large n these choices satisfy its hypotheses. Since $\varpi^3/(r_n \ell(r_n)) \ge K^3 \log(e + r_n)$, the first term in the theorem is at most $(e + r_n)^{-cK^3}$, where c is the constant of the theorem, and this is at most n^{-2} once K is large. Since $\varpi \ge 1$, we have $m \ge r_n^{(d-1)/d}$, so the second term decays faster than every power of n. The Borel–Cantelli lemma therefore shows that, almost surely, $\mathrm{B}^{::}_{r_n - \varpi} \subset A_n \subset \mathrm{B}^{::}_{r_n + m}$ for all large n, that is, $\min_{x \notin A_n} |x| \ge r_n - \varpi$ and $\max_{x \in A_n} |x| < r_n + m$.

It remains to compare ϖ and m with the widths in the statement. We have $\varpi \le (K+1)(r_n \ell(r_n) \log(e + r_n))^{1/3}$, and since $\ell(r_n) \le \log(e + r_n)$, it follows that

$$m \le ((K(K+1))^{1/d} + 1) r_n^{1-2/(3d)} (\log(e + r_n))^{2/(3d)}.$$

Taking $C := (K(K+1))^{1/d} + K + 1$ proves the inclusions. □

Both widths in Corollary 5.10 are small compared with r_n, so A_n/r_n converges almost surely to the closed unit ball in Hausdorff distance. The true fluctuations are much smaller still: in Figure 5.1 the cluster lies within a few lattice spacings of the circle of radius r, Figure 5.9 shows the same for the clusters $A_{kn/8}$ that it passes through, and the Notes describe the logarithmic bounds that hold in general.

Figure 5.9: The cluster of Figure 5.1, with the sites of particles $(k-1)n/8+1$ to $kn/8$ shaded alike, alternately dark and light, for $k = 1, \dots, 8$. The edges of the bands are the clusters $A_{kn/8}$, and each lies between the circles of radii $r_k - 5$ and $r_k + 5$, where $\pi r_k^2 = kn/8$.

EXERCISES

Exercise 5.1. This exercise bounds the discrepancy (5.1) of the independent uniform stacks in Section 5.1. Let $d \geq 2$ be an integer, and for every $x \in \mathbb{Z}^d$ and $j \geq 1$, let $\rho_j(x)$ be independent and uniform among the $2d$ directed edges leaving x.

(a) Show that for every pair of real numbers $a, b > 0$ there is a constant $C = C(a, b, d)$ such that, for every real number $R \geq 2$,

$$\mathbb{P}\left(\bigcup_{x \in \mathrm{B}_R^{::}} \bigcup_{y \sim x} \bigcup_{m=1}^{\lfloor R^a \rfloor} \left\{ |\mathcal{D}_\rho(x, y, m)| \geq C \sqrt{m \log R} \right\} \right) \leq R^{-b} .$$

(b) Deduce from part (a) that there is a constant $C = C(d)$ such that, almost surely, for all sufficiently large integers m, every $x \in \mathrm{B}_m^{::}$, every neighbor $y \sim x$ and every integer $1 \leq j \leq m$ satisfy $|\mathcal{D}_\rho(x, y, j)| \leq C \sqrt{m \log(e + m)}$.

Exercise 5.2. This exercise complements Section 5.2. Let $d \geq 1$ be an integer, and let $B \subset \mathbb{Z}^d$ be finite. For $y, z \in B$, prove that

$$\mathbb{P}_y\{\tau_z < \tau_{\partial B}\} = \frac{g_B^{::}(y, z)}{g_B^{::}(z, z)} .$$

Exercise 3.27 is the case $z = \mathbf{0}$ for a ball.

Exercise 5.3. Let $d \geq 3$ be an integer, let $r \geq 1$ be a real number, and let $n := \lfloor \omega_d r^d \rfloor$. Let A_n be the IDLA cluster formed by n particles started at $\mathbf{0}$. Show that there is a constant $c = c(d) > 0$ such that, for all sufficiently large r and every $z \in \mathbb{Z}^d$ satisfying $|z - (r/2)\mathbf{e}_1| \leq 1$,

$$\mathbb{P}\{z \notin A_n\} \geq \exp(-cr^2)\,.$$

By taking r to be an even integer and $z = (r/2 - 1)\mathbf{e}_1$, conclude that the power r^2 in (5.3) cannot be increased.

Exercise 5.4. This exercise complements the lower bound in Lemma 5.6. Let $d \geq 2$ be an integer, let h be a positive integer, let $y \in \mathbb{Z}^d$ satisfy $y_1 = h$, and let $\Pi := \{x \in \mathbb{Z}^d : x_1 = 0\}$. Prove the upper bound

$$\mathbb{P}_y\{X_{\tau_\Pi} = z\} \leq Ch^{1-d} \qquad \text{for } z \in \Pi \cap Q^{::}(y, h)\,,$$

where $C = C(d)$.

Exercise 5.5. This exercise gives a coupling version of the successive conditioning in Step 1 of the proof of Lemma 5.8. Let n be a positive integer, and let $E_1, \dots, E_n$ be events adapted to a filtration $(\mathcal{F}_j)_{j=0}^n$. Let $p \in [0, 1]$, and suppose $\mathbb{P}(E_j \mid \mathcal{F}_{j-1}) \leq p$ for $1 \leq j \leq n$. Construct, on a possibly enlarged probability space, independent Bernoulli(p) random variables $Y_1, \dots, Y_n$ such that $\mathbf{1}\{E_j\} \leq Y_j$ almost surely. Conclude that $\sum_{j=1}^n \mathbf{1}\{E_j\}$ is stochastically dominated by a Binom(n, p) random variable.

Exercise 5.6. Verify the geometric estimates used in the proof of Theorem 5.9. Let $d \geq 2$ be an integer, let n be a positive integer, and let $r := (n/\omega_d)^{1/d}$. Let $\eta \geq 2$ be a real number, and let ϖ be a positive integer satisfying $\eta \leq \varpi \leq r/2$. Suppose $I \subset \mathrm{B}^{::}_{r-\eta}$ contains $\mathrm{B}^{::}_{r-\varpi}$. Prove that $n - \#I \leq C\varpi r^{d-1}$ for a constant $C = C(d) < \infty$.

Let m be a positive integer satisfying $m \geq 12\sqrt{d}$. Let $z \in \partial \mathrm{B}^{::}_{r+m}$, and let $s := \lfloor (m-1)/(3\sqrt{d}) \rfloor$. Verify that $s \geq m/(6\sqrt{d})$ and that $Q^{::}(z, 3s)$ is disjoint from $\mathrm{B}^{::}_{r-\eta} \cup \partial \mathrm{B}^{::}_{r-\eta}$. For every fixed $b > 0$, show that there is a constant $C = C(b, d)$ such that $n - \#I \leq bs^d$ whenever $m^d \geq C\varpi r^{d-1}$.

NOTES

The internal DLA growth model appears on pp. 96–97 of Diaconis and Fulton (1991), and the instruction-stack game appears on pp. 97–98. Their Proposition 4.1 on p. 109 proves termination under an escape condition on the stacks, and that the terminal configuration and the number of moves at each site do not depend on the choice of legal moves; Corollary 5.2 gives the second statement in the form used in this book. The model of Meakin

and Deutch (1986, p. 2320) is a related diffusion-limited erosion process with different dynamics. The almost-sure convergence of A_n/r to $\overline{\mathrm{B}}(\mathbf{0}, 1)$ is Theorem 1 of Lawler, Bramson, and Griffeath (1992, p. 2119), where $r = (n/\omega_d)^{1/d}$. The coloring of Figure 5.1 by early and late sites follows Figure 4 of Levine and Peres (2017).

Theorem 1 of Lawler (1995, p. 72) shows that in every dimension $d \geq 2$, almost surely, for all large integers r the cluster A_n with $n = \lfloor \omega_d r^d \rfloor$ contains $\mathrm{B}^{::}(\mathbf{0}, r - r^{1/3}(\log r)^2)$ and is contained in $\mathrm{B}^{::}(\mathbf{0}, r + r^{1/3}(\log r)^4)$. Its outer bound is much sharper than that of Theorem 5.9. The proof of Theorem 5.9 combines Theorem 5.5 with Lemmas 5.10–5.12 of Levine and Peres (2010), which give the cubical-shell estimates in Lemmas 5.6–5.8.

Lemma 2 of Jerison, Levine, and Sheffield (2012) rules out thin tentacles in $\mathbb{Z}^2$. It is a variant of Lemma 5.12 of Levine and Peres (2010), on which Lemma 5.8 is based. For every real $m > 0$ and every site z with $|z| \geq m$, the probability that z belongs to A_n while at most a small absolute constant times m^2 sites at distance less than m from z do is exponentially small in m, uniformly in n. Lemma A in their Appendix A.1 improves the exponent to order $m^2/\log m$ in $\mathbb{Z}^2$. It also proves the analogous bound in $\mathbb{Z}^d$ for $d \geq 3$, with m^d in place of m^2 in the count and an exponent of order m^2. Almost surely, for all sufficiently large n, the inner and outer errors of A_n relative to $\mathrm{B}_r^{::}$ are at most $C \log r$ when $d = 2$, by Theorem 1 of Jerison, Levine, and Sheffield (2012) and the Borel–Cantelli lemma, and at most $C\sqrt{\log r}$ when $d \geq 3$, by Theorem 1 and Corollary 2 of Jerison, Levine, and Sheffield (2013), where C depends only on d. Asselah and Gaudillière (2013a, Theorem 1.2) obtained the inner bound $C \log r$ and the outer bound $C(\log r)^2$ in every dimension $d \geq 2$, and Proposition 1.1 and Remark 1.2 of Asselah and Gaudillière (2013b) give, by a different method, the bound $C \log r$ on both errors when $d = 2$ and the bound $C\sqrt{\log r}$ on both errors when $d \geq 3$. Theorem 1.4 of Asselah and Gaudillière (2014) shows that when $d \geq 3$ the inner error exceeds a constant times $\sqrt{\log r}$ infinitely often, almost surely, so this order is sharp.

Equations (2)–(4) of Friedrich and Levine (2013) give the instruction-stack mass-balance identity. The identity used in the proof of Proposition 13 of Holroyd and Propp (2010) gives the corresponding formula for one rotor walk. The obstacle problem for aggregation with general sources is developed in Theorems 3.3 and 3.9 of Levine and Peres (2010).

Chapter Six

Rotor-Router Aggregation in $\mathbb{Z}^d$: Single Source

Rotor-router aggregation is the deterministic counterpart of internal DLA. A **rotor configuration** assigns to every site $x \in \mathbb{Z}^d$ a directed edge from x to one of its $2d$ neighbors. A **rotor mechanism** specifies, at every site, a cyclic order of the $2d$ edges that leave it. In a **rotor walk**, the walker at a site first advances the rotor there to the next edge in this order and then follows that edge. Release particles one at a time from the origin, and let each perform rotor walk until it first reaches an unoccupied site, where it stays, as for the first few particles in Figure 1.7. The rotors are not reset between particles, so every site sends the particles of the whole process to its neighbors in strict rotation. Let A_n^{R} be the set of occupied sites after n particles have settled. Each particle settles after finitely many steps. Indeed, suppose that the walk of a particle stayed forever in the finite set of sites occupied when it was released. Then a site would be visited infinitely often, its rotor would send the walker to every neighbor infinitely often, and since $\mathbb{Z}^d$ is connected every site would be visited infinitely often, contradicting the finiteness of the occupied set.

Simulations in the plane produce nearly circular aggregates, and Figure 6.1 shows one with a million particles. The main result of this chapter, Theorem 6.6 in Section 6.3, proves that the aggregate is close to a ball in every dimension $d \geq 2$. Let ω_d be the volume of the unit ball in $\mathbb{R}^d$. For large $r = (n/\omega_d)^{1/d}$, the aggregate A_n^{R} contains the ball of radius $r - C \log r$ and is contained in the ball of radius $r + C(\log r)^2$, whatever the initial rotors and the rotor mechanisms, so the rescaled aggregates converge to the unit ball. The boundary in Figure 6.1 lies much closer to the circle than these bounds guarantee, within one lattice spacing of it. Internal DLA, the random model of Chapter 5, has the same limit shape (Figure 1.9 compares the two), and Theorem 5.9 proves bounds of the same kind for it that hold with high probability.

The proof compares rotor walk with random walk, in two parts. First, a summation identity (Proposition 6.1), tested against the Green function of a ball, bounds the number of moves made at each site from below by a function that grows quadratically with the distance to the sphere of radius r (the same function controls the inner bound for internal DLA in Chapter 5), up to an error controlled by the stack discrepancies. The error grows only linearly, up to a logarithm, and this shows that every site deeper than $C \log r$ inside the

sphere is occupied. This inner bound, Theorem 6.5 in Section 6.2, holds for every family of arrow stacks with bounded discrepancy. Second, we freeze the particles on a sphere and enlarge the sphere one unit at a time. The same summation identity, tested against the Green function of the occupied set with its pole at a frozen pile, shows that each pile of frozen particles has at most $C \log r$ particles, so the particles that leave the ball of radius r cannot travel far in a thin tentacle.

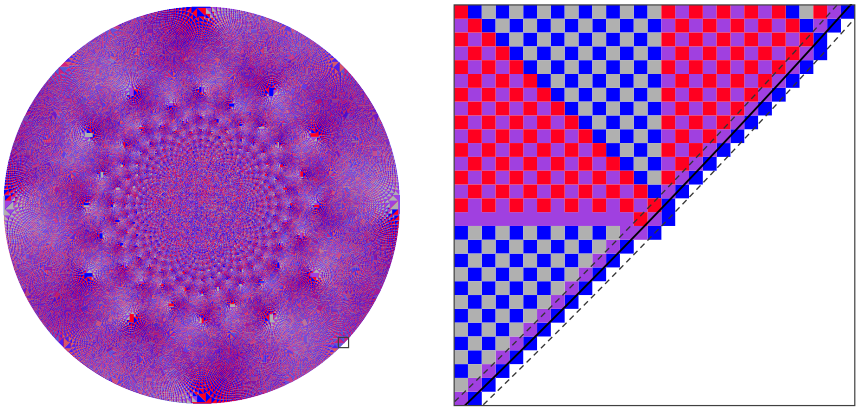


Figure 6.1: The rotor-router aggregate of 10^6 particles started at the origin of $\mathbb{Z}^2$ with the clockwise mechanism and all rotors initially North (left), and the sites in the marked square (right), colored by the final direction of each rotor as in Figure 1.1 (North blue, East purple, South red, West gray). Every site within distance 563.5 of the origin is occupied and none lies beyond 565.1 (dashed circles), while $r = (10^6/\pi)^{1/2} \approx 564.2$ (solid).

6.1 BOUNDED STACK DISCREPANCY

Both halves of the proof compare rotor walk with random walk, and the comparison is simplest when the arrows that route the particles at every site are written down in advance. We use the instruction stacks and discrepancy from Section 5.1. The stack at a site x is a sequence of arrows $\rho_1(x), \rho_2(x), \ldots$, each pointing from x to one of its neighbors, as in the left half of Figure 6.2, and the discrepancy

$$\mathcal{D}_\rho(x, y, m) = \#\{1 \le j \le m : \rho_j(x) = (x, y)\} - \frac{m}{2d}$$

is the number of the first m arrows at x that point to y, minus the expected number for a random walk. A stack move sends one particle along the next unused arrow, the top one in Figure 6.2, so the jth particle to leave x follows $\rho_j(x)$. For aggregation, a move is legal when its site contains at least two particles, and a finite legal sequence is complete when every site contains at most one particle. The odometer records the number of moves at each site.

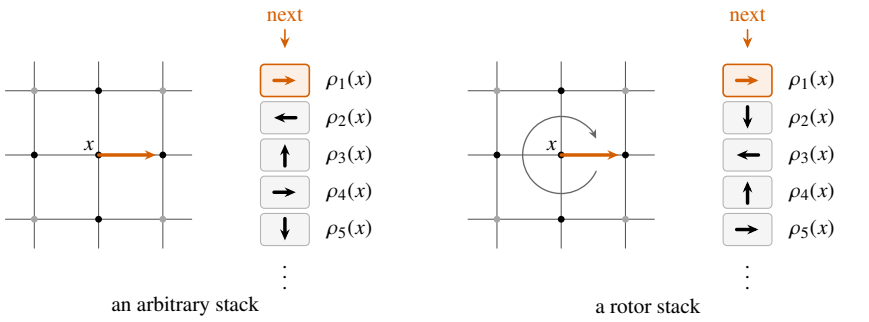


Figure 6.2: Left: the stack of arrows at a site x of $\mathbb{Z}^2$, whose top arrow (highlighted) is the one that the next particle to leave x follows. Right: a rotor stack, whose arrows run clockwise through the four neighbors, so that each block of four consecutive arrows uses every neighbor once.

Rotor-router aggregation is the special case in which each stack is periodic and cycles once through the $2d$ neighbors, as in the right half of Figure 6.2. For a rotor stack at x and an integer $m \geq 0$, write $m = 2dq + k$ with integers $q \geq 0$ and $0 \leq k < 2d$. Then k of the neighbors of x have received $q+1$ of the first m particles to leave x and the others q. Hence $\mathcal{D}_\rho(x, y, m)$ equals $1 - \frac{k}{2d}$ for those k neighbors and $-\frac{k}{2d}$ for the others, and since $0 \leq k \leq 2d-1$, every rotor stack satisfies

$$|\mathcal{D}_\rho(x, y, m)| \leq 1 - \frac{1}{2d} \qquad \text{for every edge } (x, y) \text{ and every integer } m \geq 0\,. \tag{6.1}$$

Exercise 6.1 shows that no stack has a smaller bound, and that for a stack of independent uniform arrows the discrepancy has standard deviation of order $\sqrt{m}$.

By Lemma 5.1, every finite legal sequence makes at most as many moves at each site as a complete one, and by Corollary 5.2, all complete legal sequences have the same odometer and the same final configuration.

We next express the change in total weight in terms of the odometer and the stack discrepancy.

Proposition 6.1 *Let $d \geq 1$ be an integer, and let $\sigma_0, \xi : \mathbb{Z}^d \to \mathbb{N}$ have finite support. Suppose that finitely many stack moves with fixed instruction stacks ρ convert σ_0 into ξ, and let v be their odometer. Then every function $f : \mathbb{Z}^d \to \mathbb{R}$ satisfies*

$$\sum_x f(x)(\xi(x) - \sigma_0(x)) = \frac{1}{2d} \sum_{x : v(x) > 0} v(x) \Delta^{::} f(x) + \sum_{x : v(x) > 0} \sum_{y \sim x} \mathcal{D}_\rho(x, y, v(x))(f(y) - f(x))\,.$$

Proof. Give a particle at z weight $f(z)$, so that a particle sent from x to y changes the total weight by $f(y) - f(x)$. The $v(x)$ moves at x use the first $v(x)$ arrows at x, and by the definition of the discrepancy they send $v(x)/(2d) + \mathcal{D}_\rho(x, y, v(x))$ particles to each neighbor y. Hence the moves at x change the total weight by

$$\sum_{y \sim x} \left(\frac{v(x)}{2d} + \mathcal{D}_\rho(x, y, v(x)) \right) (f(y) - f(x)) .$$

Summing over the sites x with $v(x) > 0$ and using $\sum_{y \sim x}(f(y) - f(x)) = \Delta^{::} f(x)$ proves the proposition. □

The first term on the right side of Proposition 6.1 is the change in weight that the divisible sandpile would produce if each site x emitted total mass $v(x)$, that is, with odometer $v/(2d)$. The second term is the error caused by routing whole particles, and it is small when the discrepancies are bounded and f varies slowly. The next section uses a killed Green function in Proposition 6.1 to bound the stack odometer from below; Exercise 6.3 compares the stack odometer with the expected local time of random walk.

6.2 INNER BOUND

In this section we prove that the aggregate fills the ball of radius $r - C b_\rho \log n$ whenever every stack has discrepancy at most b_ρ (Theorem 6.5). We test the stack moves against the killed Green function of the ball $\mathrm{B}^{::}(\mathbf{0}, r - 2)$. The main term is then a function whose quadratic growth was proved in Lemma 5.3 of Chapter 5 for the inner bound of internal DLA, while the stack discrepancy produces an error controlled by the gradient of the killed Green function.

Fix integers $d \geq 2$ and $n \geq 2$, and a family ρ of instruction stacks on $\mathbb{Z}^d$. Let A_n^{R} and v_n be the aggregate and odometer obtained by complete stack aggregation from $n\delta_{\mathbf{0}}$ using ρ. Let ω_d be the volume of the unit ball in $\mathbb{R}^d$, and let $r := (n/\omega_d)^{1/d}$.

For the comparison, suppose that $b_\rho \geq 1$ is real and that

$$|\mathcal{D}_\rho(x, y, m)| \leq b_\rho \qquad \text{for every edge } (x, y) \text{ and every integer } m \geq 0 .$$

Bounded discrepancy ensures that a particle routed along the successive arrows of the stacks leaves every finite set. If a routed particle remained forever in a finite set, then its infinitely visited sites would form a nonempty finite set closed under taking neighbors. No such set exists in $\mathbb{Z}^d$. Indeed, if a neighbor-closed set contains x, then it also contains $x + k\mathbf{e}_1$ for every integer $k \geq 1$.

Testing Proposition 6.1 against a killed Green function produces an error term bounded by b_ρ times the sum of the absolute differences of the Green function along edges. We first bound the differences across a coordinate plane by reflection, and then sum over planes using the probability that the walk reaches each one. Throughout the rest of the chapter, $(X_t)_{t\ge 0}$ denotes simple random walk on $\mathbb{Z}^d$, and for a set $A \subset \mathbb{Z}^d$ we write $\tau_{\partial A}$ for the first time the walk hits the outer boundary ∂A.

The next lemma bounds the sum of the absolute differences along edges starting in a ball by a constant times its radius, for the Green function $g^{::}_A$ of a walk killed on leaving an arbitrary finite set A, defined in (3.18).

Lemma 6.2 *For each integer $d \ge 2$, there is a constant $C = C(d) < \infty$ with the following property. Let $A \subset \mathbb{Z}^d$ be finite, let $x \in A$, let $p \in \mathbb{Z}^d$, and let $s \ge 1$ be real. Then*

$$\sum_{y \in \mathrm{B}^{::}(p,s)} \sum_{z \sim y} |g^{::}_A(x,y) - g^{::}_A(x,z)| \le Cs\,. \tag{6.2}$$

Proof. The idea is to reflect the Green function across a plane halfway between two lattice layers. Summing the Laplacian of the difference against its sign on one side of the plane bounds the total gradient across the plane by the total absolute value of the Laplacian of the Green function, which equals two. A ball of radius s meets at most $2s+1$ parallel planes.

Step 1: We compute the sum of the absolute values of the Laplacian. Let $f(v) := g^{::}_A(x,v)$ for $v \in \mathbb{Z}^d$, and let $\nu := -\Delta^{::} f$ on $\mathbb{Z}^d$. By (3.19), $\nu = \delta_x$ on A. Outside A we have $f = 0$, and $f \ge 0$ everywhere, so $\nu(v) = -\sum_{w\sim v} f(w) \le 0$ for $v \notin A$. Since f has finite support, each edge contributes two terms of opposite signs to $\sum_v \Delta^{::} f(v)$, so $\sum_v \nu(v) = 0$. Hence

$$\sum_{v \in \mathbb{Z}^d} |\nu(v)| = 2\,.$$

Fix $1 \le i \le d$ and an integer k. We show that

$$\sum_{v \in \mathbb{Z}^d :\, v_i = k} |f(v) - f(v + \mathbf{e}_i)| \le 1\,. \tag{6.3}$$

Let $\mathcal{R}v := v + (2k + 1 - 2v_i)\mathbf{e}_i$ be the reflection of v in the plane $\{v_i = k + \frac{1}{2}\}$ (Figure 6.3), let $Y := \{v \in \mathbb{Z}^d : v_i \le k\}$, and let $\varphi(v) := f(v) - f(\mathcal{R}v)$. Reflection commutes with $\Delta^{::}$, so $-\Delta^{::}\varphi = \nu - \nu \circ \mathcal{R}$, and $\varphi \circ \mathcal{R} = -\varphi$. Multiply $-\Delta^{::}\varphi(v)$ by $\operatorname{sgn}\varphi(v)$, with $\operatorname{sgn} 0 = 0$, and sum over $v \in Y$; all these sums have finitely many nonzero terms. Each edge $\{v, w\}$ with both endpoints in Y contributes

$$(\varphi(v) - \varphi(w))(\operatorname{sgn}\varphi(v) - \operatorname{sgn}\varphi(w)) \ge 0\,.$$

Every other edge at a site $v \in Y$ joins v, with $v_i = k$, to $v + \mathbf{e}_i = \mathcal{R}v$, and contributes $\operatorname{sgn}\varphi(v)(\varphi(v) - \varphi(\mathcal{R}v)) = 2|\varphi(v)|$. Therefore

$$2\sum_{v_i=k} |\varphi(v)| \le \sum_{v\in Y} \operatorname{sgn}\varphi(v)(\nu(v) - \nu(\mathcal{R}v)) \le \sum_{v\in Y} (|\nu(v)| + |\nu(\mathcal{R}v)|) = 2\,,$$

where the equality uses that Y and its reflection partition $\mathbb{Z}^d$. Since $\varphi(v) = f(v) - f(v + \mathbf{e}_i)$ when $v_i = k$, this proves (6.3).

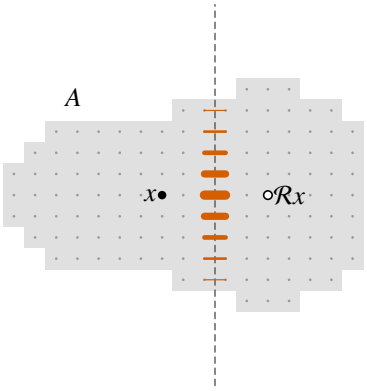


Figure 6.3: Lemma 6.2 in $\mathbb{Z}^2$: for $f = g_A^{::}(x,\cdot)$ on the set A (shaded), the differences $|f(v) - f(v + \mathbf{e}_1)|$ across the line $\{v_1 = k + \frac{1}{2}\}$ (dashed) are drawn as segments of width proportional to them, and the reflection $\mathcal{R}$ in this line maps x to $\mathcal{R}x$ (hollow dot). The proof bounds the sum of the differences by 1, as in (6.3); here it is 0.22.

The ball $\mathrm{B}^{::}(p, s)$ meets at most $2s + 1$ of the planes $\{v_i = k\}$, so (6.3) gives

$$\sum_{v\in \mathrm{B}^{::}(p,s)} |f(v) - f(v + \mathbf{e}_i)| \le 2s + 1\,.$$

Replacing v by $v - \mathbf{e}_i$ and p by $p - \mathbf{e}_i$ gives the same bound for the differences $f(v) - f(v - \mathbf{e}_i)$. Summing over the $2d$ neighbors of v shows that the left side of (6.2) is at most $2d(2s + 1) \le 6ds$, since $s \ge 1$. This proves the lemma with $C = 6d$. □

The next lemma controls the sum over all edges when A lies in a ball. The half-space bounded by the tangent plane to the sphere at the point nearest x contains the ball, and it converts the depth of x in the ball into an exit-probability bound.

Lemma 6.3 *For each integer $d \ge 2$, there is a constant $C = C(d) < \infty$ with the following property. Let $R > 0$ be real, let $A \subset \mathrm{B}^{::}(\mathbf{0}, R)$, let $x \in A$, and let $s := 1 + R - |x|$. Then*

$$\sum_{y\in A} \sum_{z\sim y} |g_A^{::}(x, y) - g_A^{::}(x, z)| \le Cs\left(1 + \log\frac{R+1}{s}\right).$$

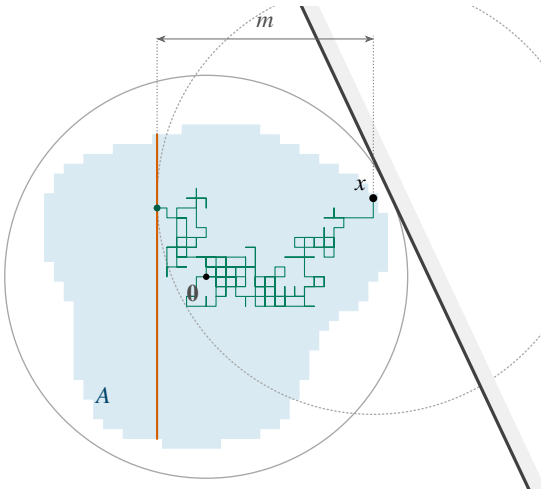


Figure 6.4: The proof of Lemma 6.3 in $\mathbb{Z}^2$: the set A (shaded) lies in the ball of radius R, hence in the half-plane bounded by the tangent line (thick) at the point of the circle nearest x, at distance $s-1$ from x. A walk from x that reaches the layer at distance m from x (vermilion) before leaving A has left $\mathrm{B}^{\cdot\cdot}(x,m)$ (dotted) by the time it leaves this half-plane, which by Lemma 3.22 has probability at most $2ds/m$.

Proof. We bound the differences across each coordinate layer by the probability that the walk from x reaches the layer before leaving A, and then sum over layers.

Fix $e \in \{\pm\mathbf{e}_1, \dots, \pm\mathbf{e}_d\}$ and an integer $m \geq 0$. Let τ_L be the hitting time of the layer $L := \{v : e \cdot (v - x) = m\}$. The coordinate $e \cdot (X_j - x)$ starts at 0 and changes by at most one at each step, so before time τ_L the walk visits neither a site v of the layer nor $v + e$. By the strong Markov property at time τ_L, for $w \in \{v, v + e\}$ the value $g^{\cdot\cdot}_A(x, w)$ is the expectation of $g^{\cdot\cdot}_A(X_{\tau_L}, w)$ restricted to the event $\{\tau_L < \tau_{\partial A}\}$, on which $X_{\tau_L} \in A$. The edges $\{v, v + e\}$ lie between two adjacent layers perpendicular to a coordinate direction, so (6.3) bounds by 1 the sum of $|g^{\cdot\cdot}_A(y, v) - g^{\cdot\cdot}_A(y, v + e)|$ over the layer, for every $y \in A$. Subtracting the identities for v and $v + e$ and using this bound with $y = X_{\tau_L}$ gives

$$\sum_{v \in L} |g^{\cdot\cdot}_A(x, v) - g^{\cdot\cdot}_A(x, v + e)| \leq \mathbb{P}_x\{\tau_L < \tau_{\partial A}\}\,.$$

Let $m \geq 1$. Choose a unit vector b with $b \cdot x = |x|$ (take $b = \mathbf{e}_1$ if $x = \mathbf{0}$). Every $y \in A$ satisfies $b \cdot (y - x) \leq |y| - |x| < s - 1$, so A lies in the half-space $\{y : b \cdot (y - x) \leq s - 1\}$. The layer is at distance at least m from x, so on $\{\tau_L < \tau_{\partial A}\}$ the walk leaves $\mathrm{B}^{\cdot\cdot}(x, m)$ by the time it leaves this half-space. By Lemma 3.22, applied with x translated to the origin, ball radius m and half-space $\{y : b \cdot y \leq s - 1\}$, this has probability at most $ds(m + 1)/m^2 \leq 2ds/m$. Figure 6.4 shows this geometry in $\mathbb{Z}^2$.

The two orientations $\{v, v + e\}$ of an edge have integer values of $e \cdot (v - x)$

with sum -1, so exactly one is nonnegative, and every edge occurs in exactly one of the sums above. Both endpoints of an edge in the sum for m are at distance at least m from x, and $|y - x| < 2R$ for $y \in A$, so the terms with $m \geq 2R$ vanish. Each edge occurs at most twice in the sum in the lemma, and there are $2d$ directions e. Since $1 < s \leq R + 1$, bounding the terms with $m \leq \lceil s \rceil$ by 1 and the others by $2ds \int_{m-1}^{m} \mathrm{d}t/t$ shows that the sum in the lemma is at most

$$4d\Big(1 + \lceil s \rceil + 2ds \log \frac{2(R+1)}{s}\Big) \leq Cs\Big(1 + \log \frac{R+1}{s}\Big) .$$

□

We now test Proposition 6.1 against the killed Green function of the ball $\mathrm{B}^{::}(\mathbf{0}, r-2)$. The resulting lower bound on the odometer applies to every finite sequence of stack moves that leaves at most one particle at each site of this ball, whether or not the moves are legal, so we state it in that generality.

Lemma 6.4 *For each integer $d \geq 2$, there is a constant $K = K(d) < \infty$ with the following property. Let $n \geq 2$ be an integer, let ρ be a family of instruction stacks on $\mathbb{Z}^d$, and let $b_\rho \geq 1$ be a real number satisfying*

$$|\mathcal{D}_\rho(x, y, m)| \leq b_\rho \qquad \text{for every edge } (x, y) \text{ and every integer } m \geq 0 .$$

Let ω_d be the volume of the unit ball in $\mathbb{R}^d$, let $r := (n/\omega_d)^{1/d}$, and suppose that $r > 2$. Suppose that finitely many stack moves using ρ convert $n\delta_{\mathbf{0}}$ into a configuration with at most one particle at each site of $\mathrm{B}^{::}(\mathbf{0}, r-2)$, and let v be their odometer. Then

$$v(z) \geq \frac{(r - |z|)^2}{4} - Kb_\rho(r - |z|) \log n \qquad \text{for every } z \in \mathrm{B}^{::}(\mathbf{0}, r-2) .$$

Proof. Let $B := \mathrm{B}^{::}(\mathbf{0}, r-2)$, let ξ be the final configuration of the moves, and fix $z \in B$. We apply Proposition 6.1 with $\sigma_0 = n\delta_{\mathbf{0}}$, with ξ, and with $f(x) := g_B^{::}(x, z)$, where $g_B^{::}$ is the killed Green function (3.18) of B. The function f is nonnegative, vanishes off B, and satisfies $\Delta^{::} f = -\delta_z$ on B. The proof of Proposition 6.1 computes the change in weight caused by the moves at each site x. If $x \notin B$, then $f(x) = 0$, so these moves can only add weight. Let

$$E_z := \sum_{x \in B} \sum_{y \sim x} |f(y) - f(x)| .$$

The moves at sites $x \in B$ contribute $-v(z)/(2d)$ in total through the first term, and at most $b_\rho E_z$ in absolute value through the second. Since $f \geq 0$ vanishes off B and $\xi \leq 1$ on B, it follows that

$$-\frac{v(z)}{2d} - b_\rho E_z \leq \sum_x f(x)(\xi(x) - n\delta_{\mathbf{0}}(x)) \leq \sum_{x \in B} f(x) - nf(\mathbf{0}) .$$

Writing $F(z) := n g_B^{::}(\mathbf{0}, z) - \sum_{x \in B} g_B^{::}(x, z)$, we conclude that

$$v(z) \geq 2d\, F(z) - 2d\, b_\rho E_z \,. \tag{6.4}$$

We bound the two terms in turn. By (3.19), the function F satisfies $\Delta^{::} F = 1 - n\delta_{\mathbf{0}}$ in B and $F = 0$ on ∂B. Lemma 5.3 of Chapter 5, applied to the ball $B = \mathrm{B}^{::}(\mathbf{0}, r - \eta)$ with $\eta = 2$, gives a constant $C_1 = C_1(d)$ such that

$$F(z) \geq \frac{(r - |z|)^2}{8d} - C_1 \,.$$

Let $\ell := r - |z|$, so that $\ell > 2$. By symmetry of $g_B^{::}$, Lemma 6.3 with $A = B$, with $r - 2$ in place of R, and with z in place of x, applies with $s = \ell - 1$ and gives

$$E_z \leq C_2(\ell - 1)\Big(1 + \log \frac{r - 1}{\ell - 1}\Big) \leq C_2\, \ell\, (1 + \log r) \leq C_3\, \ell \log n \,,$$

because $\ell - 1 > 1$ and $n \geq 2$, where the last step uses $r = (n/\omega_d)^{1/d}$. Substituting both bounds into (6.4) gives $v(z) \geq \ell^2/4 - 2dC_1 - 2dC_3 b_\rho \ell \log n$. Since $b_\rho \geq 1$ and $\ell > 2$, and since $\log n \geq \log 2$, the constant $2dC_1$ is at most a constant times $b_\rho \ell \log n$, and this proves the lemma. □

Away from the sphere of radius r, the quadratic main term in Lemma 6.4 dominates the error. Let $K = K(d)$ be the constant of the lemma, let $C := \max\{4K, 2/\log 2\}$, and consider finitely many stack moves using ρ that convert $n\delta_{\mathbf{0}}$ into a configuration with at most one particle at each site of $\mathrm{B}^{::}(\mathbf{0}, r - 2)$, with odometer v. If $z \in \mathrm{B}^{::}(\mathbf{0}, r - Cb_\rho \log n)$, then $\ell := r - |z| > Cb_\rho \log n \geq 2$, so $r > 2$ and $z \in \mathrm{B}^{::}(\mathbf{0}, r - 2)$, and the lemma gives $v(z) \geq \ell(\ell/4 - Kb_\rho \log n) > 0$. Hence

$$v(z) > 0 \qquad \text{for every } z \in \mathrm{B}^{::}(\mathbf{0}, r - Cb_\rho \log n) \,. \tag{6.5}$$

We now deduce the inner bound.

Theorem 6.5 *Let $d \geq 2$ and $n \geq 2$ be integers. Let ρ be a family of arrow stacks on $\mathbb{Z}^d$, and let $b_\rho \geq 1$ be a real number satisfying*

$$|\mathcal{D}_\rho(x, y, m)| \leq b_\rho \qquad \textit{for every edge } (x, y) \textit{ and every integer } m \geq 0 \,.$$

Let A_n^{R} be the aggregate obtained from $n\delta_{\mathbf{0}}$ using ρ. Let ω_d be the volume of the unit ball in $\mathbb{R}^d$, and let $r := (n/\omega_d)^{1/d}$. Then there is a constant $C = C(d) < \infty$ such that

$$\mathrm{B}^{::}(\mathbf{0}, r - Cb_\rho \log n) \subset A_n^{\mathrm{R}} \,.$$

Proof. Complete stack aggregation is a finite sequence of stack moves using ρ whose final configuration has at most one particle at each site, so (6.5) gives $v_n(z) > 0$ for every $z \in \mathrm{B}^{::}(\mathbf{0}, r - Cb_\rho \log n)$. A stack move occurs at each such site, and a site at which a legal move occurs is occupied in the final aggregate, so $\mathrm{B}^{::}(\mathbf{0}, r - Cb_\rho \log n) \subset A_n^{\mathrm{R}}$. □

It remains to bound the aggregate from outside.

6.3 OUTER BOUND

For rotor stacks, which have discrepancy less than one, Theorem 6.5 fills the ball of radius $r - C \log r$, so at most $Cr^{d-1} \log r$ of the n particles settle outside $\mathrm{B}^{::}(\mathbf{0}, r)$. These particles could still form a long thin tentacle. We rule this out by freezing the particles on a sphere and enlarging the sphere one unit at a time. The summation identity of Proposition 6.1, tested against the Green function of the occupied set with its pole at a frozen pile, has no odometer term, and comparing it with the divisible sandpile of the same mass, whose odometer vanishes near the sphere, shows that each pile of frozen particles has at most $C \log r$ particles. Each time the sphere grows by one unit, at least one particle of each nonempty pile stays behind, so the number of frozen particles is at most $1 - 1/(C \log r)$ times the preceding number, and after $C(\log r)^2$ steps none is left.

We now prove the shape theorem. Its inner inclusion follows from Theorem 6.5. For the outer inclusion, we use Proposition 6.1, the Green-function gradient bound of Lemma 6.3, and the shape of the divisible sandpile from Theorem 4.7.

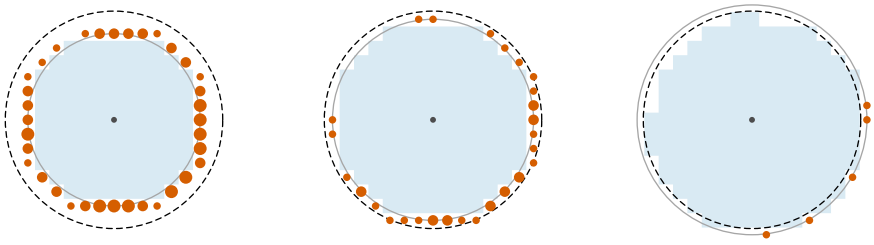
Theorem 6.6 *For each integer $d \geq 2$, there are constants $C = C(d) < \infty$ and $r_0 = r_0(d) < \infty$ with the following property. Let $n \geq 1$ be an integer, let ω_d be the volume of the unit ball in $\mathbb{R}^d$, let $r := (n/\omega_d)^{1/d}$, suppose that $r \geq r_0$, and fix an initial rotor configuration and a rotor mechanism on $\mathbb{Z}^d$. Let A_n^{R} be the rotor-router aggregate formed from n particles started at $\mathbf{0}$. Then*

$$\mathrm{B}^{::}(\mathbf{0}, r - C \log r) \subset A_n^{\mathrm{R}} \subset \mathrm{B}^{::}(\mathbf{0}, r + C(\log r)^2)\,. \tag{6.6}$$

Proof. The inner inclusion follows from Theorem 6.5, because each rotor stack has discrepancy less than one and $\log n = d \log r + O(1)$. We prove the outer inclusion in three steps. Throughout, C denotes a constant that depends only on d and may change from line to line, r is large, and $B_s := \mathrm{B}^{::}(\mathbf{0}, s)$.

Step 1: We freeze the particles on spheres of increasing radius. Let $c = c(d)$ be the constant in Theorem 4.7, and let $t_0 := \lceil r + c \rceil$. Starting from $n\delta_{\mathbf{0}}$, for each integer s with $t_0 \leq s \leq 2r$ in turn, make legal moves at sites of B_s until every site of B_s has at most one particle. This is possible: route

the particles one at a time, and note that a routed particle leaves every finite set, as shown in Section 6.2. Let η_s be the resulting configuration, and let N_s be the number of particles outside B_s, all of which lie on ∂B_s, because every move so far is made at a site of B_s. A legal move never empties a site, so every site at which a move has been made is occupied in η_s. Figure 6.5 shows the configurations η_s for 180 particles in $\mathbb{Z}^2$, with the freezing started at $s = 6$, inside the sphere of radius r, so that the piles are visible.



(a) (b) (c)

Figure 6.5: The configurations η_s in the proof of Theorem 6.6 for $s = 6, 7, 8$, for $n = 180$ particles in $\mathbb{Z}^2$ with the clockwise mechanism and all rotors initially North, and $r \approx 7.57$ (dashed circle): the occupied sites of B_s are shaded, and the frozen particles sit in piles on ∂B_s, just outside the circle of radius s, drawn as dots of area proportional to their size. Each pile leaves at least one particle behind when s grows by one, as in Step 3, so the 71 frozen particles become 35 and then 5.

Step 2: We show that $\eta_s(y) \le C \log r$ for every $y \in \partial B_s$. Let u and σ be the odometer and the stable configuration of the divisible sandpile started from $n\delta_{\mathbf{0}}$. By Theorem 4.7(i) and Lemma 4.4(i), the odometer u vanishes outside B_{r+c}, and $0 \le \sigma \le 1$.

Fix $y \in \partial B_s$ with $\eta_s(y) \ge 1$, let U be the set of sites x with $\eta_s(x) \ge 1$, and let $G := g^{\ddot{}}_U(y, \cdot)$, which is nonnegative and vanishes outside U (Figure 6.6). By (3.19), $\Delta^{\ddot{}} G = -\delta_y$ in U. Every site at which a move has been made lies in U and differs from y, because $y \notin B_s$, so $\Delta^{\ddot{}} G$ vanishes at each such site. Since $U \subset B_{s+1}$ and $s \le |y| < s + 1$, Lemma 6.3 with $A = U$, with $R = s + 1$, and with y in place of x gives

$$\sum_{x \in U} \sum_{z \sim x} |G(x) - G(z)| \le C(1 + \log(s + 2)) \le C \log r \,.$$

The rotor stacks have discrepancy less than one, so Proposition 6.1, applied to the initial configuration $n\delta_{\mathbf{0}}$, the configuration η_s and the function G, has no odometer term and gives

$$\sum_x G(x)\eta_s(x) \le nG(\mathbf{0}) + C \log r \,.$$

On the other hand, $\sigma = n\delta_{\mathbf{0}} + \Delta^{\ddot{}} u$ and u has finite support, so summation by

parts gives

$$\sum_x G(x)\sigma(x) = nG(\mathbf{0}) + \sum_x u(x)\Delta^{::}G(x)$$
$$= nG(\mathbf{0}) - u(y) + \sum_{x\notin U} u(x)\Delta^{::}G(x) \geq nG(\mathbf{0})\,,$$

because $u(y) = 0$ by $|y| \geq s \geq r + c$, and because $u \geq 0$ and $\Delta^{::}G(x) \geq 0$ for $x \notin U$, where $G(x) = 0$. Subtracting the two displays gives

$$\sum_x G(x)(\eta_s(x) - \sigma(x)) \leq C \log r\,.$$

Every term of this sum is nonnegative, because G vanishes outside U and $\eta_s \geq 1 \geq \sigma$ on U. The term at y therefore gives $(\eta_s(y) - 1)G(y) \leq C \log r$. The visit at time zero in (3.18) gives $G(y) \geq 1/(2d)$. Hence $\eta_s(y) \leq C_1 \log r$ for a constant $C_1 = C_1(d)$.

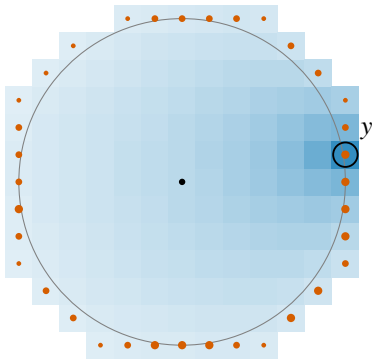


Figure 6.6: Step 2 of the proof of Theorem 6.6 for the configuration η_6 of Figure 6.5: the sites of $U = \{\eta_6 \geq 1\}$ are shaded by $G = g^{::}_U(y, \cdot)$ for a pile y (ringed) on ∂B_6. Since G is harmonic at every site where a move was made, Proposition 6.1 tested against G has no odometer term, and this bounds the pile at y.

Step 3: We show that the number of frozen particles decreases geometrically. Let P_s be the number of sites of ∂B_s at which η_s is positive, so that $N_s \leq C_1 P_s \log r$ by Step 2. Let $s+1 \leq 2r$. The configuration η_{s+1} is obtained from η_s by legal moves at sites of B_{s+1}, and $\partial B_s \subset B_{s+1}$. A legal move never empties a site, and $\eta_s \leq 1$ on B_s, so η_{s+1} is positive at $n - N_s$ sites of B_s and at P_s sites of ∂B_s. Since $\eta_{s+1} \leq 1$ on B_{s+1}, at least $n - N_s + P_s$ particles lie in B_{s+1}, and therefore

$$N_{s+1} \leq N_s - P_s \leq \Big(1 - \frac{1}{C_1 \log r}\Big) N_s\,.$$

Let $k := \lceil C_1 \log r \log n \rceil + 1$ and $t_1 := t_0 + k$, so that $t_1 \le 2r$ for large r. Iterating the last inequality gives

$$N_{t_1} \le n\Big(1 - \frac{1}{C_1 \log r}\Big)^k \le n e^{-k/(C_1 \log r)} < 1\,,$$

so $N_{t_1} = 0$. Then every site has at most one particle, so the moves made form a complete legal sequence, whose final configuration is $\mathbf{1}_{A_n^{\mathrm{R}}}$ by Corollary 5.2. Hence $A_n^{\mathrm{R}} \subset B_{t_1}$. Since $t_1 \le r + c + 3 + C_1 \log r \log n \le r + C(\log r)^2$, this proves the outer inclusion. □

After scaling A_n^{R} by r^{-1}, the inclusions (6.6) show that the aggregates converge to the closed unit ball in Hausdorff distance, with relative error $O((\log r)^2/r)$, and Figure 6.7 shows how close to a ball the aggregate is in $\mathbb{Z}^3$. In Chapter 7 we start particles at two sources in $\mathbb{Z}^2$, where the limit shape need not be a disk.

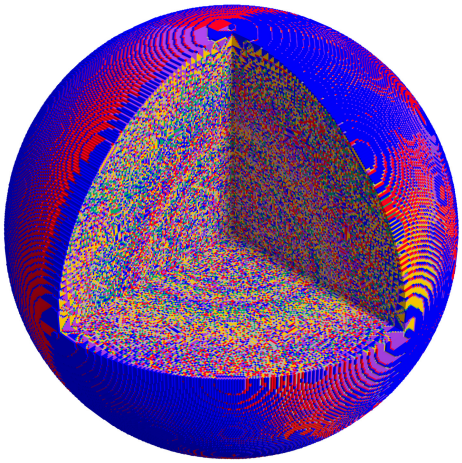


Figure 6.7: Rotor-router aggregation of $n = 10^7$ particles started at the origin of $\mathbb{Z}^3$, with the octant $\{x_1, x_2, x_3 > 0\}$ removed and each site colored by the final direction of its rotor (North, East, South, West, Up, Down: blue, purple, red, gray, gold, green); the rotors turn in the order North, South, East, West, Up, Down, and all start North. The aggregate lies between the spheres of radii 132.6 and 134.7 about the origin, close to the sphere of radius $r = (n/\omega_3)^{1/3} \approx 133.7$.

EXERCISES

Exercise 6.1. Let $d \ge 1$ be an integer, let $x \in \mathbb{Z}^d$, let $\rho_1(x), \rho_2(x), \ldots$ be a stack of arrows at x, each a directed edge from x to one of its neighbors, and let $\mathcal{D}_\rho(x, y, m)$ be its discrepancy, defined in (5.1).

(a) Show that $\max_{y\sim x} \mathcal{D}_\rho(x, y, 1) = 1 - 1/(2d)$ for every stack.

(b) Show that a stack that runs through the $2d$ neighbors of x in a fixed cyclic order minimizes $\sup_{m\geq 1} \max_{y\sim x} |\mathcal{D}_\rho(x, y, m)|$ among all stacks.

(c) For a stack of independent arrows, each pointing to a uniformly random neighbor, and for every neighbor y of x, show that $\mathbb{E}\,\mathcal{D}_\rho(x, y, m) = 0$ and

$$\mathbb{E}[\mathcal{D}_\rho(x, y, m)^2] = \frac{m(2d-1)}{(2d)^2}$$

for every integer $m \geq 1$, and that $\sup_{m\geq 1} |\mathcal{D}_\rho(x, y, m)| = \infty$ almost surely.

Exercise 6.2. This exercise spells out, for rotor stacks, the termination step behind Corollary 5.2. Let $d, n \geq 1$ be integers. Fix rotor stacks on $\mathbb{Z}^d$, and start n particles at $\mathbf{0}$. Use Lemma 5.1 to show that the aggregate and the final rotor configuration do not depend on the order in which the particles are released or on which active particle is advanced at each step.

Exercise 6.3. Let $d \geq 1$ be an integer, let $A \subset \mathbb{Z}^d$ be finite, and let $\sigma_0 : \mathbb{Z}^d \to \mathbb{N}$ be supported in A. Fix instruction stacks ρ on A. Route each particle along the successive unused arrows of the stacks until it first leaves A, suppose that this routing terminates after finitely many moves, and let $v_A^\rho(z)$ be the number of moves made at $z \in A$. Let $v_A(z)$ be the expected number of visits to z when the particles of σ_0 instead perform independent simple random walks stopped on leaving A. Use Proposition 6.1 to show that, for every $z \in A$,

$$v_A^\rho(z) - v_A(z) = 2d \sum_{x\in A} \sum_{y\sim x} \mathcal{D}_\rho(x, y, v_A^\rho(x))(g_A^{\cdot\cdot}(y, z) - g_A^{\cdot\cdot}(x, z)) .$$

Thus the stack odometer differs from the expected local time of random walk by a sum of discrepancies against the gradient of the killed Green function.

Exercise 6.4. Theorem 6.6 assumes $d \geq 2$; in one dimension the aggregate can be found exactly. Consider rotor-router aggregation on $\mathbb{Z}$ with rotor mechanism (East, West) and with all rotors initially pointing West. For every integer $n \geq 1$, show that A_n^{R} is an interval containing the origin, and determine A_n^{R} and the odometer v_n. How do the formulas change if all rotors initially point East?

NOTES

Rotor walk appeared independently in the rule ANT-WALK-1 of Wagner, Lindenbaum, and Bruckstein (1996, Section 2.1), in which a robot leaves

each vertex along its least recently used edge, and in the Eulerian walkers of Priezzhev, Dhar, Dhar, and Krishnamurthy (1996). The balancers in the load balancing circuits of Rabani, Sinclair, and Wanka (1998) send successive tokens alternately along two wires, as a rotor with two outgoing edges does, and the round-robin "whirling tours" on finite trees of Dumitriu, Tetali, and Winkler (2003, Theorem 9.2) have length equal to an expected hitting time of random walk.

Jim Propp proposed rotor-router aggregation in $\mathbb{Z}^2$ around 2001 as a deterministic version of internal DLA, as recounted by Levine and Peres (2008), who define it in $\mathbb{Z}^d$ by cycling each rotor through the $2d$ directions; the date is from Section 3 of Jerison, Levine, and Sheffield (2012). Levine and Peres (2009) recount that Propp observed from simulations that the two-dimensional aggregates are extremely close to circular, and asked why; Kleber (2005) shows such simulations. The first shape theorem is Theorem 1.1 of Levine and Peres (2008): the rescaled unions of the closed unit cubes centered at the sites of A_n^{R} converge to the ball in the sense that the volume of the symmetric difference tends to zero. Its proof uses an isoperimetric inequality for the expected exit time of random walk. Corollary 4.9 of Fey and Redig (2008), specialized to the rotor-router aggregate studied here, gives a diamond contained in the aggregate.

Theorem 1.1 of Levine and Peres (2009) gives the inner inclusion of Theorem 6.6 and an outer error of order $r^{1-1/d}\log r$. They remark, after their Theorem 1.1 and after their Lemma 5.1, that the theorem also holds for stacks of bounded discrepancy; Theorem 6.5 makes the dependence of the inner bound on the discrepancy explicit. Levine and Peres (2009) credit Matt Cook with the idea of studying the shape through the odometer. Theorem 4.1 of Levine and Peres (2017) improves the outer error to order $\log r$. Its proof first shows that for every occupied site z and every real t with $0 < t < |z|$, at least ct^d sites within distance t of z are occupied (their Proposition 4.2). It then tests the aggregate against a discrete harmonic function that is nonnegative near the ball and decays rapidly away from a given site just outside it, a variant of the functions of Lemmas 7–8 of Jerison, Levine, and Sheffield (2012) and Lemmas 5–7 of Jerison, Levine, and Sheffield (2013), and combines the inner bound, the mean value property of that function, and the Holroyd–Propp bound (their Lemma 4.11): for a function h harmonic on A_n^{R}, the sum of h over A_n^{R} differs from $nh(\mathbf{0})$ by at most the sum of $|h(x) - h(y)|$ over $x \in A_n^{\mathrm{R}}$ and $y \sim x$.

Proposition 6.1 is a many-particle form, for simple random walk on $\mathbb{Z}^d$, of the identity for a single stack walk in Proposition 21 of Holroyd and Propp (2010).

Section 5.2 of Levine and Peres (2009) proves the outer bound by showing that the number of particles that reach successive spheres halves at most

$C \log r$ times before it drops below a constant. The inner bound shows that at most $Cr^{d-1} \log r$ sites outside the sphere of radius r are occupied, and Jensen's inequality then bounds the total number of shells between successive halvings, which gives their outer error $r^{1-1/d} \log r$. Our proof of the outer bound in Theorem 6.6 instead bounds each pile of frozen particles by $C \log r$, by testing Proposition 6.1 against the Green function of the occupied set, with its pole at the pile, and comparing with the divisible sandpile of the same mass. When the sphere grows by one unit, at least one particle from each nonempty pile stays inside, so the number of frozen particles decreases geometrically and reaches zero after $C(\log r)^2$ steps.

Lemmas 6.2 and 6.3 extend Lemmas 5.3 and 5.6 of Levine and Peres (2009), which concern the Green function of a ball. The randomized stacks of Friedrich and Levine (2013, Section 6.4) on $\mathbb{Z}^2$ use independent uniform permutations of the four neighbors in successive blocks of length four, and therefore have bounded discrepancy.

Friedrich and Levine (2013) asked whether the difference between the largest distance from the origin of an occupied site and the smallest distance of an unoccupied site stays bounded as n grows. Their simulations in two dimensions, with the counterclockwise mechanism, found it between 1.3 and 1.7 for $n = 2^{10}, 2^{12}, \ldots, 2^{36}$ (their Section 6.2 and Table 1). Monte Carlo simulations of a single Eulerian walker on $\mathbb{Z}^2$, started from random initial rotors, appear in Kapri and Dhar (2009).

Chapter Seven

Internal Aggregation in $\mathbb{Z}^2$: Two Sources

Start n particles at the origin and another n at a lattice site at distance of order $\sqrt{n}$ from it. Each source alone would build a disk of area about n, so when the two disks overlap the particles must spread beyond their union, while disks that are far apart stay apart. The first main result of this chapter, in Section 7.1, is that the divisible sandpile, rotor-router aggregation and internal DLA all have the same limit shape after scaling by $\sqrt{n}$: a disk of area two if the sources coincide, two disks of area one if they are far apart, and in between an oval bounded by a quartic curve.

The second main result, in Section 7.2, makes this shape explicit: the logarithmic potential of the two sources minus that of the uniform measure on the shape is positive exactly inside the shape and vanishes together with its gradient outside. (In the language of complex analysis, the limit shape is a quadrature domain: the integral over it of every function harmonic near its closure is the sum of the function's values at the two sources.) We prove the shape theorem in three steps. First, we compute the potential by matching complex moments. Second, we compare it with the odometer of the divisible sandpile (Section 7.3). Third, we turn the odometer bounds into bounds for rotor-router aggregation (Section 7.4) by a deterministic argument and for internal DLA (Section 7.5) by a probabilistic one. The result extends the disks of the preceding three chapters to two sources and connects lattice growth with planar potential theory.

7.1 THE SHAPE THEOREM

We identify $\mathbb{R}^2$ with $\mathbb{C}$ and write $|E|$ for the area of a set E and $\#A$ for the number of lattice sites in a set A. Let $r_0 := \pi^{-1/2}$ be the radius of a disk of area one. For two sources at $\pm a$ on the real axis, with $a \geq 0$, the candidate shape is

$$O_a := \begin{cases} \{re^{\mathbf{i}\theta} : \theta \in \mathbb{R} \text{ and } 0 \leq r < \sqrt{2r_0^2 + 2a^2\cos 2\theta}\}, & 0 \leq a < r_0, \\ \mathrm{B}(-a, r_0) \cup \mathrm{B}(a, r_0), & a \geq r_0 \end{cases}.$$

Figure 7.1 compares O_a with the disks of area one about the two sources. For sources at $\mathbf{0}$ and $q \in \mathbb{C} \setminus \{\mathbf{0}\}$, rotate and translate this shape by writing

$\Omega(q) := q/2 + (q/|q|)O_{|q|/2}$, and let $\Omega(\mathbf{0}) := \mathrm{B}(\mathbf{0}, \sqrt{2}r_0)$. These are open sets, so the point of contact of tangent disks is excluded. For an open set Ω and $\epsilon > 0$, write

$$\Omega_\epsilon := \{z \in \Omega : \mathrm{dist}(z, \mathbb{R}^2 \setminus \Omega) > \epsilon\} \quad \text{and} \quad \Omega^\epsilon := \{z : \mathrm{dist}(z, \Omega) < \epsilon\}.$$

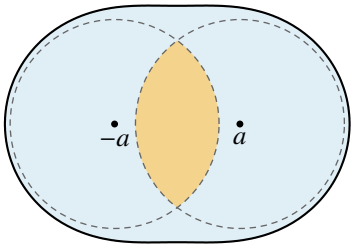


Figure 7.1: The limit shape O_a for $a = 0.6r_0$ (black) and the disks of radius r_0 about the sources $\pm a$ (dashed), which it contains by Lemma 7.4(ii). It has area two, so it exceeds their union by the area of their overlap (orange).

For each positive integer n, start all three models from $\sigma_n := n\delta_{\mathbf{0}} + n\delta_{b_n}$, where $b_n \in \mathbb{Z}^2$. Write u_n for the odometer of the divisible sandpile (the mass sent to each neighbor), $\xi_n := \sigma_n + \Delta^{::} u_n$ for its final mass, and $S_n := \{u_n > 0\}$ for its toppled set. Recall from Lemma 4.4 that S_n is finite and $\xi_n = 1$ on S_n; moreover $0 \le \xi_n \le 1$, since the final configuration is stable and $\sigma_n \ge 0$. Write A_n^{R} for the rotor-router aggregate and A_n for the internal DLA cluster. Every rotor stack is periodic and cycles once through the four neighbors, as in Section 6.1, in an order and from a first arrow that may vary from site to site. The abelian property, Corollary 5.2, makes the final aggregate independent of the order of legal moves. Both particle clusters contain exactly $2n$ sites.

The following theorem identifies the common limit, including the transition from an oval to two disks.

Theorem 7.1 *Let $p \in \mathbb{R}^2$, and let $b_n \in \mathbb{Z}^2$ satisfy $b_n/\sqrt{n} \to p$ as $n \to \infty$. Then the following hold.*

(i) For every $\epsilon > 0$ and all sufficiently large n,

$$\sqrt{n}\Omega(p)_\epsilon \cap \mathbb{Z}^2 \subset \{\xi_n = 1\} \subset \{\xi_n > 0\} \subset \sqrt{n}\Omega(p)^\epsilon.$$

(ii) For every $\epsilon > 0$ and all sufficiently large n, uniformly over the rotor stacks,

$$\sqrt{n}\Omega(p)_\epsilon \cap \mathbb{Z}^2 \subset A_n^{\mathrm{R}} \subset \sqrt{n}\Omega(p)^\epsilon.$$

(iii) Under every coupling of the clusters $(A_n)_{n\ge1}$, with probability one, for every $\epsilon > 0$ and all sufficiently large n,

$$\sqrt{n}\Omega(p)_\epsilon \cap \mathbb{Z}^2 \subset A_n \subset \sqrt{n}\Omega(p)^\epsilon.$$

If $p = \mathbf{0}$, the limit is the disk of area two, and if $|p| \geq 2r_0$, it consists of two disks of area one. For $0 < |p| < 2r_0$ its boundary is the quartic curve (7.1) of Section 7.2 with $a = |p|/2$, in coordinates centered at $p/2$ with the real axis through both sources. Figure 7.2 shows internal DLA in this regime, Figure 1.13 shows all three models, and Figure 7.3 compares their boundaries with that of the limit shape.

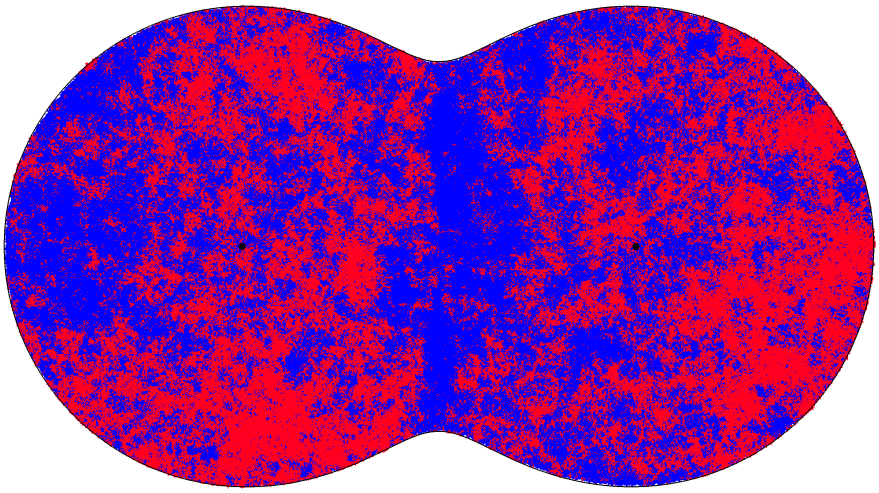


Figure 7.2: Internal DLA from $n = 282{,}743$ particles at each of the sites $(\pm 250, 0)$ (black dots), with the site of the jth particle red if it lies outside $\sqrt{t}\,O_{250/\sqrt{t}}$, the limit shape of $t = j/2$ particles at each source (Theorem 7.2), and blue otherwise. The black curve is the boundary of the limit shape for $t = n$, the quartic $(x^2 + y^2)^2 - \frac{2n}{\pi}(x^2 + y^2) - 2 \cdot 250^2(x^2 - y^2) = 0$ in lattice coordinates, which is (7.1) with $a = 250/\sqrt{n}$ scaled by $\sqrt{n}$.

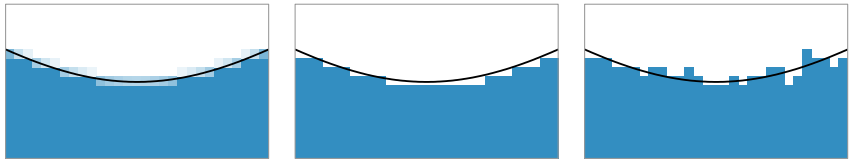


Figure 7.3: The clusters from $n = 70{,}002$ particles at each of $(\pm 127, 0)$ near the top of the waist of their limit shape: (a) the divisible sandpile (light squares hold mass strictly between 0 and 1), (b) rotor-router aggregation and (c) internal DLA, with the boundary of the limit shape (black). Over each whole cluster, every site farther than 0.6, 0.8 and 3, respectively, inside this boundary holds a particle or mass, and every site farther than that outside it is empty.

The proof starts from the logarithmic potential of the limit shape, computed in the next section: its positivity inside the shape gives the inner bounds, and its vanishing outside gives the outer bounds.

7.2 THE POTENTIAL OF THE LIMIT SHAPE

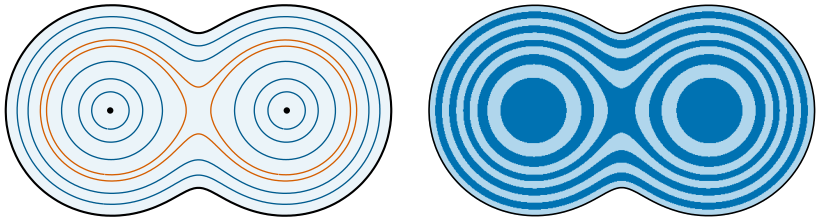
Recall the Green function $\mathcal{G}(z) = -(2\pi)^{-1}\log|z|$ from (2.14), with $\mathcal{G}(\mathbf{0}) = +\infty$, and the potential $\mathbf{G}\sigma(z) = \int \mathcal{G}(z-w)\sigma(w)\,\mathrm{d}w$ from (2.13). For the shapes $\mathcal{O}_a$ of Section 7.1, write

$$w_a(z) := \mathcal{G}(z-a) + \mathcal{G}(z+a) - \mathbf{G}\mathbf{1}_{\mathcal{O}_a}(z)\,.$$

The polar formula for the oval comes from matching moments: a boundary $r^2 = A + B\cos 2\theta$ encloses area πA and second complex moment $\pi AB/2$. Matching these with the area 2 and the second moment $2a^2$ of the two sources gives $A = 2r_0^2$ and $B = 2a^2$, where $r_0 = \pi^{-1/2}$. The next theorem checks that all the moments match and that the potential has the sign we need.

Theorem 7.2 *Let $a \geq 0$. Then $\mathcal{O}_a$ has area two, and w_a is nonnegative, positive exactly on $\mathcal{O}_a$, and continuously differentiable away from the two sources. Moreover w_a and ∇w_a vanish on $\mathbb{C}\setminus\mathcal{O}_a$. If $a < r_0$, then the boundary of $\mathcal{O}_a$ consists of the points $x + \mathbf{i}y \neq \mathbf{0}$ with $x, y \in \mathbb{R}$ satisfying*

$$(x^2+y^2)^2 - 2r_0^2(x^2+y^2) - 2a^2(x^2-y^2) = 0\,. \tag{7.1}$$



(a) (b)

Figure 7.4: (a) Level curves of the potential w_a for $a = 0.85r_0$: two loops about the sources join through the saddle at the origin (the vermilion curves lie just below and just above its value there), and w_a vanishes with its gradient on and outside $\partial\mathcal{O}_a$ (black). (b) The odometer of the divisible sandpile started from $n = 70{,}002$ at each of the two sources $(\pm 127, 0)$, in ten bands of values with equal numbers of sites, alternately light and dark, with the boundary of $\sqrt{n}\,\mathcal{O}_{127/\sqrt{n}}$ (black), where $127/\sqrt{n} \approx 0.85r_0$.

Figure 7.4(a) draws the level curves of w_a for $a = 0.85r_0$. For $a \geq r_0$ the theorem is the disk potential of Lemma 2.37 applied to each disk. Its differentiability assertion needs only the local integrability of the gradient of the logarithm, as the next lemma shows.

Lemma 7.3 *Let σ be a bounded measurable function on $\mathbb{R}^2$ with bounded support. Then $\mathbf{G}\sigma$ is continuously differentiable, and*

$$\nabla\mathbf{G}\sigma(x) = \int \nabla\mathcal{G}(x-y)\sigma(y)\,\mathrm{d}y\,.$$

Proof. The kernel satisfies $|\nabla \mathcal{G}(z)| = (2\pi|z|)^{-1}$, which is locally integrable. Write $F(x) := \int \nabla \mathcal{G}(x-y)\sigma(y)\,\mathrm{d}y$. Splitting this integral into a small disk and its complement, as in Remark 2.5, shows that F is continuous: the disk contributes at most $C\|\sigma\|_\infty$ times its radius, and the kernel is uniformly continuous on the bounded remainder. Integrating along a segment and using Fubini's theorem, which applies because the integral of $|\nabla \mathcal{G}|$ over a bounded set is bounded uniformly over translates in a bounded set, gives

$$\mathbf{G}\sigma(x+h) - \mathbf{G}\sigma(x) = \int_0^1 h \cdot F(x+th)\,\mathrm{d}t = h \cdot F(x) + o(|h|)\,,$$

where the last equality uses the continuity of F. Thus F is the continuous gradient of $\mathbf{G}\sigma$. □

Proof of Theorem 7.2. We compute the potential for disks, then prove vanishing and positivity for the oval.

Step 1: We treat $a \geq r_0$, where the two disks are disjoint and each has area one. Write $W := \mathcal{G} - \mathbf{G}\mathbf{1}_{\mathrm{B}(\mathbf{0},r_0)}$. By Lemma 2.37, W vanishes for $|z| \geq r_0$ and satisfies

$$W(z) = \frac{|z|^2 - r_0^2}{4} - \frac{r_0^2}{2}\log\frac{|z|}{r_0} \qquad (0 < |z| < r_0)\,.$$

The radial derivative of W is $(|z|^2 - r_0^2)/(2|z|) < 0$, so $W > 0$ inside. Hence $w_a(z) = W(z-a) + W(z+a)$ has the required sign and zero boundary gradient, including when the disks are tangent.

Step 2: Now let $a < r_0$. We show that w_a and ∇w_a vanish outside O_a. The positive radial graph encloses both sources, as its radius on the real axis exceeds a. For integers $j \geq 0$, polar coordinates give, with area measure $\mathrm{d}A$,

$$\int_{O_a} z^{2j}\,\mathrm{d}A(z) = \frac{1}{2j+2}\int_0^{2\pi} e^{2j\mathbf{i}\theta}(2r_0^2 + a^2e^{2\mathbf{i}\theta} + a^2e^{-2\mathbf{i}\theta})^{j+1}\,\mathrm{d}\theta\,.$$

The integral selects the coefficient of $e^{-2j\mathbf{i}\theta}$ in the power. If a contributing term has n_- negative exponent factors, n_+ positive exponent factors and n_0 constant factors, then $n_- - n_+ = j$ and $n_- + n_+ + n_0 = j+1$. Thus $2n_+ + n_0 = 1$, so the coefficient is $(j+1)2r_0^2a^{2j}$. Odd moments vanish by symmetry. Consequently

$$\int_{O_a} z^\ell\,\mathrm{d}A(z) = a^\ell + (-a)^\ell \qquad (\ell \geq 0)\,, \tag{7.2}$$

where the zeroth moment is 2, also when $a = 0$. For $|z| > \sup_{w\in O_a}|w|$, the uniformly convergent expansion

$$\log|z-w| = \log|z| - \mathrm{Re}\sum_{\ell\geq 1}\frac{w^\ell}{\ell z^\ell}\,,$$

and (7.2) give $w_a(z) = 0$. The exterior of $\overline{O_a}$ is connected, since the boundary is a positive radial graph. There w_a is harmonic, and $\partial_x w_a - \mathbf{i}\partial_y w_a$ is holomorphic by Subsection 2.2.1. The identity theorem extends the vanishing of this function from near infinity to the entire exterior, so w_a is constant there, hence zero. Lemma 7.3 extends the vanishing of w_a and of its gradient to the boundary.

Step 3: We keep $a < r_0$ and prove positivity by differentiating vertically: $\partial_y w_a$ will be negative in the upper half of the oval, while w_a vanishes on the boundary. Corollaries 2.39 and 2.40 give $\Delta w_a = 1$ inside the oval away from the sources. More precisely, $w_a - |z|^2/4$ is harmonic there by Theorem 2.29, and hence smooth by Theorem 2.3. Write $V := \partial_y w_a$ and $O^+ := O_a \cap \{y > 0\}$. Then V is harmonic on the connected set O^+. On the boundary of O^+ we have $V = 0$ except at the sources: on the curved boundary this follows from Step 2, and on the real axis from reflection symmetry. For a source $s \in \{-a, a\}$, write m_s for its mass, which is one, or two if $a = 0$. The function $w_a - m_s \mathcal{G}(\cdot - s) - |z - s|^2/4$ is harmonic and even in y on a small disk about s, so

$$V(x, y) = -\frac{m_s y}{2\pi((x - s)^2 + y^2)} + O(y)\,.$$

Therefore $V \le 0$ on sufficiently small upper half-circles about the sources. Applying the minimum principle, Lemma 2.12(a), to $-V$ with these half-disks removed and letting their radii decrease to zero gives $V \le 0$ on O^+. The strict inequality near each source and the strong minimum principle, Lemma 2.8, give $V < 0$ throughout O^+. From a point $(x, y) \in O^+$, follow the upward vertical segment to its first boundary point (x, t_1) (Figure 7.5). Since $w_a(x, t_1) = 0$,

$$w_a(x, y) = -\int_y^{t_1} V(x, t)\,\mathrm{d}t > 0\,.$$

The same integration proves positivity on the real axis away from the sources; at the sources the value is infinite. Reflection gives positivity in the lower half-plane. Finally, multiplying the polar boundary equation by r^2 gives (7.1); the only added point is the origin, which lies inside the oval. □

Figure 7.6 shows how the oval turns into two disks: as a increases to r_0, the polar radii of O_a converge uniformly to $2r_0|\cos\theta|$, the polar equation of the two tangent circles.

For sources at $\mathbf{0}$ and $q \in \mathbb{C}$, write

$$U_q(z) := \mathcal{G}(z) + \mathcal{G}(z - q) - \mathbf{G}\mathbf{1}_{\Omega(q)}(z)\,.$$

Rotating and translating Theorem 7.2 shows that U_q is nonnegative and that

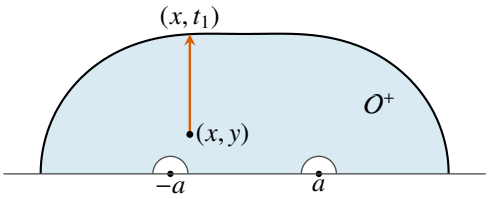


Figure 7.5: Step 3 of the proof of Theorem 7.2: $V = \partial_y w_a$ is harmonic on O^+ with the half-disks about $\pm a$ removed and is negative there, so integrating V along the vertical segment from (x, y) to the boundary point (x, t_1), where $w_a = 0$, gives $w_a(x, y) > 0$.

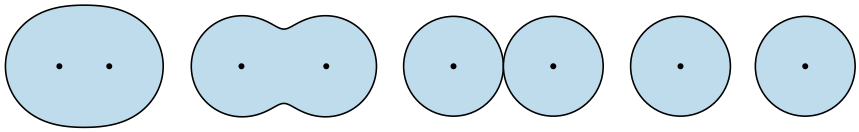


Figure 7.6: The limit shape O_a of two sources of mass one at $\pm a$, for $a/r_0 = 0.5$, 0.85, 1 and 1.25, drawn to the same scale. Every shape has area two, and the oval pinches into two tangent disks of area one when $a = r_0 = \pi^{-1/2}$.

U_q and ∇U_q vanish outside $\Omega(q)$. Before passing to the lattice we also need continuity in q and a quadratic bound near the boundary.

Lemma 7.4 *Let $q \in \mathbb{C}$.*

(i) If $q_n \to q$ in $\mathbb{C}$, then for every $\eta > 0$ and all sufficiently large n we have $\Omega(q)_\eta \subset \Omega(q_n) \subset \Omega(q)^\eta$, and $U_{q_n} \to U_q$ uniformly on compact sets disjoint from $\{\mathbf{0}, q\}$.

(ii) The set $\Omega(q)$ contains the disks of radius r_0 about $\mathbf{0}$ and q.

(iii) For every $w \in \partial\Omega(q)$ and every z with $|z - w| < r_0/4$,

$$0 \le U_q(z) \le 6|z - w|^2 . \tag{7.3}$$

Proof. (i) Uniform convergence of the radial graphs gives the inclusions on the oval side, including at tangency: compact subsets of the tangent disks avoid their contact point. Continuity of disk centers gives the inclusions on the separated side. Translations and rotations preserve these conclusions. At $q = \mathbf{0}$, the radii tend uniformly to $\sqrt{2}r_0$, regardless of rotation. Polar integration also gives

$$|O_a \triangle O_b| = \frac{1}{2}\int_0^{2\pi} 2|a^2 - b^2||\cos 2\theta|\, \mathrm{d}\theta = 4|a^2 - b^2| ,$$

for $0 \le a, b \le r_0$, which tends to zero as $a - b \to 0$; at r_0 the contact point does not affect area. For disks the same conclusion follows by enclosing the symmetric difference in annuli of vanishing width. Continuity of translations and rotations in L^1 now gives $|\Omega(q_n) \triangle \Omega(q)| \to 0$. To compare potentials, split their integrals at $|z - y| = t$ with $0 < t < 1$. The near part is at most $Ct^2(1 + |\log t|)$; on a fixed bounded remainder the kernel is bounded, so the far part tends uniformly to zero. The source terms converge away from the limiting sources.

(ii) In the oval case (Figure 7.1), put $z = x + \mathbf{i}y$ and $r = |z| > 0$. If $|z-a| < r_0$, then the left side of (7.1) is strictly less than $-(r^2 - 2ax)^2$, hence negative. The origin also belongs to O_a. Reflection treats the other disk; for $a \ge r_0$ the inclusion is part of the definition. Thus every boundary point is at distance at least r_0 from the sources.

(iii) Fix $w \in \partial\Omega(q)$. By part (ii), no source lies in the disk of radius $r_0/2$ centered at w, so $\Delta U_q = \mathbf{1}_{\Omega(q)}$ there distributionally. Hence $U_q + |z - w|^2/4$ is subharmonic and $U_q - |z - w|^2/4$ is superharmonic, by Theorem 2.29. The quadratic bound at a zero, Proposition 2.32, with $\lambda = 1/4$ and $\beta = 1/2$, proves (7.3). □

We next discretize this potential and compare it with the divisible sandpile odometer.

7.3 THE DIVISIBLE SANDPILE

Throughout the rest of the chapter, we work in the setting of Theorem 7.1 and write $p_n := b_n/\sqrt{n}$, so that $p_n \to p$ and the sets $\Omega(p_n)$ lie in a fixed disk. Recall that $\Delta^{::} f(x) = \sum_{y \sim x}(f(y) - f(x))$, and that $g^{::}$ (minus the potential kernel of the plane) satisfies $\Delta^{::} g^{::} = -\delta_{\mathbf{0}}$ and $g^{::}(w) = \mathcal{G}(w) - \kappa + O(|w|^{-2})$ for $w \ne \mathbf{0}$ by (4.6), where κ is the constant of Theorem 3.9.

Spread uniform mass on $\sqrt{n}\Omega(p_n)$ over lattice cells (Figure 7.7) by writing

$$Q_y := y + [-\tfrac{1}{2}, \tfrac{1}{2})^2 \quad \text{and} \quad \mu_n(y) := |Q_y \cap \sqrt{n}\Omega(p_n)|\,.$$

Then $0 \le \mu_n \le 1$ and $\sum_y \mu_n(y) = 2n$. Define the comparison function

$$F_n(x) := ng^{::}(x) + ng^{::}(x - b_n) - \sum_y \mu_n(y) g^{::}(x - y)\,.$$

It satisfies $\Delta^{::} F_n = \mu_n - \sigma_n$ exactly. The next estimate compares it with the explicit continuum potential U_{p_n} of Section 7.2, including near the sources; as Figure 7.4(b) suggests, the odometer u_n itself stays close to $nU_{p_n}(x/\sqrt{n})$.

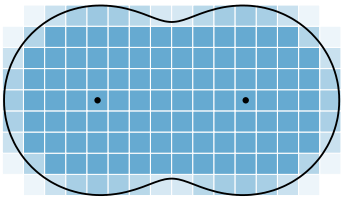


Figure 7.7: The cells Q_y shaded by $\mu_n(y)$, the area of Q_y inside $\sqrt{n}\,\Omega(p_n)$ (black), for $n = 60$ and $b_n = (7, 0)$: the full cells (dark) form the set E_n of Step 2 of the proof of Theorem 7.1(i), and the values of μ_n sum to $2n$.

Lemma 7.5 *There is a constant C, independent of n, with the following properties.*

(i) For every fixed $\eta > 0$, uniformly over sites x at distance at least $\eta\sqrt{n}$ from both sources,

$$F_n(x) = nU_{p_n}(x/\sqrt{n}) + O_\eta(\sqrt{n})\,. \tag{7.4}$$

(ii) Let x be a site at distance $r \le \sqrt{n}$ from the nearer source. Then

$$F_n(x)/n \ge \frac{1}{2\pi}\log\frac{\sqrt{n}}{1+r} - C\,. \tag{7.5}$$

(iii) The function F_n satisfies $F_n \ge -C\sqrt{n}$ everywhere.

Proof. (i) We compare cell integrals first and then handle the source terms. If $|x - y| \ge 2$, then the kernel asymptotic and $|\nabla\mathcal{G}(z)| = (2\pi|z|)^{-1}$ give

$$\left|\mu_n(y)(g^{::}(x-y)+\kappa) - \int_{Q_y\cap\sqrt{n}\Omega(p_n)} \mathcal{G}(x-w)\,\mathrm{d}w\right| \le \frac{C}{1+|x-y|}\,.$$

The same bound, after increasing C, covers $|x - y| < 2$, by integrability of the logarithm. Summing over cells in a disk of radius $C\sqrt{n}$ gives error at most $C\sqrt{n}$, uniformly in x. For x in the disk of twice that radius, sum over shells of integer radii j about x, each containing $O(j + 1)$ sites. For more distant x, all denominators are at least a constant times $|x|$, giving a bound $Cn/|x| \le C\sqrt{n}$. The source kernel errors are $O_\eta(1)$ under the separation assumption. The constants κ cancel because the masses agree. After the change of variables $w = \sqrt{n}z$, the terms involving $\log n$ cancel for the same reason. This proves (7.4).

(ii) The kernel asymptotic also gives, for every lattice point w,

$$g^{::}(w) + \kappa + \frac{\log n}{4\pi} \ge \frac{1}{2\pi}\log\frac{\sqrt{n}}{1+|w|} - C\,.$$

Since $r \le \sqrt{n}$, the point $x/\sqrt{n}$ stays bounded. After adding $\kappa + (\log n)/(4\pi)$ to every kernel in F_n/n, the term of the other source is bounded below, and the sum over y, divided by n, is bounded above by the estimate of part (i) and local integrability of the logarithm, which proves (7.5).

(iii) Choose $\eta > 0$ small enough that the lower bound (7.5) is positive whenever $r \le \eta\sqrt{n}$, for large n. Elsewhere (7.4) and $U_{p_n} \ge 0$ give $F_n \ge -C\sqrt{n}$. Enlarging C covers the finitely many remaining n, since each F_n tends to zero at infinity by cancellation of its total mass. □

For the outer bound we also need the fact that toppling far from a source forces a large odometer nearby. We prove it with an error term in the Laplacian, so that it also serves rotor-router aggregation in the next section.

Lemma 7.6 *There is an absolute constant $C_0 \ge 1$ with the following property. Let $z \in \mathbb{Z}^2$, let $r \ge 1$ and $\lambda \ge 0$, and let e be a function on ordered pairs of neighboring sites with $|e| \le \lambda$ and $\sum_{w\sim v} e(v,w) = 0$ for every site v. Let $f : \mathbb{Z}^2 \to \mathbb{R}$ satisfy $f(z) > C_0\lambda r$ and*

$$\Delta^{::} f(v) = 1 - \sum_{w\sim v} e(w,v) \qquad \text{for every } v \in \mathrm{B}^{::}(z,r) \text{ with } f(v) > 0\,.$$

Then there is a site $y \in \partial\mathrm{B}^{::}(z,r)$ with $f(y) > r^2/4$.

Proof. We compare f with $|v-z|^2/4$, whose Laplacian is one, by testing against a Green function. Let S be the set of sites $v \in \mathrm{B}^{::}(z,r)$ with $f(v) > 0$, and let $\varphi := g^{::}_S(z,\cdot)$. Then $\Delta^{::}\varphi = -\delta_z$ on S, and on ∂S the values $\Delta^{::}\varphi(v) = \sum_{w\sim v}\varphi(w)$ are nonnegative with sum one, since $\sum_v \Delta^{::}\varphi(v) = 0$. The function $h(v) := f(v) - |v-z|^2/4$ has $\Delta^{::}h(v) = -\sum_{w\sim v} e(w,v)$ on S, so summation by parts and $\sum_{w\sim v} e(v,w) = 0$ give

$$\sum_{v\in\partial S} h(v)\Delta^{::}\varphi(v) - h(z) = \sum_v \varphi(v)\Delta^{::}h(v) = \sum_v \sum_{w\sim v} e(v,w)(\varphi(v) - \varphi(w))\,.$$

Every edge with an endpoint in S has $v \in \mathrm{B}^{::}(z, r+1)$, so the gradient bound of Lemma 6.2 bounds the right side by $C\lambda(r+1) \le C_0\lambda r$ in absolute value, with $C_0 := \max\{1, 2C\}$. Hence some $y \in \partial S$ has $h(y) \ge f(z) - C_0\lambda r > 0$. As $f(y) > 0$ and $y \notin S$, it lies outside $\mathrm{B}^{::}(z,r)$ and next to it, so $f(y) > |y-z|^2/4 \ge r^2/4$. □

The least action principle bounds u_n above, giving the outer inclusion of part (i); the minimum principle on the fully occupied cells bounds u_n below, giving the inner inclusion and a margin of order n for the particle models.

Proof of Theorem 7.1(i). **Step 1:** We bound the odometer from above. The function $F_n + C\sqrt{n}$ is nonnegative by Lemma 7.5(iii), and its Laplacian plus σ_n equals $\mu_n \le 1$, so Theorem 4.3 gives $u_n \le F_n + C\sqrt{n}$. Outside $\sqrt{n}\Omega(p_n)$,

both sources are at distance at least $r_0\sqrt{n}$ by Lemma 7.4(ii), and $U_{p_n} = 0$ there by Theorem 7.2, so (7.4) gives $u_n \le C\sqrt{n}$. If $x \in S_n$ is at distance $r > 4$ from $\sqrt{n}\Omega(p_n)$, then apply Lemma 7.6 with $f = u_n$ and $e = \lambda = 0$ on the source-free disk $\mathrm{B}^{::}(x, r/2)$, where $\Delta^{::}u_n = 1$ on $\{u_n > 0\}$ by Lemma 4.4. Its boundary remains outside $\sqrt{n}\Omega(p_n)$, so $r^2/16 < C\sqrt{n}$. Consequently

$$S_n \subset \{x : \operatorname{dist}(x, \sqrt{n}\Omega(p_n)) \le Cn^{1/4}\} \subset \mathrm{B}^{::}(\mathbf{0}, C\sqrt{n}). \tag{7.6}$$

Step 2: We bound the odometer from below. Let $E_n := \{x : \mu_n(x) = 1\}$. On this finite set, $\Delta^{::}(u_n - F_n) = \xi_n - 1 \le 0$, since $\mu_n = 1$ there. A site on ∂E_n lies within distance 2 of $\partial(\sqrt{n}\Omega(p_n))$: its cell has positive area outside, and a neighboring cell has full area inside, so it is at distance at least $r_0\sqrt{n} - 2$ from both sources. The quadratic bound (7.3) and the comparison (7.4) give $|F_n| \le C\sqrt{n}$ there. Since $u_n \ge 0$, the minimum principle for the superharmonic function $u_n - F_n$ gives $u_n \ge F_n - C\sqrt{n}$ on E_n.

Let $K \subset \Omega(p)$ be compact. The inclusions of Lemma 7.4(i), applied to a small closed neighborhood of K, put $\sqrt{n}K \cap \mathbb{Z}^2$ inside E_n for large n. Away from small fixed neighborhoods of the limiting sources, U_{p_n} converges uniformly to the strictly positive function U_p. Near the sources choose those neighborhoods small enough that (7.5) is at least one, for large n. We conclude that there is $c_K > 0$ such that

$$u_n(x) \ge c_K n \qquad (x \in \sqrt{n}K \cap \mathbb{Z}^2, n \text{ sufficiently large}). \tag{7.7}$$

Step 3: We pass to the fixed domain $\Omega(p)$. By (7.7) with $K = \overline{\Omega(p)_\epsilon}$, every site of $\sqrt{n}K$ topples and hence has final mass one. For $n > 1$ the sources topple, so every site carrying final mass belongs to S_n or its outer boundary. The bound (7.6) and $\Omega(p_n) \subset \Omega(p)^{\epsilon/2}$ for large n put this support inside $\sqrt{n}\Omega(p)^\epsilon$. □

The particle comparisons use only the bounds (7.7) and (7.6). We start with rotor-router aggregation.

7.4 ROTOR-ROUTER AGGREGATION

Write ρ for the rotor stacks and $\mathcal{D}_\rho$ for their discrepancy (5.1). For a finite lattice set S, recall that the killed Green function $g^{::}_S$ is symmetric, vanishes when either argument is outside S, and satisfies $\Delta^{::}_y g^{::}_S(x, y) = -\delta_x(y)$ on S, by (3.19). For $n > 1$ both sources belong to S_n, so the sandpile odometer has the representation

$$u_n(z) = n g^{::}_{S_n}(\mathbf{0}, z) + n g^{::}_{S_n}(b_n, z) - \sum_{y \in S_n} g^{::}_{S_n}(y, z). \tag{7.8}$$

Indeed, both sides vanish off S_n and have Laplacian $1 - \sigma_n$ on S_n, so the minimum principle makes them equal. This identity gives the inner bound.

Lemma 7.7 *Let K be a compact subset of $\Omega(p)$. Then $\sqrt{n}K \cap \mathbb{Z}^2 \subset A_n^{\mathrm{R}}$ for all sufficiently large n, uniformly over the rotor stacks.*

Proof. Let m be the odometer of rotor-router aggregation from σ_n, which counts the moves at each site. We compare m with u_n by testing the summation identity, Proposition 6.1, against a killed Green function of S_n.

Step 1: We show that $m(z)/4 \geq u_n(z) - C\sqrt{n}$ for every $z \in S_n$. Let $\varphi := g^{::}_{S_n}(\cdot, z)$. Proposition 6.1 for the aggregation gives

$$\sum_x \varphi(x)(\mathbf{1}_{A_n^{\mathrm{R}}}(x) - \sigma_n(x)) = -\frac{m(z)}{4} + \sum_{x \in S_n} \sum_{y \sim x} \mathcal{D}_\rho(x, y, m(x))(\varphi(y) - \varphi(x)) + \sum_{x \notin S_n} \sum_{y \sim x} \Big(\frac{m(x)}{4} + \mathcal{D}_\rho(x, y, m(x))\Big)\varphi(y)\,.$$

The last sum is nonnegative, since each bracket counts the particles sent from x to y. The middle sum has absolute value at most $C\sqrt{n}$, by (6.1), symmetry of $g^{::}_{S_n}$ and the gradient bound of Lemma 6.2, applied in the disk $\mathrm{B}^{::}(\mathbf{0}, C\sqrt{n}) \supset S_n$ of (7.6). So the left side is at least $-m(z)/4 - C\sqrt{n}$. Since $\mathbf{1}_{A_n^{\mathrm{R}}} \leq 1$ and $\varphi \geq 0$, it is also at most $\sum_{x \in S_n} \varphi(x) - n\varphi(\mathbf{0}) - n\varphi(b_n) = -u_n(z)$ by (7.8).

Step 2: We deduce the inclusion. For $z \in \sqrt{n}K \cap \mathbb{Z}^2$, the bound (7.7) gives $u_n(z) \geq c_K n > 0$ for large n, so $z \in S_n$, and Step 1 gives $m(z)/4 \geq c_K n - C\sqrt{n} > 0$ uniformly over the rotor stacks. A site that makes a legal move retains a particle forever, so $z \in A_n^{\mathrm{R}}$. □

To prove the outer inclusion we must rule out long thin tentacles of occupied sites. The next lemma shows that an occupied site with no source nearby has a quadratic number of occupied sites around it.

Lemma 7.8 (No Thin Tentacles) *There is an absolute constant $c > 0$ with the following property. Let $\sigma : \mathbb{Z}^2 \to \mathbb{N}$ have finite support, let A^{R} be the rotor-router aggregate of σ for fixed rotor stacks, and let $x \in A^{\mathrm{R}}$ and $s \geq 1$ satisfy $\sigma = 0$ on $\mathrm{B}^{::}(x, 2s)$. Then*

$$\#(A^{\mathrm{R}} \cap \mathrm{B}^{::}(x, s)) \geq cs^2\,.$$

Proof. We find a large odometer value near x, then use it to force an occupied disk (Figure 7.8). We may take s larger than a fixed absolute constant, because decreasing c covers bounded s using the occupied site x. Let m be the rotor odometer and $f := m/4$. For neighbors v, w, let $e(v, w) := \mathcal{D}_\rho(v, w, m(v))$, so that $|e| \leq 3/4$ by (6.1), and $\sum_{w \sim v} e(v, w) = 0$

since v sends $m(v)$ particles in all. Since w sends $m(w)/4 + e(w,v)$ particles to v, counting arrivals and departures at v gives

$$\Delta^{::} f(v) = \mathbf{1}_{A^{\mathrm{R}}}(v) - \sigma(v) - \sum_{w \sim v} e(w,v)\,.$$

Hence $|\Delta^{::} f| \le 4$ where $\sigma = 0$, and $f = 0$ off A^{R}, since a site that makes a legal move retains a particle. So Lemma 7.6, with $\lambda = 3/4$, applies to f on every disk without sources; let C_0 be its constant.

Step 1: We find a site y near x with $f(y) > s^2/64$. Let T be the maximum of m on $\mathrm{B}^{::}(x, s/4)$, and for integers $0 \le j < s/4$ let a_j count the occupied sites at distance at most j from x. Let $1 \le j < s/4$. Since $\sigma = 0$ on $\mathrm{B}^{::}(x, 2s)$, every particle that ends within distance $j-1$ of x entered that ball by a move from an occupied site at distance in $(j-1, j]$, and each such site makes at most T moves. Hence $a_{j-1} \le T(a_j - a_{j-1})$. Since $a_0 = 1$, this gives $T \ge 1$, and multiplying the inequalities $a_j/a_{j-1} \ge 1 + 1/T$ gives $(1+1/T)^j \le a_j \le (2j+1)^2$. Taking $j = \lceil s/4 \rceil - 1 \ge s/8$ and using $\log(1+1/T) \ge 1/(2T)$, we get $T \ge s/(32\log(2+s))$. Let z_1 attain T, and let $r_1 := s/(512 C_0 \log(2+s))$. Then $f(z_1) \ge 4C_0 r_1$, and for large s we have $1 \le r_1 \le s/4 - 2$ and $r_1^2 > C_0 s$. Lemma 7.6 with center z_1 and radius r_1 gives a site y_1 at distance less than $r_1 + 1$ from z_1 with $f(y_1) > r_1^2/4 > C_0 s/4$. The lemma with center y_1 and radius $s/4$ then gives a site y with

$$|y - x| < s/2 + r_1 + 2 \le 3s/4 \qquad \text{and} \qquad f(y) > s^2/64\,. \tag{7.9}$$

Both disks used here lie in $\mathrm{B}^{::}(x, 2s)$, where $\sigma = 0$.

Step 2: We show that $\mathrm{B}^{::}(y, \theta s) \subset A^{\mathrm{R}}$ for $\theta := 1/(8\sqrt{153})$. Suppose that a vacant site w satisfies $|w - y| < \theta s$, and let $B := \mathrm{B}^{::}(w, 2\theta s)$. Since $\theta < 1/8$, the bound (7.9) gives $B \subset \mathrm{B}^{::}(x, 2s)$, so $|\Delta^{::} f| \le 4$ on B. The function $g := f + \sum_{v \in B} g_B^{::}(\cdot, v)\Delta^{::} f(v)$ equals f off B and, by symmetry and (3.19), is harmonic in B, so $g \ge 0$ by the minimum principle, Lemma 3.2. By the normalization (3.18) and the exit-time bound, Lemma 3.4, for $z \in B$

$$|g(z) - f(z)| \le 4 \sum_{v \in B} g_B^{::}(z, v) = \mathbb{E}_z[\tau_{\partial B}] \le (2\theta s + 1)^2 \le 9\theta^2 s^2\,.$$

As $f(w) = 0$, we get $g(w) \le 9\theta^2 s^2$. For large s, the discrete Harnack inequality, Lemma 3.16, gives $g \le 16 g(w) \le 144\theta^2 s^2$ on $\mathrm{B}^{::}(w, \theta s) \ni y$. Hence $f(y) \le 153\theta^2 s^2 = s^2/64$, contradicting (7.9). So $\mathrm{B}^{::}(y, \theta s)$ is occupied. It lies in $\mathrm{B}^{::}(x, s)$ and contains the lattice points v with $|v - y|_\infty \le \theta s/2$, at least $(\theta s)^2/4$ of them once $\theta s \ge 2$. □

We now prove part (ii), using that each aggregate has $2n$ sites.

Proof of Theorem 7.1(ii). The inner inclusion follows from Lemma 7.7 with $K = \overline{\Omega(p)_\epsilon}$. For the outer inclusion, let c be the constant in Lemma 7.8.

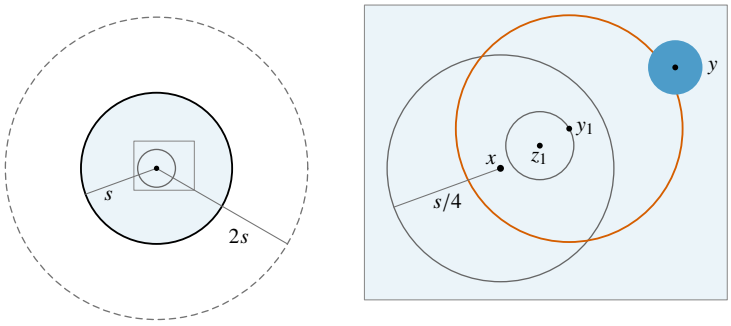


Figure 7.8: The disks in the proof of Lemma 7.8, with the small ones magnified on the right and r_1 and θs enlarged: $\mathrm{B}^{::}(x, 2s)$ (dashed) contains no source, and Step 1 takes $z_1 \in \mathrm{B}^{::}(x, s/4)$ where the rotor odometer is largest there and finds y_1 just outside $\mathrm{B}^{::}(z_1, r_1)$ and y just outside $\mathrm{B}^{::}(y_1, s/4)$ (vermilion). Step 2 shows that $\mathrm{B}^{::}(y, \theta s)$ (blue) is occupied, so $\mathrm{B}^{::}(x, s)$ (shaded) contains at least cs^2 occupied sites.

Choose $\delta > 0$ so that $|\Omega(p)_{2\delta}| > 2 - c\epsilon^2/128$. This is possible by continuity of area from below, since as $\delta \downarrow 0$ the sets $\Omega(p)_{2\delta}$ increase to $\Omega(p)$, which has area two by Theorem 7.2. For large n, each point of $\sqrt{n}\Omega(p)_{2\delta}$ lies in a cell centered in $\sqrt{n}\Omega(p)_{\delta}$, so Lemma 7.7 with $K = \overline{\Omega(p)_{\delta}}$ accounts for at least $n|\Omega(p)_{2\delta}|$ occupied sites. If an occupied site x has $\mathrm{dist}(x, \sqrt{n}\Omega(p)) \geq \epsilon\sqrt{n}$, then $\mathrm{B}^{::}(x, \epsilon\sqrt{n}/4)$ contains no source for large n, since the rescaled sources tend to points in $\Omega(p)$. Lemma 7.8 with $s = \epsilon\sqrt{n}/8$ adds at least $c\epsilon^2 n/64$ occupied sites in $\mathrm{B}^{::}(x, \epsilon\sqrt{n}/8)$, which lies outside $\sqrt{n}\Omega(p)$, giving more than $2n$ sites. This contradiction proves the outer inclusion. All constants are independent of the rotor stacks. □

Figure 7.9 shows the rotor-router aggregate from two sources, whose boundary follows the quartic (7.1). In the next section we treat internal DLA, where a probability bound replaces the deterministic discrepancy estimate.

7.5 INTERNAL DLA

We again use the toppled set S_n and the representation (7.8) of u_n. The abelian property for instruction stacks, Section 5.1, allows us to pause particles when they leave S_n and to restart them later. (It also permits the two-stage construction of Figure 7.10.)

Proof of Theorem 7.1(iii). We first bound the probability of an interior vacancy, then restart the few particles that leave S_n to obtain the outer bound.

Step 1: List the $2n$ starting sites, with multiplicity, as $s_1, \ldots, s_{2n}$. Release particles in this order, settling and pausing them as in the product bound for vacancy probabilities, Lemma 5.4, with $B = S_n$, and let I_{2n} be the set of sites where particles settle. Restarting the paused walks and completing

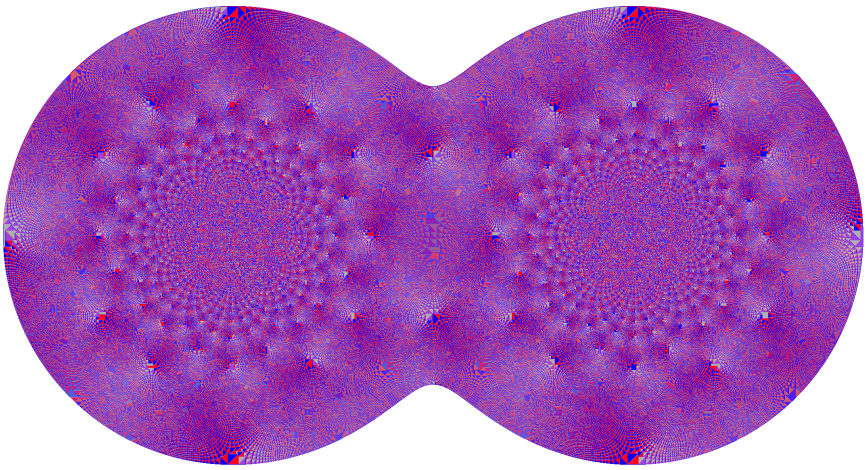


Figure 7.9: Rotor-router aggregation of $n = 10^6$ particles from each of the sites $(\pm 500, 0)$, colored by the final direction of each rotor as in Figure 1.1. Its boundary lies within half a site of the quartic $(x^2+y^2)^2 - \frac{2n}{\pi}(x^2+y^2) - 2\cdot 500^2(x^2-y^2) = 0$ in lattice coordinates, which is (7.1) with $a = 500/\sqrt{n}$ scaled by $\sqrt{n}$.

aggregation gives A_n, so $I_{2n} \subset A_n$ in this coupling.

Fix $z \in S_n$, and let $\varphi(y) := g^{\vdots}_{S_n}(y,z)$ and $L := \max_{y\in S_n} g^{\vdots}_{S_n}(y,y)$. The containment in (7.6), domain monotonicity of the killed Green function, and Lemma 3.12(b) (the value of the killed Green function of a disk at its center), applied in a disk of radius $2C\sqrt{n}$ centered at each $y \in S_n$, give $L \le C\log n$. Symmetry and the normalization (3.18) give

$$\sum_{y\in S_n} \varphi(y) = \tfrac{1}{4}\mathbb{E}_z[\tau_{\partial S_n}] \le Cn\,,$$

where the last bound is Lemma 3.4 on the containing disk.

Step 2: We bound the probability that z is vacant. For $n > 1$ both sources lie in S_n, so Lemma 5.4 applies with $d = 2$, $B = S_n$, $N = 2n$, the starting sites of Step 1, and $0 < \lambda \le 1/(2L)$. By (7.8), $\sum_{i=1}^{2n}\varphi(s_i) - \sum_{y\in S_n}\varphi(y) = u_n(z)$, so the bounds of Step 1 give

$$\log \mathbb{P}\{z \notin I_{2n}\} \le -\lambda u_n(z) + C\lambda^2 n\log n\,.$$

For a compact set $K \subset \Omega(p)$, the bound (7.7) permits a choice $\lambda = c_K/(2C\log n)$, after increasing C, and yields

$$\mathbb{P}\left\{\sqrt{n}K \cap \mathbb{Z}^2 \not\subset I_{2n}\right\} \le Cn\exp(-c_K^2 n/(4C\log n))\,. \tag{7.10}$$

Here the factor Cn counts the possible sites. The bound is summable, and it gives the inner inclusion.

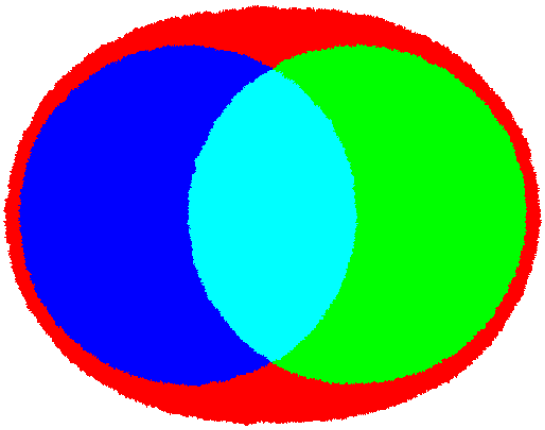


Figure 7.10: Internal DLA from 8×10^4 particles at each of the sites $(\pm 80, 0)$, built in two stages: the particles of each source perform internal DLA alone (blue and green, overlapping in cyan), and then one particle from each cyan site walks until it reaches an unoccupied site (red). The result has the law of internal DLA from both sources.

Step 3: We restart the paused particles to prove the outer inclusion. Fix $\epsilon > 0$, and let b, c_0, c_1 be the constants in the cube estimate, Lemma 5.8, for $d = 2$. Choose $\delta > 0$ with $|\Omega(p)_{2\delta}| > 2 - b\epsilon^2/400$. On the event $G_n := \{\sqrt{n}\Omega(p)_\delta \cap \mathbb{Z}^2 \subset I_{2n}\}$, counting as in the proof of part (ii) shows that at most $b\epsilon^2 n/400$ particles are paused. The complementary probabilities are summable by (7.10) with $K = \overline{\Omega(p)_\delta}$.

Write $\varpi := \lfloor \epsilon\sqrt{n}/10 \rfloor$. For a site x at distance at least $\epsilon\sqrt{n}$ from $\sqrt{n}\Omega(p)$, the cube $Q^{\ddot{}}(x, 3\varpi)$ of sites y with $|y - x|_\infty \le 3\varpi$ lies within distance $\epsilon\sqrt{n}/2$ of x. By (7.6) and the convergence of $\Omega(p_n)$ to $\Omega(p)$ from Lemma 7.4(i), this cube contains neither settled nor paused particles for large n. On G_n there are at most $b\varpi^2$ paused particles. Conditionally on the stopped walks, their continuations are independent simple random walks. Lemma 5.8, translated to x, therefore bounds the probability that a restarted particle reaches x by $c_0 e^{-c_1\varpi}$. If there are no paused particles, the probability is zero. Every occupied site is joined to a source by a path of at most $2n$ occupied sites, since a newly occupied site is a source or adjacent to an earlier occupied site. Hence $A_n \subset \mathrm{B}^{\ddot{}}(\mathbf{0}, Cn)$, and a union bound over the at most Cn^2 sites of $\mathrm{B}^{\ddot{}}(\mathbf{0}, Cn)$ gives

$$\mathbb{P}\left\{A_n \not\subset \sqrt{n}\Omega(p)^\epsilon\right\} \le \mathbb{P}(G_n^c) + Cn^2 c_0 e^{-c_1\varpi}.$$

This bound is summable as well, so the Borel–Cantelli lemma gives both inclusions eventually almost surely for each fixed ϵ, using no independence between clusters with different n. Intersecting these probability-one events for $\epsilon = 1/j$ with $j \ge 1$ and using monotonicity of the neighborhoods gives

every $\epsilon > 0$ simultaneously. □

The three models now have the same limit. We finish by applying its explicit formula to the way the two sources approach each other and to a consequence for harmonic functions.

7.6 SEPARATION AND COALESCENCE

The waist of the oval O_a, its width along the imaginary axis, is $2\sqrt{2r_0^2 - 2a^2}$, obtained by setting $\theta = \pi/2$ in its polar equation. It shrinks to zero at $a = r_0$, exactly when two disks of area one become tangent (Figure 1.12 shows rotor-router aggregation as the sources separate). Thus if $b_n/\sqrt{n} \to p$ with $|p| = 2r_0$, then Theorem 7.1 gives the same pair of tangent disks whether $|b_n|/\sqrt{n}$ stays below $2r_0$ or above it, with no condition on the rate of approach. Likewise, if $|b_n| = o(\sqrt{n})$, then the limit is the disk of area two even though the two sources $\mathbf{0}$ and b_n may remain distinct.

Theorem 7.2 also gives an integration formula: Exercise 7.2 asks you to prove that the integral over $\Omega(q)$ of every function h harmonic near the closure of $\Omega(q)$ is $h(\mathbf{0}) + h(q)$. A domain on which the integral of every integrable harmonic function is a combination of its values at finitely many points is called a **quadrature domain**, and such domains link these aggregation models to complex analysis. In Chapter 8, particles arrive from far away instead of from interior sources, and the clusters grow branching arms.

EXERCISES

Exercise 7.1. Let $q \in \mathbb{C}$, and recall the potential U_q of Section 7.2. Use its positivity on $\Omega(q)$ and its vanishing on the boundary of $\Omega(q)$, from Theorem 7.2, rather than the explicit boundary, to prove that every connected component of $\Omega(q)$ contains $\mathbf{0}$ or q. Conclude that $\Omega(q)$ has at most two components.

Exercise 7.2. Let $q \in \mathbb{C}$, and let h be harmonic near $\overline{\Omega(q)}$. Prove from Theorem 7.2 that

$$\int_{\Omega(q)} h(z)\, \mathrm{d}A(z) = h(\mathbf{0}) + h(q)\,.$$

Hint: test the distributional Laplacian of U_q against h multiplied by a smooth cutoff equal to one near $\overline{\Omega(q)}$.

NOTES

Theorem 7.1 is the planar case, with two equal sources, of Levine and Peres (2010, Theorem 1.4). Their theorem allows finitely many sources with positive masses in every dimension $d \geq 2$ and places the sources at the lattice points nearest to fixed locations, whereas Theorem 7.1 allows an arbitrary second source b_n with $b_n/\sqrt{n} \to p$. Their almost sure statement for internal DLA assumes that the nth lattice spacing is at most $1/n$; here the spacing is $n^{-1/2}$, and the summable bounds of Section 7.5 give convergence along the whole sequence under every coupling. Our sandpile proof compares the odometer directly with the potential of Theorem 7.2, and our particle comparisons use the killed Green function of the toppled set. The cube estimate, Lemma 5.8, is based on Lemma 5.12 of Levine and Peres (2010). Lemma 7.8 takes its name from Proposition 4.2 of Levine and Peres (2017), which proves in $\mathbb{Z}^d$, for n particles started at the origin, every initial rotor configuration and every simple rotor mechanism, that at least a constant times s^d of the sites within distance s of each occupied site x with $|x| > s$ are occupied; the shell-counting step of its proof follows Lemmas 4.9 and 4.10 of Levine and Peres (2010). Our lemma is the planar version for initial configurations vanishing on $\mathrm{B}^{::}(x, 2s)$. Its internal DLA analogue is Lemma 2 of Jerison, Levine, and Sheffield (2012), described in the Notes to Chapter 5.

Theorem 7.2 gives the logarithmic potential of Neumann's oval; the survey Gustafsson and Shapiro (2005) traces the theory of quadrature domains back to papers of Neumann on this potential from the beginning of the twentieth century. Equation (11) of Levine and Peres (2010) gives the quartic (7.1) for the smash sum of two equal disks and credits it to Shapiro, and their Figure 3 is reproduced in Figure 7.9. Our proof checks the moments and the sign of the potential by direct computation. The two-stage construction of Figure 7.10 is the smash sum introduced by Diaconis and Fulton (1991), also described in Section 1 of Levine and Peres (2010). The quadrature identity of Exercise 7.2 is the harmonic case of Proposition 6.6 of Levine and Peres (2010), a quadrature inequality for superharmonic functions on smash sums of balls.

For finitely many planar point sources with distinct locations and positive masses, the boundary of the common limit lies, up to finitely many points, on an algebraic curve of degree $2k$ for k sources. This follows from the quadrature identity and the work of Aharonov and Shapiro (1976) and of Gustafsson on quadrature domains, discussed in Sections 3 and 4 of Gustafsson and Shapiro (2005) and applied to aggregation in Section 1 of Levine and Peres (2010).

Chapter Eight

External DLA in $\mathbb{Z}^d$

External diffusion-limited aggregation builds a random cluster $\mathcal{X}_n \subset \mathbb{Z}^2$ one particle at a time. Starting from $\mathcal{X}_0 = \{\mathbf{0}\}$, we release a random walk far from the cluster and add the first boundary site it visits; the limit as the starting point recedes to infinity defines the attachment probabilities. Its clusters develop long branching arms separated by deep gaps, much like patterns in electrodeposition, viscous fingering, and dielectric breakdown (Figure 8.1).

How quickly does the radius grow? Off-lattice simulations suggest that a cluster of radius r has about r^{d_f} particles, with fractal dimension $d_f \approx 1.71$, so n particles have radius about n^{1/d_f} with $1/d_f \approx 0.585$. On the square lattice, large clusters align their arms with the axes, and this anisotropy may change how the radius grows (Figures 8.2, 8.3 and 8.4). No proof is known that the radius grows like a power of n.

The main result of this chapter is an upper bound on the radius of the planar cluster: with high probability, $\mathcal{X}_n$ lies in a ball of radius of order $n^{2/3}$. This exponent exceeds the off-lattice value 0.585, but the bound already requires a sharp estimate for planar random walk.

The argument separates growth from random walk: if every boundary site of a cluster of radius r receives conditional probability at most $Cr^{-\beta}$, then with high probability a cluster of n particles has radius at most a constant times $n^{1/(1+\beta)}$, and the discrete Beurling estimate gives $\beta = \frac{1}{2}$.

The attachment probabilities come from the harmonic measure from infinity of Section 3.7 in Chapter 3: recall that the harmonic measure $\mathcal{H}_K$ of a finite set $K \subset \mathbb{Z}^2$ from infinity is the limiting hitting distribution on K of a walk started far away, and that it is the unique probability measure on K whose potential, formed with the potential kernel, is constant on K. Section 8.1 proves the radius bound for general β. Section 8.2 proves the discrete Beurling estimate and shows that $\beta = \frac{1}{2}$ is sharp on a horizontal segment. Section 8.3 combines them to bound the radius by a constant times $n^{2/3}$ (Theorem 8.8). In dimensions $d \geq 3$ a walk from far away may miss the cluster, so the starred Section 8.4 conditions it to hit, and the capacity of the cluster takes the place of the Beurling estimate (Theorems 8.14 and 8.16).

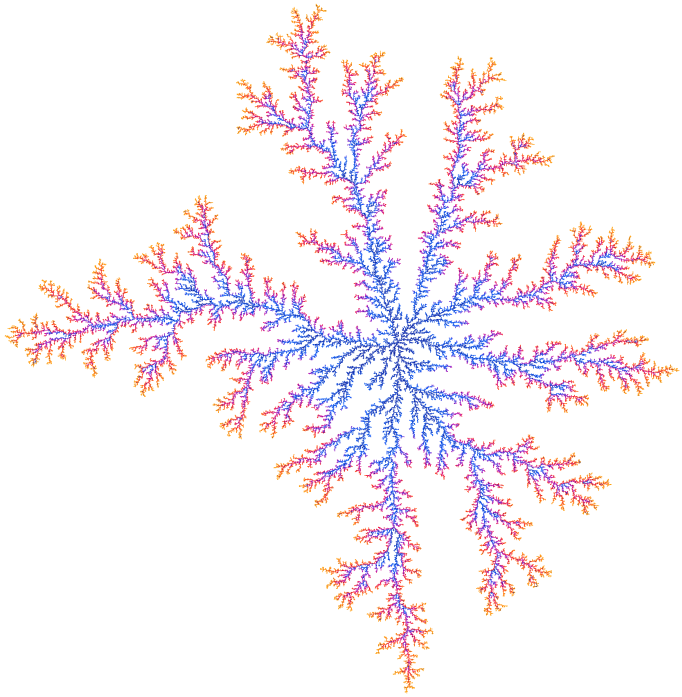


Figure 8.1: An external DLA cluster of 150,000 sites on $\mathbb{Z}^2$, each site colored by the rank of its harmonic measure from infinity, the probability that a walk from far away first hits the cluster there, from navy (smallest, including zero) through blue, magenta and red to orange (largest). Harmonic measure is largest at exposed tips and small deep between two arms.

Internal DLA, by contrast, has a spherical limit shape. Chapter 5 proves inner and outer bounds on its fluctuations, and in the plane they are only logarithmic (see the Notes). The proof uses martingales, which exist because every particle starts at the origin, and the nearly spherical shape of the cluster. External DLA has neither feature: its particles arrive from infinity, and its clusters have no such shape.

8.1 RADIUS FROM ATTACHMENT PROBABILITIES

Recall from Definition 3.28 that H_A is the harmonic measure from infinity of the outer boundary ∂A of a finite set $A \subset \mathbb{Z}^2$. Until Section 8.4, X is simple random walk on $\mathbb{Z}^2$, $\mathbb{P}_y$ and $\mathbb{E}_y$ denote its law and expectation started at y, and τ_E is its hitting time of $E \subset \mathbb{Z}^2$, as in Section 3.7.

The harmonic measure $H_{\mathcal{X}_n}$ of the outer boundary of $\mathcal{X}_n$ now defines the next attachment. A walk released from infinity first visits $\partial\mathcal{X}_n$ at the site added to the cluster, as in Figure 8.5.

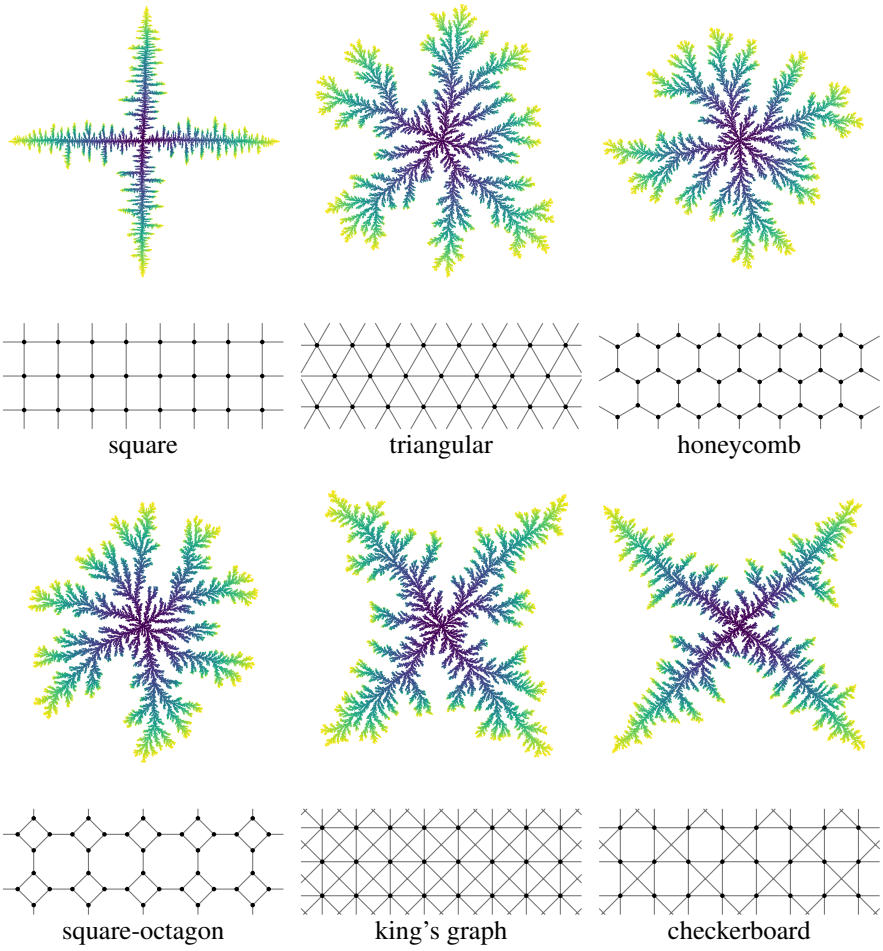


Figure 8.2: External DLA clusters above a few cells of their lattices, colored from dark (early) to light (late) by time of addition: 10^9 sites on the four planar lattices and 10^8 on the two lattices whose edges cross. The arms grow along the axes on the square lattice and along the diagonals on the king's graph and the checkerboard lattice.

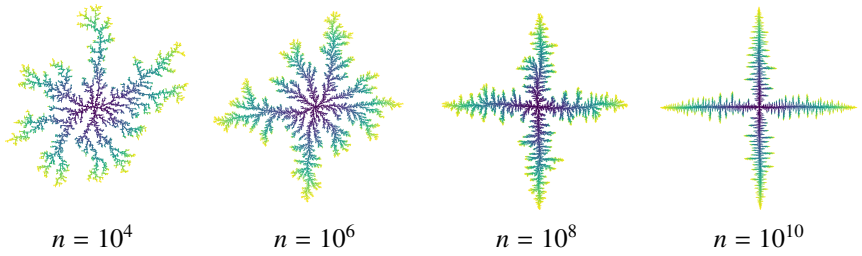


Figure 8.3: External DLA clusters on the square lattice with 10^4, 10^6, 10^8 and 10^{10} sites, each scaled to the same width; the first two are one cluster at two times. The small clusters branch in every direction, but as the cluster grows, four arms along the axes take over and their side branches shrink.

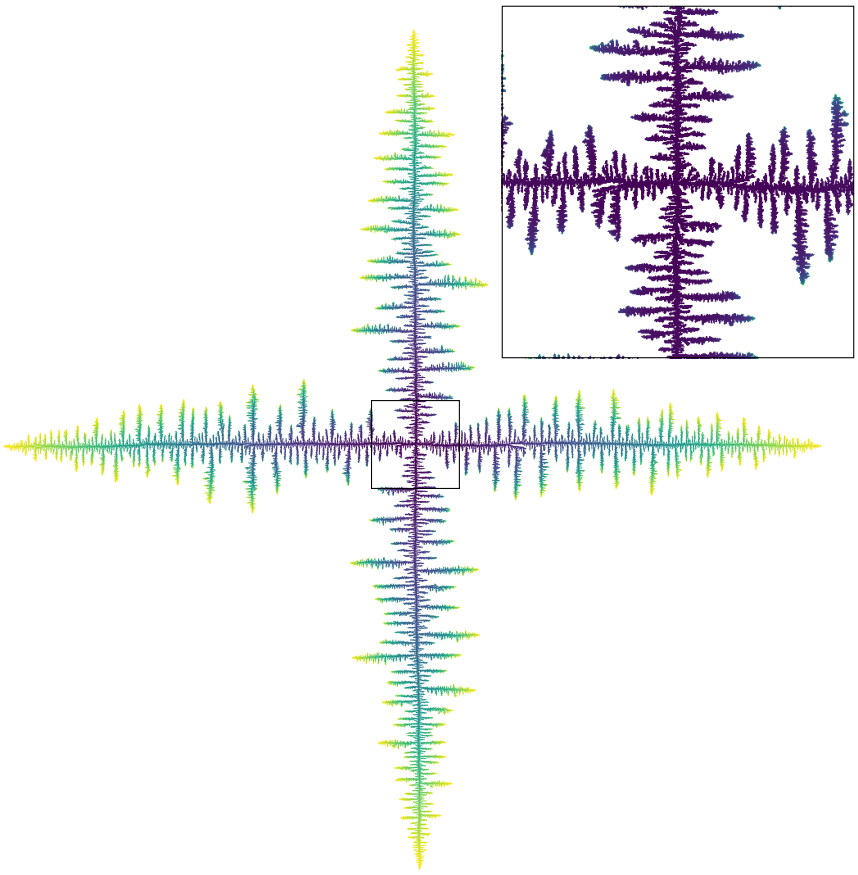


Figure 8.4: The external DLA cluster of 10^{10} sites of Figure 8.3, drawn in blocks of 512×512 sites: each block that meets the cluster is colored from dark to light by the time at which the first of its sites joined. Its arms reach about 1.07×10^6 sites from the origin, and the inset magnifies the small central square fourfold.

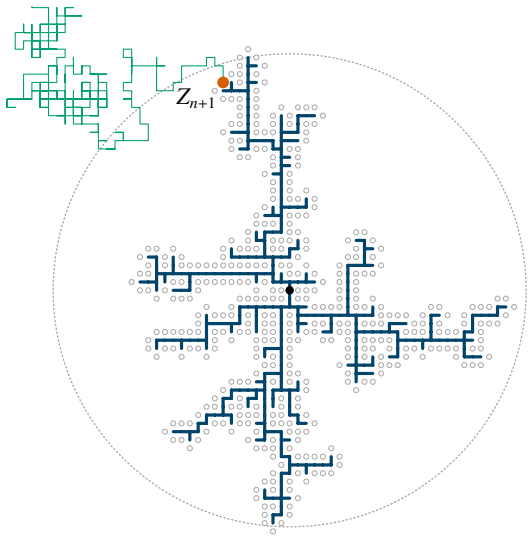


Figure 8.5: One step of external DLA: a walk arriving from far away (green, last steps only) first visits the outer boundary (open dots) of a cluster $\mathcal{X}_n$ of 320 sites, drawn as a tree, at the site Z_{n+1} that joins the cluster. Before then it avoids the connected set $\mathcal{X}_n \cup \{Z_{n+1}\}$, which reaches the dotted circle of radius $\mathrm{rad}(\mathcal{X}_n)$, so by the Beurling estimate each attachment has probability at most $C(1 + \mathrm{rad}(\mathcal{X}_n))^{-1/2}$, as in (8.4).

For every integer $d \geq 1$ and every finite nonempty set $A \subset \mathbb{R}^d$, write $\mathrm{rad}(A) := \max\{|u| : u \in A\}$ for its radius.

Definition 8.1 (External DLA) Let $\mathcal{X}_0 = \{\mathbf{0}\}$, and let $\mathcal{F}_n$ be the σ-algebra generated by $\mathcal{X}_0, \dots, \mathcal{X}_n$. Given $\mathcal{X}_n$, choose $Z_{n+1} \in \partial\mathcal{X}_n$ with conditional law

$$\mathbb{P}\{Z_{n+1} = z \mid \mathcal{F}_n\} = H_{\mathcal{X}_n}(z) \qquad \text{for every } z \in \partial\mathcal{X}_n\,,$$

and let

$$\mathcal{X}_{n+1} := \mathcal{X}_n \cup \{Z_{n+1}\}\,.$$

The process $(\mathcal{X}_n)_{n\geq 0}$ is **external DLA** on $\mathbb{Z}^2$.

The bound on the radius requires only an upper bound on the probability of each attachment, so we state it for a general nearest-neighbor growth process in $\mathbb{Z}^d$.

Proposition 8.2 *Let $d \geq 1$ be an integer, let $\beta > 0$, and let $0 < J < \infty$. Let $(\mathcal{X}_n)_{n\geq 0}$ be a process of finite subsets of $\mathbb{Z}^d$ with $\mathcal{X}_0 = \{\mathbf{0}\}$, and let $(Z_n)_{n\geq 1}$ be random sites such that, for every $n \geq 0$,*

$$Z_{n+1} \in \partial\mathcal{X}_n \qquad \text{and} \qquad \mathcal{X}_{n+1} = \mathcal{X}_n \cup \{Z_{n+1}\}\,.$$

Let $\mathcal{F}_n$ be the σ-algebra generated by $\mathcal{X}_0, \dots, \mathcal{X}_n$. Suppose that, for every $n \geq 0$ and every $z \in \partial\mathcal{X}_n$,

$$\mathbb{P}\{Z_{n+1} = z \mid \mathcal{F}_n\} \leq J(1 + \mathrm{rad}(\mathcal{X}_n))^{-\beta}\,.$$

Then there exist $C_0, C, c > 0$, depending only on d, β, J, such that, for every $n \geq 1$ and every $R \geq C_0 n^{1/(1+\beta)}$,

$$\mathbb{P}\left\{\mathcal{X}_n \not\subset \mathrm{B}^{\ddots}(\mathbf{0}, R)\right\} \leq C\exp(-cR)\,.$$

We count sites beyond successive integer radii. Their expected counts satisfy a recursion solved by Pascal's identity, giving a tail bound by Markov's inequality.

Proof of Proposition 8.2. Let $C_1 := \max\{1, 2dJ\}$. For integers $t \geq 0$ and $k \geq 1$, write $N_k(t)$ for the number of sites $x \in \mathcal{X}_t$ with $|x| \geq k$. We first show that

$$\mathbb{E}N_k(t) \leq \frac{C_1^k}{(k!)^\beta}\binom{t}{k}, \tag{8.1}$$

where $\binom{t}{k} = 0$ when $k > t$.

We start with a single step. For $k \geq 2$, each boundary site z with $|z| \geq k$ has a cluster neighbor of norm at least $k-1$ (Figure 8.6). There are therefore at most $2dN_{k-1}(t)$ such sites. For each, $1 + \mathrm{rad}(\mathcal{X}_t) \geq |z| \geq k$, so its conditional attachment probability is at most $Jk^{-\beta}$. Since $N_k(t+1) - N_k(t) = \mathbf{1}\{|Z_{t+1}| \geq k\}$, summing these probabilities gives

$$\mathbb{E}[N_k(t+1) - N_k(t) \mid \mathcal{F}_t] \leq \frac{C_1}{k^\beta}N_{k-1}(t)\,.$$

We now prove (8.1) by induction on t. For $t = 0$ both sides vanish, since $\mathcal{X}_0 = \{\mathbf{0}\}$. For $k = 1$, we have $N_1(t) \leq t \leq C_1 t$. Assume the bound at time t for every $k \geq 1$. For $k \geq 2$, taking expectations in the previous display and using the induction hypothesis gives

$$\mathbb{E}N_k(t+1) \leq \frac{C_1^k}{(k!)^\beta}\left[\binom{t}{k} + \binom{t}{k-1}\right] = \frac{C_1^k}{(k!)^\beta}\binom{t+1}{k}.$$

Pascal's identity gives the equality and completes the induction.

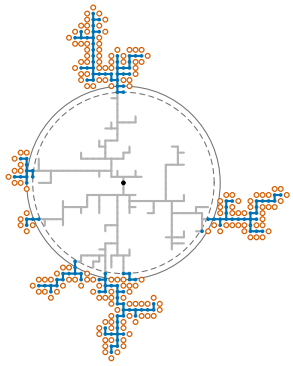


Figure 8.6: The cluster $\mathcal{X}_{320}$ of Figure 8.5 and the circles of radii $k-1$ (dashed) and k, for $k = 16$. Each site $z \in \partial\mathcal{X}_{320}$ with $|z| \geq k$ (vermilion) has a neighbor in the cluster of norm at least $k-1$ (blue), so there are at most $4N_{k-1}(320)$ of them, and each is added next with conditional probability at most $Jk^{-\beta}$.

Next we bound the tail. Fix an integer $n \geq 1$ and a real $R \geq 2$. Let $m := \lfloor R \rfloor$, so $R/2 \leq m \leq R$. Leaving $\mathrm{B}^{::}(\mathbf{0}, R)$ forces $N_m(n) \geq 1$. Markov's inequality, (8.1), $\binom{n}{m} \leq n^m/m!$ and $m! \geq (m/e)^m$ (from $e^m \geq m^m/m!$) give

$$\mathbb{P}\{\mathcal{X}_n \not\subset \mathrm{B}^{::}(\mathbf{0}, R)\} \leq \mathbb{E}N_m(n) \leq \frac{C_1^m n^m}{(m!)^{1+\beta}} \leq \left(\frac{(2e)^{1+\beta} C_1 n}{R^{1+\beta}}\right)^m , \tag{8.2}$$

where the last step also uses $m \geq R/2$.

Finally, choose $C_0 \geq 2$ so that $C_0^{1+\beta} \geq e(2e)^{1+\beta} C_1$. If $R \geq C_0 n^{1/(1+\beta)}$, then the base in (8.2) is at most e^{-1}, giving a probability at most $e^{-m} \leq e^{-R/2}$. Thus $C = 1$ and $c = 1/2$ prove the proposition. □

For R much larger than $n^{1/(1+\beta)}$, Exercise 8.1 asks you to deduce the sharper tail $C\exp[-cR(1 + \log(R/n^{1/(1+\beta)}))]$ from (8.2); the applications here use only the proposition.

For planar external DLA, the attachment bound of Proposition 8.2 holds with $\beta = \frac{1}{2}$, which gives the radius scale $n^{2/3}$. The bound comes from a sharp estimate for the harmonic measure of a connected set, the discrete Beurling estimate. The next section proves it with the lattice potential kernel and shows that the exponent $\frac{1}{2}$ is sharp at the endpoint of a horizontal segment.

8.2 THE DISCRETE BEURLING ESTIMATE

How likely is a random walk started far from a finite connected set K to hit K first at a given site x? Write $\mathrm{diam}(K) := \max\{|y - z| : y, z \in K\}$ for the diameter of K. The set K contains a nearest-neighbor path from x to a point at distance at least $\mathrm{diam}(K)/2$ from x, and a walk that hits K first at x must

avoid every other site of this path. The discrete Beurling estimate says that this is unlikely when the path is long: the harmonic measure from infinity of Definition 3.28 satisfies

$$\mathcal{H}_K(x) \le C(1 + \operatorname{diam}(K))^{-1/2} .$$

The exponent $\frac{1}{2}$ is the best possible, since it is attained at the endpoint of a straight segment, as Proposition 8.7 shows.

8.2.1 The Upper Bound

Our proof is the lattice version of the proof of the Beurling estimate for Brownian motion, Theorem 2.44 in the starred Section 2.6, with the potential kernel a in place of the logarithm, but it does not use that section. It uses the potential U_A of Lemma 3.29 in Chapter 3 and the asymptotic expansion $a(x) = \frac{1}{2\pi}\log|x| + \kappa + O(|x|^{-2})$ of (3.34), where κ is a constant. By part (b) of the lemma, the harmonic measure of a finite set A is the Laplacian of U_A; the function U_A vanishes on A, is harmonic off A, and grows like a at infinity. If F is harmonic off A and $F - a$ is bounded, then by (3.37) we have

$$U_A \le F - \min_A F \qquad \text{and} \qquad \mathcal{H}_A(x) \le \sum_{u\sim x} (F(u) - \min_A F) \quad \text{for } x \in A .$$

We build this comparison function from the arcsine law $\text{Arcsine}_{0,n}$ on $[0, n]$, whose logarithmic potential is constant on this interval by Lemma 2.42(a). For a set with a point at each distance between $j - 1$ and j from x, for $1 \le j \le n$, we give that point the mass that $\text{Arcsine}_{0,n}$ gives to the interval $[j - 1, j]$. Replacing $a(z)$ by $\frac{1}{2\pi}\log r + \kappa$, for r within two of $|z|$, costs at most a constant times $\log(1 + 1/r)$, and the arcsine law averages this error to order $n^{-1/2}$.

The first lemma bounds this average, using only Lemma 2.42(b).

Lemma 8.3 *Let $n > 0$ be real, and let $\text{Arcsine}_{0,n}$ be the arcsine law on $[0, n]$. Then, for every real number s,*

$$\int \log\Bigl(1 + \frac{1}{|s - t|}\Bigr) \text{Arcsine}_{0,n}(\,\mathrm{d}t) \le \frac{4}{\sqrt{n}} .$$

Proof. We parametrize the arcsine law by an angle, which turns $|s - t|$ into the geometric mean of two variables with the arcsine law. Moving s to the nearest point of $[0, n]$ can only increase the integrand, so we may assume that $s = n\sin^2\varphi$ with $0 \le \varphi \le \pi/2$. Let Θ be uniform on $[0, \pi]$. The substitution $t = n\sin^2\theta$ shows that $T := n\sin^2\Theta$ has law $\text{Arcsine}_{0,n}$, the projection of the uniform distribution on a circle in Figure 2.13, and so do $T_\pm := n\sin^2(\Theta \pm \varphi)$, because $\sin^2$ has period π. The

identity $\sin^2\theta - \sin^2\varphi = \sin(\theta+\varphi)\sin(\theta-\varphi)$ gives $|T-s| = \sqrt{T_+T_-}$, and expanding shows that $(1+1/\sqrt{T_+T_-})^2 \le (1+1/T_+)(1+1/T_-)$ almost surely, the difference being $(T_+^{-1/2} - T_-^{-1/2})^2$. Taking logarithms and expectations bounds the integral in the lemma by $\int \log(1+1/t)\,\mathrm{Arcsine}_{0,n}(\,\mathrm{d}t)$, which is at most $2/\sqrt{n}$ by Lemma 2.42(b) with $q=1$. □

The second lemma compares the potential kernel with the logarithm of an approximate distance.

Lemma 8.4 *There is a constant $C_1 < \infty$ with the following property. Let $r > 0$ be real, and let $z \in \mathbb{Z}^2$. If $r \le |z| + 2$, then*

$$a(z) \ge \frac{1}{2\pi}\log r + \kappa - C_1 \log\Bigl(1+\frac{1}{r}\Bigr).$$

If $\bigl||z| - r\bigr| \le 2$, then

$$\Bigl|a(z) - \frac{1}{2\pi}\log r - \kappa\Bigr| \le C_1 \log\Bigl(1+\frac{1}{r}\Bigr).$$

Proof. Since $\log(2+|z|) - \log|z| \le 2/|z|$ for $z \ne \mathbf{0}$, the expansion (3.34) gives a constant C_0 with $\bigl|a(z) - \frac{1}{2\pi}\log(2+|z|) - \kappa\bigr| \le C_0/(2+|z|)$ for every $z \in \mathbb{Z}^2$ (the finitely many sites near the origin only enlarge C_0). Both assertions follow with $C_1 = 2C_0 + 2/\pi$: if $r \le |z|+2$, then $\log r \le \log(2+|z|)$ and $1/(2+|z|) \le 2/(1+r) \le 2\log(1+1/r)$, and if moreover $|z| \le r+2$, then $\log(2+|z|) \le \log(r+4) \le \log r + 4\log(1+1/r)$. □

The next theorem bounds the harmonic measure at a point x of a finite set that has a point at every distance from x up to $n-1$, within one unit. Only these distances matter, and the set need not be connected.

Theorem 8.5 (Discrete Beurling Estimate) *There is a constant $C < \infty$ with the following property. Let $n \ge 1$ be an integer, let $K \subset \mathbb{Z}^2$ be finite, and let $x \in K$. Suppose that, for every integer $1 \le j \le n$, the set K contains a point whose distance from x is at least $j-1$ and less than j. Then*

$$\mathcal{H}_K(x) \le \frac{C}{\sqrt{n}}.$$

Proof. We give n points of K, one at each distance, the masses of the arcsine law, and compare the resulting potential with the potential U_A of the set A of selected points, as in Lemma 3.29. By translation, we may take $x = \mathbf{0}$. For $1 \le j \le n$, choose $y_j \in K$ with $j-1 \le |y_j| < j$. These points are distinct, and $y_1 = \mathbf{0}$. Let $A := \{y_1, \dots, y_n\}$, let $\mathrm{Arcsine}_{0,n}$ be the arcsine law on $[0,n]$, and let

$$F(w) := \sum_{j=1}^{n} \mathrm{Arcsine}_{0,n}([j-1,j])\, a(w-y_j) \qquad \text{for } w \in \mathbb{Z}^2.$$

The masses add up to one, so F is harmonic off A and $F - a$ is bounded, and (3.37) applies to F. Write $y(t) \coloneqq y_j$ for $j - 1 < t < j$. Since $\mathrm{Arcsine}_{0,n}$ has no atoms, $F(w) = \int a(w - y(t)) \, \mathrm{Arcsine}_{0,n}(\,\mathrm{d}t)$, and $\big||y(t)| - t\big| < 1$ for $\mathrm{Arcsine}_{0,n}$-almost every t.

We first bound F from below on A. Let $v \in A$. For $\mathrm{Arcsine}_{0,n}$-almost every t with $0 < t < n$, we have $t \neq |v|$ and $\big||v| - t\big| \le |v - y(t)| + 1$, so the first assertion of Lemma 8.4, with $r = \big||v| - t\big|$, bounds $a(v - y(t))$ from below. Since $0 \le |v| < n$, integrating this bound against $\mathrm{Arcsine}_{0,n}$ and using Lemma 2.42(a) and Lemma 8.3 gives

$$F(v) \ge \frac{1}{2\pi} \log \frac{n}{4} + \kappa - \frac{4C_1}{\sqrt{n}} \, .$$

Next we bound F from above at the neighbors of the origin. Let $u \sim \mathbf{0}$. For $\mathrm{Arcsine}_{0,n}$-almost every t with $0 < t < n$, we have $\big||u - y(t)| - t\big| < 2$, because $|u| = 1$. The second assertion of Lemma 8.4 with $r = t$, Lemma 2.42(a) at $s = 0$, and Lemma 8.3 therefore give

$$F(u) \le \frac{1}{2\pi} \log \frac{n}{4} + \kappa + \frac{4C_1}{\sqrt{n}} \, .$$

By (3.37), it follows that $U_A(u) \le F(u) - \min_A F \le 8C_1/\sqrt{n}$. By part (b) of Lemma 3.29, $\mathcal{H}_A(\mathbf{0})$ is the sum of U_A over the four neighbors of the origin, so $\mathcal{H}_A(\mathbf{0}) \le 32C_1/\sqrt{n}$.

Finally, since $\mathbf{0} \in A \subset K$, a walk that first hits K at $\mathbf{0}$ also first hits A at $\mathbf{0}$. Letting its starting point tend to infinity gives $\mathcal{H}_K(\mathbf{0}) \le \mathcal{H}_A(\mathbf{0}) \le 32C_1/\sqrt{n}$. □

For a connected set, the points required by Theorem 8.5 lie on a path to a farthest point: one step of the path changes the distance from x by at most one, so the path cannot jump over a range of distances of length one. This gives the discrete Beurling estimate in the form stated at the start of the section. Figure 8.7 shows how the points are selected.

Corollary 8.6 *There is a constant $C < \infty$ such that, for every finite connected set $K \subset \mathbb{Z}^2$ and every $x \in K$,*

$$\mathcal{H}_K(x) \le C(1 + \mathrm{diam}(K))^{-1/2} \, .$$

Proof. Choose $w \in K$ with $|w - x| \ge \mathrm{diam}(K)/2$; it exists because two points of K at distance $\mathrm{diam}(K)$ cannot both lie within distance less than $\mathrm{diam}(K)/2$ of x. Let $n \coloneqq \lfloor |w - x| \rfloor + 1$. Since K is connected, it contains a nearest-neighbor path from x to w. For $1 \le j \le n$, the path has a first point at distance at least $j - 1$ from x, because $|w - x| \ge n - 1$, and this point lies at distance less than j from x, because one step changes the distance

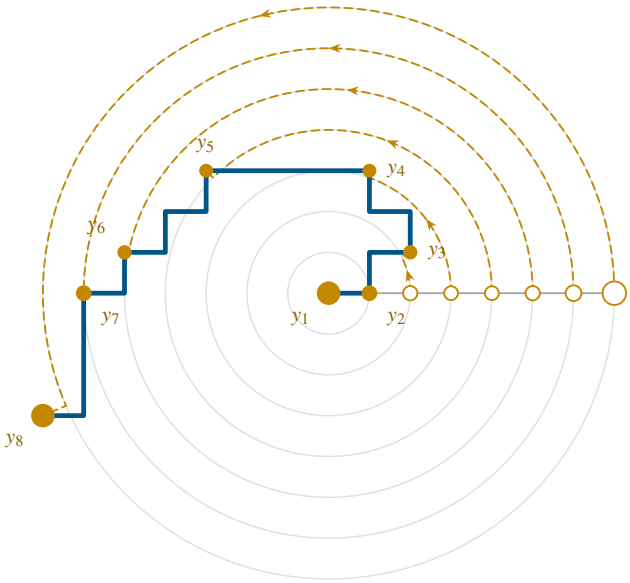


Figure 8.7: After translating x to the origin, the proof of Theorem 8.5 assigns to each selected point $y_j \in K$ the mass that the arcsine law on $[0, n]$ gives to $[j-1, j]$. Here y_j is the first point of a path in K at distance at least $j-1$ from the origin, so its distance is less than j. Each dot has area proportional to its mass, and the circles have radii $1, 2, \dots, 7$.

from x by at most one. Theorem 8.5 therefore gives $\mathcal{H}_K(x) \le Cn^{-1/2}$. Since $n \ge \max\{1, \operatorname{diam}(K)/2\} \ge \frac{1}{4}(1 + \operatorname{diam}(K))$, this proves the corollary after increasing C. □

8.2.2 Sharpness on a Horizontal Segment

The harmonic measure at the endpoint of a straight segment shows that the exponent $\frac{1}{2}$ in Theorem 8.5 cannot be improved (Figure 3.16 in Chapter 3 shows this harmonic measure for a segment of 11 sites). We put the masses of the arcsine law on the segment, and this time we bound the potential of the segment from below.

Proposition 8.7 *There are absolute constants $c, C > 0$ with the following property. For an integer $n \ge 1$, let $\mathcal{J}_n := \{\mathbf{0}, \mathbf{e}_1, \dots, (n-1)\mathbf{e}_1\}$. Then*

$$\frac{c}{\sqrt{n}} \le \mathcal{H}_{\mathcal{J}_n}(\mathbf{0}) = \mathcal{H}_{\mathcal{J}_n}((n-1)\mathbf{e}_1) \le \frac{C}{\sqrt{n}}.$$

Proof. We use the potential of the proof of Theorem 8.5, now on the segment

itself. Let $\mathrm{Arcsine}_{0,n}$ be the arcsine law on $[0, n]$, and let

$$F(w) := \sum_{j=1}^{n} \mathrm{Arcsine}_{0,n}([j-1, j])\, a(w - (j-1)\mathbf{e}_1) \qquad \text{for } w \in \mathbb{Z}^2 .$$

As there, F is harmonic off $\mathcal{J}_n$ and $F - a$ is bounded. Let k be an integer. For $j - 1 < t < j$ we have $\big||k - (j-1)| - |k - t|\big| < 1$, so the second assertion of Lemma 8.4 with $r = |k - t|$, integrated against $\mathrm{Arcsine}_{0,n}$, and Lemma 8.3 give

$$\Big| F(k\mathbf{e}_1) - \frac{1}{2\pi} \int \log|k - t|\, \mathrm{Arcsine}_{0,n}(\,\mathrm{d}t) - \kappa \Big| \le \frac{4C_1}{\sqrt{n}} . \tag{8.3}$$

For $0 \le k \le n - 1$, the integral equals $\log(n/4) = \int \log t\, \mathrm{Arcsine}_{0,n}(\,\mathrm{d}t)$, by Lemma 2.42(a). We now bound the potential from below at a fixed distance to the left of the origin. Let $M \ge 2$ be an integer.

By (3.37), and by (8.3) at $k = -M$ and on $\mathcal{J}_n$,

$$\begin{aligned} U_{\mathcal{J}_n}(-M\mathbf{e}_1) &\ge F(-M\mathbf{e}_1) - \max_{\mathcal{J}_n} F \\ &\ge \frac{1}{2\pi} \int \log\Big(1 + \frac{M}{t}\Big)\, \mathrm{Arcsine}_{0,n}(\,\mathrm{d}t) - \frac{8C_1}{\sqrt{n}} \\ &\ge \frac{\log(1+M)/\pi^2 - 8C_1}{\sqrt{n}} . \end{aligned}$$

The last step keeps only $0 < t < 1$, on which the density $1/(\pi\sqrt{t(n-t)})$ of $\mathrm{Arcsine}_{0,n}$ is at least $1/(\pi\sqrt{nt})$ and $\log(1 + M/t) \ge \log(1 + M)$, and uses $\int_0^1 t^{-1/2}\,\mathrm{d}t = 2$. Fix M so that the numerator is at least one.

Since $U_{\mathcal{J}_n}$ is nonnegative and harmonic off $\mathcal{J}_n$, its value at $-k\mathbf{e}_1$ is at least a quarter of its value at $-(k+1)\mathbf{e}_1$, for every $k \ge 1$. By part (b) of Lemma 3.29, $\mathcal{H}_{\mathcal{J}_n}(\mathbf{0})$ is the sum of $U_{\mathcal{J}_n}$ over the four neighbors of the origin, so iterating for $1 \le k < M$ gives

$$\mathcal{H}_{\mathcal{J}_n}(\mathbf{0}) \ge U_{\mathcal{J}_n}(-\mathbf{e}_1) \ge 4^{1-M} U_{\mathcal{J}_n}(-M\mathbf{e}_1) \ge \frac{4^{1-M}}{\sqrt{n}} .$$

The reflection through the midpoint of the segment exchanges its endpoints, which gives the equality of their harmonic measures. The upper bound follows from Theorem 8.5, since $\mathcal{J}_n$ has a point at distance $j - 1$ from the origin for every $1 \le j \le n$. □

Thus the endpoint of a straight segment with n points has harmonic measure of order $n^{-1/2}$, the largest value that Corollary 8.6 allows for a connected set of this diameter.

8.2.3 Other Lattices*

The proofs of Theorem 8.5 and Corollary 8.6 use very little about the square lattice: only recurrence, the identity $\Delta^{::} a = \delta_{\mathbf{0}}$, the symmetry $a(-x) = a(x)$, the expansion (3.34) with an error of order $|x|^{-1}$, and the bounded length of a step. Hence the discrete Beurling estimate extends to every irreducible symmetric random walk with bounded steps on a two-dimensional lattice, such as simple random walk on the triangular lattice, with $|x|$ replaced in the proof by the norm in which the covariance matrix of a step of the walk is a multiple of the identity. In this norm, let $L > 0$ bound the step lengths. Up to its farthest point, a path of allowed steps cannot skip an annulus of width L. Select points in these annuli and replace the errors of one and two units in the proof by L and $2L$; the constants may depend on the walk. This book proves recurrence and these properties of the potential kernel only for simple random walk on $\mathbb{Z}^2$; for the other walks, including the expansion of the potential kernel, they are proved in the reference given in the Notes.

The discrete Beurling estimate also applies to the hexagonal lattice, with a different constant. Let K be a finite connected set of vertices of the hexagonal lattice, let $x \in K$, and let K_0 be the set of points of K in the same class of the bipartition as x. This class is a triangular lattice, in which two points are neighbors when they have a common neighbor (Figure 8.8). A walk started in this class and observed at even times is the random walk on this triangular lattice that stays put with probability $\frac{1}{3}$ and moves to each of the six neighbors with probability $\frac{1}{9}$. A walk that first hits K at x also first hits K_0 at x. The points of a path in K that lie in the class of x form a path in this triangular lattice, so K_0 is connected there and has a point at distance at least $\operatorname{diam}(K)/2 - 1$ from x. The estimate on the triangular lattice therefore bounds the harmonic measure of K from infinity at x, since a walk started in the other class enters this class after one step. Next we apply the estimate on $\mathbb{Z}^2$ to external DLA.

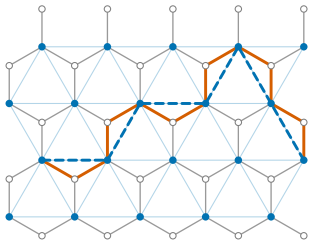


Figure 8.8: The vertices of one class of the hexagonal lattice (blue) form a triangular lattice (light blue), in which two vertices are neighbors when they have a common neighbor. The vertices of a path (vermilion) that lie in this class form a path of this triangular lattice (dashed).

8.3 THE RADIUS OF THE CLUSTER

We now return to external DLA and combine the Beurling estimate with the radius bound of Section 8.1. Let $A \subset \mathbb{Z}^2$ be finite and connected with $\mathbf{0} \in A$, and let $z \in \partial A$. A walk from far away must visit ∂A before it enters A, so a walk whose first visit to ∂A is at z first hits the connected set $A \cup \{z\}$ at z, as for the walk in Figure 8.5. Since $\operatorname{diam}(A \cup \{z\})$ is at least the radius $\operatorname{rad}(A)$ of A, Corollary 8.6 gives

$$H_A(z) \le \mathcal{H}_{A\cup\{z\}}(z) \le C(1 + \operatorname{rad}(A))^{-1/2} . \tag{8.4}$$

Combining this bound with Proposition 8.2 proves the main result of the chapter.

Theorem 8.8 *Let $(\mathcal{X}_n)_{n\ge 0}$ be external DLA on $\mathbb{Z}^2$. Then there are constants $C_0, C, c > 0$ such that, for every integer $n \ge 1$ and every real number $R \ge C_0 n^{2/3}$,*

$$\mathbb{P}\{\mathcal{X}_n \not\subset \mathrm{B}^{::}(\mathbf{0}, R)\} \le C \exp(-cR) .$$

Proof. Each site added to the cluster is adjacent to an earlier site, so every $\mathcal{X}_n$ is connected and contains $\mathbf{0}$. By Definition 8.1 and (8.4),

$$\mathbb{P}\{Z_{n+1} = z \mid \mathcal{F}_n\} = H_{\mathcal{X}_n}(z) \le C(1 + \operatorname{rad}(\mathcal{X}_n))^{-1/2} \qquad \text{for every } z \in \partial\mathcal{X}_n .$$

This is the hypothesis of Proposition 8.2 with $d = 2$ and $\beta = \frac{1}{2}$, and its conclusion is the theorem, since $1/(1+\beta) = 2/3$. □

With $R = C_0 n^{2/3}$, the probabilities $C\exp(-cC_0 n^{2/3})$ are summable in n, so the Borel–Cantelli lemma shows that, almost surely, $\mathcal{X}_n \subset \mathrm{B}^{::}(\mathbf{0}, C_0 n^{2/3})$ for all sufficiently large n. In other words, the radius of the cluster grows at most like $n^{2/3}$, while the off-lattice simulations described at the start of the chapter suggest the smaller exponent $1/d_f \approx 0.585$. In dimensions $d \ge 3$ a walk from far away may miss the cluster altogether, and the next section shows how the capacity of the cluster then takes the place of the Beurling estimate.

8.4 EXTERNAL DLA IN HIGHER DIMENSIONS*

Throughout this section, X is simple random walk on $\mathbb{Z}^d$ with $d \ge 3$, and $\mathbb{P}_y$ and $\mathbb{E}_y$ denote its law and expectation started at y. For $E \subset \mathbb{Z}^d$, the hitting time τ_E is as in (3.4), and $\tau_E^+ := \inf\{n \ge 1 : X_n \in E\}$ is the return time. The constants c and C depend only on d, and their values may change from line to line.

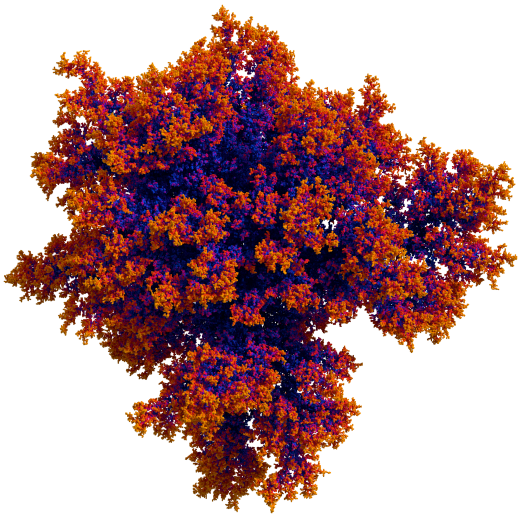


Figure 8.9: An external DLA cluster of 10^6 sites in $\mathbb{Z}^3$, each site colored by the rank of its harmonic measure from infinity, estimated from 4×10^7 walks, from navy (smallest, including zero) through blue, magenta and red to orange (largest). As in the plane, harmonic measure is largest at the exposed tips of the arms.

In dimensions $d \geq 3$ simple random walk is transient (Lemma 3.6), so a walk released far from a finite set will probably never hit it, and the limit in Definition 3.28 is zero. We therefore condition the walk to hit the set before letting its starting point recede to infinity. The resulting harmonic measure of a site is its escape probability divided by the capacity of the set (Proposition 8.9 and Figure 8.9). We prove that, almost surely, the radius of $\mathcal{X}_n$ is at most $Cn^{2/(d+1)}$ when $d \geq 4$ (Theorem 8.14) and at most $Cn^{1/2}(\log n)^{1/4}$ when $d = 3$ (Theorem 8.16), for all large n. Both proofs follow an arm through the history of the process instead of bounding each attachment separately.

8.4.1 Harmonic Measure and Capacity

Recall from Section 3.3 that $2d\, g^{::}(y - x)$ is the expected number of visits to y of a walk from x, so $g^{::} \leq g^{::}(\mathbf{0})$ by the strong Markov property at the first visit. Since $g^{::}$ is even and positive, its expansion $g^{::}(x) = a_d|x|^{2-d} + O(|x|^{-d})$, with the constant $a_d > 0$ of (3.16), gives, for all $x, v \in \mathbb{Z}^d$,

$$c(1+|x|)^{2-d} \leq g^{::}(x) \leq C(1+|x|)^{2-d} \qquad \text{and} \qquad \lim_{|y|\to\infty} \frac{g^{::}(y-v)}{g^{::}(y)} = 1\,. \tag{8.5}$$

As in Exercise 3.46 of Chapter 3, the **escape probability** of a site x of a finite nonempty set $K \subset \mathbb{Z}^d$ is $\mathrm{Es}_K(x) := \mathbb{P}_x\{\tau_K^+ = \infty\}$, and the **capacity** of K

is $\mathrm{cap}(K) := \sum_{x\in K} \mathrm{Es}_K(x)$. We set $\mathrm{Es}_K(x) := 0$ for $x \notin K$.

The next proposition, the main tool of this section, expresses hitting probabilities through escape probabilities (part (a), the last-exit decomposition) and identifies the limiting conditioned hitting distribution.

Proposition 8.9 *Let $d \geq 3$, and let $K \subset \mathbb{Z}^d$ be finite and nonempty.*

(a) For every $y \in \mathbb{Z}^d$,

$$\mathbb{P}_y\{\tau_K < \infty\} = 2d \sum_{x\in K} g^{::}(x-y)\ \mathrm{Es}_K(x)\,. \tag{8.6}$$

In particular, $\mathrm{cap}(K) \geq 1/(2d\, g^{::}(\mathbf{0}))$.

(b) For every $x \in K$,

$$\lim_{|y|\to\infty} \mathbb{P}_y\{X_{\tau_K} = x \mid \tau_K < \infty\} = \frac{\mathrm{Es}_K(x)}{\mathrm{cap}(K)}\,.$$

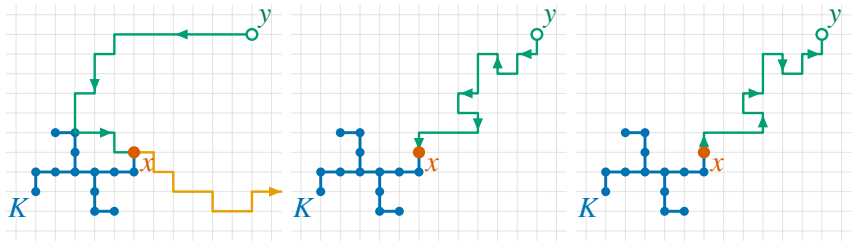
Proof. (a) The expected number of visits to K of a walk from y is finite, so on the event $\{\tau_K < \infty\}$ the walk makes a last visit to K (Figure 8.10(a)). By the Markov property, the walk is at $x \in K$ at time j and never returns to K afterwards with probability $\mathbb{P}_y\{X_j = x\}\ \mathrm{Es}_K(x)$. Summing over j and x gives (8.6). For $y \in K$ the left side equals 1, and $g^{::} \leq g^{::}(\mathbf{0})$ gives the bound on the capacity.

(b) Fix $x \in K$ and $y \notin K$. Reversing a path from y that first enters K at x, at time j, gives a path from x that is at y at time j and avoids K at the times $1, \dots, j$, and both paths have probability $(2d)^{-j}$ (Figure 8.10(b) and (c)). Summing over these paths, and then splitting the visits to y at the return time τ_K^+ by the strong Markov property, we obtain

$$\begin{aligned}\mathbb{P}_y\{X_{\tau_K} = x,\ \tau_K < \infty\} &= \mathbb{E}_x \sum_{0\leq j<\tau_K^+} \mathbf{1}\{X_j = y\} \\ &= 2d\, g^{::}(y-x) - 2d\, \mathbb{E}_x\big[g^{::}(y - X_{\tau_K^+});\ \tau_K^+ < \infty\big]\,.\end{aligned}$$

Divide by $2d\, g^{::}(y)$ and let $|y| \to \infty$. The last expectation is a finite sum over the sites of K, so by the limit in (8.5) the right side tends to $1 - \mathbb{P}_x\{\tau_K^+ < \infty\} = \mathrm{Es}_K(x)$. By (8.6) and (8.5), $\mathbb{P}_y\{\tau_K < \infty\}/(2d\, g^{::}(y))$ tends to $\mathrm{cap}(K)$, which is positive by (a). Dividing the two limits proves (b). □

We call the limit $\mathcal{H}_K(x)$ in part (b) the harmonic measure of K from infinity. In the plane the walk hits K almost surely, so this agrees with Definition 3.28. As there, $H_A(z) := \mathcal{H}_{\partial A}(z)$ for $z \in \partial A$ is the harmonic measure of the outer boundary of a finite nonempty set $A \subset \mathbb{Z}^d$, and external DLA on $\mathbb{Z}^d$, with its added sites Z_{n+1} and σ-algebras $\mathcal{F}_n$, is the process of Definition 8.1 with $\mathbb{Z}^d$



(a) (b) (c)

Figure 8.10: (a) A path from y that visits K for the last time at x, after which (orange) it never returns. (b) A path from y that first enters K at x, and (c) its reversal, a path from x to y that does not return to K, which has the same probability.

in place of $\mathbb{Z}^2$. A walk started outside the closure $\bar{A} = A \cup \partial A$ of Section 3.1 can enter $\bar{A}$ only through ∂A, so it hits $\bar{A}$ and ∂A at the same time and at the same site. Hence, by Proposition 8.9(b),

$$H_A(z) = \mathcal{H}_{\bar{A}}(z) = \frac{\mathrm{Es}_{\bar{A}}(z)}{\mathrm{cap}(\bar{A})} \qquad \text{for every } z \in \partial A\,. \tag{8.7}$$

Every neighbor of a site of A lies in $\bar{A}$, so $\mathrm{Es}_{\bar{A}}$ vanishes on A.

The next estimate, on the potential of sites with first coordinates $0, 1, \dots, r$, differs for $d = 3$ and $d \geq 4$. Let $r \geq 1$ be an integer, and let $x_0, \dots, x_r \in \mathbb{Z}^d$ have first coordinates $(x_i)_1 = i$. For every $z \in \mathbb{Z}^d$ we have $|x_i - z| \geq |i - z_1|$, so (8.5) gives

$$2d \sum_{i=0}^{r} g^{::}(x_i - z) \leq C \sum_{i=0}^{r} (1 + |i - z_1|)^{2-d} \leq \begin{cases} C \log(r+2), & d = 3, \\ C, & d \geq 4. \end{cases} \tag{8.8}$$

Summing (8.6) over such sites bounds the capacity of a connected set from below.

Proposition 8.10 *Let $d \geq 3$, and let $K \subset \mathbb{Z}^d$ be a finite connected set with at least two sites. Then*

$$\mathrm{cap}(K) \geq \begin{cases} c\ \mathrm{diam}(K)/\log(\mathrm{diam}(K)+2), & d = 3, \\ c\ \mathrm{diam}(K), & d \geq 4, \end{cases}$$

Proof. Choose a coordinate in which the sites of K take the largest range of values, and let r be the length of this range. Then $r \geq \mathrm{diam}(K)/\sqrt{d}$, and $r \geq 1$. After a symmetry of the lattice, the first coordinates of the sites of K range over $\{0, 1, \dots, r\}$. The first coordinate changes by at most one along a path in K, so the connected set K contains sites $x_0, \dots, x_r$ with $(x_i)_1 = i$;

they need not be neighbors (Figure 8.11). Summing (8.6) over $y = x_0, \dots, x_r$, where its left side equals one, gives

$$r + 1 = 2d \sum_{z \in K} \mathrm{Es}_K(z) \sum_{i=0}^{r} g^{\cdot\cdot}(x_i - z) \le \mathrm{cap}(K) \max_{z \in \mathbb{Z}^d} 2d \sum_{i=0}^{r} g^{\cdot\cdot}(x_i - z) \,.$$

By (8.8), $\mathrm{cap}(K) \ge cr/\log(r+2)$ when $d = 3$ and $\mathrm{cap}(K) \ge cr$ when $d \ge 4$. Since $t \mapsto t/\log(t+2)$ increases on $[0, \infty)$ and $r \ge \mathrm{diam}(K)/\sqrt{d}$, the proposition follows. □

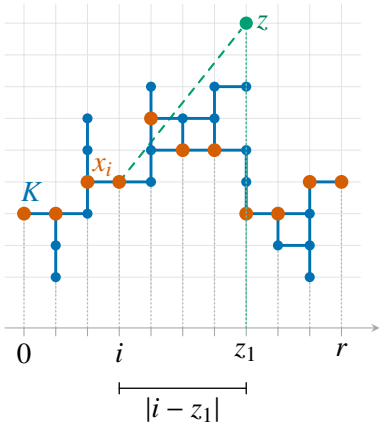


Figure 8.11: A connected set K contains a site x_i with first coordinate i for each $0 \le i \le r$, and every site z satisfies $|x_i - z| \ge |i - z_1|$.

Since $\mathrm{Es}_K \le 1$, Proposition 8.9(b) gives $\mathcal{H}_K \le 1/\mathrm{cap}(K)$. The connected set $\overline{\mathcal{X}}_n$ contains the neighbors of $\mathbf{0}$ and a site at distance $\mathrm{rad}(\mathcal{X}_n)$ from $\mathbf{0}$, so its diameter is at least $(1 + \mathrm{rad}(\mathcal{X}_n))/2$. For $d \ge 4$, (8.7) and Proposition 8.10 therefore give the hypothesis of Proposition 8.2 with $\beta = 1$, and hence radius at most of order $n^{1/2}$. Exercise 8.9 bounds the attachment probabilities by the number of sites instead, which gives order $n^{2/d}$. The next two subsections do better.

8.4.2 The Radius in Dimensions Four and Higher

We measure the cluster in the maximum norm $|x|_\infty := \max_i |x_i|$, and for an integer $r \ge 1$ we let T_r be the first time n at which $\mathcal{X}_n$ contains a site of maximum norm r. Each new site is a neighbor of the cluster, so $T_1 < T_2 < \cdots$, and no site of $\mathcal{X}_{T_r}$ has maximum norm larger than r. The T_r sites of $\mathcal{X}_{T_r - 1}$ lie in a cube of $(2r-1)^d$ sites, so T_r is a bounded stopping time. A site is added at time t if it equals Z_t. Each site of $\mathcal{X}_n$ other than $\mathbf{0}$ was added next to an earlier site, so following earlier sites back from $w \in \mathcal{X}_n$ gives a

nearest-neighbor path in $\mathcal{X}_n$ from $\mathbf{0}$ to w whose sites were added in increasing order of time.

The next lemma finds r sites added in order in each arm that crosses from maximum norm r to $2r$.

Lemma 8.11 *Let $(\mathcal{X}_n)_{n\ge 0}$ be external DLA on $\mathbb{Z}^d$ with $d \ge 3$, and let $r \ge 1$ be an integer. Then there is a nearest-neighbor path $(v_0, v_1, \dots, v_r)$ with $|v_0|_\infty = r$ and $|v_i|_\infty > r$ for $1 \le i \le r$, whose sites $v_1, \dots, v_r$ are added at times $t_1 < t_2 < \cdots < t_r$ with $T_r < t_1$ and $t_r \le T_{2r}$. At most $Cr^{d-1}(2d)^r$ nearest-neighbor paths $(v_0, \dots, v_r)$ have $|v_0|_\infty = r$.*

Proof. At time T_{2r} the cluster contains a site w with $|w|_\infty = 2r$. Follow a path from $\mathbf{0}$ to w whose sites were added in order, and let v_0 be its last site of maximum norm at most r (Figure 8.12). Each step changes the maximum norm by at most one, so $|v_0|_\infty = r$, and at least r sites follow v_0 on the path; let $v_1, \dots, v_r$ be the first r of them. Their maximum norms exceed r, so they are not in $\mathcal{X}_{T_r}$, and they were added after time T_r, in order, by time T_{2r}. For the count, v_0 has at most $(2r+1)^d - (2r-1)^d \le 2d(2r+1)^{d-1}$ choices, and each later site at most $2d$. □

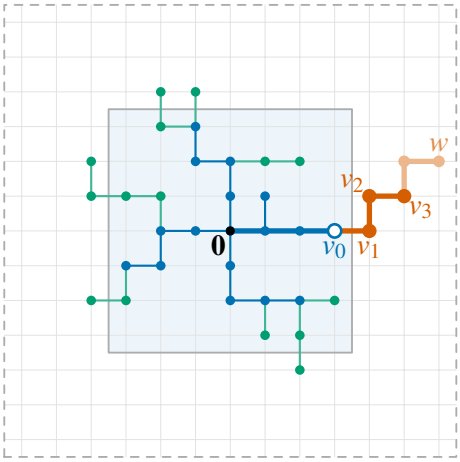


Figure 8.12: The proof of Lemma 8.11 in the plane, with $r = 3$: a path from $\mathbf{0}$ to w whose sites were added in order leaves the shaded square $\{|x|_\infty \le r\}$, which contains $\mathcal{X}_{T_r}$ (blue), for the last time at v_0. The next r sites were added after time T_r.

At time T_r the connected set $\mathcal{X}_{T_r}$ contains $\mathbf{0}$ and a site of maximum norm r. As in the proof of Proposition 8.10, it therefore contains sites $x_0, \dots, x_r$ with $(x_i)_k = \pm i$ for one coordinate k and one sign; we choose them by a fixed rule, so that they are determined by $\mathcal{X}_{T_r}$, and to simplify notation we take

$k = 1$ and the sign positive. For $m \geq T_r$, let

$$\Psi_m := 2d \sum_{i=0}^{r} \sum_{y \in \mathcal{X}_m \setminus \mathcal{X}_{T_r}} g^{::}(x_i - y)$$

be the potential at $x_0, \dots, x_r$ of the sites added after time T_r. We use two consequences of (8.7): for every $n \geq 0$, every site v and every $x \in \mathcal{X}_n$,

$$\mathbb{P}\{Z_{n+1} = v \mid \mathcal{F}_n\} \leq \frac{1}{\operatorname{cap}(\overline{\mathcal{X}}_n)} = 2d\, \mathbb{E}[g^{::}(x - Z_{n+1}) \,\big|\, \mathcal{F}_n]\,. \tag{8.9}$$

The inequality follows from $\mathrm{Es}_{\overline{\mathcal{X}}_n} \leq 1$, and the equality is (8.6) at the site $y = x$ of $\overline{\mathcal{X}}_n$, where its left side equals one. Thus no prescribed attachment is more likely than the mean potential of one particle at an occupied site, so an arm that needs r prescribed attachments to cross from maximum norm r to $2r$ forces potential of order r, on average, at each x_i.

Proposition 8.12 *Let $(\mathcal{X}_n)_{n\geq 0}$ be external DLA on $\mathbb{Z}^d$ with $d \geq 4$. Then there are constants $c_0 > 0$ and C such that, for every integer $r \geq 1$,*

$$\mathbb{P}\{\Psi_{T_{2r}} < c_0 r(r+1) \,\big|\, \mathcal{F}_{T_r}\} \leq C r^{d-1} e^{-r}\,.$$

Consequently, almost surely $\Psi_{T_{2r}} \geq c_0 r(r+1)$ for all sufficiently large r.

The proof combines the path count of Lemma 8.11 with an exponential supermartingale in the number of prescribed attachments and the potential.

Proof. The increments $\xi_m := \Psi_{m+1} - \Psi_m$, which equal $2d \sum_i g^{::}(x_i - Z_{m+1})$, satisfy, by (8.9) and (8.8),

$$0 \leq \xi_m \leq C_1 \qquad \text{and} \qquad \mathbb{E}[\xi_m \mid \mathcal{F}_m] = \frac{r+1}{\operatorname{cap}(\overline{\mathcal{X}}_m)} \qquad \text{for } m \geq T_r\,.$$

Let $\lambda := \log(2d) + 2$ and $\theta := (e^\lambda - 1)/((1 - e^{-1})(r+1))$. We assume that r is so large that $\theta C_1 \leq 1$; a large C covers the finitely many other values of r.

Fix a nearest-neighbor path $(v_0, \dots, v_r)$ with $|v_0|_\infty = r$. Let $N_{T_r} := 0$, and for $m \geq T_r$ let $N_{m+1} := N_m + 1$ if $N_m < r$ and $Z_{m+1} = v_{N_m+1}$, and $N_{m+1} := N_m$ otherwise. Thus N_m counts how many of the sites $v_1, v_2, \dots$ have been added, in this order, by time m. By (8.9),

$$\mathbb{E}[e^{\lambda(N_{m+1} - N_m)} \,\big|\, \mathcal{F}_m] \leq 1 + \frac{e^\lambda - 1}{\operatorname{cap}(\overline{\mathcal{X}}_m)}\,.$$

Since $e^{-t} \leq 1 - (1 - e^{-1})t$ for $0 \leq t \leq 1$, and $0 \leq \theta\xi_m \leq 1$,

$$\mathbb{E}[e^{-\theta\xi_m} \,\big|\, \mathcal{F}_m] \leq 1 - (1 - e^{-1})\frac{\theta(r+1)}{\operatorname{cap}(\overline{\mathcal{X}}_m)} = 1 - \frac{e^\lambda - 1}{\operatorname{cap}(\overline{\mathcal{X}}_m)}\,.$$

If $u \ge 1$ and $0 < v \le 1$, then $uv \le u + v - 1$, since $(u-1)(v-1) \le 0$. We apply this with $u = e^{\lambda(N_{m+1}-N_m)}$ and $v = e^{-\theta\xi_m}$ and add the last two displays. Setting $N_m := \Psi_m := 0$ for $m < T_r$, this shows that $Y_m := \exp(\lambda N_m - \theta\Psi_m)$ is a supermartingale with $Y_m = 1$ for $m \le T_r$. Since $Y_m \le e^{\lambda r}$ and T_{2r} is bounded, summing the increments $Y_{m+1} - Y_m$ over $T_r \le m < T_{2r}$, each with conditional mean at most zero given $\mathcal{F}_m$, gives $\mathbb{E}[Y_{T_{2r}} \mid \mathcal{F}_{T_r}] \le 1$.

Let $c_0 := (1 - e^{-1})/(e^{\lambda} - 1)$, so that $\theta c_0(r+1) = 1$. On the event that $N_{T_{2r}} = r$ and $\Psi_{T_{2r}} < c_0 r(r+1)$,

$$Y_{T_{2r}} > e^{\lambda r - \theta c_0 r(r+1)} = e^{(\lambda-1)r} = (2d)^r e^r \,,$$

so Markov's inequality bounds the conditional probability of this event by $(2d)^{-r}e^{-r}$. By Lemma 8.11, $N_{T_{2r}} = r$ for at least one of at most $Cr^{d-1}(2d)^r$ paths. Summing over these paths proves the estimate, and since $\sum_r r^{d-1}e^{-r} < \infty$, the Borel–Cantelli lemma proves the final assertion. □

The remaining step is deterministic: the pairs at distance less than a length L of order $\sqrt{r}$ carry at most half of the potential, and each particle contributes little to the rest.

Lemma 8.13 *Let $d \ge 4$ and $c_0 > 0$. There are constants $c > 0$ and r_0, depending only on d and c_0, with the following property. Let $r \ge r_0$ be an integer, let $x_0, \dots, x_r \in \mathbb{Z}^d$ have first coordinates $(x_i)_1 = i$, and let $F \subset \mathbb{Z}^d$ be a finite set with*

$$2d \sum_{i=0}^{r} \sum_{y \in F} g^{::}(x_i - y) \ge c_0 r(r+1) \,.$$

Then $|F| \ge cr^{(d+1)/2}$.

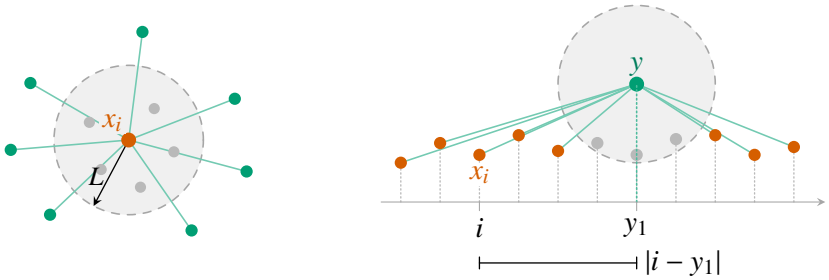
Proof. Summing (8.5) over spherical shells gives $2d \sum_{|z|<L} g^{::}(z) \le C_2 L^2$ for $L \ge 1$. Let $L := (c_0 r/(2C_2))^{1/2}$, and let r_0 be so large that $L \ge 1$. Even if F contains every site within distance L of x_i, these sites contribute at most $c_0 r/2$ to the potential at x_i (Figure 8.13(a)). Therefore

$$2d \sum_{y \in F} \sum_{i \colon |x_i - y| \ge L} g^{::}(x_i - y) \ge c_0 r(r+1) - (r+1)\frac{c_0 r}{2} = \frac{c_0}{2} r(r+1) \,. \quad (8.10)$$

Now fix $y \in F$. If $|x_i - y| \ge L$, then $|x_i - y| \ge \max\{L, |i - y_1|\}$, and at most $2L+1$ indices i have $|i - y_1| < L$. Hence (8.5) gives

$$2d \sum_{i \colon |x_i - y| \ge L} g^{::}(x_i - y) \le C(2L+1)L^{2-d} + 2C \sum_{k \ge L} k^{2-d} \le C_3 L^{3-d} \,,$$

where the last sum is at most a constant times L^{3-d} because $d \geq 4$ (Figure 8.13(b)). Summing over $y \in F$ and comparing with (8.10) gives $c_0 r(r+1)/2 \leq C_3 |F| L^{3-d}$, so $|F| \geq c r^2 L^{d-3} \geq c r^{(d+1)/2}$. □



(a) (b)

Figure 8.13: (a) The sites within distance L of x_i (gray) contribute at most CL^2 there, so the joined pairs give the rest. (b) Since $|x_i - y| \geq \max\{L, |i - y_1|\}$ for each x_i joined to y, the site y contributes at most CL^{3-d} to these pairs.

The last two results bound the number of particles needed to double the maximum norm of the cluster, which controls its Euclidean radius $\mathrm{rad}(\mathcal{X}_n)$ of Section 8.1.

Theorem 8.14 *Let $(\mathcal{X}_n)_{n\geq 0}$ be external DLA on $\mathbb{Z}^d$ with $d \geq 4$. Then there is a constant C such that, almost surely, $\mathrm{rad}(\mathcal{X}_n) \leq C n^{2/(d+1)}$ for all sufficiently large n.*

Proof. The set $F := \mathcal{X}_{T_{2r}} \setminus \mathcal{X}_{T_r}$ has $T_{2r} - T_r$ sites, and its potential at $x_0, \ldots, x_r$ is $\Psi_{T_{2r}}$. By Proposition 8.12 and Lemma 8.13, almost surely $T_{2r} \geq |F| \geq c r^{(d+1)/2}$ for all sufficiently large r. Let $n \geq T_2$, and let r be the integer with $T_{2r} \leq n < T_{2r+2}$; it tends to infinity with n. Every site of $\mathcal{X}_n$ has maximum norm at most $2r + 1 \leq 3r$, and $n \geq T_{2r} \geq c r^{(d+1)/2}$ once n is large, so $\mathrm{rad}(\mathcal{X}_n) \leq 3\sqrt{d}\, r \leq C n^{2/(d+1)}$. □

8.4.3 The Radius in Dimension Three

In $\mathbb{Z}^3$, (8.8) bounds the increments $\xi_m = \Psi_{m+1} - \Psi_m$ only by $C \log(r+2)$, and the argument of Section 8.4.2 loses a logarithm. Instead, we control all the attachment probabilities along an arm at once. An attachment at a site of harmonic measure p lowers the reciprocal of the capacity of the cluster by at least p^2, so by Proposition 8.10 the squares of the attachment probabilities after the cluster reaches diameter r add up to at most $C \log(r+2)/r$. The next lemma proves the first of these statements.

Lemma 8.15 *Let $d \geq 3$, and let $A \subset E$ be finite nonempty subsets of $\mathbb{Z}^d$. Then*

$$\operatorname{cap}(E) - \operatorname{cap}(A) = \sum_{z \in E \setminus A} \operatorname{Es}_E(z)\, \mathbb{P}_z\{\tau_A = \infty\} \geq 0\,.$$

If, moreover, $E = A \cup \{z\}$ for a site $z \notin A$, then

$$\frac{1}{\operatorname{cap}(A)} - \frac{1}{\operatorname{cap}(E)} \geq \mathcal{H}_E(z)^2\,.$$

Proof. Each $w \in A$ lies in E, so the left side of (8.6) for E at $y = w$ equals one. Applying (8.6) to E at $y = w$, exchanging the sums, and applying (8.6) to A at $y = z$, we obtain, since $g^{::}$ is even,

$$\operatorname{cap}(A) = 2d \sum_{w \in A} \operatorname{Es}_A(w) \sum_{z \in E} g^{::}(z - w)\ \operatorname{Es}_E(z) = \sum_{z \in E} \operatorname{Es}_E(z)\, \mathbb{P}_z\{\tau_A < \infty\}\,.$$

Subtracting this from $\operatorname{cap}(E) = \sum_{z \in E} \operatorname{Es}_E(z)$, and noting that $\mathbb{P}_z\{\tau_A < \infty\} = 1$ for $z \in A$, proves the identity. If $E = A \cup \{z\}$, then a walk from z that never returns to E never hits A, so $\mathbb{P}_z\{\tau_A = \infty\} \geq \operatorname{Es}_E(z)$. Together with $\operatorname{cap}(A) \leq \operatorname{cap}(E)$ and $\mathcal{H}_E = \operatorname{Es}_E / \operatorname{cap}(E)$ from Proposition 8.9(b), the identity gives

$$\frac{1}{\operatorname{cap}(A)} - \frac{1}{\operatorname{cap}(E)} = \frac{\operatorname{Es}_E(z)\, \mathbb{P}_z\{\tau_A = \infty\}}{\operatorname{cap}(A) \operatorname{cap}(E)} \geq \frac{\operatorname{Es}_E(z)^2}{\operatorname{cap}(E)^2} = \mathcal{H}_E(z)^2\,.$$

□

For $j \geq 1$, let $p_j := H_{\mathcal{X}_{j-1}}(Z_j)$. Since $\mathcal{X}_j \subset \overline{\mathcal{X}}_{j-1}$, Lemma 8.15 gives $\operatorname{cap}(\mathcal{X}_j) \leq \operatorname{cap}(\overline{\mathcal{X}}_{j-1})$, and $\operatorname{Es}_{\overline{\mathcal{X}}_{j-1}}(Z_j) \leq \operatorname{Es}_{\mathcal{X}_j}(Z_j)$, so by (8.7), $p_j \leq \mathcal{H}_{\mathcal{X}_j}(Z_j)$. Lemma 8.15 with $A = \mathcal{X}_{j-1}$ and $E = \mathcal{X}_j$, summed over $j > k$, therefore gives

$$\sum_{j > k} p_j^2 \leq \frac{1}{\operatorname{cap}(\mathcal{X}_k)} \qquad \text{for every } k \geq 0\,. \tag{8.11}$$

To bound the radius, we count the paths of Lemma 8.11 over their possible attachment times, and we use (8.11) to bound the product of the attachment probabilities along each path.

Theorem 8.16 *Let $(\mathcal{X}_n)_{n \geq 0}$ be external DLA on $\mathbb{Z}^3$. Then there is a constant C such that, almost surely, $\operatorname{rad}(\mathcal{X}_n) \leq C n^{1/2} (\log n)^{1/4}$ for all sufficiently large n.*

Proof. We start with a deterministic bound. Let $r \geq 1$, and recall from Section 8.4.2 that T_r is the first time at which $\mathcal{X}_n$ contains a site of maximum norm r. The connected set $\mathcal{X}_{T_r}$ contains $\mathbf{0}$ and a site of maximum

norm r, so its diameter is at least r, and Proposition 8.10 and (8.11) give $\sum_{j>T_r} p_j^2 \le C\log(r+2)/r$. By the inequality between the arithmetic and geometric means, all times $T_r < t_1 < \cdots < t_r$ satisfy

$$\prod_{i=1}^{r} p_{t_i} \le \Big(\frac{1}{r}\sum_{i=1}^{r} p_{t_i}^2\Big)^{r/2} \le \Big(\frac{C_4\sqrt{\log(r+2)}}{r}\Big)^r . \tag{8.12}$$

Next we fix a path and times. Let $m \ge 1$ be an integer. If $T_{2r} \le T_r + m$, then Lemma 8.11 provides a path $(v_0,\dots,v_r)$ with $|v_0|_\infty = r$ whose sites $v_1,\dots,v_r$ are added at times $T_r + j_1 < \cdots < T_r + j_r$, where $1 \le j_1 < \cdots < j_r \le m$. Fix such a path and such offsets, and for $1 \le i \le r$ let E_i be the event that $Z_{T_r+j_i} = v_i$ and $q_i := \mathbb{P}(E_i \mid \mathcal{F}_{T_r+j_i-1})$. Since $\mathbb{E}[\mathbf{1}_{E_i}/q_i \mid \mathcal{F}_{T_r+j_i-1}] \le 1$, with $0/0 := 0$, conditioning successively on $\mathcal{F}_{T_r+j_r-1},\dots,\mathcal{F}_{T_r+j_1-1}$ shows that $W := \prod_i \mathbf{1}_{E_i}/q_i$ has $\mathbb{E}[W \mid \mathcal{F}_{T_r}] \le 1$. On the intersection E of the events E_i we have $q_i = p_{T_r+j_i}$, so (8.12) gives $\mathbf{1}_E \le W(C_4\sqrt{\log(r+2)}/r)^r$, and therefore $\mathbb{P}(E \mid \mathcal{F}_{T_r}) \le (C_4\sqrt{\log(r+2)}/r)^r$.

Finally we sum over paths and offsets. There are at most $Cr^2\,6^r$ paths and $\binom{m}{r} \le (em/r)^r$ sets of offsets, so

$$\mathbb{P}\{T_{2r} - T_r \le m \mid \mathcal{F}_{T_r}\} \le Cr^2\Big(\frac{C_5 m\sqrt{\log(r+2)}}{r^2}\Big)^r .$$

For $m := \lfloor r^2/(2C_5\sqrt{\log(r+2)})\rfloor$ the right side is at most $Cr^2\,2^{-r}$. By the Borel–Cantelli lemma, almost surely $T_{2r} > m \ge cr^2/\sqrt{\log(r+2)}$ for all sufficiently large r. As in the proof of Theorem 8.14, let r be the integer with $T_{2r} \le n < T_{2r+2}$. Then $\mathrm{rad}(\mathcal{X}_n) \le 3\sqrt{3}\,r$, and since $r \le T_{2r} \le n$, we have $n \ge cr^2/\sqrt{\log(n+2)}$ once n is large. Hence $\mathrm{rad}(\mathcal{X}_n)^2 \le Cn\sqrt{\log(n+2)}$ for all sufficiently large n. □

The simulations described in the Notes suggest that neither exponent is sharp.

EXERCISES

Exercise 8.1. This exercise sharpens the tail bound of Proposition 8.2. In the setting of that proposition, show that there are constants $C_0, C, c > 0$, depending only on d, β and J, such that, for every integer $n \ge 1$ and every real number $R \ge C_0 n^{1/(1+\beta)}$,

$$\mathbb{P}\{\mathcal{X}_n \not\subset \mathrm{B}^{::}(\mathbf{0},R)\} \le C\exp\Big[-cR\Big(1+\log\frac{R}{n^{1/(1+\beta)}}\Big)\Big] .$$

Hint: Use (8.2).

Exercise 8.2. This exercise computes the harmonic measure of a diagonal segment exactly, using the values of the potential kernel on the diagonal in Lemma 3.7. For $k \geq 0$, let $b_k := 4^{-k}\binom{2k}{k}$. Let $n \geq 1$ be an integer, and for $1 \leq j \leq n$ let $z_j := (j-1)(\mathbf{e}_1 + \mathbf{e}_2)$.

(a) Show that the rational function

$$P(s) := \frac{\prod_{r=0}^{n-2}(s - r - \frac{1}{2})}{\prod_{k=0}^{n-1}(s-k)}$$

satisfies

$$P(s) = \sum_{k=0}^{n-1} \frac{b_k b_{n-1-k}}{s-k},$$

and deduce that $\sum_{j=1}^{n} b_{j-1} b_{n-j} = 1$.

(b) Show that the function $x \mapsto \sum_{j=1}^{n} b_{j-1} b_{n-j}\, a(x - z_j)$ takes the same value at each of the points $z_1, \ldots, z_n$.

(c) Deduce that the harmonic measure of $\{z_1, \ldots, z_n\}$ from infinity gives mass $b_{j-1}b_{n-j}$ to z_j, and that its mass at the endpoint z_1 is between two constant multiples of $n^{-1/2}$.

Hint for (b): By Lemma 3.7 and the evenness of a,

$$a((r+1)(\mathbf{e}_1 + \mathbf{e}_2)) - a(r(\mathbf{e}_1 + \mathbf{e}_2)) = \frac{1}{\pi(2r+1)} \qquad \text{for every integer } r\,.$$

Evaluate P at $s = i - \frac{1}{2}$ for $1 \leq i \leq n-1$.

Exercise 8.3. This exercise gives the probabilistic meaning of the weights of Exercise 8.2, a discrete counterpart of the arcsine law of Lemma 2.42. For $k \geq 0$, let $b_k := 4^{-k}\binom{2k}{k}$, and let $n \geq 1$ be an integer. Let $(S_m)_{m \geq 0}$ be simple random walk on $\mathbb{Z}$ started at 0, and let T be the largest time $m \leq 2n-2$ with $S_m = 0$.

(a) Show that the probability that S does not return to 0 during its first 2ℓ steps equals b_ℓ, for every integer $\ell \geq 0$. Hint: Use the reflection principle to count the paths that stay positive after time 0.

(b) Deduce that $\mathbb{P}\{T = 2j-2\} = b_{j-1}b_{n-j}$ for $1 \leq j \leq n$, and hence that $\sum_{j=1}^{n} b_{j-1}b_{n-j} = 1$.

Exercise 8.4. This exercise shows that Corollary 8.6 fails without connectedness, and that the hypothesis of Theorem 8.5 cannot be weakened to the existence of a single point of K at distance at least $n-1$ from x. For an integer $L \geq 2$, let $K_L := \{\mathbf{0}, L\mathbf{e}_1\} \subset \mathbb{Z}^2$.

(a) Show that $\mathcal{H}_{K_L}(\mathbf{0}) = \frac{1}{2}$.

(b) Conclude that there are no constants $C < \infty$ and $\beta > 0$ such that $\mathcal{H}_K(x) \le C(1 + \mathrm{diam}(K))^{-\beta}$ for every finite set $K \subset \mathbb{Z}^2$ and every $x \in K$.

Hint for (a): Two-dimensional simple random walk is recurrent, and K_L is invariant under reflection across the perpendicular bisector of the segment from $\mathbf{0}$ to $L\mathbf{e}_1$.

Exercise 8.5. This exercise proves the exact independence used in an alternative proof of Theorem 8.5, described in the Notes. Let a simple random walk start at $\mathbf{0}$, let $0 < \lambda < 1$, and let T be a geometric random variable, independent of the walk, with

$$\mathbb{P}\{T = k\} = \lambda(1-\lambda)^k \qquad \text{for } k = 0, 1, 2, \dots .$$

Write

$$\Lambda_+ := \{k\mathbf{e}_1 : k \ge 0\} \qquad \text{and} \qquad \Lambda_- := \{k\mathbf{e}_1 : k < 0\}.$$

Let V_+ and V_- be the events that the walk killed at time T avoids Λ_+ and Λ_-, respectively, after time 0 and up to time T. Prove that V_+ and V_- are independent. Hint: Condition on the last visit to the rightmost point of Λ_- that the walk visits, and reverse the path up to that visit.

Exercise 8.6. This exercise applies the proof of Proposition 8.2 on the triangular lattice. Let $\mathbb{T} \subset \mathbb{R}^2$ be the lattice with six nearest neighbors per site, generated by unit vectors at angles 0 and $\pi/3$. Define external DLA on $\mathbb{T}$ as in Definition 8.1, using simple random walk on $\mathbb{T}$ to define harmonic measure, and assume that the limits defining harmonic measure from infinity exist on $\mathbb{T}$ for every finite nonempty set. Assume also that there is a constant $0 < J < \infty$ such that every finite connected set $A \subset \mathbb{T}$ containing $\mathbf{0}$ satisfies

$$H_A(z) \le J(1 + \mathrm{rad}(A))^{-1/2} \qquad \text{for every } z \in \partial A .$$

Section 8.2.3 explains why the discrete Beurling estimate holds on $\mathbb{T}$, and the argument before (8.4) then gives this bound. Show that there are constants $C_0, C, c > 0$ such that, for every $n \ge 1$ and every $R \ge C_0 n^{2/3}$,

$$\mathbb{P}\{\mathcal{X}_n \not\subset \mathbb{T} \cap \mathrm{B}(\mathbf{0}, R)\} \le C \exp(-cR) .$$

Hint: Repeat the proof of the proposition with six neighbors per site.

Exercise 8.7. This exercise characterizes the capacity of Section 8.4.1 as a maximal charge. Let $d \ge 3$, and let $K \subset \mathbb{Z}^d$ be finite and nonempty. Show that $\mathrm{cap}(K)$ is the maximum of $\sum_{x \in K} \mu(x)$ over $\mu : K \to [0, \infty)$ with $2d \sum_{x \in K} g^{::}(y - x)\mu(x) \le 1$ for $y \in K$, and that Es_K attains it.

Hint: Multiply the constraint by $\mathrm{Es}_K(y)$ and sum over y.

Exercise 8.8*. In the setting of Exercise 8.7, let

$$\mathcal{I}(\mu) := 2d \sum_{x,y\in\mathbb{Z}^d} \mu(x) g^{::}(x-y)\mu(y)$$

for finitely supported $\mu : \mathbb{Z}^d \to \mathbb{R}$. Show that $\mathcal{I}(\mu) \geq 0$, and that $\mathrm{Es}_K / \mathrm{cap}(K)$ minimizes $\mathcal{I}$ over the probability measures on K, with minimum $1/\mathrm{cap}(K)$.

Hint: Write $2d\, g^{::}$ as a sum over times $n \geq 0$, and group the times $2j$ and $2j+1$.

Exercise 8.9. This exercise bounds harmonic measure by the number of sites, and deduces a growth bound for the external DLA of Section 8.4. Let $d \geq 3$, and let $K \subset \mathbb{Z}^d$ be finite and nonempty with $|K| = n$.

(a) Show that $2d \sum_{x\in K} g^{::}(x-y) \leq Cn^{2/d}$ for every $y \in \mathbb{Z}^d$, and deduce that $\mathcal{H}_K(x) \leq 1/\mathrm{cap}(K) \leq Cn^{-(d-2)/d}$ for every $x \in K$.

(b) Show that $\mathbb{P}\{Z_j = z \mid \mathcal{F}_{j-1}\} \leq C j^{-(d-2)/d}$ for every $j \geq 1$ and every site z, and that $\mathbb{P}\{\mathrm{rad}(\mathcal{X}_n) \geq \ell\} \leq (Cn^{2/d}/\ell)^\ell$ for all integers $n, \ell \geq 1$. Conclude that, almost surely, $\mathrm{rad}(\mathcal{X}_n) \leq Cn^{2/d}$ for all sufficiently large n.

Hint: For (a), split the sum according to whether $|x-y| < n^{1/d}$. For (b), the sum of $\prod_i j_i^{-(d-2)/d}$ over $1 \leq j_1 < \cdots < j_\ell \leq n$ is at most $(\sum_{j\leq n} j^{-(d-2)/d})^\ell / \ell!$.

Exercise 8.10. This exercise shows that, when $d \geq 4$, the exponent $(d-2)/d$ in Exercise 8.9(a) cannot be improved, even for connected sets. For integers $s, L \geq 2$, let $K := \mathrm{B}_s^{::} \cup \{j\mathbf{e}_1 : 0 \leq j \leq Ls\}$ and $x := Ls\,\mathbf{e}_1$.

(a) Show that $\mathrm{cap}(\mathrm{B}_s^{::}) \leq Cs^{d-2}$, and that $\mathrm{cap}(E \cup F) \leq \mathrm{cap}(E) + |F \setminus E|$ for all finite nonempty sets $E, F \subset \mathbb{Z}^d$.

(b) Show that there is L, depending only on d, such that $\mathrm{Es}_K(x) \geq c$ for every s. Conclude that $\mathcal{H}_K(x) \geq c|K|^{-(d-2)/d}$.

Hint for (b): The last $d-1$ coordinates of the walk perform a lazy random walk on $\mathbb{Z}^{d-1}$, which is transient because $d - 1 \geq 3$.

NOTES

Diffusion-limited aggregation was introduced by Witten and Sander (1981) as a model for irreversible aggregation under diffusion-limited transport, closely related to dendritic growth, and studied further by Witten and Sander (1983). Off-lattice simulations consistently estimate the fractal dimension d_f of the planar cluster near 1.71; Tolman and Meakin (1989) obtained

1.715 ± 0.004 from off-lattice clusters of up to 10^6 particles, refining earlier estimates of Meakin (1983). The rigorous results for planar DLA concern mainly the growth of the cluster radius, the subject of this chapter. On the square lattice the measurement is further complicated by anisotropy: a small cluster grows nearly isotropically, whereas a large cluster aligns its arms with the lattice axes. The crossover between the two growth regimes shifts numerical estimates of d_f. For this reason, numerical estimates of d_f usually use off-lattice clusters. Halsey (2000), a survey of DLA, notes that the fractal dimension is sensitive to the lattice. Grebenkov and Beliaev (2017) simulated square-lattice clusters of up to 1.45×10^8 particles and measured the growth of the radius of gyration, the root-mean-square distance of the particles from the origin. For one hundred clusters of 4.1×10^7 to 5.2×10^7 particles, the exponent of the radius of gyration had mean 0.596 and standard deviation 0.004. Restricted to the particles in narrow sectors around the lattice axes, the radius of gyration grew with exponent about 0.612, and around the diagonals with exponent about 0.535. On the basis of these simulations they argued, without proof, that a square-lattice cluster grows fastest along the axes, so that, rescaled by its diameter, it approaches a deterministic cross.

The discrete Beurling estimate, Corollary 8.6, is equivalent, by translation and comparison of radius with diameter, to the planar part of the theorem on page 167 of Kesten (1987a). The proof of Theorem 2.5.2 in Lawler (1996) selects points of the connected set at regularly spaced radii, as in the proof of Corollary 8.6, and compares the Green function killed on leaving a large ball, between the selected points, with the same Green function between the points of a horizontal segment, both weighted by the escape probabilities of the segment, which are estimated in Section 2.4 of the same book. Our proof keeps the selection of points but works in infinite volume with the potential kernel, and it compares with the arcsine law, the equilibrium measure of a segment for the logarithmic kernel, whose density has a square-root singularity at the endpoints of the segment. It is the lattice version of the proof of the Beurling estimate for Brownian motion in Section 2.6. Since the proof uses only the potential theory of the walk, as Section 8.2.3 explains, it applies to every irreducible symmetric random walk with bounded steps on a two-dimensional lattice: Chapter 4 of Lawler and Limic (2010) proves the recurrence of these walks and the properties of their potential kernels used here, and its Theorem 4.4.4 gives the required expansion. For the diagonal segment, the exact values of Lemma 3.7 and a partial-fraction identity give the harmonic measure explicitly, as Exercise 8.2 shows. These weights also give the law of the time of the last visit to zero of a one-dimensional simple random walk of length $2n - 2$, as Exercise 8.3 shows.

Another route to the endpoint estimate for a horizontal segment, the one taken by Kesten (1987a) on pages 172–175, follows the walk only at its

successive visits to the horizontal axis, where it performs a one-dimensional random walk. The Wiener–Hopf identities and the half-line Green-function representation for this walk are developed in Sections 17–19 of Spitzer (1976), and Kesten bounds the harmonic measure of the segment with Propositions 19.3 and 19.5 there. Section 5.2 of Lawler and Limic (2010) proves an exact independence for a one-dimensional walk with a symmetric step distribution that is killed with a fixed probability at each step (their Proposition 5.2.1): the events that the walk stays nonnegative and that it stays negative, at all positive times before it is killed, are independent. Section 5.3 applies it to the walk on the axis, killed at a geometric time, to bound the probability of avoiding a half-line up to a fixed time (their Proposition 5.3.1), and killed when the planar walk leaves a horizontal strip, to bound it up to the exit from a large square (their Proposition 5.3.2). Proposition 2.4.6 of Lawler (1996) proves the same independence for the planar walk by the path reversal of Exercise 8.5, and Section 2.4 of that book turns the resulting bound for the walk killed at a geometric time into a bound for the walk stopped on leaving a ball. Section 6.8 of Lawler and Limic (2010) then proves the Beurling estimate as a bound on the probability that a walk started at the origin reaches distance n before returning to an infinite connected set that contains the origin (their Theorem 6.8.1), also by selecting one point of the set in each annulus of unit width (their Theorem 6.8.2).

Proposition 8.7 proves the matching lower bound on a horizontal segment by placing the arcsine law on the segment and bounding its potential from below. For a classical proof, see Section 2.4 of Lawler (1996), which gives upper and lower bounds of order $n^{-1/2}$ for the harmonic measure $\mathcal{H}_{\mathcal{J}_n}(\mathbf{0})$ at the endpoint of the segment.

The bound of Theorem 8.8 is due to Kesten (1987b), whose theorem on page L29 states that, almost surely, $\mathrm{rad}(\mathcal{X}_n) \le Cn^{2/3}$ for all sufficiently large n on $\mathbb{Z}^2$. He deduced it from the planar Beurling estimate, whose proof appeared in Kesten (1987a). He derived the radius bound from the attachment bound by counting paths: he bounded the probability that the sites of a fixed path are added in order during a dyadic block of times by an exponential martingale inequality, obtained with the Borel–Cantelli lemma a bound on the growth of the radius over each dyadic block, and deduced the bound $n^{2/3}$ by a deterministic recursion over the blocks, which is written out in Section 4 of Kesten (1987a) (pages 181–182). The proof of Proposition 8.2 instead applies Pascal's identity to a recursion for expected counts beyond integer radii, then Markov's inequality for a tail bound at each fixed n. The Beurling estimate enters the proof only through the attachment bound (8.4). In every dimension, Proposition 8.2 shows that a bound $J(1 + \mathrm{rad}(\mathcal{X}_n))^{-\beta}$ on the conditional probability of each attachment gives an exponential tail for the radius above the scale $n^{1/(1+\beta)}$.

Jerison, Levine, and Sheffield (2012, Theorem 1) proved that the fluctuations of internal DLA in $\mathbb{Z}^2$ are logarithmic, as described in the Notes to Chapter 5. In Section 3 of the same paper, they observe that the martingales behind their logarithmic fluctuation bound remain martingales for a version of external DLA grown from the origin. In this version the initial cluster is the set of sites outside a large ball, and each particle, released at the origin, attaches to the cluster from inside. These martingales do not carry over to the model of this chapter, whose particles arrive from infinity. Jerison, Levine, and Sheffield (2012) add that a closely related martingale underlies the growth bound of Kesten (1990), described below, and that for external DLA the difficulty is to control the quadratic variation of these martingales, since the cluster has no deterministic limit shape.

Theorem 1 of Kesten (1990) concerns external DLA on $\mathbb{Z}^d$, where for $d \geq 3$ each new site is drawn from the conditioned hitting distribution (8.7). It shows that, eventually almost surely, the radius of the cluster of n sites is $O(n^{2/(d+1)})$ for $d \geq 2$ with $d \neq 3$, and $O((n \log n)^{1/2})$ for $d = 3$; Theorem 8.14 is the case $d \geq 4$. The earlier bound $O(n^{2/d})$ of Exercise 8.9 is the theorem on p. L29 of Kesten (1987b), proved on pp. 181 and 183 of Kesten (1987a). Kesten's proof for $d \geq 4$ combines an exponential bound for a martingale formed from Green potentials and reciprocal capacities (Lemma 1 there), an adaptation of the path count of Kesten (1987b) showing that the reciprocal capacities add up to at least a constant times r while the radius grows from r to $2r$ (Lemma 2 there), and a deterministic selection of dyadic shells around a path from $\mathbf{0}$ to distance r. In our proof the supermartingale of Proposition 8.12 replaces both lemmas at sites with first coordinates $0, \ldots, r$, and Lemma 8.13 replaces the selection of shells. Whether the exponent $2/(d+1)$ is sharp is not known, and the simulations of Meakin (1983) in dimensions two through six suggest a smaller exponent. Theorem 8.16 is Theorem 2.6.1 of Lawler (1996), whose proof uses Theorem 2.5.2 there: a site of a connected set of radius $n \geq 2$ in $\mathbb{Z}^3$ that contains $\mathbf{0}$ has harmonic measure at most a constant times $(\log n)^{1/2}/n$. Lemma 2.5.4 there, the analogue of Proposition 8.10, loses a factor $(\log n)^{1/2}$ when the escape probabilities are bounded by one. Section 8.4.3 avoids this loss by following the proof of Theorem 4 of Losev and Smirnov (2025), described at the end of these Notes; Lemma 8.15 proves, with constant one, the form of the lower bound in their Lemma 10 that the proof needs. The last-exit decomposition (8.6) is Proposition 4.6.4 of Lawler and Limic (2010), and Exercise 8.7 is their Proposition 6.5.5, which imposes the constraint on all of $\mathbb{Z}^d$.

The dielectric breakdown model, introduced by Niemeyer, Pietronero, and Wiesmann (1984), is a one-parameter family of lattice growth models. In the form of Definition 1 of Losev and Smirnov (2025), a finite cluster $A \subset \mathbb{Z}^2$ grows by one site $z \in \partial A$, chosen with probability proportional to $H_A(z)^\eta$ for

a parameter $\eta \geq 0$, with the convention $0^0 = 1$. The case $\eta = 1$ is external DLA. The case $\eta = 0$ is the Eden model, in which every site of ∂A is equally likely, including a site enclosed by the cluster, where H_A vanishes. Niemeyer, Pietronero, and Wiesmann (1984) put these probabilities on the bonds that join the cluster to its outer boundary, in proportion to the η-th power of the local electric field, so at $\eta = 0$ their model favors a boundary site with several neighbors in the cluster.

Hastings and Levitov (1998) introduced a family HL(α) of off-lattice aggregation models built by composing conformal maps. At each step a small particle is attached to the unit circle at a uniformly random angle and carried to the cluster by the current aggregate map, so its position on the cluster is distributed according to harmonic measure from infinity. The particle on the unit circle has linear size of order the square root of a size parameter, which Hastings and Levitov take to be a fixed constant times the power $-\alpha$ of the modulus of the derivative of the aggregate map at the attachment angle. Near that angle the aggregate map stretches lengths by this modulus, so to leading order the new particle has linear size proportional to the modulus raised to the power $1-\alpha/2$. For $\alpha = 2$ the particles therefore have approximately constant size, and HL(2) is the proposed off-lattice version of DLA. Carleson and Makarov (2001) defined a deterministic analogue of planar DLA through the Loewner equation and proved, in their Theorem A, a deterministic counterpart of the growth bound of Theorem 8.8; their proof also uses a form of the Beurling estimate.

Rohde and Zinsmeister (2005) proved that normalized fixed-particle HL(0) clusters have a scaling limit whose boundary has Hausdorff dimension 1. Norris and Turner (2012) proved that small-particle HL(0) clusters converge to a disk, and Silvestri (2017) obtained Gaussian fluctuations around the disk limit. Sola, Turner, and Viklund (2019) introduced the aggregate Loewner evolution ALE(α, η). Its size parameters are chosen as in HL(α), but the attachment angle is no longer uniform: its density is proportional to the power $-\eta$ of the modulus of the derivative of the aggregate map, evaluated at a small distance $\varrho > 0$ outside the unit circle. On the boundary of the cluster, harmonic measure from infinity has density with respect to length proportional to the reciprocal of this modulus. Formally, with $\varrho = 0$, ALE(α, η) therefore chooses its attachment point by the rule of the dielectric breakdown model with parameter $\eta + 1$, and ALE($2, \eta$), whose particles have approximately constant size, is its formal off-lattice analogue. For HL(α), which is ALE($\alpha, 0$), Hastings and Levitov (1998) instead match the expected local rate of area growth with that of the lattice model, which gives the parameter $\alpha - 1$, and they expect that the local area growth law determines the large-scale behavior.

The rigorous results on these models concern the small-particle limit, and

they depend on how fast ϱ tends to zero compared with the particle size, which is of order $\lambda^{1/2}$ when the conformal map that attaches one particle to the unit disk has capacity e^{λ}. For $\eta > 1$, slit-shaped particles, and ϱ below a power of λ whose exponent exceeds $5/4$ and depends on η, Sola, Turner, and Viklund (2019) proved that, for each fixed $T > 0$, with probability tending to one each of the first $\lfloor T/\lambda \rfloor$ particles attaches to the previous one, so that the ALE$(0, \eta)$ cluster after these particles converges to the unit disk with a single radial slit of logarithmic capacity T attached; for $\eta < 1$ and every ϱ, the probability that each of the first $N \geq 2$ particles attaches to its predecessor tends to zero. For ϱ at least $\lambda^{1/2-\epsilon}$, Norris, Silvestri, and Turner (2023) proved that ALE$(0, \eta)$ clusters converge to disks for every $\eta < 1$, and for ϱ at least $\lambda^{1/4-\epsilon}$ that their fluctuations are Gaussian; at $\eta = 1$ they need ϱ at least $\lambda^{1/5-\epsilon}$ and $\lambda^{1/6-\epsilon}$. Norris, Silvestri, and Turner (2024) extended these limits to ALE(α, η) with $\alpha + \eta \leq 1$, which includes regularized HL(α) for $\alpha \leq 1$. The model HL(2) proposed for off-lattice DLA lies outside the range of all these theorems, and none of them concerns the fractal dimension $d_f \approx 1.71$.

Losev and Smirnov (2025) proved growth bounds of Kesten's type for the dielectric breakdown model with parameter η. On $\mathbb{Z}^2$ with $0 \leq \eta < 2$, their Theorem 3 shows that, eventually almost surely, the radius of the cluster of n sites is at most $Cn^{2/(4-\eta)}$ times a power of $\log n$ that is absent at $\eta = 1$. On $\mathbb{Z}^3$, with H_A given by (8.7), their Theorem 4 gives $Cn^{\eta/(1+\eta)}(\log n)^{\eta/(2(1+\eta))}$ for $\eta \geq 1$ and $Cn^{1/2}(\log n)^{1/4}$ for $\eta < 1$. At $\eta = 1$ these recover the bounds of Theorems 8.8 and 8.16, by a proof that does not use the Beurling estimate: it bounds the geometric mean of the harmonic measures of the added sites through the growth of the capacity of the cluster, and then counts the paths from radius R to radius $2R$ over their possible attachment times.

PART 3

Sandpiles and Rotor Walks

Chapter Nine

Combinatorics of the Abelian Sandpile

The abelian sandpile models how a pile of grains relaxes on a graph. Place a nonnegative integer number of grains at each vertex. Whenever a vertex holds at least as many grains as its out-degree, the vertex topples and sends one grain along each outgoing edge. A toppling can render neighboring vertices unstable, so adding one grain may cause an avalanche before the sandpile becomes stable (Figure 9.7 in Section 9.5 shows one). The model was introduced as a prototype of **self-organized criticality**: local relaxation drives the system toward a critical state without tuning an external parameter, and the avalanche distributions were predicted to follow power laws (see the Notes). Defined by this one local rule, the model nevertheless produces intricate self-similar patterns that still call for an explanation. Combinatorialists discovered the same dynamics independently and call it **chip-firing**. The graph Laplacian describes the relaxation and connects the model to random walk, to the arithmetic of the graph, and, through the matrix tree theorem of Chapter 11, to spanning trees.

We first treat general graphs and then specialize to the cubic lattice $\mathbb{Z}^d$. The abelian property makes the stabilization and the number of topplings independent of the legal toppling order (Section 9.1). On a finite digraph with a global sink, the recurrent sandpiles form the sandpile group, and Dhar's burning algorithm recognizes recurrence on Eulerian digraphs (Sections 9.4–9.6). The least action principle characterizes the odometer (Section 9.7). On $\mathbb{Z}^d$, monotonicity and the least action principle bound the region affected by a finite addition and distinguish robust backgrounds, on which every finite addition stabilizes, from explosive backgrounds, on which some finite addition makes every site topple (Sections 9.9–9.10). The least action principle has counterparts for the divisible sandpile (Theorem 4.3) and for the instruction stacks of internal DLA and rotor-router aggregation (Lemma 5.1); we used these counterparts in the shape theorems of Chapters 4 to 6.

9.1 TOPPLING AND THE ABELIAN PROPERTY

We begin by defining a sandpile, a legal toppling, and stabilization.

Let $G = (V, E)$ be a finite or infinite directed graph, or **digraph**. A directed

edge $e \in E$ points from the vertex tail(e) to the vertex head(e). Self-loops, where head(e) = tail(e), and multiple edges, meaning distinct edges with the same head and the same tail, are permitted. The **out-degree** $\deg(v)$ of a vertex v is the number of edges e with tail(e) = v, and the **in-degree** $\mathrm{indeg}(v)$ of v is the number of edges e with head(e) = v. We assume that $\deg(v)$ is finite for every $v \in V$. A vertex is a **sink** if its out-degree is zero. A **global sink** is a sink s such that from every other vertex there is a directed path leading to s. A global sink is unique. The **adjacency matrix** $A = (a_{vw})_{v,w\in V}$ of G is the $V \times V$ matrix with a_{vw} equal to the number of edges from v to w. The **graph Laplacian** is the matrix $\mathsf{L} = \mathrm{diag}(\deg(v)) - A$. Thus

$$\mathsf{L}_{vw} = \begin{cases} -a_{vw}, & v \neq w, \\ \deg(v) - a_{vv}, & v = w\,. \end{cases}$$

Every row of L sums to zero, and the row corresponding to a sink is zero.

A **sandpile** σ on G is a function from the set of non-sink vertices of G to the integers, where $\sigma(v)$ represents the number of grains, or **height**, at vertex v. A sandpile σ is **stable** if $\sigma(v) < \deg(v)$ for every non-sink vertex v. We call a vertex v **unstable** in σ if v is not a sink and $\sigma(v) \geq \deg(v)$. An unstable vertex v can **topple**, resulting in a new sandpile $\widetilde{\sigma}$ obtained by moving one grain along each of the $\deg(v)$ edges emanating from v. Thus, for non-sink vertices w,

$$\widetilde{\sigma}(w) = \begin{cases} \sigma(w) + a_{vw}, & w \neq v, \\ \sigma(v) - \deg(v) + a_{vv}, & w = v\,. \end{cases}$$

Grains sent to a sink disappear. We call the sandpile $\widetilde{\sigma}$ a **successor** of σ. A toppling at v is **legal** if v is unstable in the current sandpile when the toppling occurs. The same formula defines a toppling at a stable non-sink vertex, which we call **illegal**.

If a finite sequence of legal topplings takes a sandpile σ to a stable sandpile, then σ **stabilizes**. A legal toppling sequence is **maximal** if it is not the initial segment of a longer legal toppling sequence, so a finite maximal sequence ends at a stable sandpile. Figures 9.1 and 9.2 illustrate stabilization and nonstabilization.

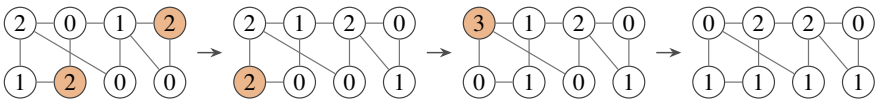


Figure 9.1: Stabilizing a sandpile on an undirected graph without sink by toppling every unstable vertex in each round. The number in a vertex is its number of grains, and a vertex is unstable, and shaded, when it holds at least as many grains as it has neighbors. The eight grains stabilize after three rounds.

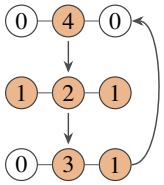


Figure 9.2: Four grains on the path with three vertices and no sink: a stable sandpile there holds at most one grain, and toppling conserves grains, so the sandpile never stabilizes. Toppling one shaded unstable vertex at a time, the middle vertex, then the left end, then the right end, returns the sandpile to its initial state.

Example 9.1 Consider the complete digraph on three vertices, without self-loops. Then placing three grains at a vertex gives a sandpile that stabilizes in one toppling, while placing four grains at a vertex gives a sandpile that never stabilizes, because topplings conserve the four grains and a stable sandpile on this graph holds at most one grain at each vertex.

The order of the legal topplings does not affect the final stable sandpile; Figure 9.3 shows an example on $\mathbb{Z}$. We derive this abelian property from a stronger statement, which compares a legal toppling sequence with an arbitrary sequence of topplings, legal or illegal, that ends at a stable sandpile. The **odometer** of a finite sequence of topplings at non-sink vertices $x_1, \ldots, x_n$ is the function $u : V \to \mathbb{N}$ given by

$$u(v) := \#\{1 \le i \le n : x_i = v\},$$

so u vanishes at every sink. A toppling at v subtracts from the sandpile the row of L indexed by v, so the sequence takes σ to $\sigma - u\mathsf{L}$.

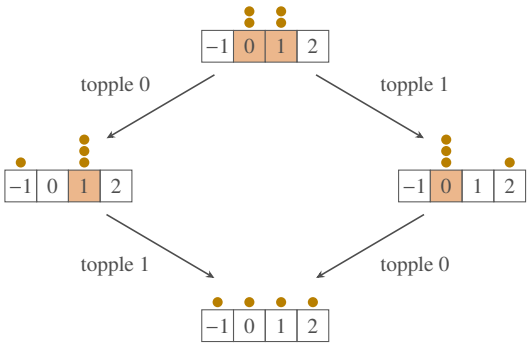


Figure 9.3: The abelian sandpile on $\mathbb{Z}$ started from two grains at each of the sites 0 and 1, with the unstable sites shaded. Both orders of toppling end in the same stable sandpile.

Lemma 9.2 (Least Action Principle) *Let G be a digraph, and let σ be a sandpile on G. Let $x_1, \dots, x_n$ be non-sink vertices whose topplings, legal or illegal, take σ to a stable sandpile. Let $y_1, \dots, y_m$ be non-sink vertices that can be toppled legally in order from σ. Then $y_1, \dots, y_m$ is a permutation of a subsequence of $x_1, \dots, x_n$.*

Proof. Let u be the odometer of $x_1, \dots, x_n$, and for $0 \le j \le m$ let w_j be the odometer of $y_1, \dots, y_j$. We must show that $w_m \le u$. Suppose not, let j be the first index with $w_j \not\le u$, and let $v := y_j$. Then $w_{j-1} \le u$ and $w_{j-1}(v) = u(v)$. Since $a_{xv} \ge 0$ for $x \ne v$, the height of v just before the jth toppling of the second sequence is

$$\begin{aligned}\sigma(v) - (\deg(v) - a_{vv})\, w_{j-1}(v) + \sum_{x \ne v} a_{xv}\, w_{j-1}(x) \\ \le \sigma(v) - (\deg(v) - a_{vv})\, u(v) + \sum_{x \ne v} a_{xv}\, u(x) = (\sigma - u\mathsf{L})(v)\,.\end{aligned}$$

The sandpile $\sigma - u\mathsf{L}$ is the stable sandpile reached by the first sequence, and v is not a sink, so this height is less than $\deg(v)$. Thus the jth toppling of the second sequence is not legal, a contradiction. Hence $w_m \le u$. □

Applied to two legal sequences, the least action principle gives the abelian property.

Lemma 9.3 *Let G be a digraph, let $\sigma_0, \sigma_1, \dots, \sigma_n$ be a sequence of sandpiles on G, each of which is a successor of the one before, and let $\eta_0, \eta_1, \dots, \eta_m$ be another such sequence with $\eta_0 = \sigma_0$.*

(a) If σ_n is stable, then each vertex topples at most as many times in $\eta_0, \dots, \eta_m$ as in $\sigma_0, \dots, \sigma_n$.

(b) If σ_n and η_m are both stable, then the two sequences topple each vertex an equal number of times.

Proof. (a) The topplings of the first sequence take σ_0 to the stable sandpile σ_n, and the topplings of the second sequence are legal. Lemma 9.2 shows that the second sequence topples each vertex at most as often as the first.

(b) Part (a), applied in both directions, shows that the two sequences have the same odometer. □

Summing over the vertices, part (a) gives $m \le n$. Under the hypotheses of part (b), the two sequences have the same odometer u, so $m = n$ and $\sigma_n = \sigma_0 - u\mathsf{L} = \eta_m$.

Definition 9.4 Let σ be a sandpile that stabilizes. We denote the resulting stable sandpile by σ° and call σ° the **stabilization** of σ.

Which sandpiles stabilize, and how long does it take? Lemma 9.3 answers neither question: it tells us what the outcome is, without promising that there is one. On a graph where grains can escape to a sink, both questions have answers in terms of random walk.

9.2 STABILIZATION TIME WITH SINK

When a digraph has a global sink, grains can leave the graph and every finite sandpile stabilizes. Expected hitting times of the sink bound the number of grain moves. We also prove that adding single grains at two vertices, each addition followed by stabilization, gives the same result in either order. On a finite digraph, stabilization also follows from the hitting-time bound of Proposition 9.7, since the expected hitting time of the sink is then finite from every vertex; the next proof covers infinite digraphs as well.

Lemma 9.5 *Let G be a digraph with a global sink, and let σ be a nonnegative sandpile on G with finitely many grains. Then σ stabilizes.*

Proof. Let s be the global sink of G. Write $h^+ := \max\{h, 0\}$ for an integer h. We prove more generally that a sandpile σ, possibly with negative entries, stabilizes whenever its positive mass $N := \sum_{v \neq s} \sigma(v)^+$ is finite. The idea is to record, for each unit of positive mass, the distance of its vertex from the sink. Every toppling moves units that sit at one distance and sends at least one of them closer to the sink, so the sorted list of these distances decreases.

For each vertex w, let $\ell(w)$ be the length of a shortest directed path from w to the sink s, so that $\ell(s) = 0$; it is finite because s is a global sink, and every vertex $v \neq s$ has an out-neighbor w with $\ell(w) = \ell(v) - 1$. A legal toppling at v removes $\deg(v)$ units of positive mass from v, because the height of v is at least $\deg(v)$, and each grain it sends raises the positive mass by at most one, so the positive mass never increases. For a sandpile η reached from σ by legal topplings, let $\vec{\ell}(\eta) \in \mathbb{Z}^N_{\geq 0}$ be the list obtained by taking $\eta(x)^+$ copies of $\ell(x)$ for each vertex $x \neq s$, adding zeros to give N entries, and sorting in nondecreasing order.

We show that each legal toppling decreases $\vec{\ell}$ in the lexicographic order. Let v topple. The toppling removes $\deg(v)$ entries equal to $\ell(v)$ from the list. For each edge out of v it inserts one entry: $\ell(w)$ if the grain arrives at a vertex $w \neq s$ and raises its positive part, and 0 otherwise. The edge to an out-neighbor w with $\ell(w) = \ell(v) - 1$ inserts an entry less than $\ell(v)$, while no entry less than $\ell(v)$ is removed. Hence the first coordinate in which the old and the new sorted lists differ has decreased.

Finally, no infinite sequence of N-tuples of nonnegative integers decreases strictly in the lexicographic order. Indeed, the first coordinates form a

nonincreasing sequence of nonnegative integers, which is eventually constant; from then on the second coordinates are eventually constant, and so on through all N coordinates, so the tuples are eventually constant. Therefore every legal toppling sequence from σ is finite, and toppling unstable vertices for as long as possible stabilizes σ. □

Finiteness of the graph also gives stabilization for sandpiles with negative entries. We will use the signed version when studying recurrent sandpiles.

Corollary 9.6 *Let G be a finite digraph with a global sink. Then every sandpile on G stabilizes.*

Proof. On a finite graph every sandpile has finite positive mass, and the proof of Lemma 9.5 shows that every sandpile with finite positive mass stabilizes. □

The next proposition bounds the number of grain moves needed to stabilize, where a toppling of a vertex v counts as $\deg(v)$ grain moves. Simple random walk on a digraph chooses an outgoing edge uniformly at each step.

Proposition 9.7 *Let G be a digraph with a global sink s. Let $(X_k)_{k\geq 0}$ be the random walk that chooses an outgoing edge uniformly at each step, and let $\tau_s := \inf\{k \geq 0 : X_k = s\}$. Let $\sigma \geq 0$ be a sandpile on G satisfying*

$$\sum_{x\in\operatorname{supp}(\sigma)} \sigma(x)\mathbb{E}_x\tau_s < \infty\,.$$

Then σ stabilizes, and the total number of grain moves needed to stabilize σ is at most

$$\sum_{x\in\operatorname{supp}(\sigma)} \sigma(x)\mathbb{E}_x\tau_s\,.$$

Proof. Let $\sigma = \sigma_0, \sigma_1, \sigma_2, \ldots$ be a sequence of sandpiles, finite or infinite, with σ_{i+1} obtained from σ_i by toppling a single unstable vertex x_i. Write $\mathrm{w}(x) := \mathbb{E}_x\tau_s$ for the weight of x, with $\mathrm{w}(s) = 0$, and write $\mathrm{w}(\sigma_i) := \sum_{x\in\operatorname{supp}(\sigma_i)} \sigma_i(x)\,\mathrm{w}(x)$ for the weight of σ_i. Conditioning on the first step of the walk gives, for $x \neq s$,

$$\mathrm{w}(x) = 1 + \frac{1}{\deg(x)} \sum_{y\in V} a_{xy}\,\mathrm{w}(y)\,.$$

Hence every vertex reached from $\operatorname{supp}(\sigma)$ has finite weight, so all the sums below are finite. Thus toppling the vertex x_i decreases the weight by $\deg(x_i)$:

$$\mathrm{w}(\sigma_i) - \mathrm{w}(\sigma_{i+1}) = \deg(x_i)\,. \tag{9.1}$$

The quantity $\mathrm{w}(\sigma_i)$ is nonnegative, and every toppling decreases $\mathrm{w}(\sigma_i)$ by at least 1. Hence the toppling sequence has at most $\mathrm{w}(\sigma) < \infty$ terms. In particular, every maximal sequence is finite and ends at a stable sandpile, so σ stabilizes. Summing (9.1) over i gives

$$\sum_i \deg(x_i) = \mathrm{w}(\sigma) - \mathrm{w}(\sigma^\circ) \le \sum_{x \in \mathrm{supp}(\sigma)} \sigma(x) \mathbb{E}_x \tau_s \,.$$

□

On a digraph with a global sink, define the **grain addition operator** E_v as the map on nonnegative sandpiles with finitely many grains that adds a single grain at vertex v and then stabilizes:

$$E_v(\sigma) := (\sigma + \delta_v)^\circ \,.$$

Lemma 9.8 *Let G be a digraph with a global sink. For every pair of non-sink vertices v and w, the grain addition operators satisfy $E_v \circ E_w = E_w \circ E_v$.*

Proof. Given a sandpile σ and two vertices v and w, every vertex unstable in $\sigma + \delta_v$ is also unstable in $\sigma + \delta_v + \delta_w$. Apply to $\sigma + \delta_v + \delta_w$ a stabilizing sequence for $\sigma + \delta_v$. The resulting sandpile is $E_v(\sigma) + \delta_w$, whose stabilization is $(E_w \circ E_v)(\sigma)$. Interchanging v and w shows that $(E_v \circ E_w)(\sigma)$ is another stabilization of $\sigma + \delta_v + \delta_w$. Lemma 9.3(b) gives $(E_w \circ E_v)(\sigma) = (E_v \circ E_w)(\sigma)$. □

Lemma 9.3, together with its consequence Lemma 9.8, forms the **abelian property**, which accounts for the word abelian in the model's name. The proof of Lemma 9.8 also gives the following formula for successive grain additions.

Corollary 9.9 *Let G be a digraph with a global sink, let $k \ge 1$ be an integer, and let $v_1, \dots, v_k$ be non-sink vertices. Let σ be a nonnegative sandpile on G with finitely many grains. Then*

$$(E_{v_k} \circ \cdots \circ E_{v_1})(\sigma) = \left(\sigma + \sum_{i=1}^{k} \delta_{v_i}\right)^\circ \,.$$

Proof. Move every grain addition to the beginning of the successive stabilizations. The topplings remain legal, and their final sandpile is stable. Hence the composition $E_{v_k} \circ \cdots \circ E_{v_1}$ produces a stabilization of $\sigma + \sum_{i=1}^{k} \delta_{v_i}$, and Lemma 9.3(b) gives the formula. □

Every bound in this section used the sink to absorb grains. Without a sink, the total number of grains is conserved, and a sandpile carrying too many grains has nowhere to put them.

9.3 STABILIZATION TIME WITHOUT SINK

We now consider connected simple graphs without a sink, first finite ones. We begin with a dichotomy between stabilization and nonstabilization.

Lemma 9.10 *Let G be a finite connected simple graph with at least two vertices, and let σ_0 be a nonnegative sandpile on G.*

(a) If σ_0 stabilizes, then there are two adjacent vertices of G that do not topple during the stabilization of σ_0.

(b) If σ_0 does not stabilize, then every vertex topples infinitely often in every maximal toppling sequence.

Proof. (a) Fix a toppling sequence that stabilizes σ_0. If every vertex toppled, choose a vertex w whose final toppling occurs first. Every neighbor of w topples afterward, so w receives at least $\deg(w)$ grains after its final toppling and ends unstable. Hence some vertex x never topples. If every neighbor of x toppled, then the nonnegativity of σ_0 would make x end with at least $\deg(x)$ grains. Thus a neighbor of x also never topples.

(b) A maximal toppling sequence cannot end at an unstable sandpile. Hence every maximal sequence is infinite when σ_0 does not stabilize. Since G is finite, some vertex topples infinitely often. If v topples infinitely often, then every neighbor w of v receives infinitely many grains. Conservation of the total number of grains forces w to topple infinitely often. Connectedness therefore forces every vertex to topple infinitely often. □

The next lemma counts edges at toppled vertices. It bounds the number of vertices that a few grains can topple, and we will apply it on finite and on infinite graphs, with and without sinks. Let G be an undirected multigraph without self-loops in which every vertex has finite degree. We regard G as a digraph by replacing each edge by two directed edges, one in each direction. Given a set S of vertices, we may also delete the directed edges leaving S, so that the vertices of S become sinks: they never topple, and grains sent to them disappear.

Lemma 9.11 *Let G be an undirected multigraph without self-loops, finite or infinite, in which every vertex has finite degree, and let S be a set of vertices of G that are made sinks. Let $\sigma_0 \geq 0$ be a sandpile on G with N grains. Fix a legal toppling sequence from σ_0 and a time $t \geq 0$, and let W be the set of vertices that topple at least once by time t. Then at most N edges of G have an endpoint in W.*

Proof. For a vertex v, let t_v be the time of its last toppling by time t, with $t_v = -\infty$ if v does not topple by time t. Let e be an edge with endpoints v and

w, at least one of which lies in W. Distinct topplings occur at distinct times, so $t_v \neq t_w$; say $t_v < t_w$. At time t_w the vertex w sent a grain along e to v, and v has not toppled since. If $v \in S$, this grain has disappeared into v. Otherwise, the height of v is nonnegative just after its last toppling by time t, or at time 0 if v does not topple by time t, and from then until time t it can only increase.

Hence at time t every vertex $v \notin S$ holds at least as many grains as there are edges at v whose other endpoint toppled more recently than v, and every $v \in S$ has absorbed at least that many grains. Each edge with an endpoint in W is counted at exactly one of its endpoints, the one whose last toppling is earlier, as Figure 9.4 illustrates. The heights at time t and the absorbed grains together total N, so at most N edges have an endpoint in W. □

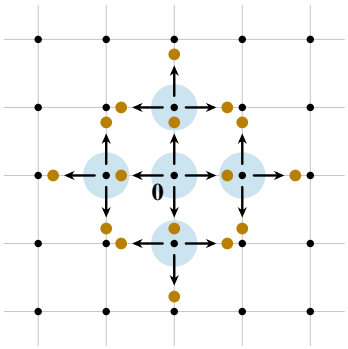


Figure 9.4: Sixteen grains at the origin of $\mathbb{Z}^2$ stabilize after the origin topples four times, each of its neighbors once, and the origin once more, so five sites topple (shaded). Each of the 16 edges with an endpoint in this set points to the endpoint whose last toppling came first, or that never toppled, and that endpoint keeps the grain sent along the edge (orange), so Lemma 9.11 holds here with equality.

If a sandpile on $G = (V, E)$ has more than $\sum_{v \in V}(\deg(v) - 1) = 2|E| - |V|$ grains, then the sandpile cannot stabilize. The next lemma shows that fewer than $|E|$ grains always stabilize, and that this threshold is sharp.

Lemma 9.12 *Let $G = (V, E)$ be a finite connected simple graph with at least two vertices.*

(a) If σ_0 is a nonnegative sandpile on G with $N < |E|$ grains, then σ_0 stabilizes.

(b) There exists a sandpile on G with $|E|$ grains that does not stabilize.

Proof. (a) We prove the contrapositive. Suppose σ_0 does not stabilize, and fix a maximal toppling sequence. By Lemma 9.10, every vertex topples. At the first time by which every vertex has toppled, every edge has an endpoint that has toppled, so Lemma 9.11 with $S = \emptyset$ gives $|E| \leq N$.

(b) Fix an ordering $v_1, \dots, v_n$ of the vertices, and define

$$\sigma_0(v_i) := \big|\{j > i : \{v_j, v_i\} \in E\}\big| .$$

Topple $v_1, \dots, v_n$ in order. When v_i topples, the grains received from earlier vertices raise its height to $\deg(v_i)$. After all n topplings, each vertex v_i has received one grain from each neighbor and lost $\deg(v_i)$ grains. The final sandpile is σ_0, so the toppling cycle can be repeated. Lemma 9.3(a) rules out stabilization. □

We next bound the number of topplings required to stabilize.

Lemma 9.13 *Let $G = (V, E)$ be a finite connected simple graph with at least two vertices. If a nonnegative sandpile σ on G with N grains stabilizes, then the number of topplings is at most $N \operatorname{diam}(G)|V|$.*

Proof. Let u be the odometer of the stabilization of σ, so that $u(v)$ is the number of times v topples. It suffices to show

$$|u(v) - u(w)| \le N \qquad \text{for all edges } \{v, w\} \in E . \tag{9.2}$$

Indeed, Lemma 9.10 gives a vertex z with $u(z) = 0$. Once we prove (9.2), it follows that $u(x) \le \operatorname{dist}(x, z)N \le N \operatorname{diam}(G)$ for every $x \in V$. Summing over V bounds the number of topplings by $N \operatorname{diam}(G)|V|$.

To prove (9.2), fix an edge $\{v, w\}$ and suppose that $u(v) > u(w)$. Consider the sublevel set $U := \{y \in V : u(y) \le u(w)\}$. Since $\sigma^\circ - \sigma = -u\mathsf{L}$, summing over U and canceling the contributions of edges internal to U gives

$$\sum_{\substack{y \in U,\ x \notin U \\ x \sim y}} (u(x) - u(y)) = \sum_{y \in U} (\sigma^\circ - \sigma)(y) . \tag{9.3}$$

Every term on the left of (9.3) is nonnegative, and the edge $\{v, w\}$ contributes $u(v) - u(w)$. The sum on the right of (9.3) is at most the total number of grains. Hence $u(v) - u(w) \le N$, which proves (9.2). □

Since stabilization preserves the number of grains and a stable sandpile has at most $2|E| - |V|$ grains, a sandpile with N grains that stabilizes has $N < 2|E|$. Thus the number of topplings in Lemma 9.13 is less than $2 \operatorname{diam}(G)|V| \cdot |E|$, a bound that depends only on G.

The next bound is useful for finitely supported sandpiles on infinite graphs. We confine the stabilization to a finite graph ball and apply the hitting-time bound in Proposition 9.7.

Lemma 9.14 *Let G be a connected locally finite simple graph. Let σ_0 be a nonnegative sandpile with N grains on G, and let B be the set of vertices at graph distance at most N from* $\mathrm{supp}(\sigma_0)$*. If B is not the whole vertex set of G, then no grain ever leaves B, the sandpile σ_0 stabilizes, and the total number of grain moves needed to stabilize σ_0 is at most*

$$\sum_{x \in B} \sigma_0(x) \mathbb{E}_x \tau_{B^c} ,$$

where $\mathbb{E}_x \tau_{B^c}$ is the expected time for a simple random walk started at x to exit B.

Proof. Local finiteness makes B finite, and $G \setminus B$ is nonempty by hypothesis. Form a finite multigraph on $B \cup \{s\}$ by retaining the edges with both endpoints in B, replacing each edge from B to $G \setminus B$ by an edge to s, and making s a sink. Connectedness makes s a global sink.

Stabilize σ_0 on this finite graph, using Lemma 9.5. We claim that no grain reaches s. Suppose otherwise, let T be the time of the first toppling that sends a grain to s, and let x be the vertex that topples at time T. A vertex outside $\mathrm{supp}(\sigma_0)$ first topples only after receiving a grain from a neighbor that toppled earlier. Following such neighbors backward from x, the first toppling times strictly decrease, so we obtain a path $x_0, x_1, \ldots, x_k = x$ of distinct vertices that topple by time T, with $x_0 \in \mathrm{supp}(\sigma_0)$. Its k edges and the edge along which x sent a grain to s are $k+1$ distinct edges with an endpoint that topples by time T. In G they form a path from x_0 to a vertex outside B, so $k + 1 \geq N + 1$. This contradicts Lemma 9.11 with $S = \{s\}$.

Thus no grain leaves B, and the finite-graph topplings stabilize σ_0 on G. Proposition 9.7, applied to the finite graph, gives the displayed grain-move bound because the sink-hitting time in the finite graph is τ_{B^c}. □

The hypothesis on B holds whenever G is infinite, since local finiteness makes B finite.

We now return to graphs with a sink. Instead of asking how long stabilization takes, we ask which stable sandpiles arise, and we find that they carry the structure of a finite abelian group.

9.4 SANDPILE GROUP

Let G be a digraph on n vertices with global sink s, and recall the graph Laplacian L from Section 9.1. The **reduced Laplacian** L' of G is obtained from L by deleting the row and column of the sink. Toppling a non-sink vertex v transforms a sandpile σ into $\sigma - \mathsf{L}'_v$, where L'_v is the row of L' corresponding to v. Since sand-

piles before and after toppling should be equivalent, we consider the group quotient $\mathbb{Z}^{n-1}/\mathbb{Z}^{n-1}\mathsf{L}'$, where $\mathbb{Z}^{n-1}\mathsf{L}'$ is the integer row-span of L'.

Definition 9.15 Let G be a digraph on n vertices with global sink s. The **sandpile group** of G is the group quotient

$$\mathcal{S}(G) := \mathbb{Z}^{n-1}/\mathbb{Z}^{n-1}\mathsf{L}'(G)\,.$$

We first show that L' is nonsingular, so that the lattice $\mathbb{Z}^{n-1}\mathsf{L}'$ is $(n-1)$-dimensional.

Lemma 9.16 *Let G be a finite digraph with a global sink s. Then the reduced Laplacian $\mathsf{L}'(G)$ is nonsingular.*

Proof. Suppose a real column vector u, indexed by the non-sink vertices, satisfies $\mathsf{L}'u = 0$, and extend it by $u(s) = 0$. If the maximum of u on V is positive, choose a maximizing vertex v with shortest directed distance to s. The identity

$$\deg(v)\,u(v) = \sum_{w \in V} a_{vw} u(w)$$

forces every out-neighbor of v to have the same value as v. The first step along a shortest path from v to s contradicts the choice of v. Thus $u \le 0$, and applying this to $-u$ gives $u \ge 0$, so $u = 0$. □

A function $f : V(G) \to [0,1)$ is **harmonic modulo 1** if for all vertices v,

$$\deg(v)\,f(v) = \sum_{w} a_{vw} f(w) \pmod 1\,.$$

Exercise 9.3 asks you to show that the sandpile group is isomorphic to the additive group of harmonic functions modulo 1 on G that vanish at the sink. For example, in the graph in Figure 9.5, write v_1 and v_2 for the two non-sink vertices. The three functions that are harmonic modulo 1 and vanish at the sink have $(f(v_1), f(v_2))$ equal to $(0,0)$, $(1/3, 2/3)$ and $(2/3, 1/3)$.

Figure 9.5: Two stable sandpiles on the complete digraph on three vertices, with one vertex made into the sink s (square) by deleting its outgoing edges. They differ by the sum of the two rows of the reduced Laplacian, so both represent the identity class; only the one on the right is recurrent.

Corollary 9.24 will show that the recurrent sandpiles, under addition followed by stabilization, form a group isomorphic to the sandpile group. For the graph in Figure 9.5, this is the cyclic group of order 3. When G is a wired tree, obtained from a finite tree by collapsing its leaves to a single sink, $\mathcal{S}(G)$ has a recursive description; see the Notes.

The reduced Laplacian also determines the number of elements in the sandpile group.

Lemma 9.17 *Let G be a finite digraph with a global sink. Then the order of $\mathcal{S}(G)$ is the absolute value of the determinant of the reduced Laplacian $\mathsf{L}'(G)$ of G.*

Figure 9.6 illustrates the proof for the graph of Figure 9.5.

Proof. Write $m := n - 1$. If $m = 0$, then both sides equal 1, so let $m \geq 1$. By Lemma 9.16, the map $x \mapsto x\mathsf{L}'$ of row vectors is a linear bijection of $\mathbb{R}^m$. Let P be the image of $[0,1)^m$, which has volume $|\det \mathsf{L}'|$. The map sends the translates $[0,1)^m + k$ with $k \in \mathbb{Z}^m$, which partition $\mathbb{R}^m$, onto the translates $P + k\mathsf{L}'$, so the translates of P by the vectors of $\mathbb{Z}^m\mathsf{L}'$ partition $\mathbb{R}^m$.

We cut P along the unit cubes $[0,1)^m + z$ with $z \in \mathbb{Z}^m$ and translate the piece in $[0,1)^m + z$ by $-z$ into $[0,1)^m$. Fix a coset of $\mathbb{Z}^m\mathsf{L}'$ in $\mathbb{Z}^m$ and a representative z_0. The sets $P - z$ with z in this coset are translates of $P - z_0$ by the vectors of $\mathbb{Z}^m\mathsf{L}'$, so they partition $\mathbb{R}^m$. Thus their intersections with $[0,1)^m$ partition the cube and have total volume 1. Summing over the cosets, the volume $|\det \mathsf{L}'|$ of P equals the number of cosets, which is the order of $\mathcal{S}(G)$. □

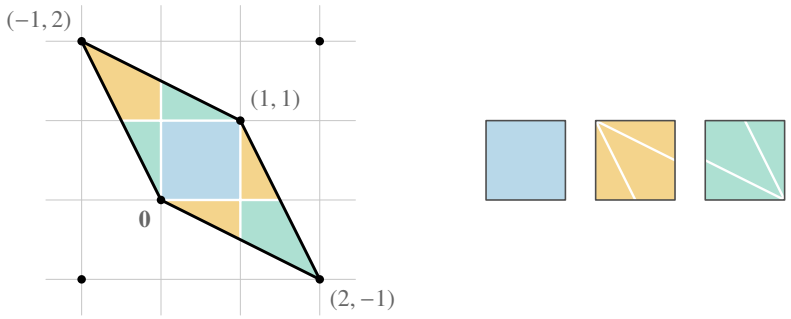


Figure 9.6: (a) For the graph of Figure 9.5 the reduced Laplacian L' has rows $(2,-1)$ and $(-1,2)$, and the unit squares cut the image P of $[0,1)^2$, of area $\det \mathsf{L}' = 3$, into pieces colored by the coset of $\mathbb{Z}^2\mathsf{L}'$ (black dots) that contains the corner z of their square $[0,1)^2 + z$. (b) Translated by $-z$ into $[0,1)^2$, the pieces of each coset tile it exactly once, so the area of P equals the number of cosets.

To find the sign of the determinant, we remove the edges between non-sink vertices continuously. Let $A_0 := (a_{vw})_{v,w\neq s}$, and write $L_t := \operatorname{diag}(\deg(v)) - tA_0$ for $0 \le t \le 1$. Every matrix L_t is nonsingular. For $t = 1$, this is Lemma 9.16. If $t < 1$ and $L_t u = 0$ for a nonzero vector u, choose a coordinate v where $|u(v)|$ is maximal. Then

$$\deg(v)\,|u(v)| = t\left|\sum_{w\neq s} a_{vw}u(w)\right| \le t\sum_{w\neq s} a_{vw}|u(w)| \le t\deg(v)\,|u(v)|\,,$$

which is impossible. Since $\det L_t$ is continuous and never vanishes, its sign is the sign of

$$\det L_0 = \prod_{v\neq s} \deg(v) > 0\,.$$

In particular, $\det \mathsf{L}'(G) > 0$, so the absolute value in Lemma 9.17 is unnecessary.

To connect the quotient with the toppling dynamics, we first show that every equivalence class contains a nonnegative stable representative.

Lemma 9.18 *Let G be a digraph on n vertices with a global sink. Then every equivalence class of $\mathbb{Z}^{n-1}/\mathbb{Z}^{n-1}\mathsf{L}'(G)$ contains at least one nonnegative stable sandpile.*

Proof. Let $\boldsymbol{\delta}$ be the sandpile given by $\boldsymbol{\delta}(v) = \deg(v)$ at every non-sink vertex, and let $\boldsymbol{\delta}^\circ$ be its stabilization, which exists by Lemma 9.5. Then $\boldsymbol{\delta}^\circ(v) < \deg(v)$ for all $v \neq s$, so every entry of $\boldsymbol{\delta} - \boldsymbol{\delta}^\circ$ is at least one, and the difference is equivalent to the zero sandpile. Given $\alpha \in \mathbb{Z}^{n-1}$, let $m \le 0$ be the minimum of 0 and the coordinates of α. Then the vector

$$\beta := \alpha + (-m)(\boldsymbol{\delta} - \boldsymbol{\delta}^\circ)$$

is nonnegative and equivalent to α. Legal topplings preserve nonnegativity, so β° is a nonnegative stable sandpile in the same equivalence class as α. □

Example 9.19 An equivalence class may contain more than one stable sandpile. In the complete digraph on three vertices with one vertex made into a sink, shown in Figure 9.5, the reduced Laplacian is

$$\mathsf{L}' = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix},$$

and the stable sandpile on the right is obtained from the one on the left by adding the two rows of L'. Hence both represent the identity class. The next section identifies which representative is recurrent.

9.5 RECURRENT SANDPILES

Throughout this section, G is a finite digraph with a global sink, with reduced Laplacian $\mathsf{L}'(G)$ and sandpile group $\mathcal{S}(G)$ from Section 9.4. An equivalence class may contain several stable sandpiles. We distinguish the stable sandpiles that can be reached from every initial sandpile.

Definition 9.20 A sandpile σ on G is **accessible** if from every sandpile on G it is possible to obtain σ by a combination of adding grains and selectively toppling unstable vertices. A sandpile that is both stable and accessible is called **recurrent**.

Lemma 9.26 below gives three equivalent reachability criteria for recurrence of nonnegative sandpiles.

Adding grains and stabilizing preserves recurrence.

Lemma 9.21 *Let G be a finite digraph with a global sink, and let σ be a recurrent sandpile on G. Then $(\sigma + \xi)^\circ$ is recurrent for every sandpile $\xi \geq 0$ on G.*

Proof. From every initial sandpile, accessibility gives a sequence that reaches σ. Adding ξ and then stabilizing gives a sequence that reaches $(\sigma + \xi)^\circ$. Hence $(\sigma + \xi)^\circ$ is stable and accessible, and therefore recurrent. □

Figure 9.7 shows the topplings caused by adding one grain to a large recurrent sandpile.

Lemmas 9.22 and 9.23 show that each element of the sandpile group has a unique recurrent representative. In Example 9.19, the all-1 sandpile is accessible: Corollary 9.6 stabilizes every sandpile, after which we may add grains to reach the all-1 sandpile. Once uniqueness is proved, the stable all-0 sandpile, which is equivalent to the all-1 sandpile, cannot also be accessible. We therefore take the all-1 sandpile as the canonical representative of the identity.

Lemma 9.22 *Let G be a digraph on n vertices with a global sink. Then every equivalence class of $\mathbb{Z}^{n-1}/\mathbb{Z}^{n-1}\mathsf{L}'(G)$ contains at least one recurrent sandpile of G.*

Proof. Write $\deg_{\max}$ for the maximum out-degree of a vertex in G. Recall from the proof of Lemma 9.18 that the sandpile $\boldsymbol{\delta} - \boldsymbol{\delta}^\circ$, where $\boldsymbol{\delta}(v) = \deg(v)$ at every non-sink vertex, is equivalent to zero and has every entry at least one. Given $\alpha \in \mathbb{Z}^{n-1}$, we may therefore choose an integer $k \geq 0$ so large that the sandpile

$$\beta := \alpha + k(\boldsymbol{\delta} - \boldsymbol{\delta}^\circ)$$

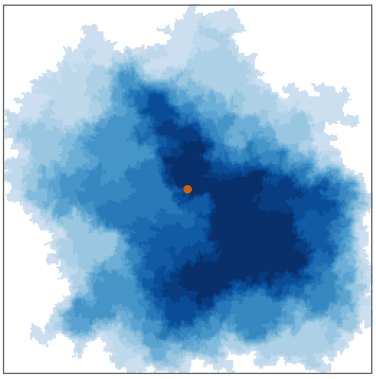


Figure 9.7: An avalanche: the sites that topple when one grain is added at the center (vermilion) of a recurrent sandpile on the 256×256 square grid with wired boundary conditions, shaded from light (once) to dark (15 times). The sandpile was reached from the all-3 sandpile by adding 2×10^6 grains at random sites, and of 2000 grains then added one at a time at the center, this one caused the most topplings.

satisfies $\beta(v) \geq \deg_{\max}$ at every non-sink vertex v. Then β is equivalent to α. We claim that β is accessible. Indeed, by Corollary 9.6, every sandpile can first be stabilized, leaving fewer than $\deg_{\max}$ grains at each vertex, and then supplemented with extra grains to obtain β. Therefore every sandpile obtained from β by toppling is also accessible. In particular, the stabilization β° is recurrent and equivalent to α. □

Accessibility and the least action principle force two equivalent recurrent sandpiles to agree.

Lemma 9.23 *Let G be a digraph on n vertices with a global sink. Then every equivalence class of $\mathbb{Z}^{n-1}/\mathbb{Z}^{n-1}\mathsf{L}'(G)$ contains at most one recurrent sandpile of G.*

Proof. We compare a legal stabilization with a possibly illegal toppling sequence using the least action principle. Let σ and ξ be recurrent and equivalent modulo $\mathbb{Z}^{n-1}\mathsf{L}'(G)$, so that $\sigma - \xi = z\mathsf{L}'$ for an integer row vector z indexed by the non-sink vertices, and let $b(v) := |z(v)| \deg(v)$. Since σ is accessible, it can be obtained from b by adding grains and legally toppling unstable vertices. Making all the additions at the start keeps every toppling legal, because the grains added early only increase the heights. So there are a sandpile $\alpha \geq b$ and a legal toppling sequence that takes α to σ; let u be its odometer, so that $\sigma = \alpha - u\mathsf{L}'$.

A toppling at v removes at most $\deg(v)$ grains from v, and topplings at other vertices only add grains there. Since σ is stable,

$$\deg(v) > \sigma(v) \geq \alpha(v) - u(v)\ \deg(v) \geq \big(|z(v)| - u(v)\big)\deg(v)\,,$$

so $u(v) > |z(v)| - 1$, and since both are integers, $u \geq |z|$. Hence $u + z \geq 0$, and

$$\alpha - (u + z)\mathsf{L}' = \sigma - z\mathsf{L}' = \xi\,.$$

Toppling each non-sink vertex v exactly $u(v) + z(v)$ times, legally or not, therefore takes α to the stable sandpile ξ, and Lemma 9.2 gives $u \leq u + z$, that is, $z \geq 0$. Exchanging the roles of σ and ξ gives $-z \geq 0$. Hence $z = 0$ and $\sigma = \xi$. □

Existence and uniqueness now identify recurrent sandpiles with the Laplacian quotient.

Corollary 9.24 *Let G be a digraph on n vertices with a global sink. Then the recurrent sandpiles on G form an abelian group under $(\sigma, \xi) \mapsto (\sigma + \xi)^\circ$. The map that sends each recurrent sandpile to its class in $\mathcal{S}(G)$ is an isomorphism.*

Proof. Every recurrent sandpile is nonnegative because it is accessible from the zero sandpile, and grain additions and legal topplings preserve nonnegativity. Lemma 9.21 therefore gives closure under the operation. Lemmas 9.22 and 9.23 give exactly one recurrent sandpile in each class, so the class map is a bijection and preserves addition. □

We henceforth identify the sandpile group with the group of recurrent sandpiles in Corollary 9.24; Figure 9.8 shows this group for a cycle with five vertices.

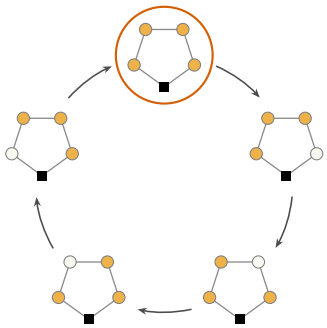


Figure 9.8: The five recurrent sandpiles on the cycle with five vertices, one of them the sink (black square), with zero and one grains in off-white and orange: adding a grain at the vertex just clockwise of the sink and stabilizing moves each to the next (arrows), so the sandpile group is cyclic of order five. Its identity element (circled) has one grain at every vertex.

The **identity element** I of the sandpile group is the unique recurrent sandpile equivalent to the all-0 sandpile. The following computation gives I. Let σ be the sandpile given by $\sigma(v) = 2\deg(v) - 2$. Since $\sigma^\circ \leq \boldsymbol{\delta} - 1$, where $\boldsymbol{\delta}(v) = \deg(v)$ as in the proof of Lemma 9.18, we have $\sigma - \sigma^\circ \geq \boldsymbol{\delta} - 1$, so

$\sigma - \sigma^\circ$ is accessible. Since $\sigma - \sigma^\circ$ is equivalent to 0, the identity element is given by $I = (\sigma - \sigma^\circ)^\circ$.

Figure 9.9 shows identity elements for the $L \times L$ square grid with wired boundary conditions, for several values of L. To be precise, the graph G is obtained by replacing each edge of the undirected square grid with a pair of directed edges, and adjoining a sink vertex s along with two edges from each of the four corner vertices to s and one edge from each of the other boundary vertices to s.

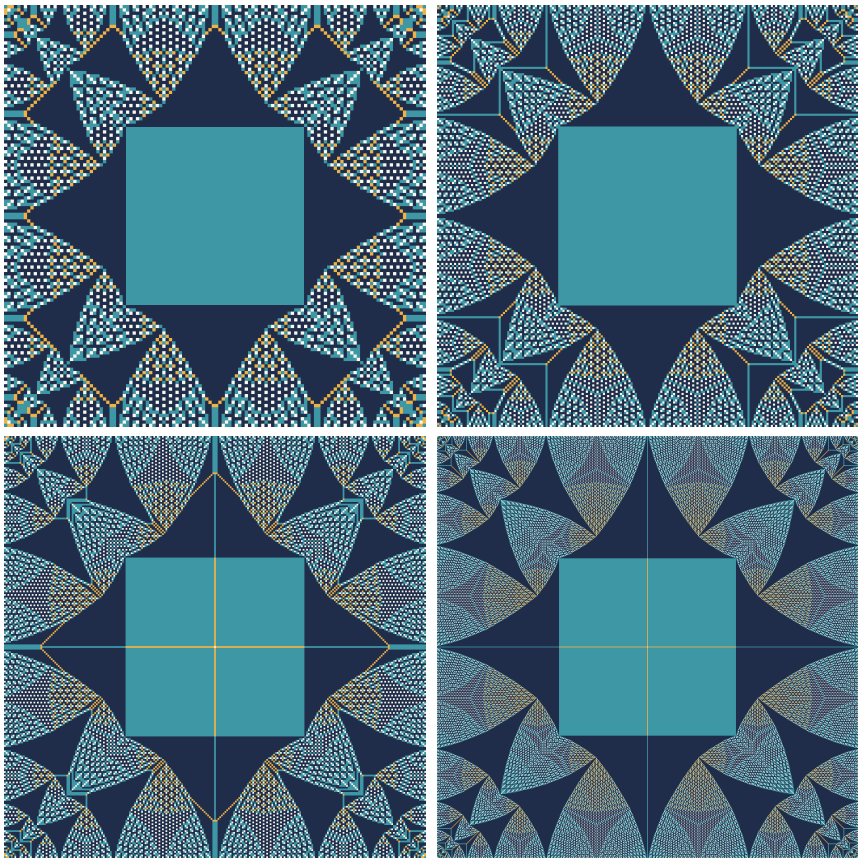


Figure 9.9: The identity element of the sandpile group of the $L \times L$ square grid with wired boundary conditions for $L = 128$ (upper left), 198 (upper right), 243 (lower left) and 521 (lower right), each grid drawn at the same size. Sites with zero, one, two and three grains are off-white, orange, teal and navy, as in the frontispiece.

We say that a sandpile σ is **reachable** from a sandpile ξ if there exists a nonnegative sandpile β such that $\sigma = (\xi + \beta)^\circ$. Every reachable sandpile is stable.

A sandpile is recurrent if and only if it is reachable from every sandpile.

Indeed, a sandpile reachable from every sandpile is stable, and it is accessible because stabilization topples only unstable vertices. Conversely, let σ be recurrent and let ξ be a sandpile. Accessibility gives a sequence of grain additions and legal topplings that takes ξ to σ. Adding grains preserves legality, so we may place all grain additions before the topplings. This gives a nonnegative sandpile β and a legal toppling sequence from $\xi+\beta$ to σ. Since σ is stable, Lemma 9.3(b) gives $\sigma = (\xi+\beta)^\circ$.

A digraph is **strongly connected** if, for every pair of distinct vertices v and w, there are directed paths from v to w and from w to v. We write $G \setminus s$ for the graph obtained from G by deleting s and all edges incident to s.

The next lemma gives a test for recurrence: it suffices that σ absorbs some sandpile whose grains can be routed to every vertex.

Lemma 9.25 *Let G be a finite digraph with a global sink, and let $\sigma \geq 0$ and $\alpha \geq 0$ be sandpiles on G with $(\sigma+\alpha)^\circ = \sigma$. Suppose that, for every non-sink vertex w, there is a directed path from* $\mathrm{supp}(\alpha)$ *to w. Then σ is recurrent.*

Proof. We first find a sandpile β with at least one grain at every non-sink vertex and with $(\sigma+\beta)^\circ = \sigma$. Among all sandpiles $\beta \geq \alpha$ with $(\sigma+\beta)^\circ = \sigma$, choose one whose support has the largest number of vertices; such sandpiles exist, since α is one of them, and the graph is finite. Suppose, for a contradiction, that $\beta(w_0) = 0$ for a non-sink vertex w_0. A directed path from $\mathrm{supp}(\alpha) \subset \mathrm{supp}(\beta)$ to w_0 contains an edge $v \to w$ with $v \in \mathrm{supp}(\beta)$ and w a non-sink vertex outside $\mathrm{supp}(\beta)$. Iterating $(\sigma+\beta)^\circ = \sigma$ gives $(\sigma + (\deg(v)+1)\beta)^\circ = \sigma$. Since $\beta(v) \geq 1$, the sandpile $\sigma + (\deg(v)+1)\beta$ has at least $\deg(v)$ grains at v, so toppling v once is legal. This toppling produces a sandpile of the form $\sigma+\beta'$ with $\beta' \geq \beta$ and $\beta'(w) \geq 1$: it removes $\deg(v)$ grains from v, where $(\deg(v)+1)\beta$ exceeds β by $\deg(v)\beta(v) \geq \deg(v)$ grains, every other vertex only gains grains, and w gains a grain along the edge $v \to w$. Lemma 9.3(b) gives $(\sigma+\beta')^\circ = \sigma$. Since $\beta' \geq \alpha$ and the support of β' strictly contains that of β, this contradicts the choice of β. Hence $\beta(w) \geq 1$ at every non-sink vertex w.

The sandpile σ equals $(\sigma+\beta)^\circ$, so it is stable. Now let ξ be a sandpile. Since β has at least one grain at every non-sink vertex, there is a positive integer ℓ such that $\xi \leq \sigma + \ell\beta$. Adding the nonnegative sandpile $\sigma + \ell\beta - \xi$ to ξ and stabilizing gives $(\sigma+\ell\beta)^\circ$, which equals σ by iterating $(\sigma+\beta)^\circ = \sigma$. Hence σ is accessible, and therefore recurrent. □

The next lemma characterizes recurrence through reversibility of reachability and through the grain addition operators $E_v(\sigma) = (\sigma + \delta_v)^\circ$ of Lemma 9.8.

Lemma 9.26 *Let G be a finite digraph with a global sink s, and let σ be a nonnegative sandpile on G. The following are equivalent.*

(1) The sandpile σ is recurrent.

(2) If ξ is a sandpile reachable from σ, then σ is reachable from ξ.

(3) The sandpile σ is reachable from every sandpile of the form $E_v(\sigma)$, where v is a non-sink vertex of G.

(4) Each strongly connected component of $G \setminus s$ contains a vertex v such that σ is reachable from $E_v(\sigma)$.

Proof. A recurrent sandpile is reachable from every sandpile, so condition (1) implies (2). Condition (2) implies (3), and condition (3) implies (4). It remains to prove that condition (4) implies (1).

Assume (4). In each strongly connected component, choose a vertex v for which σ is reachable from $E_v(\sigma)$. Reachability gives a nonnegative sandpile ξ_v such that $(E_v(\sigma) + \xi_v)^\circ = \sigma$. Since $E_v(\sigma) = (\sigma + \delta_v)^\circ$, the abelian property gives $(\sigma + \delta_v + \xi_v)^\circ = \sigma$. Let α be the sum of $\delta_v + \xi_v$ over the chosen vertices v. Repeated stabilization gives $(\sigma + \alpha)^\circ = \sigma$, and α is positive at one vertex of every strongly connected component.

Each strongly connected component of $G \setminus s$ contains a vertex of $\mathrm{supp}(\alpha)$, so every non-sink vertex can be reached by a directed path from $\mathrm{supp}(\alpha)$. Lemma 9.25 shows that σ is recurrent. □

Deciding whether a sandpile is recurrent directly from the definition means searching over every sandpile that might lead to it. On Eulerian graphs there is a much faster test, one that inspects the sandpile alone and runs in time proportional to the number of edges.

9.6 EULERIAN GRAPHS AND THE BURNING ALGORITHM

A locally finite digraph $G = (V, E)$ is **Eulerian** if it is strongly connected and the in-degree and out-degree of each vertex are equal. We call G an **Eulerian digraph with sink** if it is obtained from a finite Eulerian digraph by deleting all outgoing edges from one vertex. Such a graph has a global sink, and every other vertex has out-degree at least as large as its in-degree. An **Eulerian tour** of a finite digraph is a closed directed walk that uses each edge exactly once. Exercise 9.7 asks you to show that a finite digraph without isolated vertices has an Eulerian tour if and only if it is Eulerian. Replacing each edge

of a connected undirected graph by two oppositely directed edges produces an Eulerian digraph.

Let G be a digraph with global sink s, and let σ be a sandpile on G. A **forbidden subconfiguration** of σ is a nonempty induced subgraph Γ of $G \setminus s$ in which every vertex v has fewer grains than the in-degree of v in Γ.

For an Eulerian digraph with sink, the following test recognizes recurrent sandpiles and specifies exactly how often each vertex topples.

Lemma 9.27 (Dhar's Burning Algorithm) *Let G be a finite Eulerian digraph with sink s. Let β be the sandpile on G given by*

$$\beta(v) := \deg(v) - \mathrm{indeg}(v) \qquad \text{for } v \neq s\,.$$

Then the following statements hold.

(a) A sandpile σ on G is recurrent if and only if $(\sigma + \beta)^\circ = \sigma$.

(b) If a sandpile σ on G is recurrent, then each non-sink vertex topples exactly once during the stabilization of $\sigma + \beta$.

Proof. We first show that a recurrent sandpile σ has no forbidden subconfiguration; this step uses only that s is a global sink. There is a nonnegative sandpile α such that $(\boldsymbol{\delta} + \alpha)^\circ = \sigma$, where $\boldsymbol{\delta}(v) = \deg(v)$. Every non-sink vertex starts with at least $\deg(v)$ grains in $\boldsymbol{\delta} + \alpha$, so each topples at least once in the stabilization. Fix a stabilizing toppling sequence for $\boldsymbol{\delta} + \alpha$. Given a nonempty induced subgraph Γ of $G \setminus s$, let v be the vertex of Γ whose final toppling is earliest. During its final toppling, v receives one grain along each self-loop, and afterward it receives at least one grain along every other incoming edge in Γ. Hence $\sigma(v)$ is at least the in-degree of v in Γ, so Γ is not a forbidden subconfiguration of σ.

Now let σ be stable, and stabilize $\sigma + \beta$. For $v \neq s$, $\beta(v)$ counts the deleted edges $s \to v$, so $\beta \geq 0$. Consider a moment at which no vertex has toppled twice, and let U be the set of non-sink vertices that have not yet toppled. For a non-sink vertex v, write $\mathrm{indeg}_U(v)$ for the number of edges from vertices of U to v. Each non-sink vertex outside U has toppled once and has sent one grain along each of its outgoing edges, so the height at v is

$$\begin{aligned} &\sigma(v) + \beta(v) - \deg(v)\,\mathbf{1}_{V\setminus U}(v) + \mathrm{indeg}(v) - \mathrm{indeg}_U(v) \\ &\qquad = \sigma(v) + \deg(v)\,\mathbf{1}_U(v) - \mathrm{indeg}_U(v)\,. \end{aligned} \tag{9.4}$$

For $v \notin U$, this height is at most $\sigma(v) < \deg(v)$, so no vertex can topple a second time. Hence the stabilization of $\sigma + \beta$ topples each non-sink vertex at most once. From now on, let U be the set of non-sink vertices that never topple. By (9.4), stability of $(\sigma + \beta)^\circ$ gives $\sigma(v) < \mathrm{indeg}_U(v)$ for every

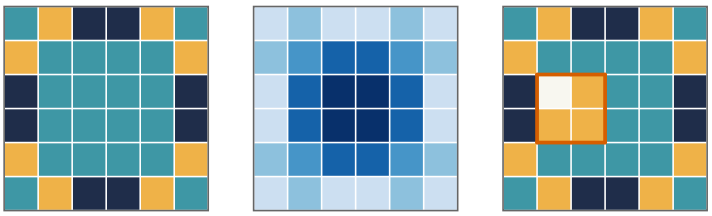
$v \in U$, so if U is nonempty, then it induces a forbidden subconfiguration of σ. Moreover, $(\sigma + \beta)^\circ = \sigma$ only if U is empty: otherwise (9.4) would give $\mathrm{indeg}_U(v) = \deg(v)$ for every $v \in U$, so no edge would leave U, which is impossible because s is a global sink. Conversely, if U is empty, then (9.4) gives $(\sigma + \beta)^\circ = \sigma$.

(a), forward direction. A recurrent sandpile σ is stable and has no forbidden subconfiguration, so U is empty and $(\sigma + \beta)^\circ = \sigma$.

(a), reverse direction. Suppose that $(\sigma + \beta)^\circ = \sigma$. Then σ is stable, so U is empty and every non-sink vertex topples. Immediately after a non-sink vertex topples, its height is nonnegative; later topplings at other vertices can only add grains there. Hence $\sigma \geq 0$. The support of β consists of the non-sink heads of the outgoing edges deleted from the sink. For each non-sink vertex v, choose a directed path from the sink to v in the original Eulerian digraph and retain the part after its last visit to the sink. The first vertex of the retained path lies in $\mathrm{supp}(\beta)$, and the rest of the path remains in $G \setminus s$. Thus every non-sink vertex can be reached by a directed path from $\mathrm{supp}(\beta)$. Since $\beta \geq 0$, Lemma 9.25 with $\alpha = \beta$ shows that σ is recurrent.

(b) For a recurrent sandpile σ, the forward direction of (a) gives $U = \emptyset$, so every non-sink vertex topples exactly once. □

Informally, the burning algorithm works as follows: to determine whether a stable sandpile σ is recurrent, first add the grains that a toppling of the sink would send in the original Eulerian digraph, which gives the sandpile $\sigma + \beta$. The proof of Lemma 9.27 shows that no vertex topples twice during the stabilization of $\sigma + \beta$, and that σ is recurrent if and only if every non-sink vertex topples. So it is enough to record which vertices topple (Figure 9.10).



(a) (b) (c)

Figure 9.10: (a) The identity element of the sandpile group of the 6×6 square grid with wired boundary conditions, with zero, one, two and three grains in off-white, orange, teal and navy, and (b) the round, light to dark, in which each of its sites topples when the burning algorithm of Lemma 9.27 stabilizes $\sigma + \beta$ by toppling every unstable site in each round. (c) With five grains removed, some sites never topple, and they (outlined in vermilion) form a forbidden subconfiguration, so this stable sandpile is not recurrent.

The burning algorithm has the following equivalent formulation.

Lemma 9.28 *Let G be a finite Eulerian digraph with sink s. A stable sandpile σ on G is recurrent if and only if σ has no forbidden subconfiguration.*

Proof. The first step of the proof of Lemma 9.27 shows that a recurrent sandpile has no forbidden subconfiguration. Conversely, suppose that σ has no forbidden subconfiguration, and let β be the sandpile defined in Lemma 9.27. The proof of that lemma shows that the non-sink vertices that never topple during the stabilization of $\sigma + \beta$ induce a forbidden subconfiguration whenever such vertices exist. Hence every non-sink vertex topples, and the same proof gives $(\sigma + \beta)^\circ = \sigma$. Lemma 9.27(a) shows that σ is recurrent. □

The proof of the forward implication uses no property of Eulerian digraphs, but the converse can fail for other digraphs with a global sink. For example, take non-sink vertices v and w and a sink s, with one edge from v to w, two edges from w to v, and one edge from w to s. Then the reduced Laplacian has $\det \mathsf{L}' = 1$, and the only recurrent sandpile has no grain at v and two grains at w, while the stable sandpile with no grain at v and one grain at w has no forbidden subconfiguration.

The sandpile group of a finite Eulerian digraph does not depend on which vertex is chosen as the sink.

Proposition 9.29 *Let G be a finite Eulerian digraph on n vertices, and for a vertex v let G_v be the Eulerian digraph with sink obtained by deleting the outgoing edges from v. Then for all vertices v and w, the sandpile groups $\mathcal{S}(G_v)$ and $\mathcal{S}(G_w)$ are isomorphic.*

Proof. By Definition 9.15, $\mathcal{S}(G_v) = \mathbb{Z}^{n-1}/\mathbb{Z}^{n-1}\mathsf{L}'(G_v)$. We show that this group is isomorphic to $\mathbb{Z}^n_0/\mathbb{Z}^n\mathsf{L}(G)$, where $\mathbb{Z}^n_0$ is the set of vectors in $\mathbb{Z}^n$ whose coordinates sum to 0. Order the coordinates so that v is last. Projection onto the first $n-1$ coordinates is an isomorphism from $\mathbb{Z}^n_0$ to $\mathbb{Z}^{n-1}$. It maps $\mathbb{Z}^n\mathsf{L}(G)$ onto $\mathbb{Z}^{n-1}\mathsf{L}'(G_v)$: the rows other than the vth project to the rows of $\mathsf{L}'(G_v)$, and the vth row is redundant because the sum of the rows of the full Laplacian $\mathsf{L}(G)$ is zero. Since $\mathbb{Z}^n_0/\mathbb{Z}^n\mathsf{L}(G)$ does not depend on the choice of v, the groups $\mathcal{S}(G_v)$ and $\mathcal{S}(G_w)$ are isomorphic. □

So far we have taken for granted that we can recognize a sandpile that stabilizes. The least action principle shows that a sandpile stabilizes as soon as one finite collection of topplings leaves a stable sandpile behind, and it bounds the odometer by every such collection. It is our main tool for the rest of the chapter.

9.7 LEAST ACTION PRINCIPLE

The least action principle compares legal relaxation with a prescribed collection of topplings that produces a stable sandpile. The principle gives both

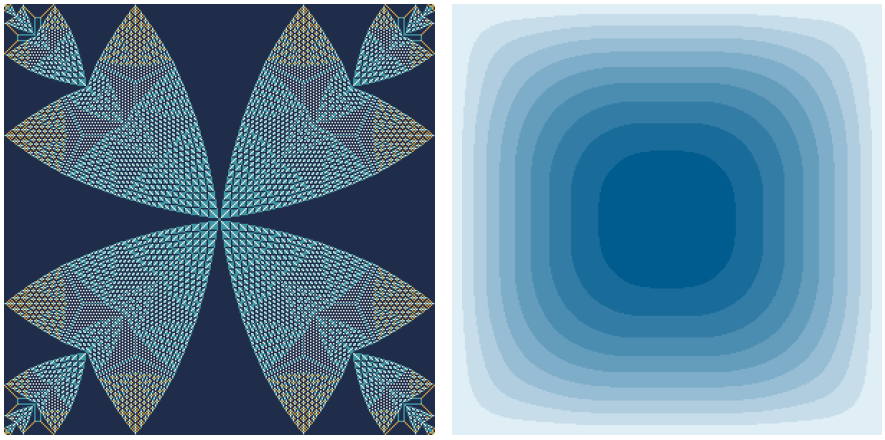


Figure 9.11: The stabilization of four grains at every site of the 300×300 square whose outside is the sink (left), with zero, one, two and three grains in off-white, orange, teal and navy, and its odometer (right), shaded in ten bands of equal height from light to dark up to its maximum of 11325 topplings at the center. The final sandpile minus 4 is the discrete Laplacian of the odometer.

a test for stabilization and a pointwise bound on the odometer. Figure 9.11 shows a stabilized sandpile and its odometer.

Recall the least action principle, Lemma 9.2: a legal toppling sequence never topples a vertex more often than a sequence of topplings, legal or illegal, that ends at a stable sandpile. A finite sequence with odometer u takes σ to $\sigma - u\mathsf{L}$, where L is the graph Laplacian, so the least action principle has the following form.

Corollary 9.30 *Let G be a digraph, and let σ be a sandpile on G. Suppose that $g : V \to \mathbb{Z}$ is nonnegative, finitely supported and zero at every sink, and that $\sigma - g\mathsf{L}$ is stable. Then every legal toppling sequence from σ is finite and has odometer $u \le g$.*

Proof. Topple each non-sink vertex v exactly $g(v)$ times. This finite sequence, which need not be legal, ends at the stable sandpile $\sigma - g\mathsf{L}$. Lemma 9.2 shows that every legal sequence contains at most $\sum_v g(v)$ topplings and has odometer at most g. □

We call a function g with the properties in Corollary 9.30 a **stabilizing toppling function** for σ. A sandpile with a stabilizing toppling function stabilizes, since by the corollary every maximal legal toppling sequence from it is finite and therefore ends at a stable sandpile.

We say that a sandpile σ on a digraph G is **robust** if $\sigma + \xi$ stabilizes for every finitely supported sandpile $\xi \ge 0$. Section 9.10 studies robustness in detail. The following example proves robustness even for sandpiles with infinite support, to which Proposition 9.7 does not apply.

Example 9.31 Let σ be a periodic sandpile on $\mathbb{Z}$ that takes values 0 or 1 and is not identically 1. Then σ is robust.

Proof. We prove more generally that a sandpile σ on $\mathbb{Z}$ with values 0 and 1 is robust when its zeros are unbounded in both directions; a periodic sandpile that is not identically 1 has this property. We first stabilize a single added grain by an explicit sequence of legal topplings.

Add one grain at a site v. If $\sigma(v) = 0$, then the result is stable. Otherwise let $x < v < y$ be the nearest zeros of σ on the two sides of v, so that $\sigma = 1$ strictly between them. Topple the sites $v, v+1, \ldots, y-1$ in this order. Each toppling is legal: v has two grains, and each later site has two grains once it has received one from its left neighbor. Each toppled site other than $y-1$ gets one grain back from its right neighbor, the site $y-1$ ends empty, and y receives a grain. The configuration is now $\sigma + \delta_{v-1} + \delta_y - \delta_{y-1}$: the extra grain has moved to $v-1$, and the nearest zero on its right has moved from y to $y-1$. While the extra grain is to the right of x, repeat the sweep with both endpoints decreased by one: the next sweep runs from $v-1$ through $y-2$. Each sweep moves both the extra grain and the zero on its right one step to the left (Figure 9.12). After $v - x$ sweeps the extra grain lands on the zero at x, and the configuration is

$$\sigma + \delta_x + \delta_y - \delta_{x+y-v}\,.$$

This configuration takes only the values 0 and 1, so it is stable, and it differs from σ only on $[x, y]$. In particular, its zeros are still unbounded in both directions.

Now let $\xi \geq 0$ be finitely supported. Adding the grains of ξ one at a time, and stabilizing after each addition as above, gives a finite sequence of legal topplings. Making all the additions at the start keeps every toppling legal, because the grains added early only increase the heights, and it leads to the same final configuration. Hence $\sigma + \xi$ stabilizes. □

Exercise 9.11 characterizes all robust sandpiles on $\mathbb{Z}$ that take only the values zero and one.

On $\mathbb{Z}^d$, the next section shows that the odometer inherits the reflection symmetries and the monotonicity of the initial sandpile.

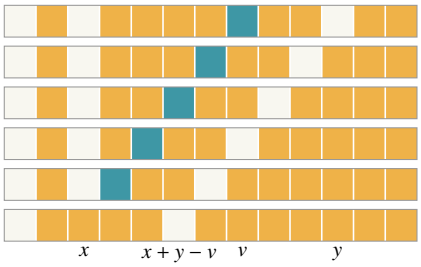


Figure 9.12: Example 9.31 with zeros at x and y and an extra grain at v, with zero, one and two grains in off-white, orange and teal: each sweep (one row to the next) moves the extra grain and the nearest zero on its right one step to the left. After $v - x$ sweeps the grain lands on the zero at x, and the only zero left in $[x, y]$ is at $x + y - v$.

9.8 MONOTONICITY OF THE ODOMETER ON THE CUBIC LATTICE

We now specialize to $\mathbb{Z}^d$. The graph Laplacian is symmetric in this case, and $-u\mathsf{L} = \Delta^{::}u$, where $\Delta^{::}$ is the discrete Laplacian of Chapter 3: for $f : \mathbb{Z}^d \to \mathbb{R}$,

$$\Delta^{::} f(x) = \sum_{y \sim x} (f(y) - f(x)) \,.$$

We will use the pointwise inequality

$$\Delta^{::} \min\{f, g\} \le \max\{\Delta^{::} f, \Delta^{::} g\} \,. \tag{9.5}$$

Indeed, if $f(x) \le g(x)$, then

$$\Delta^{::} \min\{f, g\}(x) = \sum_{y \sim x} (\min\{f(y), g(y)\} - f(x)) \le \Delta^{::} f(x) \,,$$

and symmetrically $\Delta^{::} \min\{f, g\}(x) \le \Delta^{::} g(x)$ when $g(x) \le f(x)$.

Next we observe that the odometer inherits some of the discrete symmetries of the initial sandpile. To see this we use the **parallel toppling procedure**, in which every unstable vertex topples once in each round. The sandpile and odometer after t rounds satisfy

$$\begin{aligned} \sigma_{t+1} &= \sigma_t + \Delta^{::} \mathbf{1}\{\sigma_t \ge 2d\}, \\ u_{t+1} &= u_t + \mathbf{1}\{\sigma_t \ge 2d\} \,, \end{aligned} \tag{9.6}$$

where $u_0 \equiv 0$ and $\sigma_0 : \mathbb{Z}^d \to \mathbb{Z}$ is the initial sandpile.

Induction using equation (9.6) gives

$$\sigma_t = \sigma_0 + \Delta^{::} u_t \,.$$

Hence the odometers alone satisfy

$$\begin{aligned} u_{t+1}(x) &= u_t(x) + \mathbf{1}\{\sigma_t(x) \ge 2d\} \\ &= \max\left\{ u_t(x), \min\left\{ \left\lfloor \frac{\sigma_0(x) + \sum_{y \sim x} u_t(y)}{2d} \right\rfloor, u_t(x) + 1 \right\} \right\} , \end{aligned} \tag{9.7}$$

where the second equality holds because $\sigma_t = \sigma_0 + \Delta^{::} u_t$, so $\sigma_t(x) \geq 2d$ exactly when the floor is at least $u_t(x) + 1$. We first show that the parallel odometer eventually dominates the odometer of every finite legal toppling sequence. The proofs of explosion in Section 9.10 will also use Lemma 9.32.

Lemma 9.32 *Let $d \geq 1$, let σ be a sandpile on $\mathbb{Z}^d$, and let $(u_t)_{t\geq 0}$ be the parallel toppling odometers started from σ. Then the following statements hold.*

(a) For every finite legal toppling sequence from σ, there is an integer $t \geq 0$ such that u_t is at least the odometer of the finite sequence.

(b) If σ stabilizes with final odometer u, then $u_t = u$ for some integer $t \geq 0$.

Proof. (a) We prove the odometer domination by induction on the length of the legal sequence. The empty sequence is dominated by u_0. For the inductive step, suppose u_t dominates the odometer r of the topplings already performed, and let x be the next vertex. If $u_t(x) > r(x)$, then u_t already accounts for the toppling at x. Otherwise $u_t(x) = r(x)$, while $u_t(y) \geq r(y)$ for every neighbor y of x. Therefore

$$\sigma(x) + \Delta^{::} u_t(x) \geq \sigma(x) + \Delta^{::} r(x) \geq 2d\,.$$

The vertex x topples in the next parallel round, which proves the inductive step.

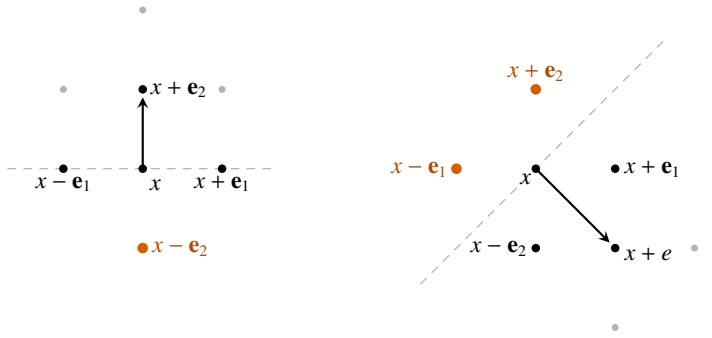
(b) Suppose σ stabilizes with odometer u. We prove by induction on t that every parallel round is finite and $u_t \leq u$. Both assertions are immediate for $t = 0$. For the inductive step, suppose the first t rounds have been performed. Appending one toppling at a currently unstable vertex is legal, so Lemma 9.2 gives $u_t + \delta_x \leq u$ for every such vertex x. Since u has finite support, the unstable set is finite. Ordering its topplings gives $u_{t+1} \leq u$. Applying part (a) to a stabilizing sequence gives an integer t with $u_t \geq u$. Hence $u_t = u$. □

Under the hypothesis of part (b), parallel toppling reaches σ°: if $u_t = u$, then after t rounds the sandpile is $\sigma + \Delta^{::} u_t$, which equals $\sigma + \Delta^{::} u = \sigma^\circ$.

Since the discrete Laplacian commutes with graph automorphisms, parallel toppling preserves every symmetry of the initial sandpile.

Lemma 9.33 *Let $d \geq 1$, let σ_0 be a sandpile on $\mathbb{Z}^d$, and let $(\sigma_t, u_t)_{t\geq 0}$ be its parallel toppling process. Let φ be a graph automorphism of $\mathbb{Z}^d$. If $\sigma_0 \circ \varphi = \sigma_0$, then $\sigma_t \circ \varphi = \sigma_t$ and $u_t \circ \varphi = u_t$ for every integer $t \geq 0$.*

Proof. We argue by induction on $t \geq 0$. The base case follows from the hypothesis and $u_0 \equiv 0$. For the inductive step, suppose $\sigma_t \circ \varphi = \sigma_t$ and $u_t \circ \varphi = u_t$. Then $\mathbf{1}\{\sigma_t \geq 2d\} \circ \varphi = \mathbf{1}\{\sigma_t \geq 2d\}$, and since the discrete



(a) $e = \mathbf{e}_2$ (b) $e = \mathbf{e}_1 - \mathbf{e}_2$

Figure 9.13: The two cases $x \cdot e = 0$ in the proof of Lemma 9.34: (a) $e = \mathbf{e}_2$ and (b) $e = \mathbf{e}_1 - \mathbf{e}_2$. The dashed line is the hyperplane $\{y : y \cdot e = 0\}$, the arrow points from x to $x + e$, the neighbors of x outside the half-space $\{y : y \cdot e \geq 0\}$ are orange, and the gray dots are the other neighbors of $x + e$.

Laplacian commutes with φ, composing both lines of (9.6) with φ gives $\sigma_{t+1} \circ \varphi = \sigma_{t+1}$ and $u_{t+1} \circ \varphi = u_{t+1}$. □

We next show that parallel toppling also preserves directional monotonicity. A function $f : \mathbb{Z}^d \to \mathbb{R}$ is **decreasing** on a set A in direction $e \in \mathbb{Z}^d$ if

$$f(x) \geq f(x + e) \qquad \text{whenever } x \in A \text{ and } x + e \in A\,.$$

For $e = \mathbf{e}_i$ with $1 \leq i \leq d$ or $e = \mathbf{e}_i - \mathbf{e}_j$ with distinct $1 \leq i, j \leq d$, write φ_e for the reflection across $\{x \in \mathbb{Z}^d : x \cdot e = 0\}$. The next lemma shows that reflection invariance and directional monotonicity of the initial sandpile pass to the parallel toppling odometer at every time.

Lemma 9.34 *Let $d \geq 1$, and let $e = \mathbf{e}_i$ for some $1 \leq i \leq d$ or $e = \mathbf{e}_i - \mathbf{e}_j$ for distinct $1 \leq i, j \leq d$. Let σ_0 be a sandpile on $\mathbb{Z}^d$, and let $(\sigma_t, u_t)_{t \geq 0}$ be its parallel toppling process. Suppose $\sigma_0 \circ \varphi_e = \sigma_0$ and σ_0 is decreasing in direction e on the half-space $\{x \in \mathbb{Z}^d : x \cdot e \geq 0\}$. Then for every integer $t \geq 0$, the odometer u_t is invariant under φ_e and is decreasing in direction e on this half-space.*

When $e = \mathbf{e}_i - \mathbf{e}_j$, the reflection φ_e exchanges the coordinates x_i and x_j.

Proof of Lemma 9.34. Lemma 9.33 gives $u_t \circ \varphi_e = u_t$. We prove the directional monotonicity by induction on the integer $t \geq 0$. The case $t = 0$ follows from $u_0 \equiv 0$.

Fix $x \in \mathbb{Z}^d$ with $x \cdot e \geq 0$. If $u_t(x) > u_t(x + e)$, then integrality and the fact that one parallel round increases each odometer by at most one give

$$u_{t+1}(x) \geq u_t(x) \geq u_t(x + e) + 1 \geq u_{t+1}(x + e)\,.$$

We may therefore assume $u_t(x) = u_t(x + e)$.

Suppose first that $x \cdot e > 0$. Every neighbor y of x satisfies $y \cdot e \geq 0$, so the induction hypothesis gives

$$\sum_{y \sim x} u_t(y) \geq \sum_{y \sim x} u_t(y + e)\,.$$

The initial monotonicity gives $\sigma_0(x) \geq \sigma_0(x+e)$. The right-hand side of (9.7) is nondecreasing separately in the three quantities $u_t(x)$, the initial height $\sigma_0(x)$, and the sum $\sum_{y \sim x} u_t(y)$. Hence $u_{t+1}(x) \geq u_{t+1}(x + e)$.

It remains to consider $x \cdot e = 0$, as illustrated in Figure 9.13. If $e = \mathbf{e}_i$, then the induction hypothesis compares every neighbor y of x except $x - \mathbf{e}_i$ with $y + \mathbf{e}_i$. For the remaining neighbor, symmetry and $u_t(x) = u_t(x + \mathbf{e}_i)$ give

$$u_t(x - \mathbf{e}_i) = u_t(x + \mathbf{e}_i) = u_t(x)\,.$$

Thus the sum of u_t over the neighbors of x is at least the corresponding sum at $x + \mathbf{e}_i$.

If $e = \mathbf{e}_i - \mathbf{e}_j$, then the induction hypothesis compares every neighbor except $x - \mathbf{e}_i$ and $x + \mathbf{e}_j$. Symmetry gives the two remaining comparisons as equalities:

$$u_t(x - \mathbf{e}_i) = u_t(x - \mathbf{e}_j),$$
$$u_t(x + \mathbf{e}_j) = u_t(x + \mathbf{e}_i)\,.$$

The sum of u_t over the neighbors of x is again at least the corresponding sum at $x+e$. In both cases, $u_t(x) = u_t(x+e)$ and $\sigma_0(x) \geq \sigma_0(x+e)$. Equation (9.7) completes the induction. □

Monotonicity describes the shape of the level sets of the odometer but says nothing about their size. How far do the topplings spread?

9.9 GROWTH RATES

We now ask what happens when n grains are added at the origin of $\mathbb{Z}^d$ to a sandpile $\eta : \mathbb{Z}^d \to \mathbb{Z}$, called the **background**. For $x \in \mathbb{Z}^d$ we write $|x|_1 := |x_1| + \cdots + |x_d|$ and $|x|_\infty := \max_i |x_i|$ for its ℓ^1-norm and ℓ^∞-norm. The next proposition shows that the toppled set has diameter at most of order $n^{1/d}$ when $\eta \leq 2d - 2$, and at least of order $n^{1/d}$ when in addition $\eta \geq 0$. Figures 9.14 and 9.15 show the toppled sets of 10^6 grains in $\mathbb{Z}^2$ and 10^7 grains in $\mathbb{Z}^3$ on the zero background. Figure 1.3 in Chapter 1 shows the planar pattern at three sizes, and Figure 4.7 in Chapter 4 places it between two balls.

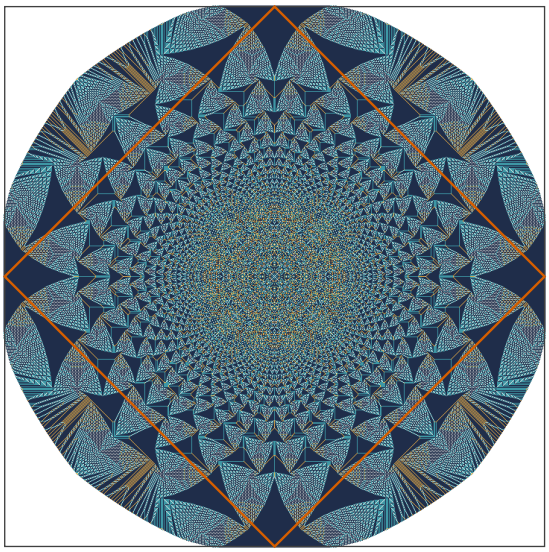


Figure 9.14: The stabilization of $10^6\delta_{\mathbf{0}}$ in $\mathbb{Z}^2$, with toppled sites colored by their final number of grains $0, 1, 2, 3$ in off-white, orange, teal and navy. Here $K = 363$ is the largest ℓ^∞-norm of a toppled site. The toppled set contains the diamond $\{|x|_1 \le K\}$ (vermilion) and lies in the square $\{|x|_\infty \le K\}$ (black), which are the inclusions (9.8) in the proof of Proposition 9.35.

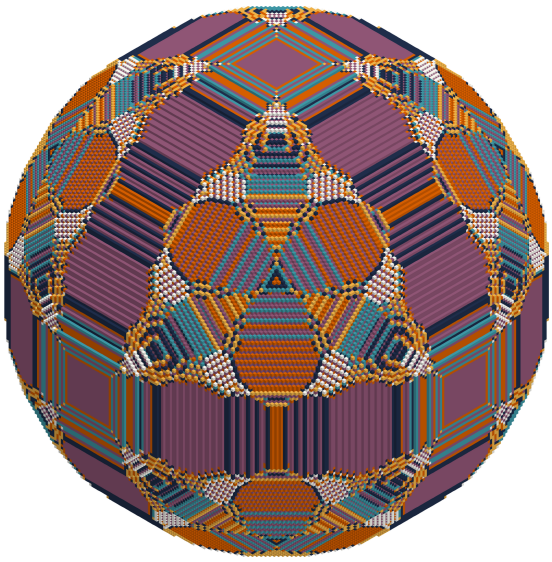


Figure 9.15: The sites of $\mathbb{Z}^3$ that topple when 10^7 grains at the origin are stabilized, each drawn as a unit cube and seen along the direction $(1, 1, 1)$. Each cube is colored by the number of grains its site keeps: off-white 0, orange 1, teal 2, navy 3, vermilion 4 and purple 5.

Proposition 9.35 *For every $d \geq 1$ and every integer b with $0 \leq b \leq 2d-2$, there are constants $c = c(d) > 0$ and $C = C(d,b) < \infty$ with the following property. Let $\eta : \mathbb{Z}^d \to \mathbb{Z}$ satisfy $\eta \leq b$. Then the following statements hold.*

(a) For every integer $n \geq 0$, the sandpile $\eta + n\delta_{\mathbf{0}}$ stabilizes.

(b) For every sufficiently large integer n, the final odometer $u_\infty^{(n)}$ of $\eta + n\delta_{\mathbf{0}}$ satisfies

$$\{u_\infty^{(n)} > 0\} \subseteq \{x \in \mathbb{Z}^d : |x|_\infty \leq Cn^{1/d}\}\,.$$

(c) If $\eta \geq 0$, then for every sufficiently large integer n, the final odometer $u_\infty^{(n)}$ of $\eta + n\delta_{\mathbf{0}}$ satisfies

$$\{x \in \mathbb{Z}^d : |x|_1 < cn^{1/d}\} \subseteq \{u_\infty^{(n)} > 0\}\,.$$

Proof. The proof has three steps. We first establish both bounds for the zero background. We next prove parts (a) and (b) by adding to the zero-background odometer a correction built from a one-dimensional sandpile. Finally, we compare nonnegative backgrounds with zero to obtain part (c).

For the zero background, the sandpile $n\delta_{\mathbf{0}}$ stabilizes by Lemma 9.14. Let u be its final odometer, which equals the parallel odometer u_t for all large t by Lemma 9.32(b). Let $U := \{u > 0\}$, and let K be the largest ℓ^∞-norm of a point in U, with $K := 0$ if U is empty. For large n, the set U is nonempty. The sandpile $n\delta_{\mathbf{0}}$ is invariant under every signed coordinate permutation, and for $e = \mathbf{e}_k$ with $1 \leq k \leq d$ and $e = \mathbf{e}_1 - \mathbf{e}_k$ with $2 \leq k \leq d$ it is decreasing in direction e on $\{x \in \mathbb{Z}^d : x \cdot e \geq 0\}$, because $x + e \neq \mathbf{0}$ there. Hence Lemmas 9.33 and 9.34 show that u is invariant under signed coordinate permutations and that

$$u(z) \geq u(z + \mathbf{e}_k) \quad \text{if } z_k \geq 0 \qquad \text{and} \qquad u(z) \geq u(z + \mathbf{e}_1 - \mathbf{e}_k) \quad \text{if } z_1 \geq z_k\,.$$

By symmetry there is $x \in U$ with $x_1 = K$ and $x_i \geq 0$ for every i. Lowering $x_2, \ldots, x_d$ to 0 one unit at a time and applying the first inequality at each step gives $u(K\mathbf{e}_1) \geq u(x) > 0$.

Now let $|y|_1 \leq K$. By symmetry we may assume $y_1 \geq y_2 \geq \cdots \geq y_d \geq 0$. Let $y' := y + (K - |y|_1)\mathbf{e}_1$, which still has nonincreasing coordinates and satisfies $|y'|_1 = K$. The first inequality gives $u(y) \geq u(y')$. We pass from $K\mathbf{e}_1$ to y' by moving one unit at a time from the first coordinate to one of the coordinates $2, \ldots, d$. Before a move $z \mapsto z - \mathbf{e}_1 + \mathbf{e}_k$, at least one unit remains to be moved and coordinate k is not yet full, so $z_1 \geq y'_1 + 1 \geq y'_k + 1 \geq z_k + 2$. The second inequality at the point $z - \mathbf{e}_1 + \mathbf{e}_k$ therefore gives $u(z - \mathbf{e}_1 + \mathbf{e}_k) \geq u(z)$. Hence $u(y) \geq u(y') \geq u(K\mathbf{e}_1) > 0$, and symmetry gives

$$\{y \in \mathbb{Z}^d : |y|_1 \leq K\} \subseteq U \subseteq \{y \in \mathbb{Z}^d : |y|_\infty \leq K\}\,. \tag{9.8}$$

A grain moves only when the current site of the grain topples. Hence every final grain lies at a site of U or at a neighbor of U. Since the final sandpile is stable and has n grains, $n \le (2d-1)(2K+3)^d$, which gives $K \ge c(d)n^{1/d}$ for all sufficiently large n.

For the reverse inequality, note that U contains $\mathbf{0}$ and is connected, because a site other than the origin first topples only after receiving a grain from a neighbor that toppled earlier. At the end of the stabilization every site of U has toppled, so Lemma 9.11 with $S = \emptyset$ shows that at most n edges have an endpoint in U. Since U is connected, at least $|U| - 1$ edges have both endpoints in U, and hence $|U| - 1 \le n$. By the first inclusion in (9.8), the set U contains the cube $\{0, \dots, \lfloor K/d \rfloor\}^d$, so

$$(\lfloor K/d \rfloor + 1)^d \le n + 1 \,.$$

Thus $K \le C(d)n^{1/d}$.

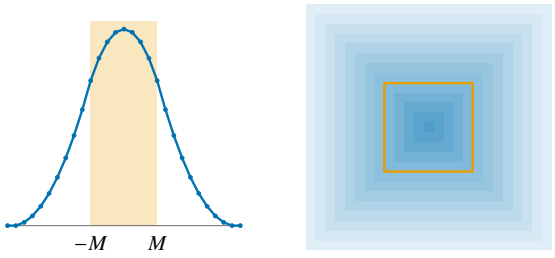
For the second step, fix $n \ge 0$ and a background $\eta \le b$. The idea is to correct the zero-background odometer u by a function whose Laplacian is at most $-b$ where grains have accumulated and at most 1 everywhere; we obtain it by stabilizing a block of $b+1$ grains per site on $\mathbb{Z}$ and taking the minimum over the coordinates. The final zero-background sandpile is $(n\delta_{\mathbf{0}})^\circ = n\delta_{\mathbf{0}} + \Delta^{::}u$. Let $M := K+1$. Every grain ends at the origin or at a neighbor of a toppled site, so $(n\delta_{\mathbf{0}})^\circ$ vanishes outside the cube $\{x \in \mathbb{Z}^d : |x|_\infty \le M\}$.

On $\mathbb{Z}$, the sandpile $(b+1)\mathbf{1}_{[-M,M]}$ has $(b+1)(2M+1)$ grains. Let $R := M + (b+1)(2M+1)$. By Lemma 9.14, this sandpile stabilizes and no grain leaves the interval $[-R, R]$. Its odometer g is therefore a nonnegative integer-valued function that vanishes outside $[-R, R]$, and stability of the final sandpile on $\mathbb{Z}$ gives

$$g(j+1) + g(j-1) - 2g(j) \le 1 - (b+1)\mathbf{1}_{\{|j| \le M\}} \qquad \text{for } j \in \mathbb{Z}\,. \tag{9.9}$$

Let $w(x) := \min_{1 \le i \le d} g(x_i)$ for $x \in \mathbb{Z}^d$, a nonnegative integer-valued function that vanishes outside the cube $\{|x|_\infty \le R\}$ (Figure 9.16). If i is an index at which the minimum is attained, then $w(x) = g(x_i)$ and $w(y) \le g(y_i)$ for every neighbor y of x, so $\Delta^{::}w(x)$ is at most $g(x_i+1) + g(x_i-1) - 2g(x_i)$, as in (9.5). By (9.9), it follows that $\Delta^{::}w \le -b$ on the cube $\{|x|_\infty \le M\}$ and $\Delta^{::}w \le 1$ everywhere.

We check that $u + w$ stabilizes $\eta + n\delta_{\mathbf{0}}$. On the cube $\{|x|_\infty \le M\}$ we have $\eta + \Delta^{::}w \le 0$ and $(n\delta_{\mathbf{0}})^\circ \le 2d-1$. Outside this cube $(n\delta_{\mathbf{0}})^\circ$ vanishes, while $\eta \le b$ and $\Delta^{::}w \le 1$, so $\eta + \Delta^{::}w \le b+1 \le 2d-1$. Hence $\eta + n\delta_{\mathbf{0}} + \Delta^{::}(u+w) = \eta + (n\delta_{\mathbf{0}})^\circ + \Delta^{::}w \le 2d-1$ everywhere, and $u+w$ is a nonnegative, integer-valued, finitely supported stabilizing toppling function for $\eta + n\delta_{\mathbf{0}}$. Corollary 9.30 shows that $\eta + n\delta_{\mathbf{0}}$ stabilizes, which proves part (a), and that its final odometer is at most $u + w$. Since U lies in the cube $\{|x|_\infty \le K\}$ and $K < R$, the final odometer vanishes outside the cube



(a) (b)

Figure 9.16: The second step of the proof of Proposition 9.35 for $d = 2$, $b = 2$ and $M = 4$: (a) the odometer g of the sandpile $3\mathbf{1}_{\{-M,M\}}$ on $\mathbb{Z}$, and (b) the level sets of $w(x) = \min\{g(x_1), g(x_2)\}$, which are squares. The discrete Laplacian of w is at most $-b$ on the square $\{|x|_\infty \le M\}$ (orange) and at most 1 everywhere.

$\{|x|_\infty \le R\}$. Now $R = (2b+3)K + 3b + 4$ and $K \le C(d)n^{1/d}$ for all large n by the first step, so $R \le C(d,b)n^{1/d}$ for all large n, which proves part (b).

For the final step, suppose $\eta \ge 0$. Every toppling used for the zero background remains legal after adding η. Since $\eta + n\delta_{\mathbf{0}}$ stabilizes, the least action principle, Lemma 9.2, shows that its final odometer is at least the zero-background odometer. The lower inclusion follows from (9.8). □

Chapter 4 obtained Euclidean ball bounds for the zero background through the divisible sandpile, with explicit leading constants (Proposition 4.11). The proof above is a different, combinatorial argument that gives bounds directly: for the zero background it places the toppled set between a diamond and a cube of the same radius, by (9.8). Its stabilization and outer bound work in every dimension and on every background at most $2d - 2$, and its inner bound on every such background that is nonnegative, whereas Chapter 4 needs $d \ge 2$ and the zero background.

Proposition 9.35 determines the size of the toppled set only up to constant factors. For the constant background $2d - 2$, the odometer support is exactly a cube, as Figure 9.17 illustrates.

Proposition 9.36 *Let $d \ge 1$ and $n \ge 2$ be integers. The sandpile $(2d-2)+n\delta_{\mathbf{0}}$ on $\mathbb{Z}^d$ stabilizes. If $u_\infty^{(n)}$ is its final odometer, then there exists a nonnegative integer K_n such that*

$$\{u_\infty^{(n)} > 0\} = \{x \in \mathbb{Z}^d : |x|_\infty \le K_n\}.$$

Proof. By part (a) of Proposition 9.35 with $b = 2d - 2$, the sandpile $(2d-2) + n\delta_{\mathbf{0}}$ stabilizes, so by the least action principle every legal toppling sequence started from it is finite. We choose a legal order in which the set of sites toppled so far is a box after each batch of topplings, so that the final toppled set is a box; symmetry then makes it a cube.

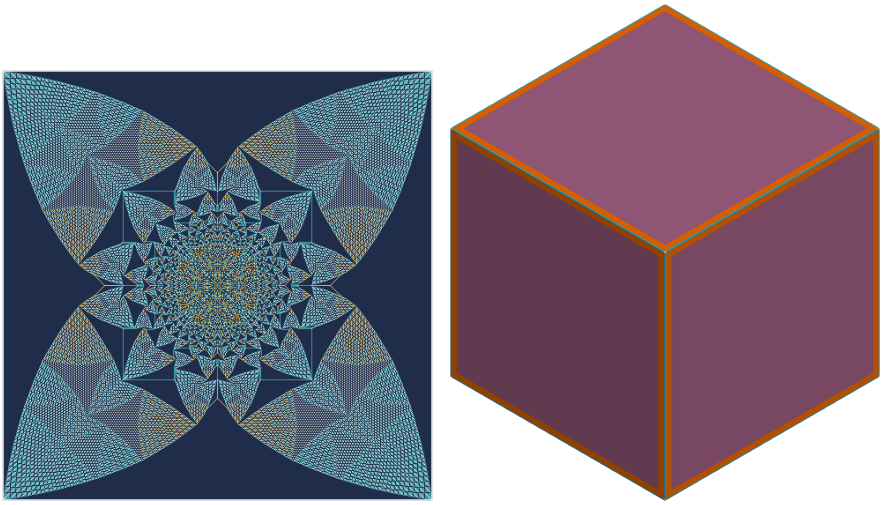


Figure 9.17: The stabilization of $(2d-2)+n\delta_{\mathbf{0}}$ for $d=2$ and $n=10^5$ (left) and for $d=3$ and $n=10^6$ (right), each toppled site colored by its final number of grains as in Figure 9.15. The sites that toppled form exactly the square $\{|x|_\infty \le 242\}$ and the cube $\{|x|_\infty \le 98\}$.

Topple the origin first, so that the toppled set is the box $\{\mathbf{0}\}$. Suppose that the set B of sites toppled so far is a box and that the current sandpile has an unstable site. If this site lies in B, then topple it; the toppled set does not change. Otherwise the unstable site v lies outside B. It was stable initially, so it has received a grain from a neighbor in B, and hence it lies on an exterior face F of B, obtained by fixing one coordinate one step beyond its interval in B and keeping the other coordinate intervals. Every site of F differs from the origin, has not toppled, and has received a grain from its neighbor in B, so it has at least $2d-1$ grains. Topple v, and then topple, one at a time, untoppled sites of F adjacent to a site of F already toppled; each has received the additional grain it needs. Since F is connected, all its sites topple, and the toppled set becomes the box $B \cup F$ (Figure 9.18).

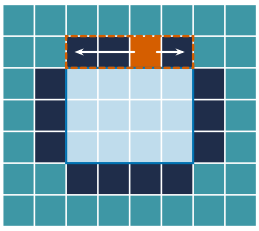


Figure 9.18: The proof of Proposition 9.36 in $\mathbb{Z}^2$: the toppled sites form a box (blue), and each site of an exterior face of the box holds three grains (navy). Once one of them is unstable (vermilion), toppling along the face (arrows) topples every site of the face (dashed), and the toppled set becomes a larger box.

This procedure consists of legal topplings, so it ends at a stable sandpile, and the final toppled set $\{u_\infty^{(n)} > 0\}$ is a box. The sandpile $(2d-2) + n\delta_{\mathbf{0}}$ is invariant under reflections and permutations of the coordinates, and the final odometer does not depend on the legal order of topplings, by Lemma 9.3(b), so this box is invariant under the same symmetries. It is therefore a cube $\{x \in \mathbb{Z}^d : |x|_\infty \le K_n\}$ with K_n a nonnegative integer. □

These bounds assumed a background of height at most $2d-2$. Raising the background even slightly can change the answer completely: on some backgrounds a single added grain sets off topplings that never stop.

9.10 ROBUST AND EXPLOSIVE BACKGROUNDS

On an infinite graph, the topplings caused by adding finitely many grains can either stop or propagate indefinitely. We now give criteria for stabilization and indefinite propagation.

Let G be a locally finite digraph. Recall from Section 9.7 that a sandpile η on G is robust if $\eta + \xi$ stabilizes for every finitely supported sandpile $\xi \ge 0$. We call η **explosive** if there exists such a ξ for which parallel toppling of $\eta+\xi$, in which every unstable vertex topples once in each round as in Section 9.8, eventually topples every vertex of G.

An infinite avalanche need not be an explosion. On $\mathbb{Z}^2$, let η equal 3 on the horizontal axis and 2 elsewhere. Adding one grain at the origin makes each site of the axis topple exactly once and no other site topple. By the least action principle, $\eta + \delta_{\mathbf{0}}$ does not stabilize, although each site topples only finitely often. Hence η is not robust, while this addition does not topple every vertex.

The next lemma gives a simple test for robustness.

Lemma 9.37 *Let G be a strongly connected, locally finite digraph, and let v be a non-sink vertex of G.*

(a) Let η be a sandpile on G, and let $\xi \ge 0$ be finitely supported. There is an integer $n \ge 0$ and a finite legal toppling sequence from $\eta + n\delta_v$ whose final sandpile is at least $\eta + \xi$.

(b) A sandpile η on G is robust if and only if $\eta + n\delta_v$ stabilizes for every integer $n \ge 0$.

Proof. (a) Let Γ be the union of $\{v\}$ and one directed path from v to each vertex of $\mathrm{supp}(\xi)$. For each vertex of Γ other than v, retain an incoming edge on a shortest path from v in Γ. These edges form a finite directed tree rooted at v. Extend ξ by zero to the other vertices of the tree. Starting at the leaves

and moving toward the root, choose nonnegative integers $q(x)$ so that, for every child y of x,

$$q(x) \geq \deg(y)\, q(y) + \max\{\xi(y), -\eta(y)\}\,.$$

For a leaf, take $q(x) = 0$. Finally, choose n so that

$$n \geq \deg(v)\, q(v) + \max\{\xi(v), -\eta(v)\}\,.$$

Starting at the root and moving outward, topple each vertex x exactly $q(x)$ times before proceeding to its children. A child y receives at least $q(x)$ grains from its parent. Before its $(k+1)$st toppling, where $0 \leq k < q(y)$, its height is at least

$$\eta(y) + q(x) - k\deg(y) \geq \deg(y)\,.$$

These topplings are therefore legal, and the final increase at y is at least $q(x) - \deg(y)\, q(y) \geq \xi(y)$. Similarly, if $0 \leq k < q(v)$, then before the root's $(k+1)$st toppling its height is at least $\eta(v) + n - k\deg(v) \geq \deg(v)$, and the final increase at v is at least $n - \deg(v)\, q(v) \geq \xi(v)$. Grains received from other edges only increase the heights. This proves part (a).

(b) The forward implication follows because $n\delta_v$ is finitely supported. Conversely, suppose $\eta + n\delta_v$ stabilizes for every $n \geq 0$, and let $\xi \geq 0$ be finitely supported. Part (a) gives a finite legal sequence from $\eta + n\delta_v$ to a sandpile β with $\beta \geq \eta + \xi$. Let r be the odometer of this sequence, and let u be the final odometer of $\eta + n\delta_v$. By the least action principle, $r \leq u$. Moreover,

$$\beta - (u - r)\mathsf{L} = \eta + n\delta_v - u\mathsf{L}\,.$$

The sandpile $\eta + n\delta_v - u\mathsf{L}$ is stable, so β stabilizes. A stabilizing toppling function for β also stabilizes the smaller sandpile $\eta + \xi$, so Corollary 9.30 proves that η is robust. □

For G a strongly connected digraph, the simplest example of an explosive sandpile is one where every vertex has $\deg(v) - 1$ grains, the maximum stable count. Adding one grain at a vertex makes that vertex topple. Each distinct out-neighbor then becomes unstable, so strong connectivity causes every vertex to topple. Figure 9.19 contrasts the confined and free spreading of four grains at every site of a bounded domain.

The growth estimate immediately gives a useful class of robust backgrounds.

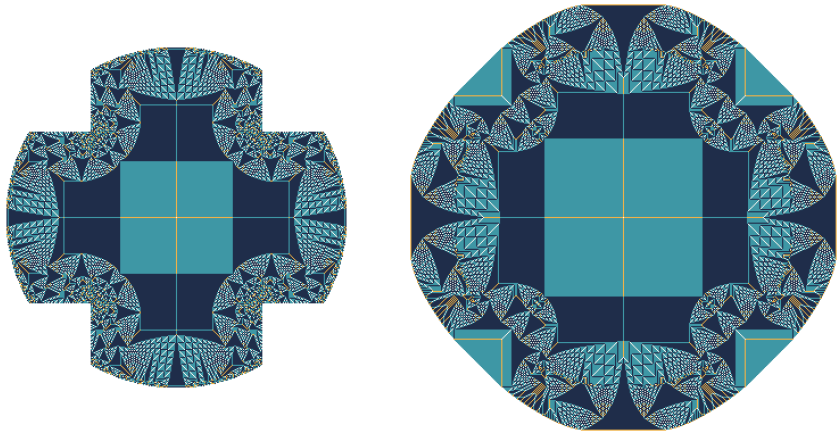


Figure 9.19: Four grains at every site of the cross $\{x \in \mathbb{Z}^2 : |x| < 150 \text{ and } \min(|x_1|, |x_2|) \le 75\}$, stabilized with the sites outside the cross as the sink (left) and with no sink (right), drawn at the same scale. Sites with zero, one, two and three grains are off-white, orange, teal and navy, and without a sink the grains spread beyond the cross.

Lemma 9.38 *Let $d \ge 1$. If a sandpile η on $\mathbb{Z}^d$ satisfies $\eta \le 2d-2$, then η is robust.*

Proof. Proposition 9.35 shows that $\eta + n\delta_{\mathbf{0}}$ stabilizes for every $n \ge 0$. Apply Lemma 9.37(b). □

Figure 9.20 shows the stabilization of a large pile over four constant backgrounds on $\mathbb{Z}^3$.

For intermediate background heights, both behaviors occur. The next lemma proves explosion in two dimensions. Figure 9.21 shows the parallel toppling process, while Figure 9.22 shows stabilizations on robust random backgrounds. Proposition 9.43 proves the corresponding statement in every dimension $d \ge 2$ for every i.i.d. background with values between d and $2d-1$ that equals $2d-1$ with positive probability; the two-dimensional argument below is shorter.

Lemma 9.39 *Let $p \in (0,1]$, and let η be the i.i.d. random background on $\mathbb{Z}^2$ given by*

$$\eta(x) := \begin{cases} 3, & \text{with probability } p, \\ 2, & \text{otherwise}. \end{cases}$$

Then η is explosive with probability one.

Proof. Let F_n be the event that every side of $[-n,n]^2$ contains a non-corner site x with $\eta(x) = 3$. We claim that, for every $k \ge 1$,

$$\bigcap_{n \ge k} F_n \subseteq \{\eta \text{ is explosive}\}.$$

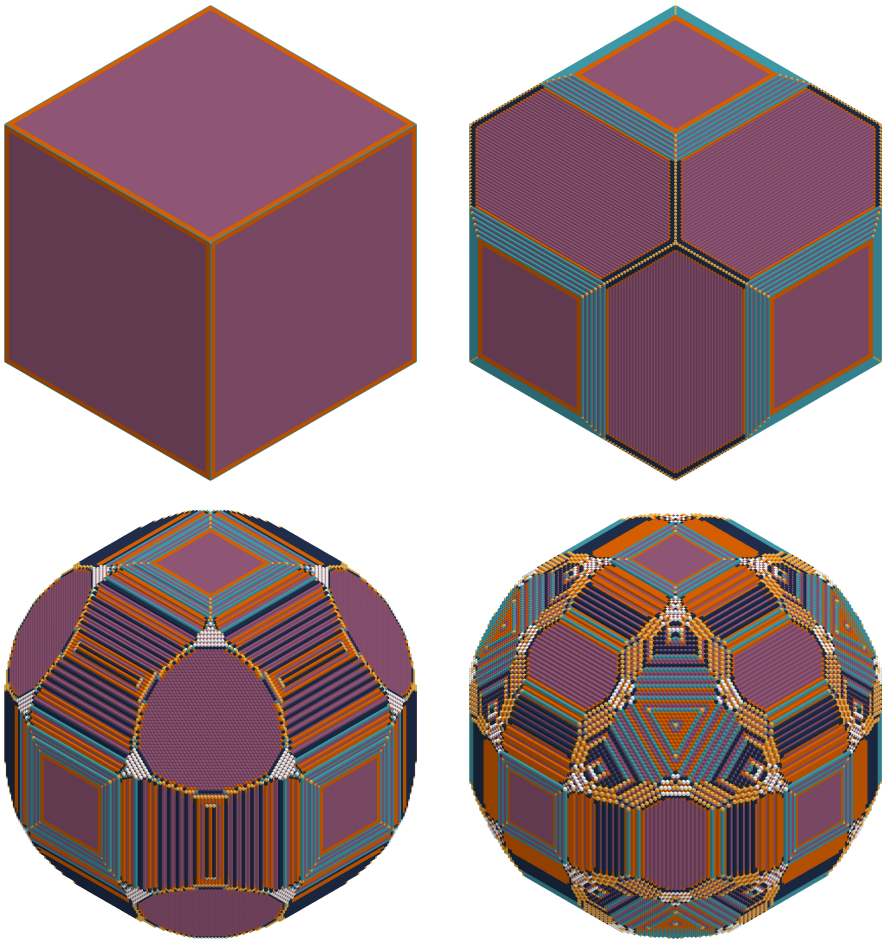


Figure 9.20: The stabilization of $4 \times 10^6 \delta_{\mathbf{0}} + (2d - k)$ on $\mathbb{Z}^3$ for $k = 2, 3, 4, 5$ (top left to bottom right), that is, on the constant backgrounds 4, 3, 2 and 1, drawn and colored as in Figure 9.15. Each solid is scaled to fill its panel: the cube for $k = 2$ has side 311, and the others are 201, 161 and 141 sites across.

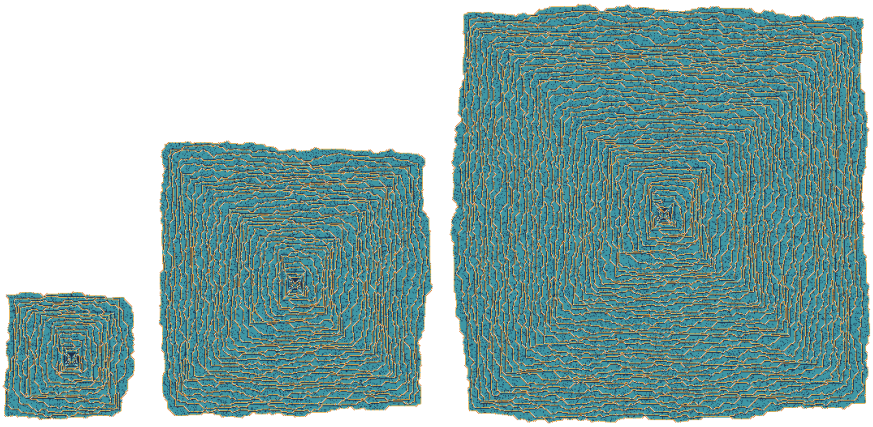


Figure 9.21: Parallel toppling on $\mathbb{Z}^2$ after 400, 800 and 1200 rounds, drawn at the same scale, started from two grains at every site of $[-3,3]^2$ added to the i.i.d. background η with $\mathbb{P}\{\eta(v) = 3\} = 1/10$ and $\mathbb{P}\{\eta(v) = 2\} = 9/10$. Sites that have toppled are colored by their height, zero, one, two and three in off-white, orange, teal and navy, unstable sites are vermilion, and the other sites are white. Lemma 9.39 shows that this background is explosive with probability one.

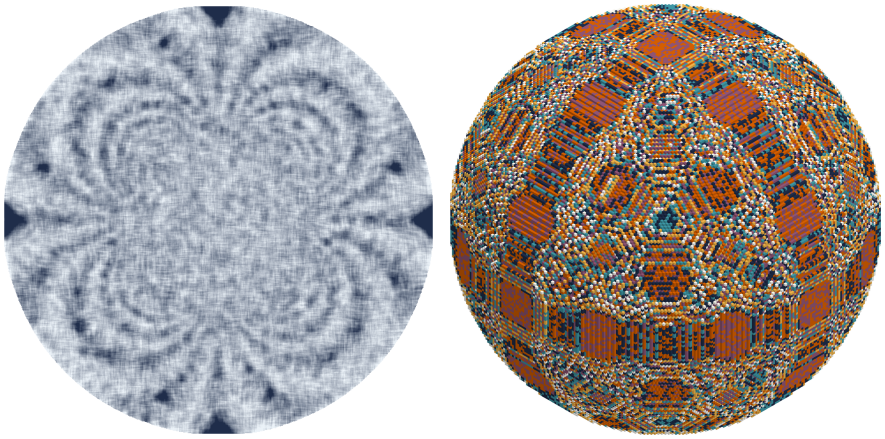


Figure 9.22: The stabilization of n grains at the origin on the i.i.d. background η with $\mathbb{P}\{\eta(v) = -1\} = \mathbb{P}\{\eta(v) = 0\} = 1/2$, for $d = 2$ and $n = 10^6$ (left) and for $d = 3$ and $n = 10^7$ (right). On the left each toppled site is shaded by the average number of grains in the 13×13 box around it, from light (2.0 or less) to navy (2.4 or more); on the right the sites are drawn and colored as in Figure 9.15. These backgrounds are robust by Lemma 9.38.

To prove the claim, add the finitely supported sandpile $2\mathbf{1}_{[-k,k]^2}$. Every site of $[-k,k]^2$ is then unstable and may topple. After these topplings, every non-corner site on the next square has at least three grains, and on F_{k+1} each side has a site with four grains. Starting from the site of height four on each side, legal topplings propagate in both directions along that side (Figure 9.23). Each corner then receives one grain from each of its two side-neighbors and topples. Repeating these legal topplings on successive squares produces a sequence in which every site of $\mathbb{Z}^2$ topples. Lemma 9.32(a) shows that every site eventually topples under parallel toppling, proving the claim.

Finally, the union bound gives

$$\mathbb{P}\Big\{\bigcap_{n\geq k} F_n\Big\} \geq 1 - \sum_{n\geq k} \mathbb{P}\{F_n^c\}\,.$$

Since $\mathbb{P}\{F_n^c\} \leq 4(1-p)^{2n-1}$, the lower bound in the display tends to one as $k \to \infty$. □

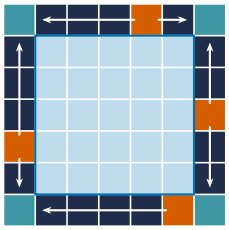


Figure 9.23: The proof of Lemma 9.39 on one square: once every site of $[-2,2]^2$ (blue) has toppled, each non-corner site of the boundary of $[-3,3]^2$ holds at least three grains (navy), and a site with background 3 on each side holds four (vermilion). Topplings spread from it along its side (arrows), and each corner (teal) topples once both of its neighbors on the square have toppled.

Infinitely many cube shells $\{x \in \mathbb{Z}^d : |x|_\infty = n\}$ on which the background is at most $2d-2$ guarantee robustness.

Lemma 9.40 *Let $d \geq 1$, let η be a stable sandpile on $\mathbb{Z}^d$, and let $1 \leq n_1 < n_2 < \cdots$ be integers. Suppose*

$$\eta(x) \leq 2d-2 \qquad \text{when } |x|_\infty = n_i \text{ for some } i \geq 1\,.$$

Then η is robust.

Proof. By Lemma 9.37(b), it is enough to prove that $\eta + k\delta_{\mathbf{0}}$ stabilizes for every $k \geq 1$. For an integer $N \geq 1$, let

$$g_N(x) := (N - |x|_\infty)^+\,.$$

We have $\Delta^{::} g_N(\mathbf{0}) = -2d$. If $0 < |x|_\infty < N$ and exactly one coordinate attains the maximum absolute value, then one inward and one outward contribution

to $\Delta^{::} g_N(x)$ cancel. If several coordinates attain the maximum absolute value, then there is no inward contribution and there are several outward ones. Thus $\Delta^{::} g_N(x) \le 0$ in either case. When $|x|_\infty = N$, at most one neighbor has positive g_N, so $\Delta^{::} g_N(x) \le 1$. Finally, $\Delta^{::} g_N(x) = 0$ when $|x|_\infty > N$.

Let $g := g_{n_1} + \cdots + g_{n_k}$. Then $\Delta^{::} g(\mathbf{0}) = -2dk$. If $|x|_\infty = n_j$ for some $j \le k$, then only g_{n_j} can contribute positively, so $\Delta^{::} g(x) \le 1$, and elsewhere $\Delta^{::} g \le 0$. Since $\eta \le 2d-1$ everywhere and $\eta \le 2d-2$ on the listed shells, it follows that $\eta + k\delta_{\mathbf{0}} + \Delta^{::} g \le 2d-1$ everywhere; at the origin, $\eta(\mathbf{0}) + k - 2dk \le 2d-1$. Corollary 9.30 shows that $\eta + k\delta_{\mathbf{0}}$ stabilizes, with odometer at most g. □

Part (b) of the next lemma shows that if a finite addition topples every vertex, then it topples each one infinitely often. Lemma 9.10 contains the finite undirected case for nonnegative sandpiles.

Lemma 9.41 *Let σ be a sandpile on a locally finite Eulerian digraph G.*

(a) If σ stabilizes, then some vertex of G never topples.

(b) If every vertex eventually topples under the parallel toppling procedure, in which every unstable vertex topples once in each round, then all vertices topple infinitely often.

Proof. (a) Suppose σ stabilizes and every vertex topples. Fix a stabilizing sequence. Since this sequence is finite, the assumption that every vertex topples implies that G is finite. Let w be the vertex whose final toppling occurs first. During that toppling, w receives one grain along each self-loop. Every other incoming edge comes from a vertex whose final toppling occurs later, so one grain arrives along each such edge after w has toppled for the last time. Since $\mathrm{indeg}(w) = \deg(w)$, the vertex w ends unstable, a contradiction.

(b) Suppose every vertex eventually topples, but a vertex topples only finitely often. For each such vertex w, let t_w be the last time at which w topples. Choose v with minimal t_v. The self-loops at v return grains during the final toppling of v. For every other incoming edge (w, v), the vertex w topples at some time at least t_v. If w topples finitely often, then minimality gives $t_w \ge t_v$; if w topples infinitely often, then w topples after time t_v. Local finiteness ensures that all these grains have arrived by a finite time. Their number is $\mathrm{indeg}(v) = \deg(v)$, so v becomes unstable after its final toppling and must topple again. This is a contradiction. □

The combinatorics in this chapter say little about the shapes in Figures 9.14 and 9.15, and describing them needs tools we do not develop here. Replacing grains by a continuous fluid loses the fractal structure entirely: Chapter 4 showed that the divisible sandpile started from n units of mass at the origin fills a Euclidean ball up to an error in the radius that does not grow with n.

The two starred sections that follow prove that certain random backgrounds are explosive, using a cellular automaton in which no sandpiles appear at all.

9.11 BOOTSTRAP PERCOLATION*

Bootstrap percolation is a cellular automaton on $\mathbb{Z}^d$. Each site is initially infected independently with probability p. An uninfected site becomes infected when at least d of its nearest neighbors in $\mathbb{Z}^d$ are infected, and infected sites remain infected. A box in $\mathbb{Z}^d$ is a product of d finite intervals of integers, and its side lengths are the numbers of integers in these intervals. We say that a box is **internally spanned** when the dynamics, restricted to the box, eventually infect every site. Enlarging a box can only add infected neighbors, so the dynamics on a box infect every site that the dynamics on a smaller box inside it infect.

The next lemma shows that a box whose side lengths are n or $n+1$ fails to be internally spanned with probability exponentially small in n. We allow both lengths because the layers added as a cube grows one coordinate at a time are boxes of this kind, one dimension lower. In the next section this estimate makes summable the failure probabilities of the layers added to a large cube.

Lemma 9.42 *For every integer $d \geq 1$ and every $p \in (0,1]$, there exists $c = c(d,p) > 0$ with the following property. Let $n \geq 1$ be an integer, and let $n_1, \ldots, n_d$ be integers in $\{n, n+1\}$. If the sites of the box $\{1, \ldots, n_1\} \times \cdots \times \{1, \ldots, n_d\}$ are initially infected independently with probability p, then the probability that this box is not internally spanned is at most* $\exp(-cn)$.

Proof. If $p = 1$, then every site is initially infected and the probability in the lemma is zero. Suppose $0 < p < 1$. For positive integers j and n, let $f_j(n)$ be the largest probability, over the boxes in $\mathbb{Z}^j$ with side lengths in $\{n, n+1\}$, that the box is not internally spanned under bootstrap percolation on $\mathbb{Z}^j$, whose threshold is j. We prove by induction on d that $f_d(n) \leq \exp(-cn)$ for all $n \geq 1$, where $c = c(d,p) > 0$. In dimension one, a box is an interval, and it is internally spanned if and only if it contains an initially infected site. Thus $f_1(n) = (1-p)^n$, which proves the base case with $c = \log(1/(1-p))$.

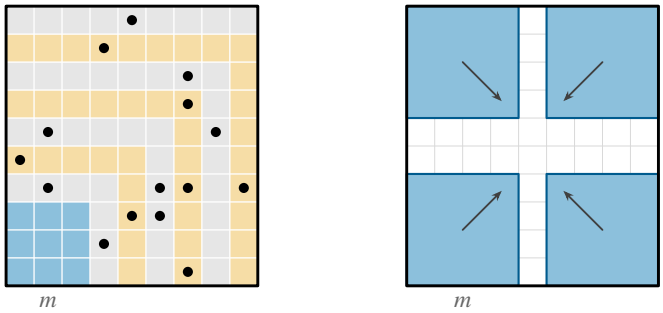
For the inductive step, suppose $d \geq 2$ and $f_{d-1}(k) \leq e^{-c_0 k}$ for every $k \geq 1$, where $c_0 > 0$. We grow an infected corner cube to a whole box one layer at a time, then bound the spanning probability away from zero, and finally amplify this bound to exponential decay.

We first grow a corner cube, as Figure 9.24(a) shows for $d = 2$. Fix a box with side lengths $n_1, \ldots, n_d$ in $\{n, n+1\}$, let $1 \leq m \leq n$, and suppose

every site of the cube of side m at one corner of the box is infected. For $k = m, m+1, \dots, n-1$ in turn, we turn the current cube of side k into a cube of side $k+1$ by extending its first, second, ..., dth coordinate interval by one integer toward the opposite corner of the box. Afterwards, we extend the ith coordinate interval once more for each i with $n_i = n+1$. Each extension adds a layer of sites with one fixed coordinate, and deleting that coordinate identifies the layer with a box in $\mathbb{Z}^{d-1}$ whose side lengths are k or $k+1$ (with $k = n$ for the last extensions). Every site of the layer has an infected neighbor in the box built so far, so a site with $d-1$ infected neighbors in the layer has d infected neighbors. Hence, if the layer is internally spanned under bootstrap percolation on $\mathbb{Z}^{d-1}$, then it fills under the dynamics on $\mathbb{Z}^d$ restricted to the box. At most d layers are added for each $k \geq m$. Let $C := d/(1-e^{-c_0})$. The union bound gives

$$\mathbb{P}\{\text{an added layer is not internally spanned}\} \leq d \sum_{k=m}^{\infty} e^{-c_0 k} = C e^{-c_0 m}. \tag{9.10}$$

The layers are disjoint from the corner cube, so if the corner cube and every added layer are internally spanned, then the box is internally spanned.



(a) (b)

Figure 9.24: (a) In $\mathbb{Z}^2$ an infected corner square of side $m = 3$ (blue) grows to a box with sides 9 and 10 by adding a column and then a row for each $k = 3, \dots, 8$ and a final row; each added layer is an interval, internally spanned when it contains an initially infected site (black). (b) The four corner squares of side $\lfloor 9/2 \rfloor = 4$ are disjoint, and each grows toward the opposite corner as in (a).

Next we bound the spanning probability away from zero. Choose an integer $K \geq 1$ with $Ce^{-c_0 K} \leq 1/2$, and let $\epsilon := p^{K^d}/2$. If $n \geq K$, then the corner cube of side K is entirely initially infected with probability p^{K^d}, independently of the added layers, which are all internally spanned with probability at least $1/2$ by (9.10). If $n < K$, then the box has at most K^d sites, so all of them are initially infected with probability at least p^{K^d}. In

both cases the box is internally spanned with probability at least ϵ, so

$$f_d(n) \le 1 - \epsilon \qquad \text{for every } n \ge 1\,. \tag{9.11}$$

It remains to obtain exponential decay. Let $n \ge 2$ and $m := \lfloor n/2 \rfloor$. Every side length of the box is at least $2m$, so the cubes of side m at distinct corners are disjoint, and since $d \ge 2$ there are at least four corners. Fix four corners, as in Figure 9.24(b). If the box is not internally spanned, then either all four corner cubes fail to be internally spanned, or a layer in one of their four enlargements fails. The four corner events are independent, so (9.10) and the union bound give

$$f_d(n) \le f_d(m)^4 + 4Ce^{-c_0 m}\,. \tag{9.12}$$

Choose an integer $M \ge 2$ with $4Ce^{-c_0 M/2} \le \epsilon$, and let

$$c := \min\left\{\frac{c_0}{6}, \frac{\log(1/(1-\epsilon))}{2M}\right\}.$$

We prove by strong induction on n that $f_d(n) \le e^{-cn}$. For $1 \le n \le 2M$, the bound (9.11) and the choice of c give $f_d(n) \le 1 - \epsilon \le e^{-2cM} \le e^{-cn}$. For the inductive step, let $n > 2M$ and assume that $f_d(k) \le e^{-ck}$ for $1 \le k < n$. Then $m \ge M$, and since $c_0 - 3c \ge c_0/2$, we have

$$4Ce^{-c_0 m} \le 4Ce^{-c_0 M/2} e^{-3cm} \le \epsilon e^{-3cm}\,.$$

We bound one factor of $f_d(m)^4$ by (9.11) and the other three by the inductive hypothesis. The recursion (9.12) then gives

$$f_d(n) \le (1-\epsilon)e^{-3cm} + \epsilon e^{-3cm} = e^{-3cm} \le e^{-cn}\,,$$

where the last inequality uses $3\lfloor n/2 \rfloor \ge n$ for $n \ge 3$. This completes the induction on n and hence on d. □

We apply Lemma 9.42 in the next section to propagate topplings across the layers of a growing cube.

9.12 EXPLOSIVE BACKGROUNDS IN HIGHER DIMENSIONS*

We now show that an i.i.d. background with values between d and $2d-1$ that equals $2d-1$ with positive probability is explosive in every dimension $d \ge 2$. Starting from a finite cube whose sites have all toppled, we enlarge the cube one layer at a time, as in the proof of Lemma 9.42 (Figure 9.24(a)). Each layer is a box of dimension $d-1$, and bootstrap percolation on the layer provides a legal order in which to topple its sites.

Recall from Section 9.10 that a background is explosive if there exists a nonnegative finitely supported sandpile whose addition makes parallel toppling eventually topple every site. The next proposition uses internally spanned layers to prove explosion for a random background.

Proposition 9.43 *Let $d \geq 2$ be an integer, let $p \in (0, 1]$, and let η be an i.i.d. random background on $\mathbb{Z}^d$ such that, for every $x \in \mathbb{Z}^d$,*

$$d \leq \eta(x) \leq 2d - 1 \qquad \text{and} \qquad \mathbb{P}\{\eta(x) = 2d - 1\} = p\,.$$

Then η is explosive almost surely.

Proof. The proof has three steps. First, we fix a sequence of growing cubes and show how topplings cross one added layer. Next, we apply Lemma 9.42 in dimension $d - 1$ to all sufficiently large layers. Finally, we start the growth by adding grains to a finite cube.

Let $J_1 := \{0\}$. For $k \geq 1$, obtain the interval J_{k+1} from J_k by adding the integer just above J_k when k is odd, and the integer just below J_k when k is even. Then J_k has k elements, and the cubes J_k^d increase to $\mathbb{Z}^d$. We enlarge J_k^d to J_{k+1}^d in d moves: for $i = 1, \ldots, d$ in turn, the ith move replaces the ith factor J_k by J_{k+1}. The sites added in the ith move form a layer. Their ith coordinate is the integer in $J_{k+1} \setminus J_k$, and deleting this coordinate identifies the layer with a box in $\mathbb{Z}^{d-1}$ whose side lengths are k or $k + 1$. Every site of the layer has a neighbor in the box before the move.

We first consider one layer. Declare a site x initially infected when $\eta(x) = 2d - 1$. Suppose every site of the box before the move has toppled once and no site of the new layer has toppled. Then every site of the layer still has its initial grains and has received at least one more. A site with $\eta(x) = 2d - 1$ therefore has at least $2d$ grains and is unstable. Every other site of the layer has at least $d + 1$ grains, so it becomes unstable after $d - 1$ of its neighbors in the layer have toppled. An order in which bootstrap percolation on $\mathbb{Z}^{d-1}$ infects the layer is therefore a legal order in which to topple its sites. Thus, if the layer is internally spanned, then every site of the box after the move has toppled once.

We next show that, almost surely, every layer added while J_k^d grows to J_{k+1}^d is internally spanned for all sufficiently large k. The events $\{\eta(x) = 2d - 1\}$ are independent, and each has probability p. For each $k \geq 1$, the d layers added while J_k^d grows to J_{k+1}^d are boxes in $\mathbb{Z}^{d-1}$ with side lengths k or $k + 1$. By Lemma 9.42 in dimension $d - 1$, the probability that one of them is not internally spanned is at most $d \exp(-ck)$, where $c = c(d-1, p)$. These bounds are summable over k. By the Borel–Cantelli lemma, almost surely there is a positive integer K such that every layer added while J_k^d grows, for every $k \geq K$, is internally spanned.

Finally, fix a realization and such an integer K, and add d grains at each site of J_K^d. Each of these sites then has at least $2d$ grains, so toppling each of them once is legal. Toppling the d layers of each enlargement as above, for $k = K, K+1, \ldots$ in turn, gives for every $k \geq K$ a finite legal sequence that topples every site of J_k^d. Since the cubes J_k^d increase to $\mathbb{Z}^d$, Lemma 9.32(a) shows that parallel toppling of $\eta + d\mathbf{1}_{J_K^d}$ eventually topples every site. Hence η is explosive almost surely. □

The exercises give further robust and explosive backgrounds.

EXERCISES

Exercise 9.1. The following reflection process is a real-valued analogue of the abelian property in Lemma 9.3. Assign a real-valued label to each vertex of a cycle with $N \geq 3$ vertices, and suppose that the sum of the labels is positive. Whenever three consecutive labels are x, y, z with $y < 0$, replace the labels by $x + y, -y, z + y$, respectively. Show that every sequence of replacements terminates with all labels nonnegative and that the final labels do not depend on the order of the replacements. Hint: each replacement sorts two adjacent partial sums of the labels.

Exercise 9.2. Let $G = (V, E)$ be a locally finite simple undirected graph. Let σ_0 be a sandpile on G, and let σ_t be the sandpile after a sequence of topplings with odometer u, so that $\sigma_t = \sigma_0 - u\mathsf{L}$. Show that for every finite set $U \subset V$,

$$\sum_{w \in U,\ v \in V \setminus U} (u(v) - u(w))\mathbf{1}\{\{v, w\} \in E\} = \sum_{w \in U} (\sigma_t - \sigma_0)(w)\,.$$

This proves the identity (9.3) used in the proof of Lemma 9.13.

Exercise 9.3. For a finite digraph G with a global sink, show that the sandpile group of Section 9.4 is isomorphic to the additive group of functions harmonic modulo 1 on G and zero at the sink.

Exercise 9.4. In the setting of Section 9.5, let $\boldsymbol{\delta}$ be the sandpile given by $\boldsymbol{\delta}(v) = \deg(v)$ at every non-sink vertex, and let $\alpha := \boldsymbol{\delta} - \boldsymbol{\delta}^\circ$. Show that every entry of α is at least 1 and that $(\sigma + \alpha)^\circ = \sigma$ for every recurrent sandpile σ on G.

Exercise 9.5. Let G be a finite digraph with a global sink. The **inverse** of a recurrent sandpile σ on G is the recurrent sandpile $\bar{\sigma}$ satisfying $(\sigma + \bar{\sigma})^\circ = I$, where I is the identity element of the sandpile group of Corollary 9.24. Show that

$$\bar{\sigma} = (\xi - \xi^\circ - \sigma)^\circ\,,$$

where ξ is a sandpile satisfying $\xi(v) \geq 3\deg(v) - 3$ at every non-sink vertex v.

Exercise 9.6. Section 9.6 defines a finite Eulerian digraph using strong connectivity. Let G be a finite digraph with $\mathrm{indeg}(v) = \deg(v)$ at every vertex v. Show that G is strongly connected if and only if its underlying undirected graph is connected. Give an infinite digraph for which the equivalence fails.

Exercise 9.7. Section 9.6 defines Eulerian tours and Eulerian digraphs. Let G be a finite digraph without isolated vertices. Show that G has an Eulerian tour if and only if G is strongly connected and $\mathrm{indeg}(v) = \deg(v)$ for every vertex v.

Exercise 9.8. In the setting of Section 9.6, consider the nearest-neighbor graph on $\{0, \ldots, k\}^d$ for integers $d \geq 1$ and $k \geq 2$. Identify the vertices with a coordinate equal to zero or k as one vertex, retain edge multiplicities, replace each edge by two oppositely directed edges, and delete the outgoing edges of the identified vertex to make it a sink. Show that the sandpile $\sigma \equiv d$ on the non-sink vertices is recurrent.

Exercise 9.9*. Let G be a finite connected plane graph, and let G^* be its plane dual. Replace every edge of G and G^* by two oppositely directed edges. The two digraphs are Eulerian in the sense of Section 9.6. Choose a vertex of each digraph and delete its outgoing edges, so that it becomes a sink. Show that, whichever vertices are chosen, the sandpile groups of the two digraphs are isomorphic. Hint: use Exercise 9.3.

Exercise 9.10. In the setting of Section 9.7, let $p \leq q$ be integers, and let $f : \mathbb{Z} \to \mathbb{Z}$ be supported on $[p, q]$ with

$$\sum_{y=p}^{q} f(y) = 0 \qquad \text{and} \qquad \sum_{y=p}^{q} y f(y) = 0\,.$$

(a) Show that there is an integer-valued function g such that $f = -g\mathsf{L}$ and $g(x) = 0$ unless $p < x < q$.

(b) Suppose that there do not exist integers $x_1 < x_2 < x_3$ with $f(x_1) < 0 < f(x_2)$ and $f(x_3) < 0$. Show that g can be chosen nonnegative.

Exercise 9.11. Section 9.7 proves robustness for zero-one sandpiles on $\mathbb{Z}$ whose zeros are unbounded in both directions. Determine all robust zero-one backgrounds as follows. (a) Let $x \leq y$ be integers, let $v \in [x, y] \cap \mathbb{Z}$, and let $\sigma = \mathbf{1}_{[x,y]}$ be a sandpile on $\mathbb{Z}$. What is the stabilization of $\sigma + \delta_v$?

(b) Let σ be a sandpile on $\mathbb{Z}$ taking only the values zero and one. Show that σ is robust if and only if its zero set is unbounded above and below.

Exercise 9.12. Generalize Lemma 9.32(a) from $\mathbb{Z}^d$ to locally finite digraphs. Let σ be a sandpile on a locally finite digraph. Under the parallel toppling procedure, every unstable non-sink vertex topples once in each round. Show

that every vertex toppled by a finite legal toppling sequence from σ is eventually toppled by the parallel toppling procedure.

Exercise 9.13. Let $d \geq 1$ and $1 \leq i \leq d$ be integers. Let σ_0 be a sandpile on $\mathbb{Z}^d$, let $(u_t)_{t\geq 0}$ be the parallel toppling odometers of Section 9.8, and let φ be the reflection across the hyperplane $x \cdot \mathbf{e}_i = 1/2$. Suppose $\sigma_0 \circ \varphi = \sigma_0$ and σ_0 is decreasing in direction $\mathbf{e}_i$ on $\{x \in \mathbb{Z}^d : x \cdot \mathbf{e}_i \geq 1/2\}$. Show that u_t is decreasing in direction $\mathbf{e}_i$ on this half-space for every integer $t \geq 0$.

Exercise 9.14. In the setting of Section 9.9, give a second proof of the case $d = 2$ of Proposition 9.36, one that does not choose a toppling order. Let $n \geq 2$ be an integer. Start an abelian sandpile on $\mathbb{Z}^2$ with $n + 2$ grains at the origin and two grains at every other site. Prove that there is a nonnegative integer k such that the set of sites that topple is

$$\{x \in \mathbb{Z}^2 : |x|_\infty \leq k\}.$$

Exercise 9.15. Apply the robustness criterion of Lemma 9.37. Let η be a background on $\mathbb{Z}^2$ with $\eta \leq 3$. Suppose the set of integers y for which $\eta \equiv 2$ on $\mathbb{Z} \times \{y\}$ is unbounded above and below. Suppose also that the set of integers x for which $\eta \equiv 2$ on $\{x\} \times \mathbb{Z}$ is unbounded above and below. Show that η is robust. Hint: dominate η by the background equal to two on these lines and three elsewhere, apply Exercise 9.11 in each direction, and take the minimum of the two final odometers.

Exercise 9.16. In the setting of Section 9.10, start a sandpile on $\mathbb{Z}^2$ with four grains at the origin and three grains at every other site, and apply parallel toppling, in which every unstable site topples once in each round. Prove that every site topples infinitely often. Use the least action principle, Lemma 9.2, to conclude that this sandpile does not stabilize.

Exercise 9.17. In the setting of Section 9.10, let $d \geq 1$ be an integer, and let $x_1, \ldots, x_d$ be linearly independent vectors in $\mathbb{Z}^d$, and write $x_i = (x_{i1}, \ldots, x_{id})$. Suppose $\gcd(x_{1j}, \ldots, x_{dj}) = 1$ for every $1 \leq j \leq d$. Let $\Lambda := \mathbb{Z}x_1 + \cdots + \mathbb{Z}x_d$. Show that the background $(2d - 2) + \mathbf{1}_\Lambda$ on $\mathbb{Z}^d$ is explosive.

NOTES

The abelian sandpile model arose independently in several communities. In combinatorics, it is commonly called chip-firing. Bak, Tang, and Wiesenfeld (1987, pp. 381–384) proposed the two-dimensional model in 1987 as a prototype of self-organized criticality and predicted that its avalanche distributions follow power laws. Dhar (1990, pp. 1613–1614) formulated the model on arbitrary finite systems with site-dependent toppling rules, proved that the final stable sandpile does not depend on the toppling order, and proved that

the grain addition operators commute (Lemma 9.8). Diaconis and Fulton (1991, Proposition 4.1) proved order independence for a related chip-moving game, following an argument of Eriksson, which shows that every legal sequence of moves is a permutation of a subsequence of each terminating legal sequence, as in Lemma 9.2. Their Lemma 2.2, a commutativity of operators, gives that the law of a related growth model does not depend on the order of the choices. Dhar and Majumdar (1990, Section 6) computed the avalanche mass and duration exponents on the Bethe lattice. Dhar and Ramaswamy (1989, pp. 1659–1661) solved the directed model and determined its critical exponents in every dimension.

Engel (1975) introduced the probabilistic abacus for computing absorption probabilities and expected absorption times in finite Markov chains; Engel (1976, pp. 59–65) gave a self-contained description and proved its basic identities, apart from the recurrence of the critical load; a note added in proof (Engel, 1976, p. 69) announced a proof of the recurrence by L. Scheller and sketched its idea. A manuscript by Snell (2009, pp. 1 and 4–6), written circa 1979 and posted in 2009, completed the proof; its argument for the recurrence of the critical load (Snell, 2009, pp. 5–6) is due to Peter Doyle. Björner and Lovász (1992, Section 6) identified the abacus as a chip-firing process and proved recurrence of the critical load. Gabrielov (1993, Sections 1–2 and 5–6) established the abelian property for continuous avalanche models and related avalanche distributions to Tutte polynomials.

The one-dimensional chip-firing process used for vector balancing appears in Section 2 of Spencer (1986) and is analyzed further in Sections 2–4 of Anderson, Lovász, Shor, Spencer, Tardos, and Winograd (1989). Lorenzini (1989, pp. 481–484) and Lorenzini (1991, pp. 277–278) studied the finite Laplacian group later identified with the sandpile group, motivated by questions from arithmetic geometry. Levine (2009, Theorems 3.3 and 1.2) gave exact sequences for wired trees and a full cyclic decomposition for wired regular trees.

Part (a) of Lemma 9.12 is Theorem 3.3(c) of Björner, Lovász, and Shor (1991), and part (b) follows from their Theorem 3.3(b), or from their Theorem 3.3(a) when G is a tree. The proof of Lemma 9.13 follows Lemmas 4 and 5 of Tardos (1988); Theorem 2 there gives the resulting bound $2|V||E|\operatorname{diam}(G)$. Theorem 4.2 of Björner, Lovász, and Shor (1991) also gives the bound $2|V|N/\lambda_1$, where N is the number of grains and λ_1 is the least positive eigenvalue of the graph Laplacian. Björner and Lovász (1992, Sections 2 and 4) studied termination and toppling bounds for digraphs.

The burning test underlying Lemma 9.27 was introduced by Dhar (1990, p. 1615). Majumdar and Dhar (1992, pp. 131–133) matched the sandpiles that pass the test with the spanning trees of the graph, for symmetric toppling matrices. Speer (1993, Theorem 3 and Section 4) showed that the test can fail

for asymmetric toppling matrices and introduced the script algorithm, which characterizes recurrence for the asymmetric toppling matrices considered there.

Dhar, Ruelle, Sen, and Verma (1995, Sections 2–3) developed the representation of the sandpile group by recurrent sandpiles. Biggs (1999, Theorems 4.2 and 7.3) and Biggs (1997, Sections 28–32) related this group to the graph Laplacian and algebraic potential theory. Our exposition in Sections 9.4–9.6 uses Corollary 2.5 of Cori and Rossin (2000) and Sections 2 and 4 of the survey by Holroyd, Levine, Mészáros, Peres, Propp, and Wilson (2008).

Fey, Levine, and Peres (2010, Lemma 2.3) proved the least action principle on $\mathbb{Z}^d$ for sandpiles that stabilize: the odometer is at most every $g : \mathbb{Z}^d \to \mathbb{Z}_{\geq 0}$ for which $\sigma - g\mathsf{L}$ is stable. Corollary 9.30 also gives stabilization, for finitely supported g, and Lemma 9.2 is a sequence form on general digraphs. Part (b) of Lemma 9.41 extends to locally finite Eulerian digraphs their argument, in Section 1, that if every site of $\mathbb{Z}^d$ topples, then every site topples infinitely often. They defined robust and explosive backgrounds on $\mathbb{Z}^d$ in Section 1, through additions of grains at the origin, and Section 9.10 adapts their criteria for explosion and robustness, Theorems 4.1 and 4.2. The background with three grains on the horizontal axis is their example, in Section 1, of an infinite avalanche in which each site topples only finitely often.

Sections 9.8 and 9.9, including the growth bounds of Proposition 9.35, use arguments from Theorems 4 and 5 of Le Borgne and Rossin (2002), Lemma 5.3 of Babai and Gorodezky (2007), and Section 2.2 and Corollary 2.1 of Bou-Rabee (2022). The upper bound in Proposition 9.35 for backgrounds up to $2d - 2$ adapts the background modification by which Fey, Levine, and Peres (2010) proved their Theorem 1.1 for constant backgrounds; Exercise 9.10 is their Lemma 3.3, and Exercise 9.17 is their Proposition 1.3. Proposition 9.36 is the part of Theorem 4.1 of Fey and Redig (2008) that identifies the toppled set as a cube. Theorem 1.1 of Pegden and Smart (2013) shows that the stabilizations of $n\delta_{\mathbf{0}}$ shown in Figures 9.14 and 9.15, rescaled by $n^{1/d}$, converge in the sense of local averages as $n \to \infty$. In $\mathbb{Z}^2$, Theorem 1.1 of Levine, Pegden, and Smart (2017) describes the constraint in the obstacle problem that determines this limit through an Apollonian circle packing, and Theorem 1.3 of Levine, Pegden, and Smart (2016b) uses it to construct piecewise quadratic solutions of the differential equation in this obstacle problem whose pieces form a fractal pattern.

Bootstrap percolation was introduced by Chalupa, Leath, and Reich (1979, p. L31). The proof of Lemma 9.42 adapts the finite-box induction for the modified basic model in Lemma 3.1 and Proposition 3.2 of Schonmann (1992); Theorem 3.1 there gives the corresponding exponential-relaxation result. In the modified basic model a site becomes infected when it has an

infected neighbor in each coordinate direction, a more restrictive rule than the one here. Lemma 9.39 is the case $d = 2$ of Proposition 1.4 of Fey, Levine, and Peres (2010), which proves that the background $2d-2$ plus an i.i.d. Bernoulli field with positive parameter is explosive almost surely in every dimension $d \geq 2$, using the criterion of their Theorem 4.1; they remark on the similarity of their argument to Straley's argument for bootstrap percolation, which van Enter (1987) made rigorous: when $d = 2$ and each site is initially infected with probability $p > 0$, the bootstrap percolation of Section 9.11 almost surely infects every site of $\mathbb{Z}^2$. For stable backgrounds, Proposition 9.43 strengthens Theorem 1.4 of Bou-Rabee (2024). That paper calls a background explosive when it is not robust, and its theorem shows that every i.i.d. background that is at least d everywhere and equals $2d-1$ with positive probability is almost surely not robust; Proposition 9.43 shows that some finite addition topples every site. Proposition 9.43 includes Proposition 1.4 of Fey, Levine, and Peres (2010) for $d \geq 2$. Our proof adapts the argument in Section 7 of Bou-Rabee (2024). The proofs of Lemma 9.42 and Proposition 9.43 given here differ from the cited proofs.

The description by functions that are harmonic modulo 1 in Exercise 9.3 appears in Section 2 of Holroyd, Levine, Mészáros, Peres, Propp, and Wilson (2008), in the paragraph after their Lemma 2.17. Theorem 2 of Cori and Rossin (2000) proves the planar-duality result in Exercise 9.9; Exercise 9.3 gives a shorter proof.

Bond and Levine (2016, Section 2 and Theorem 4.6) gave axioms under which the abelian distributed processors of Dhar (1999, Section 5) have the abelian property, and called the resulting systems abelian networks; Chan and Levine (2022, Chapters 4–7) developed their nonhalting dynamics. Reflection processes were studied earlier in Proposition 2.1 of Mozes (1990) and Propositions 1–3 of Alon, Krasikov, and Peres (1989), motivated by a 1986 IMO problem. Bond and Levine (2016, Section 5.2) list reflection processes as a possible example for an extension of abelian networks with shared memory. A reflection process appears in Exercise 9.1; its hint follows the solution on pp. 311–312 of Winkler (2024).

Chapter Ten

The Divisible Sandpile

The divisible sandpile replaces grains by real mass, and Section 4.1 of Chapter 4 proved that its odometer and final configuration do not depend on the order of sequential topplings. We now return to the model on general graphs. Throughout this chapter, graphs $G = (V, E)$ are undirected, locally finite, connected and simple, and have at least two vertices.

The quadrature inequalities of Section 10.1 characterize the final configuration by its integrals against superharmonic functions. Section 10.2 extends the least action principle to general toppling procedures, including parallel toppling. In Section 10.3, we identify the odometer with the value of an optimal stopping problem for simple random walk.

This representation gives a phase transition for bounded i.i.d. masses on an infinite connected graph of bounded degree (Section 10.4): the configuration stabilizes below mean one and explodes above it, and at mean one it explodes unless the masses are constant. At mean one, the proof combines a ball of small masses with a variance bound and a zero-one law. On infinite vertex-transitive graphs, Section 10.5 obtains the conclusions away from mean one from conservation of expected mass under the weaker assumption that the initial mass is integrable. The point-source ball of Chapter 4 (Figures 4.3 and 4.4) is one outcome of the same dynamics; here the question is whether redistribution on an infinite graph ever reaches a stable configuration.

10.1 DISCRETE QUADRATURE

The least action principle characterizes the odometer. We next characterize the final configuration by discrete quadrature inequalities, analogous to the discrete inequality (1.1) and its continuum counterpart in Chapter 1.

For $\sigma, h : V \to \mathbb{R}$ with σ having finite support, write

$$\langle \sigma, h \rangle := \sum_{v \in V} \sigma(v) h(v) .$$

If σ is a sandpile configuration, then $\langle \sigma, h \rangle$ is its total weight when a unit of mass at v has weight $h(v)$. Exercise 10.2 shows that toppling at x does not increase this quantity when h is superharmonic at x, and preserves it when h is harmonic there.

Lemma 4.4 gives finite odometer support and mass one wherever the odometer is positive. These facts allow us to characterize the final configuration without reference to a toppling order. For uniqueness, we test the inequalities of two candidate configurations against the solution of a finite Dirichlet problem and its negative.

Theorem 10.1 (Quadrature Inequalities) *Let $G = (V, E)$ be an infinite locally finite connected graph, let $\sigma : V \to \mathbb{R}$ be a finitely supported configuration, and let σ_∞ be its final configuration. Then, among all functions $\eta : V \to (-\infty, 1]$ for which $\eta - \sigma$ has finite support, σ_∞ is the unique function satisfying*

$$\langle \eta - \sigma, h \rangle \le 0 \qquad \text{for every } h : V \to \mathbb{R} \text{ such that } \Delta h \le 0 \text{ on } \{\eta = 1\}\,. \tag{10.1}$$

Proof. We first verify the quadrature inequality for σ_∞, and then compare two candidate configurations using a finite Dirichlet problem.

Step 1: We verify the inequality for σ_∞. By Lemma 4.4(ii), the odometer u_∞ has finite support. Consequently $\sigma_\infty - \sigma = \Delta u_\infty$ has finite support. Pairing the two orientations of every edge gives

$$\langle \sigma_\infty - \sigma, h \rangle = \langle \Delta u_\infty, h \rangle = \langle u_\infty, \Delta h \rangle \le 0\,,$$

whenever $\Delta h \le 0$ on $\{\sigma_\infty = 1\}$, because $u_\infty \ge 0$ and $\{u_\infty > 0\} \subseteq \{\sigma_\infty = 1\}$. Thus $\eta = \sigma_\infty$ satisfies (10.1).

Step 2: We prove uniqueness. Suppose that distinct functions $\eta, \xi : V \to (-\infty, 1]$ satisfy (10.1) and that $\eta - \sigma$ and $\xi - \sigma$ have finite support. Interchanging them if necessary, choose z with $\eta(z) > \xi(z)$. Since σ has finite support, so does η, and $A := \{\eta = 1\} \cup \{z\}$ is a nonempty finite proper subset of V.

Let h solve the finite Dirichlet problem

$$\Delta h = -\delta_z \quad \text{on } A \qquad \text{and} \qquad h = 0 \quad \text{outside } A\,.$$

This solution exists uniquely: Lemma 3.1, applied to both signs, shows that the homogeneous system has only the zero solution, so its square matrix is invertible. Applying the same lemma to $-h$ gives $h \ge 0$. The equation at z gives $\deg(z)h(z) = 1 + \sum_{y \sim z} h(y) \ge 1$, so $h(z) > 0$. Moreover, $\Delta h \ge 0$ outside A, because $h \ge 0$ and vanishes there.

Thus h is superharmonic on $\{\eta = 1\}$, and $-h$ is superharmonic on $\{\xi = 1\}$, since $\Delta h \ge 0$ off z and $\xi(z) < \eta(z) \le 1$. Applying (10.1) to these two test functions gives

$$\langle \eta - \sigma, h \rangle \le 0 \le \langle \xi - \sigma, h \rangle\,.$$

But h vanishes outside A, and $\eta = 1 \geq \xi$ on $A \setminus \{z\}$, so

$$\langle \eta - \xi, h \rangle \geq (\eta(z) - \xi(z))h(z) > 0\,.$$

This contradiction proves uniqueness, and Step 1 identifies the unique function as σ_∞. □

Both characterizations so far concern the sequential toppling procedure of Section 4.1. The random walk representation of Section 10.3 comes from parallel toppling, and other natural procedures topple asynchronously or in an order chosen by an adversary. We want the same odometer in every case.

10.2 GENERAL TOPPLING PROCEDURES

For a locally finite connected graph $G = (V, E)$ and a configuration $\sigma : V \to \mathbb{R}$, recall from Theorem 4.3 the family

$$\mathcal{F}_\sigma = \{f : V \to \mathbb{R} \mid f \geq 0 \text{ and } \sigma + \Delta f \leq 1\}\,.$$

With $\inf \varnothing = \infty$, write

$$u_\infty^\sigma(v) := \inf\{f(v) \mid f \in \mathcal{F}_\sigma\} \qquad \text{for } v \in V\,. \tag{10.2}$$

Figure 4.1(b) in Chapter 4 shows members of $\mathcal{F}_\sigma$ above the odometer of a point source on $\mathbb{Z}$. We now allow several vertices to topple at prescribed times, possibly simultaneously.

Example 10.2 (Parallel Toppling) At each integer time $n \geq 1$, all unstable vertices topple their excess mass simultaneously. This procedure commutes with graph automorphisms. Thus an automorphism-invariant law of the initial configuration gives an automorphism-invariant law of the configuration at each time n.

Example 10.3 (Asynchronous Toppling) Assign each $v \in V$ an independent rate-1 **Poisson clock** $(t_k^v)_{k \geq 0}$. Here $t_0^v = 0$, and the increments are i.i.d. exponential random variables of mean 1. At time t_k^v for $k \geq 1$, the vertex v distributes its excess immediately before that time. Example 10.13 verifies that this procedure is well defined when G has bounded degree.

A general procedure records, for each vertex v and time t, the mass that v has sent to each neighbor by time t, and two requirements make such a record a toppling procedure. First, every increase of this mass over a time interval comes with a jump inside the interval, as in both examples (condition (ii) below). Second, the topplings must have a beginning. How much v emits at time t depends on the mass it received before t from its neighbors, so

to determine it we follow the jumps of neighbors backward in time, and this search must end (condition (iii)). Jump times may accumulate from below: nested-volume toppling, in Example 10.12, uses the times $n - 1/k$ for $k \geq 1$, which increase to n. They may not accumulate from above along adjacent vertices: a path $v_1, v_2, \dots$ whose vertices jump at the times $1, \frac{1}{2}, \frac{1}{3}, \dots$ is excluded, because following the received mass backward along it never reaches a first toppling. Figure 10.1 shows this chain beside the backward search among the rings of Poisson clocks, which ends after finitely many steps. Together, the two conditions make mass leave each vertex only in jumps (Exercise 10.9). The proof of the least action principle for these procedures, Proposition 10.9, follows such backward chains. Sections 10.3–10.5 use only parallel toppling, which never moves more than the excess mass (Example 10.11) and whose odometer is u^σ_∞ (Lemma 10.15).

Definition 10.4 Let $G = (V, E)$ be a locally finite connected graph. A **toppling procedure** is a Borel function

$$U : [0, \infty) \times V \times \mathbb{R}^V \longrightarrow [0, \infty) .$$

We write its value at (t, v, σ) as $U^\sigma_t(v)$. We require that $U^\sigma_0(v) = 0$ and that the following conditions hold for every $v \in V$ and $\sigma \in \mathbb{R}^V$.

(i) The function $t \mapsto U^\sigma_t(v)$ is right-continuous and nondecreasing.

(ii) If $0 \leq s < r$ and $U^\sigma_r(v) > U^\sigma_s(v)$, then the function $t \mapsto U^\sigma_t(v)$ is discontinuous at a time in $(s, r]$.

(iii) There do not exist vertices $(v_i)_{i \geq 1}$ and strictly decreasing times $(t_i)_{i \geq 1}$ such that

- $t \mapsto U^\sigma_t(v_i)$ is discontinuous at t_i;
- $v_{i+1} = v_i$ or $v_{i+1} \sim v_i$.

The quantity $U^\sigma_t(v)$ is the mass sent from v to each neighbor by time t. The configuration at that time is $\sigma^U_t = \sigma + \Delta U^\sigma_t$.

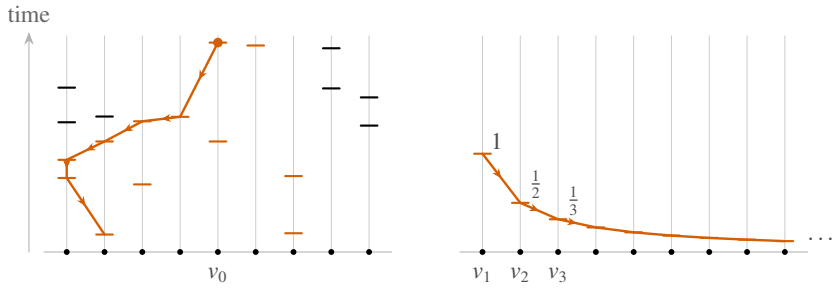
Definition 10.5 Let U be a toppling procedure and let $\sigma : V \to \mathbb{R}$. Its **odometer** is

$$U^\sigma_\infty(v) := \lim_{t \to \infty} U^\sigma_t(v) \qquad \text{for } v \in V ,$$

where the limit may equal ∞.

If $U^\sigma_\infty < \infty$ on V, then local finiteness gives

$$\sigma^U_\infty := \lim_{t \to \infty} \sigma^U_t = \sigma + \Delta U^\sigma_\infty .$$



(a) (b)

Figure 10.1: (a) Rings of independent rate-1 Poisson clocks (ticks) at the vertices of a path, with time increasing upward: the mass emitted at the top ring at v_0 is determined by the initial masses and the rings reached from it by repeatedly stepping to an earlier ring at the same or an adjacent vertex (vermilion), and every such backward chain, like the one drawn, ends after finitely many steps. (b) Jumps at the times $1, \frac{1}{2}, \frac{1}{3}, \ldots$ at the adjacent vertices $v_1, v_2, v_3, \ldots$ form a backward chain that never ends, which condition (iii) of Definition 10.4 excludes.

Definition 10.6 A toppling procedure U is **stabilizing** for $\sigma : V \to \mathbb{R}$ when U^σ_∞ is finite and $\sigma^U_\infty \le 1$ on V.

A stabilizing procedure has odometer in $\mathcal{F}_\sigma$, so Theorem 4.3 implies that σ is stabilizable. The next proposition proves the converse.

Proposition 10.7 *Let $G = (V, E)$ be a locally finite connected graph and let $\sigma : V \to \mathbb{R}$. Then σ is stabilizable if and only if there exists a stabilizing toppling procedure for σ.*

Proof. By Theorem 4.3(iii), σ is stabilizable if and only if $\mathcal{F}_\sigma$ is nonempty, so it suffices to show that a stabilizing toppling procedure for σ exists if and only if $\mathcal{F}_\sigma \neq \varnothing$. If U is stabilizing, then $U^\sigma_\infty \in \mathcal{F}_\sigma$. Conversely, for $f \in \mathcal{F}_\sigma$, define a procedure, independently of its input, by the single jump

$$U^\eta_t(v) = f(v)\mathbf{1}_{[1,\infty)}(t) \qquad \text{for } \eta \in \mathbb{R}^V .$$

This procedure stabilizes σ. □

Definition 10.8 Let U be a toppling procedure and let $\sigma : V \to \mathbb{R}$. We call U **legal** for σ when, for every $v \in V$ and $t > 0$,

$$U^\sigma_t(v) - U^\sigma_{t^-}(v) \le \frac{(\sigma^U_{t^-}(v) - 1)_+}{\deg(v)} ,$$

where $\sigma^U_t = \sigma + \Delta U^\sigma_t$ and $U^\sigma_{t^-}(v)$ denotes the left limit at t.

The least action principle extends from sequential toppling to every legal procedure. In particular, all legal procedures that stabilize have the same odometer.

Proposition 10.9 (Least Action Principle and Abelian Property) *Let σ be a real-valued configuration on a locally finite connected graph $G = (V, E)$, and let L be a legal toppling procedure for σ. Then the following statements hold.*

(i) For every $f \in \mathcal{F}_\sigma$,

$$L^\sigma_\infty \le f\,.$$

(ii) If L is stabilizing for σ, then $L^\sigma_\infty = u^\sigma_\infty$.

Proof. (i) Fix $f \in \mathcal{F}_\sigma$. For $v \in V$, let

$$t_v := \inf\{t \ge 0 : L^\sigma_t(v) > f(v)\}\,.$$

Suppose $t_v < \infty$. Conditions (ii) and (iii) of Definition 10.4 imply that $L^\sigma(v)$ jumps across $f(v)$ at time t_v. Otherwise, condition (ii) would produce jump times decreasing strictly to t_v, contrary to condition (iii). Let $g := f - L^\sigma_{t_v^-}$. Legality gives

$$\sigma^L_{t_v^-}(v) \ge 1 + \deg(v)(L^\sigma_{t_v}(v) - L^\sigma_{t_v^-}(v)) > 1 + \deg(v)g(v)\,.$$

On the other hand,

$$\begin{aligned}\sigma^L_{t_v^-}(v) &= (\sigma + \Delta f)(v) - \Delta g(v)\\ &\le 1 + \deg(v)g(v) - \sum_{w\sim v} g(w)\,.\end{aligned}$$

Thus $g(w) < 0$ for a neighbor w of v, so $t_w < t_v$. Conditions (ii) and (iii) of Definition 10.4 show that $L^\sigma(w)$ jumps at t_w. Iteration produces adjacent vertices with strictly decreasing jump times, contrary to Definition 10.4(iii). Hence $L^\sigma_\infty \le f$.

(ii) If L is stabilizing for σ, then $L^\sigma_\infty \in \mathcal{F}_\sigma$, and part (i) gives $L^\sigma_\infty \le f$ for every $f \in \mathcal{F}_\sigma$. The definition (10.2) therefore gives $L^\sigma_\infty = u^\sigma_\infty$. □

If U is a stabilizing toppling procedure for σ, legal or not, then $U^\sigma_\infty \in \mathcal{F}_\sigma$, so part (i) gives $L^\sigma_\infty \le U^\sigma_\infty$ for every procedure L that is legal for σ. By part (ii), all legal toppling procedures that stabilize σ yield the same final configuration $\sigma_\infty = \sigma + \Delta u^\sigma_\infty$.

Increasing the initial mass can only increase the odometer.

Lemma 10.10 *Let $G = (V, E)$ be a locally finite connected graph, and let $\sigma, \eta \in \mathbb{R}^V$ satisfy $\sigma \le \eta$. Then*

$$u^\sigma_\infty(v) \le u^\eta_\infty(v) \qquad \text{for every } v \in V\,.$$

Proof. Recall from (10.2) that u^σ_∞ and u^η_∞ are the pointwise infima over $\mathcal{F}_\sigma$ and $\mathcal{F}_\eta$, with $\inf \varnothing = \infty$. Since $\sigma \le \eta$, every $f \in \mathcal{F}_\eta$ satisfies $\sigma + \Delta f \le \eta + \Delta f \le 1$, so $\mathcal{F}_\eta \subseteq \mathcal{F}_\sigma$. Taking the infimum over the smaller family gives $u^\sigma_\infty \le u^\eta_\infty$. □

By Theorem 4.3 and Lemma 4.1, u^σ_∞ is also the odometer of every thorough toppling sequence started from σ, so the lemma compares these odometers as well.

Parallel toppling is the procedure used in the rest of this chapter. Nested-volume toppling, asynchronous toppling, and a procedure that stabilizes two parts of an initial configuration in succession are further examples.

Example 10.11 (Parallel Toppling as a Toppling Procedure) At each integer time $k \ge 1$, every unstable vertex topples its excess:

$$U^\sigma_k(v) - U^\sigma_{k-1}(v) = \frac{(\sigma^U_{k-1}(v) - 1)_+}{\deg(v)} \qquad \text{for } v \in V\,.$$

Here $\sigma^U_k = \sigma + \Delta U^\sigma_k$, and $U^\sigma_t = U^\sigma_{\lfloor t \rfloor}$ between integer times. This is a legal toppling procedure.

Recall $\zeta(x) = (\sigma(x) - 1)/\deg(x)$ from Section 4.1. Since $\sigma^U_k = \sigma + \Delta U^\sigma_k$ and all vertices topple at once, the update rule (4.2) gives

$$U^\sigma_{k+1}(x) = \max\Big\{U^\sigma_k(x),\ \frac{1}{\deg(x)} \sum_{y \sim x} U^\sigma_k(y) + \zeta(x)\Big\} \qquad \text{for } x \in V\,. \tag{10.3}$$

For every $f \in \mathcal{F}_\sigma$, induction from $U^\sigma_0 = 0$ and the monotonicity of (4.2) give $U^\sigma_k \le f$ for every k, so $U^\sigma_\infty \le f$. In particular, if σ is stabilizable, then $\mathcal{F}_\sigma \ne \varnothing$ by Theorem 4.3(iii), and U^σ_∞ is finite. Conversely, if

$$U^\sigma_\infty(v) = \frac{1}{\deg(v)} \sum_{k=0}^{\infty} (\sigma^U_k(v) - 1)_+ < \infty \qquad \text{for } v \in V\,,$$

then $(\sigma^U_k(v) - 1)_+ \to 0$ for every v, so the procedure stabilizes σ.

In each round a vertex emits its excess and receives the excesses of its neighbors, so

$$\sigma^U_{k+1}(v) = \min\{\sigma^U_k(v), 1\} + \sum_{y \sim v} \frac{(\sigma^U_k(y) - 1)_+}{\deg(y)} \ge \min\{\sigma^U_k(v), 1\}\,. \tag{10.4}$$

Hence once v has emitted positive mass, $\sigma_j^U(v) \geq 1$ at all later times j, and induction gives $\sigma_k^U \geq \min\{\sigma, 1\}$ for every $k \geq 0$.

Example 10.12 (Nested-Volume Toppling) Let $G = (V, E)$ be infinite, locally finite, and connected. Let $\varnothing \neq V_1 \subset V_2 \subset \cdots$ be finite proper subsets whose union is V. Between times $n-1$ and n, topple in parallel only at unstable vertices in V_n until all vertices in V_n are stable. Formally,

$$U^{\sigma}_{n-\frac{1}{k+1}}(v) - U^{\sigma}_{n-\frac{1}{k}}(v) := \frac{(\sigma^U_{n-\frac{1}{k}}(v) - 1)_+ \mathbf{1}_{V_n}(v)}{\deg(v)} \qquad \text{for } n, k \geq 1\,.$$

The odometer is constant between these times, and $U_n^{\sigma}(v)$ is the limit of $U^{\sigma}_{n-1/k}(v)$ as $k \to \infty$. This procedure is legal.

To justify each stage, fix a finite proper set $\varnothing \neq A \subset V$ and topple a configuration η in parallel inside A. Since vertices outside A never topple, replacing η by $\eta\mathbf{1}_A$ leaves the odometers unchanged. By Lemma 4.4(ii) and Theorem 4.3(ii), $\eta\mathbf{1}_A$ has a finite stabilizing odometer. Each finite sequence of rounds, run at integer times, is legal for $\eta\mathbf{1}_A$, so the least action principle, Proposition 10.9(i), bounds its odometer by this stabilizing odometer. Thus the increasing odometer converges, and its increments, hence the excesses on A, tend to zero. Local finiteness gives convergence of the configurations to a limit stable on A. Apply this with $A = V_n$ and $\eta = \sigma^U_{n-1}$.

Example 10.13 (Asynchronous Toppling as a Toppling Procedure) Suppose every vertex of G has degree at most an integer $\deg_{\max} \geq 1$, and use the independent Poisson clocks from Example 10.3. For $v \in V$ and $t > 0$, let

$$U_t^{\sigma}(v) := \frac{1}{\deg(v)} \sum_{k \geq 1} (\sigma^U_{(t_k^v)^-}(v) - 1)_+ \mathbf{1}\{t_k^v \leq t\}\,.$$

Fix $v_0 \in V$ and $t_0 > 0$, and let $N \geq 1$. Let $A_N(v_0, t_0)$ be the event that there are clock rings $t_0 > t_1 > \cdots > t_N > 0$ at vertices $v_1, \ldots, v_N$, where $v_j = v_{j-1}$ or $v_j \sim v_{j-1}$ for $1 \leq j \leq N$.

A union bound over the possible vertex sequences gives

$$\mathbb{P}\{A_N(v_0, t_0)\} \leq \frac{(\deg_{\max} + 1)^N t_0^N}{N!} \longrightarrow 0\,.$$

Taking a countable union over $v_0 \in V$ and positive integer upper bounds for t_0 shows that condition (iii) in Definition 10.4 holds almost surely. Almost surely no two clocks ring simultaneously. Trace backward the rings needed to determine the mass at a fixed ring. Each ring has only finitely many earlier

rings at its vertex and its neighbors. If infinitely many rings were needed, then repeatedly choosing an earlier ring with infinitely many predecessors would produce an infinite backward chain. Thus only finitely many rings are needed, and starting from the earliest ones defines the masses recursively. Each clock rings only finitely often in a bounded interval, which verifies conditions (i) and (ii) of Definition 10.4. Each value is obtained from finitely many coordinates of the initial configuration by finitely many sums and positive-part operations, so the resulting map is Borel. Thus the displayed formula defines a toppling procedure almost surely. Exercise 10.10 gives a sufficient condition for unbounded-degree graphs.

Example 10.14 (Stabilization in Two Steps) Suppose $\sigma = \eta + \xi$, where η stabilizes and $\xi \geq 0$. First stabilize η in parallel at times $1 - \frac{1}{k+1}$ for $k \geq 1$. These topplings remain legal after adding ξ. Starting from $\eta + \Delta u_\infty^\eta + \xi$, continue parallel toppling at times $2 - \frac{1}{k+1}$ for $k \geq 1$. If the second procedure stabilizes, then Proposition 10.9(ii) shows that the combined odometer is u_∞^σ.

The least action principle defines the odometer as an infimum over $\mathcal{F}_\sigma$. The next lemma shows that parallel toppling attains it, so the odometer is also the supremum of the mass that legal procedures emit from each vertex to each of its neighbors.

Lemma 10.15 *Let $G = (V, E)$ be a locally finite connected graph and let $\sigma : V \to \mathbb{R}$. Then*

(i) either u_∞^σ is finite on V, or $u_\infty^\sigma \equiv \infty$;

(ii) parallel toppling has odometer u_∞^σ.

Proof. (i) If $\mathcal{F}_\sigma \neq \emptyset$, then every $f \in \mathcal{F}_\sigma$ is finite and bounds u_∞^σ by (10.2). Otherwise $u_\infty^\sigma \equiv \infty$, because $\inf \emptyset = \infty$.

(ii) Let P be parallel toppling. By Example 10.11, $P_\infty^\sigma \leq f$ for every $f \in \mathcal{F}_\sigma$, so $P_\infty^\sigma \leq u_\infty^\sigma$. If P_∞^σ is finite on V, then Example 10.11 also gives $\sigma + \Delta P_\infty^\sigma \leq 1$, so $P_\infty^\sigma \in \mathcal{F}_\sigma$ and $P_\infty^\sigma \geq u_\infty^\sigma$. Otherwise $P_\infty^\sigma(w) = \infty$ for a vertex w. If $v \sim w$, then the parallel update gives

$$P_{k+1}^\sigma(v) \geq \frac{\sigma(v) - 1 + \sum_{y \sim v} P_k^\sigma(y)}{\deg(v)} \geq \frac{P_k^\sigma(w) + \sigma(v) - 1}{\deg(v)},$$

so $P_\infty^\sigma(v) = \infty$. Connectivity gives $P_\infty^\sigma \equiv \infty$, and then $P_\infty^\sigma \leq u_\infty^\sigma$ gives $u_\infty^\sigma \equiv \infty$. In both cases $P_\infty^\sigma = u_\infty^\sigma$. □

Proposition 10.9(i) and the definition (10.2) give $U_\infty^\sigma \leq u_\infty^\sigma$ for every procedure U that is legal for σ, and parallel toppling is legal and attains equality by Lemma 10.15(ii). Hence, for every $v \in V$,

$$u_\infty^\sigma(v) = \sup\{U_\infty^\sigma(v) : U \text{ is legal for } \sigma\}.$$

We next use parallel toppling to identify the odometer of a single excess mass. For simple random walk $(X_n)_{n\ge 0}$ on G, write $\mathbb{P}_w$ for its law when $X_0 = w$. Its Green function is

$$g^{::}(w,v) := \frac{1}{\deg(v)} \sum_{n\ge 0} \mathbb{P}_w\{X_n = v\} \qquad \text{for } w, v \in V\,, \tag{10.5}$$

where the sum may be infinite. Summing the first-step identity

$$\mathbb{P}_w\{X_{n+1} = v\} = \frac{1}{\deg(w)} \sum_{y\sim w} \mathbb{P}_y\{X_n = v\}$$

over $n \ge 0$ shows that, when $g^{::}$ is finite, $\Delta_w g^{::}(w,v) = -\delta_v(w)$ for $w, v \in V$. Reversing paths gives $\deg(w)\mathbb{P}_w\{X_n = v\} = \deg(v)\mathbb{P}_v\{X_n = w\}$, so $g^{::}$ is symmetric: $g^{::}(w,v) = g^{::}(v,w)$.

Proposition 10.16 *Let $G = (V,E)$ be a locally finite connected graph, let $o \in V$, let $\beta > 0$, and let $\sigma = 1 + \beta\delta_o$. Then the following statements hold.*

(a) If simple random walk on G is transient, then σ stabilizes and

$$u^{\sigma}_{\infty}(v) = \beta\, g^{::}(o,v) \qquad \textit{for } v \in V\,.$$

(b) If simple random walk on G is recurrent, then σ does not stabilize.

Proof. Let U be the parallel procedure. For $j \ge 0$ and $v \in V$, write $q_j(v) := \mathbb{P}_o\{X_j = v\}/\deg(v)$. A one-step computation gives

$$\begin{aligned} \Delta q_j(v) &= \sum_{w\sim v} \left(\frac{\mathbb{P}_o\{X_j = w\}}{\deg(w)} - \frac{\mathbb{P}_o\{X_j = v\}}{\deg(v)} \right) \\ &= \deg(v)\big(q_{j+1}(v) - q_j(v)\big)\,. \end{aligned} \tag{10.6}$$

We claim that

$$U^{\sigma}_k(v) = \beta \sum_{j=0}^{k-1} q_j(v) \qquad \text{for } k \ge 0 \text{ and } v \in V\,. \tag{10.7}$$

The case $k = 0$ follows from the empty sum. For the inductive step, suppose (10.7) holds at time k. Then (10.6) gives

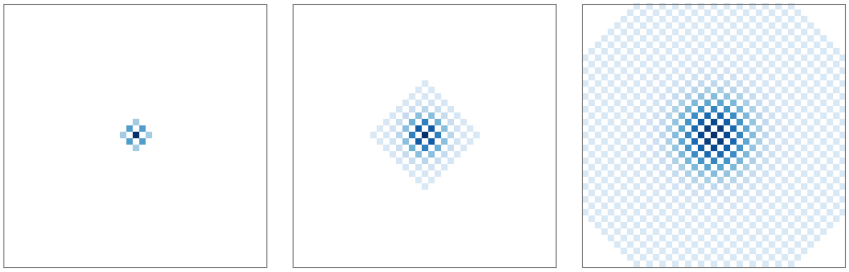
$$\sigma^{U}_k(v) = \sigma(v) + \Delta U^{\sigma}_k(v) = 1 + \beta \deg(v) q_k(v)\,.$$

Hence $U^{\sigma}_{k+1}(v) - U^{\sigma}_k(v) = \beta q_k(v)$, which proves the claim by induction (Figure 10.2).

Letting $k \to \infty$ in (10.7) gives $U^{\sigma}_{\infty}(v) = \beta g^{::}(o,v)$.

(a) If the walk is transient, then this is finite, so Lemma 10.15(ii) gives $u^\sigma_\infty = U^\sigma_\infty = \beta g^{\cdot\cdot}(o,\cdot)$. In particular $\mathcal{F}_\sigma \neq \varnothing$, and σ stabilizes by Theorem 4.3(iii).

(b) If the walk is recurrent, then $U^\sigma_\infty \equiv \infty$, so $u^\sigma_\infty \equiv \infty$ by Lemma 10.15(ii). Then $\mathcal{F}_\sigma = \varnothing$, and σ explodes by Theorem 4.3(iii). □



$k = 2$ $k = 8$ $k = 32$

Figure 10.2: The configuration $1 + \beta\delta_o$ on $\mathbb{Z}^2$ after $k = 2$, 8 and 32 rounds of parallel toppling: the excess $\sigma^U_k - 1$ (shaded, darker for larger, each panel scaled to its maximum) is β times the distribution at time k of simple random walk from o, as in the proof of Proposition 10.16.

Exercise 10.14 asks for a proof, using killed Green functions, that stabilization of $1 + \beta\delta_o$ implies transience, and Exercise 10.12 treats finitely many excess masses. The next section extends this walk interpretation from one excess mass to an arbitrary initial configuration.

10.3 RANDOM WALK REPRESENTATION

Throughout the section, $G = (V, E)$ is a locally finite connected graph, $\sigma : V \to \mathbb{R}$ is fixed, and $(X_k)_{k\geq 0}$ is simple random walk. We write $\mathbb{P}_x$ and $\mathbb{E}_x$ for its law and expectation when $X_0 = x$. All stopping times are for the natural filtration of the walk.

For a single excess mass, parallel toppling follows the transition probabilities of simple random walk. For a general initial configuration, the odometer is the value of an optimal stopping problem: the walk collects the local excess at each step and chooses when to stop.

Recall the excess per neighbor $\zeta(x) = (\sigma(x) - 1)/\deg(x)$ from Section 4.1, and write

$$\mathsf{B}_n := \sum_{k=0}^{n-1} \zeta(X_k) \qquad \text{for } n \geq 1, \qquad \text{with } \mathsf{B}_0 := 0\,. \tag{10.8}$$

Recall from Example 10.11 that parallel toppling produces $\sigma^U_n = \sigma + \Delta U^\sigma_n$,

where $U_0^\sigma = 0$ and

$$U_{n+1}^\sigma(x) - U_n^\sigma(x) = \frac{(\sigma_n^U(x) - 1)_+}{\deg(x)} \qquad \text{for } x \in V \text{ and } n \geq 0\,. \tag{10.9}$$

By Lemma 10.15(ii), these partial odometers increase to the full odometer, so $U_n^\sigma \uparrow u_\infty^\sigma$.

The parallel update becomes the Bellman recursion for this stopping problem.

Lemma 10.17 *Let $G = (V, E)$ be a locally finite connected graph, let $\sigma : V \to \mathbb{R}$, and let $(U_n^\sigma)_{n\geq 0}$ be the odometers produced by parallel toppling. Then, for every $n \geq 0$ and $x \in V$,*

$$U_{n+1}^\sigma(x) = \left(\frac{1}{\deg(x)} \sum_{y\sim x} U_n^\sigma(y) + \zeta(x) \right)_+ . \tag{10.10}$$

Proof. For $n \geq 0$ and $x \in V$, write

$$b_n(x) := \frac{1}{\deg(x)} \sum_{y\sim x} U_n^\sigma(y) + \zeta(x)\,,$$

so that (10.3) reads $U_{n+1}^\sigma(x) = \max\{U_n^\sigma(x), b_n(x)\}$. Parallel increments are nonnegative, so U_n^σ, and hence $b_n(x)$, is nondecreasing in n. Since $U_0^\sigma = 0$, we have $U_1^\sigma = \max\{0, b_0\}$. If $U_n^\sigma = \max\{0, b_{n-1}\}$ for some $n \geq 1$, then

$$U_{n+1}^\sigma = \max\{0, b_{n-1}, b_n\} = \max\{0, b_n\}\,.$$

By induction, $U_{n+1}^\sigma = (b_n)_+$ for every $n \geq 0$, which is (10.10). □

The recursion identifies the finite-time odometer exactly.

Theorem 10.18 (Random Walk Representation) *Let $G = (V, E)$ be a locally finite connected graph, let $\sigma : V \to \mathbb{R}$, and let $(U_n^\sigma)_{n\geq 0}$ be the odometers produced by parallel toppling. Then, for every $x \in V$ and $n \geq 0$,*

$$U_n^\sigma(x) = \max_{T \leq n} \mathbb{E}_x[\mathsf{B}_T]\,,$$

where the maximum is over stopping times bounded by n.

Proof. Write $w_n(x) := \max_{T\leq n} \mathbb{E}_x[\mathsf{B}_T]$. We prove by induction that $w_n = U_n^\sigma$ and that the maximum is attained. The claim follows for $n = 0$ by stopping at once.

For the inductive step, suppose the claim holds at time n. A stopping time $T \leq n + 1$ either equals 0 or is at least 1. In the second case, the remaining stopping time is bounded by n, so the Markov property gives

$$\mathbb{E}_x[\mathsf{B}_T] \leq \zeta(x) + \frac{1}{\deg(x)} \sum_{y\sim x} w_n(y)\,.$$

Including the option to stop at once gives

$$w_{n+1}(x) \le \left(\frac{1}{\deg(x)} \sum_{y \sim x} w_n(y) + \zeta(x) \right)_+ = U^{\sigma}_{n+1}(x)\,,$$

by the induction hypothesis and Lemma 10.17.

If $U^{\sigma}_{n+1}(x) = 0$, then stopping at once gives equality. Otherwise, for each neighbor y of x, choose an optimal stopping time $T_y \le n$ for a walk started at y. Take one step and, conditional on $X_1 = y$, apply the rule T_y to the future walk. The expected payoff of this rule is

$$\zeta(x) + \frac{1}{\deg(x)} \sum_{y \sim x} w_n(y) = U^{\sigma}_{n+1}(x)\,,$$

by the Markov property. Thus the rule is optimal, which completes the induction. □

Exercise 10.13 identifies an optimal rule: the walk stops upon reaching a vertex from which no further gain is possible.

Letting the number of parallel rounds tend to infinity gives the infinite-horizon representation.

Corollary 10.19 *Let $G = (V, E)$ be a locally finite connected graph, let $\sigma : V \to \mathbb{R}$, and let u^{σ}_{∞} be the divisible-sandpile odometer. Then, for every $x \in V$,*

$$u^{\sigma}_{\infty}(x) = \sup_{T} \mathbb{E}_x[\mathsf{B}_T]\,,$$

where the supremum is over bounded stopping times.

Proof. Theorem 10.18 and monotone convergence of the parallel odometers show that both sides are the supremum over n of the corresponding finite-horizon values. □

The stopping problem describes the odometer but does not evaluate it. For the point source on $\mathbb{Z}^d$, and more generally whenever the walk is killed on leaving a finite set, the supremum can be expressed through killed Green functions.

10.3.1 Green Function Representation

For a nonempty finite proper subset $A \subsetneq V$, the exit time from A is the hitting time $\tau_{A^c} = \inf\{k \ge 0 : X_k \notin A\}$ of its complement $A^c := V \setminus A$. Generalizing the killed Green function (3.18) on $\mathbb{Z}^d$, write

$$g^{::}_A(y, v) := \frac{1}{\deg(v)} \sum_{k \ge 0} \mathbb{P}_y\{X_k = v \text{ and } k < \tau_{A^c}\} \qquad \text{for } y, v \in V\,. \tag{10.11}$$

Equivalently, $g_A^{\cdot\cdot}(y,v)$ is $1/\deg(v)$ times the expected number of visits to v before leaving A. Reversing paths gives

$$\deg(y)\mathbb{P}_y\{X_k = v \text{ and } k < \tau_{A^c}\} = \deg(v)\mathbb{P}_v\{X_k = y \text{ and } k < \tau_{A^c}\},$$

so $g_A^{\cdot\cdot}(y,v) = g_A^{\cdot\cdot}(v,y)$. A first-step decomposition gives $\Delta_y g_A^{\cdot\cdot}(y,v) = -\delta_v(y)$ for $y \in A$ and $v \in V$, and by symmetry

$$\Delta_v g_A^{\cdot\cdot}(y,v) = -\delta_y(v) \qquad \text{for } y \in V \text{ and } v \in A, \tag{10.12}$$

where both sides vanish when $y \notin A$. Thus $g_A^{\cdot\cdot}(y,\cdot)$ vanishes outside A and is harmonic on $A \setminus \{y\}$.

Fix $o \in V$. For every finite proper set A containing o, summing the payoff by the visited vertex gives

$$\mathbb{E}_o[\mathsf{B}_{\tau_{A^c}}] = \sum_{v \in A} g_A^{\cdot\cdot}(o,v)(\sigma(v) - 1). \tag{10.13}$$

We first solve the stopping problem before the walk leaves a fixed finite set.

Proposition 10.20 *Let $G = (V,E)$ be a locally finite connected graph, let $\sigma : V \to \mathbb{R}$, let $o \in V$, and let $K \subset V$ be a finite proper set containing o. For $x \in V$, let*

$$w_K(x) := \sup_{T \le \tau_{K^c}} \mathbb{E}_x[\mathsf{B}_T].$$

Then $w_K(x) < \infty$ for every $x \in V$, and

$$w_K(o) = \max\left\{0, \sup_{\substack{A \subseteq K \text{ with } o \in A \\ A \text{ connected}}} \sum_{v \in A} g_A^{\cdot\cdot}(o,v)(\sigma(v) - 1)\right\}. \tag{10.14}$$

Proof. Since K is finite and proper, $\mathbb{E}_x\tau_{K^c} < \infty$. The payoff per step is bounded on K, so $w_K < \infty$; stopping at 0 gives $w_K \ge 0$, and $w_K = 0$ outside K. For every stopping time $T \le \tau_{K^c}$ and integer $n \ge 0$,

$$|\mathsf{B}_{T\wedge n}| \le \tau_{K^c} \max_{v \in K} \left|\frac{\sigma(v) - 1}{\deg(v)}\right|,$$

so dominated convergence gives $\mathbb{E}_x[\mathsf{B}_{T\wedge n}] \to \mathbb{E}_x[\mathsf{B}_T]$ as $n \to \infty$, and hence

$$w_K(x) = \sup_{\substack{T \le \tau_{K^c} \\ T \text{ bounded}}} \mathbb{E}_x[\mathsf{B}_T] \qquad \text{for } x \in V. \tag{10.15}$$

Next, a rule $T \le \tau_{K^c}$ started at $x \in K$ either stops at 0 or takes one step. In the latter case, given $X_1 = y$, the remaining rule is admissible from y and pays at most $w_K(y)$ in expectation. Hence

$$w_K(x) \le \left(\frac{\sigma(x) - 1}{\deg(x)} + \frac{1}{\deg(x)} \sum_{y \sim x} w_K(y) \right)_+ \qquad \text{for } x \in K\,. \tag{10.16}$$

For every connected $A \subseteq K$ containing o, the stopping time τ_{A^c} is admissible. Hence (10.13) gives

$$\sum_{v \in A} g_A^{::}(o, v)(\sigma(v) - 1) \le w_K(o)\,.$$

Taking the supremum over A and then the maximum with 0 shows that the right-hand side of (10.14) is at most $w_K(o)$. When $w_K(o) = 0$, this proves equality.

Suppose $w_K(o) > 0$, and let D be the component of o in $\{w_K > 0\} \subseteq K$. On D, where $w_K > 0$, inequality (10.16) gives $\Delta w_K \ge 1 - \sigma$. Let $g := g_D^{::}(o, \cdot)$. By (10.12), $\Delta g = -\delta_o$ on D. Since g is nonnegative and vanishes outside D, while $w_K = 0$ at every vertex outside D adjacent to D, summation by parts gives

$$\begin{aligned} \sum_{v \in D} g(v)(\sigma(v) - 1) &\ge -\sum_{v \in V} g(v) \Delta w_K(v) \\ &= -\sum_{v \in V} w_K(v) \Delta g(v) = w_K(o)\,. \end{aligned}$$

All sums are finite, and the middle equality follows by interchanging endpoints in the neighbor sum. Since D is admissible in (10.14), this proves the reverse inequality. □

Letting K grow gives a representation of the infinite-horizon odometer as a supremum of Green-function sums.

Theorem 10.21 *Let $G = (V, E)$ be an infinite locally finite connected graph, let $\sigma : V \to \mathbb{R}$, let $o \in V$, and let u_∞^σ be the odometer of parallel toppling started from σ. Then*

$$u_\infty^\sigma(o) = \max \left\{ 0, \sup_{\substack{A \subset V \text{ finite with } o \in A \\ A \text{ connected}}} \sum_{v \in A} g_A^{::}(o, v)(\sigma(v) - 1) \right\}.$$

Proof. For $m \ge 1$, let K_m be the finite proper graph-distance ball of radius $m - 1$ about o. Since $\tau_{K_m^c} \ge m$ from o, the union over m of the bounded

stopping times $T \le \tau_{K_m^c}$ is the family of all bounded stopping times. Equation (10.15) and Corollary 10.19 therefore give

$$u_\infty^\sigma(o) = \sup_{m \ge 1} w_{K_m}(o)\,.$$

Every finite connected set containing o lies in a K_m, so Proposition 10.20 proves the theorem. □

Figure 10.3 shows the set A that attains the supremum for an instance of i.i.d. masses on $\mathbb{Z}^2$, with a walk stopped on leaving it.

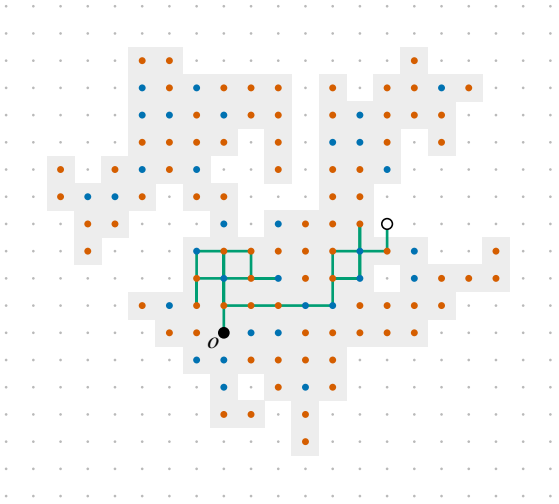


Figure 10.3: A component D (shaded) of the set where the odometer is positive, for i.i.d. masses uniform on $[0, 1.7]$ in $\mathbb{Z}^2$, and a simple random walk from o stopped at its first site outside D (hollow dot). The walk gains $\zeta(X_k)$ at each step, positive at the sites where $\sigma > 1$ (vermilion) and negative where $\sigma < 1$ (blue), and its expected gain is $u_\infty^\sigma(o)$, so $A = D$ attains the supremum in Theorem 10.21.

We now use the random-walk representation to study i.i.d. initial masses.

10.4 RANDOM SANDPILES ON BOUNDED-DEGREE GRAPHS

Throughout this section, $G = (V, E)$ is infinite and connected, with all degrees at most $\deg_{\max}$, and the initial masses $(\sigma(v))_{v \in V}$ are bounded i.i.d. real random variables with common mean μ. We use the random-walk representation of Section 10.3: $(X_k)_{k \ge 0}$ is simple random walk, independent of the masses, and $\mathbb{P}_x$ and $\mathbb{E}_x$ are its law and expectation when $X_0 = x$. Recall that $\zeta(v) = (\sigma(v) - 1)/\deg(v)$, that $\mathsf{B}_n = \sum_{k<n} \zeta(X_k)$ is the payoff collected before time n, and that U_n^σ is the odometer after n rounds of parallel toppling (Example 10.11). We write $\mathbb{P}$ and $\mathbb{E}$ for probability and expectation over the

masses. For $n \geq 1$ and $x, v \in V$, let

$$g^{::}_{<n}(x,v) := \frac{1}{\deg(v)} \sum_{k=0}^{n-1} \mathbb{P}_x\{X_k = v\} \quad \text{and} \quad \|g^{::}_{<n}(x,\cdot)\|_2^2 = \sum_{v \in V} g^{::}_{<n}(x,v)^2 \,. \tag{10.17}$$

By monotone convergence, $g^{::}_{<n}(x,v)$ increases to the Green function $g^{::}(x,v)$ of (10.5), and $\|g^{::}_{<n}(x,\cdot)\|_2$ increases to $\|g^{::}(x,\cdot)\|_2$.

For nonconstant bounded i.i.d. masses on a graph of bounded degree, the mean alone decides whether the sandpile stabilizes, and the random-walk representation shows why. The simplest strategy for the walk is to stop at a fixed time, and its payoff is a weighted sum of the masses whose mean has the sign of $\mu - 1$ and, when $\mu \neq 1$, grows linearly in absolute value with time. Above mean 1 this drift forces explosion. Below mean 1 it forces stabilization, because sufficiently many visits to a single site are unlikely. At mean 1 there is no drift, and the fluctuations of the masses, helped by a ball of small masses, force explosion. Lemma 10.23 bounds each value of the truncated Green function by $2\sqrt{n}$, giving the control needed in all three cases.

Theorem 10.22 (Phase Transition on Bounded-Degree Graphs) *Let $G = (V, E)$ be an infinite connected graph of bounded degree, and let $(\sigma(v))_{v \in V}$ be bounded i.i.d. real random variables with common mean μ. Then the following statements hold.*

(a) If $\mu > 1$, then the sandpile explodes almost surely.

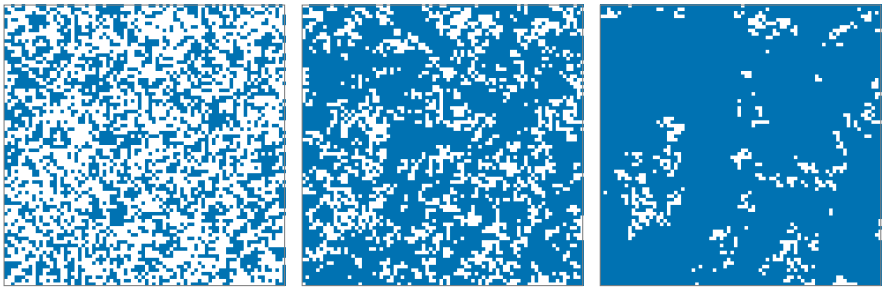
(b) If $\mu = 1$ and $\sigma(v)$ is not almost surely constant, then the sandpile explodes almost surely.

(c) If $\mu < 1$, then the sandpile stabilizes almost surely.

In other words, bounded i.i.d. masses that are not almost surely constant stabilize exactly when their mean is less than 1. Under the hypotheses of part (c), the proof gives more: the expected odometer is bounded, $\sup_{x \in V} \mathbb{E}[u^{\sigma}_{\infty}(x)] < \infty$. Boundedness is far from necessary. Part (a) extends to every law of mean in $(1, \infty]$. Part (b) extends to every law of mean 1 with finite positive variance, and to every nonconstant law of mean 1 that is symmetric about 1. For part (c), it is enough that the positive part of $\sigma(v)$ have a finite pth moment for some $p > 3$, and for every $p < 3$ there are graphs of bounded degree on which it fails. The Notes give the references. Figure 10.4 shows the set of sites that topple for three means below 1.

Let

$$Z_n(x) := \mathbb{E}_x[\mathsf{B}_n] = \sum_{v \in V} g^{::}_{<n}(x,v)(\sigma(v) - 1) \,.$$



$\mu = 0.75$ $\mu = 0.90$ $\mu = 0.97$

Figure 10.4: The sites that topple (blue) when i.i.d. masses uniform on $[0, 2\mu]$ are stabilized on the 256×256 torus, shown in a window of 80×80 sites, for $\mu = 0.75$, 0.9 and 0.97. The three use the same uniform variables, so the sets increase with μ by Lemma 10.10, and they fill more of the torus as μ approaches the critical mean 1.

Stopping the walk at time n is one of the strategies in Theorem 10.18, and stopping at time 0 is another, so

$$U_n^{\sigma}(x) \geq \max\{0, Z_n(x)\}\,. \tag{10.18}$$

The mean of $Z_n(x)$ is $(\mu-1)\sum_v g_{<n}^{::}(x,v)$. Since $n/\deg_{\max} \leq \sum_v g_{<n}^{::}(x,v) \leq n$, its absolute value is of order n when $\mu \neq 1$, and it is zero when $\mu = 1$. Its variance is $\mathrm{Var}(\sigma(v))\,\|g_{<n}^{::}(x,\cdot)\|_2^2$. Everything therefore depends on how large $\|g_{<n}^{::}(x,\cdot)\|_2^2$ can be. The next lemma shows that it is much smaller than n^2 on every graph of bounded degree: the largest value of the truncated Green function is at most of order $\sqrt{n}$, so the expected number of visits by the walk to its starting point before time n is at most of order $\sqrt{n}$.

Lemma 10.23 *Let $G = (V, E)$ be an infinite connected graph with all degrees at most $\deg_{\max}$, let $x \in V$, let $n \geq 1$ be an integer, and let $M_n(x) := \max_{v\in V} g_{<n}^{::}(x,v)$. Then*

$$M_n(x)^3 \leq 4\|g_{<n}^{::}(x,\cdot)\|_2^2 \leq 4nM_n(x)\,.$$

In particular, $M_n(x) \leq 2\sqrt{n}$, and the expected number of visits to x before time n by the walk started at x is at most $2\deg_{\max}\sqrt{n}$.

Proof. The idea is to compare the maximum of $h := g_{<n}^{::}(x,\cdot)$ with its energy, along a path from a maximizer to a site where h vanishes. Each of the first n steps of the walk contributes total probability one to $\sum_v \deg(v)h(v)$, so $\sum_v \deg(v)h(v) = n$. In particular h has finite support, $\sum_v h(v) \leq n$, and $\|h\|_2^2 \leq M_n(x)\sum_v h(v) \leq nM_n(x)$, which is the second inequality.

For the first inequality, we bound the energy of h. Decomposing the visits to v according to the previous step gives $\Delta h(v) = \mathbb{P}_x\{X_n = v\} - \delta_x(v)$ for every

$v \in V$. Since h has finite support, summation by parts over the unoriented edges gives

$$\sum_{\{v,w\}\in E} (h(v) - h(w))^2 = -\sum_{v\in V} h(v)\Delta h(v) \le h(x) \le M_n(x)\,.$$

Now let $v_0, v_1, \dots, v_L$ be a simple path from a maximizer v_0 of h to a vertex v_L outside the support of h; it exists because the support is finite and G is infinite and connected. We telescope h^2 along the path and apply the Cauchy–Schwarz inequality:

$$\begin{aligned} M_n(x)^2 &= \sum_{i<L} (h(v_i) - h(v_{i+1}))(h(v_i) + h(v_{i+1})) \\ &\le \sqrt{M_n(x)}\Big(\sum_{i<L} (h(v_i) + h(v_{i+1}))^2\Big)^{1/2} \le 2\sqrt{M_n(x)\,\|h\|_2^2}\,, \end{aligned}$$

where the last step uses $(\alpha+\beta)^2 \le 2\alpha^2 + 2\beta^2$ and the fact that each vertex lies on at most two edges of the path. Squaring gives $M_n(x)^3 \le 4\|h\|_2^2$. The two inequalities together give $M_n(x)^2 \le 4n$, and the expected number of visits to x is $\deg(x)h(x) \le \deg_{\max} M_n(x) \le 2\deg_{\max}\sqrt{n}$. □

Proof of Theorem 10.22. Let $\deg_{\max}$ bound the degrees of G, and choose $C_0 \ge 1$ such that $|\sigma(v) - 1| \le C_0$ and $|\sigma(v) - \mu| \le C_0$ almost surely; then $|\zeta| \le C_0$. Throughout the proof, c and C denote positive constants that depend only on $\deg_{\max}$ and on the law of $\sigma(v)$.

(a) Suppose that $\mu > 1$. Then $Z_n(x)$ has mean of order n and fluctuations of order at most $n^{3/4}$. Indeed, since $\sum_v \deg(v) g^{::}_{<n}(x,v) = n$, independence of the masses and Lemma 10.23 give

$$\mathbb{E}Z_n(x) \ge \frac{\mu - 1}{\deg_{\max}}\, n \qquad \text{and} \qquad \mathrm{Var}Z_n(x) \le C_0^2 \|g^{::}_{<n}(x,\cdot)\|_2^2 \le 2C_0^2 n^{3/2}\,.$$

By Chebyshev's inequality, $\mathbb{P}\{Z_n(x) \le L\} \to 0$ as $n \to \infty$, for every L. Since $u^\sigma_\infty(x) \ge U^\sigma_n(x) \ge Z_n(x)$ for every n, by Lemma 10.15(ii) and (10.18), the odometer is infinite almost surely, and the sandpile explodes almost surely.

(b) Suppose that $\mu = 1$ and $s^2 := \mathrm{Var}(\sigma(v)) > 0$. Now $Z_n(x)$ has mean zero, and explosion has to come from the fluctuations. We show in three steps that the expected odometers grow without bound, that the odometer at a fixed vertex is at least half its mean with probability bounded below, and that the zero-one law turns this into explosion.

Step 1: We show that the expected odometers tend to infinity. Let $m_n(x) := \mathbb{E}U^\sigma_n(x)$, which is at most $C_0 n$, and let $a_n := \inf_{x\in V} m_n(x)$. The parallel odometers increase in n, so a_n is nondecreasing. Suppose, for a contradiction, that a_n converges to a finite limit a. The idea is that a ball of

small masses around x makes the payoff of the first N steps negative, by a fixed amount, while the odometer stays nonnegative; this raises the mean odometer by a fixed amount. Since $\sigma(v)$ is not almost surely constant and has mean 1, there is $\epsilon > 0$ with $p_\epsilon := \mathbb{P}\{\sigma(v) \le 1 - \epsilon\} > 0$. Choose an integer N with $t := \epsilon N / \deg_{\max} > a$. For $x \in V$, let A_x be the event that every mass at graph distance less than N from x is at most $1 - \epsilon$. This ball has at most $N \deg_{\max}^N$ vertices, so $\mathbb{P}(A_x) \ge q := p_\epsilon^{N \deg_{\max}^N} > 0$ for every x.

We use two facts about the event A_x. First, conditioning on A_x lowers the masses in the ball in the stochastic order and leaves the other masses unchanged; coupling the conditional and unconditional laws through their quantile functions, we use monotonicity of the finite-time odometers in the initial masses, which follows by induction in the Bellman recursion (10.10), to obtain $\mathbb{E}[U_n^\sigma(y) \mid A_x] \le m_n(y)$ for all y and n. Second, for the walk started at x, X_k lies in the ball for every $0 \le k < N$ (Figure 10.5), so on A_x we have $Z_N(x) \le -\epsilon \sum_v g_{<N}^{::}(x, v) \le -t$; since $\mathbb{E}Z_N(x) = 0$, it follows that $\mathbb{E}[Z_N(x); A_x^c] = -\mathbb{E}[Z_N(x); A_x] \ge t\mathbb{P}(A_x)$.

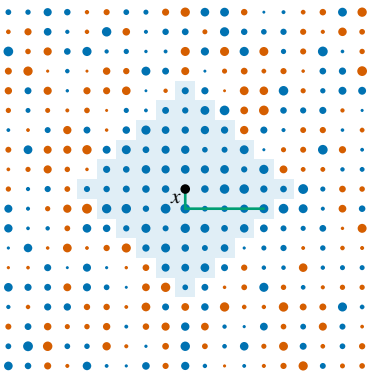


Figure 10.5: Step 1 of the proof of Theorem 10.22(b), with masses above 1 in vermilion and below 1 in blue, larger dots farther from 1: on the event A_x, every mass at graph distance less than N from x (shaded) is at most $1 - \epsilon$, so the payoff B_N of the first N steps of the walk from x (green) is at most $-\epsilon N / \deg_{\max}$.

Write $\mathsf{P}f(x) := \deg(x)^{-1} \sum_{y \sim x} f(y)$ for the transition operator of simple random walk. Dropping the positive part in the Bellman recursion (10.10) gives $U_{k+1}^\sigma \ge \zeta + \mathsf{P}U_k^\sigma$, and iterating this N times gives $U_{n+N}^\sigma \ge Z_N + \mathsf{P}^N U_n^\sigma$. We use this inequality on A_x^c, and $U_{n+N}^\sigma \ge 0$ on A_x:

$$\begin{aligned} m_{n+N}(x) &\ge \mathbb{E}[Z_N(x); A_x^c] + \mathbb{E}[\mathsf{P}^N U_n^\sigma(x); A_x^c] \\ &\ge t\mathbb{P}(A_x) + (1 - \mathbb{P}(A_x))\mathsf{P}^N m_n(x) \ge a_n + q(t - a_n)\,. \end{aligned}$$

The second inequality uses the two facts above, and the third uses $\mathsf{P}^N m_n \ge a_n$ and $t > a \ge a_n$. Taking the infimum over x and letting $n \to \infty$ gives $a \ge a + q(t - a) > a$, a contradiction. Hence $m_n(x) \to \infty$ for every x.

Step 2: We show that the odometer at a fixed vertex is at least half its mean with probability bounded below. Fix $o \in V$. We first compare $m_n(o)$ with $\|g^{::}_{<n}(o,\cdot)\|_2$. The payoff $Z_n(o)$ is a weighted sum of the independent centered variables $\sigma(v) - 1$. Expanding its second and fourth powers, the terms in which some index occurs exactly once vanish, so

$$\mathbb{E}Z_n(o)^2 = s^2\|g^{::}_{<n}(o,\cdot)\|_2^2 \qquad \text{and} \qquad \mathbb{E}Z_n(o)^4 \le 3C_0^4\|g^{::}_{<n}(o,\cdot)\|_2^4 .$$

Since $Z_n(o)$ is centered, (10.18) and the inequality $\mathbb{E}Z^2 \le (\mathbb{E}|Z|)^{2/3}(\mathbb{E}Z^4)^{1/3}$, a case of Hölder's inequality, give

$$m_n(o) \ge \mathbb{E}[Z_n(o)_+] = \frac{1}{2}\mathbb{E}|Z_n(o)| \ge \frac{(\mathbb{E}Z_n(o)^2)^{3/2}}{2(\mathbb{E}Z_n(o)^4)^{1/2}} \ge c\|g^{::}_{<n}(o,\cdot)\|_2 .$$

Next, the odometer depends little on each single mass. Changing $\sigma(v)$ by δ changes $\mathbb{E}_o[\mathsf{B}_T]$ by at most $g^{::}_{<n}(o,v)|\delta|$ for every stopping time $T \le n$, and two possible values of a mass differ by at most $2C_0$, so changing one mass changes $U_n^\sigma(o)$ by at most $2C_0 g^{::}_{<n}(o,v)$. The odometer $U_n^\sigma(o)$ depends only on the finitely many masses within distance n of o. Revealing them one at a time defines a martingale with increments bounded by $2C_0 g^{::}_{<n}(o,v)$, and orthogonality of its increments gives $\mathrm{Var}U_n^\sigma(o) \le 4C_0^2\|g^{::}_{<n}(o,\cdot)\|_2^2 \le Cm_n(o)^2$. Finally, since $\mathbb{E}[U_n^\sigma(o); U_n^\sigma(o) \ge m_n(o)/2] \ge m_n(o)/2$, the Cauchy–Schwarz inequality gives

$$\mathbb{P}\{U_n^\sigma(o) \ge m_n(o)/2\} \ge \frac{m_n(o)^2}{4\mathbb{E}[U_n^\sigma(o)^2]} \ge c .$$

Step 3: We conclude with the zero-one law. Let E be the event that $\limsup_n U_n^\sigma(o)/m_n(o) \ge 1/2$. By Step 2 and Fatou's lemma applied to the indicators of the complementary events, $\mathbb{P}(E) \ge c > 0$. Changing the masses at a finite set F of sites changes $U_n^\sigma(o)$ by at most $2C_0\sum_{v\in F} g^{::}_{<n}(o,v) \le 2C_0|F|M_n(o)$, where $M_n(o) = \max_v g^{::}_{<n}(o,v)$ as in Lemma 10.23. Since $M_n(o)^3 \le 4\|g^{::}_{<n}(o,\cdot)\|_2^2 \le Cm_n(o)^2$ by that lemma and Step 2, and $m_n(o) \to \infty$ by Step 1, this change of $U_n^\sigma(o)/m_n(o)$ tends to zero, and it does not affect whether E occurs. Hence E is a tail event of the independent masses, and Kolmogorov's zero-one law gives $\mathbb{P}(E) = 1$. On E, we have $U_n^\sigma(o) \ge m_n(o)/4$ for infinitely many n. Since $m_n(o) \to \infty$ and $u_\infty^\sigma(o) \ge U_n^\sigma(o)$, the odometer at o is infinite almost surely, and the sandpile explodes almost surely.

(c) Suppose that $\mu < 1$. Now the drift pushes the payoff down by at least $(1-\mu)/\deg_{\max}$ per step, and we show that the payoff is positive at time n only with probability that decays like $e^{-cn^{1/4}}$, so its expected maximum is finite. A few masses could outweigh the drift only if the walk spent a long time at a few sites, and Lemma 10.23 rules this out.

Step 1: We show that the walk rarely spends more than $n^{3/4}$ steps at one site. Let $\ell_n(v) := \sum_{k<n} \mathbf{1}\{X_k = v\}$ be the number of visits to v before time n. By Lemma 10.23, $\mathbb{E}_v \ell_n(v) \le 2 \deg_{\max} \sqrt{n}$. With $r := \lceil 4 \deg_{\max} \sqrt{n} \rceil$, Markov's inequality gives $\mathbb{P}_v\{\ell_n(v) \ge r\} \le 1/2$, and the strong Markov property at successive rth visits gives $\mathbb{P}_v\{\ell_n(v) \ge jr\} \le 2^{-j}$ for every integer $j \ge 1$. Let $\tau_v := \inf\{k \ge 0 : X_k = v\}$. The strong Markov property at τ_v gives

$$\mathbb{P}_x\{\ell_n(v) \ge jr\} \le \mathbb{P}_x\{\tau_v < n\}\, 2^{-j} \qquad \text{for every integer } j \ge 1 .$$

The sum of the hitting probabilities is at most n:

$$\sum_{v \in V} \mathbb{P}_x\{\tau_v < n\} = \mathbb{E}_x \big| \{X_0, \dots, X_{n-1}\} \big| \le n .$$

Thus the union bound gives $\mathbb{P}_x\{\max_{v \in V} \ell_n(v) \ge jr\} \le n 2^{-j}$ for $j \ge 1$. Taking $j = \lfloor n^{3/4}/r \rfloor$ and using $r \le (4 \deg_{\max} + 1)\sqrt{n}$, we obtain

$$\mathbb{P}_x\Big\{ \max_{v \in V} \ell_n(v) > n^{3/4} \Big\} \le 2n \cdot 2^{-n^{1/4}/(4 \deg_{\max} + 1)} .$$

If this choice gives $j = 0$, then the right-hand side exceeds one, so the same bound is immediate.

Step 2: We bound the probability that the payoff is positive at time n. If Y is centered with $|Y| \le C_0$ and λ is real, then convexity of $y \mapsto e^{\lambda y}$ on $[-C_0, C_0]$ gives $\mathbb{E} e^{\lambda Y} \le \cosh(\lambda C_0) \le e^{\lambda^2 C_0^2/2}$. Hence, for independent such variables Y_i, real weights a_i with $\sum_i a_i^2 > 0$, and $b > 0$, the exponential Markov inequality with $\lambda = b/(C_0^2 \sum_i a_i^2)$ gives

$$\mathbb{P}\Big\{ \sum_i a_i Y_i > b \Big\} \le \exp\Big(- \frac{b^2}{2 C_0^2 \sum_i a_i^2} \Big) .$$

Fix the path of the walk, and let $a_v := \ell_n(v)/\deg(v)$. Since $\sum_v \ell_n(v) = n$, we have

$$\mathsf{B}_n = \sum_v a_v(\sigma(v) - 1) \le \sum_v a_v(\sigma(v) - \mu) - \frac{1-\mu}{\deg_{\max}} n .$$

If $\max_v \ell_n(v) \le n^{3/4}$, then $\sum_v a_v^2 \le n^{3/4} \sum_v \ell_n(v) = n^{7/4}$, so the exponential inequality above with $Y_v := \sigma(v) - \mu$ shows that the conditional probability of $\{\mathsf{B}_n > 0\}$ given the path is at most $e^{-cn^{1/4}}$. Averaging over the walk and using Step 1, we obtain $\mathbb{E}\mathbb{P}_x\{\mathsf{B}_n > 0\} \le Cne^{-cn^{1/4}}$ for every $x \in V$ and $n \ge 1$.

Step 3: We bound the odometer. Each increment of the payoff is at most C_0 and $\mathsf{B}_0 = 0$, so the running maximum of the payoff increases only at times n with $\mathsf{B}_n > 0$, and then by at most C_0. Hence $\sup_n \mathsf{B}_n \le C_0 \sum_{n \ge 1} \mathbf{1}\{\mathsf{B}_n > 0\}$. By Corollary 10.19, $u_\infty^\sigma(x)$ is the supremum of $\mathbb{E}_x[\mathsf{B}_T]$ over bounded stopping

times T, and each of these is at most $\mathbb{E}_x[\sup_n \mathsf{B}_n]$. Taking expectations over the masses and using Tonelli's theorem and Step 2,

$$\sup_{x\in V} \mathbb{E}u^{\sigma}_{\infty}(x) \le C_0 \sup_{x\in V} \sum_{n\ge 1} \mathbb{E}\mathbb{P}_x\{\mathsf{B}_n > 0\} < \infty\,.$$

In particular, the odometer is finite almost surely, and the sandpile stabilizes almost surely. □

Part (c) of the proof bounds the expected odometer uniformly over the vertices. Under transitivity, the next section obtains the subcritical and supercritical conclusions assuming only that the initial mass is integrable.

10.5 RANDOM SANDPILES ON TRANSITIVE GRAPHS

On vertex-transitive graphs, conservation of mean mass determines stabilization away from mean 1, for every integrable invariant and ergodic law of the initial masses. At the critical mean, conservation alone does not decide; Theorem 10.22(b) covers bounded i.i.d. masses, and the Notes describe what is known for other laws. A graph $G = (V, E)$ is **vertex-transitive** when it has a group Γ of automorphisms such that, for all vertices x and y, there is $\varphi \in \Gamma$ with $\varphi x = y$. We call the law of σ **Γ-invariant** when

$$(\sigma(\varphi^{-1}v))_{v\in V} \overset{d}{=} (\sigma(v))_{v\in V} \qquad \text{for every } \varphi \in \Gamma\,.$$

We call it **Γ-ergodic** when every Γ-invariant event has probability 0 or 1.

Parallel toppling preserves the expected mass of every invariant integrable configuration.

Proposition 10.24 *Let $G = (V, E)$ be a locally finite connected graph, let Γ be a transitive group of automorphisms of G, and let $o \in V$. Let σ be an $\mathbb{R}^V$-valued random configuration whose law $\mathbb{P}$ is Γ-invariant and satisfies $\mathbb{E}|\sigma(o)| < \infty$. Let $(\sigma^U_k)_{k\ge 0}$ and $(U^{\sigma}_k)_{k\ge 0}$ be the configurations and odometers under parallel toppling. Then the following statements hold.*

(a) For every $k \ge 0$, the value $\mathbb{E}U^{\sigma}_k(v)$ does not depend on $v \in V$.

(b) For every $k \ge 0$, we have $\mathbb{E}\sigma^U_k(o) = \mathbb{E}\sigma(o)$.

(c) If $\mathbb{P}\{\sigma \text{ stabilizes}\} = 1$, then $\mathbb{E}\sigma_{\infty}(o) = \mathbb{E}\sigma(o)$.

(d) If $\mathbb{E}\sigma(o) = 1$ and $\mathbb{P}\{\sigma \text{ stabilizes}\} = 1$, then almost surely $\sigma_{\infty}(v) = 1$ for every $v \in V$ and $\Delta u^{\sigma}_{\infty} = 1 - \sigma$.

Proof. Transitivity makes G regular, so the parallel update (10.4) reads

$$\sigma^U_{k+1}(v) = \min\{\sigma^U_k(v), 1\} + \frac{1}{\deg(o)} \sum_{y\sim v} (\sigma^U_k(y) - 1)_+ .$$

The right side is nondecreasing in each of the values $\sigma^U_k(y)$, so parallel toppling preserves the pointwise order of configurations, and it maps the constant configuration M to itself whenever $M \geq 1$. Parallel toppling also commutes with every graph automorphism, so for each k the law of σ^U_k is Γ-invariant, and by transitivity the law of $\sigma^U_k(v)$ does not depend on v. Since $|\min\{x, 1\}| \leq |x| + 1$ and $(x-1)_+ \leq |x| + 1$, induction on k shows that every $\sigma^U_k(v)$ is integrable.

(a) Since $U^\sigma_k(v) = \frac{1}{\deg(o)} \sum_{j=0}^{k-1} (\sigma^U_j(v) - 1)_+$ is integrable and its law does not depend on v, neither does its expectation.

(b) Taking expectations in the update at $v = o$, and using that each of the $\deg(o)$ neighbors y of o has $\mathbb{E}(\sigma^U_k(y) - 1)_+ = \mathbb{E}(\sigma^U_k(o) - 1)_+$, gives

$$\mathbb{E}\sigma^U_{k+1}(o) = \mathbb{E}\min\{\sigma^U_k(o), 1\} + \mathbb{E}(\sigma^U_k(o) - 1)_+ = \mathbb{E}\sigma^U_k(o) .$$

Induction on k proves $\mathbb{E}\sigma^U_k(o) = \mathbb{E}\sigma(o)$.

(c) Suppose $\mathbb{P}\{\sigma \text{ stabilizes}\} = 1$, so that $\sigma^U_k(o) \to \sigma_\infty(o) \leq 1$ almost surely. The idea is to compare σ with $\min\{\sigma, M\}$, whose parallel evolution stays below M. Fix $M \geq 1$, and let $(\eta^U_k)_{k\geq 0}$ be the configurations under parallel toppling started from $\eta := \min\{\sigma, M\}$. Since $\eta \leq \sigma$ and $\eta \leq M$, preservation of order gives $\eta^U_k \leq \sigma^U_k$ and $\eta^U_k \leq M$, that is, $\eta^U_k \leq \min\{\sigma^U_k, M\}$. The law of η is Γ-invariant and $\mathbb{E}|\eta(o)| < \infty$, so part (b), applied to η and to σ, gives

$$\mathbb{E}\min\{\sigma(o), M\} = \mathbb{E}\eta^U_k(o) \leq \mathbb{E}\min\{\sigma^U_k(o), M\} \leq \mathbb{E}\sigma^U_k(o) = \mathbb{E}\sigma(o) .$$

By (10.4) and Example 10.11, $\sigma^U_k \geq \min\{\sigma, 1\}$, so $\min\{\sigma^U_k(o), M\}$ lies between the integrable variable $\min\{\sigma(o), 1\}$ and M, and it tends to $\sigma_\infty(o)$ because $\sigma_\infty(o) \leq 1 \leq M$. By the dominated convergence theorem, letting $k \to \infty$ gives $\mathbb{E}\min\{\sigma(o), M\} \leq \mathbb{E}\sigma_\infty(o) \leq \mathbb{E}\sigma(o)$. Since $|\min\{\sigma(o), M\}| \leq |\sigma(o)|$, the dominated convergence theorem again shows that the left side tends to $\mathbb{E}\sigma(o)$ as $M \to \infty$. Hence $\mathbb{E}\sigma_\infty(o) = \mathbb{E}\sigma(o)$.

(d) By part (c), $\mathbb{E}\sigma_\infty(o) = 1$. Since $\sigma_\infty(o) \leq 1$, we have $\sigma_\infty(o) = 1$ almost surely. Invariance gives the same conclusion at every vertex. Intersecting these countably many probability-one events yields $\sigma_\infty \equiv 1$, and therefore $\Delta u^\sigma_\infty = 1 - \sigma$. □

Conservation and ergodicity now give the phase transition away from the critical mean.

Proposition 10.25 *Let $G = (V, E)$ be a locally finite connected graph, let Γ be a transitive group of automorphisms of G, and let $o \in V$. Let σ be an $\mathbb{R}^V$-valued random configuration whose law $\mathbb{P}$ is Γ-invariant and Γ-ergodic and satisfies $\mathbb{E}|\sigma(o)| < \infty$. Then the following statements hold.*

(a) The probability that σ stabilizes is 0 or 1.

(b) If $\mathbb{E}\sigma(o) > 1$, then $\mathbb{P}\{\sigma \text{ stabilizes}\} = 0$.

(c) If $\mathbb{E}\sigma(o) < 1$, then $\mathbb{P}\{\sigma \text{ stabilizes}\} = 1$.

Proof. (a) The event that σ stabilizes is Γ-invariant, so it has probability 0 or 1 by ergodicity.

(b) If σ stabilized almost surely, then Proposition 10.24 would give $\mathbb{E}\sigma_\infty(o) = \mathbb{E}\sigma(o)$. Since $\sigma_\infty(o) \le 1$, this would imply $\mathbb{E}\sigma(o) \le 1$. Part (a) therefore forces the stabilization probability to be 0.

(c) By part (a), it suffices to prove the contrapositive. Suppose $\mathbb{P}\{\sigma \text{ stabilizes}\} = 0$, and use parallel toppling. Lemma 10.15 gives $U_n^\sigma(o) \uparrow \infty$, so $U_n^\sigma(o) > 0$ for all sufficiently large n. Once o has emitted mass, its mass remains at least 1, by (10.4). Therefore

$$\liminf_{n\to\infty} \sigma_n^U(o) \ge 1 \qquad \text{almost surely}\,.$$

Moreover, $\sigma_n^U(o) \ge \sigma(o) \wedge 1$ by the same inequality. Applying Fatou's lemma to $\sigma_n^U(o) - (\sigma(o) \wedge 1)$ and using Proposition 10.24(b), we obtain

$$1 \le \mathbb{E}\Big[\liminf_{n\to\infty} \sigma_n^U(o)\Big] \le \mathbb{E}\sigma(o)\,.$$

This proves the contrapositive. □

Remark. Parts (b) and (c) need only one-sided integrability. Suppose the law of σ is Γ-invariant and Γ-ergodic. If $\mathbb{E}[\sigma(o)_-] < \infty$ and $\mathbb{E}\sigma(o) \in (1, \infty]$, then apply (b) to $\sigma \wedge M$. If $\mathbb{E}[\sigma(o)_+] < \infty$ and $\mathbb{E}\sigma(o) \in [-\infty, 1)$, then apply (c) to $\sigma \vee (-M)$. The truncation is integrable, and by monotone convergence its mean tends to $\mathbb{E}\sigma(o)$, so for large M it lies on the same side of 1. Truncating coordinatewise preserves invariance and ergodicity. Since $\sigma \wedge M \le \sigma \le \sigma \vee (-M)$, Lemma 10.10 transfers both conclusions to σ. Exercise 10.19 shows that Proposition 10.24 can fail without transitivity.

EXERCISES

Exercise 10.1. Verify the summation identities used in the proof of Theorem 10.1. Let $G = (V, E)$ be a locally finite graph.

(a) Suppose $f : V \to \mathbb{R}$ has finite support. Show that

$$\sum_{v \in V} \Delta f(v) = 0 \,.$$

(b) Let $f, h : V \to \mathbb{R}$, and suppose at least one of them has finite support. Show that both sums below converge absolutely and that

$$\sum_{v \in V} (\Delta f)(v) h(v) = \sum_{v \in V} f(v)(\Delta h)(v) \,.$$

Hint: Pair the two orientations of each edge.

Exercise 10.2. Prove a toppling-by-toppling version of the quadrature inequality in Theorem 10.1. This extends Exercise 4.2 from $\mathbb{Z}^d$ to graphs and to functions superharmonic only where the odometer is positive. Let $G = (V, E)$ be a locally finite connected graph, and let $\sigma : V \to \mathbb{R}$ have finite support. Recall from Section 10.1 that $\langle \sigma, h \rangle = \sum_{v \in V} \sigma(v) h(v)$.

(a) Let $x \in V$, and let $h : V \to \mathbb{R}$ satisfy $\Delta h(x) \le 0$. Show that if σ_1 is obtained from σ by toppling at x, then

$$\langle \sigma_1, h \rangle \le \langle \sigma, h \rangle \,.$$

(b) Deduce that if σ stabilizes with odometer u_∞ and final configuration σ_∞, then every $h : V \to \mathbb{R}$ satisfying $\Delta h \le 0$ on $\{u_\infty > 0\}$ obeys

$$\langle \sigma_\infty, h \rangle \le \langle \sigma, h \rangle \,.$$

Exercise 10.3. Use the least action principle, Theorem 4.3, to characterize stabilization on a finite graph. Let $G = (V, E)$ be a finite connected graph with at least two vertices, and let $\sigma : V \to \mathbb{R}$. Show that σ is stabilizable if and only if

$$\sum_{v \in V} \sigma(v) \le |V| \,.$$

Hint: For the converse, solve $\Delta f = 1 - \sigma - c$, where $c := 1 - |V|^{-1} \sum_v \sigma(v)$, and add a constant to make $f \ge 0$.

Exercise 10.4. Modify Definition 4.2 by allowing the threshold to vary between vertices. Let $G = (V, E)$ be a locally finite connected graph, and let $\sigma, \theta : V \to \mathbb{R}$. Modify the divisible sandpile so that each vertex v has **threshold** $\theta(v)$ and distributes its excess above $\theta(v)$ equally among its neighbors. Show that σ stabilizes in this model if and only if $\sigma - \theta + 1$ stabilizes in the model with threshold 1.

Exercise 10.5. Explore the divergent alternative in Lemma 4.1. Let $G = (V, E)$ be a locally finite connected graph.

(a) Suppose $\sigma : V \to \mathbb{R}$ does not stabilize, but its configurations σ_k of (4.1) converge pointwise along a thorough toppling sequence. Prove that the limit is identically 1.

(b) Show that there is a toppling sequence as in part (a) when $G = \mathbb{Z}$ and

$$\sigma(v) = \begin{cases} 1, & v \neq \mathbf{0}, \\ 2, & v = \mathbf{0}. \end{cases}$$

Hint: Use the bipartition of $\mathbb{Z}$ to simulate parallel toppling by a thorough toppling sequence.

(c) Describe a thorough toppling sequence for the initial configuration σ in part (b) such that $\max_{v\in\mathbb{Z}}(\sigma_k(v) - 1) = O(k^{-\alpha})$ for a constant $\alpha > 0$.

(d) For the same initial configuration, find a thorough toppling sequence for which σ_k does not converge.

Exercise 10.6. Use the least action principle, Theorem 4.3, to study the constant configuration 2 on infinite graphs.

(a) Let G be an infinite connected graph of bounded degree. Show that the initial configuration $\sigma \equiv 2$ does not stabilize on G.

(b) Show that the bounded-degree hypothesis cannot be removed by exhibiting an infinite locally finite tree on which $\sigma \equiv 2$ stabilizes.

(c) Let G be an infinite locally finite connected graph. Show that $\sigma \equiv 2$ does not stabilize whenever, along every infinite path in G, the sum of $\deg(v)^{-1}$ diverges.

Exercise 10.7. Derive a normalization of the odometer from Theorem 4.3. Let $G = (V, E)$ be a locally finite connected graph. Prove that if $\sigma : V \to \mathbb{R}$ stabilizes with odometer u^σ_∞, then $\inf_{x\in V} u^\sigma_\infty(x) = 0$. Is the infimum necessarily attained?

Exercise 10.8. Let $G = (V, E)$ be a locally finite connected graph, let $\sigma_0 : V \to \mathbb{R}$, and let $(v_i)_{i\geq 1}$ be a toppling sequence as in (4.1). Suppose the set V_* of vertices that occur infinitely often is finite and is not all of V.

(a) Show that the resulting odometer u_∞ is finite on V. Prove the same conclusion for a sequence of parallel rounds in which the set of vertices toppled in infinitely many rounds is V_*.

(b) Suppose only vertices in V_* are toppled. Show that u_∞ does not depend on their order and agrees with the odometer from parallel toppling on V_*.

Exercise 10.9. Let U be a toppling procedure on a locally finite connected graph $G = (V, E)$. Show that Definition 10.4 implies that $t \mapsto U_t^\sigma(v)$ is a pure-jump function for every $\sigma \in \mathbb{R}^V$ and $v \in V$. Thus there are a countable set $J \subset (0, \infty)$ of jump times and positive jump sizes $(a_s)_{s \in J}$ such that

$$U_t^\sigma(v) = \sum_{s \in J} a_s \mathbf{1}\{s \le t\} \qquad \text{for } t > 0\,.$$

Exercise 10.10. The goal is to relax the bounded-degree assumption in Example 10.13. Let $G = (V, E)$ be a locally finite connected graph, and equip every vertex with an independent rate-1 Poisson clock. Suppose that there are a vertex $o \in V$ and a constant $C_o < \infty$ such that the number of neighbor-or-stay walks $o = v_0, v_1, \dots, v_n$ is at most $C_o^n n!$ for every $n \ge 1$. Verify Definition 10.4(iii) for asynchronous toppling.

Hint: Fix one vertex o and transfer its walk count to an arbitrary vertex by prepending a fixed path from o; then the argument of Example 10.13 works on every time interval of a fixed length less than $1/C_o$. An infinite decreasing sequence of ring times has a limit, so one of its tails lies in such an interval.

Exercise 10.11. Let $G = (V, E)$ be a directed graph in which every vertex x has finite in-degree and finite positive out-degree $\deg(x)$. Extend the divisible sandpile and Proposition 10.9 to G.

Hint: Use

$$\Delta f(x) := \sum_{\substack{y \in V \\ (y,x) \in E}} f(y) - \deg(x)\, f(x)\,,$$

and let a toppling send equal mass along each outgoing edge. In Definition 10.4(iii), follow incoming edges backward.

Exercise 10.12. Extend Proposition 10.16 from one excess mass to finitely many excess masses. Let $G = (V, E)$ be an infinite locally finite connected graph and let $\sigma : V \to \mathbb{R}$. Suppose $\sigma \ge 1$ and $\sigma - 1$ has finite nonempty support. Show that σ stabilizes if and only if simple random walk on G is transient.

Exercise 10.13. Identify an optimal stopping time in Theorem 10.18. Let $G = (V, E)$ be a locally finite connected graph, let $\sigma : V \to \mathbb{R}$, let $x \in V$, and let $n \ge 0$ be an integer. Let $(X_k)_{k \ge 0}$ be simple random walk started at x. Let U_n^σ be the odometer after n rounds of parallel toppling. For $0 \le k \le n$, let

$$\mathsf{B}_k := \sum_{j=0}^{k-1} \frac{\sigma(X_j) - 1}{\deg(X_j)}\,.$$

Show that $\max_{T \le n} \mathbb{E}_x[\mathsf{B}_T]$ is attained by

$$T_n^* := \min\{0 \le k \le n : U_{n-k}^\sigma(X_k) = 0\}\,.$$

Here the maximum is over stopping times for the natural filtration of the walk that are bounded by n.

Exercise 10.14. Use killed Green functions to prove that stabilization implies transience in Proposition 10.16. Let $G = (V, E)$ be an infinite locally finite connected graph, let $o \in V$ and $\beta > 0$, and suppose that $\sigma = 1 + \beta\delta_o$ stabilizes. Prove that simple random walk on G is transient.

Hint: Compare $\beta^{-1}u_\infty^\sigma$ with the Green function killed on leaving a finite graph-distance ball about o.

Exercise 10.15. This exercise extends the Green-function construction of Section 10.3 to a walk stopped when it hits a single vertex. Let $G = (V, E)$ be a locally finite connected recurrent graph, and let $y, z \in V$ be distinct. Let $(X_j)_{j\ge 0}$ be simple random walk, and write $\mathbb{P}_x$ for its law when $X_0 = x$. For $v \in \{y, z\}$, write τ_v for the hitting time of v and τ_v^+ for its first positive hitting time. Let

$$f(x) := \frac{\mathbb{P}_x\{\tau_y < \tau_z\}}{\deg(y)\mathbb{P}_y\{\tau_z < \tau_y^+\}}\,.$$

(a) Show that

$$f(x) = \frac{1}{\deg(y)} \sum_{j=0}^{\infty} \mathbb{P}_x\{X_j = y \text{ and } \tau_z > j\}\,.$$

Thus $f(x)$ is the normalized expected occupation time at y before hitting z.

(b) Show that $\Delta f = \delta_z - \delta_y$.

Hint for (b): Use a first-step decomposition and reverse paths between successive visits to $\{y, z\}$.

Exercise 10.16. Prove the abelian-sandpile analogue of Proposition 10.24(b). Let $d \ge 1$, and let $(\sigma_n)_{n\ge 0}$ be the configurations produced by parallel toppling (9.6) for the abelian sandpile on $\mathbb{Z}^d$. Suppose that the law of σ_0 is translation invariant and $\mathbb{E}|\sigma_0(\mathbf{0})| < \infty$. Show that

$$\mathbb{E}\sigma_n(\mathbf{0}) = \mathbb{E}\sigma_0(\mathbf{0}) \qquad \text{for } n \ge 0\,.$$

Exercise 10.17. The proof of Proposition 10.24(c) uses dominated convergence. The following exercise records a related uniform-integrability fact. Let $(Z_n)_{n\ge 1}$ be identically distributed random variables with $\mathbb{E}|Z_1| < \infty$. Show that $(Z_n)_{n\ge 1}$ is uniformly integrable. Deduce that for every $\epsilon > 0$ there is $\delta > 0$ such that $\sup_n \mathbb{E}[|Z_n|\mathbf{1}_A] < \epsilon$ whenever $\mathbb{P}\{A\} < \delta$.

Exercise 10.18. Verify the form of Fatou's lemma used in the proof of Proposition 10.25(c). Let $(Z_n)_{n\ge1}$ be random variables, and let Y be integrable. Show that if $Z_n \ge Y$ for every n, then

$$\mathbb{E}\left[\liminf_{n\to\infty} Z_n\right] \le \liminf_{n\to\infty} \mathbb{E}Z_n\,.$$

Exercise 10.19. The **comb** C has vertex set $\mathbb{Z}^2$. For $x, y \in \mathbb{Z}$, its edges are

$$(x,0) \sim (x\pm1,0) \qquad \text{and} \qquad (x,y) \sim (x,y\pm1)\,.$$

The **spine** vertices $(x,0)$ have degree 4, while the **tooth** vertices (x,y) with $y \ne 0$ have degree 2. This exercise shows that Proposition 10.24 can fail without transitivity.

Let $(\sigma(v))_{v\in\mathbb{Z}^2}$ be i.i.d. real-valued random variables with finite mean $\mathbb{E}\sigma(v) = \mu$ and

$$q := \mathbb{E}[(\sigma(v)-1)_+] > 0\,.$$

Apply one round of parallel toppling to obtain $\sigma_1^U = \sigma + \Delta U_1^\sigma$, where $U_1^\sigma(v) = (\sigma(v)-1)_+/\deg(v)$.

(a) Show that, for every $x \in \mathbb{Z}$,

$$\mathbb{E}\sigma_1^U(x,0) = \mu + \frac{q}{2} \qquad \text{and} \qquad \mathbb{E}\sigma_1^U(x,\pm1) = \mu - \frac{q}{4}\,,$$

and $\mathbb{E}\sigma_1^U(x,y) = \mu$ for every $x \in \mathbb{Z}$ and $|y| \ge 2$.

(b) Deduce that the identity $\mathbb{E}\sigma_1^U(v) = \mathbb{E}\sigma(v)$ does not hold for every $v \in \mathbb{Z}^2$, and neither does $\mathbb{E}[\sigma_1^U(v)/\deg(v)] = \mathbb{E}[\sigma(v)/\deg(v)]$ for every $v \in \mathbb{Z}^2$.

NOTES

The divisible sandpile was introduced by Levine and Peres (2009), who proved the bounded-width point-source shape theorem of Chapter 4 (Levine and Peres, 2009, Theorem 1.3); the Notes to that chapter give the sources of the least action principle and of the description in Section 4.1 of the odometer as a least superharmonic majorant minus an obstacle. The multiple-source scaling limit is Theorem 1.4 of Levine and Peres (2010); its planar case with two equal sources is Theorem 7.1, and the limit is a quadrature domain (Levine and Peres, 2010, Proposition 6.6); Theorems 1.2–1.4 there also cover rotor-router aggregation and internal DLA, and Theorem 3.9 allows real-valued densities for the divisible sandpile. For an overview, see (Levine and Peres, 2017, Sections 2 and 5).

The toppling procedures in Definition 10.4 are adapted from the infinite-volume abelian-sandpile framework of Fey, Meester, and Redig (2009, Definition 2.1). Proposition 10.9 extends Proposition 2.5 of Levine, Murugan, Peres, and Ugurcan (2016a), which assumes well-ordered toppling times; the proof of part (i) adapts the backward-chain argument for the abelian property in (Fey et al., 2009, Theorem 2.8). The supremum formula after Lemma 10.15 and Example 10.14 are Lemma 2.7 and Example 3 of Levine, Murugan, Peres, and Ugurcan (2016a). Fey, Meester, and Redig (2009, Lemma 2.10) proved conservation of density for translation-invariant abelian sandpiles on $\mathbb{Z}^d$ by an ergodic argument, and Levine, Murugan, Peres, and Ugurcan (2016a, Proposition 3.1 and Lemma 4.3) gave an elementary proof of Proposition 10.24. Lemma 10.17 is (Bou-Rabee et al., 2026, Lemma 3.1), which identifies parallel toppling as a projected Jacobi iteration for the discrete obstacle problem (Bou-Rabee et al., 2026, Section 1.2).

The random-walk representation in Theorem 10.18, Proposition 10.20, the variational formula in Theorem 10.21 and Corollary 10.19 are Theorem 3.2, Proposition 3.3, Theorem 3.4 and Corollary 3.5 of Bou-Rabee, Peres, and Sava-Huss (2026, pp. 11–12). Theorem 10.22 is the case of bounded masses of Theorems 1.2 and 1.3(i) there (Bou-Rabee et al., 2026, pp. 3–4). Part (a) is their Proposition 5.2, which needs only a mean in $(1, \infty]$. In part (b) they treat infinite and finite $\sum_v g^{\cdot\cdot}(o, v)^2$ separately (Bou-Rabee et al., 2026, Propositions 6.4(b) and 6.6); Lemma 10.23 and Kolmogorov's zero-one law avoid this split and their general zero-one law (Bou-Rabee et al., 2026, Proposition 4.1). Their symmetric case reduces to bounded laws by a convexity argument (Bou-Rabee et al., 2026, Proposition 6.7).

Boundedness of the masses is far from necessary in Theorem 10.22(c): Bou-Rabee, Peres, and Sava-Huss (2026, Theorem 1.3(i), pp. 3–4) proved that if the mean lies in $[-\infty, 1)$ and the positive part of $\sigma(v)$ has a finite pth moment for some $p > 3$, then the qth moment of the odometer is bounded uniformly in the vertex for every $1 \le q < (p-1)/2$. Their proof (Bou-Rabee et al., 2026, Proposition 7.6, p. 24) uses the Fuk–Nagaev inequality (Bou-Rabee et al., 2026, Lemma 7.3, p. 23), of which inequality (1.7) of Rio (2017) gives a form attributed there to Fuk and to Nagaev, and moment bounds for local times (Bou-Rabee et al., 2026, Lemma 7.2, p. 22) from the on-diagonal bound $\sup_x \mathbb{P}_x\{X_n = x\} \le Cn^{-1/2}$ (Bou-Rabee et al., 2026, p. 4), whose summed form is Lemma 10.23. The exponent 3 cannot be lowered: for every $p < 3$, Bou-Rabee, Peres, and Sava-Huss (2026, Theorem 8.9, pp. 33–34) constructed a transient bounded-degree graph on which, for every $\mu < 1$, i.i.d. masses bounded below, with mean μ and a finite centered pth moment, explode almost surely, and Bou-Rabee, Peres, and Sava-Huss (2026, Theorem 8.12, pp. 34–38) constructed, for every $d_f \in (\max\{p, 2\}, 3)$, a recurrent bounded-degree graph whose balls of radius r have at most a

constant times r^{d_f} vertices, on which such masses explode, whereas on such graphs finite pth moments of the positive parts with $p > d_f$ suffice for stabilization below mean 1 (Bou-Rabee et al., 2026, Theorem 1.3(ii), p. 4). They left open whether i.i.d. masses of mean below 1 stabilize on every bounded-degree graph for $p = 3$ (Bou-Rabee et al., 2026, Remark 8.11, p. 34).

The stabilization criterion for one excess mass (Proposition 10.16) is (Levine et al., 2016a, Lemma 2.9). For integrable initial masses on an infinite vertex-transitive graph, with law invariant and ergodic under a transitive group of automorphisms, Levine, Murugan, Peres, and Ugurcan (2016a) proved explosion above mean 1 and stabilization below mean 1 (Levine et al., 2016a, Lemmas 4.1 and 4.2), parts (b) and (c) of Proposition 10.25. On such graphs, for i.i.d. masses of mean 1, they proved explosion when the variance is positive and finite (Levine et al., 2016a, Theorem 1.1) and, on recurrent graphs, for every nonconstant law (Levine et al., 2016a, Lemma 4.4); the comment after that lemma extends this to invariant ergodic laws whose single-site conditional laws are not point masses. Some such condition is needed even for bounded, finitely dependent masses: if $(\eta(x))_{x\in\mathbb{Z}^d}$ are i.i.d. Bernoulli$(1/2)$ variables, then $\sigma := 1 - \Delta(\eta/(2d))$ is stationary, ergodic and finitely dependent, with values in $[0, 2]$, mean 1 and positive variance, yet σ stabilizes. This is a variant of (Bou-Rabee et al., 2026, Example 8.6, p. 30), where η is uniform on $[0, 1]$ and $\sigma = 1 - \Delta\eta$. On a finite connected graph, Levine, Murugan, Peres, and Ugurcan (2016a) computed the exact law of the odometer of the critical configuration

$$\sigma(v) = 1 + \xi(v) - \frac{1}{|V|}\sum_{w\in V}\xi(w)\,,$$

where the $\xi(v)$ are i.i.d. standard Gaussians. It stabilizes to the constant 1, and its odometer is a discrete bi-Laplacian Gaussian field shifted to have minimum 0 (Levine et al., 2016a, Proposition 1.3).

The preprint of Bou-Rabee and Panagiotis (2026) measured how fast the sandpile on $\mathbb{Z}^d$ explodes at mean 1. For i.i.d. masses of mean 1 with positive finite variance and an exponential moment, under parallel toppling the expected mass that the origin sends to each neighbor during the first t rounds is of order $t^{(4-d)/4}$ for $d \le 3$ and $\log t$ for $d = 4$, and at least a constant times $(\log t)^{2/d}$ for $d \ge 5$. For $d \ge 5$ the order depends on the lower tail of the masses: if $-\log \mathbb{P}\{\sigma(\mathbf{0}) \le 1 - s\}$ is of order s^ν as $s \to \infty$ for some $\nu \ge 1$ other than $d/2$, then the expected mass is of order $(\log t)^{1/\min\{\nu, d/2\}}$ (Bou-Rabee and Panagiotis, 2026, Theorem 1.3, pp. 5–6). The same paper showed that, for $d \ge 2$, the set of vertices that topple can contain an infinite connected component below mean 1: if the masses are i.i.d. with a law that

depends on their mean and, uniformly over the means in an interval with right endpoint 1, the variance is at least a positive constant and a fixed exponential moment of the deviation from the mean stays bounded, then at every mean sufficiently close to 1 from below the set of vertices that topple contains an infinite connected component almost surely (Bou-Rabee and Panagiotis, 2026, Theorem 1.1, p. 3). This answers, for the divisible sandpile, a variant of a question that Fey, Meester, and Redig (2009, Section 5) raised for the abelian sandpile.

The variance bound in Step 2 of part (b) of the proof of Theorem 10.22 is a bounded-differences form of the Efron–Stein inequality. This is inequality (2.1), p. 754, of Steele (1986), for arbitrary functions of i.i.d. random variables; Efron and Stein (1981, inequality (1.6), p. 587) had proved a form of it for symmetric functions. Exercise 13.12 of Levin and Peres (2017), p. 199, derives it for independent variables with values in a finite set from the spectral gap of the chain that resamples one coordinate at a time.

Aharonov and Shapiro (1976, Sections 1.1–1.2, pp. 39–40) studied domains on which the integrals of analytic functions satisfy quadrature identities, and Sakai (1982, Chapters I–II) constructed quadrature domains for subharmonic functions, for which the identity becomes an inequality as in Theorem 10.1, and proved their existence and uniqueness; Gustafsson and Shapiro (2005, Sections 1, 2 and 5) surveyed both notions. Aleksanyan and Shahgholian (2019) studied a perturbed divisible sandpile started from mass n at the origin of $\mathbb{Z}^d$ with $d \geq 2$. A site with mass above a threshold m topples its excess above m, and a site with positive mass at most m topples all of its mass once its odometer exceeds $n^{2/d}/(2dm)$. (Their odometer counts the total mass emitted, which is $2d$ times ours, so their threshold is $n^{2/d}/m$.) They proved that, for a suitable m growing slowly with n, the rescaled odometers converge to a spherically symmetric limit (Aleksanyan and Shahgholian, 2019, Theorem 4.3).

For the adjusted configuration above with i.i.d. weights of mean 0 and variance 1 in place of Gaussians, Cipriani, Hazra, and Ruszel (2018b) proved on discrete tori that the rescaled odometer, viewed modulo additive constants, converges in law in a negative Sobolev space to the bi-Laplacian Gaussian field (Cipriani et al., 2018b, Theorems 1–2). Cipriani, Hazra, and Ruszel (2018a) proved that i.i.d. masses of mean 1 whose excesses are symmetric and in the domain of normal attraction of a stable law explode on infinite vertex-transitive graphs, under conditions on sums of powers of the Green function (Cipriani et al., 2018a, Theorems 2–3), and they found the scaling limit of the odometer for such weights in the adjusted configuration on discrete tori (Cipriani et al., 2018a, Theorem 5). For the adjusted configuration with i.i.d. weights of variance 1, redistributed on discrete tori by a long-range random walk, Chiarini, Jara, and Ruszel (2021) proved that the

rescaled odometer converges in law to a fractional Gaussian field, with a logarithmic factor in the normalization when the jump law has tail exponent 2 (Chiarini et al., 2021, Theorem 3.4). In dimension two and for $1 < \alpha < 2$, Frómeta and Jara (2018) obtained the scaling limit of a divisible sandpile in which mass is redistributed by a truncated α-stable random walk with a nearest-neighbor correction, and identified the limiting odometer through an obstacle problem for a truncated fractional Laplacian (Frómeta and Jara, 2018, Theorems 1.2–1.3). Cipriani, de Graaff, and Ruszel (2020) replaced the i.i.d. weights by correlated centered Gaussian weights whose covariance on each torus is translation invariant, with positive Fourier transforms that are uniformly bounded and converge pointwise to a positive limit. They proved that the rescaled odometer on discrete tori, viewed modulo additive constants, converges in law to a Gaussian field determined by this limit (Cipriani et al., 2020, Theorem 1); suitable covariances give fractional Gaussian fields smoother than the bi-Laplacian field (Cipriani et al., 2020, Proposition 14).

For a uniformly bounded stationary ergodic initial configuration on lattice approximations of a bounded Lipschitz domain, Bou-Rabee (2021) proved that, almost surely, the sup-norm distance between the rescaled odometer and that of the constant configuration with the same mean tends to 0 as the lattice spacing tends to 0 (Bou-Rabee, 2021, Theorem 8.2). Huss and Sava-Huss (2019) proved that on the doubly infinite Sierpinski gasket graph, for the divisible sandpile started from mass m at the origin, the set of vertices that topple lies between two consecutive graph-metric balls centered at the origin (Huss and Sava-Huss, 2019, Theorem 1.1), and computed the odometer exactly when $m = 3^{k+1}$ (Huss and Sava-Huss, 2019, Theorem 5.12). On the two-dimensional comb, Huss and Sava (2012) proved that the divisible sandpile started from mass n at the origin is close to an explicit anisotropic shape, and used it to obtain inner bounds for internal DLA and rotor-router aggregation (Huss and Sava, 2012, Theorems 3.5, 4.2 and 5.3).

For surveys, see Járai (2018, Sections 2–6) on abelian sandpiles and Ruszel (2020) on divisible-sandpile odometers, obstacle problems, and internal DLA (Ruszel, 2020, Sections 2.2, 4, and 6).

Chapter Eleven

The Rotor Model

To visualize a rotor walk, label each vertex of a graph with an arrow (a **rotor**) pointing toward one of its neighbors. A walker standing at a vertex first advances that vertex's rotor to the next neighbor in a prescribed cyclic order, and then steps to the neighbor the rotor now points at. Each vertex therefore serves its neighbors in rotation, and after every finite number of departures the numbers of particles sent along any two outgoing edges differ by at most one.

Compare this with toppling, where a vertex also sends grains equally along its outgoing edges, but only after waiting to become unstable. Rotor-routing achieves the same balance without the wait, at the cost of remembering where each vertex last sent a particle. Rotor walk arose, under the name Eulerian walkers, as a model of self-organized criticality. It was also found independently as a model of mobile agents exploring a territory and as a way of balancing loads among processors, and it was proposed as a derandomization of random walk.

Like the sandpile, rotor walk has the abelian property: when finitely many particles are routed until they reach an absorbing set, neither the final rotors nor the number of departures from each vertex depends on the order in which the particles are moved. On a finite graph with a sink, this lets the sandpile group act on the rotor configurations that form spanning trees oriented toward the sink. The action is free and transitive, so there are exactly as many such trees as group elements, and this proves the matrix tree theorem (Section 11.2).

On an infinite graph, the first question is whether rotor walk keeps returning to its starting point. Unlike random walk, the answer depends on the initial rotors, and Section 11.3 shows how to choose them so that the walk is recurrent on $\mathbb{Z}^d$ in every dimension. When each vertex has as many incoming as outgoing edges, the walk splits into circuits (the stretches between successive full turns of the rotor at the starting vertex), and no circuit uses an edge twice. This alone forces rotor walk on $\mathbb{Z}^d$ to visit at least order $t^{d/(d+1)}$ sites in its first t steps, whatever the initial rotors (Section 11.5). With independent uniform initial rotors, rotor walk is recurrent on the Manhattan and F-lattices, by a coupling with critical percolation in which closed cycles force the walk back to its start (Section 11.6). On a transient graph, rotor walk escapes to infinity at most as often as random walk does (Section 11.7).

The last three sections read the walk off its initial rotors. A path is live if each of its internal vertices sends the walk onward before sending it back. If no live path is infinite, then every circuit is complete and the walk is recurrent (Section 11.8). With independent uniform initial rotors, this criterion applies to every graph of maximum degree three and to $\mathbb{Z}^2$ with the clockwise mechanism (Section 11.9). On the comb in $\mathbb{Z}^d$ for $d \geq 2$, a finite tree traversal identifies the circuit ranges. With independent uniform initial rotors and suitable cyclic orders, these ranges are close to diamonds, and after t steps the walk has visited order $t^{d/(d+1)}$ sites, the smallest order that the bound of Section 11.5 allows (Section 11.10).

Chapter 6 used rotor walk on $\mathbb{Z}^d$ to grow a nearly spherical aggregate (Figure 6.1), releasing particles one at a time from the origin and letting each stop at the first unoccupied site. Here we study rotor walk itself, on general digraphs.

11.1 ROTOR WALK AND THE ABELIAN PROPERTY

We extend the rotor walk of Chapter 6 from $\mathbb{Z}^d$ to digraphs, prove that it is abelian, and compare it with random walk by counting the complete turns of each rotor. Throughout this chapter, digraphs are **simple**: they have no loops, and at most one edge joins each ordered pair of vertices.

Let $G = (V, E)$ be a digraph with finite out-degree $\deg(v)$ at every vertex v. As on $\mathbb{Z}^d$, a rotor configuration assigns to each vertex v of positive out-degree an outgoing edge $\rho(v)$, and a rotor mechanism assigns to each such vertex v a cyclic permutation π_v of its outgoing edges. Vertices of out-degree zero carry no rotors. On $\mathbb{Z}^2$ we usually take π_v to be the clockwise order north, east, south, west.

Start a walker at a vertex X_0, with initial rotor configuration $\rho_0 = \rho$. At each step, the rotor walk first advances the rotor at its current vertex and then follows it:

$$\rho_{t+1}(v) = \begin{cases} \pi_v(\rho_t(v)), & v = X_t, \\ \rho_t(v), & v \neq X_t, \end{cases}$$

and X_{t+1} is the head of the edge $\rho_{t+1}(X_t)$. The walk stops when it reaches a vertex of out-degree zero. Figure 11.1 shows one step on $\mathbb{Z}^2$. The **rotor path** from a vertex x instead follows the edges of a fixed configuration ρ without advancing them, and it stops at a vertex with no rotor.

Our first lemma says that on a finite digraph, rotor walk cannot avoid forever a set that every vertex can reach. It guarantees that each routing procedure in this section stops.

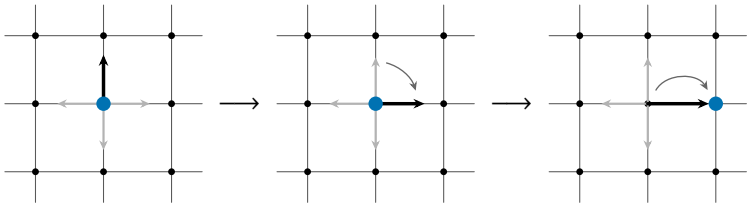


Figure 11.1: One step of the clockwise rotor walk on $\mathbb{Z}^2$. The rotor at the walker turns from north to east, and the walker then follows it.

Lemma 11.1 *Let $G = (V, E)$ be a finite digraph, and let $Z \subseteq V$ be nonempty. Suppose that every vertex outside Z has a directed path to Z. Then every rotor walk on G, stopped when it first enters Z, stops after finitely many steps.*

Proof. Suppose, for a contradiction, that the walk never enters Z. Since G is finite, the walk visits a vertex v infinitely often. Each visit to v advances the rotor at v by one position, so the walk leaves v along each of its outgoing edges infinitely often, and it visits every out-neighbor of v infinitely often. Repeating this argument along a directed path from v to Z shows that the walk enters Z, a contradiction. □

Until the end of the proof of Proposition 11.3, fix a finite digraph $G = (V, E)$ and a nonempty set $Z \subseteq V$ such that every vertex outside Z has a directed path to Z. Vertices in Z are **absorbing**: they carry no rotors, and they keep every particle that reaches them. A global sink, as in Chapter 9, is the case of a single absorbing vertex with no outgoing edges.

A **particle-and-rotor state** is a pair (σ, ρ), where $\sigma : V \to \mathbb{Z}_{\geq 0}$ records the number of particles at each vertex and ρ is a rotor configuration on $V \setminus Z$. To **actuate** a vertex $v \notin Z$ with $\sigma(v) \geq 1$, advance its rotor and move one particle from v along the new rotor edge. The resulting state is a **successor** of (σ, ρ). A state is **stable** if all its particles lie in Z. If we route the particles one at a time, each until it enters Z, then Lemma 11.1 shows that we reach a stable state after finitely many actuations.

The order of the actuations does not matter, because the sequence of edges along which a vertex sends particles is fixed in advance by its initial rotor and the mechanism, whatever the other vertices do. The next lemma makes this precise. It is the rotor analogue of Lemma 9.3, the abelian property of the sandpile, and its proof follows that of the least action principle, Lemma 9.2.

Lemma 11.2 *Let $G = (V, E)$ be a finite digraph, and let $Z \subseteq V$ be a nonempty set such that every vertex outside Z has a directed path to Z. Fix a rotor mechanism on $V \setminus Z$. Let $(\sigma_0, \rho_0), \dots, (\sigma_n, \rho_n)$ and $(\widehat{\sigma}_0, \widehat{\rho}_0), \dots, (\widehat{\sigma}_m, \widehat{\rho}_m)$ be sequences of particle-and-rotor states, each state a successor of the one before it, with $(\widehat{\sigma}_0, \widehat{\rho}_0) = (\sigma_0, \rho_0)$.*

(a) *If (σ_n, ρ_n) is stable, then each vertex is actuated at most as many times in the second sequence as in the first.*

(b) *If (σ_n, ρ_n) and $(\widehat{\sigma}_m, \widehat{\rho}_m)$ are both stable, then each vertex is actuated equally often in the two sequences, and $(\sigma_n, \rho_n) = (\widehat{\sigma}_m, \widehat{\rho}_m)$.*

Proof. (a) For a vertex $v \notin Z$, a vertex x and an integer $k \geq 0$, let $\nu_k(v, x)$ be the number of the first k departures from v that use the edge (v, x), with $\nu_k(v, x) = 0$ if there is no such edge. These departures are determined by the initial rotor at v and the mechanism, and $\nu_k(v, x)$ is nondecreasing in k. Let $N(v)$ be the number of actuations of v in the first sequence. Suppose, for a contradiction, that the second sequence actuates a vertex more often, and consider the first step at which it actuates a vertex x for the $(N(x) + 1)$st time. Just before this step, each vertex v has been actuated at most $N(v)$ times in the second sequence, and x has been actuated exactly $N(x)$ times. The number of particles at x is then at most

$$\sigma_0(x) + \sum_{v \notin Z} \nu_{N(v)}(v, x) - N(x) = \sigma_n(x) = 0\,.$$

The first equality is particle conservation at x in the first sequence, and $\sigma_n(x) = 0$ because (σ_n, ρ_n) is stable and $x \notin Z$. Thus x holds no particle and cannot be actuated, a contradiction.

(b) Applying (a) in both directions shows that each vertex is actuated equally often in the two sequences. The numbers of actuations determine how many particles cross each edge and where each rotor points at the end. Particle conservation then determines the final particle configuration, so $(\sigma_n, \rho_n) = (\widehat{\sigma}_m, \widehat{\rho}_m)$. □

Each step of the two sequences in Lemma 11.2 actuates one vertex, so their lengths n and m are the sums of the actuation counts over the vertices. Hence part (a) gives $m \leq n$, and part (b) gives $m = n$.

Rotor walk imitates random walk, and the next estimate measures how closely. Recall from Chapter 9 that simple random walk on a digraph chooses an outgoing edge uniformly at each step. We compare the number of particles that rotor walk sends into a target set with the expected number for independent simple random walks. The error does not depend on the initial rotors or on the number of particles.

Proposition 11.3 *Let $G = (V, E)$ be a finite digraph, and let $Y \subseteq Z \subseteq V$, where Z is nonempty and every vertex outside Z has a directed path to Z. Fix a rotor mechanism and an initial rotor configuration ρ on $V \setminus Z$, and let $\sigma : V \to \mathbb{Z}_{\geq 0}$ be a particle configuration. Let $H(\sigma, Y)$ be the expected number of particles that first reach Z in Y when the particles of σ perform independent simple random walks on G. Let $H_\rho(\sigma, Y)$ be the number of*

particles that first reach Z in Y when the particles of σ are routed by rotor walk from the rotors ρ. For $v \in V$, let $h(v)$ be the probability that simple random walk started at v first reaches Z in Y. Then

$$\left|H_\rho(\sigma, Y) - H(\sigma, Y)\right| \le \frac{1}{2} \sum_{\substack{(w,v)\in E \\ w\notin Z}} |h(w) - h(v)| \,. \tag{11.1}$$

Proof. We compare the total weight of the particles before and after the routing. Give each particle at a vertex v the weight $h(v)$. By Lemma 11.1, every particle reaches Z. The particles of σ have total weight $H(\sigma, Y)$ at the start and, once all of them have entered Z, total weight $H_\rho(\sigma, Y)$, because h equals 1 on Y and 0 on $Z \setminus Y$. A move along an edge (w, v) changes the total weight by $h(v) - h(w)$, so $H_\rho(\sigma, Y) - H(\sigma, Y)$ is the sum of $h(v) - h(w)$ over all the moves, each counted with the edge (w, v) that it uses.

Fix $w \notin Z$. The function h is harmonic at w for simple random walk, so

$$\sum_{v:(w,v)\in E} (h(w) - h(v)) = 0 \,. \tag{11.2}$$

The rotor at w sends the successive particles leaving w along its outgoing edges in cyclic order. Group these departures into consecutive blocks of $\deg(w)$. Each complete block uses every outgoing edge of w once, so it contributes zero by (11.2). The remaining departures use distinct outgoing edges, so they contribute the sum of a subset of the numbers $h(v) - h(w)$ with $(w, v) \in E$. These numbers sum to zero, so the sum of the positive ones and the absolute sum of the negative ones are both equal to half of $\sum_{v:(w,v)\in E} |h(w) - h(v)|$. Every subset sum is therefore at most this half in absolute value. Summing over $w \notin Z$ proves (11.1). □

Recall from Chapter 9 that a digraph is Eulerian if it is strongly connected and each vertex has equal in-degree and out-degree. The next lemma shows that on an Eulerian digraph, the first edge that rotor walk traverses twice starts at the starting vertex. We use it to study both circuits and recurrence.

Lemma 11.4 *Let G be a locally finite Eulerian digraph, and fix a rotor mechanism and an initial rotor configuration on G. If the rotor walk started at a vertex x traverses a directed edge twice, then the first edge that it traverses a second time has tail x.*

Proof. Let (v, w) be the first edge that the walk traverses a second time. The rotor at v turns through all $\deg(v)$ outgoing edges before it repeats one, so this traversal is at least the $(\deg(v) + 1)$st departure from v. Before it, no edge has been traversed twice, so the walk has entered v at most once along each incoming edge, and hence at most $\deg(v)$ times, because the in-degree of

v equals $\deg(v)$. A walk that has entered v at most $\deg(v)$ times can depart from v more than $\deg(v)$ times only if it started at v. Hence $v = x$. □

On a strongly connected digraph with finite out-degrees, a rotor walk is **recurrent** if it visits every vertex infinitely often, and **transient** if it visits every vertex only finitely often. Every rotor walk on such a digraph is one or the other. Indeed, if the walk visits a vertex infinitely often, then, as in the proof of Lemma 11.1, it visits each out-neighbor of that vertex infinitely often, and strong connectivity carries this to every vertex. The next section applies the abelian property on a finite graph with a sink, where the sandpile group acts on rotor configurations.

11.2 RELATING ROTOR WALK AND SANDPILES

Throughout this section, $G = (V, E)$ is a finite digraph with a global sink s, and we fix a rotor mechanism on $V \setminus \{s\}$. Recall the sandpile group $\mathcal{S}(G)$ and its recurrent sandpiles from Chapter 9. We use rotor walk to put $\mathcal{S}(G)$ in bijection with the oriented spanning trees of G rooted at s, and thereby prove the matrix tree theorem. A first hint of such a bijection is a count: there are $\prod_{v \neq s} \deg(v)$ rotor configurations, exactly as many as nonnegative stable sandpiles.

For $v \neq s$, the **particle addition operator** $\mathcal{E}_v$ maps a rotor configuration ρ to the rotor configuration left after one particle is added at v and routed to s from the rotors ρ. More generally, for a particle configuration σ on $V \setminus \{s\}$, we write $\sigma(\rho)$ for the rotor configuration left after all particles of σ are routed to s from the rotors ρ. By Lemma 11.2(b), $\sigma(\rho)$ does not depend on the order of the actuations. Consequently, $\mathcal{E}_v \circ \mathcal{E}_w = \mathcal{E}_w \circ \mathcal{E}_v$ for all $v, w \neq s$, the map $\rho \mapsto \sigma(\rho)$ is obtained by applying each $\mathcal{E}_v$ exactly $\sigma(v)$ times in an arbitrary order, and

$$\sigma_2(\sigma_1(\rho)) = (\sigma_1 + \sigma_2)(\rho)$$

for all particle configurations σ_1 and σ_2 and every rotor configuration ρ.

Commutativity depends on stopping each particle only when it reaches the sink. If instead each particle stops after a prescribed number of steps, then the particles become distinguishable, and the order of the moves can matter, as Figure 11.2 shows.

A rotor configuration is **acyclic** if its edges contain no directed cycle. Every vertex other than s carries exactly one rotor edge, so a rotor configuration is acyclic exactly when its edges form an oriented spanning tree rooted at s. The next lemma detects cycles through the last departure from each vertex.

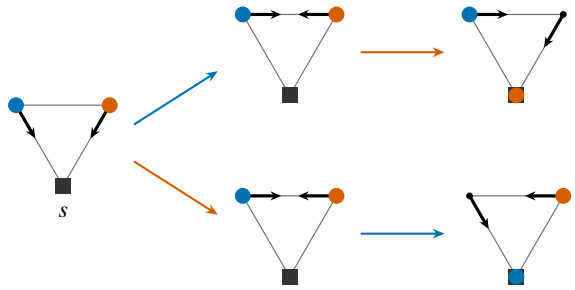


Figure 11.2: Two particles, blue and vermilion, on the triangle with sink s (square), each taking two steps of rotor walk or stopping earlier at s; the bold arrows are the rotors, and each arrow between configurations is colored by the particle that moves. Moving blue first (upper row) or vermilion first (lower row) passes through the same configuration but ends in different ones.

Lemma 11.5 *Let $G = (V, E)$ be a finite digraph with a global sink s, and fix a rotor mechanism. Let ρ be a rotor configuration and σ a particle configuration on G. Then every directed cycle of the rotor configuration $\sigma(\rho)$ consists of vertices from which no particle departs while σ is routed.*

Proof. Suppose that a directed cycle of $\sigma(\rho)$ contains a vertex that sends a particle during the routing. Among such vertices of the cycle, choose the vertex x whose last departure is latest. The last departure from x uses the final rotor at x, so it sends a particle to the next vertex y on the cycle. At the end no particle remains at $y \neq s$, so y sends a particle after this time, and its last departure is later than that of x, contrary to the choice of x. □

A cycle of $\sigma(\rho)$ whose vertices never send a particle is also a cycle of ρ. Consequently, if ρ is acyclic, then so is $\sigma(\rho)$. Moreover, if every vertex other than s sends at least one particle while σ is routed, then $\sigma(\rho)$ has no cycle at all.

To show that each particle addition operator permutes the acyclic configurations, we temporarily give the sink an outgoing edge. The rotors then contain exactly one cycle, the walker lies on it, and the edge along which the walker arrived is determined, so that each step of rotor walk can be undone.

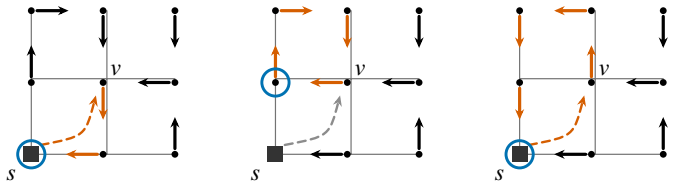
Lemma 11.6 *Let $\widehat{G}$ be a finite digraph in which every vertex has positive outdegree, and fix a rotor mechanism on $\widehat{G}$. Consider the pairs (ρ, x) in which ρ is a rotor configuration on $\widehat{G}$ whose edges contain exactly one directed cycle and x is a vertex of that cycle. Then one step of rotor walk from x permutes these pairs.*

Proof. We first show that the step maps a pair of this kind to a pair of this kind. Let (ρ, x) be such a pair. Removing the rotor edge at x from ρ leaves an oriented spanning tree rooted at x, because x lies on the unique cycle. If the advanced rotor at x is the edge (x, y), then adding this edge to the tree

creates exactly one directed cycle, and this cycle passes through y. Thus the step maps (ρ, x) to a pair of the same kind.

The set of pairs is finite, so it suffices to show that the step is injective. Suppose that it maps (ρ, x) to (ρ_1, y). The edge $\rho_1(x) = (x, y)$ lies on the cycle of ρ_1, so x is the vertex preceding y on that cycle. Moreover, ρ agrees with ρ_1 except at x, where $\rho(x)$ is the edge preceding $\rho_1(x)$ in the cyclic order at x. Thus (ρ, x) is determined by (ρ_1, y). □

For each $v \neq s$, adding the edge (s, v) makes particle addition at v the first-return map to the pairs with the walker at the sink. Figure 11.3 follows one particle addition on the 3×3 grid in this way.



(a) (b) (c)

Figure 11.3: Particle addition at v on the 3×3 grid with sink s (square) and the clockwise mechanism, run as rotor walk with the added edge (s, v) (dashed): the rotors always contain exactly one cycle (vermilion), and the walker (circled) lies on it. The walk starts from an acyclic ρ with the walker at s (a), passes after two steps through a state whose cycle avoids s (b), and returns to s after seven steps with the acyclic rotors $\mathcal{E}_v(\rho)$ (c).

Lemma 11.7 *Let $G = (V, E)$ be a finite digraph with a global sink s, and fix a rotor mechanism. For every vertex $v \neq s$, the particle addition operator $\mathcal{E}_v$ permutes the acyclic rotor configurations on G.*

Proof. Add the edge (s, v) to G, and let $\widehat{G}$ be the resulting digraph, in which every vertex has positive out-degree; the rotor at s is always this edge. A rotor configuration ρ on G, together with this rotor at s, contains exactly one cycle, passing through s, if and only if ρ is acyclic: the rotor paths of an acyclic configuration all end at s, and an additional cycle of ρ would avoid s. Thus the pairs of Lemma 11.6 on $\widehat{G}$ with the walker at s are exactly the acyclic configurations of G with the walker at s. By that lemma, rotor walk on $\widehat{G}$ permutes its pairs. Started from an acyclic ρ with the walker at s, the walk steps along (s, v) and then moves exactly as a particle routed from v in G, until it first returns to s with the rotors $\mathcal{E}_v(\rho)$. Hence $\mathcal{E}_v$ is the first-return map of this permutation to the pairs with the walker at s. In each cycle of a permutation of a finite set, the first-return map to a subset follows the elements of the subset in their cyclic order, so it permutes the subset. Therefore $\mathcal{E}_v$ permutes the acyclic configurations. □

A full turn of the rotor at v sends one particle along each outgoing edge of v, just as a toppling of v sends one grain along each outgoing edge. Consequently, particle configurations that differ by topplings act in the same way on acyclic configurations. Recall from Chapter 9 that two configurations on $V \setminus \{s\}$ are equivalent modulo the reduced Laplacian L' if their difference is an integer combination of rows of L'.

Lemma 11.8 *Let $G = (V, E)$ be a finite digraph with a global sink s, fix a rotor mechanism, and let ρ be an acyclic rotor configuration on G. If particle configurations σ_1 and σ_2 on $V \setminus \{s\}$ are equivalent modulo the reduced Laplacian L', then $\sigma_1(\rho) = \sigma_2(\rho)$.*

Proof. By Lemma 11.7, the maps $\mathcal{E}_v$ are commuting permutations of the finite set of acyclic configurations. Their action therefore extends to integer-valued configurations, with negative values acting through inverse maps, and it suffices to show that each row of L' acts as the identity. Fix $v \neq s$, and add $\deg(v)$ particles at v. Moving each of them one step turns the rotor at v once around, back to its initial position, and sends one particle along each outgoing edge of v; a particle sent to s stays there. Routing the remaining particles to s and applying Lemma 11.2(b) shows that $\deg(v)\mathbf{1}_v$ acts in the same way as the configuration with one particle at each out-neighbor of v other than s. The difference of these two configurations is the row of L' indexed by v, so this row acts as the identity. Integer combinations of rows therefore also act as the identity, which proves the lemma. □

Lemma 11.8 shows that $\mathcal{S}(G)$ acts on the acyclic configurations. The next two lemmas show that this action is transitive and free; together they identify the group with the set of oriented spanning trees rooted at s.

Lemma 11.9 *Let $G = (V, E)$ be a finite digraph with a global sink s, and fix a rotor mechanism. For all acyclic rotor configurations ρ and ρ_1 on G, there is a particle configuration σ on $V \setminus \{s\}$ with $\sigma(\rho) = \rho_1$.*

Proof. We first turn the rotors to ρ_1 and then cancel the action of the remaining particles. For each $v \neq s$, choose an integer $0 \leq \alpha(v) < \deg(v)$ such that $\pi_v^{\alpha(v)}(\rho(v)) = \rho_1(v)$, where π_v is the rotor mechanism at v. Place $\alpha(v)$ particles at each $v \neq s$, and move each of these particles once. The rotors are now ρ_1; let β be the remaining particle configuration off the sink. Completing the routing and using Lemma 11.2(b) gives $\alpha(\rho) = \beta(\rho_1)$.

By Lemma 9.18, choose a nonnegative stable sandpile σ equivalent to $\alpha - \beta$ modulo L'. Then $\sigma + \beta$ is equivalent to α, so Lemma 11.8 and the composition rule give

$$\beta(\sigma(\rho)) = (\sigma + \beta)(\rho) = \alpha(\rho) = \beta(\rho_1)\,.$$

By Lemma 11.7, β acts as a permutation on the acyclic configurations. Cancelling it gives $\sigma(\rho) = \rho_1$. □

Transitivity has a converse, which we state after naming the relation involved.

Definition 11.10 Let G be a finite digraph with a global sink, and fix a rotor mechanism. A rotor configuration ρ on G is **reachable** from a rotor configuration ρ_0 if there is a particle configuration σ with $\sigma(\rho_0) = \rho$.

By Lemma 11.9, every acyclic configuration is reachable from every acyclic configuration. The next lemma shows that the acyclic configurations are exactly the configurations reachable from every configuration.

Lemma 11.11 *Let G be a finite digraph with a global sink, and fix a rotor mechanism. A rotor configuration on G is reachable from every rotor configuration on G if and only if it is acyclic.*

Proof. Suppose first that ρ is reachable from every rotor configuration. There is an acyclic configuration, because every vertex has a directed path to the sink. In particular, ρ is reachable from an acyclic configuration, and Lemma 11.5 shows that routing preserves acyclicity, so ρ is acyclic.

Conversely, suppose that ρ is acyclic, and let ρ_0 be a rotor configuration. Route one particle from each vertex other than the sink, starting from ρ_0. Every such vertex sends a particle, so the resulting configuration is acyclic by Lemma 11.5. By Lemma 11.9, ρ is reachable from this acyclic configuration, and hence from ρ_0. □

Acyclicity can also be recognized by reachability: a configuration is acyclic exactly when, wherever a particle is added and routed to the sink, the configuration can be reached again. The next proposition records several forms of this criterion.

Proposition 11.12 *Let $G = (V, E)$ be a finite digraph with a global sink s, fix a rotor mechanism, and let ρ be a rotor configuration on G. The following are equivalent.*

(1) The configuration ρ is acyclic.

(2) The configuration ρ is reachable from every rotor configuration.

(3) For every rotor configuration ρ_0 that is reachable from ρ, the configuration ρ is reachable from ρ_0.

(4) For every vertex $v \neq s$, the configuration ρ is reachable from $\mathcal{E}_v(\rho)$.

(5) Each strongly connected component of the subgraph induced by $V \setminus \{s\}$ contains a vertex v such that ρ is reachable from $\mathcal{E}_v(\rho)$.

Proof. Lemma 11.11 proves that (1) implies (2), and the next three implications follow from the definitions. We prove that (5) implies (1).

Assume (5), and suppose that ρ has a directed cycle. Its vertices are mutually reachable, so the cycle lies in one strongly connected component of the subgraph induced by $V \setminus \{s\}$. By (5) there are a vertex v of this component and a particle configuration σ with $\sigma(\mathcal{E}_v(\rho)) = \rho$, so the composition rule gives $(\sigma + \mathbf{1}_v)(\rho) = \rho$. Route $\sigma + \mathbf{1}_v$ from ρ. The rotors end where they started, so each rotor makes a whole number of turns, and a vertex w that sends a particle sends at least $\deg(w)$ of them, one along each of its outgoing edges. The vertex v sends a particle, because the particle added at v must leave it, and every particle that reaches a vertex other than s leaves it again. Following a path in the component from v therefore shows that every vertex of the component sends a particle. By Lemma 11.5, every directed cycle of $(\sigma + \mathbf{1}_v)(\rho) = \rho$ avoids the vertices that send particles, so the cycle above is not a cycle of ρ, a contradiction. □

An action is free if no group element other than the identity fixes a point. Here freeness comes from particle conservation. Two routings that end with the same rotors differ at each vertex by a whole number of full turns, and each full turn is a toppling.

Lemma 11.13 *Let $G = (V, E)$ be a finite digraph with a global sink s, fix a rotor mechanism, and let ρ be an acyclic rotor configuration on G. If σ_1 and σ_2 are recurrent sandpiles on G with $\sigma_1(\rho) = \sigma_2(\rho)$, then $\sigma_1 = \sigma_2$.*

Proof. For $i = 1, 2$, let $n_i(v)$ be the number of departures from v while σ_i is routed from ρ. Both routings start and end with the same rotors, so the rotor at v makes a whole number of extra turns in one routing compared with the other: $n_1(v) - n_2(v) = k_v \deg(v)$ for an integer k_v. Each full turn sends one particle along each outgoing edge, so the numbers of particles sent along an edge (v, x) in the two routings differ by k_v. At each $x \neq s$, the particles present at the start plus those received equal those sent, because no particle remains at x. Subtracting these balance equations for the two routings gives

$$\sigma_1 - \sigma_2 = \sum_{v \neq s} k_v \mathsf{L}'_v \,,$$

where L'_v is the row of L' indexed by v. Thus σ_1 and σ_2 are equivalent modulo L'. By Lemma 9.23, an equivalence class contains at most one recurrent sandpile, so $\sigma_1 = \sigma_2$. □

Because the action is free and transitive, the number of oriented spanning trees equals the order of the sandpile group, which Lemma 9.17 computes as the determinant of the reduced Laplacian. Recall from Section 9.1 that the

Laplacian $\mathsf{L}(G)$ is the diagonal matrix of out-degrees minus the adjacency matrix.

Proposition 11.14 (Matrix Tree Theorem) *Let G be a finite digraph, and let v be a vertex of G. Then the number of oriented spanning trees of G rooted at v equals the determinant of the matrix obtained by deleting the row and the column indexed by v from the Laplacian $\mathsf{L}(G)$.*

Proof. Suppose first that there is a vertex with no directed path to v. Then no spanning tree is rooted at v. Let S be the nonempty set of vertices with no directed path to v. No edge leaves S, since its head would then have a directed path to v. Listing the vertices of S first puts the reduced matrix in the block form

$$\begin{pmatrix} L_S & 0 \\ * & * \end{pmatrix},$$

where L_S is the square block indexed by S, and each row of L_S sums to zero. Hence the determinant vanishes, and the two sides of the identity agree.

Suppose now that every vertex has a directed path to v. Deleting the outgoing edges of v makes v a global sink, and it changes neither the reduced matrix nor the spanning trees rooted at v. Fix one such tree ρ, viewed as an acyclic rotor configuration. By Lemma 11.8, each equivalence class acts on ρ through its recurrent sandpile, which exists and is unique by Lemmas 9.22 and 9.23, so Lemmas 11.9 and 11.13 make $\sigma \mapsto \sigma(\rho)$ a bijection from the recurrent sandpiles to the spanning trees rooted at v. The recurrent sandpiles are in bijection with $\mathcal{S}(G)$, whose order is the determinant of the reduced Laplacian by Lemma 9.17. This proves the proposition. □

Choosing one rooted spanning tree therefore identifies the recurrent sandpiles with all the rooted spanning trees. Figure 11.4 shows this for the cycle with five vertices, whose recurrent sandpiles are in Figure 9.8.

The rest of the chapter concerns infinite graphs, where the first question is whether rotor walk returns to its starting point.

11.3 RECURRENCE OF ROTOR WALK

Throughout this section, $G = (V, E)$ is a connected, locally finite undirected graph, with each edge replaced by its two orientations. This digraph is Eulerian, so Lemma 11.4 applies to it. If rotor walk on G visits a vertex infinitely often, then it also visits every neighbor of that vertex infinitely often, because the rotor there turns through all its outgoing edges. Thus a walk is recurrent as soon as it visits one vertex infinitely often. In contrast

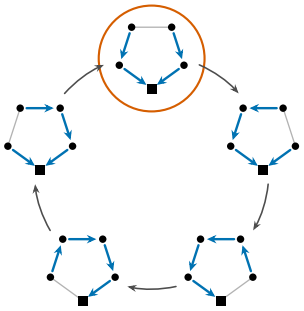


Figure 11.4: The five oriented spanning trees of the cycle with five vertices rooted at the sink (black square), in the positions of the recurrent sandpiles of Figure 9.8: the particle addition operator $\mathcal{E}_v$ at the vertex v just clockwise of the sink moves each tree to the next (gray arrows). Thus $\sigma \mapsto \sigma(\rho)$, with ρ the circled tree, matches each recurrent sandpile of Figure 9.8 with the tree in the same position, as in the proof of Proposition 11.14.

with random walk, whether the walk is recurrent depends on the initial rotors. We show that it does not depend on the starting vertex or on finitely many rotors, and then we construct, on $\mathbb{Z}^d$ in every dimension, rotor configurations from which rotor walk is recurrent.

Lemma 11.15 *Let G be a connected, locally finite undirected graph, with each edge replaced by its two orientations, and fix a rotor mechanism and an initial rotor configuration on G. Then the rotor walk is either recurrent from every starting vertex or transient from every starting vertex.*

Proof. Since G is connected, it suffices to show that if the walk from a vertex x is recurrent, then so is the walk from each neighbor y of x. We show that the walk from y visits x at least m times, for every integer $m \geq 0$.

Fix m. The walk from x is recurrent, so its first $\deg(x) + m$ returns to x occur within a finite set S of vertices. Split x into two copies: a source, which carries the rotor and the outgoing edges of x, and an absorbing copy, which receives the incoming edges of x. Also make absorbing every vertex outside S adjacent to S. Routing $\deg(x) + m$ particles one at a time from the source reproduces the original walk up to its $(\deg(x) + m)$th return, so every particle is absorbed at the copy of x.

We now route the same particles in a different order. First let $\deg(x)$ particles take one step each from the source. These steps return the rotor at x to its initial position and place one particle at each neighbor of x. Next route the particle at y until it is absorbed, then route the m particles still at the source one at a time, and finally route the remaining particles. By Lemma 11.2(b), every particle is again absorbed at the copy of x. When the two copies of x are identified, the route of the particle from y, followed by the routes of the m particles from the source, forms an initial segment of the

rotor walk from y in the original configuration. This segment visits x at least m times, as claimed. □

Changing finitely many initial rotors does not affect recurrence either.

Lemma 11.16 *Let G be a connected, locally finite undirected graph, with each edge replaced by its two orientations, and fix a rotor mechanism on G. If two rotor configurations on G differ at only finitely many vertices, then the rotor walks from them are either both recurrent or both transient.*

Proof. Every finite change of the rotors, in either direction, is a finite sequence of single rotor advances, so it suffices to consider one advance. Suppose that the walk from the configuration ρ is recurrent, and advance the rotor of ρ at a vertex v once to obtain the configuration $\widetilde{\rho}$. The walk from v in configuration ρ takes one step, to a neighbor w of v, and leaves the configuration $\widetilde{\rho}$. The rest of this walk is the walk from w in configuration $\widetilde{\rho}$, and it is recurrent. By Lemma 11.15, the walk from $\widetilde{\rho}$ is recurrent from every starting vertex. Conversely, ρ is obtained from $\widetilde{\rho}$ by $\deg(v) - 1$ further advances at v, so the same argument shows that recurrence of $\widetilde{\rho}$ implies recurrence of ρ. Since every rotor walk is either recurrent or transient, this proves the lemma. □

Exercise 11.2 asks you to show that recurrence is also unchanged when the rotor mechanisms differ at finitely many vertices.

To construct rotor configurations from which rotor walk is recurrent, we use boundaries that turn the walk back. For $S \subseteq V$, the **outer boundary** of S is

$$\partial S := \{y \in V \setminus S : (x, y) \in E \text{ for an } x \in S\}. \tag{11.3}$$

The set S has a **reflecting boundary** if, at each $y \in \partial S$, the rotor mechanism, started from the initial rotor at y, sends particles to all neighbors of y in S before it sends a particle to a neighbor outside S. Figure 11.5 illustrates this condition on the square lattice.

A reflecting boundary that the walk has not yet reached forces one more return to the start, as the next lemma shows.

Lemma 11.17 *Let $G = (V, E)$ be a connected, locally finite undirected graph, with each edge replaced by its two orientations, and fix a rotor mechanism and an initial rotor configuration on G. Let $S \subset V$ be a finite set with reflecting boundary, and let $x \in S$ satisfy $\deg(x) \geq 1$. Then the rotor walk started at x returns to x before it takes a step from ∂S to $V \setminus S$.*

Proof. Suppose first that the walk takes a step from a vertex $y \in \partial S$ to $V \setminus S$, and consider the first such step. Before it, the walk has left y toward every

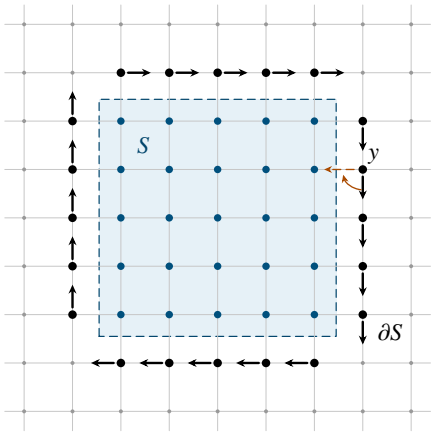


Figure 11.5: A square S (shaded) with a reflecting boundary for the clockwise mechanism: each vertex of ∂S (black) has one neighbor in S, and the next advance of its rotor points to that neighbor, as shown at y. The four corners of the surrounding ring have no neighbor in S, so they are not in ∂S; their rotors, like all rotors off ∂S, are unconstrained and not drawn.

neighbor of y in S, and it has entered y only from S. The walk did not start at y, since $y \notin S$, so it has entered y at least one more time than y has neighbors in S. Hence it has traversed an edge from S into y twice. By Lemma 11.4, the first edge traversed twice has tail x, so the walk has returned to x before this step.

If the walk never takes such a step, then it stays in the finite set $S \cup \partial S$, so it eventually traverses an edge twice, and Lemma 11.4 again gives a return to x. □

On an infinite graph, if every finite set is contained in a finite set with reflecting boundary, then the lemma gives a return after every visit to the start, so the walk is recurrent.

Proposition 11.18 *Let $G = (V, E)$ be an infinite, connected, locally finite undirected graph, with each edge replaced by its two orientations, and fix a rotor mechanism and an initial rotor configuration on G. Suppose that every finite subset of V is contained in a finite set with reflecting boundary. Then the rotor walk is recurrent from every starting vertex.*

Proof. Fix $x \in V$, and suppose for a contradiction that the walk from x visits x for the last time at time t. Choose a finite set S with reflecting boundary that contains every vertex visited by time t. The walk has not visited ∂S by time t, so the rotors on ∂S are still in their initial positions, and S also has a reflecting boundary for the rotor configuration at time t. Since G is infinite and connected, $\deg(x) \geq 1$. At time t the walk is at x, so Lemma 11.17, applied to the walk from time t on, gives a later return to x, a contradiction.

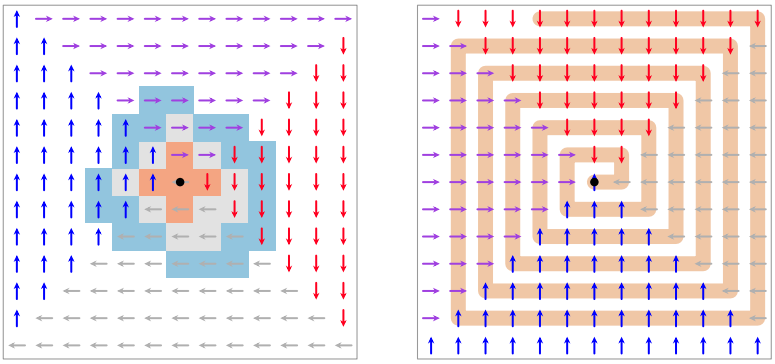
Hence the walk visits x infinitely often. Every rotor walk on G is either recurrent or transient, so the walk is recurrent. □

On $\mathbb{Z}^d$, the cubes centered at the origin provide such sets for a suitable choice of initial rotors, whatever the mechanism.

Lemma 11.19 *Let $d \geq 1$ be an integer, and fix a rotor mechanism on $\mathbb{Z}^d$. Then there is a rotor configuration on $\mathbb{Z}^d$ for which rotor walk is recurrent.*

Proof. At each vertex x, choose the initial rotor so that its first advance points to a neighbor of x with the smallest ℓ^∞ norm. For an integer $k \geq 0$, let $S_k := \{x \in \mathbb{Z}^d : |x|_\infty \leq k\}$. Each vertex of ∂S_k has exactly one coordinate equal to $\pm(k+1)$, so it has a unique neighbor in S_k, and this neighbor is its unique neighbor of smallest ℓ^∞ norm. Thus every S_k has a reflecting boundary, and Proposition 11.18 shows that the walk is recurrent. □

Figure 11.6(a) shows these rotors on $\mathbb{Z}^2$ with the clockwise mechanism. Advancing each of them once makes every rotor path lead to the origin, and the walk then never returns (Figure 11.6(b)); Theorem 11.30(i) will show that such a walk is always transient.



(a) (b)

Figure 11.6: (a) The initial rotors of Lemma 11.19 near the origin **0** (dot), colored by direction (North blue, East purple, South red, West gray), and the sites visited by the first three circuits, shaded by the circuit of first visit; these circuits end after 8, 44 and 140 steps. (b) The same rotors, each advanced once, so that every rotor path leads to **0**: the first 140 steps of the walk (shaded path) spiral outward and visit a new site at every step, and the walk is transient by Theorem 11.30(i).

Reflecting boundaries force returns because the walk cannot reuse an edge before returning. The next section develops this observation on Eulerian digraphs by cutting the walk at its returns.

11.4 CIRCUITS IN EULERIAN DIGRAPHS

On an Eulerian digraph, we cut rotor walk at the times when the rotor at the starting vertex completes a full turn. Each resulting segment, called a circuit, uses each directed edge at most once. We use this to study both recurrence and the number of visited sites.

Throughout this section, $G = (V, E)$ is a locally finite Eulerian digraph, and we fix a rotor mechanism, an initial rotor configuration ρ, and a starting vertex o with $\deg(o) \geq 1$. For the resulting rotor walk $(X_t)_{t\geq 0}$ and an integer $t \geq 0$, the **odometer** is

$$u_t(x) := \#\{0 \leq r < t : X_r = x\}, \tag{11.4}$$

the number of departures from x before time t. For integers $n \geq 0$, let

$$T(n) := \inf\{t \geq 0 : X_t = o \text{ and } u_t(o) \geq n \deg(o)\},$$

where $\inf \varnothing = \infty$. Thus $T(n)$ is the time at which the rotor at o completes its nth full turn and the walk is back at o. If $T(n-1) < \infty$, then the **nth circuit** is the segment of the walk from time $T(n-1)$ to time $T(n)$. The circuit is **complete** if $T(n) < \infty$ and incomplete otherwise. Its departures are the steps at the times r with $T(n-1) \leq r < T(n)$; when $T(n) = \infty$, they form the entire remaining walk. Figure 11.7 shows a first circuit.

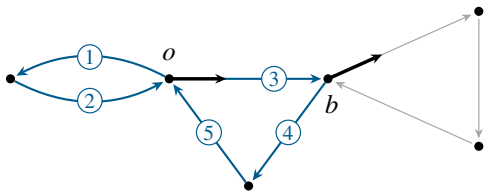


Figure 11.7: The first circuit of the rotor walk from o on an Eulerian digraph, from the initial rotors at o and b (black); the edges are numbered in the order of traversal. The walk returns to o at steps 2 and 5, so the circuit ends at $T(1) = 5$ because $\deg(o) = 2$, and it uses no directed edge twice and none of the gray edges.

Our first lemma bounds the number of departures from each vertex in a circuit.

Lemma 11.20 *Let $G = (V, E)$ be a locally finite Eulerian digraph, and fix a rotor mechanism and an initial rotor configuration. Let o be a vertex with $\deg(o) \geq 1$. Then every circuit of the rotor walk from o, complete or incomplete, departs from each vertex x at most $\deg(x)$ times.*

Proof. It suffices to show that no directed edge is traversed twice within a circuit, because a circuit that departs x more than $\deg(x)$ times uses an

outgoing edge of x twice. Consider the walk from the start of a circuit, which is again a rotor walk from o. By Lemma 11.4, the first edge that it traverses twice has tail o. Such a traversal is at least the $(\deg(o)+1)$st departure from o after the start of the circuit, so it happens after the walk has returned to o following its $\deg(o)$th departure from o, that is, after the circuit has ended. □

The initial rotor paths, which follow the initial rotors without advancing them, identify vertices at which a complete circuit makes a full turn.

Lemma 11.21 *Let $G = (V, E)$ be a locally finite Eulerian digraph, and fix a rotor mechanism and an initial rotor configuration ρ. Let o be a vertex with $\deg(o) \geq 1$, and suppose that the first circuit of the rotor walk from o is complete. If the rotor path of ρ from a vertex x reaches o, then the first circuit departs from x exactly $\deg(x)$ times.*

Proof. We prove the claim by induction on the number of steps of the rotor path from x to o. If this number is zero, then $x = o$, and the first circuit departs from o exactly $\deg(o)$ times by definition.

For the inductive step, let $\rho(x) = (x, y)$, where the rotor path from y reaches o in one step fewer. By the inductive hypothesis, the first circuit departs from y exactly $\deg(y)$ times. The circuit starts and ends at o, so it also enters y exactly $\deg(y)$ times. By Lemma 11.20, no edge is used twice, so every incoming edge of y is used, including (x, y). The rotor at x advances before each departure, so the edge $(x, y) = \rho(x)$ is used for the first time at the $\deg(x)$th departure from x. Hence the circuit departs from x at least $\deg(x)$ times, and Lemma 11.20 gives equality. □

On a finite graph, every circuit is complete, because Lemma 11.20 allows only finitely many steps in a circuit. On an infinite graph, an infinite chain of initial rotors pointing toward o prevents the first circuit from completing.

Lemma 11.22 *Let $G = (V, E)$ be a locally finite Eulerian digraph, and fix a rotor mechanism and an initial rotor configuration ρ. Suppose that there are distinct vertices $x_0, x_1, x_2, \ldots$ with $\deg(x_0) \geq 1$ and $\rho(x_k) = (x_k, x_{k-1})$ for every $k \geq 1$. Then the first circuit of the rotor walk from x_0 is incomplete.*

Proof. For every $k \geq 1$, the rotor path from x_k reaches x_0. If the first circuit were complete, then Lemma 11.21 would make it depart from each of the infinitely many vertices x_k, which a finite segment of walk cannot do. □

Let $\mathsf{A}_0 := \{o\}$. Whenever $T(n-1) < \infty$, let A_n be the set of vertices visited by the nth circuit, including both of its endpoints if it is complete and its entire infinite tail otherwise. We call A_n the nth **circuit range**. The next

lemma shows that each complete circuit makes a full turn at every vertex visited by the circuit before it.

Lemma 11.23 *Let $G = (V, E)$ be a locally finite Eulerian digraph, and fix a rotor mechanism and an initial rotor configuration. Let o be a vertex with $\deg(o) \geq 1$, and let $n \geq 0$ be an integer such that the $(n+1)$st circuit of the rotor walk from o is complete. Then:*

(i) the $(n+1)$st circuit departs from every $x \in V$ at most $\deg(x)$ times;

(ii) the $(n+1)$st circuit departs from every $x \in \mathsf{A}_n$ exactly $\deg(x)$ times.

Proof. (i) This is Lemma 11.20.

(ii) For $n = 0$, we have $\mathsf{A}_0 = \{o\}$, and the circuit departs from o exactly $\deg(o)$ times by definition. Let $n \geq 1$. At time $T(n)$, the rotor at each $x \in \mathsf{A}_n \setminus \{o\}$ points along the last departure from x in the nth circuit. Following these rotors from x leads to o, since the times of the last departures from the successive vertices increase strictly until the path reaches o. Lemma 11.21, applied to the walk restarted at time $T(n)$, shows that the $(n+1)$st circuit departs from x exactly $\deg(x)$ times. □

By part (ii), the $(n+1)$st circuit uses every outgoing edge of every vertex of A_n, so it visits A_n and all heads of edges leaving A_n. Recalling the outer boundary (11.3), we conclude that

$$\mathsf{A}_{n+1} \supseteq \mathsf{A}_n \cup \partial \mathsf{A}_n \qquad \text{whenever the } (n+1)\text{st circuit is complete}\,. \tag{11.5}$$

Figure 11.8 shows the first three circuit ranges of a rotor walk on $\mathbb{Z}^2$ and the rotors used in the proof of Lemma 11.23(ii).

Summing the departures in Lemma 11.23 gives the length of each circuit in terms of the degrees of the vertices it visits.

Proposition 11.24 *Let $G = (V, E)$ be a locally finite Eulerian digraph, and fix a rotor mechanism and an initial rotor configuration. Let o be a vertex with $\deg(o) \geq 1$, and suppose that every circuit of the rotor walk from o is complete. Then, for every integer $n \geq 0$,*

$$\sum_{x \in \mathsf{A}_n} \deg(x) \leq T(n+1) - T(n) \leq \sum_{x \in \mathsf{A}_{n+1}} \deg(x)\,.$$

Proof. The length $T(n+1) - T(n)$ of the $(n+1)$st circuit is its number of departures. By Lemma 11.23(ii), it departs from each $x \in \mathsf{A}_n$ exactly $\deg(x)$ times, and by Lemma 11.23(i), it departs from each $x \in \mathsf{A}_{n+1}$ at most $\deg(x)$ times and from no other vertex. □

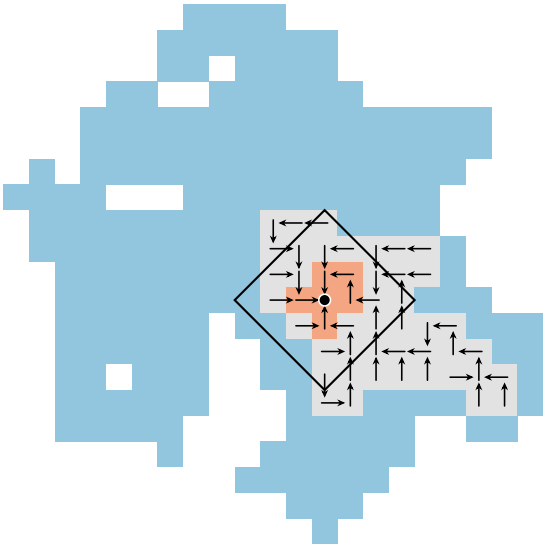


Figure 11.8: The circuit ranges $\mathsf{A}_1 \subseteq \mathsf{A}_2 \subseteq \mathsf{A}_3$ of rotor walk on $\mathbb{Z}^2$ from $\mathbf{0}$ (dot) with the clockwise mechanism and independent uniform initial rotors, each site colored light red, gray or blue by the first of them that contains it; the black outline is $B(\mathbf{0}, 3)$, which lies in A_3 by Lemma 11.25. At time $T(2)$ the rotors on A_2 (arrows) lead to $\mathbf{0}$, and the third circuit visits A_2 and its outer boundary, as (11.5) asserts.

Summing the bounds of Proposition 11.24 over the first n circuits gives

$$\sum_{j=0}^{n-1} \sum_{x \in \mathsf{A}_j} \deg(x) \le T(n) \le \sum_{j=1}^{n} \sum_{x \in \mathsf{A}_j} \deg(x) \,.$$

Suppose now that there are real numbers $\beta > 0$ and $q \ge 0$ such that

$$\sum_{x \in \mathsf{A}_n} \deg(x) = \beta n^q + o(n^q) \qquad \text{as } n \to \infty \,.$$

Since $\sum_{j=1}^{n} j^q = n^{q+1}/(q+1) + o(n^{q+1})$, both sides of the previous bound on $T(n)$ then equal $\beta n^{q+1}/(q+1) + o(n^{q+1})$, and hence

$$T(n) = \frac{\beta}{q+1} n^{q+1} + o(n^{q+1}) \qquad \text{as } n \to \infty \,.$$

The circuit ranges A_n grow at least as fast as graph balls. For $x \in V$ and an integer $r \ge 0$, let $B(x, r)$ be the set of vertices reachable from x by a directed path with at most r edges.

Lemma 11.25 *Let $G = (V, E)$ be a locally finite Eulerian digraph, and fix a rotor mechanism and an initial rotor configuration. Let o be a vertex with $\deg(o) \ge 1$, and let $n \ge 0$ be an integer with $T(n) < \infty$. Then $B(o, n) \subseteq \mathsf{A}_n$.*

Proof. We argue by induction on n. For $n = 0$, both sets equal $\{o\}$. For the inductive step, let $n \geq 1$ and suppose that $B(o, n-1) \subseteq \mathsf{A}_{n-1}$. The nth circuit is complete, so (11.5) gives $\mathsf{A}_n \supseteq \mathsf{A}_{n-1} \cup \partial\mathsf{A}_{n-1}$. Since $B(o, n-1) \subseteq \mathsf{A}_{n-1}$, the head of every edge leaving $B(o, n-1)$ lies in A_{n-1} or in $\partial\mathsf{A}_{n-1}$, so $\mathsf{A}_{n-1} \cup \partial\mathsf{A}_{n-1}$ contains $B(o, n-1) \cup \partial B(o, n-1)$, which equals $B(o, n)$. □

If every circuit is complete, then the walk is recurrent: by Lemma 11.25, every vertex lies in A_n for all large n, and by Lemma 11.23(ii), each later circuit departs from it. The same lemma shows that the rotors inside a circuit range return to the same positions at the end of every later circuit.

Proposition 11.26 *Let $G = (V, E)$ be a locally finite Eulerian digraph, and fix a rotor mechanism and an initial rotor configuration. Let o be a vertex with $\deg(o) \geq 1$, suppose that the rotor walk from o completes every circuit, and write ρ_t for its rotor configuration at time t. Then, for all integers $0 \leq n \leq N$,*

$$\rho_{T(N)}(x) = \rho_{T(n)}(x) \qquad \text{for every } x \in \mathsf{A}_n \,.$$

Proof. By Lemma 11.23(ii) and the inclusions $\mathsf{A}_n \subseteq \mathsf{A}_{n+1} \subseteq \cdots$, each $x \in \mathsf{A}_n$ makes exactly one full turn in each of the circuits $n+1, \ldots, N$, so its rotor returns to its position at time $T(n)$. □

Conversely, on an undirected graph we can prescribe the circuit ranges by placing reflecting boundaries. Recall the outer boundary ∂S and reflecting boundaries from Section 11.3.

Proposition 11.27 *Let $G = (V, E)$ be a connected, locally finite undirected graph, with each edge replaced by its two orientations. Let $S_1, S_2, \ldots \subset V$ be finite connected sets with $S_{n+1} \supseteq S_n \cup \partial S_n$ for every $n \geq 1$, and let $o \in S_1$ satisfy $\deg(o) \geq 1$. Then there are a rotor mechanism and an initial rotor configuration such that, for every $n \geq 1$, the nth circuit of the rotor walk from o traverses every edge incident to S_n exactly once in each direction and no other edge.*

Proof. We choose the rotors so that each S_n has a reflecting boundary and the rotors inside S_n point toward o; then we route the circuits one at a time.

Step 1: We choose the rotors. The sets ∂S_n are pairwise disjoint, because $\partial S_n \subseteq S_{n+1}$ while ∂S_m is disjoint from $S_m \supseteq S_{n+1}$ for $m > n$. At each $y \in \partial S_n$, place the edges from y into S_n consecutively in the cyclic order, and let the initial rotor at y be the edge just before this block. Let the rotors on $S_1 \setminus \{o\}$ form a directed forest whose paths lead to o, and let the rotors on $S_{n+1} \setminus (S_n \cup \partial S_n)$ form a directed forest whose paths lead to $S_n \cup \partial S_n$. The connectivity of S_{n+1} allows this choice. Choose all remaining rotors and cyclic orders arbitrarily.

Step 2: We prove by induction on n that, at time $T(n-1)$, the rotors on $S_n \setminus \{o\}$ form paths leading to o and the rotors on ∂S_n are in their initial positions, and that the nth circuit then uses exactly the asserted edges. For $n = 1$, the first part follows from Step 1, since $T(0) = 0$. Assume the first part at time $T(n-1)$. As in the proof of Lemma 11.17, a step from ∂S_n to $V \setminus S_n$ requires an edge to be traversed twice, so by Lemma 11.4 the walk returns to o for the $\deg(o)$th time, completing the nth circuit, before such a step. The circuit therefore stays in the finite set $S_n \cup \partial S_n$, and by Lemma 11.20 it is complete. By Lemma 11.21, it departs from every vertex $x \in S_n$ exactly $\deg(x)$ times, so it uses every edge with tail in S_n once. Each vertex of ∂S_n is entered once from each of its neighbors in S_n and, because of the reflecting order, the walk leaves it once toward each. Hence the circuit uses exactly the edges incident to S_n, once in each direction.

At time $T(n)$, the rotors on S_n are back in their positions at time $T(n-1)$, and the rotors on ∂S_n point into S_n. Together with the unvisited forest on $S_{n+1} \setminus (S_n \cup \partial S_n)$, they form paths from $S_{n+1} \setminus \{o\}$ leading to o. The circuit stayed in $S_n \cup \partial S_n \subseteq S_{n+1}$, so the rotors on ∂S_{n+1} are still in their initial positions. This completes the induction. □

Complete circuits thus make each vertex of the current range turn once per circuit, which slows the walk down. The next section turns this into a lower bound on the number of visited sites.

11.5 RANGE OF ROTOR WALK

Throughout this section, $G = (V, E)$ is an infinite, locally finite Eulerian digraph, and we fix a rotor mechanism, an initial rotor configuration ρ, and a starting vertex o with $\deg(o) \geq 1$. The **range** of the rotor walk at time t is the set $\mathsf{R}_t := \{X_0, \dots, X_t\}$ of vertices visited by time t. Recall from Section 11.4 the odometer u_t of (11.4), the completion time $T(n)$ of the nth circuit, the circuit ranges A_n, and the set $B(x, r)$ of vertices reachable from x by a directed path with at most r edges. Each complete circuit departs from every vertex x of the previous circuit range a full $\deg(x)$ times, so the walk returns to its start only after spending a long time in ever larger sets. On graphs of polynomial growth, this gives a lower bound on the range, valid for every initial rotor configuration.

Theorem 11.28 *Let $c, d > 0$ and $M \geq 1$ be real numbers. Let $G = (V, E)$ be an infinite Eulerian digraph in which every vertex has degree at most M, and suppose that*

$$\#B(x, r) \geq c r^d \qquad \text{for every } x \in V \text{ and every integer } r \geq 1 .$$

Then there is a constant $c_1 = c_1(c, d, M) > 0$ with the following property. For every starting vertex, every rotor mechanism, every initial rotor configuration and every integer $t \geq 1$, the range of the rotor walk satisfies

$$\#\mathsf{R}_t \geq c_1 t^{d/(d+1)} . \tag{11.6}$$

The exponent $d/(d+1)$ is smaller than 1, and for $d = 2$ it is $2/3$. Exercise 11.6 asks you to construct initial rotors on $\mathbb{Z}^2$ for which the range grows like $t^{2/3}$, so the exponent cannot be increased. Proposition 11.27, applied on $\mathbb{Z}^d$ with $S_n = [-n, n]^d \cap \mathbb{Z}^d$, shows the same in every dimension: the nth circuit then traverses each of the $2d(n+1)(2n+1)^{d-1}$ edges incident to S_n once in each direction, so the range after t steps has order $t^{d/(d+1)}$. We first prove a bound that involves the ball volumes and the degrees around the starting vertex only; the theorem then follows by substituting its hypotheses.

Theorem 11.29 *Let $G = (V, E)$ be an infinite, locally finite Eulerian digraph, and fix a rotor mechanism and an initial rotor configuration. Let o be a vertex with $\deg(o) \geq 1$, and consider the rotor walk from o. For integers $r \geq 0$, let*

$$w(r) := \sum_{x \in B(o,r)} \deg(x) \qquad \text{and} \qquad W(r) := \sum_{n=0}^{r-1} w(n) .$$

For integers $t \geq 1$, let $W^{-1}(t)$ be the smallest integer $r \geq 0$ with $W(r) > t$, and let $\deg_{\max}(t)$ be the largest degree of a vertex of $B(o, t)$. Then, for every integer $t \geq 1$:

(i) $u_t(o) \leq \deg(o) W^{-1}(t)$;

(ii) $u_t(x) \leq \deg(x) W^{-1}(t)$ *for every* $x \in V$;

(iii) the range satisfies

$$\#\mathsf{R}_t \geq \frac{t}{W^{-1}(t)\, \deg_{\max}(t)} . \tag{11.7}$$

Proof. We first bound the number of circuits completed by time t. If the walk completes r circuits, then, by Lemmas 11.25 and 11.23(ii), for each $0 \leq n < r$ the $(n+1)$st circuit departs from every vertex x of $B(o, n) \subseteq \mathsf{A}_n$ a full $\deg(x)$ times, so $T(r) \geq W(r)$. Hence the walk has completed fewer than $W^{-1}(t)$ circuits by time t.

(i) and (ii) By Lemma 11.20, each completed circuit and the unfinished circuit depart from a vertex x at most $\deg(x)$ times. Before time t, the walk has taken part in at most $W^{-1}(t)$ circuits, so $u_t(x) \leq \deg(x) W^{-1}(t)$. Taking $x = o$ gives (i).

(iii) By time t, the walk has visited only vertices of $\mathsf{R}_t \subseteq B(o,t)$. Summing (ii) over these vertices gives

$$t = \sum_{x \in \mathsf{R}_t} u_t(x) \le \#\mathsf{R}_t \, W^{-1}(t) \, \deg_{\max}(t) \,,$$

which proves (11.7). □

Proof of Theorem 11.28. Fix a starting vertex o. The volume hypothesis gives $w(n) \ge \#B(o,n) \ge cn^d$ for $n \ge 1$, and $w(0) \ge 1$. Comparing the sum with an integral gives a constant $c_2 = c_2(c,d) > 0$ with $W(r) \ge c_2 r^{d+1}$ for every integer $r \ge 1$, and hence a constant $C = C(c,d)$ with $W^{-1}(t) \le Ct^{1/(d+1)}$ for every integer $t \ge 1$. Substituting this bound and $\deg_{\max}(t) \le M$ into (11.7) gives

$$\#\mathsf{R}_t \ge \frac{t^{d/(d+1)}}{CM} \,,$$

which is (11.6) with $c_1 = 1/(CM)$. □

The same argument applies to graphs of exponential growth. If there are constants $c_0 > 0$ and $a > 1$ with $\#B(o,r) \ge c_0 a^r$ for every $r \ge 1$, then there is a constant C, depending only on c_0 and a, with $W^{-1}(t) \le C \log t$ for $t \ge 2$. If moreover every vertex has degree at most M, then this gives $\#\mathsf{R}_t \ge t/(CM \log t)$.

A transient walk eventually stays in one incomplete circuit, where it cannot repeat an edge, and this gives a much larger range.

Theorem 11.30 *Let $G = (V,E)$ be an infinite, locally finite Eulerian digraph, and fix a rotor mechanism and an initial rotor configuration ρ. Let o be a vertex with $\deg(o) \ge 1$, consider the rotor walk from o, and let $\deg_{\max}(t)$ be the largest degree of a vertex of $B(o,t)$.*

(i) If there are distinct vertices $x_0 = o, x_1, x_2, \ldots$ with $\rho(x_k) = (x_k, x_{k-1})$ for every $k \ge 1$, then the walk is transient.

(ii) If the first circuit is not complete by an integer time $t \ge 1$, then $\#\mathsf{R}_t \ge t/\deg_{\max}(t)$.

(iii) If the walk is transient, then there is an integer $t_0 \ge 0$ such that

$$\#\mathsf{R}_t \ge \frac{t - t_0}{\deg_{\max}(t)} \qquad \text{for every integer } t \ge 1 \,.$$

Proof. (i) Lemma 11.22 shows that the first circuit is incomplete (Figure 11.6(b) shows an example). By Lemma 11.20, the walk then departs from each vertex x at most $\deg(x)$ times in total, so it is transient.

(ii) Before time t, the walk is still in its first circuit, so Lemma 11.20 gives $u_t(x) \le \deg(x) \le \deg_{\max}(t)$ for every $x \in \mathsf{R}_t$. Summing over $x \in \mathsf{R}_t$ gives $t \le \#\mathsf{R}_t \, \deg_{\max}(t)$.

(iii) If every circuit were complete, then the walk would be recurrent, as noted after Lemma 11.25. Hence there is an incomplete circuit; let t_0 be the time at which it starts, when the walk is at o. Apply (ii) to the walk from time t_0 on, which is a rotor walk from o whose first circuit never completes, and use $B(o, t - t_0) \subseteq B(o, t)$. For $t \le t_0$, the asserted lower bound is at most zero. □

These bounds apply to every locally finite Eulerian digraph, whatever the initial rotors. The next section chooses the initial rotors at random on two directed lattices and proves that the walk is then recurrent.

11.6 ROTOR WALK ON DIRECTED LATTICES

Random walk on $\mathbb{Z}^2$ is recurrent. In this section we prove that rotor walk with random initial rotors is also recurrent on two directed versions of the square lattice, by coupling it with critical percolation. The **F-lattice** is $\mathbb{Z}^2$ with outgoing vertical edges at (x, y) when $x + y$ is even and outgoing horizontal edges when $x + y$ is odd. The **Manhattan lattice** is $\mathbb{Z}^2$ with even rows directed east, odd rows directed west, even columns directed south and odd columns directed north. Figure 11.9 shows both. They are Eulerian digraphs in which every vertex has two incoming and two outgoing edges, so each vertex has only one rotor mechanism, which alternates between the two outgoing edges. In **uniform rotor walk**, the initial rotors are independent, and each is uniform on the outgoing edges of its vertex.

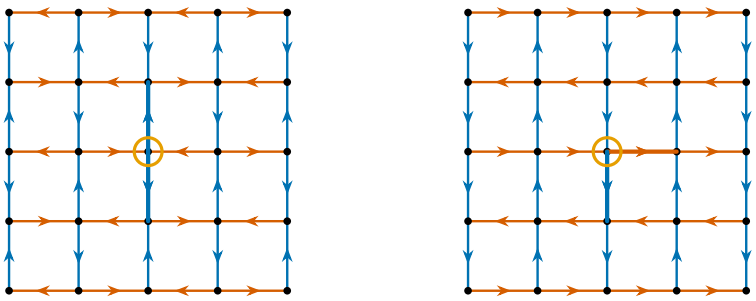


Figure 11.9: The F-lattice (left) and the Manhattan lattice (right), with vertical edges blue and horizontal edges vermilion. Each vertex has two incoming and two outgoing edges. The outgoing edges, thickened at one vertex of each lattice, are parallel in the F-lattice and perpendicular in the Manhattan lattice.

The coupling uses mirrors. The **half-dual lattice** $\mathbb{L}$ has vertex set

$$\{(x+\tfrac{1}{2}, y+\tfrac{1}{2}) : x, y \in \mathbb{Z} \text{ and } x+y \text{ is odd}\},$$

and its edges join vertices at Euclidean distance $\sqrt{2}$. Declare each edge of $\mathbb{L}$ open or closed with probability $\frac{1}{2}$ each, independently. Each $v \in \mathbb{Z}^2$ is the midpoint of a unique edge e_v of $\mathbb{L}$, and we use the state of e_v to place a two-sided mirror at v:

- on the Manhattan lattice, a closed edge e_v gives a mirror parallel to e_v, and an open edge gives no mirror;
- on the F-lattice, the mirror is parallel to e_v if e_v is closed, and perpendicular to e_v if e_v is open.

The **first-glance mirror walk** leaves $\mathbf{0}$ along an outgoing edge chosen uniformly at random, independently of the mirrors. When it visits a vertex $v \neq \mathbf{0}$ for the first time, it is reflected through a right angle by the mirror at v, or it continues straight if there is no mirror. Each first departure from a vertex sets the rotor there to the edge used, and later departures follow the rotor mechanism. Figure 11.10 shows the mirror assignments, and Figure 11.11 shows a sample path.

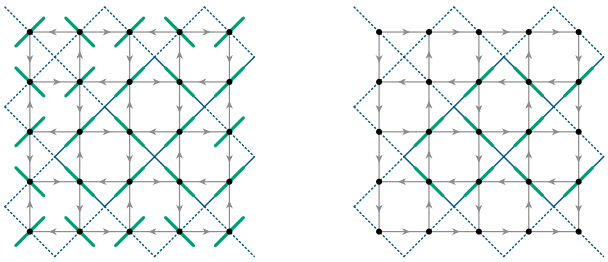


Figure 11.10: A sample of the percolation on the half-dual lattice $\mathbb{L}$ (closed edges solid, open edges dotted) and the mirrors (green) that it places on the F-lattice (left) and on the Manhattan lattice (right). Each vertex v is the midpoint of the edge e_v; on the F-lattice its mirror is parallel to e_v if e_v is closed and perpendicular if it is open, and on the Manhattan lattice only closed edges carry mirrors, parallel to them.

The mirrors are placed so that a first visit is equally likely to leave along either outgoing edge. The first-glance mirror walk is therefore uniform rotor walk in disguise.

Lemma 11.31 *On the Manhattan lattice and on the F-lattice, the first-glance mirror walk has the law of uniform rotor walk started at* $\mathbf{0}$.

Proof. When the walk first visits a vertex $v \neq \mathbf{0}$, the edge e_v has not been inspected before. Its two states send the walk along the two different outgoing edges of v, so the first departure from v is uniform given the past. Applying

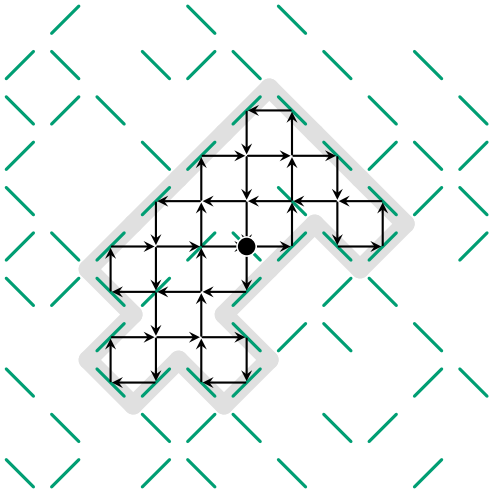


Figure 11.11: The first circuit of a first-glance mirror walk on the Manhattan lattice, which ends at the origin (large dot) after $T(1) = 44$ steps: first visits follow the two-sided mirrors (green), and later visits follow the rotor mechanism. The mirrors along a cycle of closed edges of $\mathbb{L}$ around the origin (shaded band) turn every first visit back inside, so the circuit ends before the walk crosses the cycle, as Lemma 11.32 asserts.

the inverse of the rotor mechanism at v to this edge gives the initial rotor at v, which is therefore uniform and independent of the past. The first departure from $\mathbf{0}$ has the same property, and later departures follow the rotor mechanism, as in uniform rotor walk. □

A cycle of closed edges around the origin acts like a reflecting boundary: the walk cannot cross it before completing a circuit. Figure 11.11 shows such a cycle around a first circuit.

Lemma 11.32 *Let $\Lambda_1, \dots, \Lambda_m$ be disjoint cycles of closed edges in $\mathbb{L}$ that surround $\mathbf{0}$, with Λ_j enclosed by Λ_{j+1} for $1 \le j < m$. If the first-glance mirror walk visits a vertex of $\mathbb{Z}^2$ in the unbounded component of $\mathbb{R}^2 \setminus \Lambda_m$, then it completes at least m circuits before that visit.*

Proof. We first show that the walk completes a circuit before it crosses a single closed cycle that it has not yet touched, and then we cross the nested cycles one at a time.

Step 1: Let Λ be a cycle of closed edges surrounding $\mathbf{0}$, and suppose that the walk has not yet visited a vertex v with $e_v \in \Lambda$. To cross Λ outward, the walk must depart from a vertex v with $e_v \in \Lambda$. The two incoming edges at v lie on opposite sides of e_v, and before the first outward crossing only the incoming edge on the bounded side can have been used. At the first visit to v, the mirror reflects the walk back inward, so an outward departure from v requires that incoming edge to be used twice. By Lemma 11.4, the walk has

then already traversed an edge from $\mathbf{0}$ twice, so it has completed a circuit before the crossing.

Step 2: For $1 \le j \le m$, let t_j be the time of the first visit to the unbounded component of $\mathbb{R}^2 \setminus \Lambda_j$. We prove by induction on j that $T(j) < t_j$, where, as in Section 11.4, $T(j)$ is the completion time of the jth circuit. For $j = 1$, this is Step 1. For the inductive step, suppose that $T(j-1) < t_{j-1}$. At time $T(j-1)$, the walk is at $\mathbf{0}$ and has not yet visited a vertex v with $e_v \in \Lambda_j$, since Λ_j lies outside Λ_{j-1}. Applying Step 1 to the walk restarted at time $T(j-1)$ gives $T(j) < t_j$. Taking $j = m$ proves the lemma. □

We use the following estimate from critical percolation: at every scale, a closed cycle surrounds the origin with probability bounded below. For $r > 0$, write $Q(\mathbf{0}, r) := [-r, r]^2$.

Theorem 11.33 *There is a constant $\delta > 0$ such that, for every integer $\ell \ge 1$, critical bond percolation on $\mathbb{L}$ contains a closed cycle that surrounds $\mathbf{0}$ and lies in $Q(\mathbf{0}, 3\ell) \setminus Q(\mathbf{0}, \ell)$ with probability at least δ.*

This is an estimate of Russo–Seymour–Welsh type, for annuli that are rotated by 45° relative to the edges of $\mathbb{L}$. Exercise 11.7 asks you to prove the theorem from the Harris inequality, planar duality and a lower bound, uniform in the size, on the probability that a rectangle of bounded aspect ratio has a path of open edges joining its two shorter sides; its solution gives the complete argument. Combined with Lemma 11.32, the theorem gives recurrence.

Theorem 11.34 *Uniform rotor walk on the Manhattan lattice and on the F-lattice is recurrent almost surely.*

The following quantitative bound on returns implies Theorem 11.34. Recall from (11.4) that $u_t(\mathbf{0})$ is the number of departures from $\mathbf{0}$ before time t.

Theorem 11.35 *There are constants $a, c > 0$ and $t_0 \ge 1$ with the following property. For uniform rotor walk started at $\mathbf{0}$ on the Manhattan lattice or on the F-lattice, and for every integer $t \ge t_0$,*

$$\mathbb{P}\{u_t(\mathbf{0}) < c \log t\} \le t^{-a}.$$

Proof. By Lemma 11.31, we may work with the first-glance mirror walk. Both lattices are infinite Eulerian digraphs of degree 2. Let $\delta > 0$ be the constant of Theorem 11.33, fix an integer $L \ge 1$ with $(1-\delta)^L \le 1/16$, and for an integer $t \ge 2$ let $J := \lfloor (\log t)/(8L \log 9) \rfloor$. We group the annuli $Q(\mathbf{0}, 3 \cdot 9^j) \setminus Q(\mathbf{0}, 9^j)$ for $1 \le j \le LJ$ into J groups of L. The annuli are

disjoint, so the events that the groups contain no closed cycle surrounding $\mathbf{0}$ are independent, and by Theorem 11.33 each has probability at most $(1-\delta)^L \le 1/16$.

Suppose that at least $J/2$ groups contain such a cycle. One cycle in each of $\lceil J/2 \rceil$ of them gives disjoint nested cycles in $Q(\mathbf{0}, 3 \cdot 9^{LJ})$, so by Lemma 11.32 the walk stays in this square until it completes $\lceil J/2 \rceil$ circuits. By Lemma 11.20, during these circuits it departs at most twice per circuit from each of the $(6 \cdot 9^{LJ} + 1)^2 \le 49 t^{1/4}$ lattice points of the square, so the circuits finish within $98 J t^{1/4} \le 98 t^{1/4} \log t < t$ steps for large t. Each completed circuit departs twice from $\mathbf{0}$, so $u_t(\mathbf{0}) \ge J$.

Hence if $u_t(\mathbf{0}) < J$, then there is a set of more than $J/2$ groups that all fail. Each such set fails with probability at most $16^{-J/2}$, and there are at most 2^J sets, so

$$\mathbb{P}\{u_t(\mathbf{0}) < J\} \le 2^J 16^{-J/2} = 2^{-J} .$$

Choose t_0 so that $98 t^{1/4} \log t < t$ and $J \ge (\log t)/(16L \log 9)$ for $t \ge t_0$. Take $c := 1/(16L \log 9)$ and $a := c \log 2$; since $J \ge c \log t$, the last bound proves the theorem. □

Proof of Theorem 11.34. For integers $m \ge 0$ and $t \ge t_0$ with $c \log t > m$, Theorem 11.35 gives

$$\mathbb{P}\Big\{\sup_{r \ge 0} u_r(\mathbf{0}) \le m\Big\} \le t^{-a} .$$

The left side is independent of t, so $t^{-a} \to 0$ forces it to be zero. By countable subadditivity over m, the walk departs from $\mathbf{0}$ infinitely often almost surely. The rotor at each vertex visited infinitely often sends the walk to every out-neighbor infinitely often, so strong connectivity gives recurrence. □

Experiments suggest that the range grows at different rates on the two lattices: of order $t^{2/3}$ on the Manhattan lattice and of order t on the F-lattice. Figure 11.12 illustrates this difference: on the Manhattan lattice the bands of equal increments of $t^{1/3}$ have nearly equal widths, as they would if the radius of the range grew like $t^{1/3}$. The difference remains unproved.

On transient graphs, the question changes from whether rotor walk returns to how often it escapes. The next section compares its escape rate with that of random walk.

11.7 ESCAPE RATE OF ROTOR WALK

Throughout this section, $G = (V, E)$ is a strongly connected digraph with at least two vertices and finite in-degree and out-degree at every vertex. We fix

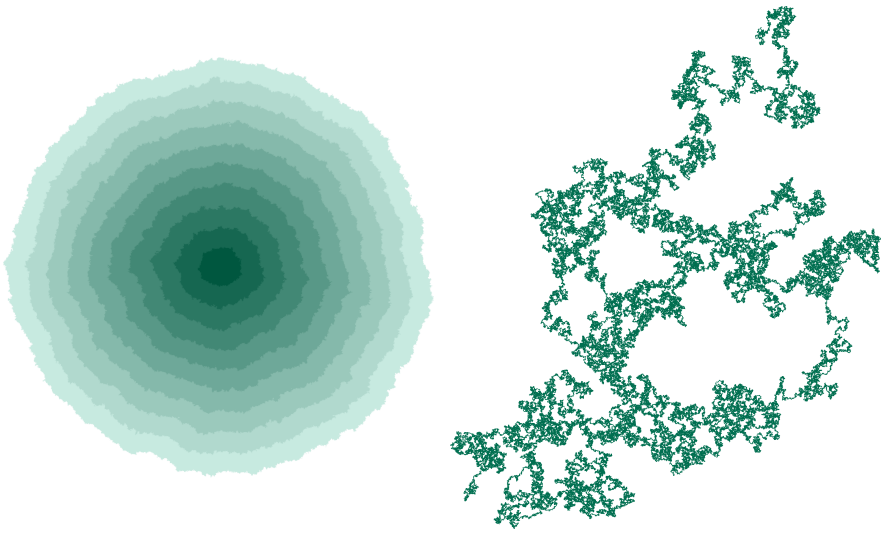


Figure 11.12: The range of uniform rotor walk on the Manhattan lattice after 5×10^{10} steps (left) and on the F-lattice after 10^7 steps (right), in windows of about $18{,}300 \times 17{,}800$ and $7{,}800 \times 9{,}500$ sites. On the left each visited site is shaded from dark to light by the cube root of its first visit time, in ten bands of equal width; on the right every visited site is dark green.

a vertex o, a rotor mechanism and initial rotors ρ. We will bound the escape rate of rotor walk by the escape probability of simple random walk, which chooses an outgoing edge uniformly at each step.

In the **infinite experiment**, release particles from o one at a time, stopping each at its first return at a positive time. If a particle visited a vertex infinitely often, then its rotor would send it along every outgoing edge infinitely often; strong connectivity would force a return to o. Thus a particle that never returns visits each vertex finitely often, so its rotors converge pointwise. Each new particle uses the final or limiting rotors of its predecessor. Let $I(\rho, n)$ count those among the first n particles that never return, and let $N_n(v)$ count all departures from $v \in V$ by the first n particles.

Theorem 11.36 *Let G be a strongly connected digraph with at least two vertices and with finite in-degree and out-degree at every vertex, and fix a vertex o, a rotor mechanism and an initial rotor configuration ρ. Let $(X_k)_{k \geq 0}$ be simple random walk on G started at o. Then*

$$\limsup_{n \to \infty} \frac{I(\rho, n)}{n} \leq \mathbb{P}_o\{X_k \neq o \text{ for every } k \geq 1\}. \tag{11.8}$$

Cyclic routing gives an inequality for the arrival counts. We compare it with the recurrence for expected visits of random walk stopped at its first return.

Proof of Theorem 11.36. For $v \in V$, let $Y_n(v)$ count arrivals at v by the first n particles, excluding their initial placements. These counts are finite by the construction above. Each particle leaves o once and leaves every other vertex whenever it arrives, so $N_n(o) = n$ and $N_n(v) = Y_n(v)$ for $v \neq o$. The arrivals at o are the returns, so $Y_n(o) = n - I(\rho, n)$.

Let $p(w, v)$ be the random-walk transition probability from w to v; sums over w below include only in-neighbors of v. Each outgoing edge of w carries $N_n(w)/\deg(w)$ departures up to an error less than one, since rotors are never reset. Summing over incoming edges at v gives

$$\left| Y_n(v) - n\,p(o, v) - \sum_{w \neq o} Y_n(w)\,p(w, v) \right| \leq \operatorname{indeg}(v)\,,$$

where $\operatorname{indeg}(v)$ is the in-degree of v. Let $\alpha(v) \coloneqq \liminf_{n\to\infty} Y_n(v)/n \in [0, \infty]$. Since the lower limit of a finite sum is at least the sum of its lower limits, dividing by n gives

$$\alpha(v) \geq p(o, v) + \sum_{w \neq o} \alpha(w)\,p(w, v) \qquad \text{for every } v \in V\,. \tag{11.9}$$

Next we compare α with expected visits up to the first return. Let $\tau_o^+ \coloneqq \inf\{k \geq 1 : X_k = o\}$, and for $m \geq 0$ let

$$g_m(v) \coloneqq \sum_{k=1}^{m} \mathbb{P}_o\{X_k = v,\ k \leq \tau_o^+\}\,.$$

For $k \geq 2$, an arrival at time $k \leq \tau_o^+$ comes from $w \neq o$. The Markov property therefore gives

$$g_{m+1}(v) = p(o, v) + \sum_{w \neq o} g_m(w)\,p(w, v)\,.$$

We prove by induction on m that $\alpha \geq g_m$ for every $m \geq 0$. The base case is $\alpha \geq 0 = g_0$. For the inductive step, if $\alpha \geq g_m$, then the nonnegativity of $p(w, v)$, the recurrence for g_{m+1}, and (11.9) give $\alpha \geq g_{m+1}$. At $v = o$ this reads

$$1 - \limsup_{n\to\infty} \frac{I(\rho, n)}{n} = \alpha(o) \geq g_m(o) = \mathbb{P}_o\{\tau_o^+ \leq m\}\,.$$

By continuity of probability along the increasing events $\{\tau_o^+ \leq m\}$, letting $m \to \infty$ proves (11.8). □

11.7.1 Comparison of the Finite and Infinite Experiments*

The proof of Theorem 11.36 did not need a finite version of the experiment. In the finite version below, the counts of escaping particles approach those of the infinite experiment as the ball grows. The rest of the chapter does not use these facts; Exercise 11.9 uses them.

We stop the particles on the boundary of a finite ball. Recall from Section 11.4 that $B(o, r)$ is the set of vertices reachable from o by a directed path with at most r edges, and from (11.3) that $\partial B(o, r)$ is the set of heads of the edges that leave $B(o, r)$. For an integer $r \geq 1$, start again from the rotors ρ and route n particles from o one at a time, each until it first returns to o or first visits $\partial B(o, r)$. We call this the **finite experiment**. Figure 11.13 shows its final rotors on $\mathbb{Z}^2$. Let $I_r(\rho, n)$ be the number of particles that visit $\partial B(o, r)$ in the finite experiment, and for $x \in V$ let $N_{n,r}(x)$ be the total number of departures from x made by its first n particles.

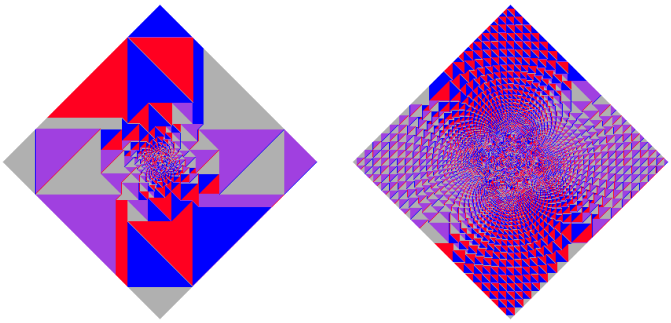


Figure 11.13: The rotors on $B(\mathbf{0}, 200) \subset \mathbb{Z}^2$ after the finite experiment with the clockwise mechanism and all initial rotors pointing west, colored by direction (North blue, East purple, South red, West gray), for $n = 3000$ particles (left), of which 694 reach $\partial B(\mathbf{0}, 200)$, and for $n = 30000$ (right), of which 7090 do.

To compare the two experiments, we count departures. A particle that never returns to o visits each vertex finitely often, so $N_n(x)$ is finite. A particle of the finite experiment that never stopped would never return to o and would stay forever in $B(o, r)$, which is finite because out-degrees are finite, so it would visit some vertex infinitely often. Hence every particle of the finite experiment stops, and $N_{n,r}(x)$ is finite.

Lemma 11.37 *Let G be a strongly connected digraph with at least two vertices and with finite in-degree and out-degree at every vertex, and fix a vertex o, a rotor mechanism and an initial rotor configuration ρ. For all integers $n, r \geq 1$, the departure counts in the finite and the infinite experiment satisfy $N_{n,r} \leq N_n$ at every vertex.*

Proof. Suppose, for a contradiction, that $N_{n,r}(x) > N_n(x)$ for a vertex x, and consider the first departure of the finite experiment after which its count

at the tail x of the departure exceeds $N_n(x)$. In both experiments each particle departs from o exactly once, so $x \neq o$. Before this departure, every vertex has departed at most as often as in the infinite experiment. Each vertex sends particles along the same sequence of edges in both experiments, so x has received at most as many particles as it receives in the infinite experiment, namely $N_n(x)$, because every particle that enters $x \neq o$ in the infinite experiment also leaves it. But x has already departed $N_n(x)$ times, so no particle is present at x, a contradiction. □

The number of particles that return to o is the number of particles sent along the edges into o. Each vertex sends particles along the same sequence of edges in both experiments, so Lemma 11.37, applied at the finitely many in-neighbors of o, shows that at most as many particles return in the finite experiment as in the infinite one. In the finite experiment each particle either returns or is counted by $I_r(\rho, n)$, and in the infinite experiment each either returns or is counted by $I(\rho, n)$. Hence

$$I(\rho, n) \leq I_r(\rho, n) \qquad \text{for all integers } n, r \geq 1 \,.$$

Two further properties describe how the finite experiment depends on r. First, for fixed n, the finite experiment agrees with the infinite one near o once the ball is large.

Lemma 11.38 *Let G be a strongly connected digraph with at least two vertices and with finite in-degree and out-degree at every vertex, and fix a vertex o, a rotor mechanism and an initial rotor configuration ρ. For every integer $n \geq 0$ and every finite set $K \subset V$, the departure counts in the finite and the infinite experiment satisfy $N_{n,r} = N_n$ on K for all sufficiently large r.*

Proof. We argue by induction on n. For $n = 0$, both counts vanish. For the inductive step, let $n \geq 1$ and assume the claim for $n - 1$ and every finite set. In the infinite experiment, the nth particle departs from K only finitely often. Choose a finite initial segment of its route that contains all these departures, and let L be the union of K and the vertices of this segment. By the inductive hypothesis, applied to L, the two experiments have made the same numbers of departures from every vertex of L after $n - 1$ particles, for all large r. Their rotors on L therefore agree at that time, and if also $L \subseteq B(o, r)$, then the nth particle of the finite experiment follows the chosen segment. Hence $N_{n,r} \geq N_n$ on K for all large r, and Lemma 11.37 gives equality. □

Applying this lemma near o shows that, for each fixed number of particles, the finite experiment counts the escapes exactly once the ball is large.

Lemma 11.39 *Let G be a strongly connected digraph with at least two vertices and with finite in-degree and out-degree at every vertex, and fix a vertex*

o, a rotor mechanism and an initial rotor configuration ρ. For every integer $n \geq 1$, the numbers of escaping particles in the finite and the infinite experiment satisfy $I_r(\rho, n) = I(\rho, n)$ for all sufficiently large r.

Proof. The number of particles that return to o is determined by the numbers of departures from the in-neighbors of o, because each vertex sends particles along a fixed sequence of edges. Lemma 11.38, applied to the finite set of in-neighbors of o, shows that these numbers agree in the two experiments for all large r. Hence the same number of particles returns, and $I_r(\rho, n) = I(\rho, n)$. □

The next two sections prove recurrence on undirected graphs by ruling out infinite live paths, whose internal vertices send the walk onward before sending it back. This gives recurrence of uniform rotor walk on graphs of maximum degree three and on $\mathbb{Z}^2$ with the clockwise mechanism. The final section studies the range on the comb.

11.8 LIVE PATHS AND RECURRENCE

Throughout this section, G is a connected, locally finite, simple undirected graph with at least two vertices, each edge carrying both orientations, and we fix a rotor mechanism and initial rotors. The **departure order** at a vertex v is the cyclic order of its outgoing edges, starting immediately after the initial rotor, and we write $r_v(w)$ for the position of the edge (v, w) in this order. The walk leaves v along these edges in departure order, one edge per visit.

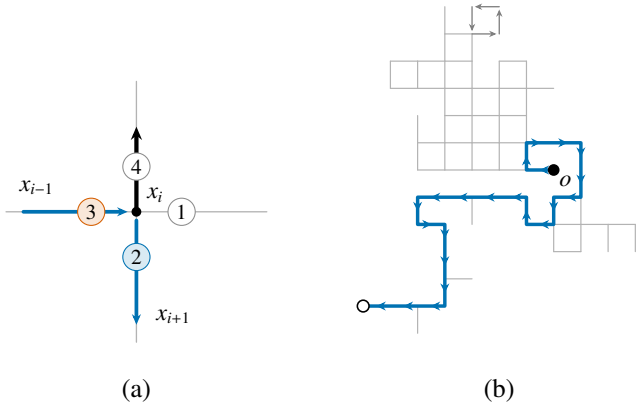
A finite or infinite path $x_0, x_1, \ldots$ of distinct vertices is **live** if

$$r_{x_i}(x_{i+1}) < r_{x_i}(x_{i-1})$$

at every internal vertex x_i. In words, each internal vertex sends the walk onward along the path before it sends it back. The absence of infinite live paths gives a criterion for recurrence. If the walk never returns to its start, then it crosses each edge at most once in each direction, and what remains after cancelling the opposite crossings contains an infinite live path. Figure 11.14 shows the live condition at one vertex, and a live path found in this way for a walk on $\mathbb{Z}^2$ stopped before it returns to its start.

Proposition 11.40 *Let G be a connected, locally finite, simple undirected graph with at least two vertices. Fix a rotor mechanism and initial rotors such that there is no infinite live path. Then rotor walk from every vertex is recurrent and completes every circuit.*

Proof. We first show that the walk returns to its starting vertex o. Suppose that it never returns. By Lemma 11.4, the first directed edge that the walk



(a) (b)

Figure 11.14: (a) With the clockwise mechanism and the initial rotor at x_i pointing North (bold), the departure order (numbers) is East, South, West, North, and the path is live at x_i because $r_{x_i}(x_{i+1}) = 2 < 3 = r_{x_i}(x_{i-1})$. (b) A rotor walk on $\mathbb{Z}^2$ from o with the clockwise mechanism and independent uniform initial rotors, stopped after 138 steps without returning to o: cancelling the edges crossed in both directions (gray) leaves crossings (arrows) that contain a live path (blue) from o to the walker (circle).

traverses twice leaves o. Since the walk never returns to o, no directed edge is traversed twice, and every vertex is visited finitely often.

Cancel the opposite traversals of each edge. In the directed graph of the remaining traversals, the number of outgoing edges minus the number of incoming edges is 1 at o and 0 at every other vertex. Infinitely many vertices are reachable from o in this graph. Otherwise, summing these differences over the finite set of reachable vertices gives 1, whereas no edge of the graph leaves that set, so the sum is at most 0.

Consequently this graph contains an infinite simple directed path from o. Indeed, the finite simple directed paths from o have unbounded lengths and finitely many extensions at each step, so we can repeatedly choose an extension that has arbitrarily long continuations. At each internal vertex of the resulting path, the walk traversed the outgoing edge of the path but not the reverse of its incoming edge. The departures from a vertex use an initial segment of its departure order, so the outgoing edge precedes the reverse of the incoming edge, and the path is live. This contradicts the hypothesis, so the walk returns to o.

Changing finitely many rotors does not create an infinite live path, because an infinite simple path has a tail that avoids the changed vertices. After each return to o, only finitely many rotors have changed, so the same argument shows that the walk returns again. Hence the walk returns to o infinitely often and completes every circuit. A vertex visited infinitely often sends the walk to each of its neighbors infinitely often, so by connectedness every vertex is visited infinitely often, and the walk is recurrent. □

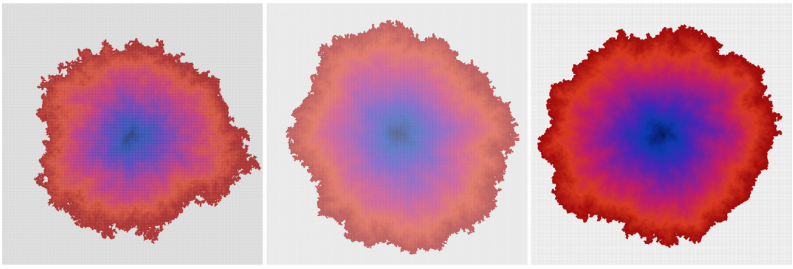


Figure 11.15: The sites visited by uniform rotor walk with the clockwise mechanism on the square, honeycomb and square–octagon lattices after 4×10^6 steps, each colored by the circuit in which it was first visited, from blue for the early circuits to red for the late ones. The last two lattices have degree three, so these walks are recurrent by Corollary 11.41, and the first is recurrent by Theorem 11.42.

Next we rule out infinite live paths for random initial rotors.

11.9 RECURRENCE OF UNIFORM ROTOR WALK

Recall from Section 11.6 that in uniform rotor walk the initial rotors are independent, and each is uniform among the outgoing edges of its vertex. We show that uniform rotor walk has almost surely no infinite live path, and is therefore recurrent, on every graph of maximum degree at most three and on $\mathbb{Z}^2$ with the clockwise mechanism. At a vertex of degree at most three, a fixed arrival edge has a uniform position in the initial departure order, so a path arriving along that edge has on average at most one live continuation. We explore the live continuations with a queue, and comparing its length with a lazy symmetric walk shows that the exploration ends. Figure 11.15 shows uniform rotor walk on the square lattice and on two lattices of degree three.

Corollary 11.41 *Let G be a connected, simple undirected graph with at least two vertices and maximum degree at most three. Fix a rotor mechanism and take independent uniform initial rotors. Then almost surely there is no infinite live path, and rotor walk from every vertex is recurrent.*

Proof. Mark a fixed edge and queue its two endpoints. To process the first vertex v in the queue, remove all its entries, examine the rotor at v if it has not been examined, and mark every unmarked edge at v that precedes a marked edge at v in the departure order. Then queue the other endpoints of the newly marked edges. A marked edge queues each of its endpoints once, except the vertex whose processing marked it, so v receives at most $\deg(v)$ entries and is processed at most $\deg(v)$ times.

Step 1: We show that the queue empties almost surely. Let $U_1, U_2, \ldots$

be independent uniform random variables on $[0, 1)$, and let $Z_i := \lfloor 3U_i \rfloor - 1$, which is uniform on $\{-1, 0, 1\}$. Let v be the ith vertex processed for the first time, and use U_i to sample the rotor at v. At this processing, the k removed entries correspond exactly to the marked edges at v. If $k = 1$, the position of the marked edge in the departure order is uniform, so we may sample it as $\lfloor \deg(v)U_i \rfloor + 1$, and the queue length changes by $\lfloor \deg(v)U_i \rfloor - 1 \le Z_i$. If $k \ge 2$, at most $\deg(v) - k$ edges are newly marked, so it changes by at most $\deg(v) - 2k \le -1 \le Z_i$. At a later processing of v, there is an old marked edge and a new one that queued v, so at most one entry is added, at least one is removed, and the queue length does not increase. Thus after m first processings the queue length is at most $2 + Z_1 + \cdots + Z_m$, a lazy symmetric walk started at 2.

For $j \ge 0$ let h_j be the probability that this walk started at j reaches 0. Then $h_0 = 1$, and conditioning on the first step gives $2h_j = h_{j-1} + h_{j+1}$ for $j \ge 1$. So h_j is affine in j, and being bounded, it is constant, equal to $h_0 = 1$. If the queue never emptied, there would be infinitely many first processings, because each vertex is processed finitely often, and the walk would stay positive forever, which has probability 0. Hence the queue empties almost surely, after finitely many edges have been marked.

Step 2: We show that live paths from the fixed edge use only marked edges. Processing v makes the marked edges at v an initial segment of its departure order, and each marking of an edge at v by a neighbor queues v. So when the queue is empty, the marked edges at every vertex form an initial segment of its departure order. Let $x_0, x_1, \ldots$ be a live path whose first edge is the fixed edge, which is marked. If $\{x_{i-1}, x_i\}$ is marked for $i \ge 1$, then so is $\{x_i, x_{i+1}\}$, because $r_{x_i}(x_{i+1}) < r_{x_i}(x_{i-1})$. By induction on i, the path uses only marked edges, so it is finite (Figure 11.16). The graph has countably many edges, since it is connected and locally finite. Hence almost surely there is no infinite live path, and Proposition 11.40 completes the proof. □

This applies to every graph of maximum degree at most three, periodic or not. On the square lattice, where vertices have degree four, a path has on average $3/2$ live continuations, and we use cycles that surround the origin instead.

11.9.1 The Square Lattice*

On $\mathbb{Z}^2$ with the clockwise mechanism, a live path cannot cross a suitable clockwise cycle from inside to outside. We find such cycles around every finite set by transferring the percolation cycles of Theorem 11.33 to the rotors.

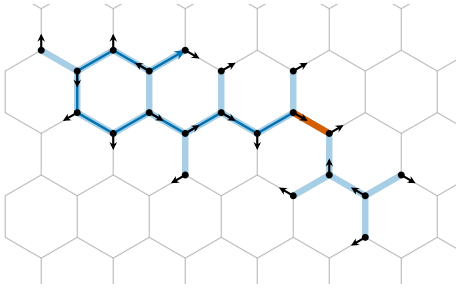


Figure 11.16: The exploration in the proof of Corollary 11.41 on the honeycomb lattice with the clockwise mechanism and independent uniform initial rotors (black arrows, at the explored vertices): from the fixed edge (vermilion), the queue empties after 20 further edges are marked (light blue). Every live path whose first edge is the fixed edge uses only marked edges, and the blue arrows continue a longest one.

Theorem 11.42 *Equip $\mathbb{Z}^2$ with the clockwise rotor mechanism and independent uniform initial rotors. Then almost surely there is no infinite live path, and rotor walk from every vertex is recurrent.*

Proof. We use only the outgoing edges in positions 1 and 4 of the departure order at each vertex. For a uniform initial rotor these are one horizontal edge and one vertical edge, each direction chosen by a fair coin, the two coins independent, and the choices at different vertices independent. We show that, almost surely, infinitely many clockwise cycles of such edges surround the origin, and that a live path cannot leave the region bounded by such a cycle.

Step 1: We find surrounding clockwise cycles of these edges. The affine map

$$(s,t) \longmapsto \Big(\frac{s-t+1}{2}, \frac{s+t}{2}\Big)$$

identifies the half-dual lattice $\mathbb{L}$ of Section 11.6 with $\mathbb{Z}^2$. Recall that $Q(\mathbf{0},\ell) = [-\ell,\ell]^2$; the map takes this square to the diamond $\{(x,y) : |x-\frac{1}{2}| + |y| \le \ell\}$. For integers $j \ge 1$, consider the image of the annulus

$$Q(\mathbf{0}, 3\cdot 9^j) \setminus Q(\mathbf{0}, 9^j)\,.$$

These annuli are disjoint, and their inner diamonds exhaust the plane. Theorem 11.33 gives a constant $\delta > 0$, independent of j. Critical bond percolation is unchanged when open and closed bonds are exchanged, so each of these annuli contains, with probability at least δ, a simple open bond cycle that surrounds its inner diamond.

Fix one annulus. We pass from fair bonds to rotor edges in two replacements. Start from independent fair bonds on the edges incident to the annulus, and orient each open bond in both directions. First, replace the two identical

bits of each bond, one for each orientation, by two independent fair bits. Then, at each vertex of the annulus, replace its two opposite horizontal bits by two complementary fair bits, and likewise its two vertical bits. After both replacements, the outgoing edges chosen at each vertex have exactly the law of its edges in positions 1 and 4.

Neither replacement decreases the probability that the annulus contains a simple clockwise cycle that surrounds its inner diamond. To see this, condition on all bits outside one pair being replaced. A simple cycle uses at most one bit of the pair, so the conditional event either is constant, or requires one particular bit, or requires at least one of the two bits. A particular bit has probability $1/2$ before and after the replacement. The probability that at least one of the two bits is set increases from $1/2$ to $3/4$ in the first replacement and from $3/4$ to 1 in the second. Hence, with probability at least δ, the annulus contains such a cycle of edges in positions 1 and 4. These events depend on the rotors in disjoint sets of vertices, so they are independent, and the Borel–Cantelli lemma gives infinitely many such cycles almost surely.

Step 2: We show that a live path cannot leave the region bounded by such a cycle. At a vertex of such a cycle, let v be its predecessor and w its successor on the cycle. The clockwise order of the directions at the vertex is

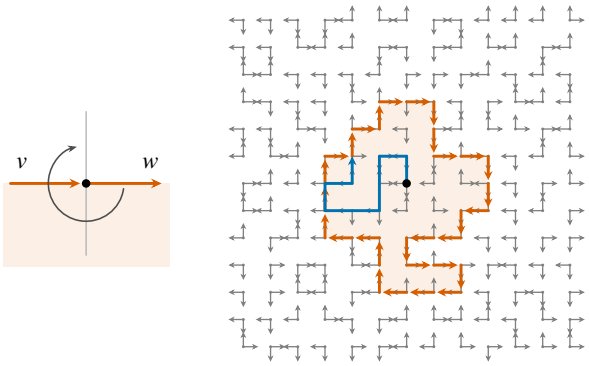
$$w, \quad \text{the directions inside the cycle}, \quad v, \quad \text{the directions outside the cycle}.$$

Since w is in position 1 or 4 of the departure order, every inside direction precedes v, and v precedes every outside direction. A live path that arrives at the vertex from inside, or from v, must therefore leave toward the inside or toward w. So a live path that starts strictly inside the cycle stays in the closed region it bounds, which contains only finitely many vertices (Figure 11.17). The cycles surround every finite set, so one of them surrounds the first edge of a proposed infinite live path, a contradiction. Hence there is almost surely no infinite live path, and Proposition 11.40 gives recurrence. □

We now turn to the comb, where a finite tree traversal and concentration give both the limit shape and the growth of the range.

11.10 UNIFORM ROTOR WALK ON THE COMB

Throughout this section, $d \geq 2$ is an integer. On the comb in $\mathbb{Z}^d$ defined below, the ball of radius r about the root contains of order r^d vertices, so Theorem 11.29 shows that rotor walk from the root visits at least of order $t^{d/(d+1)}$ sites in t steps. We show that, with suitable cyclic orders, uniform rotor walk attains this order and its range has a diamond limit shape; Exercise 11.11 treats other cyclic orders. The comb is a tree, and on a tree the walk is easy to



(a) (b)

Figure 11.17: (a) In Step 2 of the proof of Theorem 11.42, at a vertex of a clockwise cycle (vermilion) with predecessor v and successor w, the directions in clockwise order (gray arrow) are w, the inside direction (shaded), v and the outside direction. (b) The edges in positions 1 and 4 (gray) of independent uniform initial rotors near the origin (dot), a clockwise cycle of 30 of them around it, and a longest live path from the origin (blue); every live path from the origin stays in the shaded region.

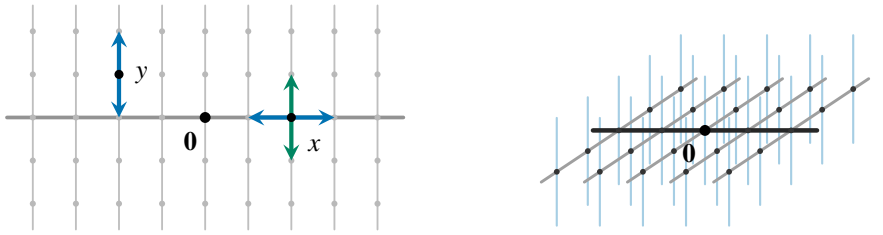
follow. A newly visited vertex sends the walk to the children that precede its parent in the initial departure order and then back to its parent; on each later visit from its parent, it sends the walk to all its children. We use this to label each vertex by the circuit in which it is first visited. Under the assumption on cyclic orders below, each nonroot label is one plus a sum of independent indicators, all but at most $d - 1$ of which are fair coins. Their concentration gives both the completion of every circuit and the limit shape.

For $x \in \mathbb{Z}^d \setminus \{\mathbf{0}\}$, let $j(x)$ be the largest index with $x_{j(x)} \neq 0$, and let

$$\mathrm{par}(x) := x - \mathrm{sgn}(x_{j(x)})\, \mathbf{e}_{j(x)} .$$

The **comb** is the graph on $\mathbb{Z}^d$ that joins each $x \neq \mathbf{0}$ to $\mathrm{par}(x)$. Each step from x to $\mathrm{par}(x)$ decreases $\|x\|_1$ by one, so following these steps from x leads to $\mathbf{0}$, and a cycle would contain a vertex of largest norm with two parents. Thus the comb is a tree with root $\mathbf{0}$ and parent map par, and its graph distance from $\mathbf{0}$ to x is $\|x\|_1$. In the plane it has a horizontal spine with a vertical tooth at each spine vertex. The children of $x \neq \mathbf{0}$ are $2x - \mathrm{par}(x)$ and $x \pm \mathbf{e}_k$ for $j(x) < k \leq d$, and the children of the root are its $2d$ lattice neighbors. Along a path away from the root, the coordinate direction changes at most $d - 1$ times. Figure 11.18 shows these edges.

Since the comb distance from $\mathbf{0}$ to x is $\|x\|_1$, the ball $B(\mathbf{0}, R)$ of Section 11.4 is the lattice diamond $\{x \in \mathbb{Z}^d : \|x\|_1 \leq R\}$, and we use this description for every real R, so $B(\mathbf{0}, R) = \varnothing$ for $R < 0$. Comparing unit cubes centered at its



(a) (b)

Figure 11.18: (a) The comb in $\mathbb{Z}^2$. At y and at x, blue arrows point toward the parent and toward the child that continues the same line; green arrows begin the tooth through the spine vertex x. (b) The comb in $\mathbb{Z}^3$, with a tooth in direction $\mathbf{e}_3$ attached to each vertex of the planar comb.

points with real diamonds whose radii differ by $d/2$ gives

$$\#B(\mathbf{0}, R) = \frac{2^d}{d!} R^d + O((R+1)^{d-1}) \qquad \text{for } R \geq 0\,,$$

since the real diamond of radius R has volume $2^d R^d/d!$. Recall from Section 11.4 that a circuit ends when the rotor at the root has made a full turn and the walker has returned to the root, that $T(n)$ is the completion time of the nth circuit, and that A_n is the set of vertices it visits, with $T(0) = 0$ and $\mathsf{A}_0 = \{\mathbf{0}\}$. The departure order at a vertex lists its outgoing edges starting just after the initial rotor and ending with it.

We now define the labels. Let $\mathsf{h}(\mathbf{0}) := 0$, and for $x \neq \mathbf{0}$ with root path $\mathbf{0} = v_0, v_1, \ldots, v_r = x$, let

$$\mathsf{h}(x) := 1 + \sum_{i=1}^{r-1} \mathbf{1}\{(v_i, v_{i-1}) \text{ precedes } (v_i, v_{i+1}) \text{ in the departure order at } v_i\}\,. \tag{11.10}$$

These numbers are read off the initial rotors, whether or not the walk ever returns. The neighbors of the root have label 1, and every other child has the label of its parent or that label plus one, so the labels are nondecreasing along paths away from the root. The next lemma shows that, as long as its sublevel sets are finite, $\mathsf{h}(x)$ is the index of the circuit in which x is first visited.

Lemma 11.43 *Let $d \geq 2$ be an integer. Fix a rotor mechanism and initial rotors on the comb in $\mathbb{Z}^d$, and let $n \geq 1$ be an integer such that the set $\{x : \mathsf{h}(x) \leq n\}$ is finite. Then the first n circuits are complete,*

$$\mathsf{A}_k = \{x : \mathsf{h}(x) \leq k\} \qquad \textit{for } 0 \leq k \leq n\,, \tag{11.11}$$

at the completion of circuit n every visited vertex other than the root has its

rotor pointing to its parent, and

$$T(n) = 2\sum_{k=1}^{n}(\#\mathsf{A}_k - 1)\,. \tag{11.12}$$

Proof. We prove by induction on k, for $0 \le k \le n$, that circuit k is complete, that it traverses each edge of the finite tree $\{\mathsf{h} \le k\}$ once in each direction and no other edge, and that at its completion every visited vertex other than the root has its rotor pointing to its parent. The case $k = 0$ is immediate, since then only the root has been visited.

For the inductive step, let $1 \le k \le n$ and assume the claim for $k-1$. To complete circuit k, the root must send the walk once to each of its children and receive it back after each departure. At a vertex $v \neq \mathbf{0}$ reached from its parent during this circuit, there are two cases. If $\mathsf{h}(v) < k$, then by the inductive hypothesis its rotor points to its parent, so its departures up to the next parent edge list all its children once, and each child has label at most $\mathsf{h}(v) + 1 \le k$. If $\mathsf{h}(v) = k$, then v has not been visited before, its rotor is untouched, and its departures before the parent edge go exactly to the children with label k; its other children have label $k+1$. In both cases the departures of v before its parent edge go exactly to its children in $\{\mathsf{h} \le k\}$, once each. We now induct on the height of the finite subtree of $\{\mathsf{h} \le k\}$ rooted at v. If v is a leaf of this subtree, then its next departure is to its parent. For the inductive step, each excursion from v into a child crosses the edge to that child and, by the height induction hypothesis, traverses the child's subtree once in each direction before returning to v. After these excursions, v sends the walk to its parent, with its rotor pointing there. Consequently the root completes its turn, circuit k is complete, its range is $\{\mathsf{h} \le k\}$, which proves the claim for k.

Finally, the finite tree A_k has $\#\mathsf{A}_k - 1$ edges, so circuit k has length $2(\#\mathsf{A}_k - 1)$, and summing over k gives (11.12). □

Figure 11.19 shows the labels for a sample of initial rotors.

To show that the sublevel sets of h are finite and close to diamonds, we use the following concentration estimate for sums of bounded independent variables.

Lemma 11.44 *Let $N \ge 1$ be an integer, and let Y be a sum of N independent random variables with values in $[0,1]$. Then, for every real $u \ge 0$,*

$$\mathbb{P}\{|Y - \mathbb{E}Y| > u\} \le 2e^{-u^2/(4N)}\,.$$

Proof. We bound exponential moments and apply Markov's inequality. Let $Z_1, \dots, Z_N$ be the independent centered summands of Y. Then $Y - \mathbb{E}Y = \sum_{i=1}^{N} Z_i$, and each Z_i has mean zero and absolute value at most one.

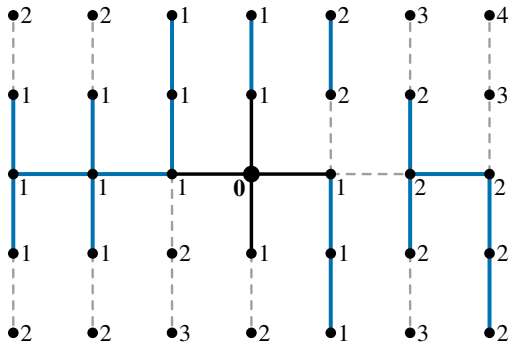


Figure 11.19: The labels $\mathsf{h}(x)$ for a sample of uniform initial rotors on the planar comb with the clockwise mechanism. Away from the root, a blue edge to a child precedes the parent edge in the departure order of its tail and gives its endpoints equal labels; a dashed edge increases the label by one. The root has label 0 and its four neighbors have label 1.

If $u \geq N$, then the left side vanishes, because $|Y - \mathbb{E}Y| \leq N$, so suppose $u < N$. For $|z| \leq 1$ and $0 \leq \lambda \leq 1$, the exponential series gives $e^{\lambda z} \leq 1 + \lambda z + \lambda^2$, because the terms of degree at least two have total absolute value at most $\lambda^2 \sum_{k \geq 2} 1/k! < \lambda^2$. Hence $\mathbb{E}e^{\lambda Z_i} \leq 1 + \lambda^2 \leq e^{\lambda^2}$. Take $\lambda := u/(2N) < 1/2$. By independence and Markov's inequality,

$$\mathbb{P}\{Y - \mathbb{E}Y > u\} \leq e^{-\lambda u} \prod_{i=1}^{N} \mathbb{E}e^{\lambda Z_i} \leq e^{N\lambda^2 - \lambda u} = e^{-u^2/(4N)} .$$

The same argument applied to $-Z_i$ bounds $\mathbb{P}\{Y - \mathbb{E}Y < -u\}$, and adding the two bounds proves the lemma. □

Unless we say otherwise, the initial rotors are independent and uniform, and at each vertex $x \neq \mathbf{0}$ the cyclic order places the parent edge half a turn from the edge $(x, 2x - \mathrm{par}(x))$. The clockwise mechanism on the planar comb has this property. The parent edge then has a uniform position k in the departure order, with $1 \leq k \leq \deg(x)$, and the edge $(x, 2x - \mathrm{par}(x))$ comes before it exactly when $k > \deg(x)/2$. Hence

$$\mathbb{P}\{(x, 2x - \mathrm{par}(x)) \text{ precedes } (x, \mathrm{par}(x))\} = \frac{1}{2} , \tag{11.13}$$

and these events are independent at distinct vertices.

Theorem 11.45 (Diamond Limit Shape on the Comb) *Let $d \geq 2$ be an integer and $a > 0$ real. Run rotor walk on the comb in $\mathbb{Z}^d$ from the root, with independent uniform initial rotors, and suppose that at each vertex x other than the root the cyclic order places the parent edge half a turn from the edge $(x, 2x - \mathrm{par}(x))$. Then:*

(a) *Almost surely every circuit is complete.*

(b) *There are constants $c, C > 0$, depending only on d and a, such that for every integer $n \geq 2$, with probability at least $1 - Cn^{-a}$,*

$$B(\mathbf{0}, 2n - c\sqrt{n \log n}) \subseteq \mathsf{A}_n \subseteq B(\mathbf{0}, 2n + c\sqrt{n \log n})$$

and

$$\left| T(n) - \frac{2^{2d+1}}{(d+1)!} n^{d+1} \right| \leq c n^{d+1/2} \sqrt{\log n}\,.$$

In other words, after n circuits the walk has visited, up to an error of order $\sqrt{n \log n}$, exactly the vertices within graph distance $2n$ of the root. This is twice the radius that Lemma 11.25 guarantees for arbitrary initial rotors. The proof reads both statements off the labels: concentration makes $\{\mathsf{h} \leq n\}$ finite and close to $B(\mathbf{0}, 2n)$, and Lemma 11.43 identifies it with A_n.

Proof. (a) **Step 1:** We estimate the labels. Fix an integer $n \geq 2$. For $x \neq \mathbf{0}$, write $r := \|x\|_1$. By (11.10), $\mathsf{h}(x)$ is a sum of r independent $[0,1]$-valued variables, counting the constant 1. Apart from this constant and at most $d - 1$ changes of coordinate direction, these variables are fair coins by (11.13), so $|\mathbb{E}\mathsf{h}(x) - r/2| \leq d$. Choose $C_1 > 0$ with $C_1^2 \geq 24(a + d)$, and let $b := C_1 \sqrt{n \log n} + d$. For $1 \leq r \leq 6n$, Lemma 11.44 gives

$$\mathbb{P}\{|\mathsf{h}(x) - r/2| > b\} \leq 2 \exp\Big(-\frac{C_1^2 n \log n}{4r}\Big) \leq 2n^{-C_1^2/24}\,.$$

The ball $B(\mathbf{0}, 6n)$ has $O(n^d)$ vertices, and the root satisfies the bound trivially, so a union bound gives

$$\mathbb{P}\Big\{ \max_{x \in B(\mathbf{0}, 6n)} \Big| \mathsf{h}(x) - \frac{\|x\|_1}{2} \Big| \leq b \Big\} \geq 1 - Cn^{-a}\,. \tag{11.14}$$

Step 2: We show that this event gives the completion of the first n circuits. For large n we have $1 \leq b \leq n$. On the event in (11.14), every vertex of norm $6n$ has $\mathsf{h}(x) \geq 3n - b \geq 2n > n$, and since the labels are nondecreasing along root paths, $\{\mathsf{h} \leq n\} \subseteq B(\mathbf{0}, 6n)$. By Lemma 11.43, the first n circuits are complete. Hence, for each fixed $m \geq 1$, the probability that $\{\mathsf{h} \leq m\}$ is infinite is at most Cn^{-a} for every large $n \geq m$, so it is zero. A countable union over m shows that almost surely every sublevel set of h is finite, and Lemma 11.43 then shows that every circuit is complete. This proves (a).

(b) **Step 1:** We obtain the ranges of the first n circuits. On the event in (11.14), the identity (11.11) and the bound on h give, for $0 \leq k \leq n$,

$$B(\mathbf{0}, \max\{2k - 2b, 0\}) \subseteq \mathsf{A}_k \subseteq B(\mathbf{0}, 2k + 2b)\,.$$

For $k < b$ the inner inclusion only says that the root is visited; otherwise both inclusions follow by comparing $\mathsf{h}(x)$ with k. The case $k = n$ is the asserted shape estimate.

Step 2: We sum the lengths of the circuits. All the radii above are at most $4n$, so the count of $\#B(\mathbf{0}, R)$ and the mean value theorem for R^d give

$$\#\mathsf{A}_k = \frac{2^{2d}}{d!}k^d + O(bn^{d-1}) \qquad \text{for } 1 \le k \le n\,.$$

By (11.12) and $\sum_{k=1}^{n} k^d = n^{d+1}/(d+1) + O(n^d)$,

$$T(n) = \frac{2^{2d+1}}{(d+1)!}n^{d+1} + O(bn^d)\,.$$

Since $b = O(\sqrt{n \log n})$, this proves both bounds for all large n, and increasing C makes the asserted probability bound trivial for the finitely many remaining $n \ge 2$. □

The theorem also determines the size of the range at arbitrary times. Recall from Section 11.5 that $\mathsf{R}_t = \{X_0, \dots, X_t\}$ is the set of vertices visited by time t.

Corollary 11.46 *Let $d \ge 2$ be an integer. Run rotor walk on the comb in $\mathbb{Z}^d$ from the root, with independent uniform initial rotors, and suppose that at each vertex x other than the root the cyclic order places the parent edge half a turn from the edge $(x, 2x - \mathrm{par}(x))$. Then, almost surely,*

$$\frac{\#\mathsf{R}_t}{t^{d/(d+1)}} \longrightarrow \frac{(2(d+1)!)^{d/(d+1)}}{d!} \qquad \text{as } t \to \infty\,.$$

Proof. Apply Theorem 11.45 with $a = 2$. The probabilities Cn^{-2} are summable, so by the Borel–Cantelli lemma, almost surely the bounds in part (b) of the theorem apply for all large n, and the count of $\#B(\mathbf{0}, R)$ gives

$$\#\mathsf{A}_n = \frac{2^{2d}}{d!}n^d(1 + o(1)) \qquad \text{and} \qquad T(n) = \frac{2^{2d+1}}{(d+1)!}n^{d+1}(1 + o(1))\,.$$

If $T(n) \le t < T(n+1)$, then $\mathsf{A}_n \subseteq \mathsf{R}_t \subseteq \mathsf{A}_{n+1}$ by (11.11). Consecutive range sizes and consecutive completion times have the same asymptotics, so squeezing $\#\mathsf{R}_t$ between $\#\mathsf{A}_n$ and $\#\mathsf{A}_{n+1}$ and dividing by $t^{d/(d+1)}$ gives the asserted limit. □

Figure 11.20 shows successive circuit ranges. The comb shows that the lower bound of order $t^{d/(d+1)}$ that Theorem 11.29 gives at the root cannot be improved; for $d = 2$, the limiting constant in Corollary 11.46 is $12^{2/3}/2 = 18^{1/3} \approx 2.62$.

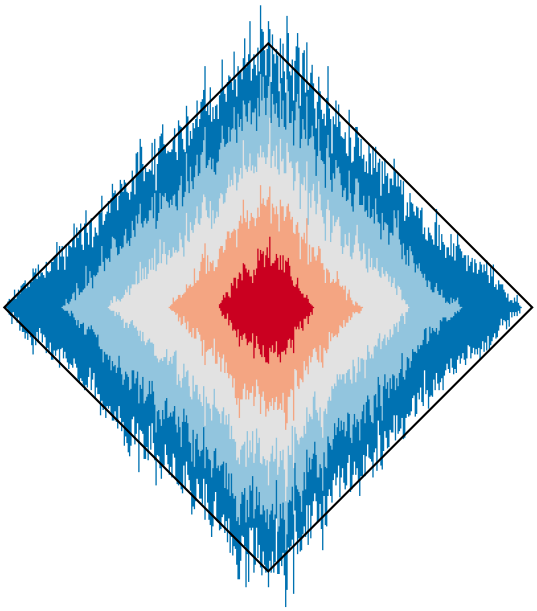


Figure 11.20: The range of a uniform rotor walk on the planar comb after 100 circuits. The vertices first visited in successive blocks of twenty circuits are colored from red (first block) to blue (last block), and the black outline is $B(\mathbf{0}, 200)$.

On a periodic graph such as $\mathbb{Z}^2$, many paths compete to reach a vertex, and the circuit in which a vertex is first visited is no longer read off a single path. A subadditivity argument still gives the range of uniform rotor walk on $\mathbb{Z}^2$ with the clockwise mechanism a deterministic limit shape, as the Notes describe.

EXERCISES

Exercise 11.1. Proposition 11.3 compares the hitting counts of rotor walk and random walk; this exercise compares visit counts. Let G be a finite connected undirected graph, regarded as a digraph by replacing each edge with its two orientations, and designate a vertex s as the sink. Fix a rotor mechanism, an initial rotor configuration ρ, and a non-sink vertex $o \in V(G)$. For every integer $n \geq 1$ and every non-sink vertex x, let $\ell_n(\rho, x)$ be the total number of visits to x made before absorption by n rotor walks starting at o, without resetting the rotors between walks. Let $(X_k)_{k \geq 0}$ be simple random

walk on G, and let τ_s be its hitting time of s. Show that

$$\left| \ell_n(\rho, x) - n\, \mathbb{E}_o \sum_{k=0}^{\tau_s - 1} \mathbf{1}\{X_k = x\} \right| \le \deg(x)\, \mathbb{E}_x \tau_s \,,$$

for every $n \ge 1$ and every $x \ne s$, and deduce that $\ell_n(\rho, x)/n$ converges to $\mathbb{E}_o \sum_{k=0}^{\tau_s - 1} \mathbf{1}\{X_k = x\}$.

Exercise 11.2. Extend Lemma 11.16 from rotor configurations to rotor mechanisms. Let G be an infinite, connected, locally finite undirected graph, regarded as a digraph by replacing each edge with its two orientations. Show that if two rotor mechanisms π and $\widetilde{\pi}$ differ at only finitely many vertices, then every rotor configuration ρ for which (π, ρ) is recurrent also makes $(\widetilde{\pi}, \rho)$ recurrent. Hint: If the two mechanisms differ only at one vertex v, then consider a multiple of $\deg(v)$ returns to v.

Exercise 11.3. Construct the boundaries used in Proposition 11.18. Let G be a locally finite undirected graph, regarded as a digraph by replacing every edge with its two orientations. Show that for every finite set of vertices $S \subset V(G)$, there exist a rotor configuration and mechanism so that S has a reflecting boundary.

Exercise 11.4. Prove a weaker form of Lemma 11.19 on every infinite, connected, locally finite undirected graph G, regarded as a digraph by replacing every edge with its two orientations: there are a rotor mechanism and a rotor configuration for which rotor walk is recurrent. It was asked whether every rotor mechanism admits such a configuration; see the Notes.

Exercise 11.5. (Eulerian Walker) Let G be a finite connected undirected graph with at least one edge, viewed as an Eulerian digraph by replacing each edge with its two orientations. Fix a rotor mechanism and an initial rotor at every vertex, and start the walk at a vertex o. Show that the rotor walk is eventually periodic and that its eventual period traverses every directed edge exactly once.

Hint: Apply Lemma 11.4 to a longest block of consecutive distinct edges and to its shifts.

Exercise 11.6. Consider rotor walk on $\mathbb{Z}^2$ started at the origin, with the clockwise rotor mechanism. Construct an initial rotor configuration for which, for every integer $n \ge 1$, the nth circuit visits exactly the diamond $B(\mathbf{0}, n) = \{x \in \mathbb{Z}^2 : \|x\|_1 \le n\}$. Let $T(n)$ be the completion time of the nth circuit, and let R_t be the range through time t. Show that

$$T(n) = \frac{8}{3} n^3 + O(n^2) \qquad \text{and} \qquad \#\mathsf{R}_{T(n)} = 2n^2 + O(n)\,,$$

and deduce that

$$\frac{\#\mathsf{R}_{T(n)}}{T(n)^{2/3}} \longrightarrow \left(\frac{9}{8}\right)^{1/3}.$$

In particular, the exponent $2/3$ in Theorem 11.28 cannot be increased. Hint: Use the proof of Proposition 11.27 with $S_n = B(\mathbf{0}, n-1)$; at a vertex $y \neq \mathbf{0}$, the edges into $B(\mathbf{0}, \|y\|_1 - 1)$ are consecutive in the clockwise order.

Exercise 11.7*. This exercise proves Theorem 11.33. Consider independent bond percolation on $\mathbb{Z}^2$ in which each edge is open or closed with probability $\frac{1}{2}$ each. A **rectangle** is the subgraph of $\mathbb{Z}^2$ consisting of all vertices and edges, boundary edges included, in a product of two closed intervals with integer endpoints. A **crossing** of a rectangle is a path of open edges of the rectangle joining two opposite sides. For integers $m, n \geq 1$, let $\Pi(m, n)$ be the probability that $[0, m] \times [0, n]$ has a crossing joining its left and right sides.

(a) Let $\beta_1, \dots, \beta_N$ be independent Bernoulli variables, and call an event determined by them **increasing** if changing a variable from 0 to 1 cannot destroy it. Prove the Harris inequality $\mathbb{P}(A \cap B) \geq \mathbb{P}(A)\mathbb{P}(B)$ for increasing events A and B. Deduce that $\mathbb{P}(A_1 \cap \cdots \cap A_k) \geq \mathbb{P}(A_1) \cdots \mathbb{P}(A_k)$ for every integer $k \geq 2$ and all increasing events $A_1, \dots, A_k$.

(b) Call an edge of the dual lattice $\mathbb{Z}^2 + (\frac{1}{2}, \frac{1}{2})$ open if the edge of $\mathbb{Z}^2$ that it crosses is closed, and define crossings in the dual lattice in the same way. For $R := [0, m] \times [0, n]$ with integers $m, n \geq 1$, let R^* be the dual rectangle $[\frac{1}{2}, m - \frac{1}{2}] \times [-\frac{1}{2}, n + \frac{1}{2}]$, keeping only the duals of edges of R. Prove that either R has a crossing joining its left and right sides or R^* has a crossing joining its bottom and top, but not both. Deduce that $\Pi(n, n) \geq \frac{1}{2}$ for every integer $n \geq 1$.

(c) Prove that $\Pi(2m - n, 2n) \geq \Pi(m, 2n)^2/32$ for all integers $n \geq 1$ and $m \geq 2n$. Deduce that there is $c > 0$ such that every rectangle whose shorter side has length at least 2 and whose longer side is at most ten times its shorter side has a crossing joining its two shorter sides with probability at least c. Hint for the inequality: Reflect in the line $y = n$ a vertical crossing of $[0, n]^2$ whose choice depends only on the edges on it and to its left.

(d) Let $\Theta(x, y) := (\frac{x-y}{2}, \frac{x+y}{2})$, which maps the half-dual lattice $\mathbb{L}$ of Section 11.6 onto a translate of $\mathbb{Z}^2$. Show that crossings of four rectangles of this translate that form a square annulus around the origin contain a simple cycle surrounding the origin. Use this and part (c) to prove Theorem 11.33, noting that percolation on $\mathbb{L}$ is unchanged when open and closed edges are exchanged.

Exercise 11.8*. Consider uniform rotor walk started at $\mathbf{0}$ on the F-lattice, with the odometer u_t of (11.4). Show that there is a constant C such that

$$\mathbb{P}\{u_t(\mathbf{0}) = 1\} \geq t^{-C} \qquad \text{for every integer } t \geq 2\,.$$

Deduce that if $a > C$ and $c > 0$, then the inequality of Theorem 11.35 fails for all large t. Hint: Put independent mirrors on a cylinder with $2t - 1$ columns and $t - 1$ rows, and note that an odd number of bottom ends cannot all be joined in pairs.

Exercise 11.9. In the setting of Section 11.7, $I(\rho, n)$ is the number of particles among the first n that never return to o, and $I_r(\rho, n)$ is the number that visit $\partial B(o, r)$ when the particles are stopped there. Consider the following incorrect "proof" of Lemma 11.39:

"Let r_1 be such that no particle that returns to o leaves $B(o, r_1)$, and let r_2 be such that no particle that escapes to infinity returns to $B(o, r_1)$ after it leaves $B(o, r_2)$. If $r \geq r_2$, then stopping the particles when they reach $\partial B(o, r)$ has no effect on the rotors inside $B(o, r_1)$, and therefore has no effect on whether later particles return to o. Hence $I_r(\rho, n) = I(\rho, n)$ for all $r \geq r_2$." Find the mistake in the quoted proof.

Exercise 11.10. Let G be an infinite, connected, locally finite undirected graph, regarded as a digraph by replacing each edge with its two orientations. Fix a rotor mechanism, a vertex o, and an initial rotor configuration ρ. Run rotor walks successively from o. After a transient walk escapes, start the next walk from the pointwise limiting rotor configuration left by the transient walk. If a walk is recurrent, then stop the experiment. For every integer $n \geq 1$, let $\ell_n(\rho, o)$ be the total number of visits to o made by the first n walks when all are transient, and let $\ell_n(\rho, o) = \infty$ otherwise. Show that

$$\liminf_{n\to\infty} \frac{\ell_n(\rho, o)}{n} \geq \mathbb{E}_o \sum_{k=0}^{\infty} \mathbf{1}\{X_k = o\}\,.$$

Here $(X_k)_{k\geq 0}$ is simple random walk on G started at o. Hint: Reformulate Theorem 11.36.

Exercise 11.11. Extend Theorem 11.45 and Corollary 11.46 to cyclic orders that do not place the parent edge $(x, \mathrm{par}(x))$ half a turn from the edge $(x, 2x - \mathrm{par}(x))$. For both parts of the exercise, let $T(n)$ be the completion time of the nth circuit, A_n the nth circuit range, and R_t the range through time t.

(a) Run rotor walk on the comb in $\mathbb{Z}^2$ from the root, with independent uniform initial rotors and cyclic order north, east, west, south at every spine vertex. For every $0 < \epsilon < 1$, show that almost surely, for all sufficiently large n, the range A_n contains every lattice point of

$(1-\epsilon)P_n$ and is contained in $(1+\epsilon)P_n$, where P_n is the quadrilateral with horizontal vertices $(4n, 0)$ and $(-\frac{4n}{3}, 0)$ and vertical vertices $(0, \pm 2n)$.

(b) Let $d \geq 2$ be an integer. Choose a cyclic order at the root, and suppose that the cyclic order at each $x \neq \mathbf{0}$ depends only on $j(x)$, the largest index with $x_{j(x)} \neq 0$, as a cyclic order of the directions $\pm\mathbf{e}_k$ with $k \geq j(x)$. Run rotor walk from the root with independent uniform initial rotors. For $1 \leq j \leq d$ and $s \in \{+, -\}$, let $m_{j,s}$ be the unique integer satisfying $1 \leq m_{j,s} < 2(d-j+1)$ such that $m_{j,s}$ advances of the rotor take $(x, \mathrm{par}(x))$ to $(x, 2x - \mathrm{par}(x))$ for every vertex x with $j(x) = j$ and x_j of sign s. Let

$$q_{j,s} := 1 - \frac{m_{j,s}}{2(d-j+1)}\,.$$

For $x \in \mathbb{R}^d$ and $1 \leq j \leq d$, let s_j be the sign of x_j. Either sign may be used when $x_j = 0$ because $q_{j,s_j}|x_j| = 0$. Let

$$K := \left\{ x \in \mathbb{R}^d : \sum_{j=1}^{d} q_{j,s_j}|x_j| \leq 1 \right\}.$$

Show that for every $0 < \epsilon < 1$, almost surely for all sufficiently large n,

$$(1-\epsilon)nK \cap \mathbb{Z}^d \subseteq \mathsf{A}_n \subseteq (1+\epsilon)nK \cap \mathbb{Z}^d\,.$$

Show also that

$$\mathcal{L}(K) = \frac{1}{d!} \prod_{j=1}^{d} \left(q_{j,+}^{-1} + q_{j,-}^{-1} \right)$$

and that, almost surely,

$$T(n) = \frac{2}{d+1}\mathcal{L}(K)n^{d+1}(1 + o(1)) \qquad \text{as } n \to \infty\,.$$

Deduce that, almost surely,

$$\frac{\#\mathsf{R}_t}{t^{d/(d+1)}} \longrightarrow \left(\frac{d+1}{2}\right)^{d/(d+1)} \mathcal{L}(K)^{1/(d+1)}\,.$$

Hint: In part (a), the east edge precedes the west edge with probability $3/4$ at a spine vertex whose parent lies to the west.

NOTES

Priezzhev, Dhar, Dhar, and Krishnamurthy (1996) introduced rotor walk in 1996 as "Eulerian walkers" in connection with self-organized criticality. In the same year, Wagner, Lindenbaum, and Bruckstein (1996, Section 2.1) introduced ANT-WALK-1, which is the same walk when there is one walker and ties are broken by a fixed rule. The algebraic relation of rotor walk with the abelian sandpile was developed further in Section II of Povolotsky, Priezzhev, and Shcherbakov (1998). The Introduction of Holroyd, Levine, Mészáros, Peres, Propp, and Wilson (2008) records the independent appearances of rotor walk in the load-balancing work of Rabani, Sinclair, and Wanka (1998), in Propp's derandomization of internal DLA, and in the whirling tours of Section 9.2 of Dumitriu, Tetali, and Winkler (2003), part of their analysis of a game on graphs. The Introduction of Florescu, Ganguly, Levine, and Peres (2014) records that Propp proposed rotor walk as a deterministic analogue of random walk in a lecture in 2003.

Sections 11.1 and 11.2 follow Sections 3–4 of Holroyd, Levine, Mészáros, Peres, Propp, and Wilson (2008): Lemma 11.2 is a form of their Lemma 3.9, and the proof of the matrix tree theorem (Proposition 11.14) through the action of the sandpile group on rotor configurations is their Corollary 3.18. Proposition 11.3 is the finite-set, many-particle form of Theorem 1 of Holroyd and Propp (2010); Proposition 3.21 of Holroyd, Levine, Mészáros, Peres, Propp, and Wilson (2008) proves the weaker bound with the sum over all edges and without the factor $\frac{1}{2}$, and Proposition 5.7 of Levine and Peres (2009), which they credit to Holroyd and Propp, gives the undirected case with the constant 1.

Section 11.3 follows Angel and Holroyd (2012). Lemma 11.15 and Lemma 11.19 correspond to Theorems 1 and 2 of Angel and Holroyd (2012), and Exercise 11.4 is their Theorem 4. For a locally finite tree, Theorem 3 of Angel and Holroyd (2011) proves that the limiting proportion of escapes equals the random-walk escape probability when only finitely many rotors are positioned so that their next advance points toward the root. Open Question (i) of Angel and Holroyd (2012) asks whether there is a graph and rotor mechanism for which every initial rotor configuration is transient; equivalently, whether every rotor mechanism admits an initial rotor configuration for which the walk is recurrent, the question mentioned in Exercise 11.4.

Section 2 of Angel and Holroyd (2011) introduces live paths, and their Proposition 8 identifies the total number of escapes of successive particles started at the root of a tree with the number of live ends, that is, infinite paths from the root whose tails beyond some vertex are live paths. Section 11.8 uses the same departure-order condition, together with the cancellation of opposite traversals, to prove recurrence.

Sections 11.4–11.6 follow Florescu, Levine, and Peres (2016); our theorem and lemma numbers for that paper refer to its long version, arXiv:1408.5533. Proposition 11.27 is their Proposition 2.7, which they credit to Proposition 11 of Angel and Holroyd (2012). Theorem 11.29 is a corrected form of their Theorem 3.1, whose part (ii), $u_t(x) \le u_t(o) + \deg(x)$, fails at vertices of degree larger than $\deg(o)$. Theorem 11.28 is the volume-growth form of their Theorem 1.1, Theorem 11.30 is their Theorem 3.3, and Theorem 11.34 is their Theorem 1.5. Section 5 of Florescu, Levine, and Peres (2016) takes the vertices of $\mathbb{L}$ with $x + y$ even; with that choice a mirror parallel to a closed edge e_v would reflect the walk on the Manhattan lattice onto an incoming edge of v, so we take $x + y$ odd. The growth rates of the range on the two directed lattices suggested by experiment, at the end of Section 11.6, are reported at the end of their Section 5. Theorem 11.35 proves the bound for one small exponent a, which suffices for recurrence. It corresponds to their Theorem 5.4, which asserts the bound for every $a > 0$ on both lattices; Exercise 11.8 shows that on the F-lattice the bound fails for large a. The precursor Florescu, Ganguly, Levine, and Peres (2014) proves that on $\mathbb{Z}^2$ at most $(\pi/2 + o(1))\, n/\log n$ of the first n particles escape, for every initial rotor configuration (their Theorem 2), and that when every vertex sends its first particle in the direction $\mathbf{e}_d$, the number of escapes is at least of order $n/\log n$ in $d = 2$ and of order n in $d \ge 3$ (their Theorems 2 and 1). Theorem 1 of Cooper and Spencer (2006) shows that in Propp's rotor-router machine on $\mathbb{Z}^d$, which moves all chips at once, the number of chips at each site after n steps differs from its expected value for independent random walks by at most a constant depending only on d, for every initial configuration of chips on the even sites and every setting of the rotors. Early results on rotor-router aggregation in one dimension appear in Sections 2–3 of Levine (2004). Theorem 11.33 adapts the open-circuit bound (9.72) on p. 220 of Grimmett (1989), which he deduces from the Russo–Seymour–Welsh (RSW) theorem (9.70)–(9.71) on p. 219, together with its restatement (9.100) with an unspecified constant and its dual closed-circuit form (9.101) on p. 230, to the annuli $Q(\mathbf{0}, 3\ell) \setminus Q(\mathbf{0}, \ell)$, which are rotated by 45° relative to the edges of $\mathbb{L}$; Exercise 11.7 carries out the adaptation. Its part (b) corresponds to Lemma 3 and Corollary 5 of Bollobás and Riordan (2006). Its proof of the RSW estimate for rectangles follows their reflection argument in Lemma 6 and Corollary 7, which treats $2\rho n$ by $2n$ rectangles for integers $\rho > 1$; the exercise extends it to every rectangle of shorter side at least 2 and longer side at most ten times the shorter. The four rectangles of part (d) are those in their proof of Theorem 8 (numbers as in arXiv:math/0410359). The planar fact (P) used in the solution is the one that Bollobás and Riordan (2006) use in the proofs of Lemma 3 and Corollary 7, remarking that only the easy part of the Jordan curve theorem is needed.

Theorem 11.36 is due to an unpublished argument of Oded Schramm. Theorem 10 of Holroyd and Propp (2010) proves the bound for irreducible Markov chains with rational transition probabilities; Theorem 11.36 is the specialization to simple random walk on a strongly connected, locally finite digraph. Theorem 1 of Angel and Holroyd (2011) also states the bound, for every graph with finite degrees. On undirected graphs, Lemma 5.1 of Chan (2019) bounds below the number of visits of successive rotor walks to each vertex, which gives another proof of the bound and implies Exercise 11.10; the convergence conclusion of Exercise 11.1 is Lemma 3.5 of Chan (2019).

For a rotor mechanism on $\mathbb{Z}^d$ with the same cyclic order at every vertex, in which, for some $i < d$, the directions $\pm\mathbf{e}_i$ separate $\mathbf{e}_d$ from $-\mathbf{e}_d$, He (2014) constructs an initial rotor configuration for which the proportion of escapes converges to the escape probability of random walk when $d \geq 3$ (his Theorem 1.3), and the number of escapes is $(\pi/2 + o(1))\, n/\log n$, the upper bound of Florescu, Ganguly, Levine, and Peres (2014), when $d = 2$ (his Theorem 1.2). Theorem 1 of Chan (2020) extends the existence of such a configuration to every connected, simple, locally finite transient graph. Theorem 1.4 of Chan (2021) proves that a rotor configuration sampled from the wired spanning forest has maximal escape rate when every tree of the wired spanning forest is almost surely one-ended.

The two-dimensional case of Theorem 11.45 is, up to the corrections below, Theorem 4.1 of Florescu, Levine, and Peres (2016), proved by viewing the walk along the spine and each tooth as rotor walks on $\mathbb{Z}$. Our proof identifies the circuit ranges by traversing the finite sublevel trees of the labels h, then uses concentration and counts their edges; it applies in every dimension. Their Theorem 4.1 states diamonds of radius $n \pm c\sqrt{n \log n}$ at time $\frac{16}{3}n^3$, and states the limit $(3/2)^{2/3}$; the calculation in their Lemma 4.3 and in the proof of their Lemma 4.4 gives radius $2n \pm c\sqrt{n \log n}$ after n circuits, as in Theorem 11.45, and the limit $18^{1/3}$ of Corollary 11.46.

Theorem 4.1 of Florescu, Levine, and Peres (2016) is stated for every rotor mechanism, but its proof uses the probability $1/2$ of (11.13) at every spine vertex, which holds only when the cyclic order at each spine vertex x places the parent edge $(x, \mathrm{par}(x))$ half a turn from the edge $(x, 2x - \mathrm{par}(x))$, as Theorem 11.45 assumes; Exercise 11.11 treats other cyclic orders.

Sections 11.8 and 11.9 follow Sections 2, 4 and 5 of Bou-Rabee and Peres (2026). Proposition 11.40 is the recurrence part of their Proposition 2.8. Theorem 11.42 is the square-lattice case of the recurrence conclusion (i) of their Theorem 1.1, which also treats doubly periodic graphs in $\mathbb{R}^2$ of maximum degree three with a doubly periodic rotor mechanism. The infinite-graph case of Corollary 11.41 follows from their Proposition 4.1, which proves recurrence without periodicity by dominating the live paths that start

with a given edge by a critical Galton–Watson tree; our exploration instead compares its queue length with a lazy symmetric walk. Figure 11.15 is reproduced from their paper.

On $\mathbb{Z}^2$ they show that there are constants $c, C > 0$ such that, for each fixed initial directed edge and every integer $R \geq 1$, the probability that a live path starting with that edge reaches graph distance R from its first vertex is at most Ce^{-cR}, by comparing a dual configuration with subcritical percolation (their Proposition 5.1); for recurrence it suffices that no live path is infinite, and we compare the fair-bond cycles of Theorem 11.33 with clockwise rotor cycles by replacing pairs of bits. Their Theorem 1.1 gives the range a deterministic limit shape: there are a compact convex set $\mathfrak{B} \subset \mathbb{R}^2$, with $\mathbf{0}$ in its interior, and a constant $\lambda > 0$ such that, almost surely, $t^{-1/3}\mathsf{R}_t$ converges to $\lambda\mathfrak{B}$ in Hausdorff distance, and $\#\mathsf{R}_t/t^{2/3}$ converges to a deterministic positive constant, so the order $t^{2/3}$ of Theorem 11.28 is attained. Their proof bounds from below the index of the circuit in which each vertex is first visited by counting the vertices of a path that send the walk back before onward, applies the subadditive ergodic theorem to obtain the circuit limit shape, and sums the circuit lengths to pass from circuit number to elapsed time (their Section 3); we do not prove these shape results here. Earlier, Monte Carlo simulations of Kapri and Dhar (2009) had indicated that the set of sites visited by this walk on $\mathbb{Z}^2$ is asymptotically a disk of radius of order $t^{1/3}$; whether $\mathfrak{B}$ is a disk is Problem 3 of Bou-Rabee and Peres (2026).

A Preview of Volume II

Every shape theorem in this volume starts from a point source: mass, particles or grains are put at the origin of $\mathbb{Z}^d$ (or at two sites of $\mathbb{Z}^2$, in Chapter 7), and the theorem says how closely the region they fill approaches a ball (or Neumann's oval). Volume II goes further in three directions. It spreads the initial mass over many sites, so that the limit shape has to be found by solving a free boundary problem; it finds the scaling limit of the abelian sandpile and proves that in the plane its limit shape is not a disk; and it shows that the clusters of internal DLA and rotor-router aggregation deviate from a ball by at most a power of the logarithm of its radius.

The first part of Volume II builds the continuum theory that the scaling limits need. Chapter 2 posed the Dirichlet problem and showed that it can fail to have a solution (on a punctured disk, no harmonic function takes the value 0 on the circle and 1 at the center). Volume II proves that Perron's method, the infimum of all superharmonic functions lying above the boundary data, always gives a harmonic function, and that a boundary point is regular (this function attains the boundary value there for all continuous data) exactly when it has a barrier, a superharmonic function that tends to zero at the point and stays positive near the rest of the boundary. By the Beurling projection theorem of Chapter 2, every boundary point of a bounded simply connected planar domain has one. Volume II also extends the convergence of discrete harmonic functions proved in Chapter 3, for limits with uniformly continuous second derivatives, to continuous boundary values on domains from which a random walk started near the boundary is unlikely to go far before leaving.

The obstacle problem comes next. In Chapter 4 the odometer of the divisible sandpile from a point source was the gap between an explicit obstacle and its least superharmonic majorant. For an initial density spread over $\mathbb{R}^d$ the same recipe defines a continuum odometer, but the set where it is positive is no longer explicit. Volume II proves that this set is bounded, and that if the density is continuous almost everywhere and takes no values strictly between a number $\lambda < 1$ and 1, then the boundary of the occupied domain has Lebesgue measure zero and mass is conserved: the volume of the occupied domain equals the initial mass inside it.

The second part starts the three models of Chapters 4–6 from a density spread over the lattice $\delta\mathbb{Z}^d$, whose spacing δ tends to zero. Their odometers

satisfy the same discrete Poisson equation up to an error (zero for the divisible sandpile, the divergence of a bounded field on the edges for rotor-router aggregation, and an error controlled by martingales for internal DLA) that is small in a negative Sobolev norm, so all three converge to the solution of the same obstacle problem; when the density is either zero or at least one at each point, the occupied sets converge as well (for internal DLA, in probability). For example, put one particle at every site of a set A and one at every site of a set B, so that their overlap starts with two: the limit is the smash sum $A \oplus B$, whose volume is the sum of the volumes of A and B. Volume II also returns to point sources in the plane. If k distinct points carry positive masses, and a bounded open set has the property that defines the limit shape of Chapter 7 (the logarithmic potential of the sources minus that of the uniform measure on the set is nonnegative, and positive exactly on the set), then its boundary lies on an algebraic curve of degree $2k$ and on none of lower degree; for two sources it is the quartic of Neumann's oval or the pair of circles bounding two disjoint disks.

Chapter 9 developed the combinatorics of the abelian sandpile, and Section 4.3 placed the sites toppled by n grains at the origin of $\mathbb{Z}^d$ between two Euclidean balls, but neither explained the patterns of the frontispiece. Volume II proves the theorem of Pegden and Smart (2013) that the odometer, scaled by $n^{1/d}$ in space and divided by $n^{2/d}$, converges, and so do the final sandpiles in the sense of local averages. The limit obeys a constraint on its curvature: wherever the limiting odometer is positive and twice differentiable, its Hessian lies on the boundary of the closed set of symmetric matrices whose quadratic forms, up to an affine function, lie below an integer-valued function on $\mathbb{Z}^d$ with Laplacian at most $2d - 1$ (the most grains a stable site can hold). Each periodic pattern of grains corresponds to a matrix on this boundary, and in the plane the boundary is described by an Apollonian circle packing, which is where the fractal patchwork of the frontispiece comes from. The same method gives the scaling limit of the identity element of the sandpile group of a large box.

Is the limit shape of the planar sandpile a disk? Look again at the frontispiece: at the top, at the bottom and on both sides the boundary runs straight for a stretch, and the triangle just inside each straight stretch holds three grains at every site. Volume II proves that both features persist in the limit. For n grains at the origin of $\mathbb{Z}^2$, with no other sand, the set where the limiting odometer is positive is convex. Near each of the four points where its boundary crosses a coordinate axis, the boundary is a segment of the line through that point perpendicular to the axis, and inside the set the limiting odometer equals three halves of the square of the distance to that line, so that the limiting density of grains there is 3. A disk has no straight pieces of boundary, so the limit shape is neither a disk nor an ellipse.

A second theorem passes from the odometer to the toppled sets themselves, for n grains added at the origin of $\mathbb{Z}^d$ to a background of h grains at every site, where $h \le 2d-2$ is an integer (a negative h is a deficit of $|h|$ grains at every site). The toppled sets, scaled by $n^{1/d}$, converge in Hausdorff distance to the set of points whose projection onto each coordinate subspace of dimension $\min(2d-1-h, d)$ lies in the projection of the support of the limiting odometer. For $h \le d-1$ this set is the support itself. For $d \le h \le 2d-2$ the boundary of the limit has flat pieces of dimension $h-d+1$, so the limit is again neither a ball nor an ellipsoid, and for $h = 2d-2$ it is a cube, as the toppled set is for every $n \ge 2$ by Proposition 9.36.

Following the topplings in time leads to a scaling limit of another kind. The parallel toppling procedure of Section 9.8 topples every unstable site once in each round. Start it from four grains at every site of the square $\{1, \dots, N\}^2$, and let grains that leave the square disappear. A site topples at most once per round, which suggests scaling time by N^2 and space by N, as for the heat equation. Volume II proves that, after this rescaling, the number of times each site has toppled, divided by N^2, converges to a unique continuous function of time and position in the unit square. Which partial differential equation the limit solves is open.

The third part sharpens the shape theorems of Chapters 5 and 6. Let r be the radius of the ball of volume n. Chapter 5 showed that the internal DLA cluster of n particles deviates from this ball by at most a power of r smaller than one. Volume II proves that, almost surely for all large n, the cluster lies between the balls of radii $r \pm C\sqrt{\log r}$ when $d \ge 3$, and between the balls of radii $r \pm C \log r \, (\log\log r)^{3/2}$ when $d = 2$, where C depends only on d. In the proof, a well-chosen function of the position of each walk is a martingale, and a large deviation estimate for the sum of these martingales controls the deviation of the cluster from the ball. For rotor-router aggregation in $\mathbb{Z}^d$ with $d \ge 2$, whatever the initial rotors and the rotor mechanism, Volume II proves that, for large n, the aggregate A_n^{R} of n particles lies between the balls of radii $r \pm C \log r$, which removes a factor $\log r$ from the outer bound of Chapter 6.

Chapter 11 showed that rotor walk on $\mathbb{Z}^d$ visits at least of order $t^{d/(d+1)}$ sites in its first t steps, whatever its initial rotors, and that rotor walk with random initial rotors is almost surely recurrent when long live paths (whose internal vertices send the walk onward before sending it back) are exponentially unlikely, as they are for independent uniform initial rotors on $\mathbb{Z}^2$ turning clockwise and on periodic graphs of maximum degree three. Volume II proves that, on a periodic graph with independent initial rotors whose laws are periodic, the same condition implies that, almost surely, the set of sites visited by time t, scaled by $t^{1/(d+1)}$, converges to a deterministic convex body, so that the lower bound is attained. (The proof follows first-passage percolation, with the travel time between two sites replaced by the number of

circuits that the walk needs to get from one to the other.) For uniform initial rotors on $\mathbb{Z}^2$ this confirms the radius of order $t^{1/3}$ seen in the simulations of Kapri and Dhar (2009), which suggested that the visited set is close to a disk; whether the limit shape is a disk is open.

A shorter chapter of Volume II surveys, without full proofs, what changes when the lattice or the rule changes. The balls of this volume come from the symmetry of the walk rather than from the rule of aggregation: on a variant of $\mathbb{Z}^2$ called the layered square lattice, internal DLA and rotor-router aggregation both grow diamonds, and with suitably chosen rotors the rotor-router aggregate is an exact lattice diamond at infinitely many times. Internal DLA driven by a walk with a constant drift, rescaled differently along the drift, fills a heat ball, the region on which solutions of the heat equation obey a mean value property. Dissipation slows the growth of the planar sandpile from radius of order $\sqrt{n}$ to radius of order $\log n$, and the Manna model, in which each toppling sends grains to random neighbors, keeps its abelian property once the random choices are fixed in advance. The chapter ends with abelian networks, in which the abelian properties of Chapters 5, 9 and 11 become instances of one local-to-global principle.

Volume II ends with open problems. One of them starts from the two sandpiles of Figure 9.17, on the background of two grains in $\mathbb{Z}^2$ and of four grains in $\mathbb{Z}^3$. A slice through the origin of the sandpile formed by n grains on the second background appears to agree, away from the origin and apart from a few sites, with a sandpile of suitably many grains on the first, each site of the slice holding two grains more; the dimensional reduction conjecture asks for a proof.

Solutions to Exercises

This chapter gives the solutions of the exercises that the text uses, and of a few others close to it, grouped by chapter. A separate document gives the solutions of all the exercises.

CHAPTER 1. MODELS OF LAPLACIAN GROWTH

1.3. (a) Given A_n, the next particle performs simple random walk from $\mathbf{0}$ until it first leaves A_n, that is, until it first visits $-a_n - 1$ or $b_n + 1$. The probability of visiting $-a_n - 1$ first is a discrete harmonic function of the starting point on A_n, equal to 1 at $-a_n - 1$ and to 0 at $b_n + 1$. On an interval of $\mathbb{Z}$ such a function is linear, so from $\mathbf{0}$ this probability is

$$\frac{b_n + 1}{a_n + b_n + 2} = \frac{b_n + 1}{n + 1} .$$

(b) Write $Y_n := a_n - b_n$. Since $a_n + b_n = n - 1$, part (a) gives

$$\mathbb{P}\{Y_{n+1} = Y_n + 1 \mid A_n\} = \frac{b_n + 1}{n + 1} = \frac{1}{2} - \frac{Y_n}{2(n + 1)} ,$$

and otherwise $Y_{n+1} = Y_n - 1$. The conditional mean of $Y_{n+1} - Y_n$ is therefore $-Y_n/(n + 1)$, and

$$\mathbb{E}[Y_{n+1}^2 \,\big|\, A_n] = Y_n^2 - \frac{2Y_n^2}{n + 1} + 1 = \frac{n - 1}{n + 1} Y_n^2 + 1 .$$

Thus $V_n := \mathbb{E}[Y_n^2]$ satisfies $V_{n+1} = \frac{n-1}{n+1} V_n + 1$. We prove by induction on n that $V_n = (n + 1)/3$ for every $n \geq 2$. For the base case, $A_1 = \{\mathbf{0}\}$ gives $V_1 = 0$, and then $V_2 = 1 = (2 + 1)/3$. For the inductive step, suppose that $n \geq 2$ and $V_n = (n + 1)/3$. By the recursion and the inductive hypothesis, $V_{n+1} = (n - 1)/3 + 1 = (n + 2)/3$, which completes the induction.

CHAPTER 2. POTENTIAL THEORY IN $\mathbb{R}^d$

2.3. Write $m := \inf_K u \in [-\infty, \infty]$, and choose $x_n \in K$ with $u(x_n) \to m$. Compactness gives a subsequence converging to a point $x \in K$, and lower semicontinuity gives

$$u(x) \le \liminf_{n\to\infty} u(x_n) = m\,.$$

Since $m \le u(x)$ by definition, equality follows; in particular $m > -\infty$.

2.4. Let $x_n \to x_0$. For each $f \in \mathcal{F}$, lower semicontinuity of f at x_0 and $f \le s$ give

$$f(x_0) \le \liminf f(x_n) \le \liminf s(x_n)\,.$$

Taking the supremum over $f \in \mathcal{F}$ gives $s(x_0) \le \liminf s(x_n)$.

2.5. One argument proves both parts without compactness. Let $\psi := \arctan \circ u$, with $\arctan(+\infty) := \pi/2$; since arctan is continuous and increasing on $(-\infty, +\infty]$, the function $\psi : K \to (-\pi/2, \pi/2]$ is lower semicontinuous in the relative topology. For $n \ge 1$, let

$$f_n(x) := \inf_{y\in K}\{\psi(y) + n|x - y|\}\,.$$

Each f_n is n-Lipschitz, and $-\pi/2 \le f_n \le \psi$. Fix $x \in K$ and $\epsilon > 0$, and choose $\delta > 0$ with $\psi \ge \psi(x) - \epsilon$ on $K \cap \mathrm{B}(x, \delta)$. The terms with $|y - x| \ge \delta$ are at least $-\pi/2 + n\delta$, so $f_n(x) \ge \psi(x) - \epsilon$ once $n\delta \ge \pi$. Hence $f_n(x) \uparrow \psi(x)$. Moreover $f_n(x) > -\pi/2$: otherwise there are points $y_k \in K$ with $\psi(y_k)+n|x-y_k| \to -\pi/2$, which forces $y_k \to x$ and $\psi(y_k) \to -\pi/2 < \psi(x)$, contradicting lower semicontinuity at x. Therefore

$$g_n := \tan(\min\{f_n, \pi/2 - 1/n\})$$

is continuous and real-valued on K, satisfies $g_n \le u$, and $g_n(x) \to u(x)$ for every $x \in K$, with $\tan(\pi/2) := +\infty$. The family $\{g_n : n \ge 1\}$ has the required property in both parts.

2.6. The function $f \circ h$ is continuous and hence locally integrable. For every ball $\mathrm{B}(y, r)$ in the neighborhood of Definition 2.2, the mean value property and Jensen's inequality give

$$(f \circ h)(y) = f(\mathbf{A}_r h(y)) \le \mathbf{A}_r(f \circ h)(y)\,,$$

so $f \circ h$ is subharmonic. The choices $f(t) = |t|$ and $f(t) = t^2$ give the two stated cases.

2.36. Fix $x_0 \in \Omega_1$. For each $x \in \overline{\Omega}_1$, choose a continuous path in Ω_2 from x_0 to x. Choose $\rho > 0$ so that the closed 4ρ-neighborhood of the compact path lies in Ω_2. By uniform continuity, we may choose finitely many successive points along the path, consecutive ones at distance less than $\rho/2$. The ball of radius 2ρ about each point lies in Ω_2, and the next point lies in its concentric ball of radius $\rho/2$. The scaled form of Theorem 2.30, with ratio $1/4$, compares h at each successive pair. One final application in the ball of radius 2ρ about x gives a neighborhood U_x of x and a constant $C_x < \infty$ such that

$$C_x^{-1} h(x_0) \le h(y) \le C_x h(x_0) \qquad \text{for } y \in U_x \,.$$

A finite collection of the neighborhoods U_x covers $\overline{\Omega}_1$. Taking the largest corresponding constant gives $C_0 < \infty$, depending only on the two domains, such that $C_0^{-1} h(x_0) \le h \le C_0 h(x_0)$ on Ω_1. Therefore $\sup_{\Omega_1} h \le C_0^2 \inf_{\Omega_1} h$.

2.39. The assertion is false. Let

$$g(x) := d\omega_d(\mathcal{G}(x + \mathbf{e}_1) - \mathcal{G}(\mathbf{e}_1)) + x_1 \,.$$

By Corollary 2.34, g is superharmonic, since the added affine function is harmonic. Also $g(\mathbf{0}) = 0$, and $g(x) \to +\infty$ as $x \to -\mathbf{e}_1$.

We first bound g below. By (2.14), the function $t \mapsto d\omega_d \mathcal{G}(\sqrt{t}\mathbf{e}_1)$ has derivative $-\frac{1}{2}t^{-d/2}$ for $t > 0$, so it is convex. Its tangent line at $t = 1$, evaluated at $t = |x + \mathbf{e}_1|^2$, gives

$$g(x) \ge -\frac{1}{2}(|x + \mathbf{e}_1|^2 - 1) + x_1 = -\frac{1}{2}|x|^2 \,.$$

The inequality is also true at the pole, where $g = +\infty$.

Now let $f(x) := |x|^2 + \min\{101, g(x)\}$. The cap makes f finite and continuous, including at $-\mathbf{e}_1$, and the lower bound gives $f(x) \ge |x|^2/2$ and $f(\mathbf{0}) = 0$. The minimum of two superharmonic functions is superharmonic, so $f(x) - |x|^2$ is superharmonic. For sufficiently small $0 < s < 1$, the point $x = -(1 - s)\mathbf{e}_1$ lies in the unit ball and satisfies $g(x) \ge 101$. Hence $f(x) = |x|^2 + 101 > 100|x|^2$, as required.

2.40. The assertion is true. Since f has a minimum at the origin, $\nabla f(\mathbf{0}) = \mathbf{0}$. Taylor's formula along the segment from $\mathbf{0}$ to x gives

$$f(x) = \int_0^1 (1 - t) x^T (D^2 f)(tx) x \, \mathrm{d}t \le |x|^2 \int_0^1 (1 - t) \, \mathrm{d}t = \frac{1}{2}|x|^2 \,.$$

2.42. Fix i, and write

$$W(x) := q_i \sum_{j \ne i} q_j \mathcal{G}(x - y_j) \,.$$

The function W is harmonic in a neighborhood of y_i. If it had a local minimum at y_i, then the strong minimum principle, Lemma 2.8, would make it constant in a ball about y_i.

The explicit kernels in (2.14) are real analytic away from their poles, so W is real analytic on the connected open set $\mathbb{R}^d \setminus \{y_j : j \neq i\}$. Being constant on a ball, W would be constant throughout this set by the identity theorem. If $q_k \neq 0$ for some $k \neq i$, then

$$W(x) = q_i q_k \mathcal{G}(x - y_k) + O(1) \qquad \text{as } x \to y_k \,.$$

This function is unbounded because $q_i q_k \neq 0$, contradicting constancy.

2.44. Let $B := \mathrm{B}(\mathbf{0}, r)$. If $|x| \geq r$, then integration in spherical shells and Lemma 2.33 give

$$\mathbf{G1}_B(x) = \int_0^r d\omega_d s^{d-1} \mathcal{G}(x)\, \mathrm{d}s = \frac{r^2}{d(d-2)} \left(\frac{|x|}{r}\right)^{2-d} .$$

If $0 < |x| < r$, then the same shell theorem gives

$$\begin{aligned} \mathbf{G1}_B(x) &= \int_0^{|x|} d\omega_d s^{d-1} \mathcal{G}(x)\, \mathrm{d}s + \int_{|x|}^r d\omega_d s^{d-1} \mathcal{G}(s\mathbf{e}_1)\, \mathrm{d}s \\ &= \frac{|x|^2}{d(d-2)} + \frac{r^2 - |x|^2}{2(d-2)} \\ &= \frac{r^2}{2(d-2)} - \frac{|x|^2}{2d} \,. \end{aligned}$$

At $x = \mathbf{0}$, direct shell integration gives $r^2/(2(d-2))$.

2.45. Let $B := \mathrm{B}(\mathbf{0}, r)$. If $|x| \geq r$, then Lemma 2.33 gives

$$\mathbf{G1}_B(x) = \pi r^2 \mathcal{G}(x) = -\frac{r^2}{2} \log |x| \,.$$

If $0 < |x| < r$, then

$$\begin{aligned} \mathbf{G1}_B(x) &= \pi |x|^2 \mathcal{G}(x) + \int_{|x|}^r 2\pi s \mathcal{G}(s\mathbf{e}_1)\, \mathrm{d}s \\ &= -\frac{|x|^2}{2} \log |x| - \int_{|x|}^r s \log s \, \mathrm{d}s \\ &= \frac{r^2}{4}(1 - 2\log r) - \frac{|x|^2}{4} \,. \end{aligned}$$

Direct integration gives the same formula at $x = \mathbf{0}$.

2.46. With y_n and m_n as in Example 2.41, write

$$v(x) := \sum_{n=1}^{\infty} m_n \mathcal{G}(x - y_n)\,.$$

The assumption there implies $\sum_n m_n < \infty$. Fix a compact ball D. Since (y_n) is bounded, there is a constant $C_D < \infty$ such that $\mathcal{G}(x - y_n) + C_D \geq 0$ for $x \in D$ and every n. Local integrability of $\mathcal{G}$ also gives

$$\sup_n \int_D (\mathcal{G}(x - y_n) + C_D)\,\mathrm{d}x < \infty\,.$$

Tonelli's theorem therefore shows that v is locally integrable. Monotone convergence applied to these nonnegative shifted kernels gives the mean value inequality term by term.

Since $\min(1, v)$ is at most 1 and at most v,

$$\mathbf{A}_r \min(1, v)(x) \leq \min(1, \mathbf{A}_r v(x)) \leq \min(1, v(x))\,,$$

so $u = \min(1, v)$ is superharmonic. Moreover, v is lower semicontinuous, which Exercise 2.33 uses: the increasing functions

$$\sum_{n=1}^{N} m_n(\mathcal{G}(\cdot - y_n) + C_D)$$

are lower semicontinuous on D. Their limit differs from v by the finite constant $C_D \sum_n m_n$, so v is lower semicontinuous. Finally, $u(y_n) = 1$ for every n, whereas $u(\mathbf{0}) < 1$, so u is discontinuous at the origin.

2.47. Let u be a lower semicontinuous superharmonic function on $\mathbb{R}$, let $a < b$, and let ℓ be affine with $\ell(a) \leq u(a)$ and $\ell(b) \leq u(b)$. The restriction of u to $[a, b]$ is lower semicontinuous on $[a, b]$ and superharmonic in (a, b), so Lemma 2.12(b) gives $u \geq \ell$ on $[a, b]$. Thus u lies above its chords, so u is concave as an $\mathbb{R} \cup \{+\infty\}$-valued function.

Local integrability makes u finite almost everywhere. Suppose that $u(x_0) = +\infty$ at some point x_0. Since u is finite almost everywhere, we may choose $x_0 < z < y$ with u finite at z and y. Concavity applied to z, viewed as a strict convex combination of x_0 and y, would give $u(z) = +\infty$, a contradiction. Thus u is finite everywhere, and every finite concave function on $\mathbb{R}$ is continuous.

The one-sided derivatives of a concave function exist and are weakly decreasing. At each point of nondifferentiability, the open interval between the right and left derivatives is nonempty. These intervals are pairwise disjoint, so assigning a rational number from each interval shows that there are at

most countably many such points. The function $u(x) = -|x|$ is superharmonic and is not differentiable at the origin.

2.48. Fix $y \in \partial B$, and for $0 < r \le 1$ let

$$h_r(x) := \mathcal{G}(x - ry) - \mathcal{G}(rx - y)\,.$$

The first term is harmonic in x away from ry. The second is harmonic in x on a neighborhood of $\overline{B}$ when $r < 1$, since its only pole y/r lies outside $\overline{B}$. Moreover $h_1 = 0$. Since

$$\nabla\mathcal{G}(w) = -\frac{w}{d\omega_d |w|^d}\,,$$

differentiating at $r = 1$ gives, for $x \in B$,

$$\left.\frac{\partial h_r(x)}{\partial r}\right|_{r=1} = -\nabla\mathcal{G}(x-y)\cdot(x+y) = \frac{|x|^2 - 1}{d\omega_d |x-y|^d} = -\frac{p(x)}{d\omega_d}\,.$$

For $r < 1$ sufficiently close to 1, the difference quotient $h_r/(r-1)$ is harmonic on each fixed compact subset of B. The formula for h_r shows that these quotients converge locally uniformly to $-p/(d\omega_d)$ as $r \uparrow 1$. Lemma 2.24, applied along a sequence $r_n \uparrow 1$ to these quotients on $\mathrm{B}(\mathbf{0}, r_n)$ and to their negatives, therefore shows that p is harmonic in B, so $\Delta p = 0$ in B.

CHAPTER 3. POTENTIAL THEORY IN $\mathbb{Z}^d$

3.5. The cross term vanishes: since $v = 0$ off A, summation by parts over the edges meeting A gives

$$\sum_{\{x,y\}} (h(x) - h(y))(v(x) - v(y)) = -\sum_{x \in A} v(x)\Delta^{::} h(x) = 0\,.$$

If v has energy zero, then it is constant along every edge meeting A, so a path from a site of A to ∂A shows that $v = 0$.

3.8. In part (a), write $p = 1/2$, and in both parts let $q := \mathbb{P}_0\{X_\tau = r\}$. In both parts, $\ell + r$ consecutive positive increments force the walk to hit r. The Markov property therefore gives

$$\mathbb{P}_0\{\tau > k(\ell + r)\} \le (1 - p^{\ell+r})^k \longrightarrow 0\,,$$

so $\tau < \infty$ almost surely.

(a) For simple random walk, the stopped process $X_{t\wedge\tau}$ is a bounded martingale. Optional stopping gives

$$0 = rq - \ell(1 - q)\,,$$

so $q = \ell/(\ell + r)$.

(b) For the biased walk, let $\rho := (1-p)/p$. The process $\rho^{X_{t\wedge\tau}}$ is a bounded martingale. Hence

$$1 = q\rho^{r} + (1-q)\rho^{-\ell}\,,$$

which gives

$$q = \frac{1-\rho^{\ell}}{1-\rho^{\ell+r}}\,.$$

3.16. Write $\mathcal{S}_n := \sum_{j=1}^{n} 1/j$. Then

$$\sum_{j=1}^{n} \frac{1}{2j-1} = \mathcal{S}_{2n} - \frac{1}{2}\mathcal{S}_n\,.$$

To obtain the required precision, write

$$r_m := \mathcal{S}_m - \log m - \frac{1}{2m}\,.$$

Taylor expansion gives $r_{m+1} - r_m = O(m^{-3})$. Since $r_m \to \gamma_E$, summing the increments from m to infinity gives

$$\mathcal{S}_m = \log m + \gamma_E + \frac{1}{2m} + O(m^{-2})\,.$$

The terms of order n^{-1} cancel, leaving

$$\mathcal{S}_{2n} - \frac{1}{2}\mathcal{S}_n = \frac{1}{2}\log n + \log 2 + \frac{\gamma_E}{2} + O(n^{-2})\,.$$

CHAPTER 4. DIVISIBLE SANDPILE IN $\mathbb{Z}^d$: SINGLE SOURCE

4.2. Suppose a toppling at x emits mass t to each neighbor, and let μ be the configuration before the toppling. The configuration after the toppling is $\mu + t\Delta^{::}\delta_x$. Directly summing the finitely many nonzero terms gives

$$\sum_{y\in\mathbb{Z}^d} h(y)\big((\mu + t\Delta^{::}\delta_x)(y) - \mu(y)\big) = t\Delta^{::}h(x) \le 0\,.$$

Thus a toppling cannot increase the total weight $\sum_y h(y)\mu(y)$.

Let u be the odometer. Lemma 4.4(ii) states that a divisible sandpile with finite initial support on an infinite graph stabilizes, with an odometer of finite support. Since $\sigma_\infty = \sigma + \Delta^{::} u$, finite summation by parts gives

$$\sum_{x\in\mathbb{Z}^d} h(x)\sigma_\infty(x) - \sum_{x\in\mathbb{Z}^d} h(x)\sigma(x) = \sum_{x\in\mathbb{Z}^d} u(x)\Delta^{::}h(x) \le 0\,.$$

The conditions $u \ge 0$ and $\Delta^{::}h \le 0$ make the final sum nonpositive. Rearranging the displayed inequality gives (1.1). If h is discrete harmonic, then h and $-h$ are both discrete superharmonic, and applying the inequality to each of them gives equality.

CHAPTER 7. INTERNAL AGGREGATION IN $\mathbb{Z}^2$: TWO SOURCES

7.2. The function h is smooth by Theorem 2.3. Choose a smooth cutoff χ supported in the neighborhood where h is harmonic, with $\chi = 1$ near $\overline{\Omega(q)}$, and put $\psi := \chi h$, extended by zero outside that neighborhood. Corollaries 2.39 and 2.40 give

$$\int U_q(z)\Delta\psi(z)\,\mathrm{d}A(z) = \int_{\Omega(q)} h(z)\,\mathrm{d}A(z) - h(\mathbf{0}) - h(q)\,.$$

All integrals exist because logarithmic singularities are locally integrable. The left side is zero: $\Delta\psi = \Delta h = 0$ near $\overline{\Omega(q)}$, and $U_q = 0$ outside $\Omega(q)$ by Theorem 7.2. This proves the identity, including when $q = \mathbf{0}$.

CHAPTER 10. THE DIVISIBLE SANDPILE

10.9. Fix σ and v, and write $F(t) := U_t^\sigma(v)$. A monotone function has at most countably many discontinuities. Let J be the set of discontinuities of F, let $a_s := F(s) - F(s^-) > 0$ for $s \in J$, and write

$$H(t) := \sum_{\substack{s\in J\\ s\le t}} a_s\,.$$

For $a < b$, the increase $F(b) - F(a)$ is at least the sum of the jumps in $(a, b]$. Hence $F - H$ is nondecreasing. Its jump at each point is zero, so $F - H$ is continuous.

Suppose that $F - H$ is not constant, and choose $a < b$ such that $(F - H)(a) < (F - H)(b)$. Let

$$t := \sup\{s \in [a, b] : (F - H)(s) = (F - H)(a)\}\,.$$

Continuity and monotonicity show that $(F-H)(t+\epsilon) > (F-H)(t)$ for every $\epsilon > 0$. Condition (ii) of Definition 10.4 gives a jump time $s_1 > t$. Given s_i, apply the same condition between t and $(t+s_i)/2$ to obtain a jump time s_{i+1} satisfying

$$s_i > s_{i+1} > t\,.$$

Taking $v_i = v$ contradicts condition (iii). Thus $F - H$ is constant. This constant is zero because $F(0) = H(0) = 0$, and therefore $F = H$.

10.10. Let o and C_o be as in the exercise. We first transfer the walk count from o to every vertex. For $w \in V$, let $k := \mathrm{dist}(o,w)$, and fix a path from o to w of length k. Prepending that path sends distinct neighbor-or-stay walks of length N from w to distinct such walks of length $N+k$ from o, so there are at most $C_o^{N+k}(N+k)!$ of them.

Let I be an interval of length $a < 1/C_o$. For a fixed walk $w = v_0, \dots, v_N$, the expected number of times $t_1 > \cdots > t_N$ in I with t_j a ring of the clock at v_{j-1} is $a^N/N!$, by the factorial moment formula for Poisson processes. Summing over walks,

$$\begin{aligned}\mathbb{P}\{\text{a decreasing chain of length } N \text{ from } w \text{ in } I\} &\le C_o^{N+k}(N+k)!\frac{a^N}{N!}\\ &= C_o^k\frac{(N+k)!}{N!}(C_o a)^N\,.\end{aligned}$$

For fixed k the ratio $(N+k)!/N!$ grows polynomially in N while $(C_o a)^N$ decays geometrically, so the bound tends to zero. Almost surely there is no infinite decreasing chain from w inside I.

Take the union over the countably many vertices w and the countably many intervals with rational endpoints of length less than $1/C_o$. An infinite strictly decreasing sequence of positive times converges, so one of its tails lies in such an interval, and that tail has its first ring at a vertex w. This proves condition (iii) of Definition 10.4 almost surely.

10.12. Suppose first that σ stabilizes. Choose w with $\beta := \sigma(w) - 1 > 0$. Since $1 + \beta\delta_w \le \sigma$, Lemma 10.10 gives $u_\infty^{1+\beta\delta_w} \le u_\infty^\sigma < \infty$, so $1+\beta\delta_w$ stabilizes by Theorem 4.3(iii). Proposition 10.16(b) therefore implies that simple random walk is transient.

Conversely, suppose the walk is transient. The Green function is finite, so

$$f(x) := \sum_{w \in \mathrm{supp}(\sigma-1)} (\sigma(w)-1)g^{::}(x,w)$$

is a finite nonnegative function. Since $\Delta_x g^{::}(x,w) = -\delta_w(x)$ for every $w \in V$, it follows that

$$\sigma + \Delta f = 1\,.$$

The least action principle now gives stabilization.

10.19. (a) We have $\mathbb{E}U_1^{\sigma}(v) = q/\deg(v)$. At a spine vertex, the two spine neighbors contribute $q/4$ each and the two tooth neighbors contribute $q/2$ each, while the vertex emits $4(q/4)$. Hence its expected mass increases by $q/2$. At $(x, \pm 1)$, the incoming expectation is $q/4 + q/2$, while the emitted expectation is $2(q/2)$, giving a change of $-q/4$. Farther along a tooth, the incoming and outgoing expectations are both q.

(b) Part (a) shows that $\mathbb{E}\sigma_1^U(v)$ is not identically $\mathbb{E}\sigma(v) = \mu$. Moreover,

$$\frac{\mathbb{E}\sigma_1^U(x,0)}{4} = \frac{\mu}{4} + \frac{q}{8} \qquad \text{and} \qquad \frac{\mathbb{E}\sigma_1^U(x,\pm 1)}{2} = \frac{\mu}{2} - \frac{q}{8}.$$

Since $q > 0$, both quantities differ from their respective time-zero values $\mu/4$ and $\mu/2$.

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Index

abelian property, 3, 248
abelian sandpile, 3
absorbing, 329
accessible, 256
actuate, 329
acyclic, 332
adjacency matrix, 243
arcsine law, 47
arrow, 157
averaging operator, 19

background, 270
bootstrap percolation, 283

capacity, 223
Cauchy–Riemann equations, 27
chip-firing, 3, 242
circuit, 343
circuit range, 344
circular projection, 54
closure, 74
comb, 322, 366
complete
  circuit, 343
  sequence of legal moves, 157
convolution, 20
Coulomb potential, 41
coupling, 98
crossing, 374

departure order, 360
digraph, 242
Dirichlet problem, 28
discrepancy, 159
discrete ball, 74
discrete Beurling estimate, 11
discrete harmonic, 12, 74
discrete Laplacian, 3
discrete subharmonic, 74
discrete superharmonic, 12, 74
distributional solution, 45
divisible sandpile, 4, 130
DLA, *see* external DLA
Doob h-transform, 78

escape probability, 223
Eulerian digraph, 261
Eulerian digraph with sink, 261
Eulerian tour, 261
explodes, 132
explosive background, 4, 276
external diffusion-limited
    aggregation, *see* external DLA
external DLA, 10, 213

F-lattice, 351
filtration, 76
finite experiment, 358
first-glance mirror walk, 352
fold, 57
forbidden subconfiguration, 262
fundamental solution, 79

Γ-ergodic, 315
Γ-invariant, 315
global sink, 243
grain addition operator, 248
graph Laplacian, 243
gravitational potential, 41
Green function, 13, 79

half-dual lattice, 352
half-space, 99
harmonic, 19
harmonic measure, 49, 78
harmonic measure from infinity, 11, 70,
    112
harmonic modulo 1, 253
Hausdorff distance, 153
height, 243

hitting time, 76

i.i.d., 76
identity element, 258
illegal, 243
in-degree, 243
increasing, 374
increments, 76
infinite experiment, 356
inner r-neighborhood, 153
inner-outer Hausdorff metric, 153
instruction stack, 157
internal diffusion-limited
    aggregation, *see* internal DLA
internal DLA, 6, 156
internally spanned, 283

killed Green function, 305

Laplacian growth, 1
lazy random walk, 98
least action principle, 4, 132, 245
least superharmonic majorant, 136
legal, 243, 297
legal move, 157
live path, 360
lower semicontinuous, 21

Manhattan lattice, 351
martingale, 76
maximal toppling sequence, 243
mean value property, 19
meeting time, 99
mollifier, 19
Moore–Penrose pseudoinverse, 125
μ-harmonic, 65
μ-superharmonic, 65

natural filtration, 76

obstacle, 135
odometer, 3, 131, 244, 296, 343
out-degree, 243
outer boundary, 74, 340
outer r-neighborhood, 153

parallel toppling procedure, 267
particle addition operator, 332
particle-and-rotor state, 329
Poisson clock, 295
Poisson kernel, 29
potential kernel, 81

quadrature domain, 207

random walk, 76
range, 348
reachable
  rotor configuration, 336
  sandpile, 259
rectangle, 374
recurrent
  rotor walk, 332
  sandpile, 4, 256
reduced Laplacian, 252
reflecting boundary, 340
reflection coupling, 98
robust background, 4, 265
rotor, 7, 327
rotor configuration, 175
rotor mechanism, 175
rotor path, 328
rotor walk, 7, 175
rotor-router aggregation, 8, 175

sandpile, 243
sandpile group, 4, 253
self-organized criticality, 242
simple digraph, 328
simple random walk, 76
sink, 4, 243
spherical averaging operator, 26
spine, 322
stabilization, 245
stabilizes, 132, 243
stabilizing, 297
stabilizing toppling function, 265
stable
  divisible sandpile, 130
  particle-and-rotor state, 329
  sandpile, 3, 243
stopping time, 76
strongly connected, 260
subharmonic, 22
submartingale, 76
successor
  particle-and-rotor state, 329
  sandpile, 243

superharmonic, 22
supermartingale, 76

thorough, 131
threshold, 318
tooth, 322
topple, 3, 130, 243
toppling procedure, 296
toppling sequence, 130
transient
  rotor walk, 332

uniform rotor walk, 351
unstable
  divisible sandpile, 130
  sandpile, 3, 243
upper semicontinuous, 22

vertex-transitive, 315